# Algorithmic Trading

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# Notation List

| | |
|---|---|
| $x \in E$, $x \notin E$ | membership / non-membership |
| $E \subseteq F$, $\varnothing$ | subset relation / empty set |
| $X \times Y$, $f : X \to Y$ | Cartesian product / function from $X$ to $Y$ |
| $\Longrightarrow$, $\Longleftrightarrow$ | implies / if and only if |
| $:=$ | defined to be equal to |
| $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ | standard number systems |
| $\mathbb{F}$ | field of scalars, usually $\mathbb{R}$ or $\mathbb{C}$ |
| $a, b, c, \lambda$ | scalars |
| $\mathbf{u}, \mathbf{v}, \mathbf{w}, 0$ | vectors and the zero vector |
| $\mathcal{U}, \mathcal{V}, \mathcal{W}$ | vector spaces or subspaces, depending on context |
| $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ | finite ordered list, often a basis |
| $\operatorname{span}(E)$, $\dim(\mathcal{V})$ | span of $E$ / dimension of $\mathcal{V}$ |
| $\mathcal{U} + \mathcal{W}$, $\mathcal{U} \oplus \mathcal{W}$, $\mathcal{V}/\mathcal{U}$ | sum, direct sum, quotient space |
| $T : \mathcal{V} \to \mathcal{W}$, $\mathcal{L}(\mathcal{V}, \mathcal{W})$ | linear map / space of linear maps |
| $\ker(T)$, $\operatorname{range}(T)$ | kernel and range of $T$ |
| $[\mathbf{v}]_\beta$, $[T]_{\gamma \leftarrow \beta}$ | coordinate vector / matrix representation |
| $A, B, C$, $\operatorname{Mat}(m, n, \mathbb{F})$ | matrices / $m \times n$ matrices over $\mathbb{F}$ |
| $A^{-1}$, $A^T$, $A^*$, $\operatorname{tr}(A), \det(A)$ | standard matrix operations |

## Declared overrides

Symbols used in a sense other than the guide's house register, with the scope of each override.

| | |
|---|---|
| $\sigma^2$ | variance of one period's value shock, not annualised; Grossman–Miller entries |
| $\sigma$ | absolute volatility of an arithmetic Brownian midprice, in price units; Avellaneda–Stoikov |
| $p^{\pm}$, $P^{\pm}(\delta)$ | arrival and fill probabilities, not prices; static limit-order depth |
| $\lambda^a$, $\lambda^b$ | Poisson execution intensities, not price-impact slopes; Avellaneda–Stoikov |
| $\phi$, $A$, $b$, $\kappa$, $\psi$; $\nu$, $\mu$ | running and terminal inventory penalties, permanent- and temporary-impact coefficients, bid–ask spread; trading rate and others' net order flow, per unit time; stochastic-control execution |
| $\lambda_{\text{jump}}$, $\lambda_{\text{invest}}$ | Poisson arrival rates of value jumps and of investors, not price-impact slopes; latency-arbitrage arms race |
| $p$, $\alpha$ | probability of the high value, not a price; fraction of informed traders; Glosten–Milgrom half-spreads and market collapse |
| $A$, $C$; $Q_t$, $q_t$ | adverse-selection and transaction-cost spread components (full-spread amounts), not matrices; trade sign, +1 for a buy, not a quantity; spread components, spread measures, Roll's estimator and component identification |

# 1 Common toolkit

## 1.1 Time value, required returns and present value

**Definition 1.1 Required-return components: real rate, expected inflation and risk premia; the Fisher relation, exact for realised inflation, approximate in its additive form**

**Statement.** A *nominal* rate is quoted in money; a *real* rate measures the growth of purchasing power. The Fisher relation, $1 + i = (1 + R)(1 + h)$, links them, with $i$ the nominal rate, $R$ the real rate and $h$ expected inflation: a 6 percent real rate with 2 percent expected inflation requires a nominal rate of 8.12 percent [principles-of-finance-2026, sec. 7.4]. The same text also uses the subtraction "real interest rate = nominal interest rate − inflation rate" (sec. 3.4), which is the first-order approximation of the exact relation. The short-term government borrowing rate is "often referred to as the risk-free rate of interest", and lenders "charge a premium to borrowers who are higher risk" (sec. 3.4).

A nominal government yield is not just the real rate plus expected inflation. In the Federal Reserve staff's term-structure model, nominal yield = real yield + expected inflation + inflation risk premium, and real yield = expected average future real short rate + real term premium; the two premia are "extra compensations bond investors demand for bearing inflation risks and real interest rate risks, respectively" [inflation-compensation-decomposition-2019, eqs. (1)–(2)]. Inflation-indexed bonds trade at a yield above the true real yield by a liquidity premium, so their breakeven (inflation compensation) equals expected inflation plus the inflation risk premium minus that liquidity premium (eqs. (3)–(4)). A borrower that may default pays a credit spread over the risk-free rate [s7-hulloptions, chap. 4, sec. 4.1, p. 98].

**Notation.** $R$ real rate, $h$ inflation, $i$ nominal rate, all for the same horizon and compounding. Interpretation: with *realised* inflation $h$, or within one specified inflation scenario, $R = (1 + i)/(1 + h) - 1$ is an identity, the realised (ex post) real return, and $i = R + h + Rh$ exactly. Approximation: $i \approx R + h$, with error $Rh$; with continuously compounded rates, $\ln(1 + i) = \ln(1 + R) + \ln(1 + h)$ holds exactly. Interpretation, a decomposition by convention: the required yield on a bond or the required return on an equity claim = nominal risk-free yield of matching maturity + premia for the claim's risks (credit or default, liquidity, and for equity an equity risk premium). Premia stated as spreads add only as an approximation to products of growth factors.

**Conditions.** When a nominal rate is agreed, future inflation is unknown, so there are two real rates: the *ex ante* real rate expected when the loan is made, $i - E\pi$, and the *ex post* real rate actually realised, $i - \pi$; the nominal rate "can adjust only to expected inflation", so the Fisher effect is written $i = r + E\pi$ [macroeconomics-2019, chap. 5, sec. 5-3]. These are the additive forms. *Uncertain inflation* (Interpretation, derived here): deflating a nominal rate by expected inflation is not the expected real return, because the realised real return $(1+i)/(1+h)-1$ is a convex function of $h$, so its mean exceeds its value at the mean (Jensen's inequality). With a fixed nominal return of 10 percent and inflation of 0 or 20 percent with equal probability, deflating by mean inflation, 10 percent, gives 0 percent, while the mean realised real return is $\frac{1}{2}\left(10 + (1.10/1.20 - 1) \times 100\right) = 0.8333$ percent. A required nominal yield under uncertainty therefore also contains compensation for inflation risk, which is what the stochastic decomposition below separates out; the teaching calculation in Practice assumes a single known inflation rate. The premia in the Federal Reserve decomposition are model estimates from an affine no-arbitrage term-structure model of U.S. Treasury and TIPS yields, not observed prices; the 2019 note updates estimates published for 1990–2013 (Kim, Walsh and Wei, introduction and model sections). Interpretation: credit, liquidity and equity premia are compensation the market requires; a component cannot be read from one yield without a model or a matched comparison, such as a breakeven or a credit spread over the same-maturity government yield.

**Practice.** Example, original, with inflation treated as known: a real rate of 1.5 percent and inflation of 2.5 percent give a nominal rate of $(1.015)(1.025) - 1 = 4.0375$ percent; the additive approximation, 4.0 percent, misses $Rh = 0.0375$ percentage points. Adding an inflation risk premium of 0.30 points gives a nominal risk-free yield of 4.3375 percent. A corporate bond with a credit spread of 1.20 and a liquidity premium of 0.30 points then needs 5.8375 percent, and an equity claim with an equity risk premium of 5.0 points needs 9.3375 percent. The premia here are illustrative, not estimates.

**Pitfalls.** The breakeven inflation rate is not the expected inflation rate; it also contains the inflation risk premium and, with the opposite sign, the indexed bond's liquidity premium (eq. (4)). Subtracting inflation from a nominal rate is an approximation; at high inflation the error is material: at 20 percent inflation and a 10 percent nominal rate, the exact real rate is −8.33 percent, the approximation −10 percent. Do not mix compounding conventions when adding components.

**See also.** return measures (real return); inflation-linked bonds and breakevens; forward rates (term-structure theories); present value of instrument cash flows.

**Method 1.2 Present value of bond and equity cash flows, and the implied yield, required return and growth rate that equate price to present value**

**Statement.** A bond's price is the present value of its coupons plus the present value of its face value, discounted at the yield to maturity per period; for semiannual coupons the annual coupon and the yield, quoted as a nominal annual rate with semiannual compounding, are halved and the number of periods doubled. The yield cannot be isolated algebraically and is found with a financial calculator or spreadsheet; a bond priced above par has a coupon above its yield, one below par a coupon below its yield [principles-of-finance-2026, sec. 10.2]. A bond can equally be priced by discounting each cash flow at the zero (spot) rate for its own maturity; its yield is the single rate that reproduces that price [s7-hulloptions, chap. 4, secs. 4.5–4.6, pp. 104–105].

A share is valued by the present value of its expected dividends. With a constant dividend $D$ forever and required return $r$, the value is the perpetuity $D/r$; with dividends growing at a constant rate $g$, it is $D_0(1+g)/(r-g) = D_1/(r-g)$, which "requires the return rate $r$ to be greater than the growth rate $g$" [principles-of-finance-2026, sec. 11.2]. The same equation, given a price, can be solved for the growth it implies (sec. 11.2, Example 2).

**Notation.** $C$ coupon per period, $F$ face value, $y$ yield per period, $N$ periods, $z_t$ zero rate for maturity $t$, annual compounding and cash flows at period ends:

$$P = \sum_{t=1}^{N} \frac{C}{(1+y)^t} + \frac{F}{(1+y)^N} = C\,\frac{1-(1+y)^{-N}}{y} + \frac{F}{(1+y)^N}, \qquad P = \sum_{t=1}^{N} \frac{CF_t}{(1+z_t)^t}.$$

Quotation (Interpretation): halving applies to a yield quoted as a nominal annual rate compounded semiannually, $y_{\text{nom}}$, whose half-year rate is $y_{\text{nom}}/2$. A yield quoted as an *effective* annual rate $y_{\text{eff}}$ has half-year rate $(1+y_{\text{eff}})^{1/2} - 1$; the two describe the same bond when $(1+y_{\text{nom}}/2)^2 = 1+y_{\text{eff}}$ (compounding conversions). Equity: $V_0 = D/r$ (constant dividend, as for non-callable fixed-rate preferred stock); $V_0 = D_1/(r-g)$, $r > g$; two-stage growth: $V_0 = \sum_{t=1}^{n} D_t(1+r)^{-t} + (1+r)^{-n} D_{n+1}/(r-g_L)$, the last term the present value of the terminal value at $n$. Interpretation, derived by rearranging: implied required return $r = D_1/P_0 + g$, dividend yield plus growth; implied growth $g = (rP_0 - D_0)/(P_0 + D_0)$.

**Conditions.** The bond formula values the cash flows on a coupon date; pricing between coupon dates needs accrued interest and day counts (quotation conventions). A single yield is a summary of the zero curve, not a rate earned on each cash flow, and realising it requires reinvestment at

that yield. The dividend models are valuation identities given their inputs: the expected dividends, the constant $g < r$ forever, and the required return, which itself contains the premia of required-return components. Value is very sensitive to $r - g$. Implied returns and growth are what the price requires under the model, not forecasts.

**Practice.** Examples, original, annual compounding. A zero-coupon bond paying 1,000 in three years at 4 percent is worth 888.9964. A three-year 5 percent annual-coupon bond, face 100, is worth 102.7751 at a 4 percent yield (premium bond, coupon above yield); at zero rates of 3.0, 3.8 and 4.1 percent it is worth 102.5709, and its yield, 4.0723 percent, lies between the zero rates and closest to the three-year rate. Priced at 98.00 the same bond yields 5.7447 percent, found by bisection.

A preferred share paying a fixed 6 a year at a required 7.5 percent is worth 80.00. A share with $D_0 = 2.00$, $g = 4$ percent and $r = 9$ percent is worth $2.08/0.05 = 41.60$, and at that price $2.08/41.60 + 0.04 = 9$ percent. With 10 percent growth for two years and 4 percent after, the dividends are 2.20 and 2.42, the terminal value at year 2 is $2.42 \times 1.04/0.05 = 50.336$, and the value is 46.4220. At a price of 52 and $r = 9$ percent, the implied constant growth is $(0.09 \times 52 - 2)/54 = 4.9630$ percent. Each figure was also obtained by summing the cash flows directly.

**Pitfalls.** The constant-growth formula fails when $g \geq r$. Use the dividend expected next period, $D_1$, not the last one paid. Match the rate to the cash-flow period and compounding: cash flows of 5 in six months and 105 in a year are worth 100.2219 at a 10 percent *effective* annual yield (half-year rate 4.8809 percent), the same as at a nominal 9.7618 percent compounded semiannually; halving the effective 10 percent instead gives 100.00, a mispricing. A higher yield means a lower price for given cash flows.

**See also.** interest-rate measures (compounding, annuities, zero and par yields); cash-flow additivity; return measures (the IRR).

#### Result 1.3 Cash-flow additivity and no arbitrage: the value of a sum of cash flows is the sum of their values, applied to implied forward rates, forward exchange rates and a replicated option

**Statement.** Two portfolios with identical payoffs in every state must have identical values today; "if this were not the case, an arbitrageur could buy the less expensive portfolio and sell the more expensive one" [s7-hulloptions, chap. 11, sec. 11.4, p. 255]. Applied to cash flows dated in the future, value is additive: a stream is worth the sum of its dated cash flows, each discounted at the rate for its date, and a position assembled from simpler instruments is worth the sum of their values. Three consequences follow.

- *Forward rates.* Investing for $T_2$ years, or for $T_1$ years and then at the forward rate from $T_1$ to $T_2$, must give the same amount; a borrower and lender at the zero rates can lock in the forward rate today: borrowing for one year and investing for two creates an outflow at year 1 and an inflow at year 2, a loan made between those dates at the forward rate, and borrowing for the longer and investing for the shorter term locks in forward borrowing [s7-hulloptions, chap. 4, sec. 4.8, p. 110]. Rolling short borrowings over at whatever rates prevail later is not a lock-in; it is a bet that rates will differ from the forwards (Business Snapshot 4.1, p. 111).
- *Forward exchange rates.* Borrowing domestic currency, converting at spot, investing the foreign currency and selling the proceeds forward must cost nothing at maturity, which gives covered interest parity, $F_0 = S_0 e^{(r - r_f)T}$ with continuous compounding (chap. 5, sec. 5.10, pp. 137–140; cost of carry).
- *Options.* When the underlying can move only to two prices, a position of $\Delta$ shares and a riskless bond can reproduce an option's payoff in both states, so the option is worth that position (chap. 13, sec. 13.1, pp. 288–290); and a call plus a zero-coupon bond paying $K$ has the same payoff as a put plus the share, so $c + Ke^{-rT} = p + S_0$ (sec. 11.4, eq. (11.6), p. 256).

**Notation.** Interpretation, the discrete-compounding forms used in the examples. Zero rates $z_1$, $z_2$ (annual): $(1 + z_2)^2 = (1 + z_1)(1 + f_{1,1})$, so $f_{1,1} = (1 + z_2)^2/(1 + z_1) - 1$. Spot $S_0$ and forward $F_0$ in domestic currency per unit of foreign currency, domestic and foreign rates $i_d$, $i_f$ for the same period: $F_0 = S_0(1 + i_d)/(1 + i_f)$. One period, share $S_0$ moving to $S_u$ or $S_d$, option payoffs $f_u$, $f_d$, rate $r$ for the period: $\Delta = (f_u - f_d)/(S_u - S_d)$, bond holding $B = (f_d - \Delta S_d)/(1 + r)$ in present value, option value $\Delta S_0 + B$; parity $c + K/(1 + r) = p + S_0$.

**Conditions.** No arbitrage among participants who can trade the component instruments without transaction costs, borrow and lend at the stated rates and, for parity, hold European options with the same strike and expiry on a share that pays no dividend before expiry (Hull, secs. 5.3 and 11.4). Additivity values claims whose cash flows are fixed or replicable; it does not say what an unreplicable payoff is worth. The two-state option model is a teaching device: real prices take many values, and the replication must be rebalanced (Hull, chap. 13). Forward rates and forwards are prices that can be locked in, not forecasts.

**Practice.** Examples, original. *Forward rate.* With annual zero rates of 3.0 and 3.8 percent, the one-year rate one year forward is $1.038^2/1.03 - 1 = 4.6062$ percent. Buying a two-year zero that pays 1,000 for 928.1225 and borrowing that amount for one year at 3.0 percent gives dated cash flows of 0 now, −955.9662 at year 1 (repaying the loan) and +1,000 at year 2: a loan of 955.9662 from year 1 to year 2 at $1{,}000/955.9662 - 1 = 4.6062$ percent, fixed today. The reverse trades, selling the two-year zero short and lending the 928.1225 for one year, give +955.9662 at year 1 and −1,000 at year 2, borrowing locked at the same rate. Repaying the year-1 debt by a new one-year loan is a different position: its rate is unknown today, and at 8 percent it would cost 1,032.4434 at year 2.

*Forward exchange rate.* At 1.1000 USD per EUR, a one-year USD rate of 4.0 percent and EUR rate of 2.5 percent give $1.1000 \times 1.04/1.025 = 1.116098$ USD per EUR, a forward premium of 160.98 pips. Borrowing USD 1,100, buying EUR 1,000, depositing to EUR 1,025 and selling that forward returns USD 1,144, the amount owed.

*Option.* A share at 50 moves to 60 or 40, and the period rate is 5 percent. A call struck at 50 pays 10 or 0, so $\Delta = 0.5$ and the bond holding is $-20/1.05 = -19.0476$: buy half a share and borrow 19.0476, at a value of 5.9524. The position pays 10 or 0, as the call does. Parity gives the put struck at 50 as $5.9524 + 50/1.05 - 50 = 3.5714$, which a separate replication of the put confirms.

**Pitfalls.** Additivity requires the same date and the same currency: amounts must be discounted to a common date before they are added. A forward exchange rate quoted the other way round (foreign per domestic) inverts the formula. The replicating $\Delta$ and bond change when the share price or time changes; a static position replicates only the one-period payoff.

**See also.** forward rates and FRAs; cost-of-carry forward price; synthetic short through options (parity with dividends); present value of instrument cash flows.

## 1.2 Descriptive statistics of returns

The entries in this section use one original series of eight monthly returns, in percent: asset A 2.0, −1.5, 3.0, 0.5, −4.0, 1.5, 2.5, −0.5 and asset B 1.0, −0.5, 1.5, 0.0, −2.5, 1.0, 2.0, 0.5. They are illustrative numbers, not data on any security.

#### Definition 1.4 Location, dispersion and quantiles of a return sample: mean, median, mode; range, absolute deviations, variance and standard deviation with population and sample denominators; percentiles and the interquartile range

**Statement.** The mean is the sum of the data divided by their number; the median is the middle value of the sorted data, or the average of the two middle values when $N$ is even; the mode is the most frequent value. With extreme values in the tails the median is a better estimate of location than the mean, and for skewed data the mean is pulled towards the long tail [statistical-methods-handbook, sec. 1.3.5.1]. Dispersion is measured by the range, the average absolute deviation from the mean (divide by $N$), the median absolute deviation from the median, the interquartile range (75th minus 25th percentile) and the variance and standard deviation, which the handbook defines with denominator $N - 1$ (sec. 1.3.5.6). For a random sample, independent observations from one distribution with finite variance $\sigma^2$, dividing the squared deviations by

$n$ gives a biased estimator of $\sigma^2$, with bias $-\sigma^2/n$; dividing by $n-1$ gives the unbiased sample variance, though its square root is still a biased estimator of $\sigma$ [probability-statistics-random-processes, sec. 8.2.2].

**Notation.** Data $x_1, \ldots, x_N$, mean $\bar{x}$. Population (descriptive, the data are the whole set): $\sigma^2 = \frac{1}{N}\sum(x_i - \bar{x})^2$. Sample (the data estimate a larger population): $s^2 = \frac{1}{N-1}\sum(x_i - \bar{x})^2$. Standard deviations are the square roots, in the units of the data. *Quantile convention used here*: the $p$th percentile sets $p(N+1) = k+d$, $k$ an integer and $0 \le d < 1$, and takes $Y_{(k)} + d\,(Y_{(k+1)} - Y_{(k)})$ from the ordered data, with $p \le 1/(N+1)$ giving the minimum and $p \ge N/(N+1)$ the maximum; other interpolations are in common use and give different answers [statistical-methods-handbook, sec. 7.2.6.2].

**Conditions.** Descriptive statistics summarise the observed data, whatever produced them; either denominator is then a choice of definition. Read as *estimates* of a distribution's moments, they need a sampling model: the unbiasedness of $s^2$ above uses independent, identically distributed observations with finite variance. A common marginal distribution is not enough. If two observations are always equal, $X_1 = X_2 = Z$ with $Z$ equally likely $-1$ or $+1$, the marginal variance is 1 but every sample of two has sample variance 0 (Interpretation). Monthly returns are dependent and their distribution shifts over time, so a sample of eight months says little about the population. Which denominator and which quantile rule a calculator or package uses must be checked before comparing figures.

**Practice.** Example (asset A). Mean 0.4375 percent; sorted data $-4.0, -1.5, -0.5, 0.5, 1.5, 2.0, 2.5, 3.0$, so the median is $(0.5+1.5)/2 = 1.0$ percent, and there is no mode. Range 7.0 points; average absolute deviation 1.8281; median absolute deviation 1.5. Variance 4.8398 with $N$ and 5.5313 with $N-1$; standard deviations 2.2000 and 2.3519 percent. Quartiles by the $p(N+1)$ rule are $-1.25$ and 2.375 (interquartile range 3.625); the interpolation rule used by many spreadsheets gives $-0.75$ and 2.125. The 10th percentile is the minimum, $-4.0$, under the first rule and $-2.25$ under the second. Interpretation: the mean lies below the median because one large loss pulls it down.

**Pitfalls.** The sample standard deviation is not unbiased for $\sigma$ even when $s^2$ is unbiased for $\sigma^2$; with positively dependent observations $s^2$ itself is biased downward. Annualising a monthly standard deviation by $\sqrt{12}$ assumes uncorrelated, identically distributed returns. With few observations, extreme percentiles are the sample extremes and are unstable.

**See also.** skewness and kurtosis; return measures (arithmetic and geometric means); moments of random variables.

### Definition 1.5 Skewness and kurtosis of a return sample: moment ratios, bias-adjusted versions, and kurtosis against excess kurtosis

**Statement.** The Fisher–Pearson coefficient of skewness is $g_1 = \frac{\sum(Y_i-\bar{Y})^3/N}{s^3}$ and the kurtosis $\frac{\sum(Y_i-\bar{Y})^4/N}{s^4}$, where in both "the $s$ is computed with $N$ in the denominator rather than $N-1$". Many programs instead compute the adjusted coefficient $G_1 = \frac{\sqrt{N(N-1)}}{N-2} g_1$. The kurtosis of a normal distribution is three; *excess* kurtosis subtracts 3 so that the normal has zero, and "which definition of kurtosis is used is a matter of convention" [statistical-methods-handbook, sec. 1.3.5.11].

**Notation.** Central moments $m_k = \frac{1}{N}\sum(x_i - \bar{x})^k$; $g_1 = m_3/m_2^{3/2}$; kurtosis $b_2 = m_4/m_2^2$; excess kurtosis $b_2 - 3$. A common bias-adjusted excess kurtosis (the one returned by scipy's `kurtosis(bias=False)`) is $G_2 = \frac{N-1}{(N-2)(N-3)}\left((N+1)(b_2-3)+6\right)$.

**Conditions.** Kurtosis "is a measure of whether the data are heavy-tailed or light-tailed relative to a normal distribution": data with high kurtosis "tend to have heavy tails, or outliers" (sec. 1.3.5.11). Interpretation: negative skewness means a longer left tail. Both statistics rest on third and fourth powers, so a few extreme observations can dominate them, and their sampling error is large in short samples. High kurtosis does not by itself say where the rest of the mass lies: probabilities 0.495 at $\pm 1$ and 0.005 at $\pm 10$ give variance 1.99 and kurtosis 25.50, yet no mass within half a standard deviation of the mean, where a normal of the same variance has 0.383. A finite sample always has a finite variance and kurtosis, which are valid descriptions of those data; what fails, when the population moment does not exist (distributions), is reading them as estimates of it.

**Practice.** Example (asset A): $g_1 = -0.7615$, $G_1 = -0.9497$; kurtosis 2.5138, excess kurtosis $-0.4862$, and the bias-adjusted excess kurtosis $+0.3789$. The sign of the excess kurtosis depends on the convention: with eight observations the data cannot tell a thin-tailed from a fat-tailed distribution. The negative skewness agrees with the mean lying below the median.

**Pitfalls.** State whether a reported kurtosis is raw or excess; "kurtosis of 3" and "excess kurtosis of 3" describe very different distributions. Monthly samples of a few years rarely pin down tail shape; heavy tails of returns are a documented feature of longer and higher-frequency data (stylised facts).

**See also.** location and dispersion; distributions used in finance.

### Definition 1.6 Covariance and correlation of two return series: definitions, units, denominators and limitations

**Statement.** The covariance of two random variables is $\mathrm{Cov}(X,Y) = E[(X-EX)(Y-EY)] = E[XY] - EX\,EY$, and the correlation coefficient $\rho = \mathrm{Cov}(X,Y)/(\sigma_X\sigma_Y)$ always lies between $-1$ and 1. Independent variables have zero covariance, but zero covariance does not imply independence. Correlation is unchanged by positive rescaling and shifting, $\rho(aX+b, cY+d) = \rho(X,Y)$ for $a, c > 0$, while covariance scales with the variables; and $\mathrm{Var}(aX+bY) = a^2\mathrm{Var}X + b^2\mathrm{Var}Y + 2ab\,\mathrm{Cov}(X,Y)$ [probability-statistics-random-processes, sec. 5.3.1].

**Notation.** Sample covariance $\frac{1}{N-1}\sum(a_i-\bar{a})(b_i-\bar{b})$, or with $N$ for the descriptive version; correlation $r = \frac{\sum(a_i-\bar{a})(b_i-\bar{b})}{\sqrt{\sum(a_i-\bar{a})^2\sum(b_i-\bar{b})^2}}$, which is the same with either denominator provided all three terms use it. Covariance has the product of the two units (percent squared for returns in percent); correlation has none.

**Conditions.** Interpretation: correlation measures *linear* association only; a perfect nonlinear relation can have zero correlation, and a few extreme observations can create or hide it. Correlations estimated over one period need not hold in another; in stress, the dependence of tails can differ from that in normal months. Large covariance matrices need special estimators (covariance estimation).

**Practice.** Example: the covariance of A and B is 2.7734 (percent squared) with $N$ and 3.1696 with $N-1$; in decimal returns it is 0.000317. The correlation is 0.9573 either way. The series $x = -2, \ldots, 2$ and $y = x^2$ are perfectly dependent yet have correlation zero.

**Pitfalls.** Mixing denominators (a covariance with $N-1$ and standard deviations with $N$) gives a correlation that is wrong by the factor $N/(N-1)$. Covariances must be compared in the same units. High correlation of two assets is not evidence that one causes the other.

**See also.** conditional moments (covariance of random variables); noise in large correlation matrices.

### Definition 1.7 Downside and relative dispersion: semi-deviation about a target, its denominators, and the coefficient of variation and its limits

**Statement.** The semideviation with respect to a chosen benchmark $B$ is $\mathrm{SSD}_B = \left\{\frac{1}{T}\sum_{t=1}^{T}\min(R_t - B, 0)^2\right\}^{1/2}$, with $T$ the number of observations: "returns below $B$ add to the calculation of the semideviation figure, but returns above it do not" [downside-risk-volatility-2026, eq. (1), p. 2]. The coefficient of variation (CV) is "the ratio of the standard deviation to the mean"; it "should typically only be used for data measured on a ratio scale", with "a meaningful zero", and "if the mean value is near zero, the coefficient of variation is sensitive to small changes in the mean" [coefficient-of-variation-reference-2017, "Coefficient of Variation"].

**Notation.** Target $B$: zero, the mean, a risk-free rate or a required return; denominator $T$ (every observation, as in Estrada), $T-1$, or the number of shortfalls only. All three conventions are in use, so the target and the denominator must be stated with the figure. CV $= s/\bar{x}$, here with the sample standard deviation.

**Conditions.** Interpretation: with denominator $N$, a semi-deviation about the mean equals $\sigma/\sqrt{2}$ for data symmetric about their mean, so a ratio to the standard deviation above $1/\sqrt{2}$ points to a heavier downside (asset A: $1.7437/2.2000 = 0.79$). Dividing by the number of shortfalls

measures their typical size, not their frequency, and is not comparable with the other two. Returns are not on a ratio scale: their mean can be zero or negative, so the CV of returns can be negative, unbounded or reversed in ranking. The CV is meaningful for positive quantities such as prices, volumes or volatility estimates.

**Practice.** Example (asset A; three of eight returns are negative): about a target of zero, the semi-deviation is 1.5207 percent dividing by $N$, 1.6257 dividing by $N-1$ and 2.4833 dividing by the three shortfalls; about the mean, 0.4375 percent, it is 1.7437 dividing by $N$. The CV is $2.3519/0.4375 = 5.38$ for A and 3.75 for B. Subtracting 0.5 points from every return of A leaves its standard deviation unchanged but turns the mean to $-0.0625$ percent and the CV to $-37.6$, which ranks nothing.

**Pitfalls.** Quoting a semi-deviation without its target and denominator. Ranking assets by CV when any mean is near zero or negative. Treating the inverse of the CV as a Sharpe ratio: it uses the total, not the excess, return.

**See also.** location and dispersion; relative performance evaluation.

## 1.3 Probability and distributions in finance

**Definition 1.8 Moments of random variables: unconditional and conditional expectation, variance and covariance, and the laws of total expectation, variance and covariance**

**Statement.** For discrete random variables the conditional expectation $E[X \mid Y=y] = \sum_i x_i P_{X|Y}(x_i \mid y)$ is a function of $y$, and $E[X \mid Y]$ is the random variable that takes that value when $Y=y$; the conditional variance is $\operatorname{Var}(X \mid Y=y) = E[X^2 \mid Y=y] - E[X \mid Y=y]^2$. The law of iterated expectations gives $E[X] = E\big[E[X \mid Y]\big]$ and the law of total variance $\operatorname{Var}(X) = E[\operatorname{Var}(X \mid Y)] + \operatorname{Var}(E[X \mid Y])$, so conditioning reduces variance on average [probability-statistics-random-processes, sec. 5.1.5].

**Notation.** Unconditional moments of a random variable with probabilities $p_j$: $E[X] = \sum p_j x_j$, $\operatorname{Var}X = \sum p_j (x_j - EX)^2 = E[X^2] - (EX)^2$, $\operatorname{Cov}(X,Z) = \sum p_j (x_j - EX)(z_j - EZ)$. Interpretation, derived from the two laws above: the law of total covariance, $\operatorname{Cov}(X,Z) = E[\operatorname{Cov}(X,Z \mid Y)] + \operatorname{Cov}(E[X \mid Y], E[Z \mid Y])$. These are *population* moments of a model; sample statistics (location and dispersion) estimate them from data.

**Conditions.** The probabilities and conditional distributions are model inputs; the decomposition is exact for the model, not evidence about markets. Expected values are probability-weighted averages, not the most likely outcome.

**Practice.** Example, original probability table (returns in percent):

| State $Y$ | Probability | Equity $X$ | Bond $Z$ |
|---|---|---|---|
| Boom | 0.125 | 12 | 0 |
| Boom | 0.125 | 8 | 2 |
| Normal | 0.25 | 6 | 2 |
| Normal | 0.25 | 4 | 3 |
| Recession | 0.125 | −6 | 6 |
| Recession | 0.125 | −10 | 8 |

Directly: $E[X] = 3.0$, $E[Z] = 3.25$, $\operatorname{Var}X = 47.0$, $\operatorname{Var}Z = 5.6875$, $\operatorname{Cov}(X,Z) = -16.25$, correlation $-0.994$. By state, $E[X \mid Y]$ is 10, 5 and $-8$ with probabilities 0.25, 0.5 and 0.25, and $\operatorname{Var}(X \mid Y)$ is 4, 1 and 4. Then $E[E[X \mid Y]] = 3.0$ and $E[\operatorname{Var}(X \mid Y)] + \operatorname{Var}(E[X \mid Y]) = 2.5 + 44.5 = 47.0$: almost all the equity risk here comes from the state of the economy, not from variation within a state. Likewise $-1.25 + (-15.0) = -16.25$ for the covariance.

**Pitfalls.** A conditional variance is not the unconditional variance with fewer data; $\operatorname{Var}(X \mid Y=y)$ can exceed $\operatorname{Var}X$ in some states, though not on average. Conditioning on information unavailable at decision time (look-ahead) gives moments no investor could have used.

**See also.** covariance and correlation; Bayesian updating.

**Definition 1.9 Distributions used in finance: normal, lognormal, Student $t$, binomial and uniform, their moments and when the moments exist**

**Statement.** *Normal* with location $\mu$ and scale $\sigma$: mean $\mu$, standard deviation $\sigma$, skewness 0, kurtosis 3 [statistical-methods-handbook, sec. 1.3.6.6.1]. *Lognormal* with shape $\sigma$ (the standard deviation of the log) and scale $m$ (the median): mean $m\,e^{\sigma^2/2}$, mode $m\,e^{-\sigma^2}$, standard deviation $m\sqrt{e^{\sigma^2}(e^{\sigma^2}-1)}$, skewness $(w+2)\sqrt{w-1}$ and kurtosis $w^4+2w^3+3w^2-3$ with $w = e^{\sigma^2}$ (sec. 1.3.6.6.9, stated for $m=1$). *Student t* with $\nu$ degrees of freedom: mean 0, undefined for $\nu = 1$; standard deviation $\sqrt{\nu/(\nu-2)}$, undefined for $\nu = 1$ or 2; skewness 0 for $\nu > 3$; kurtosis $3(\nu-2)/(\nu-4)$, undefined for $\nu \le 4$ (sec. 1.3.6.6.4, stated for integer $\nu$). *Binomial* with $n$ trials and success probability $p$: $P(x) = \binom{n}{x} p^x (1-p)^{n-x}$, mean $np$, standard deviation $\sqrt{np(1-p)}$, skewness $(1-2p)/\sqrt{np(1-p)}$, kurtosis $3 - 6/n + 1/(np(1-p))$ (sec. 1.3.6.6.18); with $n=1$ it is the Bernoulli distribution. *Uniform* on $[A,B]$: density $1/(B-A)$, mean $(A+B)/2$, standard deviation $\sqrt{(B-A)^2/12}$, skewness 0, kurtosis 9/5 (sec. 1.3.6.6.2).

In finance: if the continuously compounded return is normal, the price is lognormal. With expected return (drift) $\mu$ and volatility $\sigma$ a year, $\ln S_T$ is normal with mean $\ln S_0 + (\mu - \sigma^2/2)T$ and variance $\sigma^2 T$, $E(S_T) = S_0 e^{\mu T}$ and $\operatorname{var}(S_T) = S_0^2 e^{2\mu T}(e^{\sigma^2 T} - 1)$; a lognormal variable lies between zero and infinity and its mean, median and mode differ [s7-hulloptions, chap. 15, sec. 15.1, pp. 339–340]. Short-horizon returns have heavier tails than the normal, often fitted by a Student $t$ with a tail parameter near 3 (stylised facts).

**Notation.** Interpretation: kurtosis here is raw; excess kurtosis subtracts 3, giving $6/(\nu-4)$ for the $t$. A $t$ variable scaled by $\sqrt{(\nu-2)/\nu}$ has unit variance for $\nu > 2$, the form used to give a fat-tailed shock the same variance as a standard normal. Moment existence: for real $\nu > 0$ the $k$th moment of the $t$ exists only for $k < \nu$, so the mean needs $\nu > 1$, a finite variance $\nu > 2$, skewness $\nu > 3$ and kurtosis $\nu > 4$; for $1 < \nu \le 2$ the variance is infinite, and for $\nu \le 1$ the mean is undefined. All moments of the normal, lognormal, binomial and uniform exist. In the lognormal price model the drift $\mu$ is the expected rate of price growth; the mean log return is $\mu - \sigma^2/2$, and the mean price adds back half the log variance: $E(S_T) = S_0 \exp\big((\mu - \sigma^2/2)T + \sigma^2 T/2\big)$. The chi-square and $F$ distributions are treated with hypothesis tests.

**Conditions.** A distribution is a modelling assumption. Sample moments of any finite set of returns are finite and remain valid descriptions of those data; they do not show that the population moments exist, and where those do not exist the sample figures estimate nothing. The lognormal price model assumes constant $\mu$ and $\sigma$ (Hull, chap. 15).

**Practice.** Examples, original. A monthly return that is normal with mean 0.8 percent and standard deviation 4.0 percent falls below $-5$ percent with probability 7.35 percent; its 5th percentile is $0.8 - 1.6449 \times 4.0 = -5.78$ percent. A price of 100 with log return normal, mean 0.05 and standard deviation 0.20 over a year, (so a drift of $0.05 + 0.02 = 0.07$) has median $100e^{0.05} = 105.13$, mean $100e^{0.05+0.02} = 107.25$ and mode $100e^{0.05-0.04} = 101.01$, standard deviation 21.67, skewness 0.614 and kurtosis 3.678. A $t$ with 5 degrees of freedom has variance 5/3 and excess kurtosis 6; with 10, variance 1.25 and excess kurtosis 1; with 3, a finite variance (3) but no kurtosis; with 2, no variance: its second moment truncated at $\pm 10$, $\pm 100$ and $\pm 1{,}000$ is 3.33, 7.90 and 12.51 and keeps growing. Twelve independent months with an up-probability of 0.55 give a binomial number of up months with mean 6.6, variance 2.97, skewness $-0.058$ and kurtosis 2.837. A return equally likely anywhere between $-3$ and $+5$ percent has

mean 1.0 percent and standard deviation $8/\sqrt{12} = 2.31$ percent, with thinner tails than a normal (excess kurtosis $-1.2$). Each moment was also obtained by numerical integration or enumeration.

**Pitfalls.** Applying a normal quantile to returns with heavy tails understates extreme losses. Reading the median of a lognormal price as its mean. Treating a sample kurtosis as an estimate of a $t$ model's kurtosis when $\nu \le 4$, or a sample variance as an estimate of its variance when $\nu \le 2$: the figures describe the sample, but there is no finite population value to estimate. Confusing the drift with the mean log return.

**See also.** skewness and kurtosis; Black–Scholes–Merton (lognormal prices); VaR and ES estimation.

#### Method 1.10 Bayesian updating of a discrete prior: prior, likelihood, normalising constant, posterior and predictive probability

**Statement.** For a partition $B_1, B_2, \ldots$ of the possibilities and observed data $A$, Bayes' rule gives $P(B_j \mid A) = \frac{P(A|B_j)P(B_j)}{\sum_i P(A|B_i)P(B_i)}$, the denominator being the probability of the data by the law of total probability. A rare condition with an accurate test still leaves a low posterior probability: with prevalence 1 in 10,000, a 99 percent true-positive and 2 percent false-positive rate, a positive result raises the probability only to about 0.0049 [probability-statistics-random-processes, sec. 1.4.3].

**Notation.** Hypotheses $h_j$ with prior $\pi_j$; likelihood $L_j = P(\text{data} \mid h_j)$; evidence $E = \sum_j \pi_j L_j$; posterior $\pi'_j = \pi_j L_j / E$, which sums to one. Predictive probability of the next observation: $\sum_j \pi'_j P(\text{next} \mid h_j)$. Interpretation: with observations independent *conditional on* the hypothesis, updating one at a time and normalising each time gives the same posterior as using all the data at once, in any order. They are not independent unconditionally: under the prior below, two right calls in a row have probability $0.25 \times 0.45^2 + 0.5 \times 0.5^2 + 0.25 \times 0.55^2 = 0.25125$, more than $0.5 \times 0.5 = 0.25$, because each success raises the probability of high-skill hypotheses.

**Conditions.** The posterior is only as good as the prior and the likelihood model; here the months are assumed independent given the up-probability, which is constant, and the true value is assumed to be one of the listed hypotheses. Interpretation: a prior concentrated on few values cannot learn values it excludes.

**Practice.** Example, original and synthetic, not evidence about any strategy. The monthly probability that a signal is right is 0.45, 0.50 or 0.55, with prior probabilities 0.25, 0.50 and 0.25. In 20 independent months it is right 14 times. The binomial likelihoods $\binom{20}{14} h^{14}(1-h)^6$ are 0.01498, 0.03696 and 0.07460; weighted by the priors they sum to the evidence 0.04088; the posterior is 0.0916, 0.4521 and 0.4562. The predictive probability of a right call next month rises from 0.5000 to 0.5182, well below the sample frequency of 0.70, because the prior allows nothing above 0.55. Updating month by month gives the same posterior.

**Pitfalls.** Omitting the denominator (the posterior then does not sum to one); confusing $P(\text{data} \mid h)$ with $P(h \mid \text{data})$; treating the sample frequency as the posterior. Choosing the prior after seeing the data.

**See also.** conditional moments; skill against luck (a frequentist test of the same question).

## 1.4 Sampling and statistical inference

The entries in this section use one invented sample: twelve monthly active returns of a manager against a benchmark, in percent, 1.2, −0.4, 0.8, 2.1, −1.5, 0.6, 1.9, −0.2, 0.3, 1.4, −0.9, 1.1, with mean 0.5333, sample standard deviation 1.1106 and standard error $1.1106/\sqrt{12} = 0.3206$. The comparison of variances uses a second, independent sample: fifteen values drawn by computer, independently of the first sample, from a normal distribution with mean 0 and standard deviation 0.5 and rounded to one decimal, −0.2, 0.1, 0.3, 0.4, 0.8, 0.2, 0.4, 0.4, 0.3, 0.1, −0.7, 0.0, −0.3, 1.5, 0.3, with sample standard deviation 0.4968. They illustrate the mechanics and are not evidence about any manager.

#### Method 1.11 Sampling, the central limit theorem and confidence intervals: estimand, sampling distribution, exact $t$ and chi-square intervals, normal approximations and the repeated-sampling meaning

**Statement.** If $X_1, \ldots, X_n$ are independent and identically distributed with finite mean $\mu$ and variance $0 < \sigma^2 < \infty$, the standardised mean $(\bar{X} - \mu)/(\sigma/\sqrt{n})$ converges in distribution to a standard normal (the central limit theorem); for finite $n$ the normal is an approximation whose quality depends on the distribution, with $n \ge 30$ a common rule of thumb [probability-statistics-random-processes, sec. 7.1.2]. For a normal sample with unknown variance, $T = (\bar{X} - \mu)/(S/\sqrt{n})$ has a $t$ distribution with $n-1$ degrees of freedom and $(n-1)S^2/\sigma^2$ a chi-square distribution with $n-1$ degrees of freedom; they give the intervals $\bar{X} \pm t_{\alpha/2,n-1} S/\sqrt{n}$ for $\mu$ and $\left[(n-1)S^2/\chi^2_{\alpha/2,n-1}, (n-1)S^2/\chi^2_{1-\alpha/2,n-1}\right]$ for $\sigma^2$, valid for any $n$ (sec. 8.3.3). A 95 percent interval means that "in the long run about 95 % of these intervals would contain the true mean"; it "does not mean that there is a 95 % probability that the interval contains the true mean" [statistical-methods-handbook, sec. 1.3.5.2].

Samples can be drawn in several ways: a *simple random sample* gives every sample of the same size "an equal chance of being selected"; a *stratified* sample divides the population into strata and takes "a proportionate number from each stratum"; a *cluster* sample randomly selects some groups; a *systematic* sample takes every $n$th item from a random start; *convenience* sampling uses "results that are readily available" and invites sampling bias, where "some members of the population are not as likely to be chosen as others". Sampling itself causes sampling error [introductory-statistics-2023, sec. 1.2]. In finance, a time series of returns is not a random sample from a population of periods, and a sample of surviving funds is a convenience sample (Interpretation).

**Notation.** *Estimand*: the population quantity sought, here the manager's mean monthly active return $\mu$. *Sampling design*: how the data arose; the formulas above assume a random sample, independent draws from one distribution. $\bar{X}$ the estimator; $S/\sqrt{n}$ its estimated standard error; $\chi^2_{q,k}$ the upper-$q$ point in the source's notation. Interpretation: *exact* means the stated coverage holds for every $n$ under the model (the $t$ and chi-square intervals for normal i.i.d. data); *approximate* means it holds only as $n$ grows (the normal interval $\bar{X} \pm 1.96 S/\sqrt{n}$ for non-normal data).

**Conditions.** Monthly returns are rarely independent or identically distributed: volatility clusters and means drift. With positive serial correlation $S/\sqrt{n}$ understates the standard error of the mean, and the interval is too narrow; time-series standard errors (Sharpe-ratio standard errors) or a block bootstrap (bootstrap) are then needed. The chi-square interval for a variance depends on normality much more than the $t$ interval for a mean does (Interpretation). The interval is about the estimand of the design: a sample of one manager's months says nothing about other managers.

**Practice.** Example. The exact 95 percent $t$ interval for $\mu$ is $0.5333 \pm 2.2010 \times 0.3206$, from −0.17 to 1.24 percent a month; the normal-approximation interval, $\pm 1.96$ standard errors, is −0.10 to 1.16 and too narrow for twelve observations. The 95 percent interval for the variance is 0.62 to 3.56, a standard deviation of 0.79 to 1.89 percent. In 40,000 simulated normal samples of twelve, 95.1 percent of the $t$ intervals covered the true mean. In AR(1) samples with coefficient 0.5 the same formula covered only 75 percent.

**Pitfalls.** Reading a computed interval as having a 95 percent probability of containing $\mu$. Drawing conclusions about a population of funds or stocks from a sample selected by survival or availability. Applying i.i.d. formulas to overlapping or autocorrelated returns. Treating a narrow interval from many highly correlated observations as precise.

**See also.** hypothesis tests; sample statistics; skill against luck.

#### Method 1.12 Hypothesis tests: null and alternative, test statistic, significance level, $p$-value, Type I and Type II errors, power; $t$, chi-square and $F$ tests

**Statement.** A test sets a null $H_0$ (for example $\mu = \mu_0$, or $\mu \le \mu_0$) against an alternative $H_1$ ($\mu \ne \mu_0$, or $\mu > \mu_0$). A Type I error rejects $H_0$ when it is true, and the test is designed so that its probability is at most $\alpha$; a Type II error accepts $H_0$ when $H_1$ is true, with probability $\beta(\mu)$

depending on the true value. For a mean, $W = (\bar{X} - \mu_0)/(S/\sqrt{n})$ is compared with $t_{\alpha/2,n-1}$ for normal data, an exact test, or with $z_{\alpha/2}$ for non-normal data with large $n$, an approximation; a two-sided test accepts exactly when $\mu_0$ lies in the corresponding confidence interval [probability-statistics-random-processes, sec. 8.4.3]. The $p$-value is "the lowest significance level $\alpha$ that results in rejecting the null hypothesis" (sec. 8.4.4). A variance is tested with $T = (N-1)(s/\sigma_0)^2$ against chi-square critical values with $N-1$ degrees of freedom, and two variances with $F = s_1^2/s_2^2$ against $F$ critical values with $N_1 - 1$ and $N_2 - 1$ degrees of freedom [statistical-methods-handbook, secs. 1.3.5.8–1.3.5.9], under the conditions that "each population is normally distributed" and the samples are "independent samples from the 2 populations" [applied-statistics-course-notes, Lesson 12, sec. 12.1, "Test to Compare Two Population Variances"]; the Levene test "is less sensitive than the Bartlett test to departures from normality" (sec. 1.3.5.10).

**Notation.** Power $= 1 - \beta$, the probability of rejecting $H_0$ when a specified alternative is true. Interpretation: for the one-sided $t$ test at level $\alpha$, with a true mean $\delta\sigma$ above $\mu_0$, the statistic has a noncentral $t$ distribution with noncentrality $\delta\sqrt{n}$, so power is $P(T_{n-1,\delta\sqrt{n}} > t_{\alpha,n-1})$, exact for normal data.

**Conditions.** The stated $\alpha$ is the Type I error rate only if the test's assumptions hold and the hypothesis was fixed before seeing the data; testing many variants and reporting the best inflates it (multiple-testing control). Failing to reject is not evidence that $H_0$ is true, particularly when power is low. The chi-square and $F$ tests for variances assume normal data and are sensitive to departures from it (Interpretation, consistent with the Levene comparison above). The $F$ test also needs the two samples to be independent of each other; normality within each sample does not supply that. Returns of two managers or strategies over the same months share market shocks and are usually correlated, and the ordinary $F$ reference distribution then does not apply. Interpretation, by simulation: with equal true variances and a correlation of 0.9 between paired months ($n = 12$), 40,000 simulated two-sided 5 percent tests reject 0.05 percent of the time; with a true ratio of standard deviations of 1.5 they reject 6 percent of the time, against 24 percent for independent samples. Paired series need a test that models the pairing, which this entry does not cover. Statistical significance is not economic size.

**Practice.** Example. $H_0 : \mu \le 0$ against $H_1 : \mu > 0$ at 5 percent: $t = 0.5333/0.3206 = 1.664$ with 11 degrees of freedom, a one-sided $p$-value of 0.062 (two-sided 0.124), so $H_0$ is not rejected, and 0 lies inside the 95 percent interval. If the true mean were half a standard deviation, the test would detect it with probability only 0.49 at $n = 12$, 0.90 at $n = 36$ and 0.99 at $n = 60$. Under a true null, 4.9 percent of 40,000 simulated tests rejected. For the first sample against the independent second sample, $F = 1.1106^2/0.4968^2 = 5.00$ on 11 and 14 degrees of freedom, beyond the upper two-sided 5 percent critical value of 3.09 ($p = 0.006$), so equal variances are rejected. The arithmetic can be checked against software, but only the sampling design justifies the $F$ distribution. Testing $\sigma = 1$ for the first gives $T = 13.57$ on 11 degrees of freedom, $p = 0.52$.

**Pitfalls.** Reading $p$ as the probability that $H_0$ is true. Treating "not significant" as "no skill" with a low-power test (skill against luck gives the data needs). Using an $F$ test on fat-tailed returns without checking normality, or on two return series from the same months as if they were independent.

**See also.** confidence intervals; nonparametric tests; distributions.

#### Method 1.13 Choosing between parametric and nonparametric tests: the sign test and the signed-rank test against the $t$ test, their assumptions and power

**Statement.** The signed-rank test is a one-sample or paired test that replaces the $t$ test's normality assumption by symmetry. Its assumptions are that each difference has a symmetric distribution, that the differences are mutually independent and share the same mean, and that they are measured on at least an interval scale. It "is not as powerful as the $t$-test when the distributional assumptions are in fact valid". It is more powerful than the sign test because it uses magnitudes as well as signs, but it assumes more, so the sign test is preferred when symmetry is doubtful. Its statistic is the smaller rank sum, with tabulated critical values for $N \le 30$ and, for larger $N$, a normal approximation with mean $n(n+1)/4$ and standard deviation $\sqrt{n(n+1)(2n+1)/24}$ [signed-rank-test-reference-2023, "Signed Rank Test"].

**Notation.** Interpretation: the *sign test* counts positive observations; under $H_0$ that the median is zero, with independent observations, the count is binomial with $p = 1/2$, so its $p$-value is an exact binomial tail. The signed-rank statistic $W^+$ sums the ranks of $|x_i|$ over positive $x_i$; its exact null distribution comes from the $2^n$ equally likely sign patterns.

**Conditions.** Nonparametric does not mean assumption-free: both tests need independent observations, the sign test tests a median, and the signed-rank test needs symmetry for its null to be about the centre. Interpretation: choose by the question and the data. Use the $t$ test for a mean when normality is plausible or $n$ is large; use the signed-rank test when tails are heavy but roughly symmetric; use the sign test when only direction is trusted or symmetry fails. None repairs serial dependence.

**Practice.** Example, the twelve active returns. Eight are positive: the exact one-sided sign-test $p$-value is $\sum_{k \ge 8} \binom{12}{k}/2^{12} = 0.194$. The ranks of the positive values sum to $W^+ = 58$; among all 4,096 sign patterns, 7.6 percent give 58 or more, the exact one-sided $p$-value; the normal approximation with continuity correction gives 0.073. The $t$ test gave 0.062. Each test uses less information than the one before it, and none rejects at 5 percent.

**Pitfalls.** Calling a rank test "distribution-free" and dropping its independence and symmetry assumptions. Switching to whichever test gives the smallest $p$-value after seeing the results. Testing a mean hypothesis with a median test.

**See also.** hypothesis tests; bootstrap (another route when formulas are doubtful).

## 1.5 Simulation and resampling

Three ways to generate outcomes are distinguished here: replaying the past (historical simulation), resampling the observed data (bootstrap) and drawing from a specified probability model (Monte Carlo). The risk-measure applications, filtered historical simulation and dynamic models are owned by VaR and ES estimation.

#### Method 1.14 Historical simulation: replaying past changes on today's position, equal and age weights, and what the method assumes

**Statement.** Historical simulation "involves using past data as a guide to what will happen in the future". With data on the most recent 501 days, each day's percentage changes in the market variables define one of 500 scenarios; scenario $i$ sets tomorrow's value of a variable to $v_n v_i / v_{i-1}$, where $v_n$ is today's value. Revaluing the current portfolio under each scenario gives a loss distribution. Hull estimates its 99th percentile, the one-day 99 percent VaR, as the fifth-highest of 500 losses, noting that "a case can be made for using the fifth-highest loss, the sixth-highest loss, or an average of the two"; the estimate holds if the past 501 days are "representative of what will happen" [s7-hulloptions, chap. 22, sec. 22.2, p. 517 and n. 2]. Expected shortfall averages the losses in the tail, the five worst in his example (p. 520). To give recent scenarios more weight, scenario $i$ can receive $\lambda^{n-i}(1-\lambda)/(1-\lambda^n)$, with $0 < \lambda < 1$; the 99 percent VaR is then "the loss for the first scenario which is such that the cumulative weight exceeds 0.01", and ES weights the losses above it by their weights and the VaR loss by the weight remaining to make up 0.01 (p. 521).

**Notation.** $n$ scenarios; losses $L_1, \ldots, L_n$ of the *current* portfolio under each historical change, with weights $w_i$ summing to one ($1/n$ for equal weights); $p = 1 - c$. This compendium uses the lower loss quantile and exact-tail-mass ES of VaR and ES estimation [acerbi-tasche-2001, Definitions 2.2 and 2.6, pp. 3 and 5; Proposition 3.2, p. 6]. Rank the losses from the largest, $\ell_{(1)} \ge \ell_{(2)} \ge \ldots$, merging ties, and let $W_k$ be the weight of the $k$ largest. Interpretation: $\mathrm{VaR}_c = \ell_{(k)}$ for the first $k$ with $W_k > p$, which is Hull's weighted rule, and

$$\mathrm{ES}_c = \frac{1}{p}\Big(\sum_{j<k} w_{(j)}\,\ell_{(j)} + (p - W_{k-1})\,\ell_{(k)}\Big),$$

the average over a tail of mass exactly $p$, taking only part of the weight at the VaR loss when needed. With equal weights and $np$ an integer, VaR is the $(np+1)$th largest loss, one rank below Hull's equal-weight choice of the $np$th, and ES is the mean of the $np$ largest; at $np = 2.5$, ES $= (\ell_{(1)} + \ell_{(2)} + 0.5\,\ell_{(3)})/2.5$. The two VaR conventions differ only in which side of a jump in the empirical distribution is reported; ES is the same under both.

**Conditions.** The method keeps the joint behaviour and fat tails of the sample but can produce no loss larger than its worst day scaled to today's position. It assumes the window is representative: a calm window understates risk, a window containing a crisis states it until the crisis drops out. Age weighting reacts faster but uses fewer effective observations. Using returns or price ratios of today's holdings, not past portfolio P&L, keeps the position current (Interpretation).

**Practice.** Example, original: a portfolio of 1,000,000, 60 percent in asset A and 40 percent in asset B, and 20 invented daily return pairs. Revalued under each day, the worst losses are 18,600 (day 7) and 12,800 (day 14). The next are 8,400 (day 2) and 7,600 (day 19). At 90 percent, $p = 0.1$ and $np = 2$. With equal weights the two worst losses carry exactly the tail mass, $W_2 = 0.10$, which does not exceed $p$, so VaR is the third-ranked loss, 8,400, and ES the average of the two worst, 15,700. Hull's equal-weight rank would report the second-worst loss, 12,800, as VaR; the ES is unchanged. With $\lambda = 0.9$ the two worst days, old in the window, carry weights 0.029 and 0.060, $W_2 = 0.089$, and the third-ranked loss (weight 0.017) takes the cumulative weight past 0.10, so VaR is again 8,400. ES becomes $(0.029 \times 18{,}600 + 0.060 \times 12{,}800 + 0.011 \times 8{,}400)/0.1 \approx 14{,}000$: down-weighting the old losses lowers ES by about 11 percent, while VaR does not move on this sample. Comparing a weighted VaR with an equal-weight VaR computed under another rank convention would wrongly attribute the convention's effect to the weighting.

**Pitfalls.** Reporting a historical VaR without its window, weighting and quantile convention, or comparing estimates computed under different conventions. Computing ES as the mean of losses strictly beyond VaR. At the VaR of 8,400 that average happens to equal the ES here, 15,700, because the two losses above it carry exactly the tail mass; it omits the boundary weight whenever the losses above VaR carry less than $p$, as under the age weights. Averaging only the losses beyond Hull's equal-weight VaR of 12,800 would instead leave the 18,600 loss alone. Expecting a loss beyond the worst day in the window. Mixing scenario returns with the past composition of the portfolio.

**See also.** VaR and ES estimation (filtered HS, backtests); percentile conventions.

#### Method 1.15 Bootstrap resampling: resampling the data to estimate a sampling distribution, the sampling unit, and why i.i.d. resampling fails for dependent returns

**Statement.** The bootstrap draws subsamples "with replacement" from the data, computes the statistic on each, and repeats, typically 500 to 1,000 times; the results "form an estimate of the sampling distribution of the statistic", and percentiles of the sorted results give confidence intervals. "The bootstrap is not appropriate for all distributions and statistics": statistics driven by the extremes, such as the range, are poorly served [statistical-methods-handbook, sec. 1.3.3.4]. Applied to dependent data, the i.i.d. bootstrap "scrambles" the observations, so "all dependence information is lost" and it "fails to give valid standard error estimates in the dependent case"; the block bootstrap instead joins blocks of consecutive observations chosen at random with replacement, and the block size must be chosen, a problem like bandwidth choice in smoothing [bootstrap-time-series-2003, secs. 2 and 2.3, pp. 221–223].

**Notation.** Data $x_1, \ldots, x_n$; $B$ resamples; the *sampling unit* is what is drawn: single observations (i.i.d. bootstrap), blocks of $b$ consecutive observations (moving-block bootstrap) or whole cross-sections of assets on a date (to keep cross-sectional correlation). Interpretation: as $B \to \infty$ the i.i.d. bootstrap variance of the mean tends to $\frac{1}{n} \cdot \frac{1}{n} \sum (x_i - \bar{x})^2$, the *ideal* bootstrap value; finite $B$ adds resampling noise to it.

**Conditions.** The bootstrap approximates the sampling distribution by treating the sample as the population; it inherits the sample's size and biases and cannot create information on tails it did not observe. The i.i.d. version needs independent observations; for serially correlated returns, blocks (or other dependent-data schemes) are needed, and they remain biased in finite samples. Seeds and $B$ should be recorded so that resampling noise can be reproduced and judged.

**Practice.** Examples, original. For the twelve active returns of confidence intervals, 10,000 i.i.d. resamples give a standard error of the mean of 0.309 against the ideal 0.307, and a 2.5–97.5 percent interval of about −0.09 to 1.13, close to the $t$ interval. *Dependence*: for 120 observations of an AR(1) series with coefficient 0.5, the exact variance of the sample mean is 0.0330. The ideal i.i.d. bootstrap variance has expectation 0.0108, a third of it; moving blocks of 12 recover 0.0261, 79 percent: better, still too low.

**Pitfalls.** Resampling single days of autocorrelated or volatility-clustered returns and reporting the resulting narrow intervals. Resampling each asset separately and destroying cross-correlation. Choosing the block size, or reseeding, until a result appears.

**See also.** sequential bootstrap (overlapping labels); resampled efficiency; Sharpe-ratio standard errors.

#### Method 1.16 Monte Carlo simulation: sampling from a specified model, the estimation target, seeds, standard error and the square-root rule

**Statement.** A Monte Carlo valuation samples paths of the variables from a specified model, computes the discounted payoff on each, and averages. With $\bar{f}$ the mean and $\omega$ the standard deviation of the discounted payoffs over $M$ trials, the standard error is $\omega/\sqrt{M}$ and a 95 percent confidence interval for the value is $\bar{f} \pm 1.96\,\omega/\sqrt{M}$; uncertainty falls with the square root of the number of trials, so doubling the accuracy needs four times the trials [s7-hulloptions, chap. 21, sec. 21.6, pp. 492–494]. In Hull's example, 1,000 trials give an option value of 4.98 with a standard error of 0.24, and the closed-form price, 4.817, lies inside the interval (Example 21.8, p. 494).

**Notation.** Target: $\theta = E[g(Y)]$ under the specified distribution of $Y$, for example a discounted payoff in a risk-neutral model or a probability $P(Y \in A) = E[1_A(Y)]$, whose standard error is $\sqrt{\hat{p}(1-\hat{p})/M}$. Seed: the starting state of the pseudo-random generator, which makes a run reproducible.

**Conditions.** The estimate is only as good as the model: Monte Carlo gives the expectation under the assumed distribution, not under the true one. The standard-error formula assumes independent trials with finite payoff variance; dependent draws, such as overlapping paths, need other error estimates. Variance-reduction methods such as antithetic variables change the precision per trial, not the target (Hull, sec. 21.7). Seeds should be fixed before the run and reported, not chosen after seeing results.

**Practice.** Example, original. A stock at 100 follows geometric Brownian motion with volatility 20 percent; the risk-free rate is 3 percent; a one-year call struck at 105 has a closed-form value of 7.1281. With 100,000 draws the estimate is 7.1161 and the standard error 0.0396, so the closed form lies 0.3 standard errors away. The probability that the price ends below 90 is 0.2821 by simulation against 0.2820 exactly (standard error 0.0014). A quarter of the draws gives about twice the standard error, and antithetic pairs reduce it further.

**Pitfalls.** Reporting a Monte Carlo figure without its standard error, number of trials and seed. Reading agreement within sampling error as validation of the model. Comparing two simulated strategies on different random draws when common random numbers would sharpen the comparison.

**See also.** distributions (lognormal prices); Black–Scholes–Merton; VaR and ES estimation (Monte Carlo for risk).

## 1.6 Simple linear regression

The entries in this section use one invented data set with a declared sampling model. Ten corporate bonds have leverage $x = 20, 25, \ldots, 65$ (debt over assets, in percent), fixed by design. Their credit spreads $y$ (basis points) were generated as $y = 40 + 5x + \varepsilon$, with independent normal errors of standard deviation 30, and rounded to whole basis points: 102, 216, 229, 240, 222, 197, 265, 326, 388, 349. The generating line is known here so that estimates can be compared with it; in practice it is unknown. The data illustrate the mechanics and are not evidence about credit markets.

**Method 1.17 Simple linear regression by least squares: the population line, the estimated intercept and slope, fitted values and residuals, coefficient units, and association against causation**

**Statement.** Simple linear regression relates a response $y$ to one predictor $x$. The *population regression line* $\mu_Y = E(Y) = \beta_0 + \beta_1 x$ connects the mean responses of the subpopulations at each $x$; a sample gives the estimated line $\hat{y}_i = b_0 + b_1 x_i$, and another sample would give a different one [regression-methods-course-notes, sec. 1.3]. The prediction error, or residual, is $e_i = y_i - \hat{y}_i$. The *least squares criterion* chooses $b_0$ and $b_1$ to "minimize the sum of the squared prediction errors" $Q = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$, which gives

$$b_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}, \qquad b_0 = \bar{y} - b_1 \bar{x}$$

(sec. 1.2). "Association is not causation": a large $R^2$ "does not imply that $x$ causes the changes in $y$" (sec. 1.5). Experiments, in which the researcher controls the predictor, support causal conclusions; observational studies, the source of most regression data, do not (sec. 1.8, Caution 4).

**Notation.** $y$ is the response (dependent variable), $x$ the predictor (independent or explanatory variable), $n$ the number of observations. $b_1$ has the units of $y$ per unit of $x$, here basis points of spread per percentage point of leverage; $b_0$ is the fitted value at $x = 0$, in units of $y$. Interpretation: the first-order conditions of the criterion make the residuals sum to zero and be uncorrelated with $x$ in the sample, and put the point $(\bar{x}, \bar{y})$ on the fitted line.

**Conditions.** The intercept and slope exist and are unique when the data are finite and $S_{xx} = \sum(x_i - \bar{x})^2 > 0$, that is, when at least two predictor values differ. If every $x_i$ is the same, every line through $(\bar{x}, \bar{y})$ minimises $Q$ and the slope is not identified. This is the full-rank condition on the design matrix [linear-regression-lectures, Normal linear regression model, "Setting" and "Assumptions"], written for one predictor (Interpretation). Fitting the line needs no assumption about the errors, normality included; the assumptions behind unbiasedness, standard errors and tests are set out in regression diagnostics.

Reading $b_1$ as the average difference in $y$ between observations one unit of $x$ apart assumes the mean of $y$ is linear in $x$ over the observed range. Reading it as the effect of *changing* $x$ needs a research design that rules out other explanations, which a cross-section of bonds does not supply: issuers with high leverage may differ in sector, size or asset risk (Interpretation). One unusual observation can move the slope substantially, so the data should be plotted [regression-methods-course-notes, sec. 1.8, Caution 3]. Regressing $x$ on $y$ gives a different line, not the inverse of this one (sec. 2.8, where the two directions give slopes 0.967 and 0.911).

**Practice.** Example, the ten bonds: $\bar{x} = 42.5$, $\bar{y} = 253.4$, $\sum(x_i - \bar{x})^2 = 2{,}062.5$ and $\sum(x_i - \bar{x})(y_i - \bar{y}) = 9{,}905$, so $b_1 = 9{,}905/2{,}062.5 = 4.80$ basis points per percentage point and $b_0 = 253.4 - 4.80 \times 42.5 = 49.3$. The bond at 45 percent leverage has a fitted spread of 265.4 and a residual of $197 - 265.4 = -68.4$. In this sample, bonds with one percentage point more leverage have spreads about 4.8 basis points wider on average, close to the generating slope of 5. The intercept, a spread at zero leverage, lies far outside the data and has no practical meaning here.

**Pitfalls.** Reading the slope as the effect of an issuer raising its leverage. Interpreting the intercept at an $x$ far from the data. Inverting a regression of $y$ on $x$ to predict $x$ from $y$. Fitting without plotting the data.

**See also.** regression diagnostics; regression predictions; covariance and correlation; applications: minimum-variance hedge ratio (an OLS slope), spread–volatility regression.

**Method 1.18 Regression assumptions and diagnostics: what estimation, standard errors and exact tests each assume, residual plots, the standard error of estimate, $R^2$, coefficient $t$ tests and the simple-regression ANOVA, with $F = t^2$ for a zero slope**

**Statement.** The simple linear regression model has four conditions: the mean of $Y_i$ is a *linear* function of $x_i$; the errors are *independent*; the errors are *normally* distributed at each $x_i$; and they have *equal* variance $\sigma^2$ [regression-methods-course-notes, sec. 1.3]. The mean square error $MSE = \sum(y_i - \hat{y}_i)^2/(n-2)$ estimates $\sigma^2$, and $S = \sqrt{MSE}$ (sec. 1.4); it needs $n > 2$, because with two observations the line passes through both, $SSE = 0$ and no degrees of freedom remain for $\sigma^2$, the standard errors or the tests (Interpretation). With $SSR = \sum(\hat{y}_i - \bar{y})^2$, $SSE = \sum(y_i - \hat{y}_i)^2$ and $SSTO = SSR + SSE = \sum(y_i - \bar{y})^2$, $R^2 = SSR/SSTO = 1 - SSE/SSTO$ (sec. 1.5). The slope test uses $t^* = (b_1 - \beta)/se(b_1)$, $se(b_1) = \sqrt{MSE}/\sqrt{\sum(x_i - \bar{x})^2}$, on $n-2$ degrees of freedom, and the interval $b_1 \pm t_{(\alpha/2, n-2)}\, se(b_1)$ (sec. 2.1).

The ANOVA table has 1, $n-2$ and $n-1$ degrees of freedom for regression, error and total; $MSR = SSR/1$, and $F^* = MSR/MSE$ is compared with an $F$ distribution with 1 and $n-2$ degrees of freedom (sec. 2.6). This $F^*$ tests $H_0 : \beta_1 = 0$. For one predictor, with $t^*$ computed for that same null, $(t^*_{(n-2)})^2 = F^*_{(1,n-2)}$: the $F$ test is "algebraically equivalent to the two-tailed t-test" and gives the same $p$-value, but only the $t$ test can test a one-sided alternative (sec. 2.7). A $t$ statistic for another value, $H_0 : \beta_1 = \beta$, squares to the $F$ statistic of that restriction in the general test of linear restrictions, not to the ANOVA $F^*$ [linear-regression-lectures, Linear regression – hypothesis testing, "Test of a restriction on a single coefficient (t test)" and "Test of a set of linear restrictions (F test)"]; the equality for one restriction is algebra (Interpretation).

Residuals estimate the unknown errors. A residuals-versus-fits plot detects non-linearity, unequal error variances and outliers; a systematic pattern suggests a non-linear regression function, and fanning suggests unequal variances (secs. 4.2 and 4.4). Standardised residuals beyond ±2 are "flagged for further investigation", not removed (sec. 4.4). A residuals-versus-order plot detects serial correlation and "is only appropriate if you know the order in which the data were collected" (sec. 4.5); a normal probability plot checks normality (sec. 4.6). Sensitivity differs by condition: tests and intervals are "very sensitive to even minor departures from independence", sensitive to moderate departures from equal variance, fairly robust to non-normality for the coefficients, while prediction intervals are "quite sensitive to departures from normality" (sec. 4.1).

**Notation.** $n-2$ degrees of freedom because two parameters are estimated. $S$ is the *standard error of the estimate* (of the regression), in units of $y$. Interpretation: $se(b_0) = S\sqrt{1/n + \bar{x}^2/\sum(x_i - \bar{x})^2}$; $F^* = R^2(n-2)/(1-R^2)$; with one predictor $R^2$ equals the squared sample correlation of $x$ and $y$.

**Conditions.** Each conclusion needs a different subset of the conditions. Computing $b_0$ and $b_1$ needs only finite data with $S_{xx} > 0$ (least squares estimation). They are unbiased when, in addition, the mean of $Y$ is linear in $x$, so that $E(\varepsilon \mid X) = 0$. Uncorrelated errors of equal variance, $\mathrm{Var}(\varepsilon \mid X) = \sigma^2 I$, add the conventional standard errors above and make least squares the best linear unbiased estimator [linear-regression-lectures, Gauss–Markov theorem, "Assumptions", "OLS is linear and unbiased" and "The covariance matrix of the OLS estimator"]. Normal errors add the exact $t$ and $F$ distributions on $n-2$ degrees of freedom (Normal linear regression model, "Distribution of the OLS estimator" and "Estimation of the variance of the error terms"; Linear regression – hypothesis testing, "Tests of hypothesis in the normal linear regression model").

Without normality the same statistics are large-sample approximations, justified when the estimator is asymptotically normal (Linear regression – hypothesis testing, "Tests of hypothesis when the OLS estimator is asymptotically normal"); ten observations give little assurance that the approximation is close. With independent, identically distributed errors of mean zero and finite variance, least squares is consistent for the coefficients whether or not the errors are normal, "if the design matrix is asymptotically nondegenerate"; normality then affects only efficiency [smearing-retransformation-1983, sec. 2, pp. 605–606]. Full rank in every sample is not enough. With $x_i = 2^{-i}$, $S_{xx}$ tends to $1/3$ as $n$ grows, so $\mathrm{Var}(b_1) = \sigma^2/S_{xx}$ tends to $3\sigma^2$, not zero; with one predictor, consistency of $b_1$ needs $S_{xx} \to \infty$ (Interpretation). Correlated errors invalidate the conventional standard errors and tests even when the coefficients remain unbiased (Interpretation).

A residual plot can reveal a violation; it cannot prove that the conditions hold. With ten points a curve, a fan or serial correlation can easily go unseen, and over-reading small plots is a known error [regression-methods-course-notes, sec. 4.2]. A large $R^2$ "should not be interpreted as meaning that the estimated regression line fits the data well", a significant slope need not be meaningfully different from zero, and a large $R^2$ does not guarantee useful intervals (sec. 1.8, Cautions 2, 6 and 7). $R^2$ and the tests describe the estimation sample; out-of-sample predictive performance

needs separate evidence (Interpretation; parameter overfitting). Bonds of one issuer or sector, or observations in time order, may have dependent errors; time-series errors need time-series methods (sec. 4.5).

**Practice.** Example, the ten bonds: $SSE = 14{,}356.4$, $SSR = 47{,}568.0$ and $SSTO = 61{,}924.4$; $S = \sqrt{14{,}356.4/8} = 42.4$ basis points and $R^2 = 0.768$. The slope has $se(b_1) = 42.36/\sqrt{2{,}062.5} = 0.933$ and $t = 4.802/0.933 = 5.15$ on 8 degrees of freedom, $p = 0.0009$; its 95 percent interval is 2.65 to 6.95. Testing $\beta_1 = 5$, the generating value, gives $t = -0.212$, not rejected; its square, 0.045, is the $F$ statistic for the restriction $\beta_1 = 5$, not the ANOVA's 26.51. The intercept has $t = 1.18$, $p = 0.27$.

| Source | DF | SS | MS | $F$ |
|---|---|---|---|---|
| Regression | 1 | 47,568.0 | 47,568.0 | 26.51 |
| Error | 8 | 14,356.4 | 1,794.5 | |
| Total | 9 | 61,924.4 | | |

$F = 47{,}568.0/1{,}794.5 = 26.51 = 5.149^2$, the square of the $t$ statistic for a zero slope, with the same $p$-value as that two-sided $t$ test. The largest standardised residual is $-1.61$, at 45 percent leverage; none exceeds 2. These checks find no violation, which with ten bonds is weak evidence that none exists. Counterexample, original: the exact curve $y = 0.2(x - 10)^2$ on the same leverages gives a straight-line $R^2$ of 0.96, yet its residuals are positive, then negative, then positive.

**Pitfalls.** Treating a clean residual plot as proof of the assumptions. Reading a high $R^2$ as a good model or as predictive power. Deleting observations because their residuals are large. Using the $F$ test for a one-sided question. Equating the ANOVA $F$ with the square of a $t$ statistic for a non-zero null value. Applying the $t$ test to serially correlated errors.

**See also.** least squares estimation; hypothesis tests; regression predictions; purged cross-validation (out-of-sample evaluation).

### Method 1.19 Regression predictions: fitted values, intervals for the mean response and for a new observation, extrapolation, and log functional forms with retransformation to geometric means, medians and means

**Statement.** At a predictor value $x_h$, the confidence interval for the mean response is

$$\hat{y}_h \pm t_{(1-\alpha/2,n-2)} \sqrt{MSE\left(\frac{1}{n} + \frac{(x_h - \bar{x})^2}{\sum (x_i - \bar{x})^2}\right)},$$

narrower when $x_h$ is near $\bar{x}$; it applies when $x_h$ lies "within the range of the x values in the data set" and the LINE conditions hold, and tolerates approximately normal errors [regression-methods-course-notes, sec. 3.2]. The prediction interval for a new response adds an extra $MSE$ under the root, $MSE\ (1+1/n+(x_h-\bar{x})^2/\sum(x_i-\bar{x})^2)$: the variance of a prediction is the variance $\sigma^2$ of a new response about its mean plus the variance of $\hat{y}_h$. The prediction interval is therefore always wider and "depends strongly on the condition that the error terms are normally distributed" (sec. 3.3).

Transforming $x$ suits non-linearity alone; transforming $y$ suits non-normality or unequal variances (secs. 9.1–9.2). Predictions from a log response estimate $E(\ln Y \mid x)$ and come back in log units. In a log–level model a unit of $x$ multiplies the back-transformed value by $e^{\beta_1}$; in a log–log model a $k$-fold change in $x$ multiplies it by $k^{\beta_1}$. The course notes read the back-transformed value as the median, because the mean of the logs "approximately equals the natural log of the median" (secs. 9.2–9.3).

Exactly, $\exp E(\ln Y \mid x)$ is the conditional *geometric mean* of $Y$. It is the conditional median only when the log-scale error has median zero, as a symmetric error does; Hyndman reads a back-transformed forecast as the median "assuming the forecast densities on the transformed scale are symmetric" [back-transformed-forecast-mean-2014, para. 1]. A skewed error with mean zero separates the two (Interpretation). When the error distribution does not change with $x$, the median and the mean change by the same factors as the geometric mean.

The mean is not the retransformed prediction: $E\,Y_0 = E\,h(x_0\beta+\varepsilon) \neq h(x_0\beta)$. With normal errors and a log transformation, $E\,Y_0 = \exp(x_0\beta+\sigma^2/2)$, estimated by $\exp(x_0\hat{\beta} + \hat{\sigma}^2/2)$. Under the design and moment conditions of regression diagnostics, whatever the error distribution, that estimate is consistent for $\exp(x_0\beta + \sigma^2/2)$, which equals $E\,Y_0$ under normal errors but otherwise "might not": in Duan's normal-mixture example it has a 41 percent asymptotic bias. Normality is sufficient for it to estimate $E\,Y_0$ consistently, not necessary; what matters is whether $E\,e^{\varepsilon} = e^{\sigma^2/2}$ (Interpretation). The *smearing estimate* replaces the error distribution by the residuals: $\frac{1}{n}\sum_{i=1}^{n} h(x_0\hat{\beta} + \hat{\varepsilon}_i)$ [smearing-retransformation-1983, secs. 2–3, eq. (3.1), pp. 605–606]. It is weakly consistent for $E\,Y_0$ when $h$ is continuously differentiable, the design contains an intercept, $X'X/n$ converges to a positive-definite matrix and a moment condition holds, which for the log transformation is $E\,e^{2\varepsilon} < \infty$ (Theorem 1 and the text after it, sec. 4, p. 607). For a lognormal variable the median is the scale parameter and the mean exceeds it by the factor $e^{0.5\sigma^2}$ [statistical-methods-handbook, sec. 1.3.6.6.9].

**Notation.** Interpretation: the *standard error of the estimate* $S = \sqrt{MSE}$ measures scatter about the line; the standard error of the fit is $S\sqrt{1/n + (x_h - \bar{x})^2/\sum(x_i - \bar{x})^2}$; the standard error of prediction is $\sqrt{S^2 + (\text{standard error of the fit})^2}$. Forms and slope units:

- level–level, $y = b_0 + b_1 x$: $b_1$ units of $y$ per unit of $x$;
- log–level, $\ln y = b_0 + b_1 x$: the geometric mean of $y$ changes by $100(e^{b_1} - 1)$ percent per unit of $x$;
- level–log, $y = b_0 + b_1 \ln x$: $y$ changes by $b_1 \ln(1.01) \approx b_1/100$ units when $x$ rises 1 percent;
- log–log, $\ln y = b_0 + b_1 \ln x$: $b_1$ is an elasticity; a 1 percent rise in $x$ changes the geometric mean of $y$ by $100(1.01^{b_1} - 1) \approx b_1$ percent, a fraction $1.01^{b_1} - 1$.

Logarithms need positive values: spreads, prices and volumes qualify, returns do not.

**Conditions.** The intervals assume the fitted model is correct where they are used. Outside the observed $x$ the formula widens only for parameter uncertainty; it says nothing about whether the line still holds, which no data in the sample can check (Interpretation). A prediction interval is for one new observation from the same population, with an error independent of the sample. $R^2$ values of models with $y$ and with $\ln y$ measure different responses and cannot rank the forms. The normal-theory mean correction needs normal, equal-variance errors on the log scale; the smearing estimate drops normality but still treats the errors as independent draws from one distribution, so neither corrects for variance that changes with $x$ (Interpretation, from Duan's model assumptions).

**Practice.** Example, the ten bonds at 45 percent leverage: $\hat{y} = 49.3 + 4.802 \times 45 = 265.4$ basis points. The standard error of the fit is 13.60 and of prediction $\sqrt{42.36^2 + 13.60^2} = 44.49$. With $t_{(0.975,8)} = 2.306$, the 95 percent interval for the mean spread of all such bonds is 234.1 to 296.8. For one new bond it is 162.8 to 368.0. In 20,000 samples simulated from the declared model, the first interval covered the true mean of 265 in 94.8 percent of samples and the second covered a new bond's spread in 95.0 percent; the mean interval misused for a new bond covered it only 49.7 percent of the time (49.9 percent exactly).

Extrapolating to 90 percent leverage gives 481.5 with a standard error of the fit of 46.3, 3.4 times that at 45 percent, and a prediction interval of 336.8 to 626.2. That interval is valid only if the straight line, its variance and its normal errors extend to leverage no bond in the sample has.

The log–level fit to the same bonds is $\ln y = 4.597 + 0.02075x$ with $S^2 = 0.0472$: each percentage point of leverage multiplies the geometric-mean spread by $e^{0.02075} = 1.021$, 2.1 percent. At 45 percent the fitted geometric mean is $e^{5.530} = 252.2$, also the median if the log errors have median zero;

the normal-theory mean is $252.2 \times e^{0.0472/2} = 258.2$ and the smearing estimate is 256.8. The log–log elasticity is 0.826: a 1 percent rise in leverage raises the geometric-mean spread by $100(1.01^{0.826} - 1) = 0.825$ percent, a fraction 0.00825. The level–log slope of 183.9 adds 1.83 basis points per 1 percent rise in leverage. The data came from the level model, so these fits show the arithmetic of each form; their $R^2$ values, 0.70 to 0.75, are not evidence against the level model's 0.77.

Counterexample, original: let the log error be $\varepsilon = 0.5(E - 1)$ with $E$ standard exponential. Its mean is zero and its variance 0.25, but it is right-skewed, with median $0.5(\ln 2 - 1) = -0.153$. Relative to $e^{x\beta}$, the geometric mean of $Y$ is 1, the median 0.858, the mean $e^{-0.5}/0.5 = 1.213$ and the normal-theory factor $e^{0.25/2} = 1.133$. Exponentiating the log prediction then overstates the median by 16.6 percent and understates the mean by 17.6 percent; the normal-theory correction is still 6.6 percent below the mean.

**Pitfalls.** Quoting a confidence interval for the mean as the range for one new outcome. Extrapolating beyond the observed predictors. Exponentiating a log prediction and reporting it as the expected value, or as the median when the log errors may be skewed. Reading the fraction $1.01^{b_1} - 1$ as a percentage. Choosing a functional form by comparing $R^2$ across different responses. Taking logs of returns or other quantities that can be zero or negative.

**See also.** regression diagnostics; confidence intervals; lognormal moments; spread–volatility regression (a log–log application with the lognormal correction).

# 2 Research engineering and validation

## 2.1 Strategy research and backtest evaluation

**Method 2.1 Hypothesis-first strategy research: observation, theory, deduced consequences and attempted falsification, with Narang's four sources of ideas**

**Statement.** Narang describes quant research as following the scientific method [s7-narang, chap. "Research", sec. 9.1, pp. 161–163]:

1. observe something in the market that might be explainable, a pattern;
2. form a theory that explains it;
3. deduce the theory's consequences, for example that a trend should give a better-than-random forecast from past returns alone;
4. test the theory by looking for evidence against those consequences (Popper's falsification), not for proof. A theory not yet disproved is accepted only for the moment. At best the researcher gains enough confidence that the tests were rigorous to "warrant risking some capital on the validity of this theory" (p. 163).

Because markets change, through regulation, investor psychology and competition for alpha, quants must keep researching rather than treat a validated theory as permanent (p. 163).

Narang's four common sources of ideas [s7-narang, sec. 9.2, pp. 163–165]:

- *observing the markets*, the source closest to the scientific method; his example is Donchian's two-week breakout rule for futures;
- *academic literature*: many quant firms spend significant time on journals, working papers and conference presentations; literature from astronomy, physics or psychology is also used;
- *migration* of researchers or portfolio managers between firms, despite noncompete and nondisclosure agreements. Investors who have seen inside one firm can also carry ideas to another;
- *lessons from discretionary traders*, such as "ride winners and cut losers" formalised as a stop-loss policy.

For machine-learning alphas he paraphrases practitioners: feature selection and engineering draw on the same sources (p. 165).

**Conditions.** A practitioner's account of how ideas arise and are screened, in a 2024 book, not a statistical protocol. It does not say how many hypotheses were tried, how significance is adjusted for them, or what evidence suffices to risk capital: those are treated in multiple-testing control.

**Practice.** Interpretation: stating the theory and its deduced consequences before testing fixes what would count as evidence against it. That is the step that separates a test from a search. Narang's out-of-sample discussion makes the same point from the other side: refitting after looking at out-of-sample data, or choosing models already known to have survived a period, "burns" that data (sec. 9.3.4, pp. 183–185; parameter overfitting).

**Pitfalls.** Hypothesis-driven research needs explicit predicted consequences and attempted falsification, which Narang treats as its core (p. 162). Of discretionary traders' rules, Narang notes that systematic versions of chart patterns have mostly failed. That may be because the theory is invalid or because human practice is less rule-based than it seems, and "not all successful traders have skill" (p. 165). An idea's source is not evidence for it. Ideas that move with staff raise contractual and legal questions; Narang describes a dispute over noncompete agreements.

**See also.** backtest evaluation diagnostics; parameter overfitting (what testing guards against); cost and short-availability assumptions.

**Method 2.2 Backtest evaluation diagnostics: the profit curve, return, variability and lumpiness, drawdowns, predictive power, win rates, risk-adjusted ratios, fit with other strategies, and time decay**

**Statement.** Narang lists statistics quants use to judge a model, in sample and out of sample, illustrated on his own daily S&P 500 strategy. Its signal is an adjustment to the equity risk premium, the S&P 500's earnings yield less the 10-year Treasury note's yield, computed each day: a higher stock yield is read as a signal to be long, a lower one as a signal to be short.

It ran on daily closes from June 1982 to December 2000, was built in the mid-1990s and was never traded [s7-narang, chap. "Research", sec. 9.3.2, p. 168]. It does not trade every day: it gives a signal only when the model judges the opportunity attractive enough to trade at all, and its signal is zero on 65 percent of days (sec. 9.3.2.6, p. 173). Narang does not state the adjustment or the threshold for a non-zero signal. Each diagnostic, with his example's value:

- *cumulative profit graph*: shows whether, how smoothly and with what downside the strategy made money. His example is "lumpy", with years of inactivity (late 1989 to early 1995), sharp losses and steep gains, which he reads as "real problems" (sec. 9.3.2.1, p. 168);
- *average rate of return*: 746 percent cumulative, 12.1 percent a year, before costs and fees (sec. 9.3.2.2, p. 168);
- *variability of returns*: "in general, the less the variability for a given level of returns, the better" (sec. 9.3.2.3, p. 169); 21.2 percent annualised. His firm also uses *lumpiness*, the share of total return earned in periods significantly above average, as a measure of consistency, while noting that consistency is not always a primary goal;
- *worst peak-to-valley drawdowns* and recovery times: lower drawdowns are better and long recoveries are disliked; several drawdowns are measured, not only the worst. His example's worst is −39.7 percent, short before the October 1987 crash (sec. 9.3.2.4, pp. 169–170);
- *predictive power*: the $R^2$ of the target on the signal, 0.0085 for his example in Exhibit 9.2 (sec. 9.3.2.5, p. 171). Returns are also bucketed by quantiles of the forecast: returns rising monotonically across buckets indicate reliable predictive power, and non-monotonic buckets may mean the strategy works by accident (p. 172);
- *percentage of winning trades or periods*, a measure of consistency. His example has a zero signal 65 percent of the time, wins 19 percent and loses 16 percent, so about 54 percent of non-zero days win (sec. 9.3.2.6, pp. 172–173);
- *ratios of return to risk*, as Narang defines them (sec. 9.3.2.7, pp. 173–174): the Sharpe ratio, average periodic return above the risk-free rate divided by its variability; the information ratio, the same with "any arbitrary benchmark return—most often zero" in place of the risk-free rate (0.57 in the example); the Sterling ratio, average return over the variability of below-average returns (0.87); the Calmar ratio, "average return divided by the worst peak-to-valley drawdown" (0.31); and the Omega ratio, "the sum of all positive returns divided by the sum of all negative returns" (1.26). Taken literally, both of Narang's quotients have a negative denominator, but he reports both ratios as positive. Interpretation: the explicit conventions are

$$\text{Calmar} = \frac{\bar{R}}{|\text{MDD}|}, \qquad \Omega(0) = \frac{\sum_t \max(R_t, 0)}{\left|\sum_t \min(R_t, 0)\right|},$$

with $\bar{R}$ the average (annualised) return, MDD the worst peak-to-valley drawdown, a negative number, and $R_t$ the periodic returns. With his example's $\bar{R} = 12.1$ percent and MDD $= -39.7$ percent, the first gives $12.1/39.7 \approx 0.3048$, which rounds to 0.30, while the literal quotient gives $-0.30$. Narang reports a Calmar ratio of 0.31 (pp. 173–174); the recomputation from his printed inputs does not reproduce that second digit, and the passage does not explain the difference. For returns of +2, +1, −1.5 and −0.5 percent, the second gives $3/2 = 1.5$, while the literal sum quotient gives $-1.5$. The Sterling and Omega values cannot be recomputed without the strategy's returns;

- *relationship with other strategies*: a correlation with the existing book, or better, the existing strategy's results with and without the new idea (sec. 9.3.2.8, p. 174);
- *time decay*: the strategy's returns when trades start one, two, three or more days after the signal. This shows its sensitivity to timely information and, Narang says, "some information about how crowded the strategy is", since crowding would mean faster moves to a new equilibrium (sec. 9.3.2.9, pp. 174–176).

**Conditions.** Practitioner account as of 2024; the S&P example covers 1982–2000. One author's list and one illustrative strategy, not a validated scorecard. Narang says "a significant amount of discretion is used" in judging models (p. 176). None of the diagnostics is adjusted for the number of strategies or parameter sets tried, and none comes with a standard error.

**Practice.** Narang's time-decay example uses analyst downgrades of Microsoft announced before the open on 28 April 2006. The stock opened about 11.1 percent down and underperformed the S&P 500 by about 11.4 percent that day. A simulation must not assume a fill before the open. Selling at the close on 28 April, 1 May or 2 May made money over two weeks; selling from 3 to 12 May lost (pp. 174–175). Delay can also help: entering his S&P strategy one day late raised the total return from 746 to 870 percent (12.9 percent a year), which he takes as a bad sign, because the signal is early (pp. 175–176).

Interpretation: most of his example figures agree within rounding: $12.1/21.2 = 0.57$, $19/35 = 0.54$, and 746 and 870 percent over about 18.6 years compound to about 12.2 and 13.0 percent a year. The Calmar ratio is the exception: $12.1/39.7 \approx 0.3048$ from his printed inputs, against his reported 0.31; the difference is unexplained.

**Pitfalls.** Interpretation, qualifying the heuristics:

- *Smooth profits.* Narang reads lumpiness as a problem in his example and says consistency is not always the goal. A smooth curve can come from stale or managed valuations rather than skill (relative performance evaluation), and from short-volatility payoffs whose losses the sample did not contain. Narang's own convertible-arbitrage example (1990–1997, then 1998) shows this (p. 170). The rule "if the profit profile is not smooth … the model may have issues" is his reading of one example, not a test.
- *Lower volatility.* His rule holds return fixed ("for a given level of returns"); the "the less variable … the better" dropped that condition. His "67 percent of the time within one standard deviation" assumes roughly normal annual returns (about 68 percent for a normal distribution).
- *Drawdowns.* A worst drawdown from a favourable sample understates risk. Narang's convertible-arbitrage example is "sample bias", and his remedy, shuffling the historical returns to get many paths, still draws only from the sample (pp. 170–171). Shuffling single returns also destroys serial dependence, which drives drawdown depth; that point is not in Narang. A synthetic check shows its size. For AR(1) monthly returns with autocorrelation 0.3 over ten years, the model's 95th percentile of maximum drawdown is 0.99 in log units. The median shuffle-bootstrap estimate from one sample is 0.53. Resampling blocks of geometric length (mean 12) gives 0.85. Politis and Romano's original stationary-bootstrap paper is not covered; the method is restated below.
  White restates it: resample blocks "of random length, distributed according to the geometric distribution with mean block length $b$", which "delivers valid bootstrap approximations for means of $\alpha$-mixing processes, provided $b$ increases appropriately with $n$" [catalogue-0731, sec. 2, p. 1104]. That validity statement covers means, and statistics built from them, not a path functional such as the maximum drawdown. So the 0.85 above is an own simulation result, not something a theorem guarantees (multiple testing).
  *Block length*. The authors of the method later corrected the automatic block-length rule of Politis and White (2004): after Nordman (2008) found an error in Lahiri's variance calculation for the stationary bootstrap, the variance constant becomes $D_{SB} = 2g^2(0)$, the circular block bootstrap's optimal mean-squared error is $(2/3)^{2/3} \approx 0.763$ of the stationary bootstrap's, and the earlier simulation tables are replaced; with the correction the estimated block sizes average 90 to 110 percent of the optimal ones in their AR(1) designs [patton-politis-white-2009, secs. 1–2, pp. 372–373]. That bears on choosing $b$ for a mean; whether the same choice suits path statistics such as the maximum drawdown is not addressed there.
- *The $R^2$ of 0.05.* "A superb $R^2$ in our industry is 0.05 (out of sample)", with two colleagues' remarks (0.15 too good to be legitimate, "rich off a 0.02"), is a practitioner's judgement with no horizon, asset, sample or design stated (p. 171). An $R^2$ depends on the horizon, on whether the regression is a time series or a cross-section, and on the target's noise. Treat it as an order of magnitude for one context, not a benchmark; his own example is a daily, single-index time series.
  *Published magnitudes*. For annual U.S. equity-premium forecasts with forecasts beginning twenty years into each sample, Welch and Goyal's out-of-sample adjusted $R^2$ (in percent) is negative for every variable in their Table 1 (p. 1470). The dividend–price ratio has in-sample 0.49 and out-of-sample −2.06, the earnings–price ratio 1.08 and −1.78, the term spread 0.16 and −2.42 [s7-welch, Table 1]. Their abstract concludes that the models "predicted poorly both in-sample (IS) and out-of-sample (OOS) for 30 years" (p. 1455). An out-of-sample 0.05 would be far above these for that target and horizon.
  *Monthly forecasts with restrictions*. Campbell and Thompson forecast monthly U.S. excess returns, evaluated from 1927 or twenty years after each series starts. Their restricted forecasts set a coefficient to zero when it has the "wrong" sign, and a forecast of the equity premium to zero when it is negative. With both restrictions, 3 of 4 valuation ratios and 5 of 7 interest-rate variables have a positive out-of-sample $R^2$, against 2 of 4 and 3 of 7 without [campbell-thompson-2005-nber, Table 1, panels A–D, pp. 6–7]. They judge such an $R^2$ against the squared Sharpe ratio: a mean–variance investor's proportional gain in expected return from the predictor is about $R^2/S^2$ (eqs. (8)–(9), pp. 9–10).
  With a monthly Sharpe ratio of 0.108, so $S^2 = 1.2$ percent, the earnings–price ratio's out-of-sample $R^2$ of 0.25 percent raises the average monthly return by $0.25/1.2 = 21$ percent. That is about 3 percent a year for unit risk aversion and 1 percent for risk aversion three (p. 10). So a monthly out-of-sample 0.05 would be twenty times their earnings–price figure, against an $S^2$ of about 0.012. The paper was published in the *Review of Financial Studies* (2008).
- *Bucket study.* On p. 172 the text gives the two most bearish quintiles average next-day returns of −2.35 and −0.19 percent. In Exhibit 9.3 those figures label the signal buckets on the horizontal axis, and the bars, the next-day returns, run from about −0.03 to +0.13 percent. The monotonic ordering holds in the figure; the text's return figures do not match it.
- *Ratios.* Narang's information ratio with a zero benchmark is return over volatility, not active return over tracking error, and his Sterling ratio, the variability of below-average returns, is a downside-deviation measure. Other sources define both differently.
- *Time decay and crowding.* A fast decay shows that the strategy depends on timely information. It does not identify crowding: in Narang's Microsoft example the move is a public announcement priced at the open, and a signal that is early improves with delay. His crowding inference is an interpretation without evidence in the chapter. The two crowding studies available at abstract level measure crowding from common holdings and infer it from fire-sale dislocations; neither uses the decay of a delayed signal (common-investor risk).

**See also.** hypothesis-first research (what is being tested); parameter overfitting (sensitivity to parameters, the next diagnostics); cost and short-availability assumptions (what the returns assume); relative performance evaluation (the same statistics applied to managers, and smoothed returns).

### Pitfall 2.3 Parameter overfitting: an isolated optimum, a model that explains its sample too closely, and the balance against underfitting

**Statement.** *The error.* A parameter set is chosen because it performed best in a sample, although nearby values perform very differently. A model with many parameters or signals explains the past "so closely … that it might be useless for predicting the future" [s7-narang, chap. "Research", secs. 9.3.2.10–9.3.3, pp. 176–179].

*How it arises.* Training searches parameters and keeps the best (p. 167). In Narang's terms, overfitting is "expecting too much from the data you have", treating an unrepresentative sample as the population. An isolated peak, his point D in Exhibit 9.5, is likely a "spurious coincidence in the

fitting period" (pp. 177–178). Re-using out-of-sample data, by refitting after seeing it or by building models already known to work in that period, turns it into in-sample data ("burning data", pp. 183–184).

*Checks.* Vary each parameter slightly. In Narang's P/E example, a cheap threshold of 12 against 11 and an expensive threshold of 50 against 49 should give similar results, and if they do not, he would distrust and use neither (sec. 9.3.2.10, p. 176). Plot outcome against parameter value (Exhibit 9.5, p. 178). Compare the out-of-sample $R^2$ with the in-sample $R^2$: a ratio of about half or better "is considered a good thing" (p. 183).

*Remedies.* Choose a value with a margin of safety on a broad plateau, his point B, over the plateau's highest point near a cliff (C) or a lonely peak (D) (pp. 177–179). Prefer parsimonious models, since models with many parameters or signals carry more risk of overfitting (p. 179). Separate research from strategy selection, or withhold data from researchers, to limit burning (pp. 184–185).

**Conditions.** Practitioner view as of 2024. In the third edition Narang presents overfitting as one side of a balance. Overfitting is a Type I error and underfitting a Type II error; as easy alpha became scarce from about the early 2010s, and as data and methods improved, narrower and more fitted models became reasonable (pp. 180–182). How much fitting is warranted depends on the competition for alpha and on the strength of the prior relative to the data; he believes the industry may overestimate the value of more fitting, especially beyond the shortest horizons (pp. 182–183). Einstein's "as simple as possible, but not simpler" is his caution against oversimplifying (p. 179).

**Practice.** Interpretation: smooth performance across nearby parameters is a sign of robustness to the parameter, not evidence that the strategy has an edge. A whole family of rules, each stable in its own parameters, can be the survivor of a search over many families. The adjustment for the number of trials is in the deflated Sharpe ratio. Narang's "flat" criterion is local: his preferred point B lies on a plateau in a curve that has cliffs, so it does not require a curve without jumps.

**Pitfalls.** Fewer assumptions do not guarantee a better model: parsimony must be balanced against underfitting and the role of priors. His claim that simpler theories are better because more easily tested is attributed to Popper (p. 179) .

**See also.** backtest evaluation diagnostics (the statistics being optimised); hypothesis-first research (theory before fitting).

#### Pitfall 2.4 Backtest cost and short-availability assumptions: costs set too high or too low, and shorts the strategy could not have borrowed

**Statement.** *The error.* A backtest assumes trading costs, or the ability to short, that the strategy would not have had. A backtest has no record of what the strategy would have cost, because it did not trade, so costs, especially market impact, must be assumed [s7-narang, chap. "Research", sec. 9.3.5, p. 185].

*How it arises, costs.* Narang's example: a model right 55 percent of the time, making \$0.01 a share, and wrong 45 percent, losing \$0.01, earns \$0.10 per 100 shares if trading is free. At an all-in cost of \$0.01 a share it breaks even on its winners and loses \$0.02 on its losers, −\$0.90 per 100 shares (pp. 185–186). Overestimating costs makes a quant hold positions longer than is likely optimal; underestimating them makes the portfolio turn over too fast and "bleed" costs (p. 186).

*How it arises, shorts.* In a U.S. market-neutral or long/short equity backtest, the best short candidates may be on hard-to-borrow lists, stocks the broker generally restricts from shorting because it cannot locate shares to borrow. The live strategy then cannot place those shorts and substitutes inferior ones, so the short book's backtested value may not exist (p. 186).

*Checks and remedies.* Narang advises erring towards overestimating costs if one must err, though "it is always preferable to get the cost estimation approximately right" (p. 186). He notes that making a model aware of past hard-to-borrow status is not trivial, because such historical data are hard to come by (p. 186). Interpretation: compare the backtest's cost assumptions with pre-trade cost forecasts and with realised costs measured after trading (transaction cost analysis). Test how the result changes as assumed costs vary and when hard-to-borrow names are excluded or charged their fees.

**Conditions.** Practitioner account as of 2024; U.S. rules as cited in the linked entry. Narang's example ignores how often the position turns over and how costs scale with size; it shows only that a thin edge per share can be smaller than the cost. His short-availability point concerns U.S. equities. He says that shorting without locating shares "would be considered a naked short sale, which is illegal in the United States" (p. 186). That is broader than the rule: Regulation SHO bars a broker-dealer from accepting a short-sale order, or selling short for its own account, without a borrow, an arrangement to borrow or reasonable grounds to believe the security can be borrowed, with exceptions such as bona-fide market making, and imposes close-out duties for fails (securities lending for short sales).

**Practice.** Interpretation: Narang discusses whether shares can be borrowed, not what borrowing costs. A short backtest also needs the lending fee or rebate, which is high for securities on special (lending fees and rebates), and allowance for recall and buy-in risk (securities lending for short sales). Neither is in the schematic comparison or in Narang's passage.

**Pitfalls.** Historical hard-to-borrow lists are difficult to obtain; a reliable short-sale backtest requires point-in-time borrow data. "Overestimation … hold positions for longer than optimal" and "underestimation … high turnover and bleed" match Narang.

**See also.** transaction cost analysis (cost forecasts and measurement); transaction costs (components); securities lending for short sales (locate and borrow); lending fees and rebates (borrow cost); backtest evaluation diagnostics.

#### Method 2.5 Standard errors of an estimated Sharpe ratio: IID returns, serial correlation and time aggregation

**Statement.** Lo derives the asymptotic distribution of the estimated Sharpe ratio $\widehat{SR} = (\hat{\mu} - R_f)/\hat{\sigma}$ by the delta method [wide-lo-sharpe, "IID Returns", eqs. (5)–(9), pp. 37–38].

*Lo's statement.* Lo asserts that under IID returns "with finite mean $\mu$ and variance $\sigma^2$" the estimated variance satisfies $\sqrt{T}(\hat{\sigma}^2 - \sigma^2) \to N(0, 2\sigma^4)$, independently of $\hat{\mu}$ (eq. (5), p. 38). From this he obtains $V_{\text{IID}} = 1 + \frac{1}{2}SR^2$ and $\text{SE}(\widehat{SR}) \approx \sqrt{(1 + \frac{1}{2}\widehat{SR}^2)/T}$, with a 95 percent interval $\widehat{SR} \pm 1.96\,\text{SE}$ (eqs. (8)–(10), p. 38).

*What the derivation needs.* Finite mean and variance are not enough for Lo's general argument, though they are for one special case.

- *Sufficient in general.* A finite *fourth* moment and $\sigma > 0$ give $\sqrt{T}(\hat{\sigma}^2 - \sigma^2)$ a normal limit with variance $\nu_4 - \sigma^4$, so the joint delta method applies to the smooth map $(\mu, \sigma^2) \mapsto (\mu - R_f)/\sigma$ at any $SR$.
- *Not necessary for every normal limit.* At $SR = 0$, $\sqrt{T}\,\widehat{SR} = \sqrt{T}(\hat{\mu} - \mu)/\hat{\sigma}$. The central limit theorem for the mean and the consistency of $\hat{\sigma}$ (the law of large numbers), both of which need only a finite variance, give $\sqrt{T}\,\widehat{SR} \to N(0, 1)$ by Slutsky's lemma. So centred IID Student-$t_3$ returns, whose fourth moment is infinite, still have a standard normal limit at $SR = 0$. Away from zero the variance term enters, and without a fourth moment the $\sqrt{T}$-scaled variance has no finite limit (the $t_3$ check below at $SR = 0.5$).
- The general IID variance is

$$V_{\text{IID}} = 1 - \gamma_3\, SR + \frac{\gamma_4 - 1}{4}\, SR^2,$$

  with $\gamma_3$ the skewness and $\gamma_4$ the kurtosis (not excess). Its middle term is the covariance of $\hat{\mu}$ and $\hat{\sigma}^2$, which is $\nu_3/T$ and so zero only when the third central moment is zero. Its last term uses $\text{Var}(\hat{\sigma}^2) \approx (\nu_4 - \sigma^4)/T$.
- Lo's $2\sigma^4$ needs $\nu_4 = 3\sigma^4$ and his asymptotic independence needs $\nu_3 = 0$. Normal moments ($\gamma_3 = 0$, $\gamma_4 = 3$) are sufficient for $V = 1 + \frac{1}{2}SR^2$ but not necessary. The reduction holds exactly when $SR = 0$ or $\gamma_4 = 3 + 4\gamma_3/SR$, so a symmetric non-normal law with kurtosis 3 gives the same $V$.

Lo himself uses the third and fourth moments $\nu_3, \nu_4$ for the robust estimator under IID returns and sets $\nu_3 = 0$, $\nu_4 = 3\sigma^4$ "now suppose that returns are normally distributed" (eqs. (25)–(26), p. 42). The finite-variance wording at eq. (5) is therefore an overstatement in the source; its formulas are

quoted unchanged above. The general expression is the one in the deflated Sharpe ratio's eq. (2). With 60 observations the standard error is 0.188 at a true Sharpe ratio of 1.50 and 0.303 at 3.00, in the units of the observation period (p. 38).

For non-IID returns (serial correlation, conditional heteroskedasticity) the variance comes from a GMM estimator, SE $\approx \sqrt{\hat{V}_{\text{GMM}}/T}$, under stationarity, using Hansen's (1982) GMM results (eqs. (13)–(15), pp. 39–40); Appendix A estimates the covariance by Newey and West's (1987) procedure. *Time aggregation*. Lo defines the $q$-period return as the *sum* of one-period returns, "ignor[ing] the effects of compounding" (eq. (16), p. 40); his note 7 (p. 51) adds that the sum is exact for continuously compounded returns, and says $SR(q) = \sqrt{q}\,SR$ "is correct only under the assumption of IID returns" (p. 40). For additive returns and a benchmark aggregated the same way, independence is not needed: a constant mean and variance with zero autocovariances suffice, since the variance of the sum is then $q\sigma^2$. For stationary returns with autocorrelations $\rho_k$,

$$SR(q) = \eta(q)\,SR, \qquad \eta(q) = \frac{q}{\sqrt{q + 2\sum_{k=1}^{q-1}(q-k)\rho_k}}$$

(eqs. (17)–(20), p. 40), with a closed form for AR(1) returns (eq. (22), p. 41). A monthly first-order autocorrelation of −20 percent gives an annual factor of 4.17, against 3.46 under IID returns and 2.88 at +20 percent (Table 2, p. 41). The scaling is exact only for these additive quantities. For compounded simple returns over a compounded benchmark it is an approximation: with IID lognormal monthly returns (log mean 0.8 percent, volatility 4.5 percent, benchmark 0.2 percent), the exact annual Sharpe ratio is 0.515, against $\sqrt{12} \times 0.155 = 0.538$.

**Notation.** $T$ the number of return observations; $SR$ the Sharpe ratio per observation period; $q$ the aggregation length (12 for monthly to annual); $\rho_k$ the $k$th autocorrelation of returns.

**Conditions.** Timeless (statistics), published 2002. Asymptotic: the approximations are large-$T$ results [wide-lo-sharpe, p. 38]. The general $V_{\text{IID}}$ is derived for IID returns with a finite fourth moment and $\sigma > 0$; at $SR = 0$ a finite variance suffices for the $N(0,1)$ limit. $1 + \frac{1}{2}SR^2$ holds under normal third and fourth moments, and more generally whenever $\gamma_4 = 3 + 4\gamma_3/SR$. The GMM version relaxes independence but needs stationarity and finite moments for its central limit theorem. The $\sqrt{q}$ rule needs additive returns, a consistently aggregated benchmark, a constant mean and variance, and zero autocovariances.

*Counterexamples*, independent of the entry's algebra:

- *Skewed returns.* For IID two-point returns with Sharpe ratio 0.22, skewness −2.7 and kurtosis 8.1, Monte Carlo gives $T\,\text{Var}(\widehat{SR}) = 1.70$, matching the general 1.68, not $1 + \frac{1}{2}SR^2 = 1.02$.
- *Infinite fourth moment.* For Student-$t_3$ returns (finite variance, infinite fourth moment) with $SR = 0.5$, $T\,\text{Var}(\widehat{SR})$ rises from 2.5 to 10.0 as $T$ goes from 250 to 16,000, so it has no finite limit. Centred at $SR = 0$, the same returns give $\text{Var}(\sqrt{T}\,\widehat{SR}) = 0.99$ and a two-sided 5 percent tail frequency of 4.8 percent at $T = 10{,}000$, as the zero-$SR$ limit predicts.
- *Normal moments not necessary.* The three-point law $\{-\sqrt{3}, 0, \sqrt{3}\}$ with probabilities $\{1/6, 2/3, 1/6\}$ (skewness 0, kurtosis 3, not normal), shifted to $SR = 0.5$, gives $T\,\text{Var}(\widehat{SR}) = 1.11$ against $1 + \frac{1}{2}(0.5)^2 = 1.125$.
- *Dependent but uncorrelated returns.* An uncorrelated but dependent process, $e_t = z_t z_{t-1}$, obeys the $\sqrt{12}$ rule.

A finite fourth moment is sufficient for the general expansion, not necessary in every special case; the expression $1 + \frac{1}{2}SR^2$ is not exclusive to normal distributions. The scale factor needs the first $q - 1$ autocorrelations, themselves estimated. $\eta(q)$ from eq. (20) and eq. (22) agree, and eq. (22) reproduces the Table 2 cells to two decimals; eq. (9) reproduces the 0.188 and 0.303.

**Practice.** Positive serial correlation makes multi-period variance grow faster than $q$, so $\sqrt{q}$ annualisation overstates the annual ratio; Lo's hedge-fund example finds annual Sharpe ratios "overstated by as much as 65 percent" and rankings that "can change dramatically" once serial correlation is accounted for (abstract, p. 36). As a fraction of the ratio, the normal-moment standard error tends to $\sqrt{1/(2T)}$ as $SR$ grows (eq. (11), p. 38); with general moments the limit is $\sqrt{(\gamma_4 - 1)/(4T)}$. Either way, a high Sharpe ratio is no more precisely known in relative terms.

*Comparing two Sharpe ratios*. Ledoit and Wolf observe that the common test of Jobson and Korkie, as corrected by Memmel, "is not valid when returns have tails heavier than the normal distribution or are of time series nature", because its delta-method variance assumes i.i.d. normal returns (abstract and secs. 1–2, PDF pp. 2–4). They work with the means and uncentred second moments of the two return series, take a HAC standard error for $\widehat{Sh}_i - \widehat{Sh}_n$, and recommend instead a studentized circular block bootstrap interval, declaring the ratios different when zero lies outside it; the block size is calibrated by simulation (their Algorithm 3.1). The bootstrap resamples the observed pairs, so no null-restricted data are needed. This is the paired test for comparing two strategies; it is not a correction for having tried many strategies.

**Pitfalls.** A standard error for one strategy is not a test of the best of many: selecting the maximum over trials needs deflation.

**See also.** backtest evaluation diagnostics (where the point estimate is reported); deflated Sharpe ratio; return measures.

**Further reading.** Ledoit and Wolf's test of Sharpe-ratio differences; the stationary bootstrap.

### Method 2.6 The deflated Sharpe ratio: the probability that the selected strategy's true Sharpe ratio is positive, given the number of trials, their dispersion and non-normal returns

**Statement.** For $N$ independent trials whose estimated Sharpe ratios $\{\widehat{SR}_n\}$ are normal with mean $E[\{\widehat{SR}_n\}]$ and variance $V[\{\widehat{SR}_n\}]$, the expected maximum for $N \gg 1$ is approximately [alpha-dsr, eq. (1), p. 7]

$$E[\max\{\widehat{SR}_n\}] \approx E[\{\widehat{SR}_n\}] + \sqrt{V[\{\widehat{SR}_n\}]}\Big((1-\gamma)Z^{-1}\big[1 - \tfrac{1}{N}\big] + \gamma Z^{-1}\big[1 - \tfrac{1}{N}e^{-1}\big]\Big),$$

with $\gamma \approx 0.5772$ the Euler–Mascheroni constant and $Z$ the standard normal distribution function. The deflated Sharpe ratio is the probabilistic Sharpe ratio evaluated at the threshold that this maximum takes when the true Sharpe ratio is zero (eq. (2), p. 8):

$$\widehat{DSR} = Z\left[\frac{(\widehat{SR} - \widehat{SR}_0)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3\widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4}\widehat{SR}^2}}\right], \quad \widehat{SR}_0 = \sqrt{V[\{\widehat{SR}_n\}]}\Big((1-\gamma)Z^{-1}\big[1 - \tfrac{1}{N}\big] + \gamma Z^{-1}\big[1 - \tfrac{1}{N}e^{-1}\big]\Big),$$

where $\widehat{SR}$, $T$, $\hat{\gamma}_3$ (skewness) and $\hat{\gamma}_4$ (kurtosis) belong to the selected strategy. "Essentially, DSR is a PSR where the rejection threshold is adjusted to reflect the multiplicity of trials" (p. 9).

**Notation.** All Sharpe ratios in the formula are per observation period, not annualised; $\hat{\gamma}_4$ is kurtosis (3 for the normal), not excess kurtosis.

**Conditions.** The threshold assumes the trials' Sharpe ratios are normally distributed around a common mean and that $N$ counts *independent* trials [alpha-dsr, p. 7]; correlated configurations must be reduced to an effective number, which the formula does not supply. *The authors' example* (pp. 9–10): $N = 100$ independent trials, $V = \frac{1}{2}$ (annualised), $T = 1250$ daily returns, skewness −3, kurtosis 10 and a selected annualised Sharpe ratio of 2.5 give $\widehat{SR}_0 \approx 0.1132$ (daily, 250 observations a year) and $\widehat{DSR} \approx 0.9004 < 0.95$; after only 46 trials it would have been 0.9505, and with normal returns 0.9505 after 88. All four numbers are reproduced.

**Practice.** For a deflated Sharpe ratio over many recorded variants, the variance across trials and the trial count therefore have to be logged from the first experiment. The authors also suggest planning trials in advance and an optimal-stopping rule, sampling about 37 percent of the justifiable configurations first (pp. 10–11), which is a heuristic, not part of the statistic. They describe Harvey and Liu's false-discovery threshold as complementary and suggest computing the statistic on both thresholds (p. 9).

**Pitfalls.** Parameter stability does not deflate anything (parameter overfitting). Interpretation: $\widehat{DSR}$ is a probability under the stated assumptions, not the out-of-sample Sharpe ratio; a large undisclosed search, or an $N$ that ignores discarded families, overstates it. The 2026 unification by López de Prado and Porcu (`w4-lopezdeprado-porcu-dsr`, *JPM*, online 26 September 2026) is not covered here. Locators refer to the author version of 31 July 2014, not the published *JPM* version.

**See also.** Sharpe-ratio standard errors; purged cross-validation (the other failure of reused test sets); tests of skill against luck.

**Further reading.** `wide-bailey-sharpe-frontier`, on the probabilistic Sharpe ratio and the minimum track-record length.

#### Method 2.7 Controlling false discoveries across a family of strategies: familywise error, Bonferroni and Holm, the Reality Check, and Romano and Wolf's stepwise method

**Statement.** Romano and Wolf set up $K$ strategies compared with a common benchmark, testing $H_{0,k} : \theta_k \leq 0$ against $\theta_k > 0$ for each, where $\theta_k$ is, for example, a difference in mean return or Sharpe ratio. They seek *strong* control of the familywise error rate, the probability of rejecting at least one true null whatever the configuration of true and false nulls [`wide-romano-wolf-stepdown`, working paper of October 2003, sec. 2.2, pp. 6–7].

- *Bonferroni* rejects $H_{0,k}$ if $p_k < \alpha/K$.
- *Holm* orders the $p$-values and rejects $H_{0,(k)}$ if $p_{(j)} < \alpha/(K - j + 1)$ for all $j \leq k$. Both control the error rate but are conservative, "assuming a worst-case dependence structure" (sec. 2.3, pp. 7–8).
- *White's Reality Check* uses the $1 - \alpha$ quantile of $\max_k(w_k - \theta_k)$, the largest deviation of the test statistics from their true values, to decide only whether the *best* strategy beats the benchmark (p. 9).
- *StepM* (Algorithm 3.1, pp. 9–10) relabels strategies by descending statistic. It rejects every $H_{0,r_k}$ whose interval $[w_{r_k} - c, \infty)$ excludes zero, with $c$ that max-quantile over the strategies not yet rejected, and repeats on the remainder until nothing more is rejected. The quantiles are estimated by a bootstrap of the data (Algorithm 3.2), which for time series is a block or stationary bootstrap (Scenario 3.2, p. 11). A studentized version (Algorithm 4.1) is advocated when feasible.

**Notation.** $w_k$ the test statistic for strategy $k$ (for example $\sqrt{T}$ times an estimated difference); $\alpha$ the error level; FWE the familywise error rate; FDR the false discovery rate.

**Conditions.** Methods published 1995–2005. StepM "asymptotically controls the FWE" and rejects every false null with probability tending to one, provided the sampling distribution of $\sqrt{T}(W_T - \theta)$ has a continuous non-degenerate limit and the bootstrap estimates it consistently (Assumption 3.1 and Theorem 3.1, pp. 10–11). With time series this needs stationarity, moment and mixing conditions, and a time-series bootstrap. The *false discovery rate* is a less strict target that will "discover" more strategies, a known fraction of them falsely, as Romano and Wolf summarise it (p. 3).

*The original papers.*

- *Benjamini and Hochberg* define $Q = V/(V + S)$, "the proportion of the rejected null hypotheses which are erroneously rejected" ($Q = 0$ when nothing is rejected), and the FDR as $E(Q)$. When all nulls are true the FDR equals the familywise error rate, so controlling it controls the FWER in the weak sense; otherwise it is no larger [`wide-benjamini-hochberg`, sec. 2.1, p. 291]. Their step-up procedure rejects $H_{(1)}, \ldots, H_{(k)}$ for the largest $k$ with $P_{(k)} \leq (k/m)q^*$. Theorem 1 proves FDR control at $q^*$ "for independent test statistics and for any configuration of false null hypotheses" (sec. 3, eq. (1), p. 293). Dependent statistics, the usual case for strategies on one market, are outside that theorem.
- *White* tests "the null hypothesis that the best model encountered in a specification search has no predictive superiority over a given benchmark model", posed as the intersection of $I$ one-sided hypotheses [`catalogue-0731`, abstract and sec. 1, pp. 1097–1098]. The statistic is $\max_k n^{1/2}\bar{f}_k$. Its null distribution is taken at the configuration "least favorable to the alternative", $Ef_k = 0$ for every $k$ (sec. 2, p. 1102). The Bootstrap Reality Check resamples with Politis and Romano's stationary bootstrap, blocks of geometric length with mean $1/q$ (p. 1104), with $q = 1$ when the loss differentials are a martingale difference sequence under the null (sec. 3, p. 1109).

Both statements match Romano and Wolf's account above. One consequence is the entry's own reading, not White's text: pricing the null at $Ef_k = 0$ for all $k$ makes the test conservative when many clearly inferior strategies are included.

**Practice.** Own Monte Carlo, with 20 equicorrelated statistics ($\rho = 0.8$), 5 false nulls and $\alpha = 5$ percent: the error rates are 0.014 (Bonferroni), 0.017 (Holm), 0.043 (single-step max) and 0.046 (StepM), all within the level, and the shares of false nulls rejected are 0.57, 0.58, 0.73 and 0.74. This is the power that dependence-aware methods gain, as the paper argues. For a research log, the family must be every strategy tried, including discarded ones, as for the deflated Sharpe ratio.

*Hansen's test of superior predictive ability.* Hansen argues that pricing the null at the least favourable point, $\mu_k = 0$ for all $k$, makes the Reality Check "an asymptotically biased test" that "is sensitive to the inclusion of poor and irrelevant models" [`hansen-spa-2001-wp`, abstract and sec. 1, PDF pp. 2–3]. His remedy keeps a model's estimated mean in the null distribution only if it is not clearly negative: $\hat{\mu}_k^c = \bar{X}_{n,k}$ if $\bar{X}_{n,k} < -A_{n,k}$ and 0 otherwise, with thresholds that vanish but more slowly than the law-of-the-iterated-logarithm rate, so that poor models drop out as $n \to \infty$ and the $p$-values are consistent (sec. "A Consistent Estimate of the Asymptotic Distribution", printed pp. 25–26). This bears on the entry's own remark above: the conservatism from including clearly inferior strategies is the bias Hansen's correction removes.

*How high a hurdle for a published factor* (`alpha-harvey`, NBER WP 20592, October 2014). Harvey, Liu and Zhu count 316 factors in published work and argue that "a newly discovered factor needs to clear a much higher hurdle, with a $t$-ratio greater than 3.0" (abstract, PDF p. 2). The hurdle depends on the method and the count: a Holm adjustment gives 3.64 at 316 factors and 3.29 if the count is cut to 113 "common" factors (sec. 4, PDF p. 26); across their correlation specifications, $t$-ratios of 3.9 and 3.0 control the familywise error rate at 5 percent and the false discovery rate at 1 percent (PDF p. 34). These are thresholds for the population of published factors, which omits the unpublished trials; they do not replace counting a researcher's own trials.

**Pitfalls.** Controlling the error rate across the strategies tested says nothing about strategies tried and not recorded. The guarantees are asymptotic and depend on the bootstrap's validity for the dependence present. Section and page numbers refer to the 2003 working paper, not the 2005 *Econometrica* version.

**See also.** deflated Sharpe ratio; Sharpe-ratio standard errors; parameter overfitting; tests of skill against luck.

**Further reading.** Romano and Wolf's *Econometrica* version of 2005 and Hansen's *JBES* version of 2005.

## 2.2 Point-in-time reference and market data

#### Data 2.8 Trading reference data: universe, symbology, corporate actions, static and exchange data, condition codes, special days, futures and options reference, event calendars, related and composite instruments

**Statement.** Velu, Hardy and Nehren call reliable reference data "the key foundation of a robust quantitative strategy development" and list what a research platform must hold [`s3-velu`, chap. 1, sec. 1.4.1, pp. 26–35]:

- the *trading universe*, which changes daily with listings and delistings; knowing when a stock stopped trading avoids survivor bias (p. 26);
- *symbology mapping* (ISIN, SEDOL, RIC, Bloomberg tickers) and *ticker changes*: the mapping "needs to persist as point-in-time data and allow for historical 'as of date' usage, requiring the implementation of a bi-temporal data structure", handling changes backward and forward (pp. 26–27);
- *corporate actions*: dividends (announcement and payment dates), splits, reverse splits, rights offers, mergers, spin-offs, changes in free float or shares outstanding, suspensions; prices and volumes must be adjusted backward for splits, and suspensions leave gaps that affect backtests (pp. 27–28);
- *static data*: country, sector (GICS, ICB, TOPIX's 33 sectors), primary exchange, quotation currency and quote factor, with USD price = local price × FX × quote factor and cent-quoted currencies such as GBp (pp. 28–29);
- *exchange data*: holiday calendars, session hours with auctions, lunch breaks and daylight-saving shifts, disrupted days, tick sizes, trade and quote lots, limit-up and limit-down rules and short-sale restrictions, kept historically "in order to run realistic historical backtests" (pp. 29–30);
- *condition codes* on trades and quotes, mapped so that, for example, off-exchange trades reported for regulatory purposes are excluded from accessible liquidity and from daily volume, high and low (p. 31);
- *special-day calendars*: half days, local holidays and exam days, month and quarter ends, index rebalance days with volume skewed to the close, and expiry days (pp. 31–32);
- *futures and options reference data*: expiry calendars, the most liquid contract, roll dates, contract size and quotation, session phases, and option chains (pp. 32–34);
- *event calendars* for macro releases and company events, *related tickers* (composite tickers, dual listings, ADRs and GDRs), *composite assets* with their constituents, cash components and divisors, and *latency tables* between data centres (pp. 34–35).

**Conditions.** Examples as of about 2018–2020; exchange rules and calendars change. Every table that changes over time must be stored with the dates on which each value applied and was known, so that a backtest sees the universe, identifiers, corporate actions, tick sizes and rules in force at each decision time. Velu, Hardy and Nehren's reasons for special days are practical: a single special day moves a 30- or 60-day ADV, and low-order autoregressive models underperform on and after such days, so modellers "often remove special days and model normal days first" and model special days separately (p. 32).

Futures liquidity moves between contracts at the roll (about 1.5 million E-mini contracts a day in the front month against 30,000 in the next before expiry, reversing in the last days, p. 33), so rolling-window statistics must blend contracts across roll dates.

**Practice.** Velu, Hardy and Nehren's ABC→DEF example (p. 27) shows the three queries a symbology table must answer: a 30-day history as of a date before the change (old ticker), after it (both tickers spliced), and a forward window for a position held across it. Examples they give: European equity volume 30–40 percent lower on U.S. holidays; Indonesia's longer Friday lunch break; U.S. daylight saving starting about two weeks before Europe's (pp. 29–30). These are illustrations as of the book's writing, not current calendars.

**Pitfalls.** Velu, Hardy and Nehren distinguish the drop in price when the dividend is paid from the volatility that may accompany the announcement (p. 27); interpretation, the price drop occurs on the ex-dividend date. Merger and spin-off data can support event-driven strategies (p. 28); exchange-rule histories must also be retained. The book describes what to collect; it does not test how much results change when it is ignored. A ticker is not an instrument key: tickers are display mnemonics, and a key needs venue, currency, instrument type, expiry, strike and option right where they apply, with corporate-action history and identifier validity dates. Validity dates are the point-in-time requirement above; the attribute list is an interpretation (compare `exp-gleif` and `s8-bitemporal`).

**See also.** market data levels; derived market statistics; financial data types; ETF trick (rolled futures series); index reconstitution impact; securities lending and short sales; CME Globex trading.

**Further reading.** See bitemporal reference tables. On delisting bias, Shumway finds that delistings "for bankruptcy and other negative reasons are generally surprises", that "correct delisting returns are not available for most of the stocks that have been delisted for negative reasons since 1962", and, from over-the-counter prices, that "the omitted delisting returns are large" [`wide-shumway-delisting`, abstract]. A universe that keeps delisted stocks but lacks their final returns is therefore still biased upward (Interpretation). Shumway and Warther's abstract, from the publisher's record as served by RePEc, quantifies it for Nasdaq. The missing returns of delisted stocks are "large and negative on average". A corrected return of $-55$ percent for missing performance-related delisting returns corrects the bias.

After the correction there is "no evidence that there ever was a size effect on Nasdaq" [`shumway-warther-1999`, abstract]. Interpretation: a substitute delisting return is an estimate for one exchange and period, to be reported as an assumption. Shumway's full text remains refused. See also CRSP files, Elton, Gruber and Blake on fund survivorship, and Ljungqvist, Malloy and Marston on vendor history rewriting.

### Data 2.9 Market data levels: Level I trades and quotes, Level II depth, Level III order messages, and the special order types that complicate them

**Statement.** Market data are sold "in various levels of granularity (and price!)", from exchanges as direct feeds or from vendors [`s3-velu`, chap. 1, sec. 1.4.2, p. 35]. *Level I* holds trades and top-of-book quotes, enough to rebuild the best bid and offer; quotes update roughly ten times as often as trades in liquid markets, and trade status and qualifiers (cancelled, late, odd lot, auction, intermarket sweep, average-price, venue) decide whether a print should update the last price and cumulative volume (pp. 35–36).

*Level II* adds every lit order-book update at every level and venue; many users keep five or ten levels (p. 36). *Level III*, message data, gives each order an identifier tracked through execution, cancellation or amendment; once messages from all exchanges are "consolidated into one single dataset ordered by timestamp", a full book can be built at any moment (pp. 36–37). Their example of a U.S. Level III record has timestamp (milliseconds after midnight), ticker, order identifier, message type (add buy, add sell, partial execution, partial cancel, full execution, delete, cross, non-displayed execution), shares, price (the last four digits decimals, so "divide by 10000"), participant identifier and market centre (Table 1.10, p. 37).

**Conditions.** U.S. feed examples as of about 2018–2020; TAQ codes as of the November 2022 specification. Exchange timestamps are matching-engine times; databases also store a reception time, and the latency between them must be accounted for when opportunities are short-lived (p. 36). Special order types change identifiers and priority in venue-specific ways: price-sliding orders ranked at the locking price as hidden and displayed one tick away; pegged and midpoint-peg orders; reserve orders whose displayed part is replenished with a new timestamp; discretionary orders; intermarket sweep orders (pp. 37–38). Velu, Hardy and Nehren advise weighing the value of raw data against simpler binned data, and note that most practitioners outside high-frequency trading build strategies on binned data (pp. 36 and 39).

*Timestamps, sequencing and availability*. Hasbrouck: when data from several markets are combined, "The quality of the time stamps depends on the accuracy of internal systems clocks, latencies in data processing and transmission, and the nominal reporting precision", and with several events in one reported second "the sequencing of reporting is suspect"; for many purposes what matters is "when knowledge of an event entered the public awareness", so the timing and sequencing of the real-time consolidated feed are the appropriate ones [`s6-hasbrouck`, chap. 10, p. 105]. Time aggregation makes events separated in time look simultaneous (p. 104).

For a feature in a backtest this gives three distinct times: when the event happened at the venue (the matching-engine timestamp), when it reached the consolidated feed, and when the strategy's own systems received it; a feature is available at a decision time only if it was received by then (Interpretation, consistent with the reception-time point above). Corrections and cancellations arrive later as separate records with their own status and qualifiers (above, pp. 35–36), so a feature computed from the corrected history uses information that was not available at the time.

*How TAQ marks them*. NYSE's Daily TAQ trades file, sorted by symbol, time and sequence number, carries a two-digit Trade Correction Indicator: 00 for a regular trade "not corrected, changed or signified as cancel or error"; 01 for an "Original trade which was late corrected", a record that "contains

the original time—HHMM and the corrected data"; 07 and 08 for originals later marked erroneous or cancelled; and 10, 11 and 12 for the cancel, error and correction records that follow them, the last containing "the correction time and the original 'incorrect' data". Late and out-of-sequence reports are flagged separately in the sale conditions (CTA and UTP codes L "Sold Last" and Z "Sold (Out of Sequence)", with T and U for extended hours) [nyse-daily-taq-spec, v4.0 of 2 November 2022, sec. 3, pp. 17–19].

*Replaying corrections*. In the trades file the *Time* field is when "the trade was published by the SIP" (field 1), and a 12 record carries the correction time [nyse-daily-taq-spec, sec. 3, PDF pp. 17–19]. A 01 record, however, places corrected data at the original time. A decision-time panel should therefore take the trade as first reported, with the original values recoverable from the 12 record, until the correction time. It should keep 07 and 08 originals until their error or cancel was published, and treat L and Z prints as known at publication, not at the trade time they report.

*Unresolved times and older editions*. Which time the 10 and 11 records carry is not stated. Field 8 defines them only as following the 07 and 08 originals, and three earlier editions use the same wording: the specification for 2 October 2006 to 31 July 2012 (V1.0c), v3.0a and v3.1 [nyse-daily-taq-spec-older, V1.0c, PDF p. 19; v3.0a, PDF p. 19; v3.1, PDF p. 17]. None of the four editions examined states it, so that question stays open. The editions differ elsewhere: V1.0c describes both 07 and 08 as originals "later signified as error", where later editions have 08 as cancelled, and v3.0a and v3.1 add "The final correction will be published".

*Holden and Jacobsen's screens*. Their sample is 100 stocks drawn at random, 20 from each quintile of trades per day, from April to June 2008 (34 million trades and 351 million quotes). The main-text screens keep quotes from 9:00 a.m. and trades from 9:30 a.m. to 4:00 p.m. They drop a venue's own crossed or locked quote, a side more than $2.50 from the previous midpoint when the spread exceeds $5, quotes with non-normal conditions, and zero or missing prices or sizes, which with the withdrawn-quote adjustment are read as withdrawn sides [wide-holden-jacobsen, published version, sec. III, p. 1758 and n. 16].

NBBO quotes and trades while the NBBO is crossed or locked are deleted (p. 1750). Their DTAQ quote file's "Quote Cancel/Correction" field was "always blank in our sample" (n. 12, p. 1754). No trade-correction filter is stated in the main text: the paper points to the Internet Appendix for detail (p. 1758).

That appendix prints the July 2013 SAS program for their *Monthly* TAQ recommendations, whose only trade filter keeps "trades in which the Correction field contains '00' and the Price field contains a value greater than zero" (`where corr eq 0 and price gt 0`) [wide-holden-jacobsen, Internet Appendix, sec. I and the program's step three, PDF pp. 1–2 and 19]: corrected and cancelled trades are dropped, not replayed. A filter on the final correction code uses after-the-fact information, appropriate for measuring liquidity but not for a decision-time replay (Interpretation). Only this printed program is covered here; their DTAQ code, linked from Holden's page, was not examined.

Their Interpolated Time assigns the $i$th of $I$ events in recorded second $s$ the time $s + (2i - 1)/(2I)$ (eq. (16), p. 1766), and they note that, like matching within the same second, it "uses future information, extending as much as one second into the future" (n. 29, p. 1767): an evaluation device for liquidity measures, not a point-in-time rule. With all three adjustments the monthly-TAQ gap to DTAQ vanished for percent quoted spread and shrank by 91 and 97 percent for percent effective spread and price impact (pp. 1750–1751); Jurkatis found the interpolation lowered trade-classification accuracy (trade-sign classification).

**Practice.** With Level III data one can model inter-arrival times, arrival and cancellation rates by distance from the touch, queue position and imbalance, and then ask about the impact of market orders, a limit order's chance of moving up the queue, the probability of earning the spread and the short-horizon direction of prices (p. 38). Interpretation: the levels above usually come with a separate trades stream (price, size, time, venue and condition codes), the industry's L1/L2/L3 labels are Levels I–III, and a sound rule is to choose the shallowest feed that identifies the event cleanly. Order entry (new, cancel, replace, reject and execution-report messages), public market data, drop copies (private execution and routing records) and clearing feeds are separate message streams.

Velu, Hardy and Nehren describe the same path: client orders reach a broker in FIX, an order management system keeps each order's state, an outbound gateway translates child orders into the venues' protocols ("Essentially all of them support a FIX protocol, but in most cases they also support a much faster 'native' protocol") and captures acknowledgements and executions, and clearing and settlement follow [s3-velu, chap. 11, sec. 11.1, pp. 383–386]. Drop copies and clearing-feed contents are an additional interpretation, not in the source; the FIX order-state and execution-report values are in the order lifecycle.

**Pitfalls.** The price field is divided by 10000, not 1000 (p. 37). Reconstructing the full book requires consolidation, and reception timestamps include latency. The Table 1.10 layout is the book's illustration; a real feed's specification, which changes, governs.

**See also.** trading reference data; order exposure decisions (hidden and reserve orders); static limit-order depth; information-driven bars.

**Further reading.** [wide-holden-jacobsen], published version and Internet Appendix (MTAQ program).

#### Data 2.10 Derived market statistics: daily activity data, intraday microstructure statistics, pre-computed aggregates and binned data

**Statement.** Velu, Hardy and Nehren group the statistics most strategies derive from binned or daily data [s3-velu, chap. 1, sec. 1.4.3, pp. 39–42]. *Daily activity*: open, high, low, close and previous close (adjusted for corporate actions), the last trade before the close, volume with its lit and dark split, auction volume and price, VWAP, short interest, days-to-cover and borrow utilisation, futures open interest and basis, index-level data, options open interest, Greeks and implied volatility, and cross-asset rates, CDS spreads and the dollar index (pp. 39–40).

*Intraday microstructure*: number and frequency of trades and quote updates, top-of-book size, depth, spread, trade size and ticking time, whose daily distributions can start an algorithm's day and be updated "through online Bayesian updates" (p. 41). *Pre-computed aggregates*: $X$-day ADV and auction volume, volatility, beta (plain or asymmetric) and correlation matrices (p. 41). *Binned data* discretise the day into bins of a few seconds to 30 minutes with the same features; volume and spread profiles "are rarely built with a granularity finer than one minute" to avoid noise from frictions (p. 42).

**Conditions.** Practitioner descriptions as of about 2018–2020. Aggregates "need to fully support the peculiarities described in the Reference Data section": special days skew distributions, and asynchronous markets distort beta and correlation (p. 41). Binned data allow discrete time-series methods and reduce dimension (390 one-minute rows per U.S. stock-day against millions of messages), but relationships stable on daily closes become noisier at higher frequency, and illiquid names leave empty bins (p. 42). Interpretation: each derived field must carry the time it became available, for example short interest only after its publication date, and pre-computed $X$-day aggregates must be computed through the decision date, not the end of the sample.

*Short-sale availability.* Nasdaq's daily Regulation SHO threshold-security files are one public point-in-time input. The files examined, for 10 January 2005, 2 January 2008 and 15 October 2025, are each a pipe-delimited list with a threshold flag and a trailing creation stamp (`20050110231003`, created after the close), and the 2025 file adds a Rule 3210 flag [nasdaq-regsho-threshold, files `nasdaqth20050110`, `nasdaqth20080102` and `nasdaqth20251015`]. A threshold security is one with substantial fails to deliver, which clearing participants must close out under Rule 203(b)(3) of Regulation SHO; the rule's compliance date was 3 January 2005 [finra-ntm-04-93, Discussion]. The list measures delivery failures, not borrow cost or availability. The sources here give no public point-in-time record of hard-to-borrow status or borrow fees, so those remain the gap Narang names.

**Practice.** Velu, Hardy and Nehren's uses are judgements to test, not results: the high–low range "usually gives a better indication of market sentiment than the simple close-to-close return"; "a large short interest usually indicates a bearish view from institutional investors"; days-to-cover sizes a potential short squeeze; futures basis against one's dividend estimates may show mispricing (pp. 39–40).

**Pitfalls.** Daily activity summaries differ from binned data (p. 42); both require consistent reference data when aggregating.

**See also.** Garman–Klass volatility (range-based estimators); spread measures; securities lending fees and rebates (utilisation, hard to borrow); activity-sampled bars; trading reference data.

### Data 2.11 Holdings and insider filings: Form 13F, Schedules 13D and 13G, Forms 3 and 4, their deadlines, and when their contents can enter a decision-time panel

**Statement.** U.S. deadlines in force on 15 October 2026 (eCFR point-in-time text):

- *Form 13F*: an institutional investment manager exercising investment discretion over section 13(f) securities with an aggregate fair market value of at least $100,000,000 "on the last trading day of any month of any calendar year" files "within 45 days after the last day of such calendar year and within 45 days after the last day of each of the first three calendar quarters of the subsequent calendar year" [ecfr-17cfr240-filing-rules, 17 CFR 240.13f-1(a)(1)]; only securities on the Commission's 13(f) list count and are reported (para. (c)).
- *Schedule 13D*: a person who becomes the beneficial owner of more than five percent of a covered class files "within five business days after the date of the acquisition" (240.13d-1(a)); a material change, including an acquisition or disposition of one percent or more of the class, requires an amendment "within two business days after the date of such change" (240.13d-2(a)).
- *Schedule 13G*: qualified institutions file within 45 days after the quarter in which they become obligated, or within five business days after the end of the first month in which ownership exceeds 10 percent (240.13d-1(b)(2)); passive investors under 20 percent within five business days after the acquisition (240.13d-1(c)).
- *Forms 3 and 4*: insiders (directors, officers and more-than-10-percent owners) file Form 3 "within 10 days after he or she becomes such beneficial owner, director, or officer" and report a change "before the end of the second business day following the day on which the subject transaction has been executed" [usc-15-78p, 15 U.S.C. 78p(a)(2)(B)–(C), 2023 edition], on Form 4 (240.16a-3(a) and (g)(1)).

Under Rule 0-3(a), papers whose last timely day "falls on a Saturday, Sunday or holiday … may be filed on the first business day following"; Regulation S-T sets the *filing date* of an electronic submission. Electronic filings may be transmitted on business days from 6 a.m. to 10 p.m. Eastern [ecfr-17cfr232-filing-date, 17 CFR 232.12(c)]. A direct transmission "commencing on or before 5:30 p.m." Eastern is "deemed filed on the same business day", and one commencing later is deemed filed the next business day (232.13(a)(2)). Forms 3, 4, 5 and 144 and Schedules 13D, 13G and 14N, with their amendments, are deemed filed the same day up to 10 p.m. (232.13(a)(4)). The filing date can be adjusted for technical difficulties (232.13(b)).

**Conditions.** *History.* The 13D and 13G deadlines above come from the Commission's 2023 amendments, effective 5 February 2024, which shortened the initial 13D deadline from "within 10 days after the acquisition" to five business days; compliance with the revised 13G deadlines was not required before 30 September 2024 [fr-2023-22678, 88 FR 76896, 7 November 2023, pp. 76897 and 76942]. A panel covering earlier years must apply the deadline in force at each filing date.

*Deemed execution dates.* For trades under a plan or instruction that the insider does not time, the date the broker or plan administrator notifies the insider counts as the execution date, but no later than the third business day after the trade (240.16a-3(g)(2)–(4)); such Form 4 filings can therefore come up to five business days after the trade. Velu, Hardy and Nehren's footnotes ("within 10 days" for 13D; Form 4 "within 2 days"; 13F managers "with over $100 million in assets" reporting quarterly) [s3-velu, chap. 1, sec. 1.4, p. 43, nn. 31–32] predate the 2024 change and simplify the 13F test, which counts 13(f) securities, not assets.

**Practice.** *Four timestamps, not one.* A decision-time panel should keep these apart for each filing:

- EDGAR's *acceptance* time of the submission;
- the official *filing date* under Rule 232.13. For a 10-K or 13F transmitted after 5:30 p.m. it is the next business day, which can make a deadline-day 13F late; a Form 4 is still filed the same day up to 10 p.m.;
- the time of *public dissemination* on EDGAR;
- the time the user's own system *first observed* the filing.

EDGAR's page *Determine the Status of My Filing* (dated 4 June 2024, last reviewed 31 October 2024) states that most submissions that begin transmission after 5:30 p.m. "will receive a filing date of 6:00 a.m. ET the next business day and will not be disseminated by EDGAR until the next business day", while Forms 3, 4, 5 and 144 (with amendments) and some MEF and POS 462B forms transmitted before 10 p.m. get that day's filing date and "will be disseminated by EDGAR until 10:00 p.m. ET on that date" [sec-edgar-filing-status, sec. *Filing Date of Electronically-transmitted Submissions*]. The page gives a day, not a time, for dissemination, and no source bounds it from above.

So the start of the next business day is not a conservative availability time: an after-cutoff 10-K or 13F cannot be public at 00:00, and may not be public until some later, unstated hour. The panel keeps two quantities. *Availability* is the recorded dissemination time, else the recorded first-observed time; with neither, it is *unknown* and the filing does not enter a decision-time panel on that record. A *lower bound* (acceptance; for an after-cutoff submission not on the same-day list, 6:00 a.m. on the next business day, after any holiday) only detects certain look-ahead: use before it is a leak, use after it is not thereby safe, and a recorded first observation earlier than the bound is a data error.

Schedules 13D, 13G and 14N are deemed filed the same day up to 10 p.m. under 232.13(a)(4), but the page's same-day dissemination list omits them, so their lower bound is acceptance alone. Any fixed delay a user adopts instead (a decision-time convention) is an assumption to state and vary, not a rule. The deadline gives only the latest date, and for 13F the reported positions can be up to about 45 days (plus a weekend or holiday roll) old when public.

Worked example: positions at 30 September 2026 are due by 14 November 2026, a Saturday, so by Monday 16 November; a Form 4 for a trade on Friday 20 November 2026 is due by Tuesday 24 November; a 13D for an acquisition on Monday 23 November 2026 is due by Tuesday 1 December, Thanksgiving (26 November) not counting. Amendments (13F/A, 13D/A) restate earlier filings and keep their own four timestamps; they never overwrite the original's (trading reference data). The same holds for information first withheld and later made public. Section 13(f)(4) lets the Commission "delay or prevent public disclosure" of 13F information [usc-15-78m, 15 U.S.C. 78m(f)(4), 2023 edition], and confidential-treatment requests follow Rule 24b-2, which requires them in paper [ecfr-17cfr240-24b-2, 17 CFR 240.24b-2, preamble].

Positions released later are known only from their release, so they enter the panel at the later filing's timestamps, not at the original period. These filing-date rules and timestamps are checked on synthetic records.

**Pitfalls.** The entry covers deadlines, not the data sets: confidential treatment of 13F positions, the structure of the SEC's 13F and N-PORT data sets and the forms' instructions are not covered. EDGAR's dissemination timing is stated here only from the status page above, which gives days, not times; the EDGAR Filer Manual is not covered. The 10 p.m. cut-off for 13D and 13G that the 2023 release mentions (88 FR 76897) is now confirmed in 232.13(a)(4). Federal holidays in the example are those of 5 U.S.C. 6103.

**See also.** trading reference data; inside information (Harris's account of the insider-reporting rules as of 2002); four types of financial data.

**Further reading.** The SEC's Form 13F frequently asked questions and the 13F list of section 13(f) securities.

### Method 2.12 Bitemporal reference tables: actual (valid) time against record (transaction) time for identifiers, corporate actions and vendor corrections

**Statement.** Fowler separates two time dimensions of a history [s8-bitemporal, sec. "The Two Dimensions"]:

- *actual* history, "what history should be given perfect transmission of information";
- *record* history, "how our knowledge of history changes".

He notes that these are also called *valid* (or effective) and *transaction* time, Snodgrass's terms used in SQL:2011 (n. 1). A value is then read as value(actual date, record date), such as "on Mar 25th, we thought Sally's salary on Feb 25th was $6500" (sec. "The Two Dimensions"). The other properties he gives:

- Retroactive changes rewrite actual history, but "record history itself is append only" (sec. "Append-only History").
- Each agent (department, vendor) has its own record times (sec. "Perspectives for Record Time").
- Storage can be a table with record and actual start and end dates, or *event sourcing*. Events must then carry both an actual date and a record date (sec. "Storing and Processing Bitemporal Histories").

**Notation.** Actual time $a$ (when a fact held in the world); record time $r$ (when this system or vendor recorded it); a query is $(a, r)$; "as known at decision time $t$" sets $r = t$.

**Conditions.** Concepts as published 2021. *Applied to trading reference data* (Interpretation, not in Fowler). Several kinds of data need both axes:

- identifier changes, such as tickers and CUSIP or ISIN reassignments;
- corporate actions and their adjustment factors;
- index membership;
- vendor corrections of prices and fundamentals;
- filings and their amendments, whose four timestamps are in holdings and insider filings.

A backtest at decision time $t$ must query $r \leq t$. Today's corrected values use $r$ = now and leak later knowledge into the past. Fowler's caution applies too. Bitemporality "does complicate a system quite significantly" and is avoidable when no action depends on retroactively changed values; but a backtest's actions do depend on what was known (sec. "Using Bitemporality").

**Practice.** Synthetic check. A vendor records a 2-for-1 split effective 2 March 2026 on 3 March and corrects it to 3-for-2 on 20 March. Querying the factor for 5 March:

- as known on 10 March, it returns 2.0, what a backtest dated then saw;
- as known now, it returns 1.5;
- as known on 2 March, it returns 1.0;
- the correction closes record ranges and appends rows, so earlier rows survive unchanged.

The check verifies this illustration's logic only; vendors' actual correction histories were not examined.

**Pitfalls.** Many vendor data sets keep only the latest actual history, and some keep no record history, so a recorded "as of" date can be a later reconstruction. Fowler's "actual time is always at or before record time" is stated for history; recorded future events, such as an announced split, reverse it (n. 2).

**See also.** trading reference data; holdings and insider filings; time-validity of labels and weights; time-valid experiment records.

**Further reading.** Snodgrass's *Developing Time-Oriented Database Applications in SQL*, which Fowler cites.

## 2.3 Operational controls for automated trading

#### Method 2.13 Operational controls for automated trading: reliability and recovery, unknown order state, audit trails, and the U.S. market-access rule

**Statement.** Harris describes the operational demands on automated trading systems of markets, "as well as brokers and dealers", as of 2002 [alpha-harris, chap. "Floor Versus Automated Trading Systems", sec. 27.7.1, pp. 549–551]. The systems must validate users, accept and process orders, report trades and report order status, reliably, securely and fast:

- *reliability*: trading stops when the computer or network is down, and traders will not use unreliable systems. Builders therefore eliminate single points of failure and, since failures are inevitable, invest in recovering from interruptions: fault-tolerant hardware; redundant computers and network connections through different vendors and physical paths; backup power; a remote "hot disaster recovery site" to which activity can switch at once; significant excess capacity for surges in demand; quality controls in software development; redundant data backups; and thorough testing by independent quality-assurance groups (pp. 549–550);
- *security*: encryption and user authentication, so that hackers cannot tamper with systems and impersonators cannot trade in others' names (p. 550);
- *speed*, because traders without the latest information decide badly (pp. 550–551).

An *audit trail* records the submission and disposition of every order, with detailed information on everything that happens to it. Regulators use it to decide whether traders have violated trading rules; floor rules require time stamps when orders are received and filled, sequential recording and immediate reporting; automated systems produce complete audit trails of all activity within them (sec. 27.1.1, pp. 544–545).

When a connection fails, Harris's examples show three distinct risks (box, p. 550). A trader whose accepted limit order may or may not have filled cannot safely trade elsewhere: if she does so without knowing, she risks trading twice. An arbitrageur who has bought one leg cannot sell the other, and is left unhedged. A speculator cannot exit a losing position; a stop order placed immediately after the purchase would have represented him while he could not communicate.

In the United States, Rule 15c3-5 requires a broker or dealer with market access, or that provides others with access to an exchange or alternative trading system, to establish, document and maintain risk-management controls and supervisory procedures [cfr-240-15c3-5, 17 CFR 240.15c3-5(b)]. These are reasonably designed, among other things:

- to prevent orders that exceed pre-set credit or capital thresholds ((c)(1)(i));
- to prevent erroneous orders exceeding price or size parameters, singly or over a short period, or "that indicate duplicative orders" ((c)(1)(ii));
- to restrict access to trading systems and technology to pre-approved, authorised persons and accounts ((c)(2)(iii));
- to assure that surveillance personnel receive immediate post-trade execution reports ((c)(2)(iv)).

The controls must be under the firm's direct and exclusive control, with a limited contractual allocation of some regulatory controls to broker-dealer customers ((d)); they must be reviewed regularly, at least annually, with an annual certification by the chief executive officer ((e)).

**Conditions.** Interpretation, a synthesis of the passages above and not a procedure stated in them: after a failure the states to separate are these.

- *Unknown order state*: an order may be unacknowledged, live, partly filled, filled, cancelled or rejected. Venue messages report these transitions (acknowledgements, rejections, cancellations and fills; automated trading architecture), and the trader's view is stale until they arrive.
- *Unknown fill state and position*: what matters for risk is the quantity filled and the resulting position, including the unhedged leg of a multi-leg trade.
- *Reconciliation*: establishing state from authoritative records, such as venue or broker execution reports and post-trade reports, before acting.
- *Retry*: resubmitting before reconciliation is how a trader trades twice. For U.S. broker-dealers with market access, rejecting orders that indicate duplicative orders is a required pre-trade control, but such a filter is a backstop, not a reconciliation.
- *Cancellation*: a cancel sent while state is unknown may cross with a fill, so its outcome also needs confirmation.
- *Recovery*: redundancy, backups and disaster-recovery sites restore service; they do not by themselves establish what happened to orders in flight.

**Practice.** Interpretation: the advice to prefer safe state reconciliation over blind resubmission follows from Harris's "trading twice" and is consistent with the duplicative-order control of Rule 15c3-5, which states the control, not the reconciliation procedure. Harris's stop-order example is a way to leave a resting instruction that protects a position while disconnected; it is not a substitute for knowing the order's state. The Knight Capital order (`wide-knight-capital`) is summarised in implementation failures.

**Pitfalls.** Keep three kinds of statement apart.

- *Description*: Harris's lists are descriptions of practice in 2002 markets, not requirements.
- *Law*: Rule 15c3-5 is U.S. law for broker-dealers with market access, as in its current eCFR text. It does not mention kill switches, reconciliation or "drop copies"; the immediate post-trade execution reports it requires are the nearest provision.
- *EU*: the algorithmic-trading obligations are in RTS 6 (`wide-mifid-rts6`). U.K. supervisory statements, industry guidance (`w2-fia-automated-controls`), Regulation SCI and audit-trail rules such as the U.S. consolidated audit trail are not covered here.

The cited passages support less than common practice:

- reliability: Harris names validating users; drop copies, replayable feeds, tamper-resistant logs and order validation are additions;
- timing: Harris states a need for speed, not bounded delays for each stage;
- market quality: Harris says only that traders will not use unreliable systems and that trading stops when systems are down;
- audit trails: Harris names front running, inappropriate order exposure, fraudulent trade assignment and prearranged trading as the floor scandals that audit trails help deter; allocation, linking private messages to public market state, and tests of best execution, front running, allocation and unauthorised trading are additions.

**See also.** trading architecture (venue and recorded messages, model checks); floor against automated systems; best execution.

**Further reading.** The adopting release of Rule 15c3-5 is `wide-sec-15c3-5`.

### Rule 2.14 EU organisational requirements for algorithmic trading (RTS 6): testing and deployment, annual self-assessment, kill functionality, pre- and post-trade controls, real-time monitoring and business continuity

**Statement.** Commission Delegated Regulation (EU) 2017/589 of 19 July 2016, supplementing Article 17 of MiFID II (Directive 2014/65/EU), sets the organisational requirements of investment firms engaged in algorithmic trading; it applies from 3 January 2018 (Art. 29) [`wide-mifid-rts6`, OJ L 87, 31.3.2017, pp. 417ff.]. Its main obligations:

- *Development and deployment*: clearly delineated methodologies to develop and test any algorithmic trading system, algorithm or strategy before deployment or substantial update, authorised by a person designated by senior management, ensuring among other things that it "does not behave in an unintended manner" (Art. 5); conformance testing with venues (Art. 6), separate testing environments (Art. 7), and predefined limits before deployment on the number of instruments, the price, value and number of orders, strategy positions and the number of venues (Art. 8).
- *Post-deployment*: an annual self-assessment and validation report drawn up by the risk-management function, audited by internal audit where it exists and approved by senior management, with deficiencies remedied (Art. 9); within it, stress tests of messaging and trade volumes at twice the highest of the previous six months, run without affecting production (Art. 10); review of material changes by a designated person (Art. 11).
- *Kill functionality*: the ability "to cancel immediately, as an emergency measure, any or all of its unexecuted orders submitted to any or all trading venues", and to identify which algorithm and which trader, desk or client is responsible for each order (Art. 12).
- *Business continuity*: documented arrangements covering adverse scenarios, back-up sites where appropriate, use of the kill functionality, shutdown of algorithms "without creating disorderly trading conditions" and management of outstanding orders and positions, reviewed and tested annually (Art. 14).
- *Pre-trade controls* for all instruments: price collars, maximum order values and volumes, and maximum message limits, with all orders sent included immediately in the limits; repeated-execution throttles after which the system is disabled until re-enabled by designated staff; market and credit risk limits; blocking of unauthorised traders and of orders that compromise risk thresholds; and an exceptional, temporary override procedure verified by risk management (Art. 15).
- *Real-time monitoring* of all algorithmic activity under the firm's trading code by the trader in charge and by an independent risk function, with alerts "generated within five seconds after the relevant event" and reachable staff (Art. 16).
- *Post-trade controls*, operated continuously, including reconciliation of the firm's "own electronic trading logs with information about its outstanding orders and risk exposures as provided by the trading venues", brokers, clearing members and data providers, "in real-time where the aforementioned market participants provide the information in real-time", and real-time calculation of outstanding exposure (Art. 17(1)–(3)).

**Conditions.** Jurisdiction and scope: EU investment firms engaged in algorithmic trading under MiFID II Art. 17; Articles 19–23 cover direct electronic access providers and Articles 24–26 general clearing members. As of October 2026 there is no amending act and one corrigendum (28 July 2017, German language version only), so the English text is the text in force. Whether Directive 2024/790 changed Article 17 itself, or ESMA has proposed revised standards, is unverified.

**Practice.** Interpretation: the regulation states *what* controls must exist and who signs them off, not their parameters; collars, limits and throttles are calibrated by the firm. Compared with the U.S. market-access rule (operational controls), RTS 6 names kill functionality, real-time reconciliation with venues and brokers, and annual stress tests explicitly; Rule 15c3-5 names none of these.

**Pitfalls.** The text is the Official Journal version; Annexes I (self-assessment criteria) and II are not covered. For national competent authorities' and ESMA's guidance and supervisory reviews see, for the U.K., `wide-fca-algo-review`. The U.K. retains its own version of this regulation after 2020.

**See also.** operational controls for automated trading; order-state reconciliation; implementation failures; circuit breakers.

**Further reading.** `w2-fia-automated-controls` (industry guidance).

### Mechanism 2.15 Reconciling order state after a failure: FIX status requests, execution reports, resend and possible-duplicate flags, and the EU reconciliation duty

**Statement.** The FIX application layer gives a client three tools for establishing what happened to its orders [`s6-fix`, FIX Latest, EP312, Orchestra repository of 8 September 2026]:

- *OrderStatusRequest* (35=H) "is used by the order submitter to generate an ExecutionReport(35=8) message with ExecType(150)=I (Order Status) back from the recipient"; OrdStatusReqID (790) can identify the request;
- *OrderMassStatusRequest* (35=AF) requests the status of all orders matching criteria in MassStatusReqType (585), and an ExecutionReport with ExecType=I is returned "for each order matching the criteria provided on the request with the order's status";
- the *ExecutionReport* (35=8) itself, which acknowledges, confirms changes, relays status and fills and rejects orders, and carries "the current state of the order as understood by the recipient (OrdStatus(39)) and the purpose of the message (ExecType(150))". Its OrdStatus precedence table ranks Pending Cancel (11) and Pending Replace (10) above Done for Day, Filled, Canceled, Partially Filled and New; Pending Cancel "DOES NOT INDICATE THAT THE ORDER HAS BEEN CANCELED"; an order "can pass from New to Rejected status"; and Pending New is reported "only … in response to an OrderStatusRequest(35=H)".

The session layer recovers lost messages: *ResendRequest* (35=2) initiates retransmission; resent messages carry PossDupFlag (43), which "indicates possible retransmission of the message with the sequence number in the MsgSeqNum(34) field", with OrigSendingTime (122); PossResend (97)

marks a message that "may have been sent earlier under another sequence number"; *SequenceReset* (35=4) with GapFillFlag (123) = Y fills gaps over messages that will not be resent [s6-fix, FIXT session layer, Orchestra file FIX.5.0SP2_EP247 of 16 May 2019]. In the EU, the firm "shall reconcile its own electronic trading logs with information about its outstanding orders and risk exposures" from venues, brokers, clearing members and data providers, in real time where they provide real-time information (RTS 6, Art. 17(3); RTS 6).

**Notation.** Tag numbers in parentheses; 35 is MsgType. ClOrdID (11) is the client's order identifier and OrderID (37) the recipient's.

**Conditions.** The documented method, assembled from those definitions (the arrangement is the entry's, each step is the source's): (1) restore the session and let the counterparty resend, treating PossDupFlag messages as possible duplicates of sequence numbers already processed and PossResend messages as possible duplicates of business content under new sequence numbers, so that a fill is applied once, keyed by its execution identifier; (2) request status, order by order (35=H) or in bulk (35=AF), and take OrdStatus and the cumulative and leaves quantities in the ExecType=I replies as the recipient's view; (3) compare with the internal order log and position, and with clearing or drop-copy records where available (the RTS 6 duty); (4) treat Pending Cancel and Pending Replace as unresolved, not as cancelled or replaced; (5) only then decide whether anything may be sent.

Step (5) needs facts that reconciliation alone does not supply. A reconciled, terminal order is not thereby a candidate for resubmission: a fully filled order is terminal and reconciled and has nothing left to send. A replacement is justified only when three conditions hold together:

- the parent instruction (the intent) is still valid, not withdrawn by the strategy or a risk control;
- every child order of that intent is reconciled and resolved;
- authorised quantity remains, after deducting filled and still-fillable quantity (live leaves, plus the open quantity of unresolved or broken orders).

*Complete for today is not ended.* "Terminal" here means that the order's lifetime has ended: Filled, Canceled, Rejected or Expired. Done for Day (OrdStatus 3) is "Order not, or partially, filled; no further executions forthcoming for the trading day". Expired is an order "canceled in broker's system due to TimeInForce(59) instructions". A GoodTillCancel order "remains in effect until it is either executed or canceled". LeavesQty "could be 0" when OrdStatus is Done for Day [s6-fix, FIX Latest, EP312, OrdStatus precedence table, codes 59 and field 151].

FIX 4.4's state-change matrices, as restated in OnixS's mirror of Appendix D, show the two cases. A day order ends Done for Day with LeavesQty 0 (A.1.b, "Assuming day order"). A GTC order reports Done for Day with 8,000 of 10,000 still leaves, and is restated the next morning (ExecType D, ExecRestatementReason "GTC renewal/restatement") before filling again (H.1.a) [onixs-fix44-appd, app. D, matrices A.1.b and H.1.a].

So Done for Day ends an order's lifetime only for a day order, and only where the venue's rules say no Expired or Canceled report follows. A good-till order (GTC, GTD, or TimeInForce unknown in the log) is dormant: it can fill in the next session for OrderQty less CumQty, whatever LeavesQty it reported. Its remaining quantity is released for replacement only after a confirmed cancellation or expiry. The replacement's size is that remaining quantity. This separation of order state, remaining authorised quantity and intent is the entry's own arrangement; FIX defines the order state, not the intent. Field-level rules for deduplication (ExecID, ExecRefID, corrections) are not covered here.

The state-change matrices are cited for Done for Day (A.1.b) and GTC renewal (H.1.a) from OnixS's FIX 4.4 restatement; the Orchestra files carry message and field definitions and the OrdStatus precedence table, not the matrices.

**Practice.** The Knight failure of 1 August 2012 was in part a state failure of this kind: "one part of Knight's order handling system recognized that the parent orders had been filled, this information was not communicated" to the router, which kept sending child orders, and Knight had no control "to compare orders leaving SMARS with those that entered it" [wide-knight-capital, paras. 16 and 21, pp. 6–7] (implementation failures). It covers a duplicate (PossDup) report applied once, a fully filled order that is *not* resubmitted, a partially filled cancelled order (180 of 300 proposed when its intent is valid, nothing when it is withdrawn), an order in Pending Cancel, a missing execution found from a status reply, and a GTC order Done for Day with zero reported leaves.

That order is dormant, not terminal: 8,000 could still fill, nothing is proposed, it is renewed and partly filled the next day, and 7,000 is released only after a confirmed cancel. A day order Done for Day is ended only under a stated venue rule. It illustrates the logic and is not a test against a venue.

**Pitfalls.** Venue behaviour varies: whether a venue supports 35=H or 35=AF, honours resend requests or answers only through drop copies is set in its rules of engagement, not by FIX. Interpretation: an ExecType=I reply is the recipient's view at the time it is sent; a fill in flight can still arrive after it.

**See also.** operational controls, whose unknown-state list this entry sources; order lifecycle; RTS 6.

**Further reading.** The Orchestra repository `github.com/FIXTradingCommunity/orchestrations`, commit `cd24169`; the FIX specification volumes and the FIX Session Layer specification.

### Evidence 2.16 Implementation failures in quantitative trading: Knight Capital (1 August 2012) and AXA Rosenberg (2007–2010) in the regulators' orders, against a practitioner's account

**Statement.** *Knight Capital.* On 1 August 2012 Knight's equity router SMARS, "while processing 212 small retail orders", routed millions of orders over 45 minutes and obtained "over 4 million executions in 154 stocks for more than 397 million shares", leaving net long positions of about $3.5 billion in 80 stocks and net short positions of about $3.15 billion in 74; Knight "lost over $460 million" [wide-knight-capital, para. 1, p. 2]. New code for the NYSE Retail Liquidity Program repurposed a flag that had activated "Power Peg", a function unused since 2003 whose cumulative-quantity check had been moved in 2005 without retesting; one of eight servers did not receive the new code, without a second technician's review or written procedures; on that server the flag triggered Power Peg, which sent child orders "without regard to the number of share executions" (paras. 13–16, pp. 5–6).

Ninety-seven automated e-mails referring to "Power Peg disabled" before the open were not designed as alerts and not acted on (para. 19, pp. 6–7); there was no control comparing orders leaving SMARS with those entering it, no firm-wide capital threshold linked to automated order blocking, a post-trade monitor (PMON) relying on human monitoring without automated alerts, and no incident-response procedure; uninstalling the new code from the seven correct servers made matters worse (paras. 21–27, pp. 7–8). The SEC found violations of Rule 15c3-5, censured Knight and imposed a $12 million penalty (paras. 9–10, pp. 3–4; sec. IV, p. 18).

*AXA Rosenberg.* An error introduced in April 2007 when linking a new risk model to the optimiser failed to scale inputs reported partly in decimals and partly in percentages, so the optimiser "did not give the intended weight to common factor risks"; found in June 2009, it was concealed on a senior official's instruction, fixed in September–November 2009, disclosed to the SEC's examiners in late March 2010 and to clients on 15 April 2010; it affected 608 of 1,421 portfolios and caused $216,806,864 of losses, which the respondents undertook to pay, with a $25 million penalty [sec-axa-rosenberg-2011, paras. 1–3, 10–11, 33; sec. IV, pp. 2–4, 10 and 16].

**Conditions.** Settled orders: findings are made "without admitting or denying" them and bind only the parties. Two cases, chosen because Narang cites them; they are case studies of controls, not estimates of how often such failures occur.

**Practice.** Narang's account [s7-narang, sec. 10.1.3, pp. 195–196] differs from the orders in detail: he gives Knight's loss as "more than $400 million in a mere 30 minutes" and attributes it to a bug that "caused a dormant piece of software to come back online, multiplying order sizes"; the order gives over $460 million in about 45 minutes and describes repeated child orders sent without regard to fills, not larger orders. His AXA figures ($217 million, $25 million) match the order; his "about three years later, in April 2010" measures from the introduction of the error (April 2007), not from its discovery (June 2009). His claims about Knight's later recapitalisation, a March 2011 ETF "process error" and AXA's assets under management are not in the orders .

For control design the orders point to: deployment review and removal of dead code, an output control on the router (orders out against orders in), automated linkage of exposure limits to order entry, alerts designed as alerts, and an incident procedure that does not change production blindly; RTS 6 names several of these as EU obligations (RTS 6, Arts. 8, 12, 15–17).

**Pitfalls.** Reading the AXA case as a mere "coding error" omits the violation the order identifies: concealment and misleading statements to clients.

**See also.** sources of strategy risk (Narang's taxonomy); operational controls; order-state reconciliation.

**Further reading.** The SEC's other technology-governance orders cited in the Knight order's n. 1 (NASDAQ 2013, NYSE 2012, EDGX 2011).

## 2.4 Sources of strategy risk and their monitoring

**Definition 2.17 Sources of risk in a quant strategy: model risk (inapplicability, misspecification, implementation errors), regime change and exogenous shocks, kept apart from funding contagion and operational failure**

**Statement.** Narang names four risks that are specific to quant strategies, or larger for them than for discretionary ones [s7-narang, chap. "Risks Inherent to Quant Strategies", secs. 10.1–10.4 and 10.6, pp. 191–211]. The first three are:

- *model risk*: "the risk that the strategy does not accurately describe, match, or predict the real-world phenomenon it is attempting to exploit". It can arise in any part of the strategy, from back-testing software and data-feed handlers to the alpha, risk, transaction-cost and portfolio-construction models and the execution algorithms. Small errors may build slowly and then "explode on a busy trading day" (sec. 10.1, pp. 191–192). Its three common sources:
  - *inapplicability of modelling*, in two forms: using quantitative modelling on a question it cannot answer, and misapplying an otherwise valid technique. His example of the second is VaR built on correlations and historical volatility, which assumes normal distributions and linear relationships that market data often lack (sec. 10.1.1, pp. 192–193);
  - *misspecification*: a model that badly describes the world. The common case is a model that works most of the time and fails in rare events. His example is quants who in August 2007 found they had modelled the liquidity risk of large-cap U.S. stocks from their own holdings alone, when the aggregate size of similar holdings being liquidated mattered more (sec. 10.1.2, pp. 193–195);
  - *implementation errors*: errors in programming or in the architecture of systems, "perhaps [the] most common" and usually the least painful (sec. 10.1.3, pp. 195–196). His examples are a reversed sign in execution software, Knight's August 2012 loss and AXA Rosenberg's risk-model coding error, which he reports without sources, and an unnamed firm whose execution server restarted before its alpha server and, seeing no signals, sold 80 percent of the portfolio, which then had to be bought back;
- *regime change*: models rest on relationships in historical data, and when a structural shift changes an instrument's behaviour or the relationships between instruments "dramatically and quickly", the strategy typically suffers (sec. 10.2, pp. 196–200). His examples are the Schwab and Merrill Lynch relationship from 1996 to 2007, and the value-against-growth spread in 2007–2008, whose reversals, he says, came with reversals in sectors, currencies, bonds and commodities, so that regime changes "tend to occur across many levels simultaneously" (p. 199);
- *exogenous shocks*: big moves driven by information outside the market, which models built on market data cannot explain. His examples are 11 September 2001, the start of the Iraq war in 2003, the Bear Stearns rescue in March 2008 and the SEC's 2008 change to short-sale rules for financial stocks. He calls the risk difficult to do anything about "other than with discretionary overrides" (sec. 10.3, pp. 200–202).

The fourth, contagion or common-investor risk, is in common-investor contagion.

**Conditions.** One practitioner's classification (third edition, 2024), whose examples run from 1996 to 2012. Narang concedes that whether a regulatory intervention or a rumour-driven collapse is endogenous or exogenous to the market is arguable (p. 201). His company figures (Knight, AXA Rosenberg) are not sourced in the chapter .

**Practice.** Interpretation: ask which of five causes is at work, because each points to a different control. The categories overlap: a regime change can expose a misspecification, forced selling by others can look like a regime change, and one loss can have several causes, so the list is a checklist of questions, not a set of exclusive labels.

- A *model error* (inapplicability or misspecification) is found by validation and by monitoring for unexpected behaviour (strategy monitoring).
- A *regime change* is a change in the world after the model was built. It is not overfitting, in which the model was fitted to an unrepresentative sample (parameter overfitting), though the two can look alike in live results.
- An *exogenous shock* is information the model never sees.
- *Funding contagion* is loss from other holders' forced selling (common-investor contagion). Narang's own value-against-growth example mixes it with regime change: he attributes the 2007 reversal partly to quant unwinds (p. 198).
- An *operational failure*, such as software, architecture or restart order, is what Narang files under implementation errors. Testing, change control and the controls in operational controls for automated trading make such failures less likely and limit their cost; they mitigate it, they do not guarantee its prevention, and better modelling does not address it.

**Pitfalls.** *Discretionary overrides.* For exogenous shocks Narang allows that discretionary overrides may be required. Interpretation: two kinds of intervention need keeping apart. An *ad hoc* intervention, improvised outside the validated design, changes the strategy's behaviour in a way that was never tested, and is in effect an unvalidated model change. An *approved contingency response*, such as a pre-specified de-risking rule, a position limit or a kill switch whose triggers, authority and effects were tested and approved before use, is part of the validated design, not a departure from it. Not every override is the first kind. Narang's own account of August 2007 says that the main discretion many quants used, unwinding when models misbehave, spread the losses among managers with overlapping positions (p. 208).

His story of one firm that, by delay, held its positions and recovered is a single anecdote, not evidence that holding is better (p. 208). Who may override, on what evidence, with what record and review is a governance question. "models loaded in a wrong sequence" compresses the server-restart example, in which servers, not models, came back in the wrong order.

**See also.** common-investor contagion (the fourth risk); strategy monitoring (how the risks are watched); operational controls for automated trading (operational failure); parameter overfitting (contrast with regime change); arbitrage risks (Harris's model risk for arbitrageurs, a special case).

**Method 2.18 Monitoring a running quant strategy: exposure, profit and loss, execution and systems monitors**

**Statement.** Narang describes four kinds of monitoring tool that quants use alongside their risk models [s7-narang, chap. "Risks Inherent to Quant Strategies", sec. 10.5, pp. 209–210]:

- *exposure monitors*: start from current positions and group them by whatever exposures concern the manager. Examples are asset classes; valuation, momentum or volatility; or, for equities, gross and net exposure by sector and industry, capitalisation bucket and style factor, with proprietary or vendor tools. VaR-based tools measure overall gross risk "at least according to that measure". Exposures out of line with limits or expectations can be an early warning of a model problem or difficult markets;
- *profit and loss monitors*: compare current prices with the previous close and chart intraday performance, so that unexpected gains or losses prompt a check. Others look at *how* money is made: whether the strategy is holding losers longer or selling winners sooner than usual, and its *hit rate*, the share of positions on which it makes money, against what its designer expects;

- *execution monitors*: orders being worked and recently completed, with sizes and prices; fill rates of limit orders, particularly for passive execution; and, at some managers, slippage and market impact against expectations;
- *systems performance monitors*: software and infrastructure errors, CPU performance, the speed of each automated stage and the latency of messages to and from exchanges. Narang calls this "perhaps the most important for sniffing out systems errors and some types of model risk" (p. 210).

**Conditions.** Practitioner account as of 2024. A description of tools, not a standard or a set of thresholds: "the art is in how to use them" (p. 209). The expected values against which deviations are judged, such as normal exposures, hit rates and slippage, come from the strategy's design and its backtest (backtest evaluation diagnostics).

**Practice.** Narang reports one manager who, in August 2007, noticed intraday profit and loss falling with nearly every tick, researched it and cut risk earlier than most (p. 210). Interpretation: each monitor watches a different source of risk (sources of strategy risk). Exposures and the pattern of profit and loss point to model error and regime change; systems monitors point to implementation and operational failure; execution monitors point to costs. A monitor only detects. The response, whether a limit, a kill switch or an approved override, belongs to operational controls for automated trading.

**Pitfalls.** Execution monitoring here is the manager's view across a strategy. The indicators for supervising many execution algorithms in flight are in decision support for trading algorithms and are not repeated here. Monitoring includes VaR, the sources of profit and early-warning indicators.

**See also.** sources of strategy risk (what is monitored); decision support for trading algorithms (execution indicators); operational controls for automated trading (the controls that act on alerts); backtest evaluation diagnostics (expected hit rates and exposures).

## 2.5 The research and production chain of a strategy

#### Method 2.19 Strategy production lifecycle: López de Prado's research stations and the embargo, paper-trading, graduation, re-allocation and decommission stages

**Statement.** López de Prado argues that a firm, not a lone researcher, should produce machine-learning strategies, under what he calls the meta-strategy paradigm: "It takes almost as much effort to produce one true investment strategy as to produce a hundred", because of data curation, high-performance computing, software, feature analysis, execution simulators and backtesting [alpha-afml, chap. 1, sec. 1.2.2, p. 5]. He describes the stations of the production chain (sec. 1.3.1, pp. 6–8):

- *data curators* collect, clean, index, store, adjust and deliver all data, as experts in market microstructure and data protocols such as FIX;
- *feature analysts* turn raw data into informative signals and catalogue findings, but do not develop strategies;
- *strategists* turn features into strategies, each strategy being "the experiment designed to test the validity" of a theory that must identify "the economic mechanism that causes an agent to lose money to us"; "features may be discovered by a black box, but the strategy is developed in a white box";
- *backtesters* assess the strategy under several scenarios and evaluate "the probability of backtest overfitting by taking into account the number of trials"; their results go to management and are not shared;
- the *deployment team* integrates the code into production, keeps it "logically identical to the prototype" and optimises it for latency;
- *portfolio oversight*, which follows a deployed strategy through five stages (pp. 8–9):
  1. *embargo*: running on data after the backtest's end date; promotion if performance is consistent with the backtest;
  2. *paper trading* on a live feed, capturing data, calculation and execution delays, "for as long as it is needed to gather enough evidence that the strategy performs as expected";
  3. *graduation*: managing a real position, alone or in an ensemble, with risk, returns and costs attributed;
  4. *re-allocation*: allocations reassessed "frequently and automatically" in a diversified portfolio, "in general" following a concave function, small at graduation, rising while the strategy performs and shrinking as performance decays;
  5. *decommission*, once performance has lagged long enough "to conclude that the supporting theory is no longer backed by empirical evidence".

He prefers releasing new variations to run in parallel with old ones (p. 9).

**Conditions.** This is one practitioner's prescription as of 2018, presented as what "every successful quantitative firm I am aware of" does (p. 5), not a regulatory standard or a tested result. None of the stages carries a quantitative criterion: how much consistency promotes a strategy from embargo, how long paper trading lasts, the shape of the concave allocation function and the length of underperformance that triggers decommission are all left to the firm. His view that "the only true alpha left is microscopic" and needs "heavy ML tools", while the chances of an individual finding macroscopic alpha with simple tools such as econometrics are "quickly converging to zero" (p. 6), is an opinion without data in the passage.

**Practice.** Interpretation: the stages map onto model-governance controls. Embargo and paper trading come after deployment in López de Prado's sequence (pp. 8–9), so they are validation before live-capital trading, that is before graduation, not before production; graduation, re-allocation and decommission are change and approval decisions that need an owner, evidence and a record (strategy monitoring supplies the evidence; operational controls the pre-trade and kill-switch layer). The backtester's count of trials is the subject of parameter overfitting and backtest diagnostics. An embargo period is one further sample, of whatever length the backtest left out; a promotion decided on it is another trial, and its strength depends on that length, which the source does not set.

**Pitfalls.** Decommissioning follows a loss of empirical support for the theory; the proposed concave allocation holds only "in general". The organisational framework is a proposal, not evidence of effectiveness. The alpha-models file's machine-learning item, on the cost of producing strategies and macroscopic against microscopic alpha, is the same author's sec. 1.2.2 and 1.3.1 (pp. 5–6), not Narang's; it is represented here.

**See also.** hypothesis-driven strategy research (theory before testing, in Narang's version); components of a quant strategy; sources of strategy risk (implementation errors in deployment).

#### Method 2.20 Promoting, re-allocating and decommissioning a strategy model: owners, records and evidence from supervisory model risk guidance, against López de Prado's unquantified stages

**Statement.** Two current supervisory texts set out who decides that a model may be used, on what evidence and with what record. Both are written for banks.

- *United Kingdom.* The PRA's SS1/23 (May 2023) applies to banks and PRA-designated investment firms with internal-model approval (paras. 1.2–1.3) [wide-pra-ss123, para. 1.2, PDF p. 3]. Development testing "should demonstrate that a model works as intended", with "clear criteria (tests)" to measure quality and choose between candidates. Developers set out the "monitoring pack" of tests for the model's use (Principle 3.3(a), PDF p. 21). A material change, including drift in models that recalibrate themselves, is tested by comparing output before and after the change with actual outcomes (Principle 3.3(c), p. 22). An independent validation function reviews and periodically re-validates models and recommends approvals "to the appropriate model approval authority" (Principle 4.1(b), p. 25).
  Validators "should therefore not be part of any model development activities, nor have a stake in whether or not a model is approved" (para. 2.14, PDF p. 8). The model-risk control functions' authority includes the power "to restrict the use of models, recommend conditional approval, or temporarily grant exceptions to model validation or approval" (para. 2.13, pp. 7–8). Firm policy names the approval authorities and the procedure for material changes (Principle 2.3(c)(viii), p. 17). The board's model risk appetite covers approving models for decision-making, limits on use and exceptions, and "thresholds for acceptable model performance" (p. 14). The rationale for *decommissioning* a model is recorded and can inform later development (n. 6, p. 11).

- *United States.* SR 26-2 of 17 April 2026, issued with the OCC and FDIC, "supersedes and replaces" SR 11-7 [sr-26-2-2026, cover letter]. Its revised guidance asks for monitoring procedures "before and after a model is approved for use". Outcomes analysis, "back-testing or outlier analysis" among its forms, is reviewed in validation. A model that "no longer performs as expected may warrant overlays, adjustment, or redevelopment" (secs. IV–V, PDF pp. 7 and 9–10). "Effective challenge" requires expertise, independence and the standing "to effect any change" (p. 5).

**Conditions.** Guidance as of 17 April 2026 (U.S.) and May 2023, effective 17 May 2024 (UK). Interpretation: neither text sets numerical promotion thresholds. Both leave the criteria and tolerances to the firm's documented risk appetite and policy, so a strategy's promotion rule (minimum embargo length, paper-trading tracking error, capacity) is the firm's to set and record in advance. Applying bank model-risk guidance to a trading strategy is by analogy: a strategy is a model in the guidance's sense only when its outputs inform decisions under the firm's model definition.

**Practice.** Mapping onto López de Prado's stages (interpretation). *Embargo and paper trading* are development and pre-approval testing against criteria and a monitoring pack fixed beforehand. *Graduation* is approval by a named authority on an independent validation recommendation, with any conditions recorded. *Re-allocation* is a material change, validated by comparing before and after against outcomes. *Decommissioning* is recorded with its reason. The record for each step: the criteria and tests set beforehand, the results, the validator's opinion, the approver and conditions, and the monitoring pack. Pre-specified criteria also protect against overfitting the promotion rule itself.

**Pitfalls.** SS1/23's scope is UK banks with internal-model approval (para. 1.2), and SR 26-2 is "most relevant" to banks above $30 billion in assets (cover letter). Neither binds an asset manager or proprietary trading firm as such; they are used here as the most specific public statements of practice. SR 11-7 (2011), which the catalogue still lists, is superseded.

**See also.** strategy production lifecycle; discretionary overrides; strategy monitoring; RTS 6 (EU algorithm testing and deployment).

**Further reading.** `w4-simonian-model-governance`; SR 26-2's attachment in full.

#### Method 2.21 Who may override a model, on what evidence, with what record and review: post-model adjustments and expert judgement under supervisory guidance, against Narang's discretionary overrides

**Statement.** SS1/23 treats adjustments to a model's input, assumptions, methodology or output by expert judgement as part of the model's governance [wide-pra-ss123, Principles 3.4 and 5.1, PDF pp. 22–23 and 28].

- Firms must show that risks from model limitations and uncertainty are understood, monitored and managed, "including through the use of expert judgement" (Principle 3.4(a)).
- A high materiality of adjustments, or recurring adjustments for the same limitation, indicates "flawed model design or misspecification" and calls for recalibration or redevelopment (Principle 3.4, p. 23).
- Post-model adjustments (PMAs) follow "a consistent firm-wide process", documented in policy, with a governance and control framework covering their use, calculation, completeness, "and when PMAs should be reduced or removed" (Principle 5.1(a)). There must be "a clear rationale for the use of PMAs to compensate for model limitations" (5.1(b)). PMAs for material models "should be documented, supported by senior management, and approved by the appropriate level of authority (eg senior management, risk committee, audit committee)" (5.1(c)).

SR 26-2 likewise lets monitoring lead to "overlays, adjustment, or redevelopment" under the firm's model risk management policy [sr-26-2-2026, sec. V, PDF p. 9].

**Conditions.** Interpretation, against Narang's discretionary overrides. An override of a trading model is an adjustment to its output. Under these texts it needs a rationale tied to a stated limitation, an authority fixed in advance, a record, and a review that decides when it ends. A pre-approved contingency rule (a de-risking trigger, a kill switch) meets that before the event; an improvised override must meet it after, and recurring ones point to redesign. Neither text says when overriding is right. The evidence Narang offers from August 2007 is one episode, in which discretionary unwinding spread losses.

**Practice.** A minimal override record (interpretation): time, model and positions affected; the stated limitation or event; who authorised it, against a policy authority; the expected effect and the exit condition; and an after-the-fact review comparing the overridden and model outputs with outcomes. That is SS1/23's parallel outcomes analysis applied to one decision.

**Pitfalls.** Scope as in strategy promotion: bank guidance applied by analogy.

**See also.** sources of risk in a quant strategy; strategy promotion; operational controls (kill functionality).

**Further reading.** `w4-simonian-model-governance`.

# 3 ML and alternative data

## 3.1 Financial data structures: bars, multi-product series and event sampling

**Method 3.1 Activity-sampled bars: time, tick, volume and dollar bars, what each corrects, and what the evidence on their returns shows**

**Statement.** *Bars* are the rows of a table into which unstructured trade data are regularised; standard bar methods "transform a series of observations that arrive at irregular frequency (often referred to as 'inhomogeneous series') into a homogeneous series derived from regular sampling" [alpha-afml, chap. 2, sec. 2.3.1, p. 26]. Each bar records a timestamp, VWAP, open, close, high, low and volume. *Time bars* sample at fixed clock intervals; *tick bars* each time a fixed number of transactions has taken place (for example 1,000); *volume bars* each time a fixed number of the security's units has traded; *dollar bars* each time a fixed market value has traded, in the security's currency (secs. 2.3.1.1–2.3.1.4, pp. 26–28).

**Conditions.** López de Prado's case against time bars is that markets do not process information at a constant rate, so time bars "oversample information during low-activity periods and undersample information during high-activity periods", and that time-sampled series "often exhibit poor statistical properties, like serial correlation, heteroscedasticity, and non-normality of returns" (p. 26, citing Easley, López de Prado and O'Hara 2012). Tick bars synchronise sampling with "a proxy of information arrival" (p. 27), but auction prints and order fragmentation distort the tick count: an opening or closing auction is published as one trade "for an outsized amount" that "could be the equivalent of thousands of ticks", and one buy of 10 lots against ten resting orders of 1 lot prints as ten ticks; matching engines may also "split one fill into multiple artificial partial fills" (p. 27).

Volume bars avoid the fragmentation problem. Dollar bars keep a constant traded value when prices move a lot: after a 100 percent rise, $1,000 buys half the shares (p. 28). After split adjustment, issuance and buybacks still change tick and volume counts, and dollar bars "tend to be robust" to those actions; the bar size can be set dynamically from free-float capitalisation or outstanding debt (p. 28).

**Practice.** The evidence cited is of different kinds and does not settle which clock is best. López de Prado quotes Mandelbrot and Taylor (1967), "Price changes over a fixed number of transactions *may* have a Gaussian distribution", says that "multiple studies have confirmed" that sampling by trading activity gives "returns closer to IID Normal" (citing Ané and Geman 2000), and reports Clark (1973) as finding volume sampling closer to IID Gaussian than tick sampling (p. 27). Ané and Geman's abstract claims that normality "can be recovered through a stochastic time change" and that "the cumulative number of trades is a better stochastic clock than the volume for generating virtually perfect normality in returns" [ane-geman-2000, abstract], which ranks trades above volume, the reverse of the ranking reported for Clark.

*A replication critique*. Murphy and Izzeldin's Monte Carlo experiments show that Ané and Geman's procedure cannot accurately recover the third and higher moments of the latent information flow, "even in very large samples of 10,000 observations". On 1997 NYSE TAQ data for the same two stocks, Cisco and Intel (Ané and Geman's own data were not available to them), and on other stocks, returns conditioned on trades or volume are "not approximately Gaussian" [murphy-izzeldin-2006, version of December 2006, abstract and secs. 1 and 4–6]. The trade-clock normality claim is therefore contested on the same stocks and year. Neither statement is about independence: "IID" is López de Prado's gloss.

His own support for dollar bars is one series: on E-mini S&P 500 futures, the exponentially weighted number of tick and volume bars per day for a fixed bar size "will vary wildly over the years", and that of dollar bars less (Figure 2.1, p. 28). His exercises ask the reader to compare bar types by weekly bar-count stability, serial correlation, variance of monthly variances and Jarque–Bera statistics (Exercise 2.1, pp. 40–41): these are the tests to run on one's own data.

**Pitfalls.** Better distributional properties are an empirical, market- and period-specific finding, not a property guaranteed by the sampling rule. Robustness of dollar bars to splits and corporate actions requires qualification; López de Prado assumes splits are already adjusted and says only that dollar bars "tend to be robust" to issuance and buybacks (p. 28). That sampling by trading activity gives returns closer to IID normal is López de Prado's summary of Ané and Geman. Their abstract claims near-normality under a trade clock, Murphy and Izzeldin dispute it on the same stocks, and neither addresses independence (Practice). Interpretation: an activity clock moves the volatility clustering into the bar durations, so a bar series has irregular calendar timing that must be kept for any time-of-day feature, for joins with other data and for the timing of decisions.

**See also.** information-driven bars; event-based sampling; derived market statistics; volatility components; intraday impact profile.

**Method 3.2 Information-driven bars: the tick rule, tick, volume and dollar imbalance bars, and tick, volume and dollar runs bars, with causal thresholds**

**Statement.** Information-driven bars "sample more frequently when new information arrives to the market", information being meant in the microstructure sense of persistent imbalances in signed volume, associated with informed traders [alpha-afml, chap. 2, sec. 2.3.2, p. 29]. For ticks $(p_t, v_t)$ the *tick rule* sets $b_t = b_{t-1}$ if $\Delta p_t = 0$ and $b_t = \Delta p_t / |\Delta p_t|$ otherwise, so $b_t \in \{-1, 1\}$, with $b_0$ "set to match the terminal value $b_T$ from the immediately preceding bar" (p. 29). Each bar closes at the first $T$ at which a statistic $\theta_T$ of the ticks since the bar opened reaches its expectation as set at the bar's start (pp. 29–32):

| Bar | $\theta_T$ | bar closes at the first $T$ with |
|---|---|---|
| tick imbalance (TIB) | $\sum_{t=1}^{T} b_t$ | $\lvert\theta_T\rvert \geq E_0[T]\,\lvert 2P[b_t = 1] - 1\rvert$ |
| volume or dollar imbalance (VIB, DIB) | $\sum_{t=1}^{T} b_t v_t$ | $\lvert\theta_T\rvert \geq E_0[T]\,\lvert 2v^+ - E_0[v_t]\rvert$ |
| tick runs (TRB) | $\max\{\sum_{t\lvert b_t=1} b_t,\ -\sum_{t\lvert b_t=-1} b_t\}$ | $\theta_T \geq E_0[T]\max\{P[b_t = 1], 1 - P[b_t = 1]\}$ |
| volume or dollar runs (VRB, DRB) | $\max\{\sum_{t\lvert b_t=1} b_t v_t,\ -\sum_{t\lvert b_t=-1} b_t v_t\}$ | $\theta_T \geq E_0[T]\max\{v^+, v^-\}$ |

with $v_t$ the units (VIB, VRB) or the currency amount (DIB, DRB) traded, $v^+ = P[b_t = 1]E_0[v_t | b_t = 1]$ and $v^- = P[b_t = -1]E_0[v_t | b_t = -1]$, so that $E_0[v_t] = v^+ + v^-$ and $E_0[\theta_T] = E_0[T](v^+ - v^-) = E_0[T](2v^+ - E_0[v_t])$ for the volume imbalance (p. 30).

**Notation.** $E_0[T]$ is the expected number of ticks per bar, $P[b_t = 1]$ the unconditional probability that a tick is a buy. Imbalance bars offset buys against sells; runs bars count each side "without offsetting them" and allow sequence breaks, so a run is the larger one-sided total in the bar, not the longest unbroken sequence (p. 31).

**Conditions.** The expectations are estimated as exponentially weighted moving averages "from prior bars": of $T$, of $b_t$ (for $2P[b_t = 1] - 1$), of $b_t v_t$, of the proportion of buy ticks and of buy and sell volumes (pp. 29–32). The threshold is therefore fixed when a bar opens and uses only closed bars, so bar formation is causal. López de Prado gives no EWMA span, no warm-up rule for the first bar and no $b_0$ for the very first tick: these are implementation choices to record.

Interpretation, derived here: the factorisation $E_0[\sum_{t \leq T} x_t] = E_0[T]\,E_0[x_t]$ needs $T$ independent of the tick signs and sizes, or a Wald-type identity for IID ticks, neither of which the text states; and for runs bars $E_0[\max\{X, Y\}] \geq \max\{E_0 X, E_0 Y\}$, so the runs "expectation" is a heuristic threshold, not the expectation of $\theta_T$.

**Practice.** López de Prado's interpretation is that TIBs "are produced more frequently under the presence of informed trading" and can be understood "as buckets of trades containing equal amounts of information" (p. 30); VIBs and DIBs, like volume and dollar bars, address tick fragmentation and outliers and, because the bar size adjusts, corporate actions (p. 31); runs bars target large traders who sweep the book, use

icebergs or slice orders (p. 31). No test of these claims is given. In a toy order flow computed for this entry, all four bar types closed more often during a one-sided burst than elsewhere, but on a perfectly balanced flow the tick-imbalance threshold $E_0[T]|2P[b_t = 1] - 1|$ is near zero, bars close after a tick or two, and the EWMA of $T$ then shrinks further: the scheme needs a floor or cap that the text does not give.

**Pitfalls.** The tick-runs statistic is $\theta_T = \max\{B, S\}$, with $B$ and $S$ the numbers of buy and sell ticks in the bar, both counted as positive (pp. 31–32). It is not their sum: $B + S = T$, the bar's tick count, which carries no information about which side dominates. Likewise the volume-runs statistic is the larger of buy and sell volume, not the total volume. The imbalance thresholds take absolute values, $E_0[T]\,|2P[b_t = 1] - 1|$, not a conditional expectation, and $v^+$ is the expected buy volume per tick. The tick rule classifies a tick by the last price change, not by the aggressor; where aggressor flags exist they are the better sign.

**See also.** activity-sampled bars; event-based sampling; informed trader types; order exposure decisions (icebergs and slicing).

**Further reading.** López de Prado's chap. 19 on microstructural features.

#### Method 3.3 The ETF trick: a basket, spread or rolled futures position as the value of $1 invested, with holdings, costs, tradable volume and the single-futures roll

**Statement.** The ETF trick models "a basket of securities as if it was a single cash product", transforming a multi-product dataset "into a single dataset that resembles a total-return ETF", so that a strategy can always assume it trades "cashlike products (non-expiring cash instruments)" [alpha-afml, chap. 2, secs. 2.4–2.4.1, pp. 32–33]. It addresses spreads whose weights change (the spread can converge with no price change), spreads that go negative, constituents that do not trade at the same times, and execution costs (p. 33). With raw opens $o_{i,t}$ and closes $p_{i,t}$, the USD value of one point $\varphi_{i,t}$ (including FX), volume $v_{i,t}$ and carry, dividend or coupon $d_{i,t}$ (which can also charge margin or funding costs), allocations $\omega_t$ rebalanced or rolled on bars $B$, and $K_0 = 1$ (p. 34):

$$h_{i,t} = \begin{cases} \dfrac{\omega_{i,t} K_t}{o_{i,t+1}\varphi_{i,t} \sum_i |\omega_{i,t}|} & t \in B \\ h_{i,t-1} & \text{otherwise} \end{cases}, \qquad \delta_{i,t} = \begin{cases} p_{i,t} - o_{i,t} & (t-1) \in B \\ \Delta p_{i,t} & \text{otherwise,} \end{cases}$$

$$K_t = K_{t-1} + \sum_i h_{i,t-1}\varphi_{i,t}(\delta_{i,t} + d_{i,t}).$$

Every instrument must have been tradable at $t - 1$ and $t$ (p. 33). With $\tau_i$ the cost of trading \$1 of instrument $i$, the rebalance cost $c_t = \sum_i (|h_{i,t-1}|p_{i,t} + |h_{i,t}|o_{i,t+1})\varphi_{i,t}\tau_i$ for $t \in B$ is kept out of $K_t$ and treated as a negative dividend; the cost of buying or selling one unit is $\tilde{c}_t = \sum_i |h_{i,t-1}|p_{i,t}\varphi_{i,t}\tau_i$; and the tradable basket volume is $v_t = \min_i v_{i,t}/|h_{i,t-1}|$ (p. 34).

**Notation.** $h_{i,t}$ holdings in securities or contracts; $\delta_{i,t}$ the change in market value per unit from $t - 1$ to $t$. Dividing by $\sum_i |\omega_{i,t}|$ "de-levers" the allocation. Interpretation, derived: $\sum_i |h_{i,t}|o_{i,t+1}\varphi_{i,t} = |K_t| \sum_i |\omega_{i,t}| \operatorname{sgn}(o_{i,t+1}\varphi_{i,t})/\sum_j |\omega_{j,t}|$, which is $|K_t|$ when every $o_{i,t+1}\varphi_{i,t} > 0$, and $K_t$ itself only when also $K_t \geq 0$. With a negative price or conversion factor the sign of a holding no longer follows the sign of its weight and the sum is not a gross exposure.

**Conditions.** *Timing.* Holdings decided at $t$ are sized at the next bar's open $o_{i,t+1}$ because at a roll "we may not know $p_{i,t}$ of the new contract" (p. 34). Interpretation: this is an accounting convention, not an order. The quantity $h_{i,t}$ depends on $o_{i,t+1}$, which is unknown when the decision is taken at $t$, so weights $\omega_t$ that use only information up to $t$ are necessary but not sufficient for the series to be executable: an order placed at $t$ cannot carry that quantity.

The convention is reproduced only by an order whose quantity is set by its own fill price, for example a currency-amount (notional) order for $\omega_{i,t}K_t/\sum_i |\omega_{i,t}|$ in each leg, filled in full in a single-price opening auction at exactly $o_{i,t+1}$ and with $\varphi_{i,t}$ known (or the FX conversion fixed at that time); whether a venue accepts such orders, and fills them without rounding to lots, partial fills or a price other than the published open, varies by instrument.

A quantity sized at $t$ on $p_{i,t}$ and filled at the open leaves a gross exposure different from $K_t$ when the open gaps (on a two-leg toy with opens 1.5 percent above and 1.6 percent below the closes: 1.0026 against $K_t = 1$), and a trader who first observes the opening print can no longer trade at it. The gap between $p_{i,t}$ and $o_{i,t+1}$ on the old holdings is not booked, which is the intent at a roll (two contracts) and an approximation at a same-instrument rebalance, and $\varphi_{i,t}$ is applied to a $t + 1$ price.

*Positivity and capital.* López de Prado says the series will be "strictly positive (at worst, infinitesimal)" because profits and losses are reinvested at rebalances (pp. 33–34). Interpretation: that holds for long-only baskets with positive prices, but a short leg can lose more than $K$ between rebalances (a short worth \$1 that rises 120 percent leaves $K = -0.2$). The recursion has no rule for exhausted capital: at $K_t = 0$ the next holdings are zero, and at $K_t < 0$ the formula for $h_{i,t}$ reverses every position (the long leg becomes short) while their gross value is $|K_t|$. A series that reaches $K_t \leq 0$ therefore stops describing the strategy; a backtest must stop it or state an explicit financing, margin or loss rule, which the method does not supply.

*Costs.* $c_t$ charges the full value of old and new holdings, so it overstates the cost when holdings change little; López de Prado notes that cost functions need not be linear and can be simulated (p. 34). $v_t$ is undefined for a zero holding, which must be skipped.

**Practice.** *Single futures roll.* For one contract the ETF trick is a one-legged spread; López de Prado's more direct method forms the cumulative series of roll gaps, each the new contract's open minus the old contract's previous close, and subtracts it from the price series, backward (the end matches the raw series) or forward (the start matches) (sec. 2.4.3, pp. 36–37, Snippet 2.2). Rolled prices serve for PnL and mark-to-market; "raw prices should still be used to size positions and determine capital consumption", and rolled prices "can indeed become negative, particularly in futures contracts that sold off while in contango" (p. 37). The non-negative alternative computes the return as the rolled price change divided by the *previous raw price* and compounds $(1 + r)$ (p. 37, Snippet 2.3).

Interpretation: the result stays non-negative while $1 + r \geq 0$ on every bar, which holds on non-roll bars when raw prices are positive but not if raw prices themselves turn negative. A two-leg toy is a consistency check only: the change in $K$ equals the holdings' profit and loss recomputed from the same prices, and gross exposure at the booked open equals $K$ (here $K > 0$ and all prices positive).

**Pitfalls.** Use the next opening price $o_{i,t+1}$ in $h_{i,t}$ and the previous raw price in the return denominator. Positivity requires the capital and exposure conditions stated above. The allocation vector itself is a separate choice; one way to set it is in PCA risk-distribution weights.

**See also.** PCA risk-distribution weights; CME Globex trading (contract months); futures hedges and basis (stack and roll); transaction costs; ETF creation and redemption (real ETFs, as against this virtual one).

#### Method 3.4 Event-based sampling: downsampling for reduction and the symmetric CUSUM filter, with its update, reset and threshold rules

**Statement.** Fitting on every bar is usually a poor idea, because some ML algorithms do not scale with sample size and because they "achieve highest accuracy when they attempt to learn from relevant examples" [alpha-afml, chap. 2, sec. 2.5, p. 38]. *Sampling for reduction* (downsampling) takes bars at a constant step (linspace sampling), which is simple but has an arbitrary step and depends on the seed bar, or uniformly at random; neither selects the most informative observations (sec. 2.5.1, p. 38). *Event-based sampling* keeps the bars at which some event occurs and lets the model learn whether a prediction function exists there; if not, the event is redefined (sec. 2.5.2, p. 38). The *CUSUM filter*, a quality-control method for detecting a shift in a mean, sets for observations $y_t$

$$S_t = \max\{0,\ S_{t-1} + y_t - E_{t-1}[y_t]\}, \quad S_0 = 0,$$

and recommends an action at the first $t$ with $S_t \geq h$, the filter size; the symmetric filter adds $S_t^- = \min\{0, S_{t-1}^- + y_t - E_{t-1}[y_t]\}$ and uses $S_t = \max\{S_t^+, -S_t^-\}$ (sec. 2.5.2.1, pp. 38–39). For sampling, "we will sample a bar $t$ if and only if $S_t \geq h$, at which point $S_t$ is reset" (p. 39).

**Conditions.** $S_t = 0$ whenever $y_t \leq E_{t-1}[y_t] - S_{t-1}$: the zero floor skips downward deviations, so the one-sided filter detects run-ups from the last reset (p. 39). Without a reset, $S_t \geq h$ if and only if some partial sum $\sum_{i=\tau}^{t}(y_i - E_{i-1}[y_i]) \geq h$ (López de Prado prints $E_{i-1}[y_t]$); interpretation, after a reset $\tau$ must start after it. López de Prado describes the inputs as "IID observations arising from a locally stationary process" (pp. 38–39), but his Snippet 2.4 sets $E_{t-1}[y_t] = y_{t-1}$ and cumulates first differences of a level series such as a price, which treats $y$ as a level with martingale-like increments; the operational definition is the cumulated deviation $y_t - E_{t-1}[y_t]$ under whatever forecast is used. In the snippet the comparisons are strict ($S^- < -h$, $S^+ > h$), only the side that triggered is reset, and at most one event fires per bar, the negative side checked first (p. 39).

**Practice.** A filter event needs "a full run of length $h$", so a series hovering near a level does not fire repeatedly, unlike Bollinger bands; $S_t$ can be built on other features, such as structural-break statistics or microstructure measures (p. 40). On toy series, a series hovering within ±0.05 of a level fires no event at $h = 0.5$, and a run of exactly $h$ fires under the text's $\geq$ but not under the snippet's $>$. López de Prado's exercises use returns, or absolute returns, as $y_t$ with $h = 0.05$ (p. 40); interpretation, with the snippet's differencing that needs a cumulative return series as input. Lam and Yam's CUSUM trading rule and Fama and Blume's filter rule are different uses: they trade on the signal, whereas here the filter only selects observations (p. 39).

**Pitfalls.** The zero condition uses $\leq$, not equality alone. The run condition must use the correct expectation index and apply the sampling and reset rule. The threshold $h$ and any volatility scaling of it (López de Prado's labelling exercise sets $h$ to the standard deviation of daily returns, p. 55) are parameters: estimating them on the full sample before a backtest uses future data; see time-validity of labels and weights.

**See also.** triple-barrier labelling (which the sampled events seed); activity-sampled bars; information-driven bars.

**Further reading.** Structural-break and entropy features (López de Prado, chaps. 17–18) are not covered.

## 3.2 Labels and sample weights for overlapping financial outcomes

#### Method 3.5 Triple-barrier labelling: the fixed-horizon method and its limits, volatility-scaled barriers, first touch, barrier configurations, and labelling side against size

**Statement.** *Fixed-horizon labelling* gives an observation $X_i$, drawn from bars $t = 1, \ldots, T$, the label $y_i = -1, 0$ or 1 as the return $r_{t_{i,0},t_{i,0}+h} = p_{t_{i,0}+h}/p_{t_{i,0}} - 1$ is below $-\tau$, within $\pm\tau$, or above $\tau$, with $t_{i,0}$ "the index of the bar immediately after $X_i$ takes place" and $\tau$ a constant [alpha-afml, chap. 3, sec. 3.2, p. 43]. López de Prado gives three reasons to avoid it: time bars have poor statistical properties; one $\tau$ is applied whatever the volatility, so a $\tau$ of 1 percent labels most observations 0 at night and few at the open; and it ignores the path, labelling as winners positions that "would have been stopped-out" (p. 44).

The *triple-barrier method* labels an observation by the first of three barriers touched: an upper (profit-taking) and a lower (stop-loss) horizontal barrier, set as multiples of a volatility estimate, and a vertical barrier, an expiration limit counted from the position's start; the upper gives 1, the lower $-1$, and the vertical either the sign of the return or 0, López de Prado preferring the sign (sec. 3.4, p. 45). With $t_{i,1} \leq t_{i,0} + h$ the first-touch time, the label's return is $r_{t_{i,0},t_{i,1}}$, which uses the prices at $t_{i,0}$ and $t_{i,1}$ and compounds the one-bar returns $r_{t-1,t}$ with $t_{i,0} < t \leq t_{i,1}$; the barriers need not be symmetric (p. 45).

**Conditions.** *Volatility.* The barrier width is a target such as an exponentially weighted standard deviation of daily returns (Snippet 3.1, default span 100 days, pp. 44–45), computed from data up to the event, so it is known at the decision time; observations whose target is below a minimum return are skipped (Snippet 3.3, p. 48).

*Configurations.* Writing $[pt, sl, t1]$ for which barriers are active, López de Prado calls $[1, 1, 1]$, $[0, 1, 1]$ and $[1, 1, 0]$ useful (the last "somewhat unrealistic", holding as long as it takes); $[0, 0, 1]$, $[1, 0, 1]$ and $[1, 0, 0]$ less realistic ($[0, 0, 1]$ is the fixed-horizon method, still useful on activity bars; $[1, 0, 0]$ "could mean being locked on a losing position for years"); and $[0, 1, 0]$ and $[0, 0, 0]$ illogical, the last producing no label (p. 46).

*Side and size.* Without a model for the side of the bet, the profit-taking and stop-loss barriers cannot be told apart, so "either there are no horizontal barriers or … the horizontal barriers must be symmetric" (sec. 3.5, p. 48); with a known side, the barriers may be asymmetric and the label becomes a meta-label (meta-labelling).

**Practice.** López de Prado's code compares bar closes with the barriers using strict inequalities and takes the earliest of the three touch times (Snippets 3.2–3.3, pp. 45–48); the label's return is measured to the close at $t_{i,1}$, so it can overshoot the barrier. His vertical barrier is set in calendar days (Snippet 3.4, p. 49). Interpretation: with closes alone a bar cannot touch both horizontal barriers, but with highs and lows one bar can, and the text gives no rule for which came first; an implementation must choose one (stop-loss first, finer data, or drop the label) and record it.

The code also starts the path at the event's own timestamp, whereas the definition starts it at the next bar: the entry price should be the first one tradable after the features are known. *Arbitrage labels*. A spread that later converges is not automatically a good label, because the interim drawdown may breach margin, borrow, inventory or risk limits, so labels should record both terminal convergence and maximum adverse excursion. This is the method's path argument applied to a spread: a fixed-horizon label ignores positions that "would have been stopped-out" (p. 44), and a stop-loss barrier set at the drawdown the capital can bear makes the label path-dependent (Interpretation; Harris's staying power in arbitrage risks).

Arbitrage event labels (entry when the spread crosses an estimated arbitrage bound after costs; confirmation from flow; failure on specific news, borrow loss, deal break or hedge-ratio change; exit on convergence, stop-out, forced liquidation, expiry or capital reallocation) map onto the events that seed the barriers (event-based sampling) and onto the three barriers; confirmation and failure are added categories, not López de Prado's, and the arbitrage bound is arbitrage's.

**Pitfalls.** The vertical barrier is a time limit. Fixed-horizon labels ignore the path; distinguish the side selected by the primary model from the size selected by meta-labelling. A label uses the prices from $t_{i,0}$ to $t_{i,1}$: labels near the end of a training sample use prices from the test period; see time-validity of labels and weights.

**See also.** event-based sampling (the events that seed the barriers); meta-labelling; label concurrency and uniqueness; sample weights.

**Further reading.** López de Prado, chap. 10 (bet sizing from predicted probabilities).

#### Method 3.6 Meta-labelling: a secondary model that decides whether to act on a primary model's side and how much to bet, and the evidence for its claimed benefits

**Statement.** When a primary model (any model: ML, econometric, technical, fundamental or a human) sets the side of a bet, *meta-labelling* trains a secondary model only to decide its size, including zero [alpha-afml, chap. 3, secs. 3.6 and 3.8, pp. 50 and 53–54]. Triple-barrier events are labelled with the primary side $s_i \in \{-1, 1\}$: the meta-label is 1 if the side-adjusted return $s_i\, r_{t_{i,0},t_{i,1}}$ is positive and 0 otherwise, and the horizontal barriers may be asymmetric because profit-taking and stop-loss are now distinguishable (Snippets 3.6–3.7, pp. 50–51). The secondary model predicts the meta-label; when it predicts 1, "we can use the probability of this secondary prediction to derive the size of the bet, where the side (sign) of the position has been set by the primary model" (p. 51).

**Conditions.** Timeless; the Hudson & Thames page as of October 2026. López de Prado's recipe is to build a primary model with high recall, even at low precision, and to apply meta-labelling to its predicted positives so as to filter out false positives (sec. 3.7, pp. 52–53). He states that

meta-labelling "is particularly helpful when you want to achieve higher F1-scores" and "will increase your F1-score" (pp. 52–53), that it lets ML sit on top of a white-box model, that "the effects of overfitting are limited" because ML decides only the size, that it allows separate long and short models, and that it concentrates ML on sizing (p. 53).

These are claims without evidence in the chapter; his Exercise 3.5 asks the reader to compare the primary and secondary models' precision, recall and F1 (p. 55). Interpretation, derived here: filtering predicted positives can only keep or reduce recall, so F1 rises only if the precision gained outweighs the recall lost; it is not guaranteed.

**Practice.** Interpretation: the meta-labels must be built from the primary model's *out-of-sample* predictions (for example walk-forward, or cross-fitted with purging), because in-sample primary signals carry the primary model's overfitted hit rate, which the secondary model would learn. López de Prado's quantamental recipe says instead to "fit your model again on the same training set, but this time using the meta-labels" (sec. 3.8, p. 53), with no out-of-sample condition. For a discretionary portfolio manager he suggests features ranging from market information to stress and sleep (p. 54). Under heavy class imbalance he drops rare labels recursively (Snippet 3.8, p. 54) or uses balanced class weights (sample weights).

**Pitfalls.** The common recipe (train a binary primary model, choose a high-recall threshold, possibly from ROC curves, concatenate its features and predictions, and call the final prediction positive "only where both are true"), and its pipeline figure, come from a Hudson & Thames blog example, not López de Prado [`ht-meta-labeling-toy`, "Model Architecture" section]. In López de Prado the primary side is ±1 and the secondary output sets a size; "both true" is the special case of sizing with a 0/1 threshold. Neither source gives specific feature groups for the secondary model or a list of limitations. That meta-labelling adds little to an overfit primary model contradicts López de Prado's own claim; that it trades recall for precision and needs many trades is consistent with his precision–recall trade-off (p. 52).

**See also.** triple-barrier labelling; time-validity of labels and weights; parameter overfitting; alpha signal combination.

#### Definition 3.7 Overlapping outcomes, label concurrency and average uniqueness

**Statement.** A label $y_i$ is a function of the prices over $[t_{i,0}, t_{i,1}]$; when $t_{i,1} > t_{j,0}$ for $i < j$, $y_i$ and $y_j$ "will both depend on a common return", and the labels are not IID whenever some consecutive outcomes overlap, $t_{i,1} > t_{i+1,0}$ [`alpha-afml`, chap. 4, sec. 4.2, p. 59]. Restricting horizons to remove overlaps ties the sampling frequency to the label horizon, or, with path-dependent labels, to the first barrier touched, which López de Prado calls "a terrible solution" (pp. 59–60). Two labels are *concurrent* at $t$ when both depend on the return $r_{t-1,t} = p_t/p_{t-1} - 1$ (sec. 4.3, p. 60). With returns indexed by their ending time $t$, a label from $t_{i,0}$ to $t_{i,1}$ depends on exactly the returns with $t_{i,0} < t \le t_{i,1}$, so

$$1_{t,i} = \begin{cases} 1 & t_{i,0} < t \le t_{i,1}, \\ 0 & \text{otherwise}, \end{cases}$$

the *concurrency* is $c_t = \sum_i 1_{t,i}$, the *uniqueness* of label $i$ at $t$ is $u_{t,i} = 1_{t,i}/c_t$, and its *average uniqueness* is

$$\bar{u}_i = \frac{\sum_t u_{t,i}}{\sum_t 1_{t,i}},$$

the reciprocal of the harmonic mean of $c_t$ over the label's life, between 0 and 1 (sec. 4.4, p. 61).

**Conditions.** $\bar{u}_i$ "requires information that is not available until a future time", the end time $t_{i,1}$; López de Prado says this is not a problem because the $\bar{u}_i$ "are used on the training set in combination with label information, and not on the testing set", and are not used to forecast the label (p. 62). The no-leakage claim therefore holds only under that use; see time-validity of labels and weights.

*Index conventions.* López de Prado writes $1_{t,i} = 1$ "if and only if $[t_{i,0}, t_{i,1}]$ overlaps with $[t-1, t]$" (p. 60); read with closed intervals that may touch at an endpoint, that would also mark the return ending at $t_{i,0}$ and the one starting at $t_{i,1}$, which his own toy matrix (p. 65) does not, so the rule above is the positive-length overlap. Any indexing that marks the same returns is equivalent: indexing returns by their starting time $s$, as $r_{s,s+1}$, a label marks $t_{i,0} \le s < t_{i,1}$.

López de Prado's toy example (labels on $r_{0,3}$, $r_{2,4}$ and $r_{4,6}$, sec. 4.5.3, p. 65) is consistent in this sense: its printed matrix has rows for the returns ending at $t = 1, \ldots, 6$, and Snippet 4.6 codes it as spans {0:2, 2:3, 4:5}, the start-time indexing with each end shifted down by one; those coded values are not the labels' end times. His production code is a different convention: Snippets 4.1 and 4.3 mark every bar from the event's timestamp $t_{i,0}$ to $t_{i,1}$ inclusive (`count.loc[tIn:tOut]`, `indM.loc[t0:t1]`), and Snippet 4.10 sums `ret.loc[tIn:tOut]` with `ret` the log return ending at each bar (pp. 60–61, 64 and 69).

Applied to event times and returns indexed by ending time, that also counts the return ending at $t_{i,0}$, from before the label starts, so a label that starts where another ends acquires a return it never uses and the two appear concurrent. Including the start timestamp is therefore not an equivalent reindexing; the inclusive code is right only when its inputs are already shifted as in Snippet 4.6.

**Practice.** On the toy example, with the rule above, concurrency $(1, 1, 2, 1, 1, 1)$ for $t = 1, \ldots, 6$ and average uniqueness $(0.833, 0.75, 1)$, López de Prado's matrix and values (pp. 65–66). The inclusive rule applied to the same events, rows $t = 0, \ldots, 6$, gives concurrency $(1, 1, 2, 2, 2, 1, 1)$ and average uniqueness $(0.75, 0.5, 0.833)$: labels 2 and 3, adjacent at $t = 4$, then share the return $r_{3,4}$, which label 3 does not use. Interpretation: the mean of $\bar{u}_i$ over the sample is the number of effectively independent outcomes per observation, the quantity behind the bootstrap and bagging choices in the sequential bootstrap.

**Pitfalls.** The return is $r_{t-1,t} = p_t/p_{t-1} - 1$. Leakage-free uniqueness calculations require using only the training set.

**See also.** triple-barrier labelling; sequential bootstrap; sample weights.

#### Method 3.8 Bootstrapping overlapping labels: redundancy, oversampling, and the sequential bootstrap

**Statement.** Drawing $I$ times with replacement from $I$ items misses a given item with probability $(1 - I^{-1})^I \to e^{-1}$, so about $1 - e^{-1} \approx 0.63$ of the items are drawn [`alpha-afml`, chap. 4, sec. 4.5, p. 62]. When the mean average uniqueness $I^{-1} \sum_i \bar{u}_i \ll 1$, bootstrapped in-bag observations are redundant with one another and similar to out-of-bag ones: bagged trees become "very similar copies of a single overfit decision tree" and out-of-bag accuracy "will be grossly inflated" (pp. 62–63). Dropping overlapping outcomes loses too much information; better is to sample only a fraction near the mean uniqueness (for example `max_samples` in scikit-learn's bagging classifier) (p. 63).

The *sequential bootstrap* draws with changing probabilities: the first draw is uniform, $\delta_i^{(1)} = I^{-1}$; after the draws $\varphi$ so far, label $j$'s uniqueness given those draws is $u_{t,j}^{(2)} = 1_{t,j}(1 + \sum_{k \in \varphi} 1_{t,k})^{-1}$, its average $\bar{u}_j^{(2)} = \sum_t u_{t,j}^{(2)} / \sum_t 1_{t,j}$, and the next draw has probabilities

$$\delta_j^{(2)} = \frac{\bar{u}_j^{(2)}}{\sum_{k=1}^{I} \bar{u}_k^{(2)}},$$

updated after every draw until $I$ draws are made; repeats remain possible but less likely (sec. 4.5.1, pp. 63–64).

**Conditions.** López de Prado says the sequential sample "will be much closer to IID", which "can be verified by measuring an increase" in mean uniqueness (p. 64). His evidence is a Monte Carlo on random spans (10 observations, 100 bars, label spans of 1 to 4 bars from Snippet 4.7's `maxH=5`,

$10^6$ runs): median mean uniqueness 0.6 for the standard bootstrap against 0.7 for the sequential one, with an ANOVA test of the difference in means (Figure 4.2, pp. 66–68). That shows higher average uniqueness on those synthetic spans, a proxy for less redundancy; it is not a guarantee for every sample, nor evidence of better out-of-sample accuracy. The cost is one pass over all labels per draw.

**Practice.** On López de Prado's toy example, after label 2 is drawn the probabilities are $(5, 3, 6)/14$, as printed (p. 66), and after labels 2 and 3 they are $(5, 3, 3)/11$ (his Exercise 4.6). A 400-run Monte Carlo with his settings gives medians 0.600 and 0.700, with the sequential sample less unique than the standard one in 22 percent of runs (an independent 3,000-run check found 23 percent).

**Pitfalls.** López de Prado writes that with at most $K \leq I$ non-overlapping outcomes the probability of missing an item is $(1-K^{-1})^I$, "approximated as $(1-I^{-1})^{I\frac{K}{I}} \approx e^{-K/I}$", so that the share of unique observations drawn is $1 - e^{-K/I} \leq 1 - e^{-1}$ (p. 62). The approximation is wrong: $(1-K^{-1})^I \approx e^{-I/K}$; for $I = 1000$, $K = 100$ the exact value is $4.3 \times 10^{-5}$ and $e^{-I/K} = 4.5 \times 10^{-5}$, against $e^{-K/I} = 0.905$. Interpretation, by simulation: treating the $K$ non-overlapping outcomes as the effective units, almost all of them appear in a bootstrap sample, but its $I$ rows hold only about $K(1-e^{-I/K})$ distinct units, 0.100 per row against 0.632 under IID draws.

The conclusion that assuming IID draws oversamples survives; the formula does not. The source omits the superscript in $u_{t,j}^{(2)}$ inside $\bar{u}_j^{(2)}$; the normaliser sums $\bar{u}_k^{(2)}$ over all labels, and the argument concerns non-overlapping outcomes.

**See also.** label concurrency and uniqueness; sample weights; parameter overfitting.

**Further reading.** Cross-validation under overlapping labels, with purging and embargo (López de Prado, chap. 7), is a separate evaluation topic; Rao, Pathak and Koltchinskii (1997) is the sequential resampling scheme López de Prado distinguishes from his (p. 63).

### Method 3.9 Sample weights for overlapping labels: return attribution, time decay on cumulative uniqueness, and class weights

**Statement.** *Return attribution.* For labels that depend on the sign of the return ($\pm 1$, or $\{0, 1\}$ for meta-labels), López de Prado weights observation $i$ by the absolute sum of the returns attributed to it,

$$\tilde{w}_i = \left| \sum_{t=t_{i,0}+1}^{t_{i,1}} \frac{r_{t-1,t}}{c_t} \right| \text{ with } r_{t-1,t} = \ln(p_t/p_{t-1}), \qquad w_i = \tilde{w}_i \frac{I}{\sum_{j=1}^{I} \tilde{w}_j}, \qquad \sum_i w_i = I,$$

(log returns, "so that they are additive", in Snippet 4.10, unlike the simple return used to define concurrency), because overlapping outcomes would otherwise be over-weighted and "labels associated with large absolute returns should be given more importance than labels with negligible absolute returns"; the weights sum to $I$ because libraries assume a default weight of 1 [alpha-afml, chap. 4, sec. 4.6, pp. 68–69, Snippet 4.10].

*Time decay.* Weights are multiplied by $d[x] = \max\{0, a + bx\}$ on $x \in [0, \sum_i \bar{u}_i]$, cumulative uniqueness rather than calendar time, with $d[\sum_i \bar{u}_i] = 1$ for the newest observation; for $c \in [0, 1]$, $d[0] = c$, giving $b = (1-c)/\sum_i \bar{u}_i$; for $c \in (-1, 0)$, $d[-c \sum_i \bar{u}_i] = 0$, giving $b = [(c+1) \sum_i \bar{u}_i]^{-1}$; in both cases $a = 1 - b \sum_i \bar{u}_i$ (sec. 4.7, p. 70). *Class weights* correct for under-represented labels, "particularly critical" when the important classes are rare, such as liquidity crises; López de Prado's default is `class_weight='balanced'`, or `'balanced_subsample'` for bagging (sec. 4.8, pp. 71–72).

**Conditions.** Timeless; scikit-learn behaviour as described in 2018. Return attribution "will not work if there is a 'neutral' (return below threshold) case", for which lower returns would need higher weights; López de Prado advises dropping neutral cases (p. 69). Decay runs on cumulative uniqueness because chronological decay "would reduce weights too fast in the presence of redundant observations" (p. 70). Cases: $c = 1$ no decay; $0 < c < 1$ linear decay with every weight positive; $c = 0$ decay to zero; $c < 0$ zero weight for the oldest observations, those with $x \leq -c \sum_i \bar{u}_i$, a share $|c|$ of cumulative uniqueness (pp. 70–71). In scikit-learn, class weights that do not sum to the number of classes act like a change in the regularisation parameter (p. 71).

**Practice.** Attributed weights normalised by $I/\sum_j \tilde{w}_j$ sum to $I$; the decay constants meet $d[\sum \bar{u}] = 1$ and $d[0] = c$ for $c \geq 0$, and $d[-c \sum \bar{u}] = 0$ for $c < 0$, for $c = 1, 0.5, 0, -0.5$. Interpretation: López de Prado's implementation evaluates $d$ at the cumulative sums $\bar{u}_1, \bar{u}_1 + \bar{u}_2, \ldots$ (Snippet 4.11), so the oldest observation sits at $x = \bar{u}_1 > 0$ and receives slightly more than $c$; and his printed sum runs from $t = t_{i,0}$ (p. 69), as does Snippet 4.10, which adds the return ending at the event's own bar; the lower limit $t_{i,0} + 1$ above sums exactly the returns the label uses, as in label concurrency. On a toy path the two give attribution sums $(0.0247, 0.0145, 0.0190)$ and $(0.0149, 0.0146, 0.0288)$ for López de Prado's three labels.

**Pitfalls.** The normalisation includes the factor $I$, so weights sum to $I$, not 1. Giving lower returns higher weights applies to the neutral class only. Two slips are López de Prado's: the text says $d[1] = c$ where its own boundary condition, $a + b \cdot 0 = c$, gives $d[0] = c$; and "the oldest portion $cT$ of the observations receive zero weight" should be a share $-c$ of cumulative uniqueness, not of $T$ (pp. 70–71). Weights built from $t_{i,1}$, $c_t$ and realised returns are functions of future outcomes and belong to training only; see time-validity of labels and weights.

**See also.** label concurrency and uniqueness; sequential bootstrap; meta-labelling.

**Further reading.** King and Zeng (2001) on rare events, cited by López de Prado (p. 71).

### Pitfall 3.10 Time-validity of labels, sample weights and fitted transforms: what is known when, and what may be used only in training

**Statement.** Several of López de Prado's constructions use information from after the observation they describe. A triple-barrier label and its return use prices up to $t_{i,1}$ [alpha-afml, chap. 3, p. 45]; average uniqueness "requires information that is not available until a future time" and is safe only because it is "used on the training set … and not on the testing set" (chap. 4, p. 62); attribution weights use realised returns over the label's life (p. 69). These are legitimate as training targets and weights.

Interpretation, derived here: they leak when (i) a training label's span reaches into the test period, so the training set contains test-period prices; (ii) concurrency $c_t$ is computed with labels from both sets; (iii) a quantity fitted on the full sample (a CUSUM threshold, a barrier multiple, a fractional-differencing order $d$, an EWMA warm-up, a meta-label primary model) is then used in a backtest of that same sample; or (iv) weights or labels are used as features.

**Conditions.** The decision-time inputs, by contrast, are causal as López de Prado defines them: bar thresholds use EWMAs of prior bars (chap. 2, pp. 29–32), and the barrier width uses volatility up to the event (chap. 3, pp. 44–45). His fractional-differencing order is chosen on the whole E-mini history (chap. 5, Snippet 5.4, p. 85), and his quantamental recipe refits the secondary model "on the same training set" as the primary (p. 53); both need re-estimation inside each training window when the result is to be evaluated out of sample.

**Practice.** Interpretation, a checklist for one train/test split: compute labels, spans, concurrency, uniqueness and weights from training data; drop (purge) training observations whose spans overlap the test period, and leave a gap (embargo) after it; fit $d$, thresholds and barrier multiples on training data and apply them unchanged to the test period; build meta-labels from out-of-sample primary predictions; and record each choice with its data window.

**Pitfalls.** Uniqueness avoids leakage only when computed within the training set; the fractional-differencing order is fitted, not a fixed property of the series. Purging and embargo are López de Prado's chap. 7. His rules, stated in purged and embargoed cross-validation, agree with the checklist's purge and embargo, applied to every split.

**See also.** label concurrency and uniqueness; sample weights; meta-labelling; fixed-width fractional differencing; strategy production lifecycle (embargo before live trading, a different use of the word); parameter overfitting.

**Further reading.** On information timing, `alpha-jansen` and point-in-time language models; López de Prado's chap. 7 on evaluation.

**Method 3.11 Purged and embargoed $k$-fold cross-validation: dropping training labels that overlap the test set, a gap after it, and why shuffled $k$-fold and out-of-bag scores are inflated**

**Statement.** López de Prado gives two reasons why $k$-fold cross-validation "fails in finance": observations "cannot be assumed to be drawn from an IID process", and the test set is reused while a model is developed, "leading to multiple testing and selection bias" [alpha-afml, chap. 7, sec. 7.3, p. 104]. Chapter 7 treats only the first. With a serially correlated feature, $X_t \approx X_{t+1}$, and labels formed on overlapping data, $Y_t \approx Y_{t+1}$, putting $t$ and $t+1$ in different sets leaks information, so a classifier can score well "even if $X$ is an irrelevant feature"; leakage requires $(X_i, Y_i) \approx (X_j, Y_j)$, and "it does not suffice that $X_i \approx X_j$ or even $Y_i \approx Y_j$" (p. 105). The remedy has two parts (sec. 7.4, pp. 105–108):

- *Purging*: with each label a function of prices on a closed interval, $Y_j = f[[t_{j,0}, t_{j,1}]]$, drop from the training set every observation $i$ whose label overlaps a test label, that is, if $t_{j,0} \le t_{i,0} \le t_{j,1}$, or $t_{j,0} \le t_{i,1} \le t_{j,1}$, or $t_{i,0} \le t_{j,0} \le t_{j,1} \le t_{i,1}$ (p. 106, Snippet 7.1);
- *Embargo*: also drop training observations that start within $h$ after the test set, $t_{j,1} \le t_{i,0} \le t_{j,1} + h$, implemented by extending each test label to $t_{j,1} + h$ before purging; training labels that end before the test set begins need no embargo, since they "contain information that was available at the testing time"; "a small value $h \approx .01T$ often suffices" (p. 107, Snippet 7.2).

Purging and embargo are to be applied "whenever we produce a train/test split, whether it is for hyper-parameter fitting, backtesting, or performance evaluation"; Snippet 7.3's `PurgedKFold` assumes contiguous, unshuffled test blocks (sec. 7.4.3, p. 109).

**Notation.** $T$ is the number of bars; $[t_{i,0}, t_{i,1}]$ the span of the prices that determine label $i$ (for a triple-barrier label, from the event to the first barrier touched; triple-barrier labelling); $h$ the embargo length in bars, `pctEmbargo` $= h/T$ in the snippets.

**Conditions.** The three overlap conditions are equivalent to the two closed intervals intersecting. The text states the purge rule as dropping $i$ when the information sets satisfy "$\Phi_i \cap \Phi_j = \varnothing$" (p. 106); the next sentence and the snippet drop overlapping labels, so the condition meant is $\Phi_i \cap \Phi_j \neq \varnothing$, an apparent misprint in the 2018 printing.

In the class (p. 109) the left training set keeps only observations whose label ends by the start of the test block, and the right set starts $h$ bars after the first observation that begins after the last test label ends; the same check reproduces these index sets on a toy series, and shows that the class keeps a training label that ends exactly at the test block's first bar, or starts exactly at the last test label's end, which the closed-interval rule would purge (one shared bar); and that its embargo counts observations, `int(X.shape[0]*pctEmbargo)`, which equals bars only when there is one observation per bar.

*The source's plateau heuristic.* López de Prado writes that with leakage "performance improves merely by increasing $k \rightarrow T$"; with purging, performance improves with $k$ because the model "recalibrate[s] more often", but beyond some $k^*$ it "will not improve, indicating that the backtest is not profiting from leaks" (pp. 106–107). That is a diagnostic heuristic stated by the source, not a proof that leakage is absent.

A plateau is consistent with leakage that does not grow with $k$:

- features computed with data released after the decision time;
- a normaliser or other transform fitted on the full sample (time-validity of labels and weights);
- an embargo too short for the feature's own serial dependence;
- reuse of the same folds for selection.

Conversely, a rising curve need not mean leakage. Purging and embargo address one channel, overlapping label spans, and the overlap and endpoint corrections above stand. Ruling out the others still needs *information provenance*, a record of when each feature, label and transform input was known (reference data; filing timestamps), and a *time-valid evaluation*, a walk-forward or a holdout that follows the development period in time.

**Practice.** *Out-of-bag scores.* For bagged trees and random forests on overlapping labels, "out-of-bag accuracy will be inflated" because sampling with replacement puts in the training set samples "very similar to those out-of-bag"; López de Prado advises unshuffled stratified $k$-fold, ignoring out-of-bag accuracy, and a low $k$ (chap. 6, sec. 6.3.3, p. 97), and repeats that random-forest out-of-bag estimates in finance "are likely to be inflated" (sec. 6.4, p. 98).

*When ordinary $k$-fold is valid.* Bergmeir, Hyndman and Koo show that for a purely autoregressive model, $y_t = g(x_t; \theta) + e_t$ with $x_t$ the lags, standard $K$-fold over the rows of the embedded matrix is valid "as long as the models considered have uncorrelated errors", as when the model nests an appropriate one; underfitting should be ruled out first by testing residuals for serial correlation, and in their simulations $K$-fold beat out-of-sample evaluation and non-dependent ($h$-block) cross-validation [rf-bergmeir-hyndman-koo-2018, working paper of 23 July 2017, abstract and secs. 1–2, pp. 1–3; sec. 7, p. 14]. Cerqueira, Torgo and Mozetič find cross-validation applicable to stationary series, while for real series with non-stationary variation "the most accurate estimates are produced by out-of-sample methods that preserve the temporal order of observations" [rf-cerqueira-torgo-mozetic-2020, arXiv v1, 28 May 2019, abstract, p. 1]. Interpretation: the conditions differ.

Bergmeir et al.'s result concerns one-step autoregressive rows with uncorrelated errors; López de Prado's labels span many bars, so neighbouring rows share label prices and the errors of a model that ignores the overlap are not uncorrelated, which is the case their result excludes. Purging removes that overlap but does not address non-stationarity, Cerqueira et al.'s reason for preferring time-ordered holdouts.

**Pitfalls.** Purging and embargo address leakage, not the second failure: choosing among many configurations on the same folds still needs a correction for the number of trials (chap. 7, p. 104; deflated Sharpe ratio). Purged $k$-fold trains on data after the test block, so it is not a walk-forward backtest; López de Prado treats backtesting separately (chaps. 10–16). Cerqueira et al. are cited from the arXiv preprint (62 real series and three synthetic ones); the 2020 *Machine Learning* version uses 174 real series. Bergmeir et al. are cited from the accepted working paper.

**See also.** time-validity of labels and weights; label concurrency and uniqueness; sequential bootstrap; parameter overfitting.

**Further reading.** The rest of chap. 7 (sec. 7.5 on scikit-learn's scoring bugs, pp. 109–110) and chap. 12's combinatorial purged cross-validation.

**Method 3.12 Recording a time-valid ML or language-model experiment: data release times, training and pretraining cutoffs, tuning trials and held-out dates**

**Statement.** The problem for language models is stated by Kelly, Malamud, Schwab and Xu in their NBER working paper of May 2026 [latest-pit-language-models, WP 35247, abstract, p. 1]. Models "trained on unrestricted internet corpora inevitably embed information from the future, introducing lookahead bias that compromises the validity of backtests". Point-in-time models, "trained exclusively on text available up to each calendar date", remove this "by construction". Their implementation shows what has to be recorded:

- *Pretraining cutoff.* Monthly checkpoints from 2013 to 2024 are trained on FineWeb text, which "is already indexed with publication timestamps allowing us to enforce a temporal cutoff directly" (sec. 3, p. 5; Table 1).
- *Fine-tuning data.* These were filtered by a classifier (gpt-5-nano) that keeps only "timeless" examples free of "real-world events, real individuals, specific dates" (App. B, p. 22).
- *Use in a forecast.* Each calendar year's news is embedded with the checkpoint of December of the year before. Because the earliest checkpoint is December 2013, out-of-sample results run from 2014 to May 2020 on Dow Jones Newswire (sec. 5, p. 9).

**Notation.** Decision time $t$; data release (record) time of each input; model cutoff $c$, the last date of text or data any trained component saw; held-out window, the dates never used for fitting or selection.

**Conditions.** *A minimal record* (Interpretation). For every experiment, log:

- each input's release time, not its reference date (bitemporal tables; filing timestamps);

- every model component's cutoff, including pretrained models and any model used to filter or label data;
- the label spans, purge and embargo (purged cross-validation);
- the number and dispersion of tuning trials, kept for deflation and multiple testing;
- the held-out dates, fixed before results are seen, with a note of any later look at them.

A cutoff is only as good as its timestamps. The paper relies on FineWeb's publication timestamps and does not test their accuracy.

**Practice.** The paper reports that portfolios built from point-in-time embeddings "achieve robust positive Sharpe ratios and perform close to full-sample counterparts that violate temporal validity" (abstract). That is a claim about its sample and design.

**Pitfalls.** Interpretation: a filter or labeller trained on unrestricted data, here the classifier used for fine-tuning data, can itself carry later knowledge into what it keeps, even when it only removes examples; the paper's design limits this to fine-tuning data, but a record should name it. Commit dates of a protocol file are not proof that it preceded the results when both are committed together. Its working-paper abstract on Chicago Booth's Fama–Miller page states that "commercial LLMs suffer from significant look-ahead bias, which may explain a large portion of their predictability in various settings", and that LLMs "exhibit extremely poor numerical reasoning" [latest-levy-caution-llm, working-paper abstract on the Fama–Miller Center page].

A contrasting estimate: He, Lv, Manela and Wu train language models for each year from 1999 on text available at the time. Next-day return forecasts from news using their outputs reach Sharpe ratios "comparable to a much larger Llama model, indicating that lookahead bias is modest" in that application. They add that it is "model and application-specific", since a prediction model on top can compensate [wide-chrono-llm, arXiv v3, 6 July 2025, abstract]. The two disagree on magnitude for different models and tasks, which is a reason to record the training cutoff and the evaluation window rather than assume either.

*A taxonomy of leakage to check against.* Kapoor and Narayanan sort leakage into three classes and eight types: [L1] no clean separation of training and test data (no test set; pre-processing, or feature selection, on the combined data; duplicates); [L2] features that "should not be legitimately available", such as proxies for the outcome; and [L3] a test set not drawn from the distribution of interest (temporal leakage, when the test set contains data "from a date before the training set"; non-independence between training and test samples; sampling bias) [wide-kapoor-narayanan-leakage, sec. 2.4, PDF pp. 5–6].

Mapped to this record (interpretation): release times and training cutoffs address [L3.1]; purging and embargoes [L3.2]; fitting scalers, imputations and feature screens inside each training window [L1.2]–[L1.3]; and a field-by-field availability check [L2]. The record above does not yet name duplicates across overlapping instruments or vendor histories [L1.4], or a test window chosen after results [L3.3].

**See also.** time-validity of labels and weights; purged cross-validation; bitemporal reference tables.

**Further reading.** `latest-levy-caution-llm`; `alpha-jansen`.

## 3.3 Fractionally differentiated features

#### Model 3.13 Fractional differencing weights: the binomial expansion of $(1-B)^d$, the recurrence, integer and fractional cases, signs and decay

**Statement.** With the backshift operator $B^k X_t = X_{t-k}$, a real order $d$ gives the formal binomial series [alpha-afml, chap. 5, sec. 5.4, p. 77]

$$(1-B)^d = \sum_{k=0}^{\infty} \binom{d}{k} (-B)^k, \qquad \tilde{X}_t = \sum_{k=0}^{\infty} \omega_k X_{t-k}, \qquad \omega_k = (-1)^k \frac{\prod_{i=0}^{k-1}(d-i)}{k!},$$

so $\omega = (1, -d, d(d-1)/2!, -d(d-1)(d-2)/3!, \dots)$, generated by $\omega_0 = 1$ and $\omega_k = -\omega_{k-1}(d-k+1)/k$ (secs. 5.4.1–5.4.2, pp. 77–78). For a positive integer $d$ the weights vanish beyond $k = d$ and "memory beyond that point is cancelled": $d = 1$ gives $(1, -1, 0, \dots)$, returns; $d = 0$ gives $(1, 0, \dots)$, the original series (p. 78).

**Conditions.** For $0 < d < 1$ every weight after $\omega_0$ lies in $(-1, 0)$; for $1 < d < 2$, $\omega_1 < -1$ and $\omega_k > 0$ for $k \geq 2$ (p. 78, Figures 5.1–5.2). For $k > d$ the ratio $|\omega_k/\omega_{k-1}| = |(d-k+1)/k| < 1$; the initial weights alternate in sign while $k < d+1$, and beyond that the weights keep one sign, negative when $\mathrm{int}[d]$ is even and positive when it is odd (sec. 5.4.3, p. 80). López de Prado concludes that the weights "converge asymptotically to zero, as an infinite product of factors within the unit circle" (p. 80).

Interpretation: the ratio tends to 1 (0.9986 at $k = 1{,}000$ for $d = 0.4$), so the product argument alone is not a proof; the decay is a power law, $|\omega_k| \approx k^{-(1+d)}/|\Gamma(-d)|$, which matches within 0.05 percent at $k = 10^3$ and to four digits at $k = 10^4$. The weights therefore fall slowly and the partial sums approach $\sum_k \omega_k = 0$ slowly (0.017 after $10^4$ terms at $d = 0.4$), which is why truncation matters (fixed-width window).

**Practice.** The recurrence equals the closed form for $d = 0, 0.4, 1, 1.5, 2$; $d = 0$ and $d = 1$ give $(1, 0, \dots)$ and $(1, -1, 0, \dots)$; all weights after the first lie in $(-1, 0)$ at $d = 0.4$. López de Prado's motivation: integer differencing makes price series stationary "at the expense of removing all memory", while "stationarity is a necessary property for inferential purposes" and memory "is the basis for the model's predictive power" (sec. 5.2, p. 75); supervised learning needs stationary features, but "stationarity does not ensure predictive power" (p. 76). His literature review attributes the idea to Hosking (1981), whose fractionally differenced processes "exhibit long-term persistence and antipersistence, hence enhancing the forecasting power compared to the standard ARIMA approach" (sec. 5.3, p. 76).

**Pitfalls.** Replacing $(d-i)$ by $(d-1)$ in the general weight is incorrect: at $d = 0.4$ it gives $1, 0.6, 0.18, \dots$ instead of $1, -0.4, -0.12, \dots$; they dropped the variable from the limits and from "$\forall k > d$"; and they turned López de Prado's "this alternation of weight signs is necessary to make $\{\tilde{X}_t\}$ stationary" (p. 80) into a sufficient condition ("makes … stationary"). For $0 < d < 1$ the weights do not alternate after $\omega_1$, and no sign pattern by itself makes a series stationary: that depends on the series (fixed-width window, Conditions).

**See also.** fixed-width fractional differencing; return measures (integer differencing of log prices).

**Further reading.** Hosking (1981) and the fractional-integration literature López de Prado lists (p. 76).

#### Method 3.14 Fractionally differentiating a finite series: expanding window and weight loss, the fixed-width window (FFD), and choosing the minimum order $d$ with a unit-root test

**Statement.** A finite series $X_1, \dots, X_T$ cannot use infinitely many weights. With an *expanding window*, $\tilde{X}_T$ uses $\omega_0, \dots, \omega_{T-1}$ and earlier points fewer, so they carry different amounts of memory; López de Prado drops the early points whose relative weight loss exceeds a tolerance $\tau$, such as 0.01 [alpha-afml, chap. 5, sec. 5.5.1, p. 80]. Even so the series drifts down, "because values … are computed on an expanding window" that keeps adding negative weights (pp. 80–82, Figure 5.3). The *fixed-width window* (FFD) drops the weights once $|\omega_k|$ falls below a threshold $\tau$: with $l^*$ the last lag with $|\omega_{l^*}| \geq \tau$, $\tilde{\omega}_k = \omega_k$ for $k \leq l^*$ and 0 beyond, and $\tilde{X}_t = \sum_{k=0}^{l^*} \tilde{\omega}_k X_{t-k}$, the same weights for every point, which avoids the drift (sec. 5.5.2, pp. 82–83).

The *minimum order* $d^*$ is the smallest $d$ for which the FFD series passes a stationarity test; López de Prado's recipe is to take the cumulative sum of a feature (so that some differencing is needed), compute FFD($d$) for $d \in [0, 1]$, and keep the smallest $d$ whose ADF $p$-value is below 5 percent (secs. 5.6–5.7, pp. 84 and 88).

**Notation.** Interpretation, with one consistent index (the source's is not): the point computed from the first $t$ observations misses the weights $\omega_t, \dots, \omega_{T-1}$, a relative weight loss $\lambda_t = \sum_{k=t}^{T-1} |\omega_k| / \sum_{k=0}^{T-1} |\omega_k|$, and is kept if $\lambda_t \le \tau$. FFD values exist for $t = l^* + 1, \dots, T$.

**Conditions.** The futures evidence as of 2018. López de Prado's evidence: E-mini S&P 500 log prices (rolled with the ETF trick, daily) have an ADF statistic of $-0.34$ undifferenced and $-46.9$ as returns; the 95 percent critical value $-2.86$ is crossed near $d = 0.35$, where the FFD series still has correlation 0.995 with the original, against 0.03 for returns (pp. 84–85, Figure 5.5); across 87 liquid futures, every log-price series passes at $d < 0.6$, most at $d < 0.3$, and some need no differencing (Table 5.1, pp. 85–87). He reads this as "most studies have over-differentiated the series" (p. 85) and the FFD series as "a driftless blend of level plus noise … no longer Gaussian … however it is stationary" (p. 83).

Interpretation: these are unit-root test outcomes on single samples, not demonstrations of stationarity. Snippet 5.4 runs the ADF test with one lag and a constant only, on FFD with $\tau = 0.01$, a short window (11 weights at $d = 0.4$); the ADF test is built against a unit root, not against fractional integration, and under fractional-integration theory a series integrated of order 1 and differenced by $d$ is integrated of order $1 - d$, covariance-stationary only for $d > 1/2$. The truncated weights also do not sum to zero, so a small multiple of the level remains (0.264 times $X_{t-l^*}$ at $d = 0.4$, $\tau = 0.01$; 0.036 at $\tau = 10^{-5}$, computed here).

**Practice.** At $d = 0.4$ the FFD window keeps 11, 55 and 1,458 weights for $\tau = 10^{-2}, 10^{-3}, 10^{-5}$; for an expanding window on 200 points, the weight loss of the point using $t$ observations falls from 0.48 at $t = 1$ to 0.0135 at $t = 100$, so $\tau = 0.01$ keeps points from $t = 118$. On 2,000 points of a simulated random walk, the one-lag ADF test on FFD($d$) with $\tau = 10^{-4}$ rejects a unit root at 5 percent already at $d = 0.2$ (statistic $-5.3$) and at $d = 0.4$ ($-10.2$), although the theory above says such a series is not stationary for $d < 1/2$: the outcome reflects the test, its lag choice and the truncation. Interpretation: choose $d$, $\tau$ and the test on training data only, and apply them unchanged out of sample; López de Prado chooses $d$ on the full E-mini history (time-validity of labels and weights).

**Pitfalls.** The source contains two indexing slips: the weight-loss sum running to $T$ rather than $T - 1$, with the dropped points named $\tilde{X}_1, \dots, \tilde{X}_{l^*}$ although $\lambda_l$ refers to point $T - l$; and the FFD range $t = T - l^* + 1, \dots, T$, which should be $l^* + 1, \dots, T$ (pp. 80 and 82). FFD output is not stationary in general; the stationarity claims rest on the minimum-$d$ procedure.

**See also.** fractional differencing weights; ETF trick (the rolled series López de Prado differentiates); time-validity of labels and weights.

*Testing against fractional alternatives.* The ADF test asks only $d = 1$ against $d = 0$, so passing it at $d^*$ says nothing about the fractional order. MacKinnon and Nielsen work in Johansen's fractional VAR, $\Delta^d X_t = \Delta^{d-b} L_b \alpha \beta' X_t + \sum_{i=1}^{k} \Gamma_i \Delta^d L_b^i X_t + \varepsilon_t$. It nests Johansen's cointegrated VAR ($d = b = 1$), the standard Dickey–Fuller test ($d = b = 1$, one variable) and the fractional Dickey–Fuller test (one variable, any $d$, $b$). Its maximum-likelihood estimates of $d$, $b$ and the other parameters are asymptotically Gaussian, so they can be tested with $\chi^2$ likelihood ratios. The likelihood-ratio test of the cofractional rank, however, has a non-standard limit when $0.5 < b \le d$, a functional of fractional Brownian motion that "depend[s] on a real-valued parameter, $b$, which must be estimated".

Its critical values must therefore be simulated for the estimated $b$, which their numerical distribution functions provide [mackinnon-nielsen-2010-wp, CREATES version of September 2010, abstract and secs. 1–2, eqs. (1)–(2), pp. 1–4]. The model as analysed "does not contain any deterministic terms" (p. 2), so a constant or trend in prices is outside it. López de Prado's minimum-$d$ search is not this procedure: it tests a transformed series against a unit root, not the fractional order itself.

**Further reading.** MacKinnon and Nielsen's published version (*J. Appl. Econometrics*, 2014) and Johansen and Nielsen's estimation papers.

## 3.4 Optimal execution as stochastic control: the benchmark for learned policies

#### Model 3.15 Optimal execution with order flow as stochastic control: the Cartea–Jaimungal objective, dynamics and dynamic-programming equation, as presented by Lehalle and Laruelle

**Statement.** A trader sells an inventory $Q_0$ by time $T$, choosing a trading rate $v_t$ [alpha-laruelle, app. A, sec. A.7.3, pp. 281–282]. The state is the midprice $S_t$, the net order flow $\mu_t$ of all other traders, the remaining inventory $Q_t$ and the cash account $X_t$, with

$$dS_t = b\,(\mu_t - v_t)\,dt + \sigma\,dW_t, \qquad \hat{S}_t = S_t - \Big(\frac{\psi}{2} + \kappa v_t\Big), \qquad dQ_t = -v_t\,dt, \qquad dX_t = \hat{S}_t\,v_t\,dt$$

(p. 282): permanent impact linear in the trading rate, and temporary impact $\kappa v_t$ with full immediate decay (p. 281). The reward of a given policy $\nu$ is

$$H_t(\nu) = \mathbb{E}_t\Big[X_T^\nu + Q_T^\nu\Big(S_T - \frac{\psi}{2} - A\,Q_T^\nu\Big) - \phi\int_t^T (Q_s^\nu)^2\,ds\Big].$$

This is the book's objective (p. 281) with the half-spread added to the terminal payoff, so that it agrees with the book's own terminal condition (Interpretation; the printed objective is discussed under Pitfalls). The value function is the best such reward over admissible policies, $H^* = \sup_\nu H(\nu)$, finite in the parameter regime given under Conditions. By dynamic programming it formally solves (Interpretation: the book's Proposition 6, p. 282, with the corrections listed under Pitfalls)

$$0 = \Big(\partial_t + \tfrac{1}{2}\sigma^2\partial_{SS} + \mathcal{L}^\mu\Big)H^* - \phi q^2 + \sup_v\Big\{v\Big(S - \frac{\psi}{2} - \kappa v\Big)\partial_x H^* + b(\mu - v)\,\partial_S H^* - v\,\partial_q H^*\Big\},$$

with $H^*(T, x, S, \mu, q) = x + q(S - \psi/2) - Aq^2$, the terminal payoff above, where $\mathcal{L}^\mu$ is the generator of the flow. Defining $h := H^* - x - q(S - \psi/2)$, so that $h(T, \mu, q) = -Aq^2$, gives

$$\partial_t h + \mathcal{L}^\mu h + b\mu q - \phi q^2 + \sup_v\{-\kappa v^2 - (bq + \partial_q h)v\} = 0, \qquad v^* = -\frac{bq + \partial_q h}{2\kappa}$$

(p. 283; the half-spread cancels in this decomposition). With $h = h_0 + qh_1 + q^2 h_2$, the coefficient $h_2$ solves a Riccati equation that does not involve $\mu$, $h_1$ carries the expected future flow, and $h_0$ follows by integration (pp. 283–284). The authors' point is that the trader can act on a view of future flows (p. 284). Interpretation: with $b + 2h_2 = -2\xi$, the optimal rate is

$$v_t^* = \frac{\xi(t)}{\kappa}\,Q_t - \frac{h_1(t, \mu_t)}{2\kappa}, \qquad h_1(t, \mu) = b\int_t^T e^{-\frac{1}{\kappa}\int_t^s \xi(u)\,du}\,\mathbb{E}_{t,\mu}[\mu_s]\,ds,$$

so the trader sells in proportion to inventory and sells more slowly when other traders are expected to keep buying.

**Notation.** Units, for a sale: $Q$ in shares; $v$ and $\mu$ in shares per unit time, $v > 0$ selling and $\mu > 0$ net buying by others; $S$, $\psi$ and $\hat{S}$ in currency per share; $b$ in currency per share$^2$, the price change per share of net flow; $\sigma$ in currency per share per square root of time; $\kappa$ in currency × time per share$^2$; $\phi$ in currency per share$^2$ per unit time; $A$ in currency per share$^2$. Then $h$ is in currency, $h_1$ in currency per share, $h_2$ and $\xi$ in currency per share$^2$, and $\gamma = \sqrt{\phi/\kappa}$ per unit time; $A - b/2$ and $\sqrt{\kappa\phi}$ are both in currency per share$^2$, as the formula for $\zeta$ requires.

Signs: the trader's own selling lowers $S$ permanently by $b$ per share; each fill is $\psi/2 + \kappa v$ below the midprice; others' net buying raises $S$. $H_t(v)$ is the reward of one fixed policy; the value function is $H^*$.

**Conditions.** Model timeless; the survey's account of RL as of February 2023. The trader observes $S$, $\mu$, $Q$ and $X$ at each time. The flow $\mu$ is Markov with a known generator and does not respond to the trader. The parameters $b, \kappa, \sigma, \psi, \phi$ and $A$ are constant: the book writes $b_t$ in the price dynamics but a constant $b$ in the equation.

*Finite value.* Interpretation: with the cash account starting at $X_0 = 0$, so that cash is measured from the start of execution (otherwise add $X_0$ to the right-hand side), integrating by parts, the realised reward of any policy is

$$\begin{aligned} &X_T + Q_T\Big(S_T - \frac{\psi}{2} - AQ_T\Big) - \phi\int_0^T Q_t^2\,dt \\ &\quad = Q_0S_0 - \frac{\psi Q_0}{2} - \frac{bQ_0^2}{2} + b\int_0^T Q_t\mu_t\,dt + \sigma\int_0^T Q_t\,dW_t - \kappa\int_0^T v_t^2\,dt - \phi\int_0^T Q_t^2\,dt - \Big(A - \frac{b}{2}\Big)Q_T^2, \end{aligned}$$

because the trader's own permanent impact contributes $b(Q_T^2 - Q_0^2)/2$ whatever the path. A simple sufficient regime for a finite value and for the formulas used here is $\kappa > 0$, $\phi > 0$ and $A > b/2$, with admissible policies progressively measurable with respect to the observed state and $\mathbb{E}\int_0^T v_t^2\,dt < \infty$, and a flow with $\mathbb{E}\int_0^T \mu_t^2\,dt < \infty$. Then $\mathbb{E}\int Q\,dW = 0$; $|\mathbb{E}\int Q\mu\,dt| \le \|Q\|\,\|\mu\|$ with $\|Q\| \le \sqrt{T}Q_0 + T\|v\|$ (norms in $L^2(\Omega\times[0,T])$), so the expected reward is bounded above by a concave quadratic in $\|v\|$; and $|\zeta| < 1$, so $\xi$ is finite and positive on $[0,T]$. The regime is sufficient, not necessary, and no verification theorem is proved here: the quadratic ansatz is a formal solution. Outside it the value can be infinite.

With $T = Q_0 = 1$, $\mu = \sigma = \psi = 0$, $b = 0.1$, $\kappa = A = 0.01$, $\phi = 0.001$ and a constant buying rate $v = -M$, the reward's $M^2$ coefficient is $b/2 - \kappa - A - \phi/3 \approx 0.0297 > 0$: buying raises the price at which the enlarged position is marked at $T$ faster than the penalties grow. Simulated rewards divided by $M^2$ converge to this coefficient.

*Constraints and completion.* The unconstrained problem is a formal benchmark. Economically valid one-directional execution requires $v \ge 0$, and $Q_t \ge 0$. The unconstrained optimum can ask the seller to buy back, for instance when strong expected buying flow makes $h_1$ large, and the half-spread term then credits the purchase (Pitfalls). Clipping $v^*$ at zero does not solve the constrained problem: the constraint changes the value function itself, whose equation then takes the supremum over $v \ge 0$ and has a different $h$. A finite $A$ charges the inventory left at $T$, which is sold at $S_T - \psi/2 - AQ_T$. That is a quadratic penalty, not mandatory completion; completion is forced only in the limit $A \to \infty$.

*Penalty and variance.* The running penalty $\phi\int Q^2$, which the book calls risk aversion (p. 282), is not in general the variance of implementation shortfall. For a deterministic schedule $Q$ the random part of the reward is $b\int Q_t\mu_t\,dt + \sigma\int Q_t\,dW_t$. Its variance is

$$b^2\operatorname{Var}\Big(\int Q_t\mu_t\,dt\Big) + \sigma^2\int Q_t^2\,dt + 2b\sigma\operatorname{Cov}\Big(\int Q_t\mu_t\,dt, \int Q_t\,dW_t\Big),$$

and the covariance is nonzero when flow and price shocks are correlated. Thus $\sigma^2\int Q^2dt$ is only the contribution of the Brownian innovations. It is the whole variance when $b = 0$ or the flow path is deterministic; only then does $\phi = \lambda\sigma^2$ reproduce a mean–variance penalty. Example: $Q_t = 1 - t$, $\mu_t = B_t$ a Brownian motion, $T = 1$, $\sigma = 0$ give variance $b^2\int_0^1\int_0^1 (1-s)(1-t)\min(s,t)\,ds\,dt = b^2/20$ (exact). For a policy that adapts to the flow, $\phi$ is a penalty on holding inventory, not a variance.

*Benchmark limit.* With $\mu \equiv 0$ and $A \to \infty$ the optimal holding is $Q_t = Q_0\sinh(\gamma(T-t))/\sinh(\gamma T)$ with $\gamma = \sqrt{\phi/\kappa}$. This is the continuous-time form of the Almgren–Chriss schedule, with $\kappa$ in the role of $\eta$ and $\phi$ of $\lambda\sigma^2$ (Interpretation; matched numerically to four decimals).

**Practice.** The parameter intuition holds under the model's assumptions, with qualifications (Interpretation; own numerical paths with $\mu \equiv 0$, $Q_0 = 1$, $T = 1$, base case $\kappa = 0.01$, $\phi = 0.001$, $A = 0.5$, in consistent units):

- a larger $\phi$ front-loads the sale: tenfold $\phi$ leaves 0.45 rather than 0.50 at $T/2$;
- a larger $A$ mostly cuts what is left at $T$, 0.2 rather than 1.9 percent for tenfold $A$, and barely changes the early path;
- a larger $\kappa$ slows and smooths trading: tenfold $\kappa$ leaves 0.58 at $T/2$, and with finite $A$ it also leaves 17 percent unsold at $T$.

With flow, the rate also depends on the expected $\mu$. Lehalle and Laruelle suggest Almgren–Chriss when price formation is well understood at a 5–30-minute scale, and this framework when one knows a continuous-time version of price formation focused on trading flows (p. 284).

As a benchmark for learned policies: stochastic control solves for a policy given a specified model. Simple models are tractable but may oversimplify the market; realistic ones are often intractable [wide-hambly-xu-yang, sec. 1, p. 2]. Reinforcement learning learns decisions from repeated interaction with a system, without exemplary supervision or a complete model of the environment (sec. 2, p. 3). When the transition law and rewards are unknown the problem becomes one of learning: *model-based* methods learn them explicitly, *model-free* methods implicitly (sec. 2.2, p. 7). The survey motivates RL for execution by Almgren–Chriss's reliance on its dynamics and linear impact, and by its pre-planned schedule, which ignores real-time conditions (sec. 4.2, p. 33).

The review also reports one study that combined the Almgren–Chriss solution with Q-learning, and one double-DQN study that added a final one-second step to force liquidation (p. 34). On data, most RL algorithms need more samples than the relevant historical data provide, and financial series are non-stationary, hence the call for sample-efficient methods or good market simulators (sec. 5, p. 49). Interpretation: training against a simulator is interaction, but the simulator must be validated. Replaying historical order books does not show how the market would have responded to the agent's own orders, so own impact must still come from a model. Learning from logged historical decisions without interaction is a different setting.

*A second review.* Kolm and Ritter frame execution as one of the intertemporal-choice problems reinforcement learning can solve "in an almost model-free way". Their Example 3.2 writes the Almgren–Chriss mean–variance objective as the value function to maximise [wide-kolm-ritter-rl, undated preprint, Internet Archive capture of 28 May 2024, abstract and Example 3.2, pp. 1 and 9]. They present learning as a replacement for the model-based solution.

**Pitfalls.** Inconsistencies in the appendix (pp. 281–284; corrections are Interpretation):

- Proposition 6 defines the value function maximising the sale as "$H^* := \min_v H(v)$" while its equation takes a sup; the value is $\sup_v H(v)$.
- The printed objective is $H_t(v) = \mathbb{E}_t[X_T^v + Q_T^v(S_T - AQ_T^v) - \phi\int_t^T (Q_s^v)^2 ds]$ (p. 281). Its terminal payoff lacks the half-spread that the book's terminal condition $x + q(S - \psi/2) - Aq^2$ (p. 282) charges, and its ansatz and $h$-equations follow the terminal condition. This entry therefore uses the corrected payoff of the Statement. There every share pays $\psi/2$, whether sold before or at $T$: the total charge is the constant $\psi Q_0/2$, and the spread does not affect the control. Under the printed payoff the natural decomposition is a different one, $g := H - x - qS$ with $g(T,\mu,q) = -Aq^2$, which gives

$$\partial_t g + \mathcal{L}^\mu g + b\mu q - \phi q^2 + \sup_v\Big\{-\kappa v^2 - \Big(bq + \partial_q g + \frac{\psi}{2}\Big)v\Big\} = 0, \qquad v^{\mathrm{pr}} = -\frac{bq + \partial_q g + \psi/2}{2\kappa}.$$

Here $g_2 = h_2$, and $g_1 + \psi/2$ solves the $h_1$ equation with terminal value $\psi/2$, so

$$v_t^{\mathrm{pr}} = v_t^* - \frac{\psi}{4\kappa}\, e^{-\frac{1}{\kappa}\int_t^T \xi(u)\,du}\,;$$

shares held to $T$ escape the half-spread, so selling slows, most near $T$. Both policies agree with a direct discretised optimisation with $\mu = \sigma = 0$. With $\kappa = 0.01$, $\phi = 0.001$, $A = 0.05$, $b = 0.02$, $\psi = 0.01$ and $T = Q_0 = 1$, the inventory left at $T$ is 0.195 under the corrected payoff and 0.245 under the printed one; direct and feedback paths agree to four decimals.

- The equation prints $-\partial_q H^*$ inside the supremum. The inventory dynamics $dQ = -v\,dt$ give $-v\,\partial_q H^*$, and the book's own next step on p. 283 is consistent only with $-v\,\partial_q H^*$.
- Notation: p. 282 says "add the bid–ask spread $\phi$" where the spread is $\psi$ and $\phi$ is the running penalty. The equation writes $k$ for $\kappa$ and $b_t$ for $b$, and the terminal conditions for $h_0$ and $h_1$ are printed at $t$ rather than $T$.
- The half-spread enters $\hat{S}$ linearly in $v$, so a negative rate (buying back) would be credited the half-spread. The spread model is economically coherent only for one-directional selling, which the formal benchmark does not impose (Conditions).
- The printed closed form $h_2 = -\xi - b/2, \xi(t) = \sqrt{\kappa\phi}\,(1+\zeta e^{2\gamma(T-t)})/(1-\zeta e^{2\gamma(T-t)}), \zeta = (A - b/2 + \sqrt{\phi\kappa})/(A - b/2 - \sqrt{\phi\kappa})$ gives $h_2(T) = A - b$, not $-A$. The Riccati solution has $\xi(t) = \sqrt{\kappa\phi}\,(1+\zeta e^{-2\gamma(T-t)})/(1-\zeta e^{-2\gamma(T-t)})$ with $\zeta = (A - b/2 - \sqrt{\kappa\phi})/(A - b/2 + \sqrt{\kappa\phi})$; equivalently, the printed expression with the sign of $\xi$ reversed. This agrees with numerical integration.
- The boxed $h_1$ "when $b = 0$" omits the factor $b$: with $b = 0$ the source term vanishes and $h_1 \equiv 0$, so the flow drops out. The general solution is the $h_1$ above.

This formulation follows Lehalle and Laruelle's presentation of the Cartea–Jaimungal model. Learning a policy or value function from simulated interaction is not the same as learning it from historical data.

**See also.** mean–variance execution (the discrete deterministic benchmark); price impact (permanent, temporary and transient impact); trading envelope (where a schedule sits in a trading algorithm); implementation shortfall.

## 3.5 Types of financial data

**Definition 3.16 Four types of financial data: fundamental, market, analytics and alternative data, with their timing, cost and uniqueness**

**Statement.** López de Prado orders four "essential types of financial data" by increasing diversity, from fundamental to alternative data [alpha-afml, chap. 2, sec. 2.2, Table 2.1, pp. 23–24]:

- *fundamental data*: assets, liabilities, sales, costs and earnings, macro variables; "mostly accounting data, reported quarterly" and "reported with a lapse" (p. 23);
- *market data*: prices, yields and implied volatilities, volume, dividends and coupons, open interest, quotes and cancellations, aggressor side; "all trading activity that takes place in an exchange … or trading venue" (p. 24);
- *analytics*: analyst recommendations, credit ratings, earnings expectations, news sentiment; "derivative data, based on an original source", processed for the user and "not readily available from an original source" (p. 25);
- *alternative data*: satellite and CCTV images, web searches, social media, metadata; "primary information, that is, information that has not made it to the other sources" (p. 25).

Following Kolanovic and Krishnamachari (2017), he divides alternative data into data from individuals, business processes and sensors (p. 25).

**Conditions.** Practitioner descriptions as of 2018–2020; the 10TB figure undated and unsourced. *Timing.* Fundamental data must be used only from the moment each point was released; López de Prado's example is Bloomberg, which indexes fundamental data by the last date in the report, "which precedes the date of the release (often by 1.5 months)" (p. 23). It is also backfilled (missing values assigned values "unknown at that time") and reinstated (corrections overwriting the first release), so a dataset that keeps only final values assigns information to dates when it was not known; some vendors keep every release, for example three values for a quarterly GDP figure (p. 24).

*Cost and use.* Analytics may be costly, their methodology "biased or opaque", and "you will not be the sole consumer"; alternative data raise cost and privacy concerns (p. 25). Velu, Hardy and Nehren add that the value of an alternative dataset depends on the user's resources and "maybe more importantly—on the uniqueness of the data set. As more investors get access to it, the harder it becomes to extract meaningful alpha" [s3-velu, chap. 1, sec. 1.4.4, pp. 43–44].

**Practice.** López de Prado's judgements are opinions, not findings: fundamental data being "extremely regularized and low frequency" leaves "rather unlikely that there is much value left", though it may help combined with other data (p. 24); FIX market data is "appealing" because it is "not trivial to process" and abundant, "over 10TB being generated on a daily basis" (p. 25, no source or market given); and "a dataset may be useful if it annoys your data infrastructure team", since competitors may not have tried, gave up or processed it wrongly (p. 25).

Velu, Hardy and Nehren list fundamental and analyst inputs with their own hedges (pp. 42–43): key ratios (EPS, P/E, P/B) for cross-sectional comparison; analyst recommendations, individually "noisy" but in aggregate a consensus whose changes "can have a direct impact on price"; earnings estimates and surprises; holdings disclosed on Form 13F, where sudden changes "might indicate changes in sentiment by sophisticated investor"; insider purchases and sales, "thought to be an indicator of future stock price moves"; and credit rating downgrades, which "generally have a negative impact on equity prices". These are claims to test, not results.

*Insider trades, tested.* Cohen, Malloy and Pomorski split insiders into "routine" traders, predictable from their own past trading pattern, and "opportunistic" ones, using Thomson Reuters insider filings for 1986–2007. Routine trades, more than half the universe, carry "essentially zero" abnormal returns. A portfolio following only opportunistic insiders earns value-weighted abnormal returns of 82 basis points a month [eq-cohen-malloy-pomorski-2012, working-paper version, abstract and introduction, pp. 5–6]. So Velu, Hardy and Nehren's hedge ("thought to be an indicator") fits the whole universe, and the signal is concentrated in a subset identifiable in advance.

*The numbers, with their uncertainty.* A routine trader is one who traded in the same calendar month for a set number of past years, classified at the start of each year from past history only; the opportunistic long–short portfolio (opportunistic buys minus sells) earns 82 basis points a month value-weighted ($t = 2.15$) and 180 equal-weighted ($t = 6.07$), against $-20$ ($t = -0.57$) and 43 ($t = 1.73$) for the routine portfolio, on open-market trades from 1986 to 2007 [eq-cohen-malloy-pomorski-2012, NBER WP 16454, sec. I, PDF p. 7]. The value-weighted figure, the investable one for large books, has a $t$ close to two; the equal-weighted figure is dominated by small firms. Its timing in a backtest still depends on when the filing became public (filing lags). Thirteen-F holdings changes and rating downgrades are not tested in these sources.

*Alternative data and forecast horizon.* Dessaint, Foucault and Frésard argue that alternative data raise the benefit of short-term over long-term information. With sell-side analysts' forecasts and a new measure of informativeness by horizon, they confirm that alternative data make "short-term forecasts more informative but long-term forecasts less so" [rf-dessaint-foucault-fresard-2024, working-paper version of 20 July 2021, not the 2024 article, abstract and sec. I]. Farboodi and Veldkamp's growth model reaches a related theoretical point: better data technology shifts traders toward extracting information from order flow, which "does crowd out production of fundamental information", yet does not lower price informativeness [rf-farboodi-veldkamp-2020, NBER WP 23457, May 2017, not the 2020 article, abstract]. Whether a given dataset's predictive value falls as access spreads (Katona et al.) is a separate question.

**Pitfalls.** The Bloomberg indexing lag is specific to that example, not a universal delay for fundamental data. The two source taxonomies differ; holdings and insider-data features must account for filing lags.

**See also.** trading reference data; market data levels; derived market statistics; activity-sampled bars.

**Further reading.** `wide-kolanovic-big-data` and `rf-monk-prins-rook-2019`, and the evidence on how alternative-data edges decay.

# 4 Rates: yield curves, term premia and derivatives

## 4.1 Interest rates, discounting and rate risk

**Definition 4.1 Interest-rate measures: reference rates, compounding conventions, zero rates, bond yields and par yields, with time-value identities derived here**

**Statement.** An interest rate states what a borrower promises to pay; the extra amount over a risk-free rate for credit risk is the *credit spread* [s7-hulloptions, chap. 4, "Interest Rates", sec. 4.1, p. 98]. Hull's types of rate (pp. 98–99):

- *Treasury rates*, at which a government borrows in its own currency; "it is usually assumed" a developed country will not default on them;
- *overnight rates*, from banks lending reserves to one another overnight; in the United States the federal funds rate, which the Federal Reserve "may intervene" to move;
- *repo rates*, secured: securities are sold and bought back at a slightly higher price, the difference being the interest; overnight and term repos.

The *n-year zero* or spot rate is earned on an investment with no intermediate payments; a bond's theoretical price discounts each cash flow at its own zero rate; its *yield* is the single rate that reprices it to its market price; its *par yield* is the coupon that prices it at par (secs. 4.5–4.6, pp. 104–106). An amount $A$ at rate $R$ compounded $m$ times a year grows to $A(1+R/m)^{mn}$ in $n$ years, and to $Ae^{Rn}$ with continuous compounding; with $m=1$ the rate is the *equivalent annual* rate (sec. 4.4, pp. 101–103).

**Notation.** $R_m$ a rate compounded $m$ times a year, $R_c$ the equivalent continuous rate: $R_c = m\ln(1+R_m/m)$ and $R_m = m(e^{R_c/m}-1)$ (eqs. (4.3)–(4.4), p. 104). Interpretation: the *effective annual rate* (EAR) is the annual-compounding equivalent, $\text{EAR} = (1+R_m/m)^m - 1 = (1+r_p)^m - 1$ for a periodic rate $r_p = R_m/m$, and $e^{R_c}-1$ in the continuous case; present value is $PV = FV_N(1+R_m/m)^{-mN}$ or $FV_N e^{-R_c N}$. For a level payment $A$ at a periodic rate $r>0$ over $N$ periods:

$$FV_N^{\text{ord}} = A\,\frac{(1+r)^N-1}{r}, \qquad FV_N^{\text{due}} = FV_N^{\text{ord}}(1+r), \qquad PV^{\text{ord}} = A\,\frac{1-(1+r)^{-N}}{r}, \qquad PV^{\text{perp}} = \frac{A}{r},$$

where an ordinary annuity pays at the end of each period (first payment one period from now) and an annuity due at the start (first payment now).

**Conditions.** Conventions timeless; the reference-rate methodology as published on 15 October 2026. The annuity factors need $r \neq 0$; as $r \to 0$ they tend to $N$ (each payment simply summed), and the perpetuity value $A/r$ is the limit of $PV^{\text{ord}}$ as $N \to \infty$, which exists only for $r>0$. At $r=5$ percent and $N=10$, the ordinary and due future values of 100 a period are 1,257.79 and 1,320.68 both by formula and by summing the payments; the future-value factor tends to 10.46, 10.0045, 10.000005 for $r = 10^{-2}, 10^{-4}, 10^{-7}$; perpetuity partial sums are 1,825.59, 1,999.88 and 2,000.00 after 50, 200 and 1,000 periods against $A/r = 2{,}000$.

Monthly compounding at 6 percent gives an EAR of 6.1678 percent, continuous 6.1837 percent. Hull's examples check: 10 percent semiannual is 9.758 percent continuous (p. 104); the two-year bond on his Table 4.2 zero curve prices at 98.39, yields 6.76 percent and has a par yield of 6.87 percent (pp. 105–106).

**Practice.** *Reference rates.* Hull's description of the effective federal funds rate as the volume-weighted *average* of brokered transactions (p. 99) is out of date. The New York Fed changed the EFFR's data source and methodology with the rate for 1 March 2016; it is now a volume-weighted *median*, the rate at the 50th percentile of dollar volume, of overnight federal funds transactions reported on the FR 2420 [exp-sofr-data, EFFR and OBFR methodology]. SOFR, "a broad measure of the general cost of financing Treasury securities overnight", is likewise a volume-weighted median, of tri-party repo, GCF Repo and FICC-cleared bilateral (DVP) Treasury repo transactions, with some specials filtered out [exp-sofr-data, SOFR description]; Hull calls it an important volume-weighted median of overnight repo rates (p. 99).

Derivatives traders do not use Treasury rates as risk-free rates, because tax and regulatory factors make them artificially low; they use rates derived from overnight rates (sec. 4.3, p. 101), and the new overnight reference rates are backward-looking (p. 101).

**Pitfalls.** EAR is the effective annual rate, a conversion of compounding, not an expected rate. Hull's EFFR definition predates the 2016 change described above. Treasury rates of developed countries are risk-free with respect to default in their own currency, as Hull states it, not with respect to price risk, and they are not the rates used to discount derivatives. Continuous-compounding future values need the continuously compounded rate, not a rate quoted with another frequency; compounding conventions must not be mixed within one calculation.

**See also.** forward rates and FRAs; duration and convexity; U.S. day count and quotation conventions; securities lending and short sales (repo and lending kept apart).

**Further reading.** Bootstrapping and curve building beyond Hull's illustration.

**Model 4.2 Forward rates and forward rate agreements: forward rates from zero rates, the instantaneous forward, FRA payoff, value and timing, and theories of the term structure**

**Statement.** With continuously compounded zero rates $R_1, R_2$ for maturities $T_1 < T_2$, the forward rate for the period between them is [s7-hulloptions, chap. 4, "Interest Rates", sec. 4.8, pp. 109–110, eqs. (4.5)–(4.6)]

$$R_F = \frac{R_2T_2 - R_1T_1}{T_2 - T_1} = R_2 + (R_2 - R_1)\frac{T_1}{T_2 - T_1}, \qquad R_F(T) = R + T\frac{\partial R}{\partial T} = -\frac{\partial}{\partial T}\ln P(0,T),$$

the second being the instantaneous forward rate, with $P(0,T) = e^{-RT}$ the zero-coupon bond price. A *forward rate agreement* exchanges a fixed rate $R_K$ for a reference rate observed at a future time, both applied to a principal $L$ for a period of length $\tau$; the principal is not exchanged (sec. 4.9, pp. 110–112). Derived from Hull's definitions (pp. 111–112), the party receiving fixed receives $\tau(R_K - R_M)L$ at the end of the period, where $R_M$ is the realised reference rate (negative when it pays), and the FRA is worth the *present value* of $\tau(R_K - R_F)L$, with $R_F$ the current forward rate; the payer of fixed holds the negative (p. 112).

**Notation.** Rates in $R_F$'s formula are continuously compounded; in the FRA, $R_K$, $R_F$ and $R_M$ are expressed with the compounding of the period $\tau$ (quarterly for a three-month rate, p. 111). $P(0,T)$ discount factor.

**Conditions.** Formulas timeless; LIBOR-based conventions historical, SOFR conventions as Hull states them in 2021. The FRA value rests on the result that an FRA can be valued by assuming the forward rate is realised; it needs the discount rate for the payment date (Hull's Example 4.3 discounts at the 2-year risk-free rate). Interpretation: this is a single-curve illustration, in which the forward reference rate and the discount rate come from consistent curves; current practice separates projection and discounting curves and, for backward-looking compounded overnight rates such as SOFR, the floating amount is known only at the end of the period (Hull p. 101). Settlement timing changes the amount: Hull's LIBOR FRA, known at the start of the period, was paid at the start as the present value of the end-of-period amount (p. 111, n. 4).

Hull presents three term-structure theories, none established as a result: *expectations* theory (forward rates equal expected future zero rates), *market segmentation* (no necessary relation between maturities) and *liquidity preference* (forward rates above expected future zero rates, consistent

with upward-sloping curves being more common); Hull calls expectations and segmentation conjectures and finds liquidity preference "most appealing", consistent with the empirical observation (sec. 4.12, pp. 117–119).

**Practice.** From zero rates of 3, 4, 4.6, 5 and 5.3 percent for 1–5 years, forward rates are 5, 5.8, 6.2 and 6.5 percent (Table 4.5, p. 109); the instantaneous forward of an illustrative curve $R(T) = 0.03 + 0.005T$ at $T = 2$ is 5.0 percent by finite difference of $-\ln P$ and by $R + TR'$. Hull's Example 4.3, receiving 5.8 against a forward of 5.0 percent (semiannual) on $100 million for six months, discounted at 4 percent continuous for 2 years, gives $369,247; Hull prints $369,200 (p. 112). Locking in forward rates by borrowing and lending at different maturities, and the Orange County yield-curve plays, are Hull's illustrations of the risk of betting against forwards (pp. 110–111).

*Market notation*: a "3 × 6" FRA starts in three months and ends in six, a three-month rate three months forward; Gliner's example is a 3 × 6 "over 15th" traded in March 2013, settling 15 June and maturing 15 September 2013 [s7-gliner, chap. 9, sec. "Forward Rate Agreements (FRAs)", PDF p. 263]. FRAs can be dated to straddle a policy meeting or a release, unlike quarterly futures bound to their expiry dates (p. 263); in U.S. dollars the reference is now SOFR rather than LIBOR (LIBOR cessation).

**Pitfalls.** Hull's own example has the payment direction reversed: Party A receives the fixed 3 percent and pays LIBOR, so with LIBOR at 3.5 percent A pays, rather than receives, $125,000 (p. 111); his formula for the FRA's value (p. 112) has the correct sign. The undiscounted amount $\tau(R_K - R_F)L$ is the payoff if the forward is realised, not the present value. That forward rates exceed expected future spot rates is what the liquidity-preference theory implies, as Hull presents it, not an observed fact.

*Term-premium evidence*. Adrian, Crump and Moench split model-fitted yields into a risk-neutral part, "the time $t$ expectation of average future short rates", and a term premium, "the difference between the risk neutral yield and model-implied fitted yield" [exp-adrian-crump-moench-term-structure, Staff Report 340, rev. April 2013, sec. 2.4, PDF p. 12]. In their five-factor model "investors on average require a positive expected excess return for holding the level portfolio", and the price of that risk "varies significantly as a function of the slope and the fifth factor" (sec. 3.3, PDF p. 21). The New York Fed's published estimates (monthly, June 1961 to September 2026, October 2026 vintage) put the 10-year term premium at +1.43 points on average. It was positive in every month from 1970 to 2014, with a maximum of +5.20 in May 1984.

It was negative in 101 of 784 months, first again in January 2015, and in 51 of the 81 months since 2020, reaching −1.38 in July 2020; it stood at +0.89 in September 2026. Read against the theories: the pure expectations theory's zero premium is rejected on average, but so is a constant positive liquidity premium. The premium is time-varying and sometimes negative. These are model estimates: they depend on the factor specification, are revised when the model is re-estimated, and are not observed prices.

*A second model*. Kim and Wright's three-latent-factor arbitrage-free model, fitted to Treasury yields from 1990 with survey forecasts of short rates in its Kalman filter, "ascribes a large portion of the decline in long-term yields and distant-horizon forward rates since the middle of 2004 to a fall in term premiums" [rt-kim-wright-2005, FEDS 2005-33, abstract and sec. 2, PDF pp. 2–12]. A comparison of the Board's published KW 10-year zero-coupon term premium (daily, month-end) with ACM's (October 2026 vintages): over 441 common months, January 1990 to September 2026, the correlation is 0.86 and KW is on average 0.24 points lower (range −2.22 to +1.18); KW was negative in 58 months and ACM in 98; in June 2020 KW was −0.61 against ACM's −1.23, and in September 2026 +1.08 against +0.89.

The two models agree on the broad path and differ by up to two points in level, which is the size of the specification risk in any single estimate. Gliner calls the fixing date "the date of the trade" (p. 263); the fixing date is when the reference rate is observed, normally shortly before the start of the period, not the trade date. His remark that Eurodollar futures "lack convexity" refers to the convexity difference between daily-settled futures and FRAs, which makes futures rates exceed forward rates (Hull's convexity adjustment).

**See also.** interest-rate measures; duration and convexity; cost-of-carry forward price (valuing a forward by assuming the forward is realised).

**Further reading.** Multi-curve FRA and swap valuation is in multi-curve valuation; exp-kiley-quasi-real-time-term-premium, Kiley's quasi-real-time term premiums.

### Model 4.3 FRA and swap valuation with separate projection and discounting curves, and the backward-looking compounded SOFR period rate

**Statement.** Hull values an interest rate swap as a portfolio of FRAs in three steps [s7-hulloptions, chap. 7, sec. 7.6, pp. 183–185]:

1. calculate forward rates for each unknown floating rate;
2. set the swap's cash flows on the assumption that those rates equal the forwards;
3. discount the net cash flows "at the risk-free rate".

The risk-free zero curve comes from overnight indexed swaps. OIS rates to one year "provide the risk-free zero rates that are equivalent to the underlying overnight rates". Longer OIS rates define par bonds, and are bootstrapped like a par-yield curve (sec. 7.2, p. 175). The two curves can differ.

- *An OIS or SOFR swap.* Projection and discounting both use the risk-free curve. The first floating rate "is not really a forward rate. It is a blend of one-day rates already observed and the zero rate that corresponds to the remaining one-day rates in the period" (p. 184).
- *A LIBOR (term-rate) swap.* The forwards are estimated from Eurodollar futures, with a convexity adjustment, and from swap rates that make par swaps worth zero, while the cash flows are still discounted at the OIS-based risk-free rate (p. 184).

In symbols, take a receive-floating contract on reference index $j$ over $[T_1, T_2]$, with accrual $\tau$, fixed rate $K$, notional $N$ and fixing $L$ observed at $T_1$.

- *End settlement (a swaplet, paid at $T_2$):* the payoff $N\tau(L - K)$ is worth $N\tau\,(F_j(T_1, T_2) - K)\,P_{\mathrm{OIS}}(T_2)$, where $F_j$ is read from index $j$'s own projection curve. The formula is exact when $F_j$ is *defined* as the expectation of $L$ under the $T_2$-forward measure of the OIS curve, which is what a projection curve calibrated to OIS-discounted swaps delivers. This is how Hull's swap cash flows are valued, each paid at the end of its period.
- *Start settlement (the conventional FRA, paid at $T_1$):* the end amount is discounted at the fixing itself, $N\tau(L - K)/(1 + \tau L)$, as in Hull's footnote: "the payment would be made at time two years and equal to the present value of $125,000 discounted for three months at 3.5%" [s7-hulloptions, chap. 4, sec. 4.9, p. 111, n. 4]. Its value is $NP_{\mathrm{OIS}}(T_1)\,\mathbb{E}^{T_1}[\tau(L - K)/(1 + \tau L)]$. If $L$ is deterministic, $L = F_j$, this is $N\tau(F_j - K)P_{\mathrm{OIS}}(T_1)/(1 + \tau F_j)$.

The two conventions coincide in the single-curve case, the formula of forward rates and FRAs: there $1/(1 + \tau L)$ is the discount factor from $T_2$ to $T_1$, so the start-settled payoff is replicated with bonds, $N[P(T_1) - (1 + \tau K)P(T_2)]$, whatever $L$ does. With separate curves they differ: by $N\tau(F_j - K)[P_{\mathrm{OIS}}(T_1)/(1 + \tau F_j) - P_{\mathrm{OIS}}(T_2)]$ even with deterministic rates. With a stochastic $L$, the deterministic substitution is no pricing theorem. The start-settled payoff is concave in $L$, and the $T_1$-forward mean of $L$ need not equal $F_j$, so its value carries a convexity adjustment that depends on $L$'s volatility and its correlation with OIS discount factors.

Neither that adjustment nor the market's FRA conventions (ISDA FRA discounting, fixing and payment lags) are covered here; see s6-brigo and wide-henrard-multicurve. Bianchetti and Carlicchi's comparison table values a FRA as $NP_d(t, T_k)\omega[\tilde{F}_{x,k}(t) - K]\tau_{x,k}$, with the FRA rate $\tilde{F}_{x,k}(t) = E^{Q_d^{T_k}}[L_x(T_{k-1}, T_k) \mid \mathcal{F}_t]$, the expectation of the fixing under the $T_k$-forward measure of the discounting curve, which "is a martingale under such measure" while the classical forward is not (p. 9, and Table 2, p. 11) [gap-rates-multicurve-sabr, arXiv 1103.2567, version of 28 March 2012, Table 2, p. 11]. That is the end-settled form above, with $F_j$ defined as that expectation; their table does not treat payment at $T_1$.

For a backward-looking rate compounded over the period, $\prod_i(1 + r_i d_i)$ of the overnight forwards implied by a curve telescopes to $P(T_1)/P(T_2)$. So its forward is the simple forward from discount-factor ratios, before convexity, day-count and lookback adjustments.

**Notation.** $P_{\mathrm{OIS}}(T)$ discount factor from the OIS curve; $F_j(T_1, T_2)$ the forward for reference index $j$ (the $T_2$-forward expectation of its fixing); $L$ the fixing at $T_1$; $\mathbb{E}^T$ expectation under the $T$-forward measure of the OIS curve; $r_i$, $d_i$ the overnight rate and its day fraction.

**Conditions.** Conventions as Hull states them in 2021; SOFR replaced USD LIBOR after June 2023. Hull's calculations "are approximate because they do not take account of holiday calendars and day count conventions" (p. 185). Discounting at the OIS rate presumes cash-collateralised trades earning the overnight rate; uncollateralised trades bring funding and credit adjustments, which are not covered here.

**Practice.** Numerical checks. Hull's Example 7.1 is reproduced: paying 3 percent semiannually against six-month SOFR on \$100 million, with floating rates 2.50, 3.36 and 3.68 percent continuous (2.516, 3.388 and 3.714 semiannual), the swap is worth +\$0.292 million (p. 185). The first rate is $0.6 \times 2.3 + 0.4 \times 2.8 = 2.50$ percent, the blend of observed and expected overnight rates. With invented curves, a term reference 30 basis points above OIS and $K = 3.4$ percent on \$100 million for a quarter starting in one year, an end-settled receive-floating contract is worth −\$20,793 on two curves and −\$20,715 discounted on the term curve. Projecting at OIS gives −\$93,604. The projection curve matters far more than the discounting curve for one period.

Settled at the start under the deterministic substitution, the same contract is worth −\$20,777.37, against −\$20,792.96 settled at the end; the conventions coincide on one curve and diverge on two. With a lognormal $L$ of 20 percent volatility, deterministic OIS rates and $\mathbb{E}[L] = F_j$ (an invented illustration), the start-settled value falls to −\$21,044.50: the concavity term is larger than the settlement-convention gap.

**Pitfalls.** Hull's 11th edition still describes LIBOR forwards from Eurodollar futures, which ended with LIBOR (LIBOR cessation). Lookback, lockout and payment-delay conventions, and the futures convexity adjustment, are not covered here (`s6-brigo`, `gap-rates-sofr-futures-hedging`).

**See also.** forward rates and FRAs; interest-rate measures; OTC clearing and collateral.

**Further reading.** `s6-brigo`; `gap-rates-sofr-futures-hedging`.

### Model 4.4 Duration and convexity: Macaulay and modified duration, convexity, the second-order price approximation, and the parallel-shift assumption

**Statement.** For a bond paying $c_i$ at $t_i$ with price $B = \sum_i c_i e^{-y t_i}$ at continuously compounded yield $y$, the duration is the present-value-weighted average time of payment [`s7-hulloptions`, chap. 4, "Interest Rates", sec. 4.10, pp. 112–115, eqs. (4.7)–(4.13)]:

$$D = \sum_i t_i \frac{c_i e^{-y t_i}}{B}, \qquad \frac{\Delta B}{B} \approx -D\,\Delta y, \qquad C = \frac{1}{B}\frac{d^2 B}{dy^2} = \frac{\sum_i c_i t_i^2 e^{-y t_i}}{B}, \qquad \frac{\Delta B}{B} \approx -D\,\Delta y + \tfrac{1}{2} C (\Delta y)^2.$$

With $y$ compounded $m$ times a year, $\Delta B \approx -B D^* \Delta y$ with modified duration $D^* = D/(1 + y/m)$ (pp. 114–115); dollar duration is $BD^*$ (p. 115). Convexity is from sec. 4.11 (pp. 116–117).

**Notation.** $D$ Macaulay and $D^*$ modified duration (years), equal under continuous compounding; $C$ convexity (years squared); $\Delta y$ a change in the bond's own yield.

**Conditions.** The duration relation is a first-order approximation, accurate for small changes; convexity adds the second-order term. For a portfolio, the weighted-average duration assumes all yields change by about the same amount, which for bonds of widely different maturity means a small parallel shift of the zero curve; matching durations protects against small parallel shifts only, and matching convexity as well against larger parallel shifts, but neither against non-parallel shifts (pp. 115–117).

**Practice.** On Hull's Table 4.6 bond (3-year 10 percent coupon, yield 12 percent continuous): $B = 94.213$, $D = 2.653$ years and $C = 7.570$. For $\Delta y = +0.001$ the exact price is 93.9634 against 93.9631 from duration, as in Hull's Example 4.4 (p. 114). For $\Delta y = +0.02$ the exact price is 89.354 against 89.214 from duration and 89.357 with convexity; for −0.02, 99.357 against 99.212 and 99.355. With the yield expressed semiannually (12.3673 percent), $D^* = 2.499$, and a 10 bp rise gives 93.978 both exactly and by modified duration (Example 4.5, p. 115).

*DV01 and the macro trader's usage*: DV01 $= D^* B/10{,}000$, the price change for 1 bp; for a 10-year 4 percent annual-coupon bond at par, $D^* = 8.111$, convexity 80.8 and DV01 0.0811 per 100 of face, and a 100 bp rise gives −7.722 percent exactly against −7.707 percent from $-D^*\Delta y + \frac{1}{2} C (\Delta y)^2$. Curve trades weight legs by DV01 to neutralise level moves (curve trades).

*What positive convexity does and does not mean*. Positive convexity makes the price *better than the linear (duration) approximation* for a move either way; it does not make an unhedged bond gain from moves either way: the same bond loses 7.722 percent for +100 bp (8.111 by duration alone, so convexity adds back 0.389 points) and gains 8.530 percent for −100 bp (0.419 more than duration). The analogy between long convexity and long volatility applies to a *duration-hedged* position: long the 10-year and short a DV01-matched amount (4.30 times the face) of a 4 percent 2-year bond, which has much less convexity, gains 0.274 per 100 of 10-year face for a parallel +100 bp move and 0.302 for −100 bp, and only 0.018 for +25 bp, since the gain grows with $(\Delta y)^2$.

Qualifications: the hedge holds only for parallel moves of both yields (a curve twist can swamp the convexity gain), and it must be rebalanced as durations change. *Carry is not determined by convexity.* In this example the curve is flat at 4 percent and both bonds are at par; if the long 10-year is financed and the short 2-year's proceeds earn the same 4 percent, the annual carry per 100 of 10-year face is $4 - 4k + 4(k - 1) = 0$ for any hedge ratio $k$, and with no yield change there is no rolldown either. Holding-period P&L then comes only from yield moves, through the $(\Delta y)^2$ gain above, and from rebalancing.

That a flat curve moving only in parallel would give a riskless gain to a long-convexity position is itself a sign that real curves cannot behave that way; where convexity is "paid for", it shows up as a curve shape in which more convex (longer) bonds yield less than they otherwise would, and so as negative carry relative to the hedge. Whether a given hedged position has positive or negative carry depends on the curve's slope and shape, on repo rates against the rebate on the short (specials, haircuts and balance-sheet costs make them differ), on the financing period and on the costs of rebalancing; positive convexity alone does not settle the sign. The analogy with a long-gamma option, which loses time value, holds only when such a carry cost is present.

*Negative convexity.* Callable bonds and MBS can be negatively convex where the issuer's call or the borrowers' prepayment option is near the money: as yields fall, prepayments or a call cap the price rise. Far from that region (a callable bond when rates are well above its coupon) the convexity can still be positive; MBS convexity depends on a prepayment model and on the path of rates, so "effective convexity" figures are model outputs, not observed quantities.

**Pitfalls.** Treating duration and convexity relations as equalities, including $\Delta B = \frac{dB}{dy}\Delta y$ omits higher-order terms; Hull marks these as approximations. A change in a bond's yield is not the same as a shift in the zero curve. Interpretation: under a parallel shift $\Delta$ of the zero curve the bond's yield moves by exactly $\Delta$, for every $\Delta$, if and only if the zero rates at all of the bond's payment dates equal its yield. A flat curve is therefore sufficient but not necessary: a zero-coupon bond has $y = R(T)$ and moves one for one on any curve, and the curve may take any shape between payment dates.

For a coupon bond on a curve that is not level across its payment dates, the yield moves by only about $\Delta$: $0.9996\Delta$ for Hull's Table 4.2 bond and $0.986\Delta$ for a 10-year 5 percent annual-coupon bond on a curve rising 40 bp a year from 1.4 percent. Duration against the bond's own yield, $-B^{-1}\,\partial B/\partial y$, and sensitivity to a parallel zero-curve shift, $-B^{-1}\,\partial B/\partial\Delta$, differ by that factor: 8.13 against 8.01 years for the 10-year bond. Modified duration applies only when $y$ uses the stated compounding; mixing a continuously compounded $D$ with an $m$-compounded $\Delta y$ misstates the change.

Writing the convexity adjustment as $\Delta P \approx -\text{Duration} \times \Delta y + \frac{1}{2} \times \text{Convexity} \times (\Delta y)^2$ drops the price: the right side is the *proportional* change $\Delta P/P$, as above; and its modified duration as "Approximate percentage price change for 1% yield change" holds only to first order.

**See also.** interest-rate measures; forward rates and FRAs; minimum-variance hedge ratio (duration-based futures hedges, Hull chap. 6).

**Further reading.** Hull's duration-based futures hedging (sec. 6.4); key-rate and factor durations for non-parallel moves.

**Convention 4.5 U.S. day count and price quotation: actual/actual, 30/360 and actual/360; Treasury bill discount quotes; Treasury bond prices in thirty-seconds; clean and dirty prices**

**Statement.** A day count $X/Y$ gives interest between two dates as the days between them, counted by $X$, over the days in the reference period, counted by $Y$, times the interest for the reference period [s7-hulloptions, chap. 6, "Interest Rate Futures", sec. 6.1, pp. 152–153]. In the United States, Hull lists actual/actual (in period) for Treasury bonds, 30/360 for corporate and municipal bonds and actual/360 for money-market instruments. Treasury bills are quoted as a discount rate on face value: $P = \frac{360}{n}(100 - Y)$, with $P$ the quoted price, $Y$ the cash price per \$100 and $n$ the days to maturity. Treasury bonds are quoted in dollars and thirty-seconds per \$100 face, and the quote is the *clean* price; the cash or *dirty* price adds interest accrued since the last coupon (p. 154).

**Conditions.** U.S. conventions as of Hull's 2021 account. These are U.S. conventions as Hull gives them; conventions differ by country and instrument (money-market instruments on actual/365 in Australia, Canada and New Zealand; euro and sterling bonds usually actual/actual, p. 153). Futures quotes have their own conventions (sec. 6.2, p. 155).

**Practice.** Hull's examples: \$4 earned March 1 to July 3 on an 8 percent \$100 bond is $\frac{124}{184} \times 4 = 2.6957$ on actual/actual and $\frac{122}{180} \times 4 = 2.7111$ on 30/360 (p. 153); a 90-day bill at a cash price of 99 is quoted at $\frac{360}{90}(100 - 99) = 4$ (p. 154); 120-05 is $120\frac{5}{32}$, \$120,156.25 for \$100,000 face; accrued interest of $\frac{54}{181} \times 5.50 = 1.64$ makes a clean price of 155.50 a dirty price of 157.14 (p. 154).

**Pitfalls.** With $P$ the quoted price and $Y$ the cash price, describing the discount rate only as "interest as a percentage of final face value"; Hull adds that the true return for the period is on the price paid, 2.064 percent rather than 2.022 percent in his example (p. 154). Hull notes that SOFR compounding assumes 360 days a year (p. 100); day counts for SOFR and other reference rates are set per product, not by the bond conventions above.

**See also.** interest-rate measures; fixed-income market structure.

**Data 4.6 The Federal Reserve's fitted Treasury curve (Gürkaynak, Sack and Wright): the Svensson form, its published parameters and how to recompute the yields**

**Statement.** Gürkaynak, Sack and Wright fit off-the-run Treasury notes and bonds each day with the Svensson extension of Nelson–Siegel: the zero yield at maturity $n$ is $y_t(n) = \beta_0 + \beta_1 \frac{1-e^{-n/\tau_1}}{n/\tau_1} + \beta_2\left[\frac{1-e^{-n/\tau_1}}{n/\tau_1} - e^{-n/\tau_1}\right] + \beta_3\left[\frac{1-e^{-n/\tau_2}}{n/\tau_2} - e^{-n/\tau_2}\right]$, continuously compounded; the parameters minimise duration-weighted squared price errors, approximately the squared yield errors (eq. (22), p. 15) [wide-gsw-yield-curve, FEDS 2006-28, sec. 3, pp. 13–15]. The Board publishes the daily parameters ($\beta_0$ to $\beta_3$, $\tau_1$, $\tau_2$) with zero yields (SVENY), par yields, instantaneous forwards and one-year forwards [fed-gsw-data, file `feds200628.csv`, 14 June 1961 to 9 October 2026].

**Notation.** $n$ maturity in years; $\beta_3 = 0$ and no $\tau_2$ in the early years, when the three-term Nelson–Siegel form is used.

**Conditions.** The file is a staff research product, "subject to delay, revision, or methodological changes without advance notice" (file header); its yields are fitted values, smooth by construction, not traded prices; the fit excludes bills and on-the-run issues, as described in the paper.

**Practice.** Recomputing the 10-year zero yield from each day's parameters reproduces the published SVENY10 on all 13,759 days with a 10-year yield to within 0.00005 points (rounding of the published value); 2,084 of those days use the three-term form. On 9 October 2026 both give 5.2514 percent. This curve is a dated, reproducible input, not a substitute for bootstrapping one's own.

**Pitfalls.** The paper describes the method as of 2006; later changes to the sample or filters, which the file header allows, may apply.

**See also.** forward rates and FRAs (term premia, ACM and KW); duration and convexity.

**Further reading.** `wide-nelson-siegel`; `wide-hagan-west-interp`.

**Mechanism 4.7 OTC derivatives clearing and collateral: central counterparties, bilateral ISDA/CSA clearing, variation and initial margin, and post-crisis reforms as Hull describes them**

**Statement.** An OTC trade can be presented to a central counterparty (CCP) or cleared bilaterally [s7-hulloptions, chap. 1, "Introduction", sec. 1.2, p. 25]. A CCP accepting a trade between A and B becomes counterparty to both and takes on the credit risk of each; its members post initial margin and daily variation margin and contribute to a guaranty fund, and non-members clear through members (chap. 2, sec. 2.5, pp. 54–55). In the CPSS–IOSCO principles a CCP "interposes itself between counterparties", ensuring the performance of open contracts through novation or an open-offer system, and has "the potential to reduce significantly risks to participants" through multilateral netting and risk controls [exp-pfmi, para. 1.13, p. 9]. Bilaterally cleared trades are covered by a master agreement, usually ISDA's, with a credit support annex (CSA) requiring collateral, typically valued daily (p. 55).

**Conditions.** *Reforms, as Hull reports them* (pp. 26 and 55–56): standardised OTC derivatives between U.S. financial institutions must, "whenever possible", trade on swap execution facilities; most parts of the world require CCP clearing of most standardised derivatives between financial institutions; trades must be reported to a repository; and from 2016 regulations required initial and variation margin for bilaterally cleared transactions between financial institutions, with initial margin held by a third party. Hull notes that "financial institutions" include banks, insurers, pension funds and hedge funds, while most non-financial corporations and some FX transactions are exempt (p. 56, n. 4). *The U.S. primary rules*:

- *Reporting.* A swap executed on a SEF or designated contract market is reported to a swap data repository by the facility, and an off-facility swap by its reporting counterparty, "as soon as technologically practicable after execution" for public dissemination [ecfr-17cfr-cftc-swaps, 17 CFR 43.3(a)]. Full creation data follow "not later than the end of the next business day" after execution, or the second business day for a reporting counterparty that is not a swap dealer, major swap participant or clearing organisation (45.3(a)–(b)).
- *Uncleared margin.* The CFTC's rules bind "covered swap entities", swap dealers and major swap participants without a prudential regulator; bank dealers fall under the prudential regulators' parallel rules (23.151). They do not apply where the counterparty qualifies for the end-user clearing exception or the cooperative exemption (23.150(b)). Variation margin is exchanged daily with swap entities and *financial end users* (23.153). Initial margin is collected from *covered counterparties*, which are swap entities and financial end users with *material swaps exposure* (an average month-end notional above \$8 billion). It is posted to financial end users with material swaps exposure (23.152, 23.151). It is held by a custodian that is neither party nor an affiliate, and may not be rehypothecated (23.157).
  The definitions set a \$50 million initial-margin threshold amount(23.154). No transfer is required below a combined minimum transfer amount of \$500,000 (23.152(b)(3), 23.153(c), 23.151).
- *Phase-in* (23.161). Both margins applied from 1 September 2016 where both groups' average daily notional exceeded \$3 trillion. Variation margin applied to all covered entities from 1 March 2017. Initial margin phased down through \$2.25 trillion (2017), \$1.5 trillion (2018), \$0.75 trillion (2019) and \$50 billion (1 September 2021), reaching all remaining covered counterparties on 1 September 2022.

Hull's "from 2016" is therefore the first phase only; initial margin reached the smallest covered counterparties six years later. His "held by a third party" matches 23.157. His exemption of "most non-financial corporations" corresponds to the end-user exception and to the definition of financial end user. *Interest.* Initial margin in cash usually earns interest; futures variation margin does not, being daily settlement, while OTC variation margin in cash does, because OTC trades are not settled daily (p. 57). Securities used as collateral are subject to haircuts (p. 57).

**Practice.** Interpretation: clearing and collateral mitigate counterparty risk rather than eliminate it. Margin is sized to cover most, not all, losses on a default (Hull's clearing-house example aims at about 99 percent certainty, p. 54), a guaranty fund mutualises the rest, and the CCP itself concentrates risk; Hull cites Duffie and Zhu for the dependence of the benefit on the number of CCPs and the share of trades cleared (p. 57, n. 5). Collateral also

creates funding and liquidity demands: LTCM had to post collateral on both legs of its trades in 1998 and, being highly leveraged, had to close out positions (Business Snapshot 2.2, p. 56).

*The uncleared-margin framework.* Covered entities must exchange full variation margin (a zero threshold) and two-way initial margin with a threshold "not to exceed €50 million" at the level of the consolidated group, with a minimum transfer amount of at most €500,000 (Requirement 2, PDF p. 14). Variation margin applied to all covered entities from 1 March 2017, and only new contracts are covered (Requirement 8, pp. 23–24) [bcbs-iosco-margin-2020, April 2020, Requirements 2 and 8, printed pp. 10 and 23–24]. These are the international standards; the U.S. prudential regulators' rule and EMIR implement them in their own terms.

Initial margin was phased in by group notional of non-centrally cleared derivatives (month-end average for March–May): above €3.0 trillion from 1 September 2016, then €2.25 trillion (2017), €1.5 trillion (2018), €0.75 trillion (1 September 2019 to 31 August 2021) and €50 billion (1 September 2021 to 31 August 2022). From 1 September 2022 the threshold is €8 billion on a permanent basis (Requirement 8).

**Pitfalls.** Clearing does not eliminate credit risk; Hull's phrase is that it does so "by requiring … margin" (p. 25), which reduces the risk without removing it. Hull's account of the reforms carries the qualifiers "whenever possible" and "in most parts of the world". Futures-specific margin rules (maintenance margin, margin calls, interest on futures balances) are in futures margin and settlement.

**See also.** futures margin and settlement; derivative contracts; common-investor contagion (margin calls in deleveraging).

**Further reading.** The BCBS–IOSCO margin framework and the prudential regulators' margin rules (12 CFR part 45 and parallel parts).

## 4.2 Central banks, the policy path and the curve

**Definition 4.8 Central-bank policy framework: mandates, conventional tools, the balance sheet and money aggregates, the zero lower bound, quantitative easing and its measured effects, crisis lending facilities, and hawks and doves**

**Statement.** Gliner's chapter on central banks [s7-gliner, chap. 11, PDF pp. 355–400]:

- *Goals* (sec. "Monetary Policy Goals", pp. 356–357): price stability above all, with the Fed's statutory goals of maximum employment, stable prices and moderate long-term rates, the Bank of England's monetary and financial stability, and the SNB's price stability under its law.
- *Tools* (sec. "Tools Used by Central Banks", pp. 358–361): the policy rate, the overnight rate at which banks lend reserves, which he calls "the single strongest action"; the reserve requirement, with a money multiplier of $1/r$ (10 at 10 percent); open market operations and repos; discount-window lending; and currency intervention, *sterilised* when the central bank offsets the effect on the monetary base and *unsterilised* when it does not (pp. 360–361).
- *Monetary base and money supply* (pp. 365–367): the base is currency plus reserves; the quantity identity $MV = PY$; aggregates M1–M3 in Table 11.1, with M2 "historically" a predictor of nominal income.
- *Zero lower bound and liquidity trap* (p. 370), and *quantitative easing* (pp. 371–374): purchases of government bonds and MBS financed with reserves; Table 11.4, from Bauer's San Francisco Fed letter of May 2012, gives the cumulative change in the 10-year Treasury yield on announcement days of $-1.00$ percentage point for QE1 (five dates, 25 November 2008 to 18 March 2009), $-0.14$ for QE2 (three dates in 2010) and $-0.08$ for the maturity extension programme (21 September 2011) (p. 373).
- *Communication* (pp. 375–379): hawks and doves, with a dove's hawkish remark moving markets more than a hawk's, and the Bank of England's minutes as the example (event queries).
- *Fed crisis programmes* (appendix, pp. 396–398): central-bank liquidity swap lines (first announced 12 December 2007); the Term Auction Facility (28-day, later 84-day, loans auctioned to avoid the discount-window stigma); the Term Securities Lending Facility (Treasuries lent to primary dealers for 28 days against other collateral); the Primary Dealer Credit Facility; the AMLF; the Commercial Paper Funding Facility; the Term Asset-Backed Securities Loan Facility; and agency MBS purchases of $1.25 trillion in QE1 (p. 371).

**Notation.** $r$ required reserve ratio; $M$ money, $V$ velocity, $P$ price level, $Y$ real output; M0 to M3 as each central bank defines them.

**Conditions.** *Announcement-day effects are not total effects.* Bauer's letter (FRBSF Economic Letter 2012-16, 21 May 2012) gives the same figures, $-1.00$, $-0.14$ and $-0.08$ for the 10-year Treasury yield, as the "Cumulative change of key interest rates in percentage points over the announcement days" of each programme, and warns that the event study "misses effects that occur on days other than those when key announcements are made" and that "other events may occur on announcement days" [bauer-2012, Table 1 and text]; Gliner reproduces the table faithfully. These figures sum one-day yield changes on announcement dates; they measure the surprise component on those days, not the cumulative effect of the purchases, and depend on which dates are chosen.

*The underlying event study.* Gagnon, Raskin, Remache and Sack's working-paper version sums eight baseline announcements from November 2008 to November 2009: the 10-year Treasury yield, 10-year agency debt yield and current-coupon agency MBS yield fell 91, 156 and 113 bp; on all other days since November 2008 the 10-year Treasury yield and the swap rate "increased nearly 100 basis points", leaving them "roughly unchanged over the entire period", which the authors attribute to other factors such as the fiscal outlook [nyfed-sr441, pp. 17–18 (PDF pp. 19–20)]. Their time-series estimate is that the purchases "reduced the 10-year term premium by between 50 and 100 basis points" (p. 21), and their summary range is "somewhere between 30 and 100 basis points" (p. 28). So the event-day sums, the net change in yields and the model-based term-premium effect are three different numbers.

*Money multiplier.* $1/r$ is a textbook upper bound; with interest on reserves and ample reserves, bank lending does not follow reserves mechanically. *Definitions change.* Gliner's U.S. M1–M3 are as of 2013; current H.6 definitions differ.

**Practice.** Three policy regimes (conventional, with the rate as the tool; unconventional, at the lower bound, working through asset prices, expectations and FX; crisis, where liquidity provision dominates), four QE channels (portfolio rebalancing, term-premium and credit-spread compression, signalling, exchange rate), and mandates that include currency stability for the SNB and PBOC are a reasonable synthesis, but an interpretive framing, not Gliner's text.

**Pitfalls.** Open-market operations are not the only mechanism affecting the monetary base; lending facilities, FX intervention and interest on reserves also change it. That at the ZLB yield curves flatten naturally, carry trades distort, volatility is suppressed, forward guidance becomes critical and fiscal policy matters more are untested assertions. Reading Bauer's smaller later numbers as declining effectiveness over successive rounds is not the only reading; smaller surprises would give the same pattern.

**See also.** Taylor rule; major central banks; policy path from OIS and futures; currency regimes and intervention; money-market reference rates.

**Further reading.** Large-scale asset purchases [rt-gagnon-lsap-2011]; its March 2010 working-paper version [nyfed-sr441] provides the event-study estimates. The published version may differ from the working paper.

**Model 4.9 The Taylor rule: Taylor's 1993 equation, its 2 percent equilibrium real rate and coefficients, the Taylor principle, and Gliner's printed variants**

**Statement.** Taylor's rule for the federal funds rate is [taylor-1993, p. 202, eq. (1)]

$$r = p + 0.5\,y + 0.5\,(p - 2) + 2,$$

where $r$ is the federal funds rate, $p$ the rate of inflation over the previous four quarters and $y$ the percent deviation of real GDP from target; the equilibrium real rate and the inflation target are both 2 percent. The general form used is $i_t = r^* + \pi_t + \alpha(\pi_t - \pi^*) + \beta(y_t - y_t^*)$, with $r^* = 2$, $\pi^* = 2$ and $\alpha = \beta = 0.5$ in Taylor's version. Gliner quotes the Kansas City Fed's verbal definition and prints the formula as an image [s7-gliner, chap. 11, sec. "Taylor Rule", PDF pp. 361–362].

**Notation.** $i_t$ (Taylor's $r$) nominal policy rate; $\pi_t$ inflation; $\pi^*$ target; $y_t - y_t^*$ output gap in percent; $r^*$ equilibrium real rate.

**Conditions.** The response of the nominal rate to inflation is $1 + \alpha = 1.5$, greater than one: the *Taylor principle*, under which a rise in inflation raises the real rate. On target with no gap the rule gives 4 percent; inflation of 4 percent gives 7 percent; a −2 percent gap gives 3 percent. Taylor fitted the rule to U.S. policy over 1987–1992 as a description and benchmark, not as a law central banks follow.

**Practice.** Gliner suggests forecasting the rule's inputs with the market's one-year CPI forecast and an output measure, and comparing the result with fed funds futures (p. 362).

**Pitfalls.** The neutral-rate estimate $r^*$ is model-dependent; Taylor used 2 percent, and current estimates vary by model. One of Gliner's two printed forms reads "$p + 1/(2y)$", a typesetting slip for $\frac{1}{2}y$: at $y = 2$ it gives 4.25 instead of 5.00 (checks, section 5). Inputs measured with revised data overstate how well the rule fits in real time.

**See also.** central-bank policy; policy path from OIS and futures.

**Further reading.** Taylor (1993).

### Rule 4.10 Major central banks as of 16 October 2026: Fed, ECB, Bank of England, Bank of Japan and SNB mandates, committees and meeting frequency; ECB crisis tools and TARGET2; communication and the Bank of England's 2012 minutes

**Statement.** *Federal Reserve.* The FOMC has twelve voting members, the seven members of the Board of Governors, the New York Fed's president and four of the other eleven Reserve Bank presidents in one-year rotation, and meets eight times a year [fomc-page, page updated 7 October 2026]. Its inflation goal is 2 percent on the annual change in the PCE price index [fomc-longer-run-goals, pp. 1–2].

*ECB.* The Governing Council comprises the six members of the Executive Board and the governors of the 21 euro-area national central banks, and takes monetary policy decisions every six weeks [ecb-governing-council, page dated 10 September 2026]; its strategy targets 2 percent HICP inflation over the medium term, symmetrically [ecb-strategy, web page]. *Bank of England.* The Monetary Policy Committee has nine members, four of them external [boe-mpc-page, web page]; "The Monetary Policy summary and minutes are published at 12 noon on a Thursday eight times a year" [boe-mps-minutes, page last updated 31 January 2023].

*Bank of Japan.* The Policy Board has nine members, the Governor, two Deputy Governors and six other members [boj-policy-board, web page]. *SNB.* Price stability means a rise in the consumer price index of less than 2 percent a year, assessed quarterly [snb-strategy, web page].

**Conditions.** *Historical, as Gliner gives them* [s7-gliner, chap. 11, sec. "Central Banks", PDF pp. 380–393]. Four to five of the eight FOMC meetings had projections and a press conference (p. 380). In 2025 every meeting had a press conference, and in 2026 the four meetings marked for a Summary of Economic Projections are those of March, June, September and December, with a press conference after each meeting held so far [fomc-calendars, 2025 and 2026 calendars]. The ECB's Governing Council had 23 members and "usually meets twice a month" (p. 383).

The three-year LTROs allotted €529.5 billion in December 2011 and €489.2 billion on 29 February 2012 (p. 384). OMT was announced on 6 September 2012, with the SMP terminated, for one- to three-year bonds of countries under strict conditionality, with "no limits to size" and full sterilisation (pp. 386–387). German credit exposure through TARGET2 to countries like Spain had risen "to around €700 billion" (pp. 387–388). The SNB introduced a minimum EUR/CHF rate of 1.20 on 6 September 2011 (p. 391).

The Bank of England raised its asset purchases by £50 billion to £375 billion on 5 July 2012 (Table 11.3, p. 372), after June minutes showing a 5–4 vote against more QE, against 8–1 in May (pp. 377–379).

**Practice.** Membership changes with euro-area enlargement: Gliner's 23 (six plus seventeen governors) and 25 (six plus nineteen) were each right for their time; 27 is current. Communication channels, list (statements after each meeting, minutes, press conferences, projections such as the Fed's dot plot, speeches), are standard; their current formats should be read from each bank's site.

**Pitfalls.** *Historical conventions.* ECB "HICP < 2%" is an earlier formulation (Gliner: the ECB "aims at inflation rates of below, but close to, 2" percent, p. 356); the current page states a 2 percent target that is "symmetric" [ecb-strategy, web page]. *Minutes.* "Released with lag (2 to 3 weeks)" fits the Bank of England only before August 2015: the Warsh review of December 2014 described minutes released "about a fortnight after each meeting" [warsh-2014, PDF p. 10]; the July 2015 decision was published on 9 July and its minutes on 22 July, and from August 2015 the Monetary Policy Summary and minutes were published together, on 6 August 2015 [boe-mpc-docs-2015, 2015 documentation page]. FOMC minutes follow about three weeks after each meeting (those of 15–16 September 2026 on 7 October 2026) [fomc-calendars, 2026 calendar].

*TARGET2.* Gliner calls TARGET2 a platform "for settling almost all heavily traded securities" (p. 387); it is the Eurosystem's large-value payment system, and securities settlement is the separate T2S service. *Later central-bank history.* The Bank of Japan ended its framework of Quantitative and Qualitative Monetary Easing with Yield Curve Control and the negative interest rate policy on 19 March 2024, guiding the overnight call rate to around 0 to 0.1 percent and discontinuing ETF and J-REIT purchases, by a 7–2 vote on the rate guideline [boj-2024-framework, statement of 19 March 2024, pp. 1–2]; its yield curve control and negative rates are therefore history, and the claim that it pioneered QE is unverified. The SNB discontinued the EUR/CHF minimum exchange rate and lowered its rate to −0.75 percent on 15 January 2015 [snb-2015-floor, press release title and date].

The ECB's asset purchase programme began purchases in October 2014, with net purchases of €60 billion a month from March 2015, and reinvestments were discontinued from July 2023 [ecb-app, APP page]; on 21 July 2022 the ECB said OMT "is part of the Eurosystem's toolkit" and that the Governing Council "retains discretion to conduct" it [ecb-tpi-2022, Transmission Protection Instrument release]. *OMT's use.* In *Gauweiler* the Court of Justice recorded that "more than two years after the programme … was announced, that programme has not been implemented, the Governing Council taking the view that its activation was not justified by the economic situation of the euro area" [cjeu-gauweiler-2015, Case C-62/14, judgment of 16 June 2015, para. 84, via CELLAR], and its press release says the programme "has still not been activated" [cjeu-gauweiler-2015, Press Release No 70/15].

That establishes non-activation to June 2015; for later years the support is the ECB's 2022 description of OMT as still "part of the Eurosystem's toolkit" (above) and the absence of any activation in the sources, so "never actually used" is well supported but rests partly on an absence. *TLTROs and the renminbi band.* The ECB announced three series of targeted longer-term refinancing operations, on 5 June 2014, 10 March 2016 (TLTRO II, "a series of four … each with a maturity of four years") and 7 March 2019 (TLTRO III, "a series of ten … each with a maturity of three years"), the third with voluntary early repayment, after one year from September 2021 under Decision ECB/2020/13 [ecb-tltro, TLTRO page].

The PBOC's Public Announcement [2014] No. 5 widened the interbank band of the renminbi against the U.S. dollar: it "is enlarged from 1 percent to 2 percent", "Effective from 17 March 2014" [pboc-2014-5, English text on the State Council's site, attributed to pbc.gov.cn]. Whether a later notice changed the band was not established. The PBOC's English regime page states only that "China pursues a managed floating exchange rate regime based on market supply and demand with reference to a basket of currencies", with no band width [pboc-en-2026, Exchange Rate Regime page].

*Reserve requirements.* The PBOC's required-reserves page lists cuts announced on 9 July 2021, 6 December 2021, 15 April 2022, 17 March 2023, 24 January 2024, 27 September 2024 and 7 May 2025, among others; its English pages give no complete history. The 2024 cut lowered the required reserve ratio by 0.5 percentage point, effective 27 September 2024, "excluding those that are already subject to an RRR of 5 percent", leaving a weighted average of "about 6.6 percent". The 2025 cut, also 0.5 point, took effect on 15 May 2025, with a further 5 points for auto-finance and financial-leasing companies [pboc-en-2026, Required Reserves page and the two announcements].

*The August 2015 fixing change* (the PBC statement as reproduced in NAFMII's weekly update): "Effective from 11 August 2015, the quotes of central parity that market makers report to the CFETS daily before market opens should refer to the closing rate of the inter-bank foreign exchange market on the previous day, in conjunction with demand and supply condition in the foreign exchange market and exchange rate movement of the major currencies" [nafmii-2015-08-14, p. 1]. The band's current width and the reserve-requirement history are unverified.

*Inflation forecasts.* The Bank of England published its first *Inflation Report* in February 1993, with a two-year inflation projection and a shaded uncertainty band built from the previous ten years' forecast errors, and its first fan chart in February 1996 [boe-qb-1998-fan-chart, pp. 30–31]. That dates the Bank's own practice; it does not establish that the Bank was the first major central bank to publish inflation forecasts, which needs other central banks' dates (the Reserve Bank of New Zealand, for example).

**See also.** central-bank policy; release calendar; currency regimes (the SNB floor).

**Further reading.** Each bank's own site for current decisions; Gliner, chap. 11, for the 2008–2013 record.

### Method 4.11 Reading the policy path: fed funds futures, overnight index swaps, meeting-dated OIS probabilities, and Eurodollar futures as a historical LIBOR contract converted to SOFR in 2023

**Statement.** *Fed funds futures* settle on 100 minus the average daily effective fed funds rate for the contract month, on a $5 million face value [s7-gliner, chap. 9, sec. "Federal (Fed) Funds", PDF pp. 257–259, Table 9.1]. An *overnight index swap* exchanges a fixed rate for the compounded overnight rate (the effective fed funds rate in the United States; in Gliner's time EONIA, SONIA and TOIS elsewhere); OIS are "great tools to hedge or speculate in central bank action", usually under two years, with customisable dates (sec. "Overnight Indexed Swaps (OIS)", pp. 260–261). Gliner's example: with Bank Rate at 0.50 percent, a December MPC-dated OIS at 0.625 prices "a 50 percent chance for a hike" of 25 bp $((0.625 - 0.50)/0.25)$ (p. 261).

*Eurodollar futures*, as Gliner describes them in 2013–2014 (sec. "London Interbank Offered Rate (LIBOR)", pp. 254–255): $1,000,000 notional on three-month LIBOR, price 100 minus the rate, one basis point worth $25, March, June, September and December months, and colour codes by year (white, red, green, blue, gold and beyond).

**Notation.** $P = 100 - R$; for a 90-day period, $\$1{,}000{,}000 \times 0.0001 \times 90/360 = \$25$ per basis point.

**Conditions.** *Probabilities from OIS.* The $(\text{OIS} - \text{current})/\text{step}$ reading holds when the swap's period starts at the meeting's effective date (a meeting-dated OIS) and only one move of one size is possible. If part of the period precedes the meeting, the average is diluted: with 60 percent of the period after the meeting, the same quote implies 83 percent (checks, section 6). Term premia, the spread between the overnight index and the policy rate, and the possibility of a larger move make the reading approximate.

**Practice.** *The end of LIBOR and of Eurodollar futures.* On 5 March 2021 ICE Benchmark Administration announced that it could not publish the main U.S. dollar LIBOR settings on a representative basis beyond 30 June 2023, and the FCA confirmed that three-month U.S. dollar LIBOR would cease to be representative immediately after that date; SOFR, selected by the ARRC in 2017 and published since April 2018, became the successor, with CME SOFR futures from May 2018; CME converted open Eurodollar futures and options into three-month SOFR futures and options in April 2023 (the CBOT filing gives "April 23, 2023", a date corrected below); synthetic one-, three- and six-month U.S. dollar LIBOR was published until 30 September 2024 and barred from new contracts and cleared derivatives [cbot-23-214, secs. 1–2, pp. 1–5].

The ARRC records that new use of U.S. dollar LIBOR had stopped after 31 December 2021 [arrc-sofr-transition, web page]; the end of the panel and €STR's replacement of EONIA are in money-market reference rates.

*The conversion date.* CME Clearing Advisory 23-082 (15 March 2023) states that "following close of business on Friday, April 14, 2023, CME will convert specific Eurodollar Futures and Options open interest to corresponding SOFR positions, at a ratio of 1:1, for Trade Date Monday, April 17, 2023", with the transformation run at about 10 a.m. CST on Sunday 16 April; it covered Eurodollar futures and options from the September 2023 expiry onwards and the one- to five-year mid-curve options (codes E0 and E2–E5, June 2023 the last listed), while the May and June 2023 contracts stayed listed to expiry (the June futures settling on 19 June 2023 against that day's LIBOR) [cme-23-082, p. 1].

The ARRC's minutes of 9 March 2023 give the same 14 April [arrc-minutes-2023-03-09, p. 1], as did the CFTC chairman's keynote in advance ("CME intends to convert all eligible Eurodollar futures and options to SOFR equivalent contracts on April 14, 2023") [cftc-behnam-libor, keynote text]. A later CME filing reports that "In April 2023, CME successfully converted 7.5 million contracts of Eurodollar futures and options open interest", citing a "CME Press Release 23 April 2023" [cme-24-271, p. 4 (PDF p. 5), n. 7]. CME's press release on the conversion is dated 24 April 2023, a Monday: CME "today announced that it successfully converted 7.5 million contracts of Eurodollar futures and options open interest and $4 trillion in cleared USD LIBOR swaps to corresponding SOFR derivatives in April", and only the May and June 2023 contracts remained; it gives no conversion date [cme-pr-2023-04-24, reprint dated 24/04/2023].

Three dates are therefore distinct: the *processing weekend* (positions taken after the close on Friday 14 April, transformation on Sunday 16 April), the *trade date* on which the converted positions first existed (Monday 17 April 2023) and the *announcement* (24 April 2023). The "April 23, 2023" in the CBOT filing and in CME's 2024 footnote, a Sunday, matches none of them; the filings do not explain it. The press release is dated 24 April.

**Pitfalls.** Eurodollar futures on 3-month LIBOR no longer trade, and EONIA no longer exists. Gliner calls Eurodollars "the most actively traded futures market in the world based on contact [sic] volume" (p. 254), as of about 2013, unverified against exchange statistics. *The conversion date.* CME Clearing Advisory 23-082 states the close of business on Friday 14 April 2023 and trade date Monday 17 April 2023. Do not read the filings' "April 23, 2023" as either the conversion date or the announcement date. Gliner's fed funds example (FFH4 at 99.79 "pricing in 0.21", computed as $100 - 99.97$) has a typo in the arithmetic (p. 257).

**See also.** money-market reference rates; forward rates and FRAs; interest-rate swaps; interest-rate measures.

**Further reading.** Market-based policy expectations [rt-kuttner-2001]; [gap-rates-sofr-futures-hedging] on pricing and hedging with SOFR futures.

### Method 4.12 Curve trades, carry and rolldown, and inversion as a recession signal: DV01-neutral steepeners and flatteners, bull and bear moves, carry signs that depend on the curve, and the Estrella–Trubin record

**Statement.** Flatteners and steepeners trade the spread between two points of the curve, "done DV01 neutral", for example 2s10s, the 10-year yield minus the 2-year [s7-gliner, chap. 9, sec. "Flatteners and Steepeners", PDF pp. 266–267]; a *bear flattener* has short rates rising more than long rates, a *bull flattener* long rates falling more than short (p. 267, Figure 9.7), and by symmetry a *bull steepener* has short rates falling more and a *bear steepener* long rates rising more. A steepener is long the short maturity and short the long maturity in DV01-matched amounts.

*Carry* is the income of a position net of financing over a holding period; *rolldown* is the price change as a bond ages along an unchanged curve. Total return ≈ carry + rolldown + the effect of yield changes.

*Inversion as a recession signal.* Estrella and Trubin's preferred spread is the 10-year constant-maturity Treasury yield minus the three-month secondary-market bill rate on a bond-equivalent basis, in monthly averages; it "has turned negative before each recession in the period from January 1968 to July 2006". All six NBER recessions since 1968 were preceded by at least three negative monthly averages in the twelve months before, with no false signals on a monthly-average basis, while negative spreads occurred on 100 days between 1968 and 2005 in months that did not have a negative average [estrella-trubin-2006, pp. 4–5]. The 2s10s spread "tends to turn negative earlier and more frequently" (p. 4).

**Notation.** DV01 price change per 1 bp yield change (duration and convexity); a DV01-neutral steepener holds $N_2\,\mathrm{DV01}_2 = N_{10}\,\mathrm{DV01}_{10}$.

**Conditions.** *Carry signs.* carry "+ for steepeners, − for flatteners in normal yield curve" is not general. For a DV01-neutral steepener (long 2-year, short 10-year), carry per unit DV01 is positive with 2-year at 1.5 percent, 10-year at 3 percent and repo at 1 percent, but negative with an inverted front (2-year 4.5, 10-year 4.0, repo 5.0) and also negative with a mildly upward curve when repo is close to the 2-year yield (2.0, 2.5, 2.4). Rolldown is deterministic only if the curve is unchanged, and carry is known at inception only for a fixed financing rate.

**Practice.** Gliner defines rolldown as rolling a futures or bond position to keep the original maturity profile (p. 268); that is the roll of a position, not rolldown in the usual sense defined above. Porter's view that a steepener is in effect a bet on a bond rally (practitioner frameworks) reflects the empirical tendency of the curve to steepen in rallies. Gliner reads an inversion far out the curve, as in Portugal and Greece in 2010–2012, as pricing default risk, and a mild inversion near the front as pricing cuts (sec. "Curve Inversion", p. 271; sovereign credit).

**Pitfalls.** *Three different statements*. (1) *A historical observation*: for each of the six NBER recessions from 1968 to 2005, the monthly-average spread was negative in at least three of the twelve months before the recession began (3 to 12 such months; minima from −0.08 to −3.51) (table, p. 6). (2) *The paper's forecast*: a probit for recession *status* twelve months ahead, $P(\text{recession in month } t+12) = F(-0.6045 - 0.7374\, s_t)$, estimated on 1959–2005 data, where recession months run from the month after an NBER peak through the trough (pp. 2–3); a zero spread gives 27 percent. (3) *Onset rules*, defined here and not in the paper, applied to negative months during expansions: a *month rule* (a recession begins in $t+1$ to $t+12$) and an *episode rule* (a run of consecutive negative months is followed by a recession beginning within twelve months of its last month).

*The episode rule is retrospective*. It is scored from an episode's *last* inverted month, which is known only after the spread has turned positive again, and against NBER peak dates announced months to more than a year after the event; at a date inside an episode neither the end of the run nor the recession's start is observable. Its hit rates below therefore describe the historical record and are not a prospective forecasting result; nor is the month rule one, since it too is scored on ex-post NBER dates, though each of its months is observable when it occurs. Neither is comparable with the probit's probability, which the paper states as a forecast made at $t$.

The paper's statement that monthly-average inversions gave "no false signals" (p. 5) matches the episode rule, not the month rule: its own footnote gives −0.16 in June 1989, "fourteen months before the recession" of August 1990 (p. 6, n. a to the table). Replicating the paper's series from the Federal Reserve's H.15 data (three-month secondary-market bill converted to a bond-equivalent yield by the paper's formula, n. 4, against the 10-year CMT; business-day averages by month) [fed-h15-data, H.15 data package, series RIFSGFSM03_N.B and RIFLGFCY10_N.B] reproduces the table's counts and minima (the 1990 minimum −0.07 against the printed −0.08) and the June 1989 value; over 1968–2005 the 54 negative expansion months include 6 month-rule misses (December 1968, November 1978 to January 1979, June and July 1989), while all 9 episodes satisfy the episode rule.

So (1) and the episode-level claim hold in sample; no general twelve-month onset law follows from six recessions. A typical lead time of 6–18 months matches none of the three rules. *After the paper's sample*, on the paper's series: the 43 complete negative expansion months from 2006 (October 2026, with data to 14 October, flagged incomplete and excluded) form five episodes, August 2006–May 2007, May–September 2019, February 2020, November 2022–November 2024 and March–April 2025. NBER peaks after 2005 are December 2007 and February 2020 (first contraction months January 2008 and March 2020), and no later peak was dated as of 16 October 2026 [nber-business-cycles, chronology table and dating page].

The month rule gives 11 hits, 5 misses (August–December 2006, 13 to 17 months before January 2008) and 27 provisional misses from November 2022; the episode rule gives three hits (the 2020 one before a pandemic-driven recession) and two provisional misses. *Recession outcomes through 17 October 2026*. Two conditions matter. *Calendar*: the twelve-month window of every one of the 27 months, the last ending in April 2026, has now passed.

*Dating*: NBER still lists February 2020 as the latest peak (chronology data last updated 14 March 2023; latest announcement 19 July 2021, the April 2020 trough) [nber-business-cycles, chronology table and announcements page]. Its peak announcements have come 4 to 12 months after the peak: January 1980 on 3 June 1980, July 1981 on 6 January 1982, July 1990 on 25 April 1991, March 2001 on 26 November 2001, December 2007 on 1 December 2008, February 2020 on 8 June 2020. So the 24 months from November 2022 to October 2024 are misses even allowing the longest lag yet seen; a recession dated in their windows would need a longer one. November 2024, March 2025 and April 2025, and with them both post-2022 episodes, are misses only if no peak up to April 2026 is announced within the lags seen so far.

Retrospective dating cannot be excluded for any of them. These outcomes stay provisional, for the dating reason only. Treasury's 3-month CMT, a proxy and not the paper's series, gives the same picture with 46 months and a further episode in June–August 2025 [ust-par-yield-curve, daily CSV files]. Both monthly series are rebuilt from daily data, with unique dates and consecutive months.

**See also.** duration and convexity; forward rates and term-structure theories; regime map; government bond futures.

**Further reading.** Term-structure decomposition [rt-gurkaynak-wright-2012].

### Definition 4.13 TIPS and breakeven inflation: index ratio and CPI-U lag, the principal floor, semiannual interest, the auction schedule then and now, and what a breakeven contains besides expected inflation

**Statement.** TIPS are Treasury securities whose principal is indexed to the CPI-U, not seasonally adjusted, through a reference index with a three-month lag; the fixed coupon rate is applied to the adjusted principal, and at maturity the holder receives the greater of the adjusted principal and the original principal [s7-gliner, chap. 9, PDF pp. 247–249]; the floor "does not apply to coupon payments" (p. 248).

TreasuryDirect describes TIPS as issued in 5-, 10- and 30-year terms, paying interest every six months, and repaying the adjusted or the original principal, whichever is greater; its current schedule has the 5-year auctioned in April and October and reopened in June and December, the 10-year in January and July and reopened in March, May, September and November, and the 30-year in February and reopened in August [treasurydirect-tips, web page]. The *breakeven inflation rate* is the nominal Treasury yield minus the TIPS real yield of the same maturity: 4.0 minus 1.5 gives 2.5 percent [s7-gliner, chap. 9, PDF p. 250].

**Notation.** Index ratio $=$ Ref $\text{CPI}_{\text{settle}}$ / Ref $\text{CPI}_{\text{dated}}$; adjusted principal $=$ face $\times$ index ratio.

**Conditions.** A breakeven equals expected inflation plus an inflation risk premium minus a liquidity premium for TIPS (and is affected by the deflation floor and the indexation lag), so breakevens reflect market-implied inflation expectations only approximately.

**Practice.** With cumulative CPI up 5 percent, the principal per 100 is 105 and coupons are paid on 105; with CPI down 2 percent, coupons are paid on 98 and the maturity payment is floored at 100. Gliner's walk-through of the reference-index calculation for a March 2012 settlement uses December 2011 CPI-U (p. 249).

**Pitfalls.** Gliner's "if CPI is 5 percent for a year, then the TIPS will trade at 105" conflates the adjusted principal with the market price, which is the index ratio times a real clean price that moves with real yields. "5Y, 10Y, and 30Y maturities issued quarterly" follows Gliner's "TIPS are typically issued on a quarterly basis" (p. 247) and does not match the current schedule above, under which the 10-year is offered six times a year. Gliner's 2014 calendar (p. 247) differs from the current one in two ways: the 5-year was then a new issue in April with reopenings in August and December, against new issues in April and October and reopenings in June and December now; and the 30-year was reopened twice, in June and October, against once, in August, now; the 10-year pattern is unchanged.

*When the calendar changed*. Treasury's refunding statement of 31 October 2018 made both changes from 2019: "Treasury will introduce a new October 5-year TIPS maturity starting in 2019", which would "replace the existing October 30-year reopening auction"; "the first reopening of the April 5-year TIPS will be moved from August to June"; and the 30-year would be reopened once instead of twice, with its reopening moved from June to August [treasury-refunding-2018-10, statement and its 2018–2019 auction table]. The 5-year TIPS itself dates from the May 2004 statement: "A new 5-year TIPS with the first auction in October 2004", then "auctioned on an April/October cycle" [treasury-refunding-2004-05, statement of 5 May 2004]; the October new issue had been dropped by Gliner's 2014 calendar.

"Short-end TIPS trade like food and energy" is Gliner's sentence (p. 248); it compresses the observation that short TIPS carry depends on headline CPI, which energy prices dominate month to month. Negative real yields mean holders accept a real loss if inflation equals the breakeven, not a "guaranteed" loss.

**See also.** inflation indicators; regime map; fixed-income universe.

**Further reading.** TIPS breakeven decomposition [gap-inflation-tips-decomposition]; [gap-inflation-tips-spec] for TreasuryDirect's CPI reference data.

**Definition 4.14 Interest-rate swaps for macro use: fixed for floating, payer and receiver, the swap curve, clearing, and uses**

**Statement.** An interest-rate swap exchanges fixed for floating interest payments on a notional; "future LIBOR expectations drive [the] swap curve", and "while most trade OTC" they are used to hedge interest-rate exposure and lock in spreads [s7-gliner, chap. 9, sec. "Interest Rate Swaps", PDF p. 256]. A *payer* pays fixed and receives floating, gaining when rates rise; a *receiver* receives fixed, gaining when rates fall. Today's U.S. dollar swaps reference SOFR, compounded in arrears (LIBOR cessation).

**Notation.** Par swap rate: the fixed rate that gives the swap zero value at inception; the floating leg follows the conventions of its reference rate.

**Conditions.** SOFR conventions as of 2026; Gliner's LIBOR account as of 2014. The payer gains from a rise in the swap rate for the remaining term, not only in the floating rate; with SOFR, the floating amount is known only at the end of each period.

**Practice.** After Dodd-Frank, many interest rate swaps became centrally cleared and some trade on swap execution facilities, rather than on exchanges; clearing is covered in OTC clearing and collateral. Interest rate derivatives as a whole are the largest derivatives market by notional: the BIS reports that interest rate derivatives, "the largest component of the global aggregate" of OTC derivatives, had $579 trillion of notional outstanding at end-June 2024 [bis-otc-2024h1, release of 21 November 2024].

*Swaps within it*: the BIS's OTC derivatives dataset gives, at end-December 2025, interest rate derivatives of $669.5 trillion (79.3 percent of all OTC notional), of which interest rate swaps $545.0 trillion (81.4 percent), forward rate agreements 10.7 percent and options 7.9 percent; at end-June 2024 swaps were $469.2 trillion, 81.1 percent of the $578.8 trillion [bis-otc-dataset, bulk file `WS_OTC_DERIV2`, net-net notional, all reporting countries, counterparties, currencies and maturities]. So interest rate swaps alone are the largest single OTC instrument by notional. Notional overstates economic exposure.

**Pitfalls.** Do not read a swap curve as pure expectations: it contains term premia and, against Treasuries, swap spreads with their own drivers.

**See also.** forward rates and FRAs; OIS; OTC clearing.

**Further reading.** The 2021 ISDA definitions [gap-rates-isda-definitions].

**Definition 4.15 The fixed-income universe for macro: money markets, government, inflation-linked, corporate, municipal, mortgage and asset-backed bonds, and fixed-income ETFs**

**Statement.** Gliner opens his fixed-income chapter with bank credit and debt capital markets funding governments, corporations and consumers, and covers money markets, LIBOR, swaps, fed funds, OIS, FRAs, futures, TIPS, sovereign credit and CDS [s7-gliner, chap. 9, opening, PDF p. 245]; he calls fixed income "the most important" product group for macro (chap. 4, PDF p. 95). The fixed-income universe comprises money markets (maturities under a year: bills, commercial paper, repo, fed funds, SOFR); government bonds; inflation-linked bonds (TIPS, U.K. index-linked gilts, French OATi); corporate bonds, investment grade and high yield; municipal bonds, often tax-exempt; agency and non-agency MBS, with prepayment risk; and ABS. Gliner's ETF list (sec. "Exchange Traded Funds (ETFs)", PDF p. 280) gives liquid access to these segments.

**Notation.** IG and HY the investment-grade and high-yield credit classes; agency MBS guaranteed by GSEs.

**Conditions.** Whether fixed income is the largest of the four product groups by notional outstanding is an ill-posed question: SIFMA's highlights give global fixed-income debt outstanding of $160.7 trillion and global equity market capitalisation of $157.8 trillion for 2025 [sifma-factbook-2026, Fact Book highlights, sec. 1, published 13 August 2026], similar magnitudes measured differently (face or market value against market value), while FX and commodities are traded mainly through derivatives, whose notionals ($579 trillion for interest rate derivatives alone at end-June 2024, interest-rate swaps) are not comparable with stocks of securities.

**Practice.** Gliner's Table 9.5 lists SHY, IEF, TLT, TBT, AGG, LQD, TIP, MBB, MUB, PCY, HYG, JNK and BKLN (p. 280); EMB is an additional example. These are funds as of 2013; check each fund's current prospectus before use. Inverse and leveraged ETFs such as TBT reset daily, so their multi-day returns are not the stated multiple of the index's.

**Pitfalls.** An ETF's price can deviate from the value of its bonds in stress; creation and redemption arbitrage is in ETF creation and redemption.

**See also.** money markets; TIPS; sovereign credit; CDS; fixed-income market structure.

**Further reading.** The section's readings on each segment.

# 5 Government bond markets and sovereign debt

## 5.1 Sovereign credit

**Definition 5.1 Sovereign credit risk: issuers and CDS reference entities, inverted sovereign curves, the 40 percent recovery convention against Moody's recoveries, and the 1998–2006 default record with later cases left to check**

**Statement.** Gliner gives ISDA's 2003 definition of a sovereign, "any state, political subdivision, or government, or any agency, instrumentality, ministry, department, or other authority (including the central bank) thereof", and notes that a country has many issuers (for Greece, the Hellenic Republic's several bond programmes and state companies) in several currencies [s7-gliner, chap. 9, sec. "Sovereign Credit", PDF pp. 269–270]. The *reference entity* is the legal name used in CDS, "Greece is the Hellenic Republic" (Table 9.3, pp. 274–275). An inverted curve "further out" typically means "the market is pricing in default risk", as in Portugal's five-year yield above its long bonds (sec. "Curve Inversion", p. 271). Loss severity is one minus recovery, "usually assumed to be 40 percent"; Moody's recoveries on defaulted sovereign bonds for 1983–2006 average 55 percent with a standard deviation of 24 percent (Figure 9.10, p. 278).

His Table 9.4 (from Moody's, *Sovereign Default and Recovery Rates, 1983–2006*, June 2007) lists defaults from 1998 to 2006, among them Russia (August 1998, $72,709 million: local-currency Treasury obligations first, then a moratorium on foreign-currency obligations, restructured in 1999–2000), Argentina (November 2001, $82,268 million, a distressed exchange with haircuts of about 70 percent), Uruguay (April 2003, $5,744 million, a distressed exchange extending maturities by five years), and Peru, Moldova, the Dominican Republic and Belize (pp. 275–278).

**Notation.** Recovery $R$ as a fraction of par; loss given default $1 - R$; amounts in the table in millions of U.S. dollars of debt affected.

**Conditions.** Default record to 2006; Greece 2012 and Argentina 2020 from IMF reports of 2012–2022; Argentine default counts as published in 2010 (Reinhart), 2020 (CRS) and 2021 (Fitch). *The recovery comparison.* Gliner concludes that a mean of 55 with a standard deviation of 24 "is not enough evidence to dispute the standard 40 percent". Whether it is depends on the number of independent defaults, which he does not give: the mean differs from 40 at the 5 percent level once there are more than about ten independent issuers. Recoveries also differ in measurement (trading prices after default against the value of exchanged bonds) and by restructuring.

**Practice.** Peripheral euro-area spreads to Bunds, which one account calls key risk indicators, are the subject of `dm-sovereign-bonds`; Gliner's account of Greece's two bailouts and of five-year CDS moving from 345 to 890 bp in April–May 2010 is in his chapter 11 (PDF p. 389).

**Pitfalls.** *Unverified examples*: "Greece 2012: PSI restructuring, largest sovereign default in history, $200B+" and "Argentina 2020: Ninth default, $65B restructured". *Against IMF reports*. Greece: the exchange subjected €197 billion of debt to the exchange, with a nominal haircut of $53\frac{1}{2}$ percent and NPV losses of 70 to 75 percent (at exit yields of 9–12 percent), "the largest write down recorded for a pre-default sovereign debt restructuring"; 85.8 percent of domestic-law holders participated before collective action clauses were used [imf-cr-12-57, PDF p. 46]; the IMF's later evaluation gives €197 billion exchanged for €62 billion of new debt and €30 billion of EFSF notes, a write-down of €106 billion or 52 percent of 2012 GDP, with 97 percent of eligible bonds tendered [imf-cr-17-44, PDF pp. 45–46].

So "$200B+" can only be the face amount exchanged converted to dollars, not the loss, and "largest" holds for the write-down. Argentina: "The September 2020 restructuring of US$82 billion in FX-denominated debt held by the private sector (US$65.5 billion of which corresponded to the foreign-law debt exchange)" gave cash-flow relief of about US$36 billion over 2020–30, a recovery value of about 52 percent at a 10 percent exit yield and 99 percent participation, using two-limb CACs and redesignation [imf-cr-22-92, Box 4, PDF p. 42]; $65B is the foreign-law part only.

"Ninth default" is the Congressional Research Service's count: Argentina "has defaulted on its external debt (debt held by foreigners) nine times since independence in 1816", and missing a $503 million interest payment on New York-law dollar bonds on 22 May 2020 put it "into its ninth default" [crs-if10991-v6, pp. 1–2]; the note gives no list of the nine. Other counts use other definitions and are not to be merged with it. Reinhart's chartbook lists seven external default episodes for 1816–2009 (1827–1857, 1890–1893, 1951, 1956–1965, 1982–1993, 1989 and 2001–2005, the 1989 episode given no separate duration and lying within 1982–1993) and, separately, five domestic default episodes (1890–1893, 1982, 1989–1990, 2001–2005 and 2007–2009), on Reinhart and Rogoff's episode definitions [reinhart-chartbook-2010, Figure 3a and table, printed p. 16, PDF p. 17]; that dataset ends in 2009, before the 2014, 2019 and 2020 events.

Fitch, which counts rating defaults, records Argentina in default in 2001 (more than US$80 billion of foreign-currency bonds), 2014 (a missed payment on foreign-law discount bonds after the grace period), 2019 (distressed debt exchanges on 30 August and 20 December through unilateral extensions of local-law bills) and 2020 (a distressed exchange from the decree of 5 April and missed interest within the 22 May grace period, rated RD) [fitch-sovereign-2020-study, pp. 2–3].

So whether May 2020 was the ninth default depends on whose list and definition is used; the CRS figure is an attributed count, not a fact independent of method. The CDS side of Greece 2012 is now dated from ISDA: a unanimous determination on 9 March 2012 that a Restructuring Credit Event had occurred, after collective action clauses reduced payments on Greek-law bonds, and an auction on 19 March 2012 [isda-greece-2012, releases of 9 and 19 March 2012] (CDS contracts).

These sources support neither a 20–80 percent range of recovery rates nor higher recovery in orderly restructurings. An inversion prices default risk only when it is credit-driven; a front-end inversion in a default-free curve prices rate cuts (curve trades).

**See also.** CDS contracts and credit events; curve inversion; FX valuation signals (sovereign risk and currencies).

**Further reading.** Restructuring practice [gap-sovereign-restructuring-playbook].

# 6 Corporate credit, hybrids and labelled bonds

## 6.1 Credit default swaps

**Mechanism 6.1 The credit default swap: premium leg paid quarterly, physical settlement against par and its cash equivalent par minus recovery, credit events for sovereigns and corporates, counterparty risk, and what has changed since Gliner's account**

**Statement.** A CDS works like insurance on a reference entity's debt: the protection buyer pays a premium, "quoted at an annual spread, but paid quarterly" (100 bp a year is 25 bp a quarter, "usually in March, June, September, and December"); being long credit risk means selling protection and receiving the spread [s7-gliner, chap. 9, sec. "Credit Default Swaps (CDS)", PDF pp. 272–273]. The basic terms are the reference obligation, notional, spread and maturity (p. 273). On a credit event, in Gliner's description, "the protection buyer stops paying the premium and delivers the reference bond to the protection seller. Then the protection seller makes the protection buyer whole by transferring par (100)" (p. 273, Figure 9.9).

His two characteristic credit events are *failure to pay* and *restructuring* (postponing, reducing or deferring obligations); for sovereigns, he writes, "CDS is not automatically triggered during a restructuring event and both protection buyer and seller have a right (but not an obligation) to trigger the CDS" (p. 273). More than one party can buy protection on the same debt, so CDS notional can exceed the reference obligations (p. 272), and the seller's own default is a counterparty risk (p. 273).

**Notation.** $s$ annual spread; $N$ notional; $R$ recovery as a fraction of par; protection payout $N(1-R)$.

**Conditions.** Gliner's account as of 2013, largely pre-2009 practice. *An illustration under one assumption, not an identity of real settlements*. If the delivered bond is worth $R$ per unit of par and cash settlement pays par minus that same $R$, the two methods give the same payout, $1-R$. In practice the values need not coincide: the buyer chooses which deliverable obligation to deliver (the cheapest), cash settlement uses a price fixed by an auction or poll at a particular time, and accrued premium and timing differ; so the payout under one method is not proof of what the other would have paid.

With a delivered bond worth 38 and an auction final price of 40, physical delivery nets 62 and cash settlement 60 per 100. A buyer of protection receives par only on delivery of the bond; "par minus recovery" paid by the seller is the cash equivalent.

**Practice.** Gliner's argument for buying protection, that on $10 million at 100 bp the investor pays $100,000 a year "with almost unlimited upside (up to $10,000,000)" (p. 279), overstates the payout: the gain is $N(1-R)$, $6 million at 40 percent recovery, less premiums paid. He notes that CDS are more often bought to hedge a bond position, the difference between the bond and CDS spreads being the *basis* (p. 279).

**Pitfalls.** *Practice has moved on, by contract coverage*. Gliner's account follows earlier contract practice. ISDA's Big Bang Protocol, open from 12 March to 7 April 2009, supplemented the 2003 Credit Derivatives Definitions with the 2009 Determinations Committees and Auction Settlement Supplement [isda-big-bang-2009, protocol page and FAQ]. What changed depends on which contracts incorporate it. *Coverage*: adhering parties amended their existing "Protocol Covered Transactions" (DC provisions effective from 8 April 2009; the 60- and 90-day backstop dates fully from 20 June 2009); the Supplement applies to later trades that incorporate it, and DTCC applied it to covered trades whether or not a party had adhered; "Excluded Transactions like LCDS and CDS on ABS" were outside it, and auction settlement did not apply to "Covered Non-Auction Transactions".

*Triggering a covered contract*: a party may ask the DC to determine a credit event, with Publicly Available Information, or "Alternatively, the party could send a Credit Event Notice and Notice of Publicly Available Information directly to its counterparty", either within 60 days of the event; if the DC accepts the question and then declines to decide it, the 60-day limit is tolled while it deliberates, and if no member accepts it within two business days it is dismissed and the limit keeps running. So bilateral notices did not end in 2009; DC resolutions bind covered trades, and bilateral notice remains a route, subject to the backstop dates.

*Settlement*: if the DC holds an auction, covered parties "will not have the option to physically settle prior to the auction, unless agreed bilaterally", but can submit physical settlement requests into it; in the 2009 FAQ the DC "will not hold auctions for Restructuring Credit Events" and may skip illiquid names, in which case the confirmation's fallback settlement method applies (restructuring auctions came with the later Small Bang protocol; the Greek restructuring of 2012 was auctioned, below).

*Standard North American contracts*: the same FAQ describes the "100/500" Standard North American Corporate transaction, with investment-grade names on a fixed 100 bp coupon and high-yield names on 500 bp quoted as points upfront, quarterly roll dates of 20 March, June, September and December, No Restructuring as the standard (Modified Restructuring electable), and auction hardwiring. The notice route, under which both buyer and seller have the right but not the obligation to trigger, therefore still applies to contracts that use it; it is not simply pre-2009 practice.

*Greece.* On 9 March 2012 the ISDA EMEA Determinations Committee "resolved unanimously" that a Restructuring Credit Event had occurred with respect to the Hellenic Republic under Section 4.7(a) of the 2003 Definitions (as amended by the July 2009 Supplement), following the exercise of collective action clauses on Greek-law bonds, and an auction was held on 19 March 2012 [isda-greece-2012, releases of 9 and 19 March 2012]; a republished copy of the results gives a final price of 21.5.

The 2014 ISDA Credit Derivatives Definitions [add-cds-contract, catalogue entry], as summarised in ISDA's release, add a new credit event "triggered by a government-initiated bail-in" for financial reference entities, delivery of the assets into which sovereign debt is converted for sovereign CDS, a standard reference obligation per entity and seniority, and amendments on successors, guarantees, contingent obligations, currency redenomination and restructuring settlement; the Definitions "do not apply automatically", the implementation date was the September 2014 roll, and they apply only if referenced in new trades or adopted for existing ones by protocol [isda-2014-release, pp. 1–2]. A press release is not the contract terms, so the provisions' wording, and later standard terms outside North America, are not covered here.

The adoption protocol itself, published by ISDA on 21 August 2014, amends adhering parties' Protocol Covered Transactions to incorporate the 2014 Definitions from an Implementation Date defined as "the later of (a) September 22, 2014 and (b) the ISDA NY Business Day after any Cut-off Date" [isda-2014-protocol, PDF pp. 1 and 13], and ISDA's protocol page records that all eight Reference Banks had adhered by 17 September 2014, satisfying the Protocol Effectiveness Condition. That dates the switch; it still does not give the Definitions' operative wording on credit events or deliverable obligations. Bankruptcy as a credit event applies to corporates, not sovereigns, which cannot file for bankruptcy.

**See also.** sovereign credit; OTC clearing and collateral; CDX high yield as an equity indicator.

**Further reading.** The ISDA CDS standard model [add-cds-model].

# 7 Financial statement analysis

## 7.1 Cash-flow analysis

**Method 7.1 Linking the three statements: deriving operating, investing and financing cash flows from the income statement and balance sheets, by the indirect and direct methods, under IFRS and US GAAP**

**Statement.** The statement of cash flows reports the period's cash flows classified by operating, investing and financing activities [eu-2023-1803-ias7, IAS 7 para. 10]. Operating activities are the principal revenue-producing activities and any activity that is neither investing nor financing; investing activities acquire and dispose of long-term assets and other investments outside cash equivalents; financing activities change the size and composition of contributed equity and borrowings (para. 6). Only expenditure that results in a recognised asset may be classified as investing (para. 16). The three sections together explain the change in cash and cash equivalents between the two balance sheets, with the effect of exchange-rate changes on foreign-currency cash shown separately (para. 28).

Operating cash flow is reported by either the *direct* method, which shows major classes of gross receipts and payments, or the *indirect* method, which starts from profit and adjusts it for non-cash transactions, deferrals and accruals of operating receipts and payments, and items whose cash effects are investing or financing (para. 18). The indirect adjustments include changes in inventories and operating receivables and payables, and non-cash items such as depreciation, provisions and deferred taxes (para. 20). The direct method is encouraged, and its gross amounts may be derived by adjusting revenue, cost of sales and other income-statement items for the same changes (para. 19). A gain on selling plant enters profit, but the proceeds are an investing inflow (para. 14).

**Notation.** Interpretation: for a balance-sheet account, closing balance = opening balance + additions − reductions. Applying this identity account by account turns accruals into cash. With Δ denoting closing minus opening balance, indirect operating cash flow is

$$\text{CFO} = \text{starting profit} + \text{non-cash charges} - \text{gains on investing disposals} - \Delta\text{operating assets} + \Delta\text{operating liabilities} \pm \text{reclassified items}.$$

Operating assets include receivables, inventory and prepaid expenses; operating liabilities include trade payables, accrued expenses and deferred revenue. Capital expenditure is the change in gross property, plant and equipment plus the cost of assets disposed of; dividends paid are opening retained earnings plus profit minus closing retained earnings when no other equity movement exists. Then $\Delta\text{cash} = \text{CFO} + \text{CFI} + \text{CFF}$.

*Line-by-line conversion* (Interpretation, from the same identity). Each income-statement line becomes a cash line by removing the change in its related operating balances. The rules below assume that those balances move only through the period's income-statement items and cash settlements: no business combinations or disposals of businesses, no foreign-currency translation of the balances, no non-cash settlements and no amounts recognised in other comprehensive income (OCI) or directly in equity. Any such movement must be removed from Δ first.

- cash from customers = revenue $-\Delta$receivables $+\Delta$deferred revenue;
- cash paid to suppliers = cost of sales $+\Delta$inventory $-\Delta$trade payables;
- cash paid for an expense = expense $+\Delta$prepaid expense $-\Delta$accrued expense;
- income tax paid = current tax recognised in the period $-\Delta$net current tax payable, where the net payable is current tax payable less current tax receivable; interest paid = interest expense $-\Delta$interest payable;
- depreciation, amortisation and gains or losses on disposals have no operating cash line.

The tax line is a roll-forward of the *current* tax balance. Current tax unpaid is a liability and an overpayment an asset [eu-2023-1803-ias12, IAS 12 para. 12], so opening net payable plus current tax recognised minus closing net payable is tax paid. Current tax recognised includes tax charged in OCI or equity as well as the current component of tax expense in profit or loss, which entities disclose separately from deferred tax (paras. 79–80, 81(a) and (ab)). Deferred tax is a non-cash charge: its profit-or-loss part is added back in the indirect method, and deferred-tax balances are not used to compute cash tax. They also move outside profit or loss, for example when a revaluation of property is recognised in OCI and its deferred tax with it (paras. 58, 61A and 62(a)).

The textbook form of the same rule is: cash paid for a prepaid item is closing prepaid plus expense less opening prepaid, and cash paid for an accrued item is opening payable plus expense less closing payable [principles-financial-accounting-2019, sec. 16.6]. Every adjustment in the indirect method appears in exactly one direct-method line, so the two totals agree.

**Conditions.** Framework and period. The paragraph references above are to IAS 7 as adopted in the EU in 2023, before the IFRS 18 amendments. IFRS 18 *Presentation and Disclosure in Financial Statements* applies to annual periods beginning on or after 1 January 2027, with earlier application permitted [eu-2026-338-ifrs18, Annex, IFRS 18 para. C1]; EU companies apply it at the latest from their first financial year starting on or after that date (Art. 2). It changes the indirect method's starting point from profit to *operating profit or loss*, with tax and other operating cash flows whose income or expense lies outside the operating category added as explicit adjustments (amended IAS 7 paras. 18(b) and 20).

IFRS 18 also removes classification choices. Under the earlier text, entities other than financial institutions could classify interest paid and interest and dividends received as operating, or as financing and investing respectively, and dividends paid as financing or operating [eu-2023-1803-ias7, IAS 7 paras. 33–34]. The amended standard requires dividends paid to be financing, and, for an entity whose main business activity is not investing in assets or financing customers, interest paid to be financing and interest and dividends received to be investing [eu-2026-338-ifrs18, Annex, amended IAS 7 paras. 33A and 34A]. Entities with such a main business follow their income-statement classification (paras. 34B–34D). Taxes on income are operating unless specifically identified with investing or financing activities [eu-2023-1803-ias7, IAS 7 para. 35].

*US GAAP* (FASB Accounting Standards Codification Topic 230). Interest and dividends received are operating inflows, and interest paid and payments to governments for taxes are operating outflows; dividends paid are financing outflows [us-cash-flow-classification-amendment-2016, ASC 230-10-45-16(b), 45-17(c)–(d) and 45-15(a), as amended, pp. 9–10 and 14–15]. Those amendments apply to public business entities for fiscal years beginning after 15 December 2017 and to other entities a year later (p. 5). The direct method is encouraged, with at least the classes of receipts and payments listed in 230-10-45-25 (p. 10). Whichever method is used, net income must be reconciled to operating cash flow, in a separate schedule when the direct method is used [cash-flow-statement-roadmap-2026, ASC 230-10-45-28 to 45-30, as quoted]; IAS 7 paragraphs 18–20 contain no such reconciliation requirement. The US GAAP text above is the 2016 Update's reprint and a 2026 secondary quotation; the current Codification was not consulted, so later amendments to these paragraphs are not reflected. In 2025–2026 the FASB staff reported stakeholder suggestions that interest need not all be operating as research feedback, not as a proposal [cash-flow-standard-setting-2026, AP35, slides 2–4; AP20A, slide 4].

| | IFRS before IFRS 18 | IFRS with IFRS 18 (from 2027) | US GAAP (Topic 230) |
|---|---|---|---|
| Indirect method starts from | profit or loss | operating profit or loss | net income |
| Interest paid | operating or financing | financing* | operating |
| Interest, dividends received | operating or investing | investing* | operating |
| Dividends paid | financing or operating | financing | financing |
| Income taxes paid | operating unless identified | operating unless identified | operating |
| Reconciliation if direct method | not required | not required | required |

*For entities whose main business activity is not investing in assets or providing financing to customers.

**Practice.** Example, original and synthetic. A distributor reports revenue 1,000, cost of sales 600, insurance expense 30, wages expense 150, depreciation 50 and a gain of 10 on equipment sold for 80 (cost 100, accumulated depreciation 30), so operating profit is 180. Interest expense is 20, all paid, and tax expense is 40; profit is 120. Receivables rise from 120 to 150, inventory falls from 200 to 180, prepaid insurance rises from 10 to 16, payables rise from 100 to 130, wages payable rise from 12 to 20, and tax payable falls from 15 to 10. Gross PP&E rises from 800 to 900, accumulated depreciation from 300 to 320, and borrowings from 300 to 350. Dividends of 60 are paid, and opening cash is 50.

| | |
|---|---|
| Operating profit | 180 |
| Depreciation, less gain on sale (50 − 10) | 40 |
| Receivables, inventory, prepaid (−30 + 20 − 6) | −16 |
| Payables, wages payable (+30 + 8) | 38 |
| Tax paid (40 + 15 − 10) | −45 |
| **Operating cash flow** | **197** |
| Capital expenditure (900 − 800 + 100); sale proceeds | −200; +80 |
| **Investing cash flow** | −120 |
| New borrowing; interest paid; dividends paid | +50; −20; −60 |
| **Financing cash flow** | −30 |
| **Change in cash** (50 → 97) | **47** |

The direct method converts each line. Cash from customers is 1,000 − 30 = 970; purchases are 600 − 20 = 580, of which 580 − 30 = 550 was paid to suppliers. Insurance paid is 16 + 30 − 10 = 36, six more than the expense because the prepayment grew; wages paid are 12 + 150 − 20 = 142, eight less than the expense because more wages were left unpaid. With tax paid of 45, operating cash flow is 970 − 550 − 36 − 142 − 45 = 197. The closing balance sheet, with cash at 97 and retained earnings rising by 120 − 60, balances at 1,023.

Classification changes the sections, not the change in cash. Under US GAAP, or under the earlier IFRS text with interest paid classified as operating, the indirect method starts from profit of 120 and gives operating cash flow of 177: 120, plus 40 of depreciation net of the gain, plus the 22 working-capital release, less the decrease of 5 in tax payable. Financing cash flow is then −10. The 20 of interest paid is the whole difference from the IFRS 18 figure of 197.

Interpretation: operating cash flow exceeds profit mainly because depreciation is non-cash and working capital released 22. Operating cash flow less capital expenditure is −3; the sale proceeds and new borrowing funded the dividend. A single year does not show whether the inventory release or the capital expenditure will recur.

**Pitfalls.** Deduct a gain on sale from profit and show the full proceeds under investing; adding the proceeds to operating cash flow counts the gain twice. An increase in an operating asset, such as a prepayment, reduces cash, and an increase in an operating liability, such as an accrual, increases it. Use tax paid, not tax expense, and interest paid, not interest expense, where payables change. Derive tax paid from current tax, not from total deferred-tax balance changes: with current tax of 30 in profit or loss, no deferred tax there, an unchanged tax payable and a revaluation that adds a deferred tax liability of 10 through OCI, tax paid is 30; subtracting the change in the deferred tax liability would give 20. Investing and financing transactions that use no cash, such as acquiring an asset by assuming a liability or converting debt into equity, are excluded from the statement and disclosed elsewhere [eu-2023-1803-ias7, IAS 7 paras. 43–44]. Before comparing operating cash flows across companies, frameworks or the 2027 transition, put interest and dividends in the same sections.

**See also.** return measures (timing of cash flows in returns).

**Further reading.** The indirect-method steps and a worked statement are taught in [principles-financial-accounting-2019, secs. 16.3–16.4], the direct method in sec. 16.6.

# 8 Systematic equities

## 8.1 Combining alpha signals

**Method 8.1 Combining alpha signals into one forecast: linear weights, conditional and rotational models, machine learning, or separate factor portfolios**

**Statement.** Narang treats blending as the problem of turning several alpha models' forecasts for an instrument into one composite forecast, which then goes to portfolio construction [s7-narang, chap. 3, "Alpha Models: How Quants Make Money", sec. 3.5, pp. 64–68]. He names four approaches:

- *linear models*, "by far the most common": each factor contributes additively, its weight usually set by multiple regression on past returns. In his example a yield factor forecasting +20 percent at weight 0.7 and a trend factor forecasting −10 percent at weight 0.3 give $0.7\times 20+0.3\times(-10)=+11$ percent (pp. 65–66). Equal weights, equal-risk weights and discretionary weights are special cases (p. 66);
- *nonlinear models*: *conditional* models, where one factor's reading sets another's weight, for example acting on value only when momentum agrees, which in his example gives no signal, or +20 percent if the trend agreed (Exhibit 3.10, pp. 66–67); and *rotation* models, whose weights are refitted on recent data and "usually" favour the factors that have done best recently, "a form of trend following in the timing of alpha factors" (p. 67);
- *machine-learning* models that choose the weights, which he calls "both more common and more successful" than machine learning used to forecast markets (pp. 67–68);
- not combining at all: building one portfolio per alpha model and combining those portfolios, as in signal-portfolio allocation (p. 64).

Signal mixing produces one forecast per instrument; portfolio construction then sizes positions from forecasts, risk and costs (p. 68).

**Conditions.** Practitioner account as of 2024; evidence 1993–2015. Narang's argument for blending is that two partly right forecasts can together be right more often (Exhibit 3.9, p. 65), and that "any reasonably intelligent combination of alphas" is likely to beat any one of them over time (p. 64). Interpretation: that holds when the forecasts each carry information and their errors are not highly correlated; combining a weak forecast with a strong one, or two near-duplicates, need not help, and weights fitted to past returns carry estimation error. His Exhibit 3.9 is a diagram, not evidence.

**Practice.** Interpretation: "linear" describes the form of the combination, not the factors. A linear model assumes each factor's contribution does not depend on the others (no interaction), which is what Narang means by inclusion being "independent" (p. 65); it does not need the factors to be statistically independent. Correlated factors can be combined linearly, though highly correlated ones make regression weights unstable. A conditional rule is an interaction; a rotation rule is a time-varying weight. Combining signals into one forecast before optimising, or optimising on the signals separately, is the choice between mixing and integrating.

*Evidence on that choice.* Fitzgibbons, Friedman, Pomorski and Serban, of AQR, contrast two long-only constructions [eq-fitzgibbons-2017, *Journal of Investing* 26(4), Winter 2017, AQR-hosted reprint, pp. 1, 7 and 9]:

- the "portfolio mix" combines single-style portfolios;
- the "integrated portfolio" first combines the style scores and builds one portfolio.

They test value and momentum on a universe similar to MSCI World, February 1993 to December 2015, rebalanced monthly. Both are held at about 4 percent tracking error, net of estimated transaction costs and gross of fees (Exhibit 5). The integrated portfolio earned 3.6 percent a year over the benchmark, against 2.5 for the mix, an information ratio of 0.9 against 0.6. Its transfer coefficient, the ex ante correlation between portfolio and model, was 0.6 against 0.3.

They attribute the gain to better capture of the styles, not to better stocks, and expect larger gains with negatively correlated styles, higher risk or more styles. The sample is the authors' own backtest, by a firm that runs such products. Their claim of "highly statistically significant" differences is not shown in the source; a later critique argues the approaches cannot be distinguished after multiple-testing adjustments.

**Pitfalls.** Narang's linear models require additivity, not statistical independence of factors. Rotation by recent performance is a bet that factor performance trends; Narang gives no evidence for it, and refitting weights on recent data counts as further trials (parameter overfitting). His claim that machine learning works better for weighting than for forecasting is a judgement without data in the chapter. He refers to Chapter 7 for portfolio construction techniques (p. 64), but they are in Chapter 6; Chapter 7 is on execution. For López de Prado's account of machine-learning strategy costs and "microscopic alpha", see the strategy production lifecycle.

**See also.** alpha models; alpha strategy specification; signal-portfolio allocation; rule-based sizing; mean–variance optimisation.

**Further reading.** Grinold's 1994 paper; Qian.

**Method 8.2 Scaling a raw signal into a return forecast: alpha = volatility × IC × score, and combining two correlated signals**

**Statement.** Grinold and Kahn's basic forecasting formula takes the refined forecast to be the change in expected return from observing the raw forecasts $g$, $E\{r \mid g\} - E\{r\} = \mathrm{Cov}\{r,g\}\,\mathrm{Var}^{-1}\{g\}(g - E\{g\})$, derived in the chapter's appendix [alpha-grinold, chap. 10, eqs. (10.1)–(10.2), pp. 262–263]. For one asset and one forecast it becomes $\phi = \mathrm{Std}\{r\}\cdot\mathrm{Corr}\{r,g\}\cdot(g - E\{g\})/\mathrm{Std}\{g\}$ (eq. (10.9), p. 265), the forecasting rule of thumb

$$\text{refined forecast} = \text{volatility}\cdot\text{IC}\cdot\text{score} \qquad \text{(eq. (10.11), p. 266)},$$

where the score standardises the raw forecast by its own mean and standard deviation and the information coefficient (IC) is the correlation between forecast and return. The same result follows from a time-series regression of $r$ on $g$, with sample estimates in place of the true moments (eqs. (10.12)–(10.15), pp. 266–267).

**Notation.** $r$ excess (or residual) return over the forecast horizon; $g$ the raw forecast in its own units; score $= (g - E\{g\})/\mathrm{Std}\{g\}$; volatility $= \mathrm{Std}\{r\}$ over the same horizon. The IC and score are dimensionless and the volatility carries the units of return (p. 268).

**Conditions.** *Projection, not in general a conditional mean.* What the appendix proves, as its Proposition 1, is that the formula is the *best linear projection*: among forecasts of the form $a + Bg$, it is the unbiased one with the smallest mean squared error [alpha-grinold, chap. 10, Technical Appendix, pp. 287–289]. This needs only finite second moments and a nonsingular $\mathrm{Var}\{g\}$. The book says it uses the notation $E\{r \mid g\}$ "somewhat loosely" (n. 1, p. 263).

Equality with the conditional expectation $E\{r \mid g\}$ needs that conditional mean to be affine in $g$. That holds when $\{r,g\}$ are jointly normal (the appendix's Proposition 2, which then also gives the conditional variance and minimum mean squared error among all unbiased estimates) or, more generally, elliptically distributed; otherwise it is an extra assumption.

The covariance conditions differ in strength. The projection's error is *uncorrelated* with $g$, so $\mathrm{Cov}\{r-\phi, g\} = 0$. The conditional mean's error is *mean-independent* of $g$, so $E\{r - E\{r \mid g\} \mid g\} = 0$, and is therefore uncorrelated with every function of $g$.

So IC $= 0$ makes the *linear* refined forecast zero (p. 266); it does not mean $g$ carries no information. Counterexample: with $g \sim N(0,1)$ and $r = g^2 - 1 + \varepsilon$, $\mathrm{Cov}\{r,g\} = E\{g^3\} = 0$, so the IC is zero, yet $E\{r \mid g\} = g^2 - 1$ explains $2/(2+\sigma_\varepsilon^2)$ of the return variance, two thirds at $\sigma_\varepsilon = 1$.

The formula also needs the forecast's mean and standard deviation, the return volatility and the IC, all estimated in practice, and sample means of returns "generally contain a great deal of sample error" (n. 2, p. 267). In the binary-model example the quarterly volatility is 9 percent and the forecast shares 3 of its 16 unit components with the return, so $\mathrm{IC} = 3/(9 \cdot 4) = 0.0833$ and the refined forecast is 0.75×score: within ±0.75 percent two quarters in three and outside ±1.50 percent one quarter in twenty, if scores are normal (eqs. (10.6)–(10.10), pp. 264–266). If $\mathrm{IC} = 0$ the linear refined forecast is zero (p. 266).

**Practice.** The scaling controls for expectations (the score subtracts the expected raw forecast), skill (the IC) and volatility: of two stocks with the same score and skill, the more volatile receives the larger alpha (p. 268). Cross-sectionally the scores should have mean near 0 and standard deviation near 1 at each date, as should each stock's scores over time (p. 268); Table 10.1 applies an IC of 0.09 to residual volatilities and simulated scores for the Major Market Index of December 1992 (p. 269).

*Two signals.* With a second forecast $g'$ ($\mathrm{Var} = 25, \mathrm{Cov}\{r, g'\} = 4, \mathrm{Cov}\{g, g'\} = 5$) the regression weights are 0.1467 and 0.1307 and the combined IC is 0.1090 (pp. 269–271); for uncorrelated forecasts the refined forecasts add and the squared ICs add, and for perfectly correlated ones the second adds nothing (p. 270). The IC, the 0.75 multiplier, the two weights, the combined IC and Table 10.1's alphas recompute to the rounding of the printed scores.

**Pitfalls.** Interpretation: "bigger signal, bigger bet" (signal combination) needs this calibration first; a raw score is not a return forecast, and forecasts on different signals or volatilities are not comparable until scaled. The rule does not by itself size positions: portfolio construction still trades forecasts against risk and costs. The source is the book's second edition (McGraw-Hill, 1999); see also Grinold's 1994 paper (`eq-grinold-1994`) and his 1989 fundamental-law paper (`s7-grinold`).

**See also.** signal combination; mean–variance optimisation (which consumes the forecasts).

**Further reading.** Chap. 11 (advanced forecasting, more assets and signals) and chap. 12 (information analysis) of the same book.

# 9 Option pricing, volatility models and hedging

## 9.1 Forwards, futures and options, and the cost of carry

**Definition 9.1 Derivative contracts: spot, forward, futures and options; payoffs, rights and obligations, exercise styles, futures specification and convergence**

**Statement.** A derivative "involves two parties agreeing to a future transaction"; its value depends on the values of underlying variables, very often traded asset prices [s7-hulloptions, chap. 1, "Introduction", p. 23]. Hull's basic contracts:

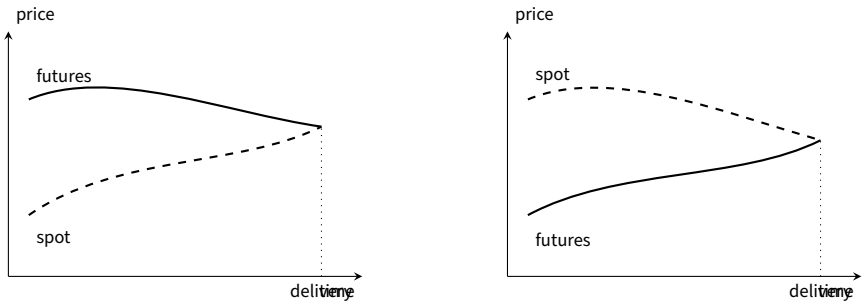
- a *spot contract* buys or sells an asset almost immediately; a *forward contract* buys or sells it at a certain future time for a certain price, the delivery price $K$, and trades over the counter. The long side agrees to buy and the short side to sell (sec. 1.3, p. 28). At maturity the payoff per unit is $S_T - K$ to the long and $K - S_T$ to the short, where $S_T$ is the spot price; because it costs nothing to enter a forward, the payoff is also the total gain or loss (p. 29);
- a *futures contract* is the same kind of agreement, standardised and normally traded on an exchange, with the exchange clearing house standing between the parties (sec. 1.4, p. 30). Most positions are closed out by the opposite trade before delivery, but "it is the possibility of final delivery that ties the futures price to the spot price" (sec. 2.1, pp. 47–48);
- an *option*: a *call* "gives the holder the right to buy" and a *put* "the right to sell" the underlying asset by a certain date for the strike price. *American* options can be exercised at any time up to expiration, *European* options only on the expiration date. The holder has a right, not an obligation, and pays for it; forward and futures holders are obliged to buy or sell and pay nothing to enter apart from margin (sec. 1.5, p. 31).

The exchange specifies a futures contract's asset (with acceptable grades), contract size, delivery place and period, last trading day and price quotation; when alternatives are allowed, the short side usually chooses and files a notice of intention to deliver (sec. 2.2, pp. 48–50). For most contracts the exchange also sets daily price limits, after which trading normally stops for the day, though the exchange can change them; it also sets position limits for speculators (p. 50).

**Notation.** $S_T$ spot price at maturity; $K$ delivery (forward) or strike (option) price; for a European call the payoff is $\max(S_T - K, 0)$ and for a put $\max(K - S_T, 0)$ (interpretation of Hull's definitions; the writer's payoffs are the negatives). *Profit* is payoff less the premium paid, with the premium's time value ignored, as Hull does in his footnote (p. 32, n. 5).

**Conditions.** *Convergence.* As the delivery period approaches, the futures price converges to the spot price of the deliverable asset (sec. 2.3, pp. 50–51; Figure 9.1). If futures traded above spot during delivery, shorting futures, buying the asset and delivering would profit; if below, buyers who want the asset would go long and take delivery. Interpretation: the argument needs the asset delivered to be the asset priced as spot, at the delivery location and grade, and delivery to be possible; cash-settled contracts converge through their final settlement rule instead, and when the short side chooses grade, place or timing, the futures price tracks the alternative it is expected to choose; Hull treats the timing choice in sec. 5.13 (p. 144).

*Forward and futures* differ in standardisation, a single date against a range, settlement at the end against daily settlement, and credit risk (Table 2.3, p. 65); a forward gain is realised at maturity, a futures gain day by day (pp. 64–65).




(a) futures above spot (b) futures below spot

Figure 9.1: Convergence of the futures price to the spot price of the deliverable asset as delivery approaches; schematic after Hull, Figure 2.1, p. 51.

**Practice.** Hull's examples: a long six-month forward on £1 million at 1.2230 is worth +$77,000 if spot ends at 1.3000 and −$23,000 at 1.2000 (p. 29). One Apple December call contract on 100 shares, strike $340, bought for $2,030, pays $100 \times (400 - 340) = \$6{,}000$ if exercised at $400, a profit of $3,970; one September put written at $12.70 ($1,270 received), strike $290, exercised at $250, costs $4,000, a loss of $2,730 (p. 32). Daily settlement and margin are in futures margin and settlement; pricing in the cost of carry.

**Pitfalls.** A call gives the right to buy and a put the right to sell (Hull, p. 31). That the "derivatives market is much larger than the stock market in terms of underlying assets" is Hull's statement (p. 23), but the size is a notional principal, not a value: Hull gives $558.5 trillion of OTC notional against a gross market value of about $11.6 trillion in December 2019 (pp. 27–28). Daily price limits, delivery terms and grades are contract-specific exchange rules, not universal features; Hull's examples are CME Group and ICE contracts as of 2021, and Hull notes that whether limits are on balance good is "controversial" (p. 50).

**See also.** cost-of-carry forward price; futures margin and settlement; synthetic short through options; OTC clearing and collateral.

**Result 9.2 Cost-of-carry forward price: no-arbitrage forward prices with income, yield and storage, the value of an existing forward, and the limits for consumption assets**

**Statement.** For an *investment asset*, one held solely for investment by at least some traders, with continuously compounded risk-free rate $r$ to maturity $T$ [s7-hulloptions, chap. 5, "Determination of Forward and Futures Prices", secs. 5.1–5.6, pp. 124–132]:

$$F_0 = S_0 e^{rT} \text{ (no income)}, \qquad F_0 = (S_0 - I)e^{rT} \text{ (income with present value } I), \qquad F_0 = S_0 e^{(r-q)T} \text{ (yield } q).$$

If $F_0$ is above the bound, buying the asset with borrowed money and selling forward locks in a profit; if below, shorting the asset (or, if shorting is impossible, selling it from holdings) and buying forward does (pp. 127–131). A long forward with delivery price $K$ agreed earlier is worth $f = (F_0 - K)e^{-rT}$, for investment and consumption assets alike (sec. 5.7, p. 133). Special cases: a stock index with dividend yield $q$ (index arbitrage, sec. 5.9, pp. 135–137); a currency, with the foreign risk-free rate $r_f$ as the yield, $F_0 = S_0 e^{(r-r_f)T}$, interest rate parity (sec. 5.10, pp. 137–140);

a commodity that is an investment asset, with storage cost net of income $U$ in present value, $F_0 = (S_0 + U)e^{rT}$, or $u$ as a proportion of price, $F_0 = S_0 e^{(r+u)T}$ (sec. 5.11, pp. 141–142).

The cost of carry $c$, storage plus financing less income, gives $F_0 = S_0 e^{cT}$ for an investment asset (sec. 5.12, pp. 143–144).

**Notation.** $S_0$ spot price; $F_0$ forward (or futures) price; $r$ the $T$-year zero rate, continuously compounded; $q$ average yield over $T$; $U$, $u$ storage cost net of income; $y$ convenience yield; $c$ cost of carry ($r$ for a non-dividend stock, $r - q$ for an index, $r - r_f$ for a currency, $r - q + u$ for a commodity with income and storage, p. 144). Here $S_0$ and $F_0$ for a currency are dollars per unit of foreign currency (p. 138).

**Conditions.** The derivation assumes that *some* participants, such as large dealers, trade without transaction costs, pay the same tax rate on all net trading profits, borrow and lend at the same risk-free rate, and take arbitrage opportunities as they arise (sec. 5.3, p. 126). For a *consumption asset*, held to be used rather than as an investment, the second arbitrage fails: owners will not sell the commodity to buy futures, so only $F_0 \le (S_0 + U)e^{rT}$ or $F_0 \le S_0 e^{(r+u)T}$ holds. The convenience yield $y$ is defined to close the gap, $F_0 e^{yT} = (S_0 + U)e^{rT}$, so that $F_0 = S_0 e^{(c-y)T}$; it is zero for investment assets (pp. 142–144).

Forward and futures prices are equal in theory when the short rate is constant or a known function of time; with random rates, futures exceed forwards when the asset is positively correlated with rates and fall below them when negatively correlated, differences Hull finds small for contracts of a few months (sec. 5.8, p. 135). Index arbitrage failed on 19–20 October 1987, when order delays and program-trading restrictions left December S&P 500 futures up to 18 percent below the index (Business Snapshot 5.4, p. 137).

**Practice.** Hull's examples: $40e^{0.05\times0.25} = 40.50$ (p. 128); $(900 - 40e^{-0.03/3})e^{0.04\times0.75} = 886.60$ (p. 131); $(50 - 2.162)e^{0.08\times10/12} = 51.14$ (p. 132); with 2 percent paid once in six months, $q = 2\ln 1.02 = 3.96$ percent and $F_0 = 25e^{(0.10-0.0396)\times0.5} = 25.77$ (p. 132); $f = (26.28 - 24)e^{-0.05} = 2.17$ (p. 133); $0.7500e^{(0.01-0.03)\times2} = 0.7206$ USD per AUD (p. 139); $(450 + 2e^{-0.07})e^{0.07} = 484.63$ (p. 141). When short sales are impossible the arbitrage against a forward that is too low still works through holders who sell the asset and buy forward (pp. 129 and 131).

**Pitfalls.** For consumption commodities only the upper arbitrage bound holds, and Hull's convenience yield, which "reflects market expectation on future availability" (p. 143), applies to consumption assets only. That a forward's value is close to zero when first entered is Hull's opening sentence (p. 133); he adds that $K$ is set equal to the forward price, so $f = 0$ at inception. In Hull both $S_0$ and $F_0$ are the domestic price of one unit of foreign currency, with $r$ domestic and $r_f$ foreign, and many market quotes run the other way (p. 138). Borrowing constraints, borrow fees and recall, which make shorting costly, are in lending fees and rebates.

**See also.** derivative contracts; arbitrage (carrying costs and bounds in Harris's terms); minimum-variance hedge ratio; futures hedges and basis.

## 9.2 Option pricing and the implied volatility surface

### Model 9.3 The Black–Scholes–Merton model: the hedging argument, the differential equation and formula, the assumptions it needs, Merton's no-early-exercise result and the Greeks

**Statement.** Black and Scholes value an option under "ideal conditions" [s6-blackscholes, "The Valuation Formula", pp. 640–641]:

- a known, constant short rate;
- a stock price following a random walk in continuous time with variance rate proportional to the squared price, so prices at any horizon are lognormal;
- no dividends;
- a European option;
- no transaction costs;
- borrowing any fraction of a security's price at the short rate, and short selling without penalty.

A position long one share and short $1/w_1$ options is then riskless over a short interval if continuously adjusted, so it must earn the short rate. That gives the differential equation and, with the payoff $w(x, t^*) = \max(x - c, 0)$, the unique solution [s6-blackscholes, eqs. (7), (8) and (13), pp. 643–644]:

$$w_2 = rw - rxw_1 - \tfrac{1}{2}v^2x^2w_{11}, \qquad w(x,t) = xN(d_1) - ce^{r(t-t^*)}N(d_2), \quad d_{1,2} = \frac{\ln(x/c) + (r \pm \tfrac{1}{2}v^2)(t^* - t)}{v\sqrt{t^* - t}}.$$

The stock's expected return does not appear (p. 644). In Hull's notation, for a stock with continuous yield $q$, $c = S_0e^{-qT}N(d_1) - Ke^{-rT}N(d_2)$ and $p = Ke^{-rT}N(-d_2) - S_0e^{-qT}N(-d_1)$ [s7-hulloptions, chap. 15, eqs. (15.20)–(15.21), p. 352, for $q = 0$; the yield version through chap. 17 and Table 19.6, p. 439].

*Merton's bound and early exercise.* Without payouts to the stock over the option's life, or with protection against them, a European call satisfies $f(S, \tau; E) \ge \max[0, S - EP(\tau)]$, with $P(\tau)$ the price of a riskless discount bond (Theorem 1). Then "an American warrant will never be exercised prior to expiration, and hence, it has the same value as a European warrant" (Theorem 2) [p2-merton, p. 144].

*Greeks* for a European call on an asset with yield $q$ [s7-hulloptions, chap. 19, Table 19.6, p. 439]:

- delta $e^{-qT}N(d_1)$;
- gamma $N'(d_1)e^{-qT}/(S_0\sigma\sqrt{T})$;
- vega $S_0\sqrt{T}N'(d_1)e^{-qT}$;
- rho $KTe^{-rT}N(d_2)$;
- theta $-S_0N'(d_1)\sigma e^{-qT}/(2\sqrt{T}) + qS_0N(d_1)e^{-qT} - rKe^{-rT}N(d_2)$.

Setting $q$ to an index yield, the foreign rate or $r$ gives index, currency and futures options (p. 439).

**Notation.** Black and Scholes: $x$ stock price, $c$ exercise price, $t^*$ maturity, $v^2$ variance rate, $w_1$, $w_2$, $w_{11}$ partial derivatives. Hull: $S_0$, $K$, $T$, $\sigma$, $q$. Merton: $f$ European and $F$ American warrant values, $\tau$ time to expiry, $E$ exercise price.

**Conditions.** Theory timeless (1973). The riskless hedge needs continuous rebalancing. Black and Scholes argue that a discretely adjusted hedge carries only diversifiable risk (p. 642), and Hull's list adds no riskless arbitrage, perfect divisibility, no taxes and a flat, constant rate. He notes that $\sigma$ and $r$ may be known functions of time, and rates stochastic if the terminal price stays lognormal (sec. 15.6, p. 348). Merton's Theorems 1–2 need no payouts and positive interest rates; with a large dividend yield the European call can fall below intrinsic value, and early exercise of an American call can then have value (for example, $S = 150$, $K = 100$, $q = 10$ percent, $r = 2$ percent, $\sigma = 20$ percent, $T = 2$ gives $c = 29.98 < 50$).

**Practice.** Numerical checks:

- Hull's Example 15.6 ($S_0 = 42$, $K = 40$, $r = 10$ percent, $\sigma = 20$ percent, $T = 0.5$) gives $c = 4.76$ and $p = 0.81$ (p. 356);
- the formula equals the discounted risk-neutral expectation computed by quadrature;
- it satisfies the differential equation by finite differences;
- each Greek of Table 19.6 matches its finite difference;
- $\Theta + (r - q)S\Delta + \tfrac{1}{2}\sigma^2S^2\Gamma = rc$, which is the differential equation restated in Greeks;
- Merton's bound holds at several volatilities, and implied volatility is recovered by bisection.

Delta and delta–vega hedges can be compared on profit and loss. That needs the discrete-hedging error, costs and smile moves, which these checks do not cover.

**Pitfalls.** The constant-volatility lognormal model is contradicted by the smile itself (implied volatility surface). The Greeks are partial derivatives at fixed $\sigma$, so a vega hedge under one implied volatility per option is a model-dependent hedge.

**See also.** derivative contracts; cost-of-carry forward price (the yield $q$); implied volatility surface.

**Further reading.** Merton's later sections (dividends, stochastic rates, the American put) and Hull's chaps. 19 and 21 on hedging practice and the Greeks of American options.

### Definition 9.4 Implied volatility, the smile and an arbitrage-free surface: total implied variance, the calendar and butterfly conditions, raw SVI and the wing bound

**Statement.** The *Black–Scholes implied volatility* $\sigma_{BS}(k,t)$ of a European option is the volatility that, put into the Black–Scholes formula, reproduces its price, with $k = \ln(K/F_t)$ the log-moneyness against the forward. Gatheral and Jacquier work with the *total implied variance* $w(k,t) = \sigma_{BS}^2(k,t)\,t$. They call the map $(k,t) \mapsto w$ the volatility surface and a fixed-$t$ curve a slice [exp-gatheral-jacquier-svi, "Notations", p. 2]. A surface is free of *static arbitrage* if and only if it is free of calendar-spread arbitrage and each slice is free of butterfly arbitrage (Definition 2.1, p. 3):

- *Calendar spreads.* With dividends proportional to the stock price, the surface is free of calendar-spread arbitrage if and only if $\partial_t w(k,t) \geq 0$ for all $k$ and $t > 0$: total variance at fixed log-moneyness cannot fall with maturity (Lemma 2.1, p. 3).
- *Butterflies.* A slice is free of butterfly arbitrage, a non-negative implied density, if and only if

$$g(k) = \Big(1 - \frac{k\,w'(k)}{2w(k)}\Big)^2 - \frac{w'(k)^2}{4}\Big(\frac{1}{w(k)} + \frac{1}{4}\Big) + \frac{w''(k)}{2} \geq 0 \text{ for all } k, \qquad \lim_{k\to+\infty} d_+(k) = -\infty$$

(eq. (2.1) and Lemma 2.2, p. 4).

*Raw SVI* parameterises a slice as $w(k) = a + b\{\rho(k-m) + \sqrt{(k-m)^2+\sigma^2}\}$, with $b \geq 0$, $|\rho| < 1$, $\sigma > 0$ and $a + b\sigma\sqrt{1-\rho^2} \geq 0$ so that $w \geq 0$ (eq. (3.1), p. 5). It is convex in $k$, but its parameters alone do not exclude butterfly arbitrage. Their Example 3.1, Vogt's parameters $(a,b,m,\rho,\sigma) = (-0.0410, 0.1331, 0.3586, 0.3060, 0.4153)$ at $t = 1$, gives a negative density (p. 10). Their SSVI family (sec. 4) gives sufficient conditions for a whole surface free of static arbitrage.

*Wings.* Lee's moment formula bounds the tails model-free: the large-strike limsup of $\sigma_{BS}^2 T/|x|$ is a number $\beta_R \in [0,2]$, with $\frac{1}{2\beta_R} + \frac{\beta_R}{8} - \frac{1}{2} = \sup\{p : E\,S_T^{1+p} < \infty\}$, and a reciprocal formula holds for small strikes [exp-lee-wings, sec. 1, eq. (1.1), p. 2; Theorems 3.2 and 3.4]. So total implied variance can grow at most linearly in $|k|$, with slope at most 2.

**Notation.** $F_t$ the forward; $k$ log-moneyness; $w$ total implied variance; $d_\pm(k) = -k/\sqrt{w} \pm \sqrt{w}/2$; raw SVI parameters $a, b, \rho, m, \sigma$ (here $\sigma$ is a shape parameter, not a volatility).

**Conditions.** The conditions assume a twice-differentiable surface. Observed quotes form a finite grid with bid–ask spreads, so "arbitrage-free" applies to the fitted surface, and the spread may admit several such fits. The calendar condition needs proportional dividends (Lemma 2.1).

**Practice.** Numerical checks. The quantity $g/\sqrt{2\pi w}\,e^{-d_-^2/2}$ matches $K\,\partial^2 C/\partial K^2$ of the undiscounted call by finite differences. An illustrative SVI slice has $g \geq 0$ on $|k| \leq 1.5$, while Vogt's slice has $\min g = -0.033$. Two slices with the same shape and higher $a$ at the longer maturity satisfy the calendar condition.

*Source discrepancy.* The proof of Lemma 2.2 (p. 5) writes the density as $p(k) = \partial^2 C/\partial K^2$ at $K = F_t e^k$ and then as $g(k)/\sqrt{2\pi w(k)}$ times $\exp(-d_-(k)^2/2)$. The second is the density per unit log-moneyness, $K$ times the first. The sign condition and the lemma are unaffected.

*Earlier versions:* arXiv versions v1 and v2 carry the same two expressions in the same proof, there numbered Lemma 4.1 (PDF p. 8 of each), so the slip predates the 2024 revision.

**Pitfalls.** Locators refer to the arXiv preprint as revised on 27 November 2024, not the 2014 journal version. No surface is fitted to market quotes here.

**See also.** Black–Scholes–Merton; the VIX and variance swaps (variance from a strip of options).

**Further reading.** s6-surface (local and stochastic volatility, surface parameterisations).

### Model 9.5 Heston's stochastic-volatility model: the closed-form price through characteristic functions, and Carr and Madan's FFT for many strikes

**Statement.** Heston lets the variance follow a square-root process correlated with the stock: $dS = \mu S\,dt + \sqrt{v}S\,dz_1$, $dv = \kappa(\theta - v)\,dt + \sigma\sqrt{v}\,dz_2$, $dz_1\,dz_2 = \rho\,dt$. With $x = \ln S$, a call is $C = SP_1 - Ke^{-r\tau}P_2$, where each $P_j$ is the probability, under a different measure, that the option ends in the money. Their characteristic functions are $f_j = \exp[C(\tau;\phi) + D(\tau;\phi)v + i\phi x]$ with $C$ and $D$ in closed form ($u_1 = \frac{1}{2}$, $u_2 = -\frac{1}{2}$, $a = \kappa\theta$, $b_1 = \kappa + \lambda - \rho\sigma$, $b_2 = \kappa + \lambda$), and $P_j = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \mathrm{Re}[e^{-i\phi \ln K} f_j/(i\phi)]\,d\phi$ (eqs. (10) and (14)–(18), p. 331) [s6-heston, *RFS* 6(2), 1993, eqs. (10)–(18), pp. 330–331].

Carr and Madan price a whole strike grid at once: they Fourier-transform the call price in log strike after multiplying by $e^{\alpha k}$, because the plain call price is not square-integrable, and invert with the FFT; a sufficient condition for the damped price to be integrable is that $E[S_T^{\alpha+1}]$ is finite, and they find "one quarter of this upper bound serves as a good choice for $\alpha$" (eqs. (5)–(7), p. 64) [s6-carrmadan, *Journal of Computational Finance* 2(4), 1999, author copy, pp. 63–64].

**Notation.** $v$ instantaneous variance; $\kappa$ speed of mean reversion; $\theta$ long-run variance; $\sigma$ volatility of variance; $\rho$ correlation; $\lambda$ the variance risk-premium parameter (zero in the checks); $\alpha$ the damping coefficient; $k = \ln K$.

**Conditions.** Heston's form of $g$ and $\log(1 - ge^{d\tau})$ can cross a branch cut of the complex logarithm at long maturities, which is the subject of vo-lord-kahl-2010; the checks here use a half-year maturity. The FFT needs a damping $\alpha$ for which the moment $E[S_T^{\alpha+1}]$ exists; in Heston that moment can explode in finite time for some parameters.

**Practice.** With a negligible volatility of variance and $v_0 = \theta$ the formula equals Black–Scholes at $\sqrt{\theta}$ to $10^{-3}$; with $\kappa = 1.5, \theta = v_0 = 0.04, \sigma = 0.5, \rho = -0.7$ and $T = 0.5$ it matches a 200,000-path Monte Carlo at strikes 90, 100 and 110 within three standard errors (5.705 against 5.708 at the money); the FFT with $\alpha = 1.5$ reproduces Black–Scholes within 0.001 and the Heston integral within 0.002. Negative $\rho$ lowers out-of-the-money call prices relative to Black–Scholes at the same variance (1.41 against 2.47 at strike 110), the skew the model exists to produce.

**Pitfalls.** The checks use illustrative parameters; the model is not calibrated to quotes.

**See also.** Black–Scholes–Merton; implied volatility surface; Merton's jump-diffusion.

**Further reading.** vo-lord-kahl-2010; audit-duffie-pan-singleton; s7-derman.

### Model 9.6 Merton's jump-diffusion option price: a Poisson mixture of Black–Scholes prices, and why jump risk cannot be hedged with the stock alone

**Statement.** Merton adds to the diffusion a Poisson jump with intensity $\lambda$ that multiplies the price by $Y$. With jumps "there is no way to construct a riskless portfolio of stock and options", so the Black–Scholes argument fails; if the jump risk is diversifiable (uncorrelated with the market), the

option price satisfies an equation in which only the interest rate appears, and "one cannot 'act as if' the jump component was not there" (secs. II–III, printed pp. 12–15). The solution is $F(S,\tau) = \sum_n \frac{e^{-\lambda\tau}(\lambda\tau)^n}{n!}\,\mathcal{E}_n[\,W(SX_n e^{-\lambda k\tau},\tau;E,\sigma^2,r)\,]$, with $W$ the Black–Scholes price and $X_n$ the product of $n$ jumps (eq. (16), printed p. 16).

With lognormal jumps of log-variance $\delta^2$ and $E[Y] = 1$, $F = \sum_n \frac{e^{-\lambda\tau}(\lambda\tau)^n}{n!} W(S,\tau;E,\sigma^2 + n\delta^2/\tau, r)$ (eqs. (18)–(19), printed p. 17); an option on a stock with jumps is worth more, other things equal, $\partial F/\partial\lambda > 0$ at $\lambda = 0$ (sec. IV, printed p. 21) [wide-merton-jumps, Sloan Working Paper 787-75, 1975, printed pp. 12–21].

**Notation.** $\lambda$ jump intensity; $Y$ the jump multiplier, $k = E[Y] - 1$; $\delta^2$ the variance of $\ln Y$; $W(S,\tau;E,\sigma^2,r)$ Black–Scholes with strike $E$.

**Conditions.** The pricing equation rests on the jump risk being diversifiable (Merton's two derivations, printed pp. 13–20); with priced jump risk it does not hold. Merton notes he knows of no empirical test of that proposition (printed p. 20).

**Practice.** With $\lambda = 0$ the series is Black–Scholes; $\partial F/\partial\lambda > 0$ at $\lambda = 0$ at strikes 80, 100 and 120; with $\lambda = 1$, $\delta^2 = 0.02$, $\sigma^2 = 0.04$ and $T = 0.5$ it matches a Monte Carlo within three standard errors (7.214 against 7.212 at the money).

**Pitfalls.** Locators refer to the 1975 MIT working paper, whose equation numbers may differ from the 1976 *Journal of Financial Economics* article.

**See also.** Heston; Black–Scholes–Merton.

**Further reading.** `wide-bates-svj` (stochastic volatility with jumps).

### Model 9.7 The SABR model and its implied-volatility expansion, in single- and multiple-curve form

**Statement.** SABR lets a forward follow $dF = VF^\beta\,dW$ with lognormal volatility $dV = \nu V\,dZ$, $V(0) = \alpha$, $dW\,dZ = \rho\,dt$, $0 \le \beta \le 1$. Hagan et al.'s expansion gives the Black implied volatility as $\sigma(K) = \frac{\nu \ln(F/K)}{x(z)}\,\frac{A}{B}$, with $z = \frac{\nu}{\alpha}(FK)^{(1-\beta)/2}\ln\frac{F}{K}$, $x(z) = \ln\frac{\sqrt{1-2\rho z+z^2}+z-\rho}{1-\rho}$, a time correction $A = 1 + [\,\frac{(1-\beta)^2\alpha^2}{24(FK)^{1-\beta}} + \frac{\alpha\beta\nu\rho}{4(FK)^{(1-\beta)/2}} + \nu^2\frac{2-3\rho^2}{24}\,]\tau$ and $B = (FK)^{(1-\beta)/2}[1 + \frac{(1-\beta)^2}{24}\ln^2\frac{F}{K} + \frac{(1-\beta)^4}{1920}\ln^4\frac{F}{K}]$. In the multiple-curve version the forward is replaced by the FRA rate and the measure by the $T_k$-forward measure of the discounting curve; "the SABR volatility formula remains unchanged" [gap-rates-multicurve-sabr, arXiv 1103.2567, version of 28 March 2012, sec. 4.5, Table 5, p. 18].

**Notation.** $F$ the forward or FRA rate; $\alpha$ initial volatility; $\beta$ the CEV exponent; $\rho$ correlation; $\nu$ volatility of volatility; $\tau$ time to expiry.

**Conditions.** Formula of 2002 as restated in 2012. It is an asymptotic expansion, accurate for small $\tau\nu^2$ and not too far from the money; at low strikes it can imply a negative density, which the arbitrage-free version (`rt-arbitrage-free-sabr-2014`) addresses.

**Practice.** The formula is continuous at the money and flat at $\alpha$ when $\nu = 0$, $\beta = 1$; with $F = 3\%$, $\beta = 0.5$, $\rho = -0.3$, $\nu = 0.4$ and one year to expiry, it is within 0.2 volatility points of a Monte Carlo of the dynamics at strikes 2.5, 3 and 3.5 percent (22.53 against 22.37 percent at 2.5). That is one parameter set, not an accuracy study.

**Pitfalls.** The formula is from Bianchetti and Carlicchi's restatement of Hagan et al. (`wide-hagan-sabr`), compared with a simulation.

**See also.** multi-curve FRA and swap valuation; implied volatility surface.

**Further reading.** `wide-hagan-sabr`; `rt-arbitrage-free-sabr-2014`; `s6-brigo`.

# 10 Volatility trading, option premia and implied information

## 10.1 VIX, variance swaps and dividend swaps

**Definition 10.1 The VIX and its futures curve, variance swaps (strike, vega notional, variance units and payoff) and dividend swaps, valuation and settlement**

**Statement.** *VIX.* Gliner describes the VIX as derived from S&P 500 options "one month out", negatively correlated with equity returns, able to move "from the mid-teens to 30 in a matter of days" in sell-offs; VIX futures "typically trade in contango" and move to backwardation when spot spikes; VIX options are in effect options on the volatility of volatility; and other indices include V2X, VXN, VXD, RVX, GVZ and OVX [s7-gliner, chap. 8, sec. "The Volatility Index (VIX)", PDF pp. 222–225].

*Variance swaps* give "pure variance exposure" without delta hedging and without the interest-rate and dividend exposure of options; they are quoted by strike, vega notional, variance units and maturity (pp. 225–226). Variance units = vega notional$/(2K)$ and payoff = $(\sigma_R^2 - K^2)\times$ variance units: with $K = 24$, vega notional \$1,000,000 and realised volatility 29, there are 20,833 units and a payoff of \$5,520,833 (p. 226).

*Dividend swaps* let one take a view that an index's dividends will be higher or lower than a fixed amount, OTC for the S&P 500 and listed for the Euro Stoxx 50 (sec. "Dividend Swaps", p. 227).

*How the VIX is computed.* Cboe's methodology defines the index as VIX $= 100 \times \sigma$, where $\sigma^2$ is an annualised 30-day variance: for each of a near and a next expiry, $\sigma_j^2 = \frac{2}{T_j}\sum_i \frac{\Delta K_i}{K_i^2} e^{R_j T_j} Q(K_i) - \frac{1}{T_j}\left(\frac{F_j}{K_0} - 1\right)^2$ over out-of-the-money SPX and SPXW option mid-quotes $Q(K_i)$, the two variances are interpolated to a constant 30 days, and "Then take the square root of that value and multiply by 100"; time is measured in minutes, with 43,200 minutes in 30 days and 525,600 in a year [gap-cboe-vix6, sec. 3.1, pp. 4–6, and app. 3, pp. 9–14].

So the VIX is quoted in volatility points (percent per year), not variance: VIX 20 means $\sigma = 0.20$ and an annualised variance of $0.20^2 = 0.04$, or $0.04 \times 30/365 \approx 0.0033$ over the 30 days. The index's first value date is January 1990 and the current methodology dates from 2003 (launch 22 September 2003), when the 1993 index on S&P 100 at-the-money implied volatility was replaced (pp. 3 and 8).

**Notation.** $K$ strike in volatility points; $\sigma_R$ realised volatility over the life, annualised from daily log returns, in points; vega notional is the approximate P&L for a one-point move in realised volatility near the strike.

**Conditions.** Gliner's examples as of 2013. *The strike is fixed at inception from option prices.* Gliner writes that variance swaps "are priced with realized volatility, which gives the buyer slightly more edge" (p. 225); the payoff depends on realised variance, but the strike is set at the start from the market's implied variance and is usually above subsequent realised variance (the variance risk premium), so the long typically pays: with strike 24 and realised 20 the long loses \$3,666,667. The payoff is convex in volatility, linear in variance.

**Practice.** *Daily moves.* With $\sqrt{252} = 15.87$, a volatility of 16 corresponds to a daily standard deviation of about 1 percent and 24 to about 1.5 percent (Gliner, p. 226; checks, section 13); that is a standard deviation, not the "expected daily move", whose mean absolute size under normality is about 0.8 of it. *Dividend swap example* (p. 227): with the 2014 S&P 500 dividend quoted at \$31 against an index of 1,350 (an implied yield of 2.3 percent), a buyer of 500,000 units gains $(36 - 31) \times 500{,}000 = \$2.5$ million if the market moves to \$36; Gliner's "baskets of 100,000" sits uneasily with that arithmetic, which counts one index unit per basket.

**Pitfalls.** Gliner's description, that the VIX "is extrapolated by using various strikes of implied volatility" (p. 222), does not match Cboe's methodology above, which aggregates option *prices* across strikes into a variance, the formula of the variance-swap replication literature (it cites Demeterfi, Derman, Kamal and Zou), and takes its square root; implied volatilities are not inputs. Squaring a VIX level gives a variance; adding or averaging VIX levels across maturities is not the same as averaging variances. The rule "daily move $\approx$ VIX$/\sqrt{252}$" mixes Cboe's calendar-time annualisation with a 252-trading-day year and treats all variance as falling on trading days; it is an approximation. Gliner's "V2X Index—Euro Stoxx 500" should read Euro Stoxx 50.

VIX options give asymmetric hedges, but VIX futures and options settle to a Special Opening Quotation computed on expiration morning from a single SPX or SPXW expiry 30 calendar days later, without the near/next interpolation of the spot index (sec. 3.4, p. 7), so an option's payoff is not tied to the spot VIX on the day it is bought.

**See also.** equity indices; AUD volatility and the VIX; volatility components.

**Further reading.** [gap-dividend-strips] on dividend strips, [wide-demeterfi-varswaps] on replicating variance swaps and [gap-vix-premium] on the VIX premium.

# 11 Multi-asset and systematic macro

## 11.1 Global macro: strategy, record, risk and practitioners

**Definition 11.1 Global macro: a top-down, unconstrained strategy; discretionary, systematic, high-frequency and CTA approaches; directional, relative-value and thematic trades; the four product groups and multi-asset expression of a view**

**Statement.** Gliner calls global macro "often considered the most flexible and opportunistic hedge fund strategy", because of the range of products and markets it trades, and gives its aim as preserving capital "using stringent risk management to limit drawdowns" [s7-gliner, chap. 1, opening, PDF p. 29]. He distinguishes four approaches (sec. "Types of Global Macro Strategies", PDF pp. 30–33):

- *discretionary*: the trader's experience and top-down analysis of industries, sectors, countries and the macroeconomy, expressed through directional and relative-value trades (pp. 30–31);
- *systematic*: model-driven positions without discretion; over several years or more "systematic funds can produce more consistent returns than discretionary strategies; however, in periods of high volatility, they tend to underperform discretionary macro, as they did in 2008"; trading many liquid markets lets them scale (pp. 31–32). He advises allocating to both, because discretionary macro "is negatively correlated during periods of stress" while systematic macro offers predictable, back-tested drawdowns (p. 32);
- *high-frequency*: holding periods from milliseconds to a few hours, not as large or scalable (p. 32);
- *CTAs*: mostly automated trend followers, whose position sizing "originated with the Turtle Traders"; they perform well over several years but suffer large drawdowns (p. 33).

*Trades.* A directional trade bets on an asset's direction (long copper on a bullish commodity view); a relative-value trade pairs assets, as in short Italian BTPs against long German Bobls in the euro crisis (p. 31). Thematic trades are long-horizon views, which Gliner sizes smaller and treats separately (chap. 2, sec. "Thematic Trade", PDF p. 60). *Product groups.* Gliner's building blocks are fixed income, FX, equities and commodities (chap. 4, sec. "The Four Product Groups", PDF pp. 92–97), which he calls among the most liquid markets; he calls fixed income "the most important" of them (p. 95).

**Notation.** *Discretionary* and *systematic* describe how positions are decided, not what is traded; *CTA* is a U.S. regulatory category (commodity trading advisor) whose members mostly follow trends.

**Conditions.** The descriptions are Gliner's, as of 2013–2014, and are characterisations, not tested results: the 2008 comparison is a statement without data in the passage, and "negatively correlated during periods of stress" is not quantified there (the record is in global macro in crises).

**Practice.** *Expressing one view several ways*: a growth view through duration, equity beta, pro-cyclical currencies and cyclical commodities; an inflation view through commodities, breakevens and TIPS; a crisis view through bonds, safe-haven currencies, volatility and gold; a policy-divergence view through rate differentials and curve spreads. This list is an interpretive synthesis, consistent in spirit with Gliner's chapter 4 on relationships between assets (pp. 98–102); the asset responses it assumes are regime-dependent (regime map).

*Liquidity.* The markets Gliner names (G10 and major EM currencies, government bond futures, equity index futures, major commodity futures) are why he calls macro "one of the most, if not the most, liquid strategies" (chap. 1, PDF p. 36). Liquidity does not guarantee scale without market impact, fast exit in stress or better redemption terms: impact grows with size even in liquid futures (square-root law), and liquidity can vanish in stress.

**Pitfalls.** *Unsourced explanations.* Explaining the strategy's existence by policy moving in jumps while markets are continuous, by sticky institutional capital or by under-arbitraged cross-asset links, and describing macro as exploiting structural inefficiency rather than informational advantage, or as trading economic regimes rather than securities, are hypotheses, not findings of Gliner or the other sources. *Sizing thematic trades.* Sizing thematic trades smaller because they take years and may carry negatively is a discretionary choice; Gliner discusses them as a separate category with their own sizing (p. 60), and the claim is advice, not a result.

**See also.** global macro in crises; macro position sizing and risk; practitioner frameworks; regime map; fixed-income universe.

**Further reading.** Carry across asset classes [alpha-carry] and the section's other readings on trend and value test what Gliner describes.

**Evidence 11.2 Global macro's return record and its behaviour in crises: Gliner's index comparisons, conditional correlation with equities, and what the cited figures do and do not show**

**Statement.** Gliner reports that the Barclays Global Macro Index "has achieved annualized returns of 10 percent from 2002 to 2012 compared to the S&P 500, which has been 2 percent" with lower annualised volatility, and infers "a higher Sharpe ratio" than other hedge fund strategies; Figure 1.1 plots the Dow Jones Credit Suisse Global Macro Hedge Fund Index against the S&P 500 from January 1995 to September 2013 [s7-gliner, chap. 1, sec. "Return Profile and Allocations", PDF p. 34]. He states that global macro "has shown a low correlation to S&P 500 returns, particularly in periods of market stress", with "a negative correlation during market collapses" because many macro traders go short in bear markets (Figure 1.2, pp. 34–35), and that its return profile and liquidity make it "the most popular hedge fund allocation by pension funds" (Figure 1.3, Barclays Prime Services, 2009–2012, pp. 36–38).

**Notation.** Hedge fund indices are averages of reporting funds' net returns; their Sharpe ratios use reported monthly returns.

**Conditions.** These are index statistics over one window, quoted without standard errors, and subject to the usual hedge-fund-index biases (self-reporting, survivorship and backfill), which Gliner does not discuss. The S&P 500's 2 percent over 2002–2012 depends on whether dividends are included, which the passage does not say. The higher Sharpe ratio than other hedge fund strategies is asserted, not shown.

**Practice.** Macro's record is often summarised as lower volatility than equities, comparable or higher long-run returns, low or negative correlation in crises, and a higher Sharpe ratio, with crises described as times when equity managers are trapped long, credit liquidity vanishes and correlations converge toward 1, while macro funds can cut gross exposure, short directly, and trade rates, FX or volatility. Those mechanisms are plausible but are an interpretive generalisation; Gliner's evidence is the index comparison and the conditional-correlation chart above, and his chapter 4 shows correlations rising around Lehman (Figure 4.5, PDF pp. 100–101).

**Pitfalls.** Do not extend a 2002–2012 or 1995–2013 record to later years: the guide's section readings and later index data are needed. *Own bounded test, 2022–2026.* The protocol predates the data acquisition. The data choices below were fixed before computing. It uses the Barclay Global Macro Index's monthly net returns as displayed on BarclayHedge's public page on 17 October 2026, excluding the months it marks as estimated, against French's Mkt-RF, January 2022 to July 2026 (55 months).

- *All months.* The index earned 0.68 percent a month at 5.6 percent annual volatility, an annualised Sharpe ratio on excess returns of 0.76 (monthly 0.22, Lo standard error 0.14).
- *Down months* (24): it averaged −0.39 percent ($t = -1.5$), with correlation +0.29 (95 percent interval −0.13 to 0.62) and beta +0.13.
- *Up months* (31): it averaged +1.51 percent.
- *The ten worst benchmark months*, which averaged −6.5 percent: the index averaged −0.79 percent.

By the pre-set rule (down-month mean positive with $t > 2$), crisis protection is not supported in this window. The low betas say the index fell much less than equities, not that it rose.

*The protocol's window, January 2014 to July 2026.* The public page's Excel download covers the whole history from 1997. Run under the protocol and the data choices above, with both date bounds enforced and the two months marked as estimated (August and September 2026) excluded, it gives for 151 months:

- *All months.* The index earned 0.46 percent a month at 5.1 percent annual volatility, an annualised Sharpe ratio on excess returns of 0.74 (monthly 0.21, Lo standard error 0.08).
- *Down months* (53): it averaged −0.52 percent ($t = -3.0$), with correlation +0.42 (95 percent interval +0.17 to +0.62) and beta +0.17 (+0.07 to +0.28).
- *Up months* (98): it averaged +1.00 percent.
- *The ten worst benchmark months*, which averaged −8.6 percent: the index averaged −1.22 percent.

The verdict is unchanged, crisis protection not supported, but the longer window strengthens it. The down-month loss is now distinguishable from zero, and the positive co-movement is significant, where the 55-month interval included zero. The 2022–2026 figures above are the recent-period comparison; the workbook reproduces them exactly.

*Limits of the index.* BarclayHedge's page says the index "is recalculated and updated real-time on this page as soon as the monthly returns for the underlying funds are recorded". Only funds that report net returns are included, and the number of funds "will continue to increase until all funds categorized within the sector have reported". So recent months are provisional, and earlier values can change as late reporters arrive. The page does not state the weighting, how newly added funds' histories are treated (backfill) or how closed funds are kept (survivorship), and no methodology document is public. The test also gives equal treatment to all down months and uses one index of self-reporting funds.

Negative correlation in crises is a conditional statistic on a few episodes, sensitive to how crises are chosen. Gliner's "most popular hedge fund allocation by pension funds" is his reading of one prime broker's survey for 2009–2012, not a market-wide statistic.

**See also.** global macro; macro position sizing and risk; relative performance evaluation.

**Further reading.** Crisis performance of trend and carry after 2013, for example [wide-harvey-worst-times].

### Method 11.3 Macro position sizing and risk limits: Turtle-style volatility units, risk/reward targets, gap risk, VaR utilisation tied to year-to-date P&L, stress tests, performance ratios and portfolio correlation, as Gliner describes them

**Statement.** Gliner sizes positions by volatility, following the Turtle Traders: $N$ is "the 20-day exponential moving average of the True Range", with true range $\max(H-L,\ H-C_{-1},\ C_{-1}-L)$, and position size is set so that a move of $N$ (or $2N$, for the stop) costs a fixed fraction of capital [s7-gliner, chap. 2, secs. "Unit Size" and "Volatility Adjusting Position Size", PDF pp. 50–53]. He asks for at least 1:3 risk to reward, for example risking 25 bp to make 75 bp, with targets set before entry and reasons logged; a trader right 40 percent of the time then earns $0.4 \times 75 - 0.6 \times 25 = 15$ bp a trade (sec. "Risk/Reward", pp. 54–55). Gap risk, a price change with no trading in between, is in his account one of the biggest risks in sizing (sec. "Gap Risk", p. 57, with Cyprus, 16 March 2013).

*Risk utilisation* (pp. 66–68): with a maximum 95 percent VaR of 5 percent, "if you have 50 percent conviction, that doesn't mean you should run 50 percent of your risk utilization"; a limit predetermined at the start of the year rises to 6 percent if the portfolio is up 5 percent and 7 percent if up 10 percent, falls to 4 percent if down 2.5 percent and 3 percent if down 5 percent, and the year starts again at 0 percent return.

*Stress tests* (p. 69): sort the last 500 days' P&L, average the five worst, and in calm periods also replay extreme days (his Table 2.1 includes the October 1987 crash, the 2011 Japanese earthquake and the U.S. downgrade).

**Notation.** $H$, $L$, $C_{-1}$ the day's high and low and the previous close; $N$ in price units; VaR as a percentage of capital at 95 percent over one day ("the worst one out of 20 days", p. 66).

**Conditions.** $N$ is an average range, not a standard deviation. Gliner's statement that a $1N$ stop is hit with about 16 percent probability and a $2N$ stop with about 2.3 percent (pp. 52–53) treats $N$ as a one-day standard deviation; for Brownian daily moves the mean true range is about $2\sqrt{2/\pi} \approx 1.6$ standard deviations, so a close-to-close move beyond $2N$ is far rarer than 2.3 percent under normality, and fat tails and gaps change both numbers. The 1:3 rule's break-even hit rate is 25 percent before costs (section 2); it says nothing about whether the targets are reachable. A 5 percent one-day VaR is Gliner's illustration and implies an annualised volatility near 48 percent at normality, far above typical macro fund risk.

**Practice.** Gliner's metrics (sec. "Sharpe Ratio" to "Value at Risk", pp. 62–66): Sharpe ratio, which he says "does assume a normal distribution"; Sortino ratio, with downside deviation; drawdowns; and VaR. The average of the 5 worst of 500 days is an empirical 1 percent expected shortfall, not a "three standard deviation" loss as he puts it (p. 69); under normality it is about 2.7 standard deviations. *Correlation.* Gliner warns that positions correlated with one another can be stopped out together (sec. "Correlation", p. 56) and that correlations rise in stress (chap. 4, pp. 100–101). His illustration that perfectly correlated assets move by equal amounts (p. 100) confuses correlation with beta: $b = 2a$ has correlation one and beta two (checks, section 3).

**Pitfalls.** *Additional scenarios.* LTCM (1998), the 2015 renminbi devaluation and the 2020 COVID crash are reasonable stress scenarios but are not on Gliner's list. Gliner says the Sharpe ratio assumes a normal distribution, but the ratio itself does not, though reading it as a probability statement does. VaR is a loss exceeded on about 1 day in 20 (or 100), not an expected loss. That position sizing matters more than knowing what to buy or sell is advice, not a result. *Raising risk after gains* is a P&L-contingent rule Gliner links to Thorp; whether it improves risk-adjusted returns is untested here.

**See also.** global macro; relative performance evaluation; practitioner frameworks; regime map (correlations by regime).

**Further reading.** Volatility targeting and its out-of-sample evidence are in volatility-managed portfolios.

### Method 11.4 Practitioner frameworks from Drobny's interviews: risk-premia allocation, long optionality, trade construction, front-end rates, commodity microeconomics, equity concentration, the anonymous currency manager, emerging-market crises, flows and contrarian signals

**Statement.** These are practitioners' opinions recorded in interviews around 2005, not tested rules; each is attributed to the interviewee who said it in [s7-drobny, Wiley 2006 printing, printed pages]:

- *Leitner*: start "with 20 percent in each" of five asset classes (equities, fixed income, currencies, commodities, real estate) to earn risk premia (p. 60); fixed income levered at the front end, where "you could be five times levered in the two-year" for the risk of a 10-year bond (p. 62); commodities by rolling the second contract, said to pay 6 percent real a year (p. 62); an absolute-return bucket for "great, out-of-the-box ideas we come across about twice a year", with up to 10 percent of the fund on one idea (p. 63); look for disconfirming evidence (p. 50); not being invested forgoes risk premia, and fighting them requires being "doubly right" (p. 54).
- *Siva-Jothy*: never short gamma, carry being synthetically short gamma (pp. 78–79); "I always structure my portfolio to be 50 to 60 percent options", with three-to-six-month options and overnight options, without saying whether the share is of notional, premium or risk; "three or four major macro opportunities each year"; "Overtrading is as bad as running losing positions for too long" (p. 85).
- *Andres Drobny*: think about positioning over "the last two, three, or five years", and a rising current-account deficit means "a long building up" (p. 109); agrees with the interviewer's "three or four good trades a year" style, looking for the few 20–30 percent events with about 80 percent of the money in carry or lower-risk positions (p. 125).
  *Stops* (pp. 131–132): after a stop on a single trade you may get back in, "but with a very tight stop", counting the loss already taken, so the risk/reward, which "should at least be 4 to 1 in any given trade", is worse the second time; after hitting the 1 "a couple of times" a 4:1 trade has become 4:3 and you should move on. Asked about a cooling-off period, he answers "On a specific trade, no"; only "On a portfolio level, if you get stopped

out of all positions due to a bad run", after big losses, should managers "get away from the markets for about a month"; he also recommends three to four weeks away once a year. *Wadhwani*: positioning matters more than the macro news (p. 163); never ignore a stop, and take a 24-hour break before re-entering (p. 177).

- *Porter*: "time horizon arbitrage" against short-term investors (p. 135); maximise current income per unit of VaR at the front end; expertise in one-year, one-year forwards, "the most volatile part of the yield curve"; "most interest rate cycles are between 16 and 24 months" (p. 139; p. 148); a steepener means "you basically think the bonds are going to rally" and a flattener that "the market is going to sell off" (pp. 147–148); central banks and magazine covers as counter-indicators (p. 152).
- *Rogers*: a Fortune cover or front-page story as a sign a move is over (pp. 217, 230).
- *Anderson*: invest from microeconomics and fundamentals, not price (pp. 249, 256); stress the thesis against adverse macro ("when we're long, we assume very low growth", pp. 256–257); "margins revert to the mean", not prices (p. 258); size by volatility (p. 259); only low-cost producers, since "you can't buy a high-cost asset cheap enough" (p. 261); research on the ground (pp. 261–262); managing off a VaR number is "incredibly dangerous" for a concentrated book (p. 250).
- *Bessent*: at Soros Fund Management "we did a lot of bottom-up research"; Druckenmiller and Soros "believed that if you can find out what's going on at a micro level, you can figure out the big picture" (p. 274); "50 percent of a stock's move is the overall market, 30 percent is the industry group, and then maybe 20 percent" stock-specific (p. 280); 8 to 14 large, unequal positions (p. 283). He does not usually use stop-losses: from Druckenmiller he learned "how to enter a trade", and he tries to make his entry point "good enough that I rarely stop myself out of a trade". He quotes Soros, "Don't try to play the game better, try to figure out when the game has changed", and says catching a paradigm shift is "where you make a lot of money" (p. 285).
- *Dimitrijević*: country limits of 10 percent in the macro fund and 15 percent in the emerging-market fund (p. 293); the clean-sheet question, would we buy at the current price (p. 294); hedge a rich asset against an attractive one in the same country, short Brazilian debt against Brazilian equities (p. 295); stops "work in continuous markets" and are dangerous in distressed or value positions, are kept internal because stops given to counterparties "invariably get hit", and are followed by a one-month cool-off (pp. 295–296); the best EM opportunities come after panics (p. 296).
- *Yra Harris*: the "matrix" of global money flows seeking the highest risk-adjusted return; "money … craves stability"; fear and greed (p. 207); the Brazilian real devaluation of 1999 and selling soybeans (p. 208); Dennis's Slower Fool Theory (pp. 204–205); never the "prayer trade" back to break-even; "clear the books, and start over" (pp. 209–210); best when focused on three or four things (p. 214); "80 percent of my day is being wrong", probing "like a reconnaissance mission" (p. 215).
- *The anonymous currency specialist*: FX as the tail of the credit curve (p. 327); three blocs, Europe, Asia and Latin America (p. 328); three or four macro calls (pp. 328–329); a gestalt shift (p. 329); no relative value, from a fear of negative gamma (p. 330); bullish means long or flat (p. 331); long gamma and vega, leverage a function of correct analysis, "75 percent of price movement" in "10 to 15 percent of the time" (p. 332); never a forced seller (p. 333); a five-month horizon to judge a view (p. 334); the "frozen ice" theory of crowded trades (p. 336).

**Notation.** Page numbers are printed pages of the 2006 printing (copyright 2006, ISBN 978-0-471-79447-9). The quotations were also searched in the revised edition the catalogue names (EPUB, ISBN 978-1-118-86572-9, published 16 November 2013, copyright 2006, 2009 and 2014); each recurs unchanged in the same interview except Andres Drobny's, whose interview is not in it, and its chapters are numbered one lower from Porter on (Bessent is chap. 12, "The Stock Operator").

**Conditions.** Opinions of successful survivors, selected by the author, with no record of how the rules performed; several conflict (Leitner's long risk premia against the anonymous manager's fear of carry as negative gamma; Andres Drobny's 4:1 against Gliner's 1:3; Bessent's practice of not usually setting stop-losses against the stop rules of Andres Drobny, Wadhwani and Dimitrijević). Bessent's entry discipline and paradigm-shift remarks are his account of his own practice and of lessons he attributes to Druckenmiller and Soros; they record no results and are not advice to trade without stops.

**Practice.** Attributions in the interviews: the claim that most price movement happens in a small share of the time, long-gamma leverage, never being a forced seller and buying options when an outcome looks "inevitable" are the anonymous manager's (pp. 331–333, 336); 4:1 and the stop rules are Andres Drobny's, and his cooling-off applies to a portfolio-wide stop-out after large losses, not to a single trade; three or four major opportunities a year and overtrading being as bad as holding losers are Siva-Jothy's (p. 85), while Andres Drobny only accepts the interviewer's three-or-four-trades framing (p. 125); Wadhwani's own re-entry rule is a 24-hour break (p. 177); time-horizon arbitrage is Porter's; contrarian front-page signals are Rogers's and Porter's.

**Pitfalls.** *Reading the interviews.* Porter says the curve "is 90 percent explained by the direction of interest rates" (pp. 147–148), which does not make a steepener a 90 percent bullish bond trade. The anonymous manager's three blocs are Europe, Asia and Latin America. Leitner's FX carry and volatility-premium trades and leveraged short-duration fixed income are examples, not rules, and his equal 20 percent risk allocation is a starting point, adjusted afterwards. The contrarian observations come from Porter, Rogers and the anonymous manager.

**See also.** global macro; macro position sizing and risk; curve trades (steepeners and direction); FX carry.

**Further reading.** Drobny's later collections and the section's systematic evidence.

## 11.2 Regimes, indicators, releases and valuation

**Definition 11.5 A regime map of asset responses: growth, recession, inflation, deflation, liquidity and geopolitical shocks across rates, credit, equities and sectors, currencies and commodities, as hypotheses to test**

**Statement.** A *regime map* lists the direction in which each asset class is expected to move under a type of macro shock. Gliner's version is qualitative: a risk-on/risk-off grouping of assets, correlations that rise in stress (Figure 4.5, around Lehman), and the need to know which relationships hold in the current environment [s7-gliner, chap. 4, sec. "Assessing the Relationships between Assets", PDF pp. 98–102]; sector rotation, defensive sectors such as consumer staples in recession, and the need to know "whether or not the economy is in contraction, recovery, expansion, or slowing growth" (chap. 8, sec. "Sector Rotation", PDF pp. 216–217, Figure 8.3). The map, as hypotheses to test:

- *Growth shock*: yields up, curve steeper, credit tighter, equities, cyclicals and industrial commodities up, growth-linked currencies (AUD, CAD, EM) up.
- *Recession*: yields down, curves bull-steepen or flatten, credit wider, equities down, USD up, commodities down except gold; defensives (staples, utilities, health care) lead.
- *Inflation shock*: nominal yields up, curve bear-steepens, TIPS beat nominals, commodities (energy, agriculture) up; equities depend on real rates; low-credibility central banks' currencies punished.
- *Deflation or disinflation*: nominal yields and breakevens down, duration rallies.
- *Liquidity shock or financial crisis*: USD up sharply, Treasuries rally, credit spreads gap wider, correlations jump, volatility spikes, the cross-currency basis widens, funding markets seize.
- *USD weakness*: commodity prices up. *Geopolitical crisis*: energy and gold up.

*Sector rotation against Gliner's Figure 8.3.* The figure lists four phases: contraction (health care, telecom, consumer staples, utilities), recovery (financials, materials, consumer discretionary, technology), expansion (industrials, energy, materials, technology) and slowing growth (financials, telecom, consumer staples, utilities). An earlier sector-rotation table in these notes matched it only in part: its recession list (staples, utilities, health care) omitted telecom; its recovery list added real estate, which the figure does not show, and omitted materials and technology; its expansion list

omitted energy; and its late-cycle "Energy, Materials (inflation hedges)" contradicts the figure's slowing-growth phase, which is defensive. Both are schematic: Gliner's figure gives no data, period or source.

**Notation.** A *bull steepener* is a steeper curve with falling yields, a *bear steepener* with rising yields (curve trades).

**Conditions.** None of these signs is a law. They depend on the monetary regime and on what kind of shock moves inflation: the stock–bond correlation, for example, has changed sign across decades. Classifying regimes needs information available at the time; a map built on ex post NBER dates or revised data is look-ahead.

**Practice.** Use the map to state a thesis's expressions and its hedges, then test each sign on dated regimes with real-time data before relying on it. Reading the curve as a summary of the macro narrative, or fixed income as telling what the market expects before it happens, rests on the expectations component of yields, which also contain term premia (forward rates and term-structure theories).

**Pitfalls.** No source here supports an ordering such as real rates falling first in an inflation shock, or the curve bull-steepening and then flattening in a recession; the ordering depends on the policy response. An inverse correlation between commodities and the dollar holds on average in some samples and fails in others. *Own test of the recession row in real time*. The hypothesis and pass rule were fixed in the script before its first run, though not preregistered. Regimes were classified with a Sahm-type rule on each quarterly vintage of the Philadelphia Fed's Real-Time Data Set, 1965–2025: the three-month average unemployment rate at least 0.5 point above its minimum over the previous twelve months. That gives 241 vintages; the three 2026 vintages are skipped because October 2025 is missing in them. Asset responses are measured over the next three months.

*Real-time test.* The rule signals in 53 quarters, from 10 episodes. None of the recession signs is supported at the pre-set $|t| > 2$ (Welch).

- The 10-year yield change is −0.05 point relative to other quarters ($t = -0.5$), and the Baa spread change −0.07 ($t = -0.8$).
- Equity excess returns are +1.2 percent ($t = +0.8$). The sign is wrong, since the rule fires late and recoveries follow.
- Defensive sectors (staples, utilities, health care) less the market return −0.7 ($t = -1.0$), also the wrong sign.
- The dollar is the closest. The Fed's broad index weighted goods trade only until its 2019 revision, which added services, simplified the weights and set the new index from January 2006 [fed-dollar-index-revisions-2019, secs. II–III]. On the goods-only broad index, a consistent definition for 1973–2019 and now the primary dollar measure, it rises +1.1 percent ($t = +1.98$, Newey–West $t = +1.87$, 6 episodes): the right sign, just short of the threshold. On the goods-and-services index from 2006 it moves −0.4 ($t = -0.3$, 3 episodes).

*Ex-post description.* NBER windows are dated afterwards, so they describe what happens inside recessions; they are not a usable classification. Under the *any-overlap* definition, a quarter counts if any of its three response months lies inside a recession. On that definition the signs are as the map says: yields −0.16, spread +0.25, equities −3.1 percent, defensives +1.1, dollar +0.8 (on a splice of the two indices). None reaches Welch $|t| = 2$ (spread 1.78), though the spread does on a Newey–West $t$ with four lags (2.46). The *all-months* definition, which requires all three months inside a recession, is a sensitivity, not chosen for significance. On it the yield (−0.41, $t = -2.2$) and spread (+0.45, $t = +2.9$) signs pass Welch $|t| > 2$; defensives (+2.1, $t = 1.9$; Newey–West 2.7), equities and the dollar do not.

Only with the any-overlap windows and Welch $t$ does no ex-post sign reach $|t| = 2$. Signal and recession quarters cluster in 6–10 episodes, so every $t$ here overstates the independent evidence. The recession row describes recessions dated afterwards; it is not a tradable real-time rule.

*A second row: inflation shock*. The protocol was written before the data were downloaded, with two source substitutions recorded before computing; it is not a preregistration. The protocol names the Philadelphia Fed's quarterly CPI vintages. The quarterly file, `cpiQvMd.xlsx`, exists [philfed-rtdsm-cpi, CPI data-files page, dated 2 October 2026; specific notes for CPI and PCPI; general notes, sec. II and IV], and its timing differs: a quarterly vintage holds what was available on the 15th of the quarter's middle month and excludes any release published after the 15th, while a monthly vintage holds what was available just after the BLS's monthly report, which may come a few days before or after the 15th. A quarter signals an inflation shock when twelve-month CPI inflation in that vintage's latest month exceeds its level a year earlier by at least one point.

The hypothesised responses over the next three months are 10-year yields up, the 10-year–3-month slope up (bear steepening) and WTI spot up. The bill rate and WTI are from H.15 and EIA. On the quarterly vintages, as the protocol specified:

- 128 vintages, 1994Q3 to 2026Q2 (2026Q3 skipped: responses incomplete); 34 signal quarters in 13 episodes.
- 10-year yield: +0.07 point against other quarters, Welch $t = 0.83$ (Newey–West 0.78).
- Slope: −0.17, $t = -1.56$ (Newey–West −1.49). The opposite sign: the curve tended to flatten after accelerations.
- WTI spot: −2.2 percent, $t = -0.60$.

*Timing sensitivity: monthly middle-month vintages*: 111 vintages, November 1998 to May 2026; 31 signal quarters in 14 episodes; 10-year yield +0.06 ($t = 0.76$), slope −0.16 ($t = -1.44$), WTI −1.0 percent ($t = -0.24$). On the 111 vintage months that have responses in both files, the latest CPI month differs in 55, always later in the monthly vintage (the BLS report had come out after the 15th), and the signal label differs in 11. The verdict does not change. No sign of the inflation row is supported in real time on this rule.

*A third row: deflation or disinflation*. The mirror-image rule, a fall of at least one point in twelve-month CPI inflation over a year in each quarterly vintage, flags 28 of 128 quarters in 12 episodes. The hypothesised fall in 10-year yields over the next three months is not supported: +0.04 point against other quarters (Welch $t = 0.51$, Newey–West 0.50), the wrong sign; the slope (+0.11, $t = 1.20$) and WTI (+5.5 percent, $t = 1.45$) also rose, reported alongside only. Limits: three rows of the map, one real-time rule each, three-month windows, U.S. assets only; CPI vintages from 1994 only; WTI spot is a price, not an investable return; TIPS, equities and currencies untested.

**See also.** global macro; macroeconomic indicators; central-bank policy; TIPS and breakevens.

**Further reading.** Regimes and the stock–bond correlation [wide-ang-timmermann-regimes].

### Definition 11.6 Macroeconomic indicators for macro trading: growth, inflation, employment, external accounts, fiscal, consumption, housing and business surveys, and demographics, with the FOMC's inflation goal and Gliner's demographic table read correctly

**Statement.** Gliner's chapter on data releases [s7-gliner, chap. 12, PDF pp. 401–428] groups the indicators a macro trader watches:

- *Growth* (sec. "Measuring Growth", p. 402): GDP, nominal and real, quarterly with revisions.
- *Inflation* (sec. "Inflation", pp. 404–405): CPI and core CPI, PCE, the euro area's HICP, and the PPI, which measures "the average change over time in the selling prices received by domestic producers".
- *Employment* (sec. "Employment and Population", pp. 406–409): the unemployment rate, the labour force and participation rate (labour force as a share of the civilian noninstitutional population), and nonfarm payrolls, "released on the first Friday of the month", with average hourly earnings.
- *External accounts* (sec. "Balance of Payments", pp. 410–412): the current account (trade balance, net income and transfers), the trade balance, and the terms of trade, export prices relative to import prices.
- *Government* (sec. "Government Indicators", pp. 413–415): the budget balance and debt-to-GDP.
- *Consumption* (sec. "Consumption Indicators", pp. 416–417): U.S. consumption "more than 70 percent of GDP in 2011" and, with the United States at about 20 percent of world GDP, the U.S. consumer at "a little more than 15 percent of global GDP" (p. 416); retail sales, the savings rate and consumer confidence (Conference Board, University of Michigan).
- *Industry and services* (sec. "Industry and Services Indicators", pp. 418–420): purchasing managers' indices, with above 50 read as expansion and new orders and inventories among their components; industrial production; building permits, which lead housing starts; and capacity utilisation.

- *Demographics* (sec. “Demographics”, pp. 421–426): working-age (15–64) and prime-saving (35–69) populations, with Table 12.2 (U.S. Census Bureau; heading and source on p. 424, data on p. 425) giving each country’s peak year for both, and Trichet’s remark that the current account “is an important summary indicator” (p. 424).

**Notation.** *Headline* and *core* inflation (core excludes food and energy); PCE the personal consumption expenditures price index; PMI a diffusion index, %better + $\frac{1}{2}$%same, scaled so that 50 means no change.

**Conditions.** *The Fed’s inflation goal.* Gliner writes that “the Fed targets core PCE” (p. 405), and more broadly that central banks target core inflation. The FOMC’s goal is 2 percent inflation “as measured by the annual change in the price index for personal consumption expenditures”, the headline index; the statement was first adopted effective 24 January 2012 and was reaffirmed effective 27 January 2026 [fomc-longer-run-goals, Statement on Longer-Run Goals and Monetary Policy Strategy, pp. 1–2]. Core measures are watched as guides to the trend, not targeted. Gliner’s further statements that PCE “targets urban consumers” and HICP “rural consumers” are wrong as descriptions: urban coverage defines the CPI-U, not PCE, and the HICP covers household consumption in general.

*Payroll dates.* The Employment Situation is usually published on the first Friday but not always; the CES FAQ puts the release “typically on the third Friday after the conclusion of the reference week” [bls-ces-faq, Data Overview, Q8], which usually, not always, falls on the first Friday of the next month; the BLS’s own schedule is the record (the next release, as of October 2026, 6 November 2026) [bls-ces-page, program page].

**Practice.** *Demographic peaks.* Gliner’s two columns are easily mixed: Netherlands 1989, U.K. 2009 and U.S. 2007 are working-age (15–64) peaks, while Germany 2006, Japan 2016, China 2032, Korea 2023, Russia 2025, Spain 2022, India, Indonesia and the Philippines 2050 are prime-saving (35–69) peaks; Nigeria shows 2050 in both (Table 12.2, heading p. 424, data p. 425). These are projections published around 2013, and they have moved.

*Against the UN’s World Population Prospects 2024* [un-wpp-2024, file `WPP2024_PopulationByAge5GroupSex_Medium`]: the 15–64 peaks are Germany 1997, Japan 1995, China 2015, Korea 2017, Russia 2010 and Spain 2009 (Gliner: 1986, 1992, 2011, 2015, 2011 and 2004), but the Netherlands 2026, the United Kingdom 2044, India 2048, Indonesia 2045 and the Philippines 2053, against his 1989, 2009, 2027, 2023 and 2037, and the United States and Nigeria are still rising at 2050 (the U.S. to 2100); the 35–69 peaks are Germany, Korea, Russia and Spain all in 2024, Japan 2010, China 2031, the Netherlands 2030, and after 2050 for the United Kingdom, the United States, India, Indonesia, Nigeria and the Philippines.

*Why the years differ is not established here*. Gliner’s table is labelled “Source: U.S. Census Bureau” (p. 424) with no vintage or release date; WPP 2024 is another publisher’s 2024 revision. Publisher, vintage, base-period estimates and, for future years, projection assumptions all differ at once, and no controlled comparison was made (the same publisher across vintages, or both publishers in one vintage; the Census Bureau’s International Database now requires an API key), so no cause is attributed. Two kinds of year should also be kept apart.

Peaks before about 2012 were historical estimates in both sources (WPP 2024 gives estimates to 2023 and projections after): there, a gap such as Germany’s 15–64 peak in 1986 against 1997, or Japan’s in 1992 against 1995, cannot come from projection assumptions and must reflect estimation, revision or territorial coverage (for example, how Germany before 1990 is defined). Peaks after about 2013 were projections in Gliner’s table, and those after 2023 are still projections in WPP 2024; revised fertility, mortality and migration assumptions may contribute there, but that too is unverified. The point that projected peaks are revised with each release matters more than any single year.

*Other claims.* Consumption “more than 70 percent of GDP” in the United States is Gliner’s (chap. 8, p. 238; chap. 12, p. 416); U.S. consumer at about 15 percent of global GDP is also his (p. 416), the product of two approximate 2011 shares rather than a measured figure. PPI leading CPI by one to three months, that the ISM new-orders-minus-inventories spread is a reliable leading indicator, the quits rate as an indicator of labour-market tightness, wage growth driving stickier services inflation, median ages, home-price indices lagging the market, household formation as the structural driver of housing demand, the dependency ratio as a measure of fiscal and savings pressure, and a positive primary balance letting debt-to-GDP stabilise when growth exceeds the interest rate are conventional practitioner statements with no source here.

**Pitfalls.** *Capacity utilisation* “over 100 percent” (Gliner, p. 420) is not a reading one sees in the Federal Reserve’s series, which measures output relative to sustainable capacity and has stayed well below 100 percent; read deviations from its own history. Debt-to-GDP above 90 percent as a threshold (Gliner, p. 414) is the contested Reinhart–Rogoff finding, not a rule.

*Nonfarm payroll scope*. The BLS Handbook of Methods states that CES data “exclude sole proprietors, the unincorporated self-employed, unpaid volunteer or family workers, farm workers, and domestic workers”, that “Salaried officers of corporations are included”, and that government employment “covers only civilian employees; military personnel are excluded” [bls-hom-ces, CES concepts, page modified 28 February 2025]. The CES FAQ lists employees of non-profits and religious organisations as in scope, the largest group not covered by unemployment-insurance records, for which CES estimates separately [bls-ces-faq, Employment Concepts, Q4 and Q6]. Nonprofit employees are therefore included; farm, private household, self-employed and military workers are excluded.

**See also.** U.S. release calendar; event queries; equity leading indicators; TIPS and breakevens; central-bank policy.

**Further reading.** Nowcasting with release vintages [wide-giannone-nowcasting].

### Method 11.7 Leading indicators for equities in Gliner’s macro toolkit: commodity prices, the CDX high-yield index, the ISM PMI, the Baltic Dry Index, consumer confidence and Australian-dollar implied volatility

**Statement.** Gliner lists indicators that lead or confirm equity moves [s7-gliner, chap. 8, sec. “Valuation Techniques for Equities”, PDF pp. 233–240]. The year-on-year change in commodity prices matters because a rise is inflationary and invites tightening (pp. 233–234). The five-year CDX high-yield index (CDX.NA.HY) rises with risk aversion and is inversely related to equities (pp. 234–235). For the ISM manufacturing PMI, when “the ISM … goes below 50, the United States typically enters recession” (p. 235). The Baltic Dry Index, a composite of Capesize, Panamax, Handysize and Supramax dry-bulk freight assessments, is watched because raw materials lead (p. 236).

His list also includes the ARMS index (pp. 237–238) and consumer confidence, since consumption is over 70 percent of U.S. GDP (p. 238). Three-month AUD/USD implied volatility is “highly correlated” with the VIX and inversely with the S&P 500, and is most useful as a rate of change, though “given the choice, using the VIX is a more effective factor”, and the two combined “smooth out any error” (pp. 239–240).

**Notation.** CDX.NA.HY the Markit North American high-yield CDS index, quoted as a price or spread; ISM the U.S. Institute for Supply Management.

**Conditions.** Gliner’s charts to 2013. These are Gliner’s chart readings (Figures 8.12–8.18), without lead-lag statistics or out-of-sample tests; the reading of ISM below 50 as recession has had false signals, since the index has fallen below 50 outside recessions.

**Practice.** Treat each as a candidate nowcasting signal: build it from real-time vintages and test its lead against equity returns.

**Pitfalls.** Neither the absence of a directly traded index future nor a slow supply response makes freight prices immune to speculation. Dry-bulk freight is traded through forward freight agreements settled on Baltic route assessments, and fleet supply changes with deliveries and scrapping.

**See also.** macroeconomic indicators; equity-index valuation; VIX.

**Further reading.** Mixed-frequency nowcasting [wide-banbura-nowcasting].

### Data 11.8 U.S. macro release calendar: the usual monthly cadence of employment, inflation, activity and survey releases, FOMC and ECB meeting frequency, and why the agencies’ own schedules are the record

**Statement.** The cadence: the Employment Situation (payrolls, unemployment, wages) on the first Friday; CPI, retail sales and industrial production mid-month; housing starts and permits in the third week; GDP (quarterly), personal income and outlays (PCE) and durable goods late in the month; the ISM manufacturing PMI on the first business day and the services PMI on the third. Of these, the first-Friday payroll convention is Gliner’s

[s7-gliner, chap. 12, PDF p. 406], and the BLS's schedule shows the next Employment Situation on Friday 6 November 2026 [bls-ces-page, program page]. The FOMC holds eight regularly scheduled meetings a year [fomc-page, page updated 7 October 2026]. The ECB's Governing Council takes monetary policy decisions every six weeks [ecb-governing-council, web page], about eight a year.

**Notation.** Release times are U.S. Eastern; each agency (BLS, BEA, Census Bureau, Federal Reserve Board, ISM) publishes its own calendar.

**Conditions.** The cadence is a pattern, not a rule: payrolls move when the first Friday falls too early in the month or on a holiday, and government shutdowns have postponed releases. Use the agencies' published calendars, archived as they stood on each date, to rebuild a release timeline without look-ahead.

**Practice.** A backtest that uses data at its reference date rather than its release time, or uses revised values, has look-ahead; keep first-release values and their timestamps (event queries).

**Pitfalls.** Do not confuse six meetings per year with six-week intervals; the ECB's own page gives decisions every six weeks. *Placements against agency calendars*. Industrial production (Federal Reserve G.17) is released mid-month, for example 16 October and 17 November 2026 [fed-g17-dates, release dates]; BEA's advance GDP for the third quarter of 2026 and personal income and outlays (PCE) for September are both due on 29 October 2026, late in the month [bea-schedule, 2026 schedule]; the Census Bureau's calendar shown (for 2027) puts retail sales near the 15th, housing starts and permits near the 20th and durable goods near the 27th of the month after the reference month [census-econ-calendar, economic indicator calendar]. These agree with the cadence.

*CPI*: the BLS schedule lists the October 2026 CPI for 10 November 2026, November's for 10 December 2026 and December's for 13 January 2027, all at 8:30 a.m., and the 2027 reference months between the 10th and the 14th of the following month [bls-cpi-schedule, CPI release schedule], so CPI comes before retail sales and well before PCE for the same month.

*ISM*. The ISM states that the Manufacturing PMI "is released on the first business day of the month at 10:00 a.m. (EST)" and the Services PMI on the third, and lists 2026 dates (Manufacturing, Services): 5 and 7 January, 2 and 4 February, 2 and 4 March, 1 and 6 April, 1 and 5 May, 1 and 3 June, 1 and 6 July, 3 and 5 August, 1 and 3 September, 1 and 5 October, 2 and 4 November, 1 and 3 December, with notes that January's Manufacturing report moves to the 5th "due to ISM holiday" and July's Services report moved for the Independence Day holiday observed on 3 July [ism-calendar-2026, 2026 calendar, p. 1].

Counting business days against U.S. federal holidays reproduces every date except January's two and April's Services date: 6 April follows from skipping Good Friday (3 April), which is not a federal holiday, so the ISM's own holidays decide the count (an inference). The rule therefore holds as a rule, with ISM-specific exceptions to be read from each year's calendar.

**See also.** macroeconomic indicators; event queries; major central banks.

**Further reading.** Announcement returns [wide-savor-wilson-announcements].

#### Method 11.9 Event queries around data releases and policy decisions: surprise against consensus, revisions, conditional back-tests of asset returns after past events, hit rates and their small-sample error

**Statement.** Gliner's *query* back-tests how assets behaved after past occurrences of an event like the one expected: choose the event and the conditioning (for an expected "no change" from the ECB, back-test meetings with no change), list the dates, and tabulate returns at fixed horizons with the share of up and down outcomes and the standard deviation [s7-gliner, chap. 3, secs. "Simple Back-Tests and Queries" and "An Example of Building a Query", PDF pp. 79–85].

His example is the Bank of England's quantitative easing in July 2012: the June minutes showed a 5–4 vote against more QE (against 8–1 in May), traders expected an expansion, and the query over the six earlier QE announcements shows the FTSE 100 up on 5 of 6 announcement days (83 percent), averaging 0.3 percent, and up on all six four days later, averaging 1.7 percent, at horizons of 1 to 5 days, two weeks and one month (Tables 3.1–3.2, pp. 80–83). He adds analogs, comparing current price paths with past ones (sec. "Historical Correlations and Analogs", p. 86).

**Notation.** *Surprise* = actual minus consensus forecast, often scaled by its standard deviation; *hit rate* = share of events in which the asset moved in the expected direction.

**Conditions.** With six events, a hit rate means little: five or more up out of six has probability 0.109 under a fair coin, and six of six 0.016; overlapping horizons from the same six dates are not independent tests. The conditioning set is chosen after the fact, which invites data mining. A query on a release also needs the surprise, not the level: markets price the consensus beforehand.

**Practice.** Practitioner principles (surprise direction and size drive the move; initial releases move markets and material revisions move them again; NFP and CPI dominate in the United States and PMIs globally; what matters depends on the regime) are standard practitioner views, consistent with the announcement literature, but not shown in Gliner's chapter. Report the number of events and a confidence interval with any hit rate.

**Pitfalls.** Cutting a position quickly when it moves against you right after an event is advice, not a result. Back-testing on revised data, or on event dates known only afterwards, adds look-ahead.

**See also.** release calendar; macroeconomic indicators; central-bank communication; tests of skill against luck.

**Further reading.** [rt-kuttner-2001] on policy surprises and [wide-savor-wilson-announcements] on announcement-day returns.

#### Method 11.10 Equity-index valuation for macro: top-down country and sector selection, price-to-book, dividend yield, P/E and forward P/E, free-cash-flow yield and market capitalisation to GDP, with Gliner's thresholds as rules of thumb

**Statement.** Gliner describes a top-down approach to equities, from a global view to countries, then sectors, then optionally companies; if the investor knows where the next opportunity lies, the reward comes from concentrated rather than index exposure [s7-gliner, chap. 8, sec. "Taking a Top-Down Macro Approach", PDF pp. 215–216, Figure 8.1].

His valuation measures (sec. "Valuation Techniques for Equities", PDF pp. 228–233) start with price-to-book, where "a P/B greater than 2.5× is generally thought to be overbought, and a P/B lower than 1.5× is oversold" (the S&P 500 near 1.5 in 2008, p. 228). They continue with dividend yield, paid while waiting (p. 229); trailing and forward P/E, with earnings yield its reciprocal (pp. 230–232); price-to-free-cash-flow and its reciprocal, a Graham-and-Dodd screen (pp. 232–233); and stock market capitalisation to GDP, "more worrisome" the further above 100 percent and quickly rising, as when Japan's passed 150 percent in 1989 (p. 233).

**Notation.** P/B price over book value per share; P/E price over earnings per share (forward: analysts' next-year estimate); FCF operating cash flow minus capital expenditure.

**Conditions.** Gliner's thresholds as of 2013. The thresholds are rules of thumb for the S&P 500's history, not levels that transfer across markets or eras: sector mix, accounting, intangible assets and interest rates move normal valuation ranges. Market capitalisation to GDP mixes a global firm's value with domestic output.

**Practice.** For `cross-asset-style-premia`, value in equity indices is measured cross-sectionally (cheap against expensive countries), not against a fixed threshold. Gliner's Tobin's $q$, which he says gives "a score between 0 and 1" (p. 228), is not bounded above by one: $q$ is market value over replacement cost and has exceeded one for long periods.

**Pitfalls.** P/B above 2.5× as "generally overbought", as a timing rule, is untested; valuation predicts long-horizon returns weakly and timing poorly.

**See also.** equity indices; equity leading indicators; regime map.

# 12 Currency markets: exchange rates, premia and hedging

## 12.1 Currencies in global macro

**Definition 12.1 The dollar in global macro: FX as shock absorber, transmission channel and expression; the dollar's share of turnover and reserves; the dollar index and its weights; and the SDR**

**Statement.** Gliner: "The U.S. dollar accounts for 85 percent of all foreign exchange (FX) transactions", the shares summing to 200 percent because each trade involves two currencies (Figure 7.1, BIS Triennial Survey, April 2013), and more than 60 percent of central banks' reserves, declining gradually [s7-gliner, chap. 7, sec. "The Role of the U.S. Dollar", PDF pp. 180–181]. The U.S. Dollar Index is the dollar against a basket changed once, in 1999, with weights EUR 57.6, JPY 13.6, GBP 11.9, CAD 9.1, SEK 4.2 and CHF 3.6 percent, traded as the DX future and the UUP ETF (p. 181).

The SDR was created by the IMF in 1969, initially defined as 0.888671 grams of fine gold, later as a basket of currencies; it "is neither a currency, nor a claim on the IMF" and serves as the IMF's unit of account (sec. "Special Drawing Rights (SDRs)", pp. 187–188). The IMF's current factsheet gives the basket as the U.S. dollar, euro, Chinese renminbi, Japanese yen and pound sterling, with the valuation reviewed every five years and the current weights effective 1 August 2022 [imf-sdr-factsheet, factsheet dated 7 October 2023].

**Notation.** DXY the ICE U.S. Dollar Index, a geometric average of six exchange rates; SDR Special Drawing Right.

**Conditions.** Gliner's 85 percent means the dollar is on one side of 85 percent of trades; he sources it to the BIS 2013 survey (Figure 7.1). In April 2025 the dollar "was on one side of 89.2% of all trades", up from 88.4 percent in 2022, with the euro at 28.9, the yen at 16.8, sterling at 10.2 and the renminbi at 8.5 percent, out of a total of 200 [bis-triennial-2025, preliminary results, sec. 2].

*Index methodology*. ICE describes the index as "a geometrically-averaged calculation of six currencies", computed as $\text{USDX} = 50.14348112 \times \text{EURUSD}^{-0.576} \times \text{USDJPY}^{0.136} \times \text{GBPUSD}^{-0.119} \times \text{USDCAD}^{0.091} \times \text{USDSEK}^{0.042} \times \text{USDCHF}^{0.036}$, the exponent negative where the dollar is the quote currency; it "originally was developed by the U.S. Federal Reserve in 1973" as a bilateral trade-weighted average, futures have traded since 1985, and the weights match Gliner's [ice-dxy-faq, FAQ, June 2015, p. 2]; later changes, if any, are not reflected. The weights are fixed, changed only once, in 1999 (Gliner); the index is therefore not a trade-weighted index against today's partners: it gives the euro over half the weight and omits the renminbi and the Mexican peso.

**Practice.** FX plays three roles in macro: shock absorber, transmission channel and expression instrument. That global macro is de facto USD-centric, that many local trades are disguised USD liquidity trades, and that stress appears first in the dollar index, the cross-currency basis and EM currencies are interpretations, not findings of Gliner's chapter.

**Pitfalls.** The SDR is not a reserve currency anyone trades; official reserve diversification is tracked by the IMF's COFER data on reserve composition (Gliner's Figure 7.2), not from the SDR basket. Weights in percent (57.6 for the euro) are exponents of a geometric average, not shares of a basket of currency amounts; a 1 percent rise in EUR/USD lowers the index by about 0.576 percent.

**See also.** spot FX market structure; currency regimes; FX carry; FX valuation signals.

**Further reading.** The dollar's role and safe-asset demand [wide-jkl-safe-asset].

**Definition 12.2 Currency regimes and intervention: fixed, managed and floating regimes in Gliner's grouping, sterilised and unsterilised intervention, the impossible trinity, and reserve depletion**

**Statement.** Gliner groups currencies by the central bank's intervention: *fixed* (HKD, DKK, LVL, AED, TRL in his 2013 table), *floating with frequent intervention* (JPY, BRL, CNY, CHF with the EUR/CHF floor) and *floating with no intervention* (USD, EUR, GBP, AUD, NZD, MXN, CAD, SEK, NOK, INR). He stresses that "no model will be of help if, for example, a currency is pegged" [s7-gliner, chap. 7, sec. "Currency Regimes", PDF pp. 185–194]. His examples include the Hong Kong dollar's peg at 7.8 (p. 190) and Brazil's repeated interventions (Table 7.2, pp. 191–192).

Intervention is *sterilised* when the central bank offsets its effect on the monetary base and *unsterilised* when it lets the base change (chap. 11, pp. 360–361). The *impossible trinity*, from Mundell's work, says a country can have only two of a fixed exchange rate, free capital flows and an independent monetary policy (chap. 11, sec. "The Impossible Trinity", p. 363).

**Notation.** Peg, crawling peg, band and currency board are forms of a fixed regime; a floor (EUR/CHF 1.20 from 6 September 2011) limits movement in one direction.

**Conditions.** Gliner's grouping is de jure and as of 2013; de facto classifications can differ, and several of his currencies have since changed regime (the EUR/CHF floor ended in 2015).

*A de facto check* uses the Ilzetzki–Reinhart–Rogoff monthly fine classification for December 2013 and December 2025 [irr-era-2025, sheets "Classification" and "Fine", last update 5 October 2026]. Its codes are 1 no separate legal tender, 2 pre-announced peg or currency board, 3 pre-announced horizontal band within ±2%, 11 moving band within ±2%, 12 de facto moving band within ±5% or managed floating, and 13 freely floating. HKD, DKK and AED are 2 in both months, and Latvia moves from 2 to 1 in January 2014, when it adopted the euro. But Gliner's "fixed" TRL is 12 in both, and his intervening floats differ: CNY is 3 (a band) and CHF 2 in 2013 (the floor), 11 in 2025; JPY moves from 13 to 12 in August 2024.

Of his "no intervention" floats only USD and AUD are 13 in both months; GBP, CAD and NZD are 11 in 2025 (since January 2020), SEK and INR 11 in both, MXN and NOK reach 13 only in 2016 and 2023. So de jure "free float" and de facto "freely floating" are different lists.

**Practice.** Standard arguments, not shown in Gliner's chapter, interpret the regimes: in a free float price discovery happens in FX; in a managed regime pressure appears in rates, reserves or capital controls; a hard peg looks stable until it breaks; and "cheap" can stay cheap if the regime prevents adjustment. The same applies to the usual intervention motives and conditions for success (reserve adequacy, credibility, consistency with monetary policy, liquidity), to the view that intervention against fundamentals typically fails, and to reserve depletion as a leading indicator. The trinity explains the ERM crisis of 1992 and the Asian crisis of 1997 in telling; Gliner gives no such cases in that section.

**Pitfalls.** The trinity is about full capital mobility and a fixed rate; intermediate regimes trade off partial independence against partial stability, so "choose two of three" is a corner-solution statement.

**See also.** central-bank tools; major central banks; the dollar and the SDR.

**Further reading.** On whether intervention works, [cur-fratzscher-et-al-2019]; for a survey, [cur-menkhoff-2013].

**Method 12.3 Gliner's FX valuation signals: equity performance, sovereign credit, risk reversals, trade and the trade-weighted index, export-partner growth, debt and the current account, reserves, PPP and GDP per capita, and policy-rate and LIBOR differentials**

**Statement.** Gliner lists factors for a currency model [s7-gliner, chap. 7, sec. "Valuation Techniques for Foreign Exchange", PDF pp. 195–209]: relative *equity index performance*, through flows (pp. 195–196); *credit*, five-year sovereign CDS (Australia against Mexico, p. 197); *sentiment*, "call-put skew on three-month risk reversals", to be "back-tested on a currency-by-currency basis" and correlating better in distress than in calm periods (p. 197); the *trade* balance and current account (pp. 198–199); the *trade-weighted index* (pp. 199–200); and *economic activity* (p. 200).

His further factors are *export-partner growth* weighted by export shares, with Australia's exports to China, Japan, Korea, India and Taiwan (pp. 200–201); *debt to GDP* (p. 202); *absolute and relative PPP*, with the Big Mac index, and Meese and Rogoff's finding that it does not forecast in the short run (pp. 203–205); *GDP per capita* (p. 205); and *carry* indicators, the policy-rate (OIS) differential and the LIBOR differential, decomposed as LIBOR = fed funds + (LIBOR − fed funds) (pp. 206–208), with risk-adjusted carry (FX carry).

**Notation.** Risk reversal: implied volatility of an out-of-the-money call minus that of an equally out-of-the-money put (usually 25-delta); TWI: a trade-weighted index of the currency against its partners.

**Conditions.** Gliner's examples as of 2012–2013; forecasts to 2021–2025 by currency, outcomes to 2026. The list is a set of candidate factors with charts, not a tested model: Gliner gives no out-of-sample results, and whether such factors beat a random walk is the open question. Several cut both ways: strong activity can attract inflows and also worsen the trade balance.

**Practice.** Build each factor from data available at the time and test against a random walk with economic as well as statistical criteria. The factors can be grouped into risk appetite, trade, sustainability, long-term anchors and carry; FX reserve adequacy (months of import cover, short-term debt coverage) fits beside them, although Gliner treats it under reserves elsewhere.

**Pitfalls.** Gliner's account of the *Balassa–Samuelson effect*, that as a country becomes richer "foreign investment flows into the country" and strengthens the currency (p. 205), is not the effect: Balassa–Samuelson says faster productivity growth in traded goods raises the relative price of non-traded goods and so the real exchange rate. GDP per capita adjusted for PPP corrects for price levels, not for inflation as he puts it. Meese and Rogoff's 1983 result concerned structural exchange-rate models against a random walk at short horizons, a wider statement than "PPP is not an accurate forecasting tool". LIBOR differentials are historical (reference rates). *Own out-of-sample test of one factor, relative PPP*. The hypothesis and pass rules were fixed in the script before its first run but were not preregistered.

- *Design.* Nine currencies against the dollar (JPY, GBP, CHF, CAD, AUD, NZD, SEK, NOK, DKK), on FRED's monthly H.10 *averages*. The signal $z_t$ is the deviation of the log real rate from its expanding mean.
- *Timing.* A month is eligible as a predictor only if its price levels were known at $t$: monthly CPI exactly two months old, or quarterly CPI four to six months past its quarter end, never older. Outcomes need only spot rates, not future CPI.
- *Forecasts.* Expanding regressions of the next 1- and 12-month spot change on $z_t$, using only outcomes already observed, were compared with a random walk from January 1985. The effective sample ends are August 2021 (JPY), May–June 2025 (the monthly-CPI currencies) and September 2025 (AUD, NZD).
- *Results.* Full-sample slopes are negative for all nine at 12 months, as PPP predicts. The forecasts lose to the random walk out of sample for every currency: pooled $R^2 = -2.0$ percent at 1 month and $-14.9$ percent at 12 (Clark–West $t = -1.3$ and $-0.2$).
- *Portfolio.* A monthly portfolio long the three cheapest and short the three dearest currencies returned +0.09 percent a year in spot terms, 1985 to May 2025 (Sharpe 0.01). Adding three-month interbank differentials gives −0.17 percent a year (2002–2025), before an assumed 5 bp per unit of turnover. Because it is computed from monthly averages, this is a statistical portfolio, not an executable trading result, and interbank rates are not the trader's funding or forward points.

Neither pass rule is met. *A second factor: carry*. Gliner's carry indicator is the short-rate differential (pp. 206–208). It was tested on the same FRED files with a protocol written before any carry result was computed. The protocol is not blind to the data, which the PPP test had used. The predictor is the previous month's three-month interbank differential against the dollar. The model is a pooled expanding regression of 1- and 12-month dollar exchange-rate changes on it, forecasting from January 1995.

- *Forecasts: not supported.* The pooled out-of-sample $R^2$ against a random walk is $-0.4$ percent at 1 month (Clark–West $t = -0.41$) and $-6.7$ percent at 12 (Clark–West $t = -1.97$), negative for every currency at 12 months. *Panel inference*: Newey–West lags must be counted in calendar months, not in stacked currency rows, since same-month currencies are contemporaneously dependent. The Clark–West scores keep their forecast-origin month, are summed across currencies within it (each currency-month keeps weight $1/N$, so the pooled $R^2$ is unchanged), and the long-run variance is taken over 11 calendar months (12 as a sensitivity: $t = -1.94$). The forecasts still lose to the random walk.
  The full-sample pooled slope is +0.0017 per percentage point at 12 months, against $-0.01$ under uncovered interest parity: high-rate currencies did not depreciate by the differential, on average. That is the forward-premium pattern, though it does not help forecast spot rates.
- *Portfolio: supported, narrowly.* Long the currencies above the cross-sectional median differential and short those below, earning spot change plus the interest accrual, net of 5 bp per unit of turnover. January 1985 to August 2026, the portfolio returned +2.07 percent a year (volatility 5.6 percent, Sharpe 0.37, Newey–West $t = 2.04$, against the threshold of 2). Its three worst months were March 1985, September 2008 and February 2020. The portfolio has one return a month, so its Newey–West $t$ uses 12 monthly lags.

So carry pays through the accrued differential, not through forecastable exchange-rate moves. The portfolio is a statistical return on monthly-average prices and interbank rates, not an executable trade: an executable return would use month-end (or trade-time) rates, forward points or funding actually paid, and spreads; none of these was measured. Its $t$ barely clears the rule. Limits: monthly averages rather than month-end rates; revised CPI, not vintages; dollar bases only; interbank rates in place of policy-rate or OIS differentials; two of Gliner's twelve factors, since the others need data (CDS, risk reversals, export shares) not used here.

**See also.** FX carry; external accounts; sovereign credit.

**Further reading.** Order flow and fundamentals in exchange-rate determination, and currency value [wide-menkhoff-value].

### Model 12.4 The FX carry trade: carry as the rate differential at a constant exchange rate, funding and target currencies, crowding and crash risk, and carry-to-risk ratios

**Statement.** "In currencies, carry is the return one makes on holding a currency, assuming that the exchange rate is held constant"; carry trades borrow in low-yielding currencies (yen, dollar, sterling in Gliner's time) and buy high-yielding ones (Australian and New Zealand dollars, Brazilian real, Turkish lira) [s7-gliner, chap. 7, sec. "Carry Indicators for Currency Valuation", PDF pp. 206–207]. "The carry trade is a very crowded trade, which means that in large risk off markets such as 2008, the strategy suffers big losses" (p. 207). *Risk-adjusted carry* divides three-month carry by 30-day volatility; Gliner's Table 7.6 (September 2013) ranks pairs within groups, NZD/EUR at 0.26 being the most attractive New Zealand cross, and he says risk-adjusted carry "tends to outperform standard carry trades" and needs frequent adjustment (pp. 208–209).

**Notation.** *Conventions*. $S_t$ is the spot rate quoted as domestic currency per unit of foreign currency; $i_d$ and $i_f$ are simple money-market rates, each quoted on its own day-count basis, and $\tau_d$ and $\tau_f$ are the accrual fractions for the same calendar period under those bases (for 90 days, 90/360 for an ACT/360 rate and 90/365 for an ACT/365 rate); $F$ is the forward for the period. *Exact relations*: covered interest parity,

$$\frac{F}{S_0} = \frac{1 + i_d \tau_d}{1 + i_f \tau_f};$$

and the funded long-foreign position (borrow one unit of domestic currency at $i_d$, buy $1/S_0$ of foreign currency, deposit it at $i_f$, convert back at $S_T$) earns, per initial domestic unit, the arithmetic P&L

$$\Pi = \frac{S_T}{S_0}(1 + i_f \tau_f) - (1 + i_d \tau_d).$$

Carry is $\Pi$ at an unchanged rate, $S_T = S_0$: exactly $i_f\tau_f - i_d\tau_d$, which equals $(i_f - i_d)\tau$ only when both rates share one basis. Hedging at $F$ gives $\Pi = (F/S_0)(1+i_f\tau_f) - (1+i_d\tau_d) = 0$ exactly, on either basis. *First-order forms*: with $s = S_T/S_0 - 1$, $\Pi = (i_f\tau_f - i_d\tau_d) + s + s\,i_f\tau_f$, so $\Pi \approx i_f\tau_f - i_d\tau_d + s$ after dropping the cross term; the forward premium $(F - S_0)/S_0 = (i_d\tau_d - i_f\tau_f)/(1 + i_f\tau_f) \approx i_d\tau_d - i_f\tau_f$.

*A different quantity*: the continuously compounded log excess return $x = \ln(S_T/S_0) + \ln(1 + i_f\tau_f) - \ln(1 + i_d\tau_d)$, equivalently $\ln(S_T/S_0) + (r_f - r_d)\tau$ with continuously compounded rates on a common $\tau$; it satisfies $e^x - 1 = \Pi/(1 + i_d\tau_d)$, a return per unit of domestic funding repaid, not the arithmetic P&L $\Pi$.

*Example* (90 days, $i_d = 5\%$ ACT/360, $i_f = 2\%$ ACT/365): $\tau_d = 0.25$, $\tau_f = 0.246575$, $F/S_0 = 1.0075314$; carry $-0.75685\%$, not the $-0.75\%$ a common basis would give; a 1 percent rise in $S$ gives $\Pi = +0.24808\%$ against $+0.24315\%$ to first order, and $x = 0.24472\%$. The carry-to-risk ratio divides annualised carry by annualised volatility.

**Conditions.** The realised arithmetic P&L is carry plus the exchange-rate change, to first order (above); uncovered interest parity predicts that the change offsets carry on average, and carry's profitability is the forward-premium puzzle. Carry-to-risk uses volatility as the risk measure and so ignores skewness and crash risk; the ranking by ratio can differ from the ranking by carry.

**Practice.** The usual carry risks (crowding, weekend gaps in EM currencies, policy regime shifts, correlated sell-offs of carry currencies against funders, illiquidity when exits are needed) are consistent with Gliner's 2008 remark but go beyond it. That exchange-rate moves often dominate carry returns, and that carry has positive average returns with significant negative skewness, are findings of the carry literature. The abstract of Brunnermeier, Nagel and Pedersen's working paper states that "exchange rate movements between high-interest-rate and low-interest-rate currencies are negatively skewed", attributes the skewness to "sudden unwinding of carry trades" when risk appetite and funding liquidity fall, and reports that carry losses reduce future crash risk but raise its price [catalogue-0241, abstract, NBER Working Paper 14473, November 2008, revised December 2013].

*The paper's body*: 3-month interbank rates and exchange rates against the dollar for eight developed currencies (AUD, CAD, JPY, NZD, NOK, CHF, GBP, EUR) from Datastream, 1986–2006 (PDF p. 8); equal-weighted portfolios long the $k$ highest-rate and short the $k$ lowest-rate currencies, rebalanced weekly (1992–2006) or quarterly (1986–2006), have skewness of $-0.717$ and $-0.700$ for $k = 1$, $-0.537$ and $-0.748$ for $k = 2$ and $-0.695$ and $-0.977$ for $k = 3$ (weekly and quarterly), with annualised Sharpe ratios between 0.59 and 0.78 and average quarterly returns of 1.6 to 2.2 percent (Table 2, PDF p. 12); the CRSP value-weighted U.S. stock index has skewness of $-0.41$ weekly and $-0.88$ quarterly over the same samples (p. 13), so carry skewness is "roughly comparable" to the stock market's, and it is not diversified away by adding currencies [catalogue-0241, working paper of November 2008, PDF pp. 8 and 12–13].

These are developed-market G10 samples to 2006, before the 2008 unwind. *A broader sample through the crisis* (authors' accepted version, February 2011). Menkhoff, Sarno, Schmeling and Schrimpf sort up to 48 currencies against the dollar (and a subsample of 15 developed ones, ten after the euro) into five portfolios by one-month forward discount each month, November 1983 to August 2009, net of bid–ask costs charged on entry and exit with about 30 percent monthly turnover [msss-2012, sec. II, pp. 8–9].

The high-minus-low carry portfolio earned 7.23 percent a year net with a standard deviation of 9.81 percent, skewness $-1.03$, kurtosis 4.79 and a Sharpe ratio of 0.74 for all countries, and 5.72 percent, 10.24, $-0.92$ and 0.56 for developed countries; skewness falls from portfolio 1 to 5 for all countries but not monotonically for developed ones (Table I, p. 45). Their pricing result is about volatility, not skewness: carry returns load negatively on innovations in global FX volatility, which price the five portfolios, while they "do not find that the skewness of a portfolio is robustly related to average excess returns" and aggregate-skewness betas are insignificant with low cross-sectional fit (sec. V.C, pp. 28–29).

So negative skewness of carry is confirmed in this sample, including 2008, but it is not what explains the cross-section there. The Sharpe ratios are consistent with the means and standard deviations (7.23/9.81 = 0.74, 5.72/10.24 = 0.56). "Measuring the standard error of the sample and finding the most attractive pairs" is Gliner's (p. 209), without method.

**Pitfalls.** Gliner's passage gives no evidence that the strategy outperforms standard carry trades. His Table 7.6 labels the carry "3M Carry (LIBOR)" and the volatility "30D Volatility", without saying that the volatility is realised.

**See also.** FX valuation signals; the dollar; gap risk.

**Further reading.** Carry crashes [catalogue-0241]; carry and global volatility [wide-menkhoff-carry]; [catalogue-0240] on common risk factors in currency returns.

#### Evidence 12.5 Deviations from covered interest parity after 2008: the cross-currency basis, its size and its quarter-end pattern

**Statement.** Du, Tepper and Verdelhan define the cross-currency basis $x_{t,t+n}$ by $(1 + y^{\$}_{t,t+n})^n = (1 + y_{t,t+n} + x_{t,t+n})^n\, S_t/F_{t,t+n}$, in logs $x_{t,t+n} = y^{\$}_{t,t+n} - (y_{t,t+n} - \rho_{t,t+n})$: the gap between the direct dollar rate and the synthetic dollar rate obtained by swapping a foreign-currency rate into dollars with forwards, zero when CIP holds (sec. 2.2, eqs. (3)–(4), printed p. 9).

For the G10 currencies, "The average annualized absolute value of the basis is 24 basis points at the three-month horizon and 27 basis points at the five-year horizon over the 2010–2016 sample"; the five-year yen basis was close to $-90$ basis points at the end of 2015, larger than the yen–dollar Libor differential (sec. 1, printed p. 2). They argue the deviations are not explained by credit risk or transaction costs, are larger for contracts that span quarter-ends, "pointing to a causal effect of banking regulation", and co-move with other fixed-income spreads and nominal rates [dtv-cip-2017, NBER WP 23170, February 2017, abstract and secs. 1–2, printed pp. 1–9].

**Notation.** $y^{\$}$ and $y$ dollar and foreign interest rates for maturity $n$; $S_t$ spot and $F_{t,t+n}$ forward in foreign currency per dollar; $\rho_{t,t+n}$ the forward premium, $\frac{1}{n}(\ln F - \ln S)$, per unit of maturity.

**Conditions.** Libor-based bases mix a bank-credit component into both legs; the paper also uses repo and OIS rates to argue that credit risk does not explain them. The sample is 2010–2016.

**Practice.** Carry or hedged-return arithmetic that uses interest differentials instead of forward points (as the FX tests in FX valuation signals do) omits the basis; with a basis of tens of basis points a year, a hedged foreign bond's return can differ from the interest-differential arithmetic by that much.

**Pitfalls.** Figures are from the working paper's abstract, introduction and definition sections, not from the 2018 *Journal of Finance* version.

**See also.** FX carry; money-market reference rates.

# 13 International and emerging equity markets

## 13.1 Equity indices and instruments for macro

**Definition 13.1 Equity indices and instruments for macro: major indices by region, price, capitalisation and equal weighting, and ETFs, ADRs, index futures and options as ways to trade them**

**Statement.** Gliner's Table 8.1 lists major indices by region: United States (S&P 500, Nasdaq, DJIA, Russell 2000, Wilshire 5000), Europe (Euro Stoxx 50, DAX, CAC 40, FTSE 100, FTSE MIB, IBEX 35, AEX), Asia ex-Japan (Shanghai Composite, Hang Seng, KOSPI, ASX 200, NZX 50), Japan (Nikkei, Topix, JASDAQ) and emerging markets (Bovespa, NIFTY, Micex and RTS, Mexico IPC, TOP 40 and JALSH, BIST 30 and 100, Jakarta Composite); indices "can be traded directionally or as a relative value trade", and their exposures differ, so "the Bovespa is more likely to be affected by commodities than the NIFTY, while the S&P 500 has more financial exposure than the Nasdaq, which is more technology-weighted" [s7-gliner, chap. 8, sec. "Equity Indices Overview", PDF pp. 214–215].

Price-weighted indices consider only prices and "can be strongly affected by the move of one" stock; capitalisation-weighted indices are dominated by the largest companies; "it is useful to know the weighting structure of each index in order to accurately interpret its meaning" (pp. 214–215). Instruments (sec. "Equity Derivatives", PDF pp. 218–221): ETFs, "a fast, cheap way to get exposures" that trade like stocks but charge fees, whose composition can differ from the index (EWZ held as much as 15 percent Petrobras, more than the Bovespa); ADRs, a depositary bank's holdings of a non-U.S. stock listed in the United States in dollars, "subject to foreign exchange risk".

**Notation.** Price-weighted: $I_t \propto \sum_i P_{i,t}$; capitalisation-weighted: $I_t \propto \sum_i N_{i,t} P_{i,t}$ (often free-float adjusted); equal-weighted: equal weights reset at each rebalance.

**Conditions.** Index lists, constituents and ETF holdings are as of 2013. *Two changes*. JASDAQ is no longer a market segment: "On April 4, 2022, TSE reorganized the former market segments (First Section, Second Section, Mothers and JASDAQ) into three new market segments: Prime, Standard and Growth" [jpx-market-restructure, market-structure page], so a JASDAQ index in Gliner's sense describes a market that no longer exists. Moscow Exchange now publishes the "MOEX Russia Index and RTS Index", listing "the MICEX Index" only as a registered trademark [moex-indices, index page]; Gliner's "Micex" is the MOEX Russia Index (the date of the renaming was not given on the page).

*Two weighting schemes*. The Nikkei 225 "is a price-weighted equity index, which consists of 225 stocks in the Prime Market of the Tokyo Stock Exchange", reviewed semi-annually and computed every five seconds [nikkei-225-profile, profile page]; TOPIX "is a free-float adjusted market capitalization-weighted index", with base value 100 on 4 January 1968 [jpx-topix, TOPIX page]. So Japan's two headline indices sit on opposite sides of the price-versus-capitalisation distinction above.

*Two more*. South Africa: Gliner's TOP 40 and JALSH are the FTSE/JSE Top 40 and All Share indices, two of seven "headline" indices in FTSE Russell's FTSE/JSE Africa series; the Top 40 is "the 40 most investable companies ranked by investable market capitalisation, in the FTSE/JSE All Share Index", and the headline indices "were harmonised effective 18 March 2024" to reflect the SWIX free float, the shareholder-weighted method of the rules' Appendix C [ftse-jse-ground-rules-2026, v9.3, September 2026, secs. 1.7.1 and 4.5.1, pp. 4 and 13]. United States: the Wilshire 5000 is now the FT Wilshire 5000, "a comprehensive, float adjusted measure of the U.S. stock market, designed to reflect the performance of all publicly traded U.S. equity securities that have readily available prices", with capitalisation-based size segments [ft-wilshire-5000-2026, WI-METH-001, January 2026, sec. 1.1, p. 4]; its name is not a constituent count.

*Two more*. India: "The Nifty 50 is a 50 stock, float-adjusted market-capitalization weighted index for India", its stocks representing "about 66%" of NSE's float-adjusted capitalisation [nse-nifty-methodology-2026, Methodology Document for Equity Indices, September 2026, PDF p. 19] (the same page's turnover sentence also says "float-adjusted market capitalization", apparently a slip).

Turkey: BIST indices weight by free-float market value, with a weighting factor below one only in capped versions; BIST 100 "automatically covers BIST 30 and BIST 50 stocks", and the capped variants bound weights at 10 or 25 percent with thresholds of 15 or 30 percent [bist-ground-rules-2020, Ground Rules, April 2020, secs. 2.2–2.3, 2.20 and 3.1, pp. 1–5], the edition current in October 2026. Hong Kong: the Hang Seng Index is "freefloat-adjusted market capitalisation weighted" with the number of constituents "fixed at 100". Individual weights are capped at 8 percent for non-foreign constituents and 4 percent for foreign ones, 10 percent in aggregate.

It is reviewed quarterly, launched 24 November 1969 with a base of 100 on 31 July 1964 [hsi-methodology-2023, Hang Seng Index and Sub-indexes methodology, version 2.3 of July 2023, PDF pp. 3, 5 and 6–7]. *Five more*. The Russell and FTSE rules came directly from FTSE Russell.

- The DJIA "is a 30-stock, price-weighted index", drawn from S&P 500 members outside the transportation and utilities groups (GICS 2030 and 55) [spdji-dj-averages-2026, May 2026, "Index Objective and Highlights" and "Eligibility Criteria", PDF pp. 4–5; capture of 27 September 2026]. It is the price-weighted case of the notation below.
- The Russell 2000 is companies ranked 1,001 to 3,000 by total market capitalisation, within "float-adjusted, market capitalization-weighted" indexes. Reconstitution is now semi-annual, "on the fourth Friday in June and the second Friday in December", with IPOs added quarterly [russell-us-construction-2026, v7.2, August 2026, secs. 1.1.1, 4.2.3 and 6.10, PDF pp. 3, 9 and 20].
- The FTSE 100 "will consist of the largest 100 UK companies by full market capitalisation (i.e. before the application of any investability weightings)". Constituents are then weighted with free-float and investability adjustments and reviewed quarterly [ftse-uk-ground-rules-2026, v17.3, September 2026, secs. 6.4 and 7.1.1, PDF pp. 14 and 16].
- The EURO STOXX 50 holds 50 Eurozone supersector leaders, weighted by free-float market capitalisation and capped at 10 percent quarterly. It is selected with a 40–60 buffer and reviewed annually in September [stoxx-index-guide-2026, Index Methodology Guide, October 2026, sec. 9.2, PDF pp. 100–101; capture of 16 October 2026].
- The DAX "includes the 40 largest companies in terms of market capitalisation on the Regulated Market of FWB" that meet quality and profitability requirements [dax-guide-2022, Guide to the DAX Equity Indices, version 11.3.0 of 3 August 2022, PDF p. 21; capture of 2 September 2025]. A later version may exist; none was captured.

*IBEX 35*: constituents carry free-float factors in bands (the notice moves Colonial from 60 to 40 percent and Indra from 80 to 100 percent), and "The individual weighting of stocks will be adjusted to fulfil the 20% limit", with a 40 percent limit in the bank, energy and construction sub-indices; changes applied from 23 December 2024, and only the Spanish text is legally binding [bme-ibex-notice-18-2024, Notice 18/2024, pp. 2–3]. So the IBEX 35 is free-float capitalisation-weighted with a 20 percent cap. The rules given above for the S&P 500, ASX 200, NZX 50, Mexico IPC, Nasdaq, CAC 40, AEX, FTSE MIB, KOSPI, Shanghai Composite, Bovespa and Jakarta Composite are unverified against current methodology documents.

**Practice.** For relative-value index trades, size legs by beta and volatility, and hedge or deliberately keep the currency exposure; for the section's outcome, compare markets in local currency and in dollars, hedged and unhedged.

**Pitfalls.** An ADR's dollar price follows the local share price converted at the exchange rate, so it carries currency risk for a dollar investor, not because the shares are held locally but because the underlying is priced in local currency. Index futures give liquid, leveraged exposure and options asymmetric payoffs; futures conventions are in CME Globex trading and ETF mechanics in ETF creation and redemption.

**See also.** equity-index valuation; VIX and variance swaps; global macro.

**Further reading.** The section's readings on market integration and cross-listing.

# 14 Portfolio construction and risk

## 14.1 The structure of a quant strategy and its alpha models

**Definition 14.1 Components of a quant trading strategy: alpha, risk and transaction cost models feeding portfolio construction, which interacts with execution, built on data and research**

**Statement.** Narang's schematic of a live, production trading strategy has three modules, "an alpha model, a risk model, and a transaction cost model", which "feed into a portfolio construction model, which in turn interacts with the execution model" [s7-narang, chap. 2, "An Introduction to Quantitative Trading", sec. 2.2, Exhibit 2.1, p. 17]:

- the *alpha model* predicts the future of the instruments considered, to generate returns;
- the *risk model* helps limit exposure to factors "unlikely to generate returns but could drive losses", for example a trend follower's limit on directional exposure to commodities (pp. 17–18);
- the *transaction cost model* estimates the cost of the trades needed to move from the current to the desired portfolio (p. 18);
- the *portfolio construction model* balances profit, risk and cost to choose the best portfolio to hold; the difference between the target and the current portfolio is the list of trades (Exhibit 2.2, p. 18);
- the *execution model* carries out those trades, using their urgency and liquidity dynamics as inputs, "in an efficient and low-cost manner" (p. 18).

The schematic covers only the production system; data, without which black boxes are "inert and useless", and research complete it (Exhibit 2.3, p. 19). Figure 14.1 redraws Exhibit 2.1.

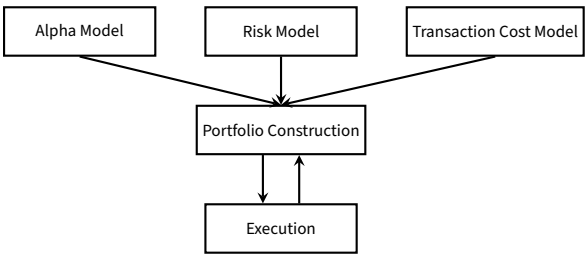



Figure 14.1: The modules of a live quant trading strategy, after Narang's Exhibit 2.1 (p. 17); data and research, his Exhibit 2.3, are not drawn.

**Conditions.** Practitioner framework as of 2024. Narang calls a trader a quant when both what to own and how much are decided systematically as standard practice, allowing emergency overrides (sec. 2.1, p. 16). The structure "is by no means universal": many strategies run without a transaction cost, portfolio construction or execution model, risk constraints can be built into the alpha model, and some links are recursive, such as executions feeding the cost model (p. 19).

**Practice.** Each module has its entries: forecasts in alpha models and their specification, combined in signal combination; risk in the factor risk model; costs in cost functions; the decision in rule-based sizing and mean–variance optimisation; execution in order exposure decisions; and the research and production chain around them in the strategy production lifecycle. Narang's Exhibit 2.2 trades follow from target minus current weights ($-25-(-30)=+5, 25-20=+5, 25-40=-15, -25-(-10)=-15$ percent).

**Pitfalls.** Narang's exhibit and text show portfolio construction and execution interacting, not a one-way flow. A directional exposure limit on an asset class is his single example of a risk model, not a definition. As in the factor risk model, the risk model informs the decision; the portfolio construction model selects and sizes.

**See also.** sources of strategy risk (model risk can arise in every module); strategy monitoring.

**Definition 14.2 Alpha models: theory-driven and data-driven, and Narang's eight classes of theory-driven alpha by input**

**Statement.** Narang defines alpha, "not conventional" by his own account, as "skill in timing the selection and/or sizing of portfolio holdings"; an alpha model is the software that does this timing systematically [s7-narang, chap. 3, "Alpha Models: How Quants Make Money", pp. 23–24]. Alpha models are *theory-driven*, starting from an economic explanation that is then tested, or *data-driven*, looking for patterns without a prior theory (sec. 3.1, pp. 25–28). Theory-driven alphas fall into eight classes, sorted by input (Exhibit 3.2, p. 29):

- *price data*: *trend* or momentum, explained by slow consensus-building or by "greater fools" (pp. 29–30); *mean reversion*, explained by temporary liquidity imbalances, such as index additions, and overshooting, with statistical arbitrage on similar stocks as its best-known form (pp. 33–35); and *technical sentiment*, from option volumes and skews, trading volume and the order book (pp. 35–37);
- *fundamental data*: *value/yield*, ratios with price in the denominator such as earnings yield, read as being paid to hold riskier instruments (pp. 38–41); *growth/sentiment*, including analysts' revisions (pp. 42–43); *supply/demand* (pp. 43–44); *quality*, from leverage, revenue diversity, management, fraud risk and quality sentiment (pp. 44–46); and *tactical/events*, one-off events such as elections or the pandemic (pp. 46–47).

Data-driven strategies are "far less widely practiced", more common at high frequency where data are abundant, and in his view mostly unsuccessful as forecasters of markets (sec. 3.3, pp. 47–52).

**Conditions.** Practitioner taxonomy as of 2024; anecdotes from 1970 to 2023. The classes are Narang's framework for organising strategies, not a result: he offers explanations for trend and reversion and cites little evidence: two papers for value/yield (Fama and French, n. 5; Ball, n. 6; pp. 37–40) and his reading of research on volume as a contrarian signal (p. 37), and otherwise examples and anecdotes. Trend and reversion coexist "largely" because they operate at different horizons, longer trends with shorter oscillations (Exhibit 3.5, pp. 35–36). He stresses that every class has "relatively long periods of poor returns" because the behaviours are "unstable and episodic" (p. 32), and that some strategies once called alpha, such as longer-term trend, are now treated by many as risk premia or "smart beta" (p. 24).

**Practice.** Interpretation: whether a class is used as an alpha to forecast, a premium to harvest or a risk to limit depends on the strategy, as in the factor risk model. His relative-value example combines value and financing: "when done on a relative basis … this strategy is also known as a carry trade", earning the yield spread (p. 40); with the euro at 4.25 percent and the dollar at 2 percent, the carry is 2.25 percent.

*Published evidence for four of the classes.*

- *Carry is not value.* Koijen, Moskowitz, Pedersen and Vrugt define an asset's carry as "its expected return assuming that its price does not change", measurable in advance without a pricing model [alpha-carry, NBER WP 19325, August 2013, abstract and sec. I, PDF pp. 2–3]. Carry predicts

returns in the cross-section and over time in global equities, bonds, currencies, commodities, Treasuries, credit and index options. A carry strategy within an asset class earns an annualised Sharpe ratio of 0.7 on average, and one diversified across classes 1.1 (p. 2).
Regressed on value, momentum and time-series momentum factors, carry keeps a significant alpha in every class. Equity carry loads positively on value without much loss of alpha. Commodity carry loads negatively on value and positively on momentum, and its alpha falls by 29 to 64 basis points a month, still significant. Currency and credit carry show no reliable loading (Table IV discussion, pp. 23–24). So Narang's "carry trade" for relative value is a distinct signal from value, though correlated with it in places.

- *Value.* Asness, Moskowitz and Pedersen find "consistent value and momentum return premia across eight diverse markets and asset classes". Value and momentum are negatively correlated with each other, within and across classes, and are partly linked to global funding liquidity risk [alpha-value, abstract, p. 929].
- *Quality and growth.* Asness, Frazzini and Pedersen define quality as characteristics investors should pay more for: profitability, growth and safety, with payout. A quality-minus-junk factor "earns significant risk-adjusted returns in the United States and across 24 countries", while high-quality stocks trade at only modestly higher prices [alpha-qmj, published version, abstract, p. 34]. Growth enters as one component of quality, not as a separate premium; the paper gives no evidence for a stand-alone growth premium.
- *Mean reversion in residuals.* Avellaneda and Lee trade mean reversion in stock residuals from PCA or sector-ETF factor models, assuming 5 basis points a trade. PCA strategies earn an annual Sharpe ratio of 1.44 over 1997–2007, but only 0.9 in 2003–2007; ETF strategies 1.1, with the same decline after 2002. Their August 2007 results agree with Khandani and Lo's unwinding account [alpha-residual, version of 15 June 2009, abstract and sec. 1, pp. 1 and 4].
*The trading rule*: the residual's distance from its fitted Ornstein–Uhlenbeck mean in equilibrium standard deviations is the *s-score*, $s_i = (X_i(t) - m_i)/\sigma_{\text{eq},i}$; they open a trade when $|s_i| > 1.25$, close longs when $s_i > -0.50$ and shorts when $s_i < 0.75$ (eqs. (15)–(16), pp. 18–19). The cutoffs were "selected … empirically, based on simulating strategies from 2000 to 2004", and the asymmetric short exit from 2000–2002, inside the 1997–2007 backtest period, so the reported Sharpe ratios include an in-sample choice of thresholds (p. 19).
*Where contrarian profits come from.* For the Lo–MacKinlay rule, which weights each stock by minus its lagged return relative to the equal-weighted market, expected profit splits into a cross-autocovariance term, an own-autocovariance term and the cross-sectional variance of mean returns. On 551 NYSE–AMEX stocks with complete weekly returns from 6 July 1962 to 31 December 1987, half the lag-1 expected profit comes from positive cross-autocovariances, about 67 percent in the central size quintile; they conclude that "less than 50 percent of the expected profits from a contrarian investment rule may be attributed to overreaction", with larger stocks leading smaller ones [wide-lo-mackinlay-contrarian, NBER WP 2977, May 1989, sec. 4, printed p. 17, and sec. 6, printed p. 22].
So residual reversal is partly a lead–lag effect, which a factor-residual construction may remove or keep depending on the factors. Gatev, Goetzmann and Rouwenhorst's distance-based pairs, 1962–1997, earn about 12 percent a year in excess for top pairs, "reduced—but still positive and significant" after bid–ask bounce and costs [alpha-gatev, NBER WP 7032, 1999, sec. I, p. 1].

**Pitfalls.** Narang's third edition distinguishes eight classes: trend, mean reversion, value/yield, growth, quality, technical sentiment, supply/demand and tactical/events. The sentence that a moving-average crossover "has less than one point of return for every point of downside risk" attaches to one rule what Narang says of "the typical successful trend follower" (p. 32), with no data.

*Evidence.* Moskowitz, Ooi and Pedersen find "significant 'time series momentum' in equity index, currency, commodity, and bond futures for each of the 58 liquid instruments", with persistence over one to 12 months that partly reverses later. Their diversified portfolio "performs best during extreme markets" [s6-tsmom, abstract, p. 228].

Each position is sized to an ex ante annualised volatility of 40 percent, $40\%/s_{t-1}$, where $s_t^2 = 261 \sum_i (1-\delta)\delta^i (r_{t-1-i} - \bar{r}_t)^2$ is an exponentially weighted variance of daily returns with centre of mass $\delta/(1-\delta) = 60$ days, the same for all assets, and the estimate at $t-1$ is applied to returns at $t$ to avoid look-ahead; the 40 percent is "inconsequential" and makes the equal-weighted factor about 12 percent volatile over 1985–2009 [s6-tsmom, secs. 2.4 and 4.1, eq. (1), pp. 233 and 236]. Hurst, Ooi and Pedersen extend an equal-weighted 1-, 3- and 12-month version, scaled to 10 percent ex ante volatility, to 67 markets from January 1880 to December 2016 [alpha-century, reprint PDF pp. 3–7, Exhibit 1 on PDF p. 4]:

- excess returns of 18.0 percent a year gross, 11.0 percent net of simulated costs, and 7.3 percent net of costs and a hypothetical 2-and-20 fee, at 9.7 percent volatility, a net Sharpe ratio of 0.76;
- positive net returns in every decade (lowest 1.6 percent, 1910–1919), and positive returns in 8 of the 10 largest drawdowns of a 60/40 portfolio;
- its own drawdowns "losing up to 25%".

Against Narang's sentence, a net 7.3 percent a year over a worst drawdown of about 25 percent is roughly 0.3 points of annual return per point of maximum drawdown (Interpretation; drawdown read from the text, not Exhibit 8's table). That is consistent with "less than one point" on that measure, while the volatility-based ratio is 0.76. Both papers are by AQR principals and are simulated, before any real fund's frictions beyond the cost and fee assumptions.

The "buying undervalued security and selling overvalued security is a carry trade" drops the context of his sentence, in which "this strategy" is the yield strategy, so that the long earns the higher yield; interpretation: value (cheapness against fundamentals) and carry (the return if prices do not change) are distinct signals in later work, though they can coincide. His performance figures for Seykota, Mint and Renaissance (pp. 31–32) are anecdotes with press or web sources. "In general, strategies that use data-mining techniques to forecast markets do not work" (p. 49) is his judgement.

**See also.** alpha strategy specification; signal combination; hypothesis-driven strategy research; parameter overfitting.

### Method 14.3 Specifying an alpha strategy: forecast target, time horizon, bet structure, investment universe, model specification, conditioning variables and run frequency

**Statement.** Narang lists the choices that turn an alpha idea into a strategy [s7-narang, chap. 3, sec. 3.4, pp. 52–63]:

1. *forecast target*: direction, magnitude and/or duration of a move, possibly with a confidence. Signal strength is the expected return and/or the confidence, and "in general, though certainly not always" a stronger signal means a bigger bet (pp. 52–53);
2. *time horizon*, from microseconds to over a year, classed as high-frequency, short-term (a day to two weeks), medium-term and long-term (pp. 53–55);
3. *bet structure*: an instrument by itself (intrinsic) or relative to others, in pairs or larger clusters, grouped statistically or heuristically (pp. 55–57);
4. *investment universe*: geography, asset and instrument class and exclusions, preferring liquidity, clean data and predictable behaviour (pp. 57–59);
5. *model specification*: the precise definition and parameters, and how often the model is refitted (pp. 59–60);
6. *conditioning variables*: modifying conditioners (stops, profit targets) and secondary conditioners requiring signals to agree (pp. 60–61);
7. *run frequency*: from monthly to continuous, trading noise and cost against large, market-moving rebalances and missed prices (pp. 61–62).

With two forecast types, four horizons, two bet structures and four asset classes, one idea gives $2 \times 4 \times 2 \times 4 = 64$ variants (pp. 62–63).

**Conditions.** Practitioner account as of 2024. Narang qualifies several items himself. Very large signals are rare, so the forecast–outcome relation may be less reliable there (p. 53). Differences between strategies are larger at short horizons, because small changes are amplified over many trades (pp. 54–55). Pairs are rarely precisely comparable, statistical groupings "can be fooled by the data" and heuristic ones are imprecise and rigid; grouping techniques suffer when regimes change (pp. 56–57). Refitting raises the risk of overfitting (p. 60). Most directional trend-following strategies make money on a minority of their trades, "often less than 40 percent" (pp. 60–61).

**Practice.** Interpretation: forecast strength becomes position size only through calibration and the portfolio decision. Sizing by strength presumes that strength is calibrated to realised returns (rank-based or equal sizing need not); size also depends on each instrument's risk, its correlation with

other positions, costs and constraints, so "bigger signal, bigger bet" holds for a calibrated forecast with other inputs equal (rule-based sizing, where Narang gives the case against unequal weights; mean–variance optimisation; cost functions). Specification choices made after looking at results are trials to be counted (parameter overfitting).

**Pitfalls.** "Relative alpha strategies tend to exhibit smoother returns during normal times than intrinsic alpha strategies, but … unique problems related to incorrect groupings during stressful periods" (p. 57) is a conditional generalisation with no data in the chapter. Narang's statement that machine-learning techniques are "designed to provide" a "scientifically valid way of testing … without overfitting" (p. 60) states a design aim, not a guarantee. Interpretation: searching many specifications is a source of overfitting, which no technique removes by its name (backtest diagnostics). His biotech example is that "some quants exclude" such stocks (p. 58); this is an example, not a general rule. The conditioning variables matter. "Relative value", he notes, is industry jargon for any relative bet structure, mostly unrelated to value (p. 57).

**See also.** alpha models; signal combination; hypothesis-driven strategy research; factor risk model (relative bets as risk control).

## 14.2 Portfolio return and risk: moments and diversification

The two entries in this section are the foundations used by the optimisers and risk models that follow. Their examples are invented, with stated units and sampling assumptions, and are not evidence about any market.

#### Method 14.4 Portfolio return moments: one-period value weights, expected return, variance, standard deviation, covariance and correlation of portfolios, from a probability model and from historical data

**Statement.** Markowitz writes the return on a portfolio as $R = \sum_i R_i X_i$, where $R_i$ is the return on security $i$ and $X_i$ "the percentage of the investor's assets which are allocated to the $i$th security". The $R_i$ are random variables; the $X_i$ are fixed by the investor, sum to one and, in his analysis, are not negative. The expected return and variance of the portfolio are then

$$E = \sum_{i=1}^{N} X_i \mu_i, \qquad V = \sum_{i=1}^{N} \sum_{j=1}^{N} \sigma_{ij} X_i X_j,$$

with $\mu_i$ the expected value of $R_i$ and $\sigma_{ij} = E\{[R_i - E(R_i)][R_j - E(R_j)]\} = \rho_{ij}\sigma_i\sigma_j$ their covariance, so that $\sigma_{ii}$ is the variance of $R_i$ [s6-markowitz, *Journal of Finance* 7(1), 1952, pp. 80–81]. The standard deviation is $\sigma = \sqrt{V}$ (p. 80).

For the covariance of two portfolios, the covariance of two weighted sums is $\mathrm{Cov}(\sum_i a_i X_i, \sum_j b_j Y_j) = \sum_i \sum_j a_i b_j \mathrm{Cov}(X_i, Y_j)$, and $\mathrm{Var}(aX + bY) = a^2\mathrm{Var}X + b^2\mathrm{Var}Y + 2ab\,\mathrm{Cov}(X, Y)$ [probability-statistics-random-processes, sec. 5.3.1, Lemma 5.3 and eq. (5.21)].

**Notation.** One period from time 0 to time 1, all amounts in one base currency. Asset $i$ has value $V_{i0}$ at the start and $V_{i1}$ plus income $D_{i1}$ at the end; its arithmetic total return is $R_i = (V_{i1} + D_{i1})/V_{i0} - 1$ (return measures). The portfolio starts at $V_0 = \sum_i V_{i0}$, and the *beginning-of-period value weights* are $w_i = V_{i0}/V_0$. In matrix form, with $\mu$ the vector of expected returns and $\Sigma$ the covariance matrix:

- $E[R_P] = w^\top \mu$ and $\mathrm{Var}(R_P) = w^\top \Sigma w$; for two assets, $w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1 w_2 \rho \sigma_1 \sigma_2$;
- $\mathrm{Cov}(R_P, R_Q) = w^\top \Sigma v$ for a second portfolio with weights $v$, and $\rho_{PQ} = w^\top \Sigma v / \sqrt{(w^\top \Sigma w)(v^\top \Sigma v)}$.

Variances are in squared return units (percent squared for returns in percent); standard deviations and expected returns are in return units per period. Interpretation: $R_P = \sum_i w_i R_i$ is exact because the end value is $\sum_i V_{i0}(1 + R_i) = V_0(1 + \sum_i w_i R_i)$.

*Model moments against estimates.* With states $s$ of probability $p_s$, the moments are probability-weighted sums (moments of random variables); they describe a model and are exact for it. From $N$ past periods, the sample mean, sample variances and covariances estimate them: dividing the sums of squares and cross-products by $N - 1$ gives the usual sample estimates, by $N$ the descriptive moments of the data themselves (location and dispersion; covariance and correlation). The identities above hold for a portfolio's sample moments as well, provided every term uses the same denominator (Interpretation).

**Conditions.** The weighted-sum formula assumes one period with no contributions or withdrawals inside it; income is reinvested or counted at the period end. All returns are measured in the base currency, so a foreign asset's return includes its currency return. Over several periods the weights drift with relative performance. A constant-weight series therefore describes a portfolio rebalanced to those weights at the start of each period, not one bought and held (Interpretation).

Log returns do not aggregate in this way: the log return of a portfolio is not the weighted sum of the assets' log returns (Interpretation). With short sales or borrowing, weights can be negative or exceed one, which Markowitz excludes (p. 81); the formulas still hold for value weights on net capital. Population moments are model assumptions; sample estimates carry estimation error that grows with the number of assets relative to the number of observations (covariance estimation).

**Practice.** Example 1, original, a one-year probability model in US dollars. A portfolio of 1,000,000 holds 600,000 in an equity fund E and 400,000 in a bond fund B, so $w = (0.6, 0.4)$. Total returns, in percent, are:

| State | Probability | E | B | Portfolio |
|---|---|---|---|---|
| Boom | 0.3 | 25 | −2 | 14.2 |
| Normal | 0.5 | 8 | 4 | 6.4 |
| Recession | 0.2 | −20 | 10 | −8.0 |

In the boom the end value is 600,000 × 1.25 + 400,000 × 0.98 = 1,142,000, a portfolio return of 14.2 percent, equal to 0.6 × 25 + 0.4 × (−2). The model gives $\mu = (7.5, 3.4)$, variances 243.25 and 17.64 (standard deviations 15.60 and 4.20), covariance −64.5 and correlation −0.985. Then $E[R_P] = 0.6 \times 7.5 + 0.4 \times 3.4 = 5.86$ percent. The variance is 87.57 + 2.8224 − 30.96 = 59.43, a standard deviation of 7.71 percent. Both equal the probability-weighted mean and variance of the portfolio column.

A second portfolio Q holds 30/70: $E[R_Q] = 4.63$, $\mathrm{Var}(R_Q) = 3.4461$, $\mathrm{Cov}(R_P, R_Q) = w^\top \Sigma v = 13.894$ and $\rho_{PQ} = 0.971$. Two portfolios of the same two assets can be highly correlated even when the assets are strongly negatively correlated.

Example 2, original, six invented months of total returns in percent: E 3.0, −1.5, 4.2, −2.8, 1.6, 0.9 and B 0.4, 0.9, −0.3, 1.2, 0.1, 0.5. The sample means are 0.90 and 0.467. Over $N - 1 = 5$, the variances are 7.048 and 0.2907 and the covariance −1.336; over $N$ they are 5.873, 0.2422 and −1.113. The correlation is −0.933 either way. Rebalanced to 60/40 each month, the portfolio returns 1.96, −0.54, 2.40, −1.20, 1.00 and 0.74. Their mean is $0.727 = w^\top \bar{x}$ and their sample variance $1.9425 = w^\top S w$, a monthly standard deviation of 1.39 percent. Bought and held instead, the weights drift to 0.606/0.394 after the first month and the second month's return is −0.555, not −0.540.

**Pitfalls.** Using end-of-period or target weights for a period in which the holdings drifted. Mixing currencies, or price and total returns, across assets. Averaging log returns across assets. Reporting a sample variance without its denominator, or computing a correlation with a covariance over $N - 1$ and standard deviations over $N$, which inflates it by $N/(N-1)$ (1.2 here). Treating six months of history as the population moments.

**See also.** diversification and correlation; covariance and correlation; mean–variance optimisation ($w^\top \Sigma w$ as an input); factor risk model.

**Mechanism 14.5 Diversification and correlation: how the portfolio standard deviation depends on correlation, when variance can reach zero, what diversification guarantees and does not, and why historical correlation need not persist**

**Statement.** For two assets with long-only weights $w_1, w_2 \geq 0$ summing to one,

$$\sigma_P^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho\sigma_1\sigma_2$$

[s6-markowitz, pp. 80–81] rises with the correlation $\rho$. At $\rho = +1$ it equals $(w_1\sigma_1 + w_2\sigma_2)^2$. For any $\rho < 1$ and both weights positive, $\sigma_P$ is below the weighted average $w_1\sigma_1 + w_2\sigma_2$ of the two standard deviations: that reduction is what imperfect correlation buys. At $\rho = -1$, $\sigma_P = |w_1\sigma_1 - w_2\sigma_2|$, which is zero at $w_1 = \sigma_2/(\sigma_1 + \sigma_2)$. At $\rho = 0$ or $+1$, no long-only portfolio of two risky assets has zero variance; at $\rho = +1$ a short position, $w_1 = -\sigma_2/(\sigma_1 - \sigma_2)$ when $\sigma_1 > \sigma_2$, is needed (Interpretation, from the formula).

Diversification does not remove all risk. Markowitz rejects the idea that the law of large numbers makes a diversified portfolio's yield almost certain: "The returns from securities are too intercorrelated. Diversification cannot eliminate all variance" (p. 79). With equal weights over $N$ assets of equal standard deviation $\sigma$ and common correlation $\bar{\rho}$, his double sum gives $\sigma_P^2 = \sigma^2[1/N + (1 - 1/N)\bar{\rho}]$, which tends to $\bar{\rho}\sigma^2$ as $N$ grows (Interpretation).

What that limit means depends on the sign of $\bar{\rho}$ (Interpretation). A fixed positive correlation leaves a positive floor $\bar{\rho}\sigma^2$, which is the situation Markowitz's sentence describes. At $\bar{\rho} = 0$ the variance is $\sigma^2/N$, which does tend to zero: uncorrelated returns would let the law of large numbers work. A fixed negative correlation cannot be kept as $N$ grows. For $N > 1$ the $N \times N$ matrix with ones on the diagonal and $\bar{\rho}$ elsewhere has eigenvalues $1 + (N - 1)\bar{\rho}$ and $1 - \bar{\rho}$, so it is a valid correlation matrix only when $-1/(N - 1) \leq \bar{\rho} \leq 1$. At the lower bound the equal-weight variance is exactly zero, and $\bar{\rho} = -0.1$, for example, is impossible for more than 11 assets.

Correlation measured in one period need not hold in another. Longin and Solnik show that sample correlations conditioned on large or small returns differ even when the true correlation is constant. Their example is a bivariate normal with correlation 0.50, split at the median absolute value of one return, 0.674: the conditional correlations are 0.21 for the small returns and 0.62 for the large. In monthly data on the five largest stock markets from 1958 to 1996 they reject multivariate normality for the negative tail but not the positive tail, and find that correlation rises in bear markets but not in bull markets [w2-longin-solnik, working-paper version of April 2000, abstract and pp. 1–2].

**Notation.** $\sigma_1, \sigma_2$ standard deviations, $\rho$ correlation, $w_i$ value weights; asset 2 is the safer one when $\sigma_2 < \sigma_1$. Interpretation: holding a little of the riskier asset lowers the standard deviation below the safer asset's only when $\rho < \sigma_2/\sigma_1$, because the derivative of $\sigma_P^2$ with respect to $w_1$ at $w_1 = 0$ is $2\sigma_2(\rho\sigma_1 - \sigma_2)$.

**Conditions.** The comparison with the weighted average holds for long-only weights; with shorting, $\sigma_P$ can exceed it. The results are statements about the model's standard deviation, a measure of dispersion; they say nothing about losses in the tails, which Longin and Solnik find are more correlated in falling markets. Common exposures, such as to the whole equity market, remain after diversification across many assets; how they are measured is in factor risk model, and whether investors are paid for bearing them belongs to asset-pricing models. Correlations estimated over one window, or conditioned on its large moves, are not the correlations of the next one (covariance and correlation).

**Practice.** Example, original: weights 0.6 in an asset with $\sigma_1 = 20$ percent and 0.4 in one with $\sigma_2 = 10$ percent, so the weighted average is 16 percent.

- $\rho = +1$: $\sigma_P = 16.0$, no reduction.
- $\rho = 0.3$: $\sigma_P = 13.74$. $\rho = 0$: $\sqrt{0.36 \times 400 + 0.16 \times 100} = 12.65$.
- $\rho = -1$: $\sigma_P = |12 - 4| = 8.0$, and a 1/3 weight in the riskier asset gives zero variance.
- At $\rho = 0$ and $\rho = 0.3$ the 60/40 mix is less risky than the weighted average but riskier than the 10 percent asset held alone. At $\rho = 0$, a 20 percent weight gives $\sqrt{80} = 8.94$ percent, below 10; here the threshold is $\rho < 0.5$.

Equal weights over $N$ assets with $\sigma = 20$ percent and $\bar{\rho} = 0.3$ give 20.0, 16.1, 12.2 and 11.1 percent for $N = 1, 2, 10$ and 100, against a floor of $20\sqrt{0.3} = 10.95$ percent.

**Pitfalls.** Reading diversification as a promise of lower risk than the safest holding. Expecting many holdings to remove risk that they share. Carrying a correlation from a calm sample into a crisis, or concluding that correlation has changed from a comparison of subsamples selected by return size. Using a negative correlation from a short sample, such as the six months of portfolio return moments, as if it were a property of the assets.

**See also.** portfolio return moments; mean–variance optimisation (instability of estimated correlations, Narang's rolling examples); noise in large correlation matrices.

## 14.3 Portfolio choice: asset classes, risk aversion and efficient portfolios

These entries build on portfolio return moments and diversification. The historical table is dated, single-country evidence. The three-fund example is invented, with one-year expected returns, standard deviations and correlations stated in US dollars; it illustrates mechanics and is not a forecast. Its inputs are $\mu = (8, 3.5, 6)$ percent and $\sigma = (18, 6, 12)$ percent for an equity fund E, a bond fund B and a property fund P, with correlations $\rho_{EB} = 0.1$, $\rho_{EP} = 0.6$ and $\rho_{BP} = 0.3$, and a risk-free rate of 3 percent where one is used.

**Definition 14.6 Asset classes and their characteristics: cash, government and corporate bonds, equities, real estate and gold; risk, return, income, liquidity and inflation exposure; dated historical returns against expected returns**

**Statement.** An asset class groups investments whose returns respond to similar risks. The SEC's investor guide names stocks, bonds and cash as the three major categories and lists real estate, precious metals and other commodities, and private equity as further categories with "category-specific risks". Stocks have historically had "the greatest risk and highest returns" of the three, and their volatility makes them "a very risky investment in the short term". Bonds are "generally less volatile than stocks but offer more modest returns", and high-yield bonds carry higher risk. Cash equivalents are the safest but offer the lowest return, and their principal risk is that inflation will "outpace and erode investment returns over time" [asset-allocation-investor-guide, pp. 1–3].

What distinguishes the classes (Interpretation, consistent with the sources cited):

- *Claim and cash flow*: cash and government bonds pay fixed nominal amounts; corporate bonds add default risk; equities are residual claims on uncertain profits; property pays rent; gold and other commodities pay no income, so their return is price change only.
- *Main risks*: inflation and reinvestment risk for cash; interest-rate (duration) risk for longer bonds; credit risk for corporate bonds; business, earnings and valuation risk for equities; local market, leverage and valuation risk for property.
- *Liquidity*: Treasury bills, Treasury bonds and large listed equities trade continuously at low cost. Direct property and private equity trade rarely and at high cost, and their reported returns rely on appraisals that can be smoothed (relative performance evaluation).

**Data.** Annual US-dollar nominal total returns, 1928–2025 (98 years), computed from Damodaran's January 2026 workbook [us-asset-returns-1928-2025, histretSP.xls, sheet "Returns by year"]. Percent a year. "Real" is the geometric mean deflated by CPI inflation, whose own geometric mean was 3.04 percent.

| | Arithmetic | Geometric | Std. dev. | Real geometric | Worst year |
|---|---|---|---|---|---|
| S&P 500 with dividends | 11.85 | 10.02 | 19.40 | 6.78 | −43.8 (1931) |
| 3-month T-bill | 3.41 | 3.37 | 3.04 | 0.33 | 0.0 (2014) |
| 10-year T-bond | 4.82 | 4.53 | 7.90 | 1.45 | −17.8 (2022) |
| Baa corporate bond | 6.90 | 6.63 | 7.65 | 3.49 | −15.7 (1931) |
| Residential real estate | 4.38 | 4.20 | 6.18 | 1.13 | −12.0 (2008) |
| Gold | 7.36 | 5.61 | 21.51 | 2.50 | −32.6 (1981) |

The arithmetic and geometric means reproduce the workbook's own summary rows. The standard deviations, real means and worst years are computed here from its annual series. The S&P 500 fell in 26 of the 98 years, and the T-bill never had a negative nominal annual return but lost to inflation in 40 years.

**Conditions.** *Construction of the series.* Damodaran's T-bill return is the average 3-month rate during the year. The T-bond and Baa returns are computed from yields on a constant 10-year maturity, and the real-estate series is the Shiller (now Case–Shiller) home-price index, which "is just price appreciation" and so understates the return by the omitted rent. Gold prices were fixed under the gold standard before 1971 [us-asset-returns-1928-2025, sheet "Explanations and FAQ" and note to "Returns by year"]. Each column is therefore a published index or a model of one, not an investable fund, and none is after costs or taxes.

*One country's realised path.* The figures describe the United States, a market that survived and prospered over the period; other countries' histories differ, and an investor's own currency changes the returns. The geometric mean is below the arithmetic mean by roughly half the variance, most for gold and equities.

*Historical evidence against expected returns.* A historical average estimates a mean return with error. The stock-minus-bond premium of 7.03 percent a year has a standard error of about 2.10 percent if annual returns are treated as independent, and the 2016–2025 means (stocks 15.75, T-bonds 1.22, gold 16.53 percent) differ greatly from the full-sample means (Interpretation). An expected return is a forecast made at a date. For bonds it is anchored by current yields. For equities it can come from an implied premium read from current prices, or from a historical premium on the assumption that the past premium is a guide to the future, with its estimation error stated (required return on equity).

Dependence also changes. In these annual data the stock–T-bond correlation was 0.15 in 1928–1999, −0.65 in 2000–2021 and 0.88 in 2022–2025, the last from only four observations (Interpretation).

**Practice.** Markowitz notes that the units of a mean–variance analysis "might represent individual securities or they might represent aggregates such as, say, bonds, stocks and real estate", and warns that relations among aggregates must be used with care [s6-markowitz, *Journal of Finance* 7(1), 1952, p. 91 and n. 15]. Interpretation: an allocation decision needs, for each class, a forward-looking expected return, volatility and correlations for the same horizon and currency, its liquidity in the investor's likely need, and its costs. The historical table informs the scale of these inputs and the depth of losses; it does not supply them.

**Pitfalls.** Reading a long-run average as next year's expected return. Comparing a price-only series (the real-estate column, or a stock index without dividends) with a total-return series. Treating cash as riskless in real terms. Reading the smooth returns of appraised assets as low risk. Taking the strong negative stock–bond correlation of 2000–2021 as permanent.

**See also.** return measures (arithmetic, geometric and real returns); risk aversion and utility; efficient frontier.

### Definition 14.7 Risk aversion and utility: expected utility, concavity, the certainty equivalent and risk premium, the Arrow–Pratt coefficient, the mean–variance utility $E(r) - \frac{1}{2}A\sigma^2$ and its units, and the consequences for portfolio selection

**Statement.** Under expected-utility theory an investor ranks uncertain outcomes by the expected value of a utility function $u$ of wealth, and "risk aversion is equivalent to the concavity of the utility function": a concave $u$ makes the utility of the expected outcome exceed the expected utility, so a fair bet is refused [risk-choice-review-2013, IDEI working paper 804, July 2013, p. 3]. The *certainty equivalent* is the sure amount with the same utility; the *risk premium* is the expected value minus the certainty equivalent. Pratt showed that for a small risk the premium is proportional to the Arrow–Pratt coefficient $r(w) = -u''(w)/u'(w)$, and that a more risk-averse decision maker pays more to eliminate a risk (pp. 13–14).

Markowitz reaches the same refusal from preferences over mean and variance. An investor whose utility $U(E, V)$ rises with $E$ and falls with $V$ "will never accept an actuarially fair bet", and a preference for the third moment would lead to accepting some fair bets [s6-markowitz, *Journal of Finance* 7(1), 1952, pp. 90–91].

**Notation.** The mean–variance utility of a portfolio return $r$ over one period is

$$U = E(r) - \tfrac{1}{2}A\,\sigma^2(r),$$

with $A > 0$ for risk aversion, $A = 0$ for risk neutrality and $A < 0$ for risk seeking. $U$ is itself a certainty-equivalent rate of return: the riskless rate the investor values equally. Returns are in decimals, so that $\sigma^2$ is in decimal squared; with returns in percent the formula becomes $U = E(r) - \frac{1}{2}A\sigma^2/100$. A dimensionless $A$ belongs to returns; for wealth $W$ with constant absolute risk aversion $a$ per currency unit, $A = aW_0$ (Interpretation).

Boyd et al.'s objective $\mu^\top w - \gamma w^\top \Sigma w$ uses $\gamma = A/2$ (mean–variance optimisation).

**Conditions.** *Why* $E - \frac{1}{2}A\sigma^2$. The representation is exact when utility has constant absolute risk aversion, $u(W) = -\exp(-aW)$, and wealth is normally distributed. Ross writes this expected utility as $-\exp\{-Aw\alpha'[E - \rho e] + \frac{\sigma^2}{2}(Aw)^2\alpha'\alpha\}$ for independent normal returns [arbitrage-theory-1976, *Journal of Economic Theory* 13, 1976, sec. I, eq. (9), p. 344], which depends only on the mean and the variance. Quadratic utility also makes expected utility a function of mean and variance alone, but with a risk penalty that changes with wealth rather than a constant $A$. Otherwise the formula is a second-order approximation that is good for small risks and degrades for large ones (Interpretation, from Pratt's small-risk result).

The model is one-period and ignores skewness and other higher moments, which Markowitz ties to gambling (p. 90). Expected-utility theory "is not unanimously accepted" as a normative model and is often violated in experiments [risk-choice-review-2013, p. 3]. $A$ is not observed: it is elicited from choices or set to match a stated tolerance for loss. The SEC guide separates risk tolerance, "your ability and willingness to lose some or all of your original investment", from the time horizon [asset-allocation-investor-guide, p. 1].

**Practice.** Examples, original. With $A = 3$, a portfolio with $E(r) = 7$ percent and $\sigma = 15$ percent has $U = 0.07 - 1.5 \times 0.0225 = 0.03625$: the investor values it like a riskless 3.625 percent. In percent units, $7 - 0.5 \times 3 \times 225/100 = 3.625$. The investor therefore prefers a riskless 4 percent; an investor with $A = 1$ (U = 5.875) prefers the portfolio.

With CARA utility, $a = 0.03$ per dollar and wealth $N(107, 15^2)$ dollars (so $A = 3$ on 100 dollars), the certainty equivalent obtained by numerical integration is $107 - 0.5 \times 0.03 \times 225 = 103.625$ dollars. For log utility ($r(w) = 1/w$, relative risk aversion 1) and a return of ±5 percent with equal probability, the exact risk premium is 0.1251 percent against the approximation $\frac{1}{2}\sigma^2 = 0.1250$; for ±40 percent it is 8.35 percent against 8.00.

*Implications for portfolio selection.* In a (standard deviation, expected return) diagram the indifference curves $E = U + \frac{1}{2}A\sigma^2$ slope upward and are convex for $A > 0$. The investor picks the attainable portfolio on the highest curve, so a more risk-averse investor holds a less risky portfolio of the same opportunity set (capital allocation line).

**Pitfalls.** Mixing percent returns with a coefficient calibrated for decimals, which changes the risk penalty a hundredfold. Equating $A$ in $E - \frac{1}{2}A\sigma^2$ with Boyd et al.'s $\gamma$. Treating the mean–variance ranking as exact for options, concentrated positions or other strongly skewed payoffs. Reading risk aversion as a fixed trait independent of wealth and horizon.

**See also.** efficient frontier; capital allocation line; asset classes.

### Method 14.8 Feasible portfolios and the efficient frontier: target-return minimum variance, the global minimum-variance portfolio, dominated portfolios, closed forms under a budget constraint, and boundary solutions and infeasible targets under short-sale limits

**Statement.** For given expected returns $\mu_i$ and covariances $\sigma_{ij}$, each choice of weights gives one $(E, V)$ combination, and the attainable combinations form a set. Markowitz's $E$–$V$ rule selects those "with minimum $V$ for given $E$ or more and maximum $E$ for given $V$ or less"; these are the efficient portfolios [s6-markowitz, *Journal of Finance* 7(1), 1952, p. 82].

With the budget $\sum X_i = 1$ and no short sales, the efficient set in weights is a series of connected line segments that runs from the portfolio of minimum variance to the portfolio of maximum expected return. Plotting $V$ against $E$ for efficient portfolios gives connected parabola segments (p. 87). Measuring risk by the standard deviation instead of the variance leaves the efficient set unchanged (p. 89).

Fama and French draw the *minimum variance frontier* for risky assets in (standard deviation, expected return) space. Only portfolios above its leftmost point are mean–variance efficient, "since these portfolios also maximize expected return, given their return variances" [capm-theory-evidence-2004, *JEP* 18(3), 2004, pp. 26–27 and Fig. 1].

**Notation.** $w$ portfolio weights, $1$ a vector of ones, $\Sigma$ the covariance matrix, $m$ a target expected return. Define $A = 1^\top\Sigma^{-1}1$, $B = 1^\top\Sigma^{-1}\mu$, $C = \mu^\top\Sigma^{-1}\mu$ and $D = AC - B^2$. Minimising $w^\top\Sigma w$ subject to $1^\top w = 1$ and $\mu^\top w = m$ by Lagrange multipliers gives (Interpretation)

$$w(m) = \Sigma^{-1}(\lambda 1 + \gamma\mu), \quad \lambda = \frac{C - Bm}{D},\ \gamma = \frac{Am - B}{D}; \qquad \sigma^2(m) = \frac{Am^2 - 2Bm + C}{D}.$$

The *global minimum-variance* (GMV) portfolio is $w_g = \Sigma^{-1}1/A$, with mean $B/A$ and variance $1/A$. The frontier is a hyperbola in (standard deviation, mean) space. Its upper branch, $m \geq B/A$, is the efficient frontier; the lower branch is dominated.

**Conditions.** The closed forms need $\Sigma$ positive definite, so that $\Sigma^{-1}$ exists and the minimiser is unique, and $\mu$ not proportional to 1, so that $D > 0$ (equal expected returns give $D = 0$ and a frontier collapsed to one mean). They allow unlimited short sales: weights may be negative and the frontier extends to every target $m$. They also assume a single period and inputs taken as known (Interpretation).

With long-only or other inequality limits the problem becomes a quadratic programme. If the unconstrained frontier portfolio for a target already satisfies the limits, it is also the constrained optimum: adding constraints that a minimiser already meets cannot produce a lower variance. When a limit binds, the solution sits on a boundary of the feasible set, with some weights held at their limits, and the constrained frontier lies to the right of the unconstrained one (Interpretation).

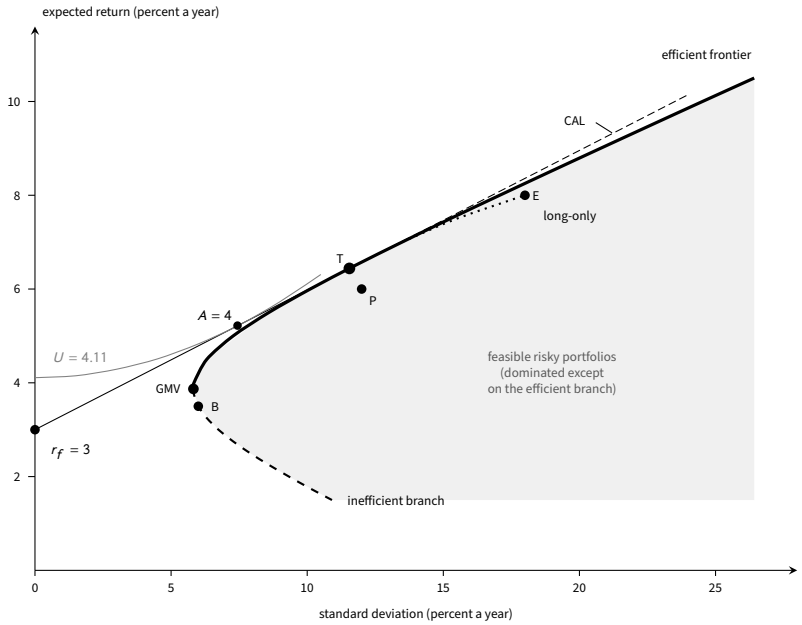
Each set of binding limits has its own closed form, valid over a range of targets. That is why Markowitz's efficient set is a series of connected segments that turn where a weight enters or leaves the solution (p. 87, n. 10). No single unrestricted formula covers these changes of the active set; constrained frontiers are computed by quadratic programming or by tracing the segments. A long-only target above the largest $\mu_i$ is infeasible, and so is any target that the limits exclude. A singular $\Sigma$ needs the treatment in mean–variance optimisation.

**Practice.** Example, original: the three funds of this section. $\Sigma$ is positive definite and $D > 0$. The GMV weights are $(0.054, 0.893, 0.053)$, with mean 3.87 and standard deviation 5.82 percent, below the bond fund's 6 percent: imperfectly correlated assets make a portfolio safer than its safest component.

| Target | $w_E$ | $w_B$ | $w_P$ | $\sigma$ | Long-only $\sigma$ |
|---|---|---|---|---|---|
| 5.0 | 0.222 | 0.578 | 0.200 | 7.28 | 7.28 |
| 6.0 | 0.372 | 0.297 | 0.331 | 10.11 | 10.11 |
| 7.0 | 0.521 | 0.017 | 0.461 | 13.48 | 13.48 |
| 7.5 | 0.596 | −0.123 | 0.527 | 15.25 | 15.49 |
| 8.0 | 0.671 | −0.263 | 0.592 | 17.07 | 18.00 |
| 9.0 | 0.821 | −0.543 | 0.723 | 20.77 | infeasible |

The closed-form weights agree with a numerical optimiser at each target. At 5 and 6 percent every weight is strictly positive, so the long-only optimum is the same portfolio and no limit binds. The bond weight is zero at a target of 7.06 percent and negative above it. With short sales excluded, the 7.5 percent solution sits on the boundary $w_B = 0$: it is $(0.75, 0, 0.25)$, with standard deviation 15.49 percent rather than 15.25. Its multiplier on $w_B \geq 0$ is positive, so the constraint binds. At 8 percent only the equity fund qualifies, and 9 percent cannot be reached at all.

*Dominated portfolios.* The frontier portfolio with mean 3 percent has standard deviation 6.74 percent; the one with mean 4.75 percent has the same standard deviation, so the first is dominated. No fund lies to the left of the frontier. The bond fund lies almost on its lower branch (a standard deviation of 6.00 against the frontier's 5.9999 percent at a mean of 3.5 percent), and the efficient portfolio with the same 6 percent standard deviation has a mean of 4.25 percent: holding the bond fund alone is dominated.




**Pitfalls.** Calling the lowest-volatility asset the minimum-variance portfolio. Choosing a portfolio on the lower branch. Quoting an unconstrained frontier for an account that cannot short, or a target beyond what its limits allow. Forgetting that the frontier is drawn from estimated inputs, so that an "efficient" portfolio is efficient only for those inputs (resampled efficiency; DeMiguel et al. in mean–variance optimisation). Markowitz plots $V$ against $E$ with $E$ horizontal (p. 82, Fig. 1) and Sharpe puts standard deviation on the vertical axis; most later diagrams, including this one, put risk on the horizontal axis.

**See also.** portfolio return moments; diversification; capital allocation line; mean–variance optimisation.

**Method 14.9 Adding a risk-free asset: the capital allocation line, the tangency portfolio, separation, the optimal allocation for a given risk aversion, borrowing and leverage assumptions, and the capital market line as an equilibrium special case**

**Statement.** A fraction $x$ in a risk-free security and $1-x$ in a risky portfolio $g$ gives $E(R_p) = xR_f + (1-x)E(R_g)$ and $\sigma(R_p) = (1-x)\sigma(R_g)$ for $x \le 1$, so such combinations plot on a straight line from $R_f$ through $g$. Points beyond $g$ represent "borrowing at the risk-free rate, with the proceeds from the borrowing used to increase investment in portfolio $g$". Swinging the line up to the tangency portfolio $T$ makes "all efficient portfolios … combinations of the risk-free asset (either risk-free borrowing or lending) and a single risky tangency portfolio", Tobin's separation theorem [capm-theory-evidence-2004, *JEP* 18(3), 2004, pp. 27–28 and n. 2].

Sharpe derives the same line and adds that when the borrowing rate equals the lending rate, choice divides into selecting the risky combination and then borrowing or lending to reach the investor's preferred point [capital-asset-prices-1964, *Journal of Finance* 19(3), 1964, pp. 432–433].

**Notation.** Write $y = 1 - x$ for the share in the risky portfolio. A *capital allocation line* (CAL) is

$$E(r_p) = r_f + \frac{E(r_g) - r_f}{\sigma_g}\,\sigma_p,$$

whose slope is the Sharpe ratio of $g$. The tangency portfolio maximises that slope. With a budget constraint only, it is $w_T = \Sigma^{-1}(\mu - r_f 1)/1^\top \Sigma^{-1}(\mu - r_f 1)$ (Interpretation). Maximising $U = r_f + y[E(r_T) - r_f] - \frac{1}{2}Ay^2\sigma_T^2$ gives

$$y^* = \frac{E(r_T) - r_f}{A\,\sigma_T^2};$$

$y^* < 1$ means lending, $y^* > 1$ borrowing (risk aversion and utility).

**Conditions.** *Tangency formula.* It needs $\Sigma$ positive definite and $r_f$ below the GMV mean $B/A$, so that $1^\top \Sigma^{-1}(\mu - r_f 1) = B - r_f A > 0$. Otherwise the normalised vector lies on the lower branch and no efficient tangency exists. Short-sale limits replace the formula by a constrained maximisation of the Sharpe ratio (Interpretation).

*Risk-free asset.* The asset must be riskless over the investor's horizon and in the investor's currency, which a nominal Treasury bill matching the horizon approximates in nominal terms only. Borrowing at the lending rate in unlimited amounts is a strong assumption. Fama and French describe unrestricted risk-free borrowing and lending as "an unrealistic assumption" (p. 29), and margin rules, financing spreads and leverage limits bound $y$ in practice. Sharpe argues that the assumption is less onerous than it seems: an investor who cannot borrow but wants a point below the tangency still chooses it and lends, and with borrowing limited only the most risk-tolerant investors choose otherwise (p. 433).

With a borrowing rate above the lending rate the efficient set has three parts. The lending line runs to its tangency; a stretch of the risky frontier follows; the borrowing line starts at a second, riskier tangency (Interpretation).

*CAL against CML.* Any line from $r_f$ through a risky portfolio is a CAL, and the best one depends on the investor's own estimates. The *capital market line* (CML) is the CAL through the market portfolio. Identifying the two needs equilibrium assumptions: a common pure rate for borrowing and lending and homogeneous expectations, which Sharpe calls "highly restrictive and undoubtedly unrealistic" (pp. 433–434). Prices then adjust until every asset enters a combination on the CML (p. 435).

Fama and French state the same assumptions as complete agreement on the true joint distribution of returns and borrowing and lending at a common risk-free rate (p. 26). With them "all investors see the same opportunity set", hold the same tangency portfolio, and market clearing makes it "the value-weight market portfolio of risky assets" (p. 28). In Black's version short sales replace the risk-free asset (pp. 29–30). Without these assumptions the market portfolio need not be any investor's tangency portfolio, and its proxies are discussed in CAPM.

**Practice.** Example, original: the three funds of this section with $r_f = 3$ percent. The GMV mean of 3.87 percent exceeds $r_f$, so the tangency exists: $w_T = (0.437, 0.175, 0.388)$, with $E(r_T) = 6.44$ and $\sigma_T = 11.55$ percent, and a Sharpe ratio of 0.298. A numerical maximisation of the Sharpe ratio returns the same weights, and the CALs through the GMV, the 7 percent frontier portfolio and a 60/20/20 mix are flatter.

| $A$ | $y^*$ | Position | $E(r_p)$ | $\sigma_p$ |
|---|---|---|---|---|
| 2 | 1.289 | borrow 28.9 percent of wealth | 7.43 | 14.89 |
| 4 | 0.645 | lend 35.5 percent | 5.22 | 7.44 |
| 8 | 0.322 | lend 67.8 percent | 4.11 | 3.72 |

All three investors hold the same risky mix and differ only in how much of it they hold. A grid search over $y$ confirms each optimum. The $A = 4$ point and its indifference curve, with $U = 4.11$ percent, are drawn in the efficient frontier figure.

*Borrowing at a higher rate.* Suppose lending is at 3 percent but borrowing costs 3.5 percent, still below the GMV mean. The borrowing tangency is $(0.951, -0.787, 0.836)$, with mean 9.87 and standard deviation 24.03 percent. The $A = 2$ investor would borrow on the lending line ($y^* = 1.29$) but would lend on the borrowing line ($y = 0.55$). That investor therefore holds a frontier portfolio with no riskless position: mean 7.18, standard deviation 14.11 percent and $U = 5.19$ percent, which here includes a small short in B. A 5 percent borrowing rate would exceed the GMV mean, and the tangency formula would return a lower-branch portfolio with mean 1.88 percent.

**Pitfalls.** Placing individual assets on the CML: only efficient portfolios lie on it, and single assets lie below it (systematic risk). Treating leverage as free or unlimited. Ranking portfolios by the Sharpe ratio for an investor who cannot borrow and wants more return than the tangency offers. Applying the tangency formula when $r_f$ is above the GMV mean. Calling a CAL built on one investor's estimates the CML.

**See also.** efficient frontier; risk aversion and utility; CAPM and the security market line; mean–variance optimisation (maximum-Sharpe objective).

## 14.4 Asset pricing: systematic risk, the CAPM and factor models

These entries take the capital allocation line to equilibrium and then to factor models. The regression example uses twelve invented months declared in CAPM regression; the cost-of-equity example uses inputs published for 1 October 2026. The other examples are invented, and none is evidence about a security or a market.

**Mechanism 14.10 Systematic and nonsystematic risk: splitting an asset's variance by its covariance with a common return, why diversification removes only the residual part, and the model-dependent claim that only systematic exposure earns a risk premium**

**Statement.** Sharpe regards the scatter of an asset's returns $R_i$ around their mean as evidence of its total risk. Part of the scatter is due to the asset's relation with the return $R_g$ of an efficient combination, measured by the slope $B_{ig}$; "it is this component of the asset's total risk which we term the *systematic* risk", and the remainder, uncorrelated with $R_g$, "is the unsystematic component". The part of an asset's risk due to its correlation with the combination "cannot be diversified away when the asset is added to the combination", so prices adjust until $B_{ig}$ is directly related to expected return [`capital-asset-prices-1964`, *Journal of Finance* 19(3), 1964, pp. 438–440 and Fig. 8].

Fama and French give the portfolio reading of beta. The market variance is a value-weighted average of the assets' covariances with the market, so "$\beta_{iM}$ is proportional to the risk each dollar invested in asset $i$ contributes to the market portfolio" [`capm-theory-evidence-2004`, *JEP* 18(3), 2004, pp. 28–29 and n. 3]. Huberman and Wang describe the covariance in both the CAPM and the APT as "a measure of risk that investors cannot avoid by diversification" [`arbitrage-pricing-review-2005`, FRBNY Staff Report 216, August 2005, sec. 1].

**Notation.** With $R_i = a_i + \beta_i R_M + \varepsilon_i$ and $\varepsilon_i$ uncorrelated with $R_M$,

$$\sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma_{\varepsilon_i}^2, \qquad \beta_i = \frac{\operatorname{Cov}(R_i, R_M)}{\sigma_M^2} = \rho_{iM} \frac{\sigma_i}{\sigma_M}.$$

The first term is systematic and the second residual (idiosyncratic, nonsystematic). The systematic share of variance is $\rho_{iM}^2$, which a regression reports as $R^2$ (Interpretation).

**Conditions.** *Why residual risk diversifies.* In an equally weighted portfolio of $n$ assets whose residuals are uncorrelated with each other and with bounded variances, residual variance falls like $1/n$ while the systematic part stays (diversification; factor risk model, where residual correlation shared by every pair behaves like an omitted factor).

*Why only systematic risk earns a premium.* The claim is a result of a pricing model, not a fact about returns. In the CAPM every investor holds the market portfolio, and an asset's contribution to that portfolio's risk is its covariance with it, so expected excess return is proportional to beta. In the APT, mispriced residual risk can be diversified away and combined into a near-arbitrage (APT). The conclusion therefore needs investors who are diversified across the relevant assets and a correct specification of the common factors (Interpretation).

Risk that is residual with respect to one market proxy may be systematic with respect to another factor that is priced. Fama and French read the size and value premia as compensation for undiversifiable covariances missed by the market return, a reading that behavioural explanations dispute (pp. 37–40). An investor who is not diversified, such as an owner-manager or a concentrated fund, bears residual risk in fact; the model says only that the market will not pay for it (Interpretation).

**Practice.** Example, original, annual percent: an asset with standard deviation 30 and beta 1.25 against a market with standard deviation 16 has systematic standard deviation $1.25 \times 16 = 20$ and residual standard deviation $\sqrt{30^2 - 20^2} = 22.36$; its correlation with the market is 2/3. Equal-weight portfolios of $n$ such assets with independent residuals have standard deviations of 30.0, 21.21, 20.12 and 20.01 for $n = 1$, 10, 100 and 1,000, approaching 20.

With the inputs of CAPM, a 5.29 percent risk-free rate and a 4.20 percent market premium, the SML gives the asset 10.54 percent. A portfolio on the capital market line with standard deviation 20, the asset's systematic part, has the same 10.54 percent. The capital market line at the asset's own 30 percent would imply 13.165 percent, which is not available for a single asset: its extra 22.36 percent of residual risk earns nothing in the model.

In the twelve-month sample of CAPM regression, 81.3 percent of the variation in the stock's excess returns is associated with the market's.

**Pitfalls.** Ranking assets for expected return by total volatility. Concluding that residual risk is irrelevant to an investor who is not diversified, or to an active manager judged against a benchmark. Measuring "systematic" risk against a narrow proxy and assuming it is the priced risk. Treating the pricing claim as an empirical finding: the CAPM's version of it fails in the tests reviewed in CAPM.

**See also.** return-generating models; CAPM and the SML; APT; factor risk model (Narang's "unintended is not the same as uncompensated").

**Model 14.11 Return-generating models: the market (single-index) model and multifactor models as statistical descriptions of returns, how they differ from equilibrium pricing restrictions, and their uses**

**Statement.** A return-generating model describes how returns vary with common factors; it does not by itself say what expected returns must be. Huberman and Wang write the factor model behind the APT as $r = \mu + \beta f + e$, with $n$ assets, $k$ factors, $E[f] = 0$ and $E[e] = 0$, so that $E[r] = \mu$. Pricing theory then adds a restriction on $\mu$, exactly $\mu = \iota\lambda_0 + \beta\lambda_1$ or approximately [`arbitrage-pricing-review-2005`, FRBNY Staff Report 216, August 2005, sec. 2, eqs. (1)–(3)].

Ross starts from the same kind of model, $\tilde{x}_i = E_i + \beta_i \tilde{\delta} + \tilde{\varepsilon}_i$ [`arbitrage-theory-1976`, *Journal of Economic Theory* 13, 1976, eq. (2), p. 342, and eq. (15), p. 347]. Sharpe's regression of an asset's return on that of an efficient combination is the single-index form, which he notes "has been called

the diagonal model" because it makes the covariance matrix easier to handle [capital-asset-prices-1964, *Journal of Finance* 19(3), 1964, pp. 438–439 and n. 23].

**Notation.** The *market model* for asset $i$ in period $t$ is

$$R_{it} = a_i + b_i R_{Mt} + \varepsilon_{it}, \qquad E(\varepsilon_{it}) = 0,\ \mathrm{Cov}(\varepsilon_{it}, R_{Mt}) = 0.$$

In the single-index version the residuals of different assets are also uncorrelated, so $\mathrm{Cov}(R_i, R_j) = b_i b_j \sigma_M^2$ for $i \neq j$. A multifactor return-generating model replaces $b_i R_{Mt}$ by $\sum_k b_{ik} F_{kt}$ (Interpretation).

**Conditions.** *Statistical against pricing.* The market model fits any intercept $a_i$. The Sharpe–Lintner CAPM restricts it: with excess returns the intercept of $R_{it} - R_{ft} = \alpha_i + \beta_{iM}(R_{Mt} - R_{ft}) + \varepsilon_{it}$ must be zero [capm-theory-evidence-2004, *JEP* 18(3), 2004, p. 32]. With a constant risk-free rate the raw-return intercept must then equal $(1 - \beta_i) R_f$ (Interpretation). A return-generating model can therefore describe returns well while the pricing restriction fails, and a pricing model can hold with residuals that are correlated across assets.

*Assumptions of the single-index model.* Residual correlation across assets, from industries for example, violates it. Markowitz notes that firms in the same industry tend to do poorly together [s6-markowitz, *Journal of Finance* 7(1), 1952, p. 89], and Fama and French name industry effects as a source of common residual variation (p. 31). Parameters estimated over one window need not hold in another, and the index used as "the market" is a proxy.

**Practice.** *Uses* (Interpretation, with the sources cited):

- Estimating beta for the CAPM (required return on equity).
- Building a covariance matrix with few parameters. For 500 assets a full covariance matrix has $500 \times 501/2 = 125{,}250$ distinct entries, while the single-index model needs 500 betas, 500 residual variances and one market variance, 1,001 in all.
- Separating market-related from residual returns, as in event studies and performance evaluation. Fama and French list three-factor intercepts used to calibrate price responses to news and to measure managers' information (p. 39).
- Hedging market exposure with the estimated beta.

In the twelve-month example of CAPM regression, the market model on raw returns gives $\hat{a} = 0.016$ and $\hat{b} = 1.257$ percent per percent, with $R^2 = 0.814$.

**Pitfalls.** Reading a market-model intercept as a CAPM alpha. Assuming residuals are independent across stocks of the same industry. Treating a fitted factor structure as evidence for a pricing model; Huberman and Wang stress that any test of the APT is a joint test of the factors and the pricing relation (sec. 6).

**See also.** systematic risk; CAPM; factor model types; factor risk model; simple regression.

### Model 14.12 The capital asset pricing model: assumptions, beta, the security market line, expected and required returns, the CML against the SML, the Black version, applications and the empirical record

**Statement.** The CAPM adds two assumptions to Markowitz's model, in which investors are risk averse and care only about the mean and variance of one-period returns. The first is complete agreement on the joint distribution of returns, which is the true one. The second is borrowing and lending at a risk-free rate that is the same for all investors and independent of the amount [capm-theory-evidence-2004, *JEP* 18(3), 2004, p. 26]. Every investor then holds the same tangency portfolio, which must be the value-weight market portfolio, and the market must lie on the minimum-variance frontier (p. 28). The Sharpe–Lintner relation follows:

$$E(R_i) = R_f + [E(R_M) - R_f]\,\beta_{iM}, \qquad \beta_{iM} = \frac{\mathrm{Cov}(R_i, R_M)}{\sigma^2(R_M)}$$

(pp. 28–29). It holds for portfolios as well as securities, since a portfolio's expected return and beta are the weighted averages of its assets' (p. 31, n. 4). Sharpe states the equilibrium conditions as a common pure rate of interest at which all investors can borrow or lend, and homogeneous expectations, assumptions he calls "highly restrictive and undoubtedly unrealistic" [capital-asset-prices-1964, *Journal of Finance* 19(3), 1964, pp. 433–434].

Black's version drops risk-free borrowing and lending and allows unrestricted short sales instead. The market is still efficient, and $R_f$ is replaced by the expected return on zero-beta assets, which need only be below the market's (Fama and French, pp. 29–30). With neither a risk-free asset nor short sales, the market portfolio is not typically efficient "and the CAPM relation between expected return and market beta is lost" (p. 30).

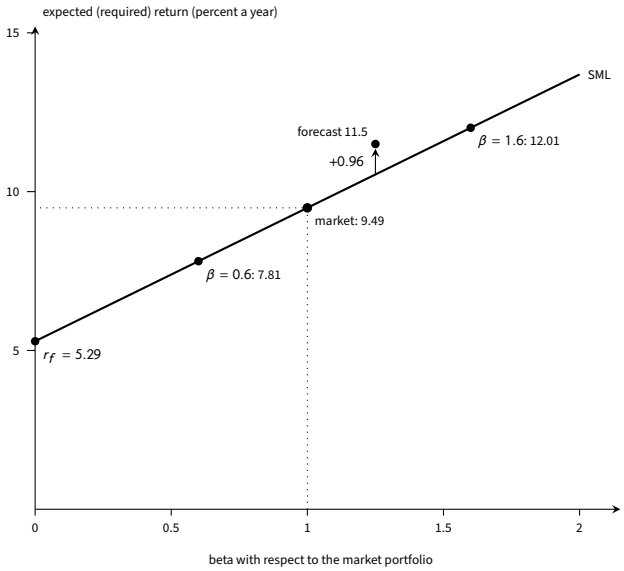
**Notation.** The *security market line* (SML) plots expected or required return against beta: intercept $R_f$, slope the market premium $E(R_M) - R_f$. The CML plots expected return against *standard deviation* and holds only for efficient portfolios. Every asset and portfolio lies on the SML if the CAPM holds; single assets lie below the CML, because part of their standard deviation is residual (systematic risk). Interpretation: a security's position against the SML is its alpha, $E(R_i) - [R_f + \beta_i(E(R_M) - R_f)]$.

**Conditions.** Usual further simplifications are a single period, no taxes or transaction costs, divisible assets, price-taking investors and a market portfolio that contains all risky assets (Interpretation). Fama and French note that the market portfolio "in principle can include not just traded financial assets, but also consumer durables, real estate and human capital" (p. 25). Roll's critique is that tests use proxies; Fama and French take the pragmatic view that a proxy on the minimum-variance frontier would serve, but no reasonable proxy has been found (pp. 41–42).

*Evidence.* The relation between average return and beta is too flat. For ten beta-sorted US portfolios over 1928–2003, the lowest-beta portfolio was predicted to earn 8.3 percent a year and earned 11.1, and the highest was predicted 16.8 and earned 13.7 (pp. 32–33). Size, earnings–price, debt–equity and book-to-market ratios add explanatory power beyond beta (pp. 35–36). For book-to-market portfolios in 1963–2003 average returns rose from 10.1 to 16.7 percent while beta did not (p. 42, Fig. 3). Their conclusion: "the CAPM's empirical problems probably invalidate its use in applications" (p. 44).

**Practice.** Example, original: forward-looking inputs of 1 October 2026, a risk-free rate of 5.29 percent and a market premium of 4.20 percent (required return on equity).

- *Required returns*: 7.81, 9.49, 10.54 and 12.01 percent for betas 0.6, 1.0, 1.25 and 1.6.
- *A portfolio* 50/30/20 in the 0.6, 1.0 and 1.6 assets has beta 0.92 and required return 9.154 percent, the same weighted average of the assets' required returns.
- *Security selection*: an analyst who forecasts 11.5 percent for the beta-1.25 stock places it 0.96 percentage points above the SML, a positive alpha relative to the CAPM. It looks underpriced only if both the forecast and the model are right.
- *Hedging and timing*: beta is the market exposure per unit of capital; selling index futures worth beta times the position removes the market component to the extent that beta is right.
- *Performance evaluation*: Jensen's alpha is the intercept of the excess-return regression (CAPM regression).
- *CML against SML*: with a market standard deviation of 16 percent, the CML would assign the stock's 30 percent volatility 13.165 percent; the SML assigns its beta 10.54.




**Pitfalls.** Putting standard deviation on the SML's axis or beta on the CML's. Reading a point above the SML as a guaranteed gain. Using the CAPM's required return for a value or small stock without regard to the evidence that it is biased (pp. 43–44). Treating a stock index as the market portfolio of the theory. Fama and French's footnote 7: with a market premium of 8.3 percent a year for 1927–2003 and a standard error of 2.4, the two-standard-error range of 3.5 to 13.1 percent makes most cost-of-equity estimates "rather meaningless" (p. 44).

**See also.** capital allocation line (CAL and CML); systematic risk; CAPM regression; required return on equity; APT and multifactor models; relative performance evaluation.

### Method 14.13 Estimating CAPM alpha and beta by regression: aligned excess returns, closed forms and an independent least-squares check, units, fit and uncertainty, raw-return against excess-return regressions, and why a fitted alpha is not skill

**Statement.** The Sharpe–Lintner CAPM implies a time-series test. When an asset's excess return is regressed on the market's excess return, "Jensen's alpha", the intercept, "is zero for each asset" [capm-theory-evidence-2004, *JEP* 18(3), 2004, p. 32]:

$$R_{it} - R_{ft} = \alpha_i + \beta_{iM}(R_{Mt} - R_{ft}) + \varepsilon_{it}.$$

Least squares estimates the coefficients as in simple regression: $\hat{\beta} = S_{xy}/S_{xx}$ and $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$, with $y$ the asset's and $x$ the market's excess return. Huberman and Wang give the same estimators for a factor regression [arbitrage-pricing-review-2005, FRBNY Staff Report 216, August 2005, sec. 7, eqs. (14)–(17)].

**Data.** Invented and declared: twelve months of total returns in percent, typed by hand, for a stock S, a market index M and a one-month Treasury bill whose rate rises through the year. Each row is the same calendar month for all three, and the bill rate is the return on a bill held over that month. S was built as $r_f + 0.1 + 1.25(R_M - r_f)$ plus hand-chosen residuals, rounded; the estimates do not use this.

| Month | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $R_S$ | 3.6 | −3.3 | 2.6 | 2.8 | −3.6 | 0.5 | 6.6 | −0.2 | −0.8 | −3.0 | 3.7 | 1.6 |
| $R_M$ | 2.1 | −1.4 | 3.0 | 0.8 | −2.6 | 1.9 | 4.2 | −0.7 | 1.1 | −3.3 | 2.5 | 0.6 |
| $R_f$ | 0.30 | 0.31 | 0.32 | 0.33 | 0.35 | 0.36 | 0.38 | 0.39 | 0.40 | 0.41 | 0.42 | 0.42 |

**Practice.** *Estimates.* $\hat{\beta} = 1.2548$ and $\hat{\alpha} = 0.1108$ percent a month. The closed forms, a matrix least-squares solution and a regression package agree, and $\hat{\beta}$ equals the sample covariance of the excess returns divided by the market's sample variance with either denominator. The residual standard deviation is 1.447 percent a month and $R^2 = 0.813$. The standard errors are 0.190 for $\hat{\beta}$ and 0.422 for $\hat{\alpha}$, with 10 degrees of freedom. The 95 percent intervals are 0.83 to 1.68 for beta and −0.83 to 1.05 for alpha ($t = 0.26$).

*Interpretation.* Beta is unit-free: in this sample a month in which the market's excess return was one percentage point higher came with a stock excess return about 1.25 points higher on average. Alpha is in percent a month; twelve times it is 1.33 percent a year, with an interval of −9.95 to 12.61. $R^2$ is the share of the sample variance of the stock's excess return associated with the market: 81 percent systematic and 19 percent residual (systematic risk).

*Raw against excess returns.* Regressing raw $R_S$ on raw $R_M$, the market model, gives $\hat{a} = 0.0158$ and $\hat{b} = 1.2574$. The intercepts differ by −0.0949, of which $(1 - \hat{\beta})\bar{R}_f = -0.0932$ comes from the risk-free rate's place in the model and −0.0017 from its variation over the year. With a constant risk-free rate the two slopes coincide and $\hat{a} = \hat{\alpha} + (1 - \hat{\beta})R_f$ exactly. A raw intercept is thus not Jensen's alpha: for a high-beta asset it is lower by about $(\beta - 1)R_f$, and the gap grows with the level of rates (Interpretation).

**Conditions.** *Alignment.* The three series must cover the same periods, use total returns in the same currency, and use a bill rate for the same holding period, known at its start. The market index is a proxy, and its choice changes beta. The standard errors assume the conditions in regression diagnostics: errors that are independent and of constant variance. Return data often violate both, and nonsynchronous trading biases betas estimated from short intervals (Interpretation).

*Uncertainty.* Twelve observations are far too few. Fama and French call individual betas "imprecise" (p. 31) and note that large standard errors of betas and of the premium make cost-of-equity estimates "rather meaningless" (p. 44, n. 7). Betas also change over time; Harris measures them concurrently for this reason (relative performance evaluation).

*Alpha is not skill.* A fitted alpha measures the average excess return not associated with the market proxy in that sample, given the model. It can reflect luck: here the interval spans more than 22 percentage points a year. It can also reflect the model's failure: funds tilted to low-beta, small or value stocks "will tend to produce positive abnormal returns" against the Sharpe–Lintner CAPM "even when the fund managers have no special talent for picking winners" (p. 44). Testing for skill needs a credible benchmark and the inference in skill against luck.

**Pitfalls.** Regressing raw returns and calling the intercept alpha. Mixing a stock's total return with a price-only index. Using an annual bill yield in a monthly regression. Annualising a monthly alpha without its uncertainty. Reporting beta without its interval or estimation window.

**See also.** return-generating models; CAPM; required return on equity; simple regression; regression diagnostics.

**Method 14.14 Estimating a required return on equity with the CAPM: a forward-looking risk-free rate, equity premium and beta, the market model's role, sensitivity, and the difference from a historical average or fitted return**

**Statement.** The textbook prescription that Fama and French describe is "to estimate a stock's market beta and combine it with the risk-free interest rate and the average market risk premium to produce an estimate of the cost of equity" [capm-theory-evidence-2004, *JEP* 18(3), 2004, p. 43]:

$$k_e = r_f + \beta_i \text{ ERP}.$$

The required return is the expected return investors demand for bearing the equity's risk. It is a forward-looking rate for discounting the firm's expected cash flows, and "the cost of capital is forward looking, so if the premiums are sample specific they are irrelevant" (p. 40).

**Data.** Inputs for 1 October 2026, in US dollars, nominal, percent a year:

- *Equity risk premium*: Damodaran's implied premium on 1 October 2026 was 4.20 percent on his base method (trailing twelve-month cash flows with adjusted payout). His four other variants gave 3.70, 5.67, 3.61 and 3.30 percent [implied-equity-premium-2026, home page, "Equity Risk Premiums", read 2026-10-18].
- *Risk-free rate*: he computed those premia against a US Treasury rate of 5.29 percent, and noted that netting out a 0.22 percent default spread would give 5.07 percent (same page). The Treasury's 10-year par yield was 5.29 percent on 30 September 2026 [ust-par-yield-curve, 2026 file, retrieved 2026-10-18].
- *Beta*: 1.15, an assumption of the example, standing for a forward-looking beta that an analyst would justify from a longer estimation window, comparable firms or an adjustment; it is not the twelve-month estimate.

**Practice.** $k_e = 5.29 + 1.15 \times 4.20 = 10.12$ percent. Across the published premia, 3.30 to 5.67 percent, it ranges from 9.09 to 11.81 percent. Holding the premium at 4.20, the twelve-month beta interval of 0.83 to 1.68 (CAPM regression) gives 8.78 to 12.34 percent.

*A historical premium instead.* The 1928–2025 arithmetic average excess return of US stocks over 10-year Treasury bonds was 7.03 percent (asset classes). Added with the same beta to the current 10-year yield, it gives $5.29 + 1.15 \times 7.03 = 13.38$ percent; the geometric-average premium of 5.48 percent gives 11.60. This is a standard CAPM estimate, not a mismatch: the premium was measured over 10-year bonds and is added to the current 10-year yield.

The two estimates differ in their assumption about the premium. The implied premium is read from current prices and expected cash flows; the historical premium assumes that the premium realised in the past is a good estimate of the premium expected now. Damodaran presents the choice as a point of view: to keep a valuation market neutral, use the current implied premium; "if you believe that numbers revert back to historical averages, use the historical premium" [corporate-finance-project-notes, "Project Emails/Information", sec. V.b "Risk Premiums", read 2026-10-18].

The historical estimate is also imprecise. Treating the years as independent, the arithmetic premium has a standard error of about 2.10 percentage points. A band of two standard errors (2.83 to 11.24 percent) moves $k_e$ from 8.54 to 18.21 percent, and the choice between arithmetic and geometric averages alone moves it by 1.8 points (Interpretation). Fama and French report a standard error of 2.4 points for the 1927–2003 market premium over Treasury bills (p. 44, n. 7).

*The market model's role.* The regression of the stock's returns on the market's supplies beta. Its intercept is not part of the required return, and the fitted return $\hat{a} + \hat{b}R_M$ reproduces the sample's average return. In the twelve-month example that average is $0.875 \times 12 = 10.50$ percent a year. It is a description of one past year, whose closeness to 10.12 is coincidence, not an estimate of what investors require (Interpretation).

**Conditions.** The rate, premium and beta must refer to the same currency, horizon and market, and the premium must be defined against the kind of risk-free rate that is added back (Interpretation). Damodaran's project notes take the 10-year government bond rate in the currency of the analysis as the riskless rate (sec. V.a), and advise against estimating a historical premium for a non-US company from its own market's history (sec. V.b).

*For a historical premium*, the matching rate is the current yield on the security over which the excess returns were measured, here the 10-year Treasury bond. Adding a bond-based premium to a bill rate, or the reverse, mixes benchmarks. The approach also assumes that past realised premia inform the expected premium. Its estimation error, and its sensitivity to the sample period, the averaging method and the country, are part of the estimate (Interpretation).

*For an implied premium*, the matching rate is the one used to solve for it. Damodaran's 4.20 percent was computed against 5.29 percent. Pairing it with another rate, or with his spread-adjusted 5.07 percent without also adding the 0.22 percent spread to the premium, changes $k_e$ mechanically with no change in view; that is a different error from choosing a historical premium. A long-term government yield suits long-lived equity cash flows better than a bill rate, and a nominal rate makes $k_e$ nominal, to be applied to nominal cash flows (Interpretation).

The estimate inherits the CAPM's empirical problems. Fama and French find CAPM costs of equity "too high (relative to historical average returns)" for high-beta stocks and too low for low-beta and value stocks (pp. 43–44). Multifactor alternatives, such as the three-factor model offered by practitioners for the cost of equity (p. 39), or the APT used in utility rate cases [arbitrage-pricing-review-2005, FRBNY Staff Report 216, August 2005, sec. 9], give different answers depending on the factors chosen.

**Pitfalls.** Treating the result as precise to a basis point. Adding a premium to a different risk-free rate from the one it was measured or solved against: a bond-based premium to a bill rate, or an implied premium to a rate other than its own. Reporting a historical premium without its period, averaging method and standard error, or as if the past premium must recur. Using the trailing sample average return as the required return. Using a raw twelve-month beta for a firm whose business or leverage has changed.

**See also.** CAPM and the SML; CAPM regression; asset classes (historical against expected returns).

**Model 14.15 Multifactor pricing and arbitrage pricing theory: factor exposures, factor risk premia, residual risk, expected returns from sensitivities, exact arbitrage among well-diversified portfolios and residual-risk-bearing approximations**

**Statement.** Ross's heuristic for the APT starts from a factor model in which residuals are independent enough for the law of large numbers to hold [arbitrage-theory-1976, *Journal of Economic Theory* 13, 1976, pp. 342–343]:

- form an *arbitrage portfolio* that uses no wealth and is well diversified;
- for large $n$ its residual return becomes negligible;
- choose it to have no systematic risk, so that its return is certain;
- to rule out arbitrarily large positions, that certain return must be zero.

Expected returns are then spanned by a constant and the factor sensitivities, $E_i = \rho + \lambda\beta_i$, where $\rho$ is the riskless rate or, without a riskless asset, the zero-beta return. The argument suggests the relation holds "in all but the most profound sort of disequilibria", and "the market portfolio plays no special role" (p. 343). With $k$ factors, an asset's premium is "the $\beta$-weighted sum of the factor risk premiums" $E^k - \rho$, where $E^k$ is the return on a portfolio with unit sensitivity to factor $k$ and none to the others (p. 353, eq. (24)).

**Notation.** In the notation of Huberman and Wang, $r = \mu + \beta f + e$, and exact arbitrage pricing is $\mu = \iota\lambda_0 + \beta\lambda_1$. $\beta$ holds the *factor exposures* (loadings or sensitivities), $\lambda_1$ the *factor risk premia*, the expected return per unit of exposure, and $e$ the *residual* return. The APT in general asserts only the approximate form $(\mu - X\lambda)' Z^{-1} (\mu - X\lambda) \leq a$ for a constant $a$, with $X = (\iota, \beta)$; exact pricing is the case $a = 0$ [arbitrage-pricing-review-2005, FRBNY Staff Report 216, August 2005, sec. 2, eqs. (1)–(3)].

**Conditions.** *Exact and approximate.* Huberman formalises arbitrage as a sequence of zero-cost portfolios whose expected income grows without bound while its variance vanishes. If residual covariances are diagonal and bounded, no arbitrage implies the approximate bound, and the exact relation holds in the limit for well-diversified portfolios, whose residual variance is zero (sec. 4). Ross proves that the sum of squared pricing deviations stays bounded as the number of assets grows (Theorem I, eq. (18), p. 347). For any finite set of assets, individual mispricings are therefore allowed.

His rough calibration for 1962–1971 put the average absolute discrepancy below 0.084 percent, "about 1% of the market return" (pp. 353–354).

*Further assumptions.* The bound needs agents whose relative risk aversion is uniformly bounded. Ross's counterexample, a single agent with constant absolute risk aversion, shows that pricing need not converge to the APT without such a condition (secs. I–II, pp. 344–347). The APT does not need mean–variance preferences, normal returns or the market portfolio. It needs less agreement than the CAPM: investors must share expectations and the factor sensitivities, but may disagree about the factors' distributions (p. 355).

The theory does not name the factors, so "any test of the APT is a joint test that the factors are correctly identified and that the linear pricing relation holds" (Huberman and Wang, sec. 8). It concerns static portfolios, so using it to evaluate dynamically managed portfolios is "contradictory to the no-arbitrage spirit of the model" (abstract; sec. 9). Multifactor pricing can also come from equilibrium: in Merton's ICAPM investors care about covariances with state variables, which requires betas on further factors as well as the market [capm-theory-evidence-2004, *JEP* 18(3), 2004, pp. 37–38].

**Practice.** *Expected return from sensitivities*, original, annual percent: with $\lambda_0 = 4$ and premia of 3 for factor 1 and 1.5 for factor 2, a stock with sensitivities $(1.2, -0.4)$ should return $4 + 3.6 - 0.6 = 7.0$ percent, a premium of 3.0 over $\lambda_0$. The negative sensitivity lowers the required return: the stock pays off when factor 2 is bad for other holdings.

*An arbitrage*, original. Well-diversified portfolios A, B and C have sensitivities $(1.0, 0.5)$, $(0.5, -0.5)$ and $(1.5, 1.0)$ and expected returns 7.75, 4.75 and 10.00 percent; solving the three pricing equations recovers $\lambda_0 = 4$, $\lambda = (3, 1.5)$. A fourth well-diversified portfolio D with sensitivities $(0.8, 0.2)$ should earn $4+2.4+0.3 = 6.70$ percent but is priced to earn 7.50. Weights $(1.0, 0.2, -0.2)$ in A, B and C have D's sensitivities, sum to one and earn 6.70.

Buying 1,000,000 of D and selling 1,000,000 of that replica costs nothing and has no factor exposure. If D and the replica have no residual risk, it earns a certain 8,000 a year, an exact arbitrage that the APT rules out; prices must move until D yields 6.70.

*Residual risk.* If D is instead one stock with a residual standard deviation of 25 percent, the same trade has expected profit 8,000 and standard deviation 250,000: it is a bet, not an arbitrage. Spread over $n$ such stocks with independent residuals and the same mispricing, the residual standard deviation of the position falls to $25/\sqrt{n}$ percent, 2.5 at $n = 100$ and 0.5 at $n = 2{,}500$, small but never zero. This is why the APT restricts prices only approximately for individual assets (Interpretation). Factor sensitivities are themselves estimated, so in practice even a well-diversified version carries model risk.

*Empirical premia.* Fama and French's three-factor model, $E(R_{it}) - R_{ft} = \beta_{iM}[E(R_{Mt}) - R_{ft}] + \beta_{is}E(\mathrm{SMB}_t) + \beta_{ih}E(\mathrm{HML}_t)$, is described as in the spirit of both the ICAPM and the APT. Over 1927–2003 the average premia were 8.3, 3.6 and 5.0 percent a year, with standard deviations of 21.0, 14.6 and 14.2: "high volatility implies substantial uncertainty about the true expected premiums" (pp. 38–39).

**Pitfalls.** Confusing a factor sensitivity with a factor premium, or a factor's realised return with its expected premium. Calling a trade with residual risk an arbitrage. Concluding from any set of factors that fit average returns that the APT is confirmed: a mean–variance efficient portfolio always exists and can serve as a "factor" (Huberman and Wang, sec. 6).

**See also.** return-generating models; factor model types; CAPM; systematic risk; factor risk model.

### Definition 14.16 Macroeconomic, fundamental and statistical factor models: inputs, estimation, outputs and interpretation, their uses in risk, attribution, pricing and construction, the value of several risk dimensions, and their limitations

**Statement.** Connor divides multifactor models of returns into three types [factor-model-types-1995, *Financial Analysts Journal*, May–June 1995, pp. 42–43 and Table 1]:

- *Macroeconomic* models use observable economic time series, such as inflation, industrial production, the excess return on long-term government bonds and the premium of low-grade over high-grade corporate bonds, as measures of the pervasive shocks. Each security's factor betas are estimated by time-series regression, which "requires a long and stable history of returns".
- *Fundamental* models use observed company attributes, such as size, dividend yield, book-to-market and industry, as the factor betas. The factors are the realised returns of mimicking portfolios, estimated each period by cross-sectional regression: the dividend-yield factor is "the realized return per extra unit of dividend yield, holding other attributes constant".
- *Statistical* models derive both factors and betas from returns alone by factor analysis or principal components. Their factors are "essentially unlabeled statistical artifacts" that can be linearly recombined.

Without estimation error and with unlimited data the three are "rotations of one another" and can all hold at once (p. 43).

Huberman and Wang describe three ways of identifying factors. The first is an algorithmic analysis of the covariance matrix. The second is a researcher's judgement guided by it, as in the size and book-to-market factors of Fama and French. The third is pure judgement, as in Chen, Roll and Ross's financial and macroeconomic variables. Only the first two respect the factor structure, and the third is "more in the spirit of Merton's (1973) intertemporal CAPM than in the spirit of the APT" [arbitrage-pricing-review-2005, FRBNY Staff Report 216, August 2005, sec. 8].

**Data.** Connor's comparison covers 779 large-capitalisation US stocks, January 1985 to December 1993, with explanatory power measured as one minus average specific variance over average total variance (pp. 44–45, Tables 2–5):

- five statistical factors explain 39.0 percent, the first alone 29.0;
- five macroeconomic factors explain 10.9 percent, needing both the default and the term-structure factors;
- the fundamental model of 55 industry dummies and 12 risk indices explains 42.6 percent, the industries alone 16.3;
- the macroeconomic model adds nothing once the fundamental model is included.

He adds that by theoretical consistency and intuitive appeal the macroeconomic model "is probably the strongest" of the three (p. 46).

**Conditions.** *Comparison* (Interpretation, from the sources cited and factor risk model):

- *Macroeconomic.* The factors are economic surprises, which must be measured as unexpected changes, and the exposures are estimated. The model is interpretable, linked to theory and useful for macro scenarios. Its limitations are low explanatory power, error in measuring surprises, and betas that need long, stable histories.
- *Fundamental.* The exposures are observed and current, and the factor returns are estimated. The model responds immediately to changes in a firm and its explanatory power is high. It depends on vendor definitions and many inputs, and its industry factors carry much of the fit.
- *Statistical.* The model needs returns data only and fits the sample best. Its factors are unlabeled, defined only up to sign and rotation, and unstable across samples.

All three are return-generating descriptions. Pricing requires the further restriction of APT or an equilibrium model, and evidence that a factor explains covariance does not show that it carries a premium.

**Practice.** *Uses and outputs.* Common uses (Interpretation):

- Risk decomposition into factor and specific parts, with contributions by factor.
- Return attribution.

- Control or hedging of unintended exposures in construction (factor risk model).
- Expected returns and costs of capital. Huberman and Wang report macroeconomic and Fama–French factor models proposed for utility rate cases in New York, where the commission used the CAPM instead (sec. 9).
- Performance evaluation with factor alphas, and asset allocation with fewer parameters, since $k$ factor portfolios span the efficient frontier when the factor model holds (sec. 9).

Example, original, monthly percent, a fundamental model with market, size and value factors. A portfolio's exposures are $(1.0, -0.3, 0.6)$ against its benchmark's $(1.0, 0, 0.1)$, so its active exposures are $(0, -0.3, 0.5)$. In a month with factor returns 2.0, −0.8 and 1.2, the factors explain $0 + 0.24 + 0.60 = 0.84$ percentage points of active return. If the realised active return was 1.10, 0.26 is specific to the holdings.

With factor covariance matrix rows $(16, 2, -1)$, $(2, 9, 0.5)$ and $(-1, 0.5, 6.25)$ and a specific variance of 4.0, the portfolio's factor variance is $x^\top F x = 16.48$ and its total 20.48, so factors account for 80.5 percent. The contributions $x_k(Fx)_k$ are 14.80 from the market, 0.12 from size and 1.56 from value. The output says the portfolio's risk is mostly market risk and its size tilt barely matters for risk, though it added 0.24 to this month's return.

*Several risk dimensions.* Two portfolios with the same market beta of 1.0 but inflation sensitivities of +0.5 and −0.5 look identical to the CAPM. A month with a market factor return of −2 and an inflation factor return of +1 moves them −1.5 and −2.5 percent. Modelling several dimensions shows such differences. It allows a risk budget or hedge per source, and it matches investors who care about covariances with state variables such as labour income or inflation, as in the ICAPM [capm-theory-evidence-2004, *JEP* 18(3), 2004, pp. 37–38].

**Pitfalls.** Naming a statistical factor after the fact and treating the name as a property of the method. Reading a factor's explanatory power as evidence that it is priced. Adding factors until a sample is explained: Huberman and Wang note that "the number of factors, as well as the methods of factor construction, is exploding" (sec. 8). Comparing exposures from different models as if their factors were the same.

**See also.** factor risk model (the covariance decomposition and economic against statistical factors); APT and multifactor models; return-generating models; covariance estimation.

## 14.5 Portfolio construction models: rules and optimisers

### Method 14.17 Rule-based position sizing: equal position, equal risk, alpha-driven and decision-tree weighting, with Narang's case against unequal weights

**Statement.** Narang divides quantitative portfolio construction models into two families: *rule-based* models, heuristics "derived from human experience, such as by trial and error", which can be "exceedingly simple or rather complex"; and *optimisers*, which search for the best value of an objective function [s7-narang, chap. 6, "Portfolio Construction Models", pp. 99–100] (the second family is in mean–variance optimisation). He names four rule-based types (sec. 6.1, p. 100):

- *equal position weighting*: every position the alpha model deems worth holding gets the same size; signal strength is ignored "except insofar as the signal is strong enough to be worthy of a position at all", and the instruments are implicitly assumed homogeneous in risk (sec. 6.1.1, pp. 100–101);
- *equal risk weighting*: sizes "inversely to their volatilities (or whatever other measure of risk, such as drawdown, is preferred)", so that each position is equalised "by the amount of risk that the allocation contributes to the portfolio" (sec. 6.1.2, p. 102);
- *alpha-driven weighting*: size grows with the forecast, under limits from the risk model on single positions and groups, for example 3 percent a position and 20 percent a sector (sec. 6.1.3, pp. 103–104);
- *decision-tree weighting*: "a defined set of rules in a particular order to determine position sizing", named on p. 100 only; the third edition has no section on it.

**Conditions.** Practitioner account as of 2024; the risk-contribution arithmetic is timeless. *Why some quants weight equally.* Narang gives two problems with *unequal* weighting, which the equal-weighters avoid (pp. 101–102). First, it assumes enough statistical power to forecast not only direction but also the relative size or probability of moves. Second, it leads to "a few large bets on the 'best' forecasts and many smaller bets on the less dramatic forecasts", and so to excess idiosyncratic risk: the strongest momentum or reversal signals often come where prices have already moved most, sometimes on real information (his "adverse selection bias"), and large forecasts can come from bad data, such as a price quoted in pence instead of pounds. Winsorising forecasts can partly address this; equal weights need no such adjustment.

*Equal risk weighting* relies on a backward-looking risk measure: his example is bank stocks, stable for a decade and then more volatile than many technology stocks in 2008 (p. 103). *Alpha-driven weighting* relies on the alpha model being right about size as well as direction; trend followers using it "can suffer sharp drawdowns relatively frequently", holding the largest position as the trend reverses (p. 104). Costs can enter any rule, for example by setting a forecast to zero when it is below its expected transaction cost (sec. 6.1.4, p. 104; cost functions). Liquidity can cap any weight (p. 102).

**Practice.** Interpretation: his Exhibit 6.1 (p. 102) gives volatilities of 2.5 and 2.0 percent and "volatility-adjusted" weights of 39 and 61 percent. Those are inverse-*variance* weights, $(1/2.5^2)/(1/2.5^2 + 1/2.0^2) = 39.0\%$; inverse-volatility weights would be 44.4 and 55.6 percent, which give each stock the same $w_i\sigma_i$ (1.11) while the table's weights do not (0.98 and 1.22). Neither scheme in general equalises the contribution to *portfolio* risk, $w_i(\Sigma w)_i / w^\top \Sigma w$, defined when the portfolio variance $w^\top \Sigma w$ is positive, and dependent on correlations. For two assets inverse-volatility weights do equalise it at any correlation, but not for more.

With three assets of 20 percent volatility, where the first two correlate at 0.8 and the third is uncorrelated, equal (inverse-volatility) weights give risk contributions of 39, 39 and 22 percent. Equal contributions need weights of about 29.93, 29.93 and 40.15 percent, $a = 1/(2 + \sqrt{1.8})$ for the first two and $a\sqrt{1.8}$ for the third. Narang's "contributes to the portfolio" therefore describes equal-risk-contribution weighting, which uses the full covariance matrix, more than the inverse-volatility rule he gives.

**Pitfalls.** Narang's concern about a few large bets on the best forecast and many smaller bets is an argument against unequal weighting: it describes concentration that equal weighting avoids. Narang's choice between rules is his observation: rule-based users are mostly futures traders with "intrinsic" forecasts, and optimiser users are mostly equity market-neutral traders with relative forecasts (sec. 6.4, p. 123). He gives no data for this.

**See also.** mean–variance optimisation (the other family); signal-portfolio allocation (rule-based signal portfolios mixed by an optimiser); cost functions (costs as inputs); common-investor contagion (risk targeting under rising volatility).

**Further reading.** Equal risk contribution and volatility targeting (equal risk contribution).

### Method 14.18 Equal risk contribution: equalising each asset's share of portfolio volatility under a full covariance matrix, against inverse-volatility weights

**Statement.** For long-only weights $x$ and covariance matrix $\Sigma$, with $\sigma(x) = \sqrt{x^\top \Sigma x}$, Maillard, Roncalli and Teïletche define the risk contribution of asset $i$ as $\sigma_i(x) = x_i\, \partial\sigma/\partial x_i = x_i(\Sigma x)_i/\sigma(x)$. Because $\sigma$ is homogeneous of degree one, the contributions sum to $\sigma(x)$ [wide-maillard-roncalli-erc, authors' version, May 2009, sec. 2.1, p. 4]. The *equal risk contribution* (ERC) portfolio is the one in $[0,1]^n$ with $\sum x_i = 1$ and $x_i(\Sigma x)_i = x_j(\Sigma x)_j$ for all $i, j$ (eqs. (1)–(2), pp. 4–5). Its weights are $x_i = \beta_i^{-1}/n$, with $\beta_i = (\Sigma x)_i/\sigma^2(x)$ the asset's beta to the portfolio. This is endogenous, so in general there is no closed form (eq. (5), p. 6). Two cases are closed-form:

- with two assets, $x_1 = \sigma_1^{-1}/(\sigma_1^{-1} + \sigma_2^{-1})$ whatever the correlation (sec. 3.1, p. 5);
- with one common correlation, $x_i = \sigma_i^{-1}/\sum_j \sigma_j^{-1}$, the inverse-volatility portfolio (eq. (3), p. 5).

Otherwise inverse-volatility weights do not equalise contributions. The ERC solution is unique for a positive-definite $\Sigma$, since it solves $\min \sqrt{y^\top \Sigma y}$ subject to $\sum \ln y_i \geq c$, $y \geq 0$, then rescaled (eq. (7), p. 7). The three heuristics compare as equal weights ($x_i = x_j$), minimum variance (equal marginal contributions) and ERC (equal total contributions), with volatilities ordered $\sigma_{mv} \leq \sigma_{erc} \leq \sigma_{1/n}$ (sec. 3.4, pp. 8–9, proof in their Appendix A.3).

*When ERC maximises the Sharpe ratio*. The source says that "the ERC portfolio is optimal if we assume a constant correlation matrix and supposing that the assets have all the same Sharpe ratio" (sec. 3.5, p. 9). Its argument checks the first-order condition $\mu - r = \frac{\mu(x)-r}{\sigma(x)} \Sigma x/\sigma(x)$, which holds for *any* common Sharpe ratio $s$, so it finds a stationary point, not a maximum. Own derivation: let $\Sigma = DCD$ with volatilities $D > 0$ and one correlation $\rho \in (-1/(n-1), 1)$, so that $\Sigma$ is positive definite, and let $\mu_i - r = s\sigma_i$. Over long-only, fully invested weights, $SR(x) = s\,DR(x)$, where $DR(x) = \sum_i x_i\sigma_i/\sigma(x) \geq 1$ is maximised by inverse volatility, which is ERC here. Then:

- if $s > 0$, ERC maximises the Sharpe ratio, and is also the tangency portfolio $\Sigma^{-1}(\mu - r)$ normalised;
- if $s = 0$, every portfolio has Sharpe ratio zero;
- if $s < 0$, ERC *minimises* it among long-only portfolios, and the maximum, $s$, is at a single asset.

For example, two uncorrelated assets of unit volatility, each with expected excess return $-1$: ERC has Sharpe ratio $-\sqrt{2}$, either asset alone $-1$. The correct condition is a constant correlation, equal individual Sharpe ratios *and* a positive common excess Sharpe ratio.

*Their backtests*. Monthly rebalancing on the last trading day, covariances from one year of daily returns, Sharpe ratios over the Fed funds rate:

- ten U.S. sector indices, January 1973 to December 2008, similar volatilities and high correlations: Sharpe ratio 0.62 (1/n), 0.77 (minimum variance), 0.65 (ERC), with ERC close to 1/n (Table 2, p. 13);
- light agricultural commodities, January 1979 to March 2008: 0.27, 0.74 and 0.49 (Table 4, p. 14);
- a global diversified set of equity, bond, credit and commodity indices, January 1995 to December 2008: 0.27, 0.49 and 0.67, with volatilities 10.9, 3.2 and 4.9 percent (Table 6, p. 17); the authors attribute much of 1/n's result to its 2008 drawdown (n. 7).

These are in-sample illustrations on three universes ending in 2008, not a test; ERC sits between the other two in volatility each time, and its Sharpe ratio is highest in only one case. *Risk parity as Qian put it* (`s16-r24`, PanAgora, September 2005). With stock and bond volatilities of 15.1 and 4.6 percent and correlation 0.2 (Russell 1000 and Lehman Aggregate excess returns, 1983–2004), "stocks contributed 93% of risk" in a 60/40 portfolio; the equal-risk mix is 23/77, with Sharpe ratio 0.87 against 0.67 for 60/40, and it must be levered to reach 60/40's volatility (pp. 1–3) [`s16-r24`, pp. 1–3]. These inputs give 92.7 percent and 23.4/76.6; and 48/24/16/12 percent for Maillard, Roncalli and Teïletche's four-asset example.

Qian's statement that risk-parity portfolios "are expected to generate superior return for a given level of targeted risk" is a claim from one 22-year sample in which bonds had the higher Sharpe ratio (0.80 against 0.55).

**Notation.** $x$ weights; $\Sigma$ covariance matrix; $\sigma_i(x)$ total and $\partial\sigma/\partial x_i$ marginal risk contribution; $\beta_i$ the beta of asset $i$ to the portfolio.

**Conditions.** Long only, as the authors restrict it: without that constraint several portfolios satisfy the ERC condition (p. 7). The risk measure is volatility from one estimated $\Sigma$, so the weights inherit its estimation error (covariance estimation). No expected returns enter.

**Practice.** Example with four assets of 10, 20, 30 and 15 percent volatility:

- contributions sum to $\sigma(x)$, and the ERC solution satisfies $x_i = \beta_i^{-1}/n$;
- with a common correlation of 0.3, ERC equals inverse volatility (40, 20, 13.3, 26.7 percent);
- with unequal correlations (0.8 between the first two, $-0.3$ between the last two), ERC is 33.9, 14.8, 16.4 and 34.9 percent. The inverse-volatility weights then put 32.8, 36.2, 17.2 and 13.8 percent of risk on the four assets;
- in both cases $\sigma_{mv} \leq \sigma_{erc} \leq \sigma_{1/n}$ (7.8, 9.2 and 11.3 percent in the second).

Narang's "equal risk weighting" is inverse volatility, or inverse variance in his exhibit (rule-based position sizing). It equals ERC only in the two closed-form cases above: two assets, or equal correlations.

**Pitfalls.** Section 4's backtests (above) are illustrations, not out-of-sample evidence across samples. The copy is the authors' version on Roncalli's site, dated May 2009, before the 2010 *Journal of Portfolio Management* article.

**See also.** rule-based position sizing; mean–variance optimisation; PCA risk-distribution weights.

**Further reading.** `wide-roncalli-risk-parity` on risk budgeting and risk parity.

### Evidence 14.19 Volatility-managed portfolios: scaling exposure by the inverse of recent variance, its in-sample alpha, its leverage, and why the gain does not survive real-time implementation

**Statement.** Moreira and Muir scale a factor's excess return by the inverse of its recent variance, $f^{\sigma}_{t+1} = \frac{c}{\hat{\sigma}^2_t(f)} f_{t+1}$, with $\hat{\sigma}^2_t$ the previous month's realised variance from daily returns and $c$ set so that the managed and unmanaged portfolios have the same unconditional volatility (sec. 2.2, eq. (1), printed pp. 5–6). The managed market portfolio has an annualised alpha of 4.86 percent and a beta of 0.6 against the market itself (printed p. 7).

Its weight has a median near one but 75th, 90th and 99th percentiles of 1.6, 2.6 and 6.4; scaling by volatility instead of variance brings the 99th percentile to about 3, and capping the weight at 1 or 1.5 leaves the Sharpe ratio unchanged but lowers the alpha (printed p. 12) [`s16-r25`, NBER WP 22208, 2016, secs. 2.2 and 5, printed pp. 5–12]. Cederburg, O'Doherty, Wang and Yan test 103 equity strategies: 77 of the scaled versions earn positive spanning-regression alphas (23 significantly positive, 3 significantly negative), but the combination of scaled and unscaled portfolios that those regressions imply cannot be formed in real time, and a real-time version earns a lower certainty-equivalent return than the original portfolio in 72 of 103 cases, which they attribute "primarily" to "structural instability in the underlying spanning regressions" [`wide-cederburg-volmanaged`, *JFE* 138, 2020, abstract and sec. 1, pp. 95–97].

**Notation.** $f_{t+1}$ the unmanaged excess return; $\hat{\sigma}^2_t$ realised variance over month $t$; $c$ a scaling constant chosen in-sample.

**Conditions.** Evidence on U.S. equity factors to about 2015–2017. The constant $c$ and the alpha come from the full sample, so the reported Sharpe-ratio gain is in-sample unless $c$ and the combination weights are estimated in real time, which is the point of Cederburg et al.'s test. A positive spanning alpha means the scaled portfolio adds to the unscaled one ex post; it does not mean the scaled portfolio beats the unscaled one on its own.

**Practice.** In a risk-parity evaluation, report the leverage distribution, the turnover and the real-time version alongside any spanning alpha. Interpretation: the leverage percentiles bear on the question of how much volatility-targeted capital levers up in calm markets and must sell when volatility jumps (`mod:volatility-targeting-deleveraging`), but they describe one strategy's rule, not the assets that follow such rules.

**Pitfalls.** Moreira and Muir are cited from the 2016 working paper (published in the *Journal of Finance*, 2017). Cederburg et al.'s strategies are equity factors; see DeMiguel, Martín-Utrera and Uppal (`ma-demiguel-martin-utrera-uppal-2024`) for a multifactor version.

**See also.** equal risk contribution; common-investor contagion (volatility targeting and deleveraging); variance forecasts.

**Further reading.**

- `ma-bongaerts-kang-van-dijk-2020`, on targeting volatility only in extreme states;
- `ma-demiguel-martin-utrera-uppal-2024`, on managing volatility jointly across factors.

**Method 14.20 Resampled efficiency: averaging the weights of mean–variance portfolios computed on resampled inputs, and what its proponents' simulation shows**

**Statement.** Richard and Robert Michaud describe Resampled Efficiency (RE, a trademark) in five steps [michaud-michaud-2008-wp, authors' version © 2007, p. 5]: (1) sample a mean vector and covariance matrix from a distribution centred at the point estimates; (2) compute the mean–variance efficient frontier for that draw; (3) repeat until step 4 converges; (4) average the portfolio weights across draws to form the RE optimal portfolio; (5) optionally apply investability constraints.

In their illustration each draw simulates 120 monthly returns per asset from a multivariate normal fitted to the data, a resample of the same length as the original history (p. 5, footnote 14). Portfolios are matched across frontiers by rank (for example the minimum-variance and maximum-return ends, each averaged across draws), so the resampled frontier is an average of "properly associated" optimal portfolios (pp. 6–7). Each asset's weight in the maximum-return RE portfolio equals the probability that it is truly the maximum-return asset (p. 9, footnote 21).

**Notation.** RE resampled efficiency; REF the resampled efficient frontier; MV mean–variance.

**Conditions.** Sign- and budget-constrained (long-only, fully invested) total-return optimisation; the authors treat benchmark-relative and long–short cases separately (p. 10, footnote 24). The REF lies below and is shorter than the classical frontier in sample, so any in-sample comparison favours the classical frontier (p. 7).

**Practice.** The evidence for out-of-sample improvement is a simulation, not a backtest. A referee takes 20 stocks' risk and return estimates (monthly, January 1997 to December 2006) as the truth, simulates histories, has an MV and an RE investor optimise on them, and scores both with the true parameters. On average the RE portfolios earn about the same return with less risk (pp. 9–10, Figure 4). The authors state that superiority cannot be proven for every simulation test, only for "financially relevant cases". A truth with negative mean returns, for example, is excluded under sign constraints (p. 9, footnote 22). They report that Markowitz and Usmen (2003) replicated the Michaud (1998) simulation (p. 10, footnote 23).

**Pitfalls.** All the evidence here is the proponents' own; the paper is marked as not to be quoted without permission, so it is paraphrased. Critics named by Michaud and Michaud include Harvey et al. (2006) on the in-sample comparison and Scherer's 2002 review; Markowitz and Usmen's experiment, as the authors describe it, raised a concern about expected-utility axioms (p. 7, footnote 17). The simulation result depends on the chosen truth, the constraints and the number of draws, so it is not evidence about any one real allocation.

**See also.** mean–variance optimisation; Black–Litterman; covariance estimation.

**Further reading.** wide-michaud-enigma (the estimation-error critique); Markowitz and Usmen, "Resampled frontiers vs diffuse Bayes: an experiment", *Journal of Investment Management*, 2003; Scherer, "Portfolio resampling: review and critique", *Financial Analysts Journal*, 2002.

**Model 14.21 Mean–variance portfolio optimisation: objectives, inputs, constraints and solvers**

**Statement.** Markowitz's problem, as written by Boyd, Johansson, Kahn, Schiele and Schmelzer, chooses the weights $w \in \mathbb{R}^n$ given expected returns $\mu$ and a covariance matrix $\Sigma$ in one of two forms [wide-boyd-markowitz-seventy, author version of 5 Jan. 2024, sec. 1.1, eqs. (1)–(2), pp. 4–5]:

$$\max_w \ \mu^\top w \ \text{ s.t. } \ w^\top \Sigma w \le \sigma_{\text{tar}}^2,\ 1^\top w = 1; \qquad \max_w \ \mu^\top w - \gamma\, w^\top \Sigma w \ \text{ s.t. } \ 1^\top w = 1.$$

Both trace the same trade-off curve of Pareto-optimal portfolios as $\sigma_{\text{tar}}$ or $\gamma > 0$ varies, and both have closed forms, though the first "can be infeasible when $\sigma_{\text{tar}}$ is too small". The closed form of the second is $w^\star = \frac{1}{2\gamma}\Sigma^{-1}(\mu + \nu^\star 1)$ with $\nu^\star = (2\gamma - 1^\top\Sigma^{-1}\mu)/(1^\top\Sigma^{-1}1)$, the optimal dual variable of the budget constraint; its $\Sigma^{-1}$ hints at sensitivity to the inputs. Adding $w \ge 0$ (long only) makes the second a quadratic programme (pp. 4–5). Interpretation: the closed form assumes $\Sigma$ positive definite, so that $\Sigma^{-1}$ exists and the objective is strictly concave with one maximiser, and $\gamma > 0$.

The term $\nu^\star 1$ shifts every expected return by the same amount, chosen so that $1^\top w^\star = 1$; with a three-asset example, $\gamma = 1$ and $\gamma = 5$, the weights sum to one and the first-order condition $\mu - 2\gamma\Sigma w^\star + \nu^\star 1 = 0$ holds to rounding. The first form is feasible only when $\sigma_{\text{tar}}^2$ is at least the minimum attainable variance under the budget constraint, $1/(1^\top\Sigma^{-1}1)$. A singular $\Sigma$, for example two perfectly correlated assets, has no inverse and leaves directions of zero variance along which the objective may be unbounded or the maximiser not unique. It needs separate treatment, such as a better-conditioned estimate or explicit constraints, chosen and justified for the problem; substituting a pseudoinverse into the formula is not a justified default (Interpretation).

Maximising the Sharpe ratio is a third objective, also solvable by convex optimisation "at least when the constraints are simple" (sec. 3.2, p. 20). Narang describes the same model in words: inputs of expected returns, volatilities and a correlation (covariance) matrix; output the efficient frontier; a common objective of "the highest possible return for a unit of risk"; and in practice also the portfolio's size in currency, a desired risk level "usually measured in terms of volatility or expected drawdown", and other constraints such as a hard-to-borrow list, which "are not required by the optimizer" [s7-narang, chap. 6, sec. 6.2, pp. 104–106].

**Notation.** $\mu$, $\Sigma$, $w$, $\gamma$ (risk aversion), $\sigma_{\text{tar}}$ (target volatility) as in Boyd et al.; the Sharpe ratio $\mu^\top w/\sqrt{w^\top\Sigma w}$ (excess returns, no benchmark) is unchanged when $w$ is scaled by any positive number.

**Conditions.** *Which objective.* Mean–variance maximises return for a variance target, or return minus a variance penalty; maximising the Sharpe ratio selects one direction on the frontier (with cash and leverage, the tangency portfolio, Narang's Exhibit 6.7, p. 122). These are different problems with different solutions. Expected drawdown is not part of the mean–variance model: Narang lists it among practical inputs that set a risk level (p. 105) and names a return-to-drawdown objective as a different choice (p. 124). Setting a desired risk level as volatility or expected drawdown while maximising return relative to volatility puts both inside one model.

*Unconstrained optimisation.* Narang calls it "a quirk of unconstrained optimizers" that they "often" put everything in the instrument with the highest risk-adjusted return (sec. 6.2.2.1, p. 114); this is an assertion, not a theorem. It is not a property of mean–variance: the unconstrained solution is proportional to $\Sigma^{-1}$ terms and generally holds every asset, and his own unconstrained two-stock example has its maximum Sharpe ratio at about 85 percent CVX and 75 percent NFLX (p. 106; Exhibit 6.2, p. 107). Interpretation: a single-asset answer arises when only expected return is maximised under a budget with no shorting, a bounded problem whose optimum is the asset with the highest expected return.

Ignoring risk without such bounds does not give one asset: with shorting allowed and expected returns not all equal, the budget-constrained return $\mu^\top w$ is unbounded, since any direction $d$ with $1^\top d = 0$ and $\mu^\top d > 0$ can be scaled without limit. Box limits on each weight make it bounded again, with the solution at the limits. Because the Sharpe ratio is scale-free, the point he marks is one of a ray of equally good portfolios within his ±100 percent search box ($\Sigma^{-1}\mu$ and three times it have the same ratio).

**Practice.** *Inputs.* Expected returns come from the alpha model; a sign-only forecast can be used by setting all positive and all negative forecasts equal (p. 108). For volatility Narang calls GARCH, with Bollerslev's 1986 paper, "the most widely used such technique (particularly in equity trading)", and adds that a simpler EWMA "may perform just about as well" (pp. 108–110); he cites no evidence for the prevalence. Boyd et al. list EWMA, dynamic conditional correlation and iterated EWMA as covariance estimators that work well in practice (sec. 3.4, p. 22).

For correlations Narang shows instability: S&P 500 against Nikkei 225 rolling one-year correlations from +0.01 to +0.66 (pp. 110–111, Exhibit 6.4), and XOM–CVX steadier than XOM–NFLX (Exhibit 6.5, p. 112), and says grouping similar instruments, "in the definition of *relative* in a relative alpha strategy and/or in the specification of the risk model", improves stability (p. 111).

*Constraints* limit positions and groups, and can end up driving the portfolio: with 100 names and a 1.5 percent cap the best position is only 1.5 times the average (sec. 6.2.2.2, p. 114). Risk penalties can rise faster than linearly, and costs enter as per-stock or generic impact functions (pp. 114–115);

Boyd et al.'s spread-plus-impact form is in cost functions. Industry-neutral constraints can turn a positive forecast into a short, and the substitution effect can prefer a cheaper correlated stock (sec. 6.2.3, pp. 120–121). The output is a target portfolio; the trades are its difference from the current one (sec. 6.3, pp. 122–123).

*Solvers.* Convex formulations, including nondifferentiable terms, are solved reliably and fast; integer shares, cardinality limits and minimum position sizes make the problem nonconvex and much slower (Boyd et al., secs. 1.4 and 4.3, pp. 9–10 and 27–28). Narang's "10,000 assets … in about one second" (p. 120) matches Boyd et al.'s Table 2 only with a factor model: 0.65–0.89 seconds for 10,000 assets and 50–100 factors with one commercial solver (sec. 5.6, p. 40).

*Size limits*. Narang's risk-model chapter classifies limits by how size is limited, how risk is measured and what is limited [s7-narang, chap. 4, "Risk Models", sec. 4.1, pp. 73–76]. A hard constraint draws a line, such as no position above 3 percent; because "why is a 3.01 percent position so much worse?", some quants use a penalty that lets a position pass the level only for a much larger expected return, increasingly hard the further past it (pp. 73–74). Boyd et al. describe the same device as softening: the hard constraint $f \leq f_{\max}$ is removed and a penalty $\gamma(f - f_{\max})_+$ is subtracted from the objective, with $\gamma$ a priority parameter; softening preserves convexity, and limits that are strict or mandated, such as long-only or no borrowing, stay hard (sec. 4.2, pp. 25–27).

A penalty therefore replaces a limit; it does not permit breaching a separate hard cap on the same quantity. Limit levels come from theory, set and adjusted by testing (his 5 percent limit revised to 3 percent), or from data (p. 74); tuning them on a backtest invites overfitting. Limits can apply to positions, groups such as sectors, factor exposures (factor risk model) and leverage (p. 75).

*The original paper and a counterpoint.* Markowitz divides portfolio selection into forming beliefs and choosing a portfolio, and treats only the second. He rejects the rule that the investor maximises discounted expected return "both as a hypothesis to explain, and as a maximum to guide investment behavior", because "the foregoing rule never implies that there is a diversified portfolio which is preferable to all non-diversified portfolios"; he proposes instead the rule that the investor considers "expected return a desirable thing *and* variance of return an undesirable thing" [s6-markowitz, *Journal of Finance* 7(1), 1952, p. 77].

The counterweight is DeMiguel, Garlappi and Uppal: of 14 models across seven datasets "none is consistently better than the 1/N rule in terms of Sharpe ratio, certainty-equivalent return, or turnover", and their analytical critical window for sample mean–variance to beat 1/N in certainty-equivalent return, with parameters calibrated to U.S. stocks, is "around 3000 months for a portfolio with 25 assets and about 6000 months for a portfolio with 50 assets", against the 60 or 120 months usually used [s7-demiguel, *RFS* 22(5), 2009, abstract, p. 1915, and sec. 1, p. 1919; authors' copy from an Internet Archive capture of an LBS faculty page]. The window depends on the number of assets and the two Sharpe ratios, so it is a calibration, not a constant.

**Pitfalls.** *Estimation error.* Boyd et al. note that the basic method is sensitive to its inputs, and answer with regularisation and robust optimisation: long-only or leverage limits, Black–Litterman's shrinkage of the mean towards market-implied returns, and covariance shrinkage (sec. 1.3, pp. 8–9; sec. 3.4, p. 22). *Black–Litterman.* Narang credits it with a default portfolio and a blend of "an investor's expectations with a degree of confidence about those expectations with the historical precedent evident in the data" (sec. 6.2.2.3, p. 117). In Boyd et al.'s account it acts on the expected returns, towards the market-implied return, not on correlations. Narang's note 5 dates Black–Litterman to 1982; his text and the catalogue give 1992, the date of Black and Litterman's paper.

*Resampled efficiency.* Narang presents Michaud's method as reordering the observed returns by Monte Carlo "to reduce the estimation error inherent in the inputs" (sec. 6.2.2.5, pp. 118–119). His own remark that average return and volatility "remain the same across all these alternate histories" concerns one series, the S&P 500: reordering a single series leaves its mean and variance unchanged. That does not show that every optimiser input is unchanged.

Interpretation: applying one common permutation to the rows of a multivariate return sample also leaves its covariances unchanged, but permuting each asset's returns independently changes them; for $x = (1, 2, 3, 4)$ and $y = (1, 2, 3, 4)$, reversing $y$ alone keeps both means (2.5) and variances (1.25) but turns their covariance from +1.25 into −1.25. What Narang describes, then, does not settle what Michaud's procedure changes. His example concerns the robustness of a trading strategy, not optimiser inputs. What Michaud's procedure draws and averages is set out in resampled efficiency; these sources give no independent out-of-sample evidence.

*Machine learning.* Narang says "we do not see many strictly 'machine learning' models for portfolio construction" and that the state of the art remains Markowitz-type optimisation (sec. 6.2.2, pp. 113–114). Interpretation: three tasks need to be kept apart.

*Forecasting* estimates inputs such as expected returns or covariances, and supervised learning is commonly used for it. *Learning allocation weights* maps data directly to portfolio weights by training a model on a portfolio objective; that is possible and has its own literature. *Optimisation* chooses weights for given inputs and constraints; for a convex problem, solvers reach the optimum, so the size of the search space is no reason to use a generic heuristic such as a genetic algorithm, though such heuristics have a place in nonconvex problems.

*Errors in Narang's details.* He describes convex functions as "smooth and continuous" with "a single highest or lowest point" (p. 120); convex functions can have kinks, such as $|x|$ or the costs above, and flat sets of minima. He calls GARCH a stochastic-volatility model and says a negative autoregression "implies … trending behavior" (pp. 108–110); neither statement is correct. His 1,998,000 distinct correlations for 2,000 instruments (p. 113) should be $2000 \times 1999/2 = 1{,}999{,}000$. His grouping evidence is two pairs over one sample, not a test of a method; factor and shrinkage estimators are in covariance estimation.

*Volatility and covariance.* Narang names two "generally accepted" measures of risk: longitudinal volatility, and dispersion, the cross-sectional standard deviation of returns or the correlation among instruments, a higher dispersion making the market "less risky" (sec. 4.1.2, pp. 74–75). Interpretation: these are not alternatives. A covariance risk model contains both volatilities, on its diagonal, and correlations, off it, and a portfolio's variance $w^\top \Sigma w$ needs both; cross-sectional dispersion in one period is a different statistic from covariance over time, and a high reading can reflect large idiosyncratic moves rather than safety.

*VaR-based leverage.* He describes VaR models as setting leverage inversely to measured risk, criticises the practice, and says "most VaR models calculate what a daily single standard deviation move in portfolio returns will be" (pp. 75–76). That last description is inaccurate: VaR is usually defined as a loss quantile at a stated confidence level. The deleveraging mechanism is in common-investor contagion.

**See also.** rule-based position sizing; signal-portfolio allocation; cost functions; mean–variance execution (the same trade-off over an execution schedule); backtest evaluation diagnostics (Sharpe and Calmar ratios as Narang defines them).

**Further reading.** Markowitz's paper, Black and Litterman, and factor models. Michaud's resampling and covariance shrinkage are in resampled efficiency and covariance estimation.

### Method 14.22 Signal-portfolio allocation: weighting simulated factor portfolios rather than positions, as Narang describes Grinold and Kahn's approach

**Statement.** Narang describes an optimiser "directly aimed at building a portfolio of signals, whereas most optimizers try to size positions", attributing it to Grinold and Kahn's *Active Portfolio Management* [s7-narang, chap. 6, sec. 6.2.2.4, pp. 117–118]:

1. build one *factor portfolio* per alpha type (momentum, value, growth), usually rule-based, "very often equal-weighted or equal risk-weighted";
2. simulate each historically to obtain a return series;
3. treat those series as the instruments of an optimiser, with the risk model, transaction cost model, portfolio size and risk target as inputs;

4. set each position as the sum, over factors, of its weight in the factor portfolio times that factor's signal times the factor portfolio's weight.

In his example (p. 118), 100 equally weighted stocks, factor weights 60 and 40 percent and one company's signals +1 and −1 give

$$1\% \times (+1) \times 60\% + 1\% \times (-1) \times 40\% = +0.2\%.$$

**Conditions.** Practitioner account as of 2024. The number of factor portfolios is "typically much more manageable, usually not more than about 20, corresponding to the number of individual factors in the alpha model" (p. 117). That number is Narang's practitioner statement, with no source; it depends on how many signals a model has. The factor-portfolio returns are simulated, so the optimiser's inputs inherit the backtests' assumptions and their overfitting risk (parameter overfitting).

**Practice.** Interpretation: the method trades a large position-level estimation problem for a small one over a handful of return series, "an easier hurdle to clear in terms of the amount of data needed" (p. 117). The position weights that result are linear in the factor weights, so position limits and costs at stock level are not enforced unless they are added back.

**Pitfalls.** The procedure is Narang's account; compare Grinold and Kahn (`alpha-grinold`). That what is optimised is then a handful of factor portfolios, not more than 20, is Narang's account, not an established fact.

**See also.** mean–variance optimisation; rule-based position sizing (how each factor portfolio is weighted); backtest evaluation diagnostics.

**Further reading.** Grinold and Kahn, *Active Portfolio Management* (`alpha-grinold`).

### Model 14.23 PCA risk-distribution weights: allocating a target variance across the principal components of a covariance matrix

**Statement.** Given an $N \times N$ covariance matrix $V$ of an invariant (returns, yield changes, volatility changes) and a target distribution of risk $R$ across $V$'s principal components, López de Prado computes the allocation $\omega$ as follows [`alpha-afml`, chap. 2, sec. 2.4.2, p. 35]. Decompose $VW = W\Lambda$ with eigenvalues in descending order. For any $\omega$, with $\beta = W'\omega$ its projection on the orthogonal basis,

$$\sigma^2 = \omega' V \omega = \beta' \Lambda \beta = \sum_{n=1}^{N} \beta_n^2 \Lambda_{n,n}, \qquad R_n = \frac{[W'\omega]_n^2 \Lambda_{n,n}}{\sigma^2}, \qquad \sum_n R_n = 1.$$

To deliver a chosen $R$ at risk $\sigma$, set $\beta_n = \sigma\sqrt{R_n/\Lambda_{n,n}}$ and $\omega = W\beta$; rescaling $\omega$ rescales $\sigma$ and keeps $R$ (p. 35). With no $R$ given, Snippet 2.1 puts all risk on the smallest-eigenvalue component, so $\omega$ is the last eigenvector scaled to $\sigma$ (p. 36).

**Notation.** $\omega$ is in the units of the invariant (for example weights on returns); $\sigma$ is the target standard deviation (`riskTarget`); $R_n \geq 0$.

**Conditions.** López de Prado frames $V$ as the covariance of an IID multivariate Gaussian process with mean $\mu$ (p. 35); interpretation, the mean plays no part, and the construction needs only $V$. It needs $\Lambda_{n,n} > 0$ for every component given risk: a zero eigenvalue cannot carry risk, and a near-zero one, where estimation error is largest, produces very large weights, which the snippet does not guard against.

The decomposition does not fix signs: each $\beta_n$ with $R_n > 0$ can be $\pm\sigma\sqrt{R_n/\Lambda_{n,n}}$, so with distinct eigenvalues $2^k$ allocations share one risk distribution, $k$ being the number of components given risk (two for the snippet's default); the snippet's positive roots, and the arbitrary signs of the computed eigenvectors, are conventions. With repeated eigenvalues the eigenvectors, and so the allocations, are not unique at all. The weights are those of an estimated $V$; with an estimated covariance, the lowest-variance component is the one most likely to be noise.

**Practice.** On a random positive-definite $4 \times 4$ matrix: the construction delivers the target $R = (0.1, 0.2, 0.3, 0.4)$ and $\sigma$ exactly, and flipping the sign of one $\beta_n$ with $R_n > 0$ leaves $R$ unchanged. López de Prado's Figure 2.2 contrasts an inverse-variance portfolio, whose risk is spread over nearly all components, with the PCA portfolio, whose risk is all in the lowest-variance component (pp. 35–36). In his chapter the weights feed the ETF trick for a spread or hedge.

**Pitfalls.** The quadratic form has only one transpose: $\sigma^2 = (\Lambda^{1/2}\beta)'(\Lambda^{1/2}\beta)$. Risk contributions use $[W'\omega]_n^2$, not $[W'n]_n^2$. Allocating all risk to the smallest eigenvalue is a hedging choice, not a return forecast; it says nothing about expected return.

**See also.** ETF trick; factor risk model; mean–variance portfolio optimisation.

**Further reading.** López de Prado cites López de Prado and Leinweber (2012) and Bailey and López de Prado (2012) for other hedging weights (p. 35).

### Model 14.24 Black–Litterman: equilibrium returns implied by market weights, combined with views and their confidence into posterior expected returns

**Statement.** In Black and Litterman's mathematical appendix [`alpha-black`, Goldman Sachs Fixed Income Research, October 1991, Appendix, points 1–8, pp. 37–38]:

- excess returns are normal with covariance $\Sigma$, and the *equilibrium* risk premiums are $\Pi = \delta\Sigma W$, with $W$ the market weights (bonds and equities by capitalisation, currencies by a universal hedging constant) and $\delta$ a proportionality constant (point 6);
- the unobservable expected return $E[R]$ has a distribution "proportional to a product of two normal distributions": one centred at $\Pi$ with covariance $\tau\Sigma$, and one expressing $k$ views $PE[R] = Q + \varepsilon$, with $\varepsilon \sim N(0, \Omega)$ and $\Omega$ diagonal (point 7);
- the result is normal with mean (point 8)

$$\overline{E[R]} = \left[(\tau\Sigma)^{-1} + P'\Omega^{-1}P\right]^{-1}\left[(\tau\Sigma)^{-1}\Pi + P'\Omega^{-1}Q\right],$$

which is then used "as the vector of expected excess returns" in the optimiser.

**Notation.** $n$ assets; $W$ market weights; $\Sigma$ the return covariance; $\tau$ a scalar for the uncertainty of the equilibrium mean; $P$ ($k \times n$) the view portfolios, $Q$ their expected returns, $\Omega$ the view variances (lower variance, more confidence).

**Conditions.** The posterior mean is the precision-weighted combination of two normal sources. Numerical checks on a three-asset example:

- it equals the conditional mean of $E[R]$ given the views by Gaussian conditioning;
- with no confidence in the views ($\Omega \to \infty$) it returns $\Pi$;
- with certainty ($\Omega \to 0$) it satisfies $P\overline{E[R]} = Q$;
- a view on one asset moves correlated assets' expected returns. A 3-point view on asset 1 with 2-point standard error raises asset 2's by 1.77 points.

$\tau$ and $\Omega$ are inputs the appendix does not calibrate.

**Practice.** Interpretation: the equilibrium gives the optimiser a neutral starting mean, so that portfolios without views are the market portfolio rather than the extreme positions an unconstrained optimiser produces from noisy means (mean–variance optimisation, whose estimation-error pitfall cites this shrinkage via Boyd et al.).

**Pitfalls.** The model neither blends views with historical precedent nor adjusts historically observed correlations. The prior is the market-implied equilibrium $\Pi = \delta\Sigma W$, not historical average returns. The formula changes the mean, not $\Sigma$, so correlations are not adjusted; correlations instead carry a view on one asset to the others. Locators refer to the October 1991 Goldman Sachs research paper, not the 1992 *Financial Analysts Journal* article.

**See also.** mean–variance optimisation; covariance estimation (the other input); forecast scaling (another way of shrinking views towards zero).

**Further reading.** He and Litterman's intuition paper and the 1992 *FAJ* article.

## 14.6 Risk models: factor exposures and their control

**Model 14.25 Factor risk model: factor returns, exposures, residuals and the factor covariance, with economic and statistical factors and their use in limiting unintended exposures**

**Statement.** In Narang's framing a *risk* is an exposure that accompanies the pursuit of alpha but is not intentionally forecast: "risk exposures are those that are not intentionally sought out by the nature of whatever forecast the quant is making in the alpha model". A risk model measures such exposures, highlights them and provides input to the portfolio construction model, which decides whether to accept, limit or eliminate each [s7-narang, chap. 4, "Risk Models", pp. 71–73]. Risk *management*, in his words, "is about the intentional selection and sizing of exposures to improve the quality and consistency of returns" (p. 71). The factor model that underlies most risk models, as Boyd, Johansson, Kahn, Schiele and Schmelzer write it, decomposes the $n$ asset returns into $k$ factor returns and idiosyncratic returns [wide-boyd-markowitz-seventy, author version of 5 Jan. 2024, sec. 3.3, eq. (14), pp. 20–21]:

$$r = Ff + \epsilon, \qquad \Sigma = F\Sigma_f F^\top + D, \qquad \sigma^2 = (F^\top w)^\top \Sigma_f (F^\top w) + w^\top D w,$$

with $F$ the $n \times k$ loading (exposure) matrix, known when the portfolio is built, $f$ the factor returns with covariance $\Sigma_f$, and $\epsilon$ the idiosyncratic returns with diagonal covariance $D$ (positive entries), uncorrelated with $f$. The means of $\epsilon$ are the assets' alphas.

**Notation.** $F$, $f$, $\Sigma_f$, $\epsilon$, $D$, $w$ as in Boyd et al.; $F^\top w$ is the portfolio's vector of factor exposures; $\sigma^2$ the portfolio variance. Usually $k \ll n$, so $\Sigma$ is low rank plus diagonal.

**Conditions.** *Assumptions.* The decomposition is exact only under its assumptions: factor and idiosyncratic returns uncorrelated, and idiosyncratic returns uncorrelated across assets ($D$ diagonal). Interpretation: under them, an equally weighted portfolio of $n$ assets with unit loading on one factor of 15 percent volatility and residuals of 30 percent has volatility 33.5, 17.8, 15.3 and 15.0 percent for $n = 1$, 10, 100 and 1,000, so the residual part diversifies away and the factor part does not. This is what Narang means by systematic risks that "cannot be diversified away" (p. 77). A diagonal $D$, with bounded residual variances and equal weights, is sufficient for this, not necessary. What matters is how much residual covariance remains when it is summed over pairs (Interpretation).

With a constant residual correlation of 0.2 between every pair and residual variance 0.09, equal-weighted residual variance is $0.09\,(0.2 + 0.8/n) = 0.018 + 0.072/n$, which stays above 0.018 however large $n$ is: that common correlation behaves like an omitted factor. With the same correlation of 0.2 only within disjoint pairs of assets, it is $0.108/n$, which still vanishes as $n$ grows. In these examples correlation confined to small groups still diversifies, while a correlation shared by every pair leaves a floor.

*Two kinds of factor.* Theory-driven or economic factors are named in advance on an economic argument: for equities the market, sectors and market capitalisation; for bonds, interest rates (pp. 77–78). Boyd et al. give market, industry and style portfolios such as Fama–French for equities and yield-curve, rate and spread factors for bonds, which are "interpretable". Empirical or statistical factors come from historical returns by methods such as principal component analysis; apart from the first, "which typically is close to the market return, these factors are less interpretable" (Boyd et al., p. 20). Narang says the first principal component of Treasury yields usually corresponds to the level of rates, and in equities the market comes first and then sectors (p. 78).

Interpretation: a principal component is defined only up to sign and can change order and composition from one sample to the next, so its economic label is an after-the-fact reading, not a property of the method. Narang notes that statistical models can be "fooled" into factors that do not persist or into spurious exposures (p. 78).

**Practice.** *Uses* (pp. 72 and 82–83): measuring which systematic exposures a portfolio takes and how large they are; explaining profits and losses; and, inside an alpha model, forecasting the residuals of returns against the risk model's factors. *Limiting exposures.* Narang's general rule is "to eliminate any unintentional exposures, since there should be no expectation of being compensated sufficiently for accepting them" (p. 76); his examples are a value strategy's incidental sector and small-cap bets (pp. 72 and 77) and a view that CVX would beat XOM, expressed as a long-only CVX trade that also bets on the market, which buying CVX and shorting an equivalent amount of XOM "substantially eliminated" (p. 76).

Interpretation: equal dollar amounts hedge the market exactly only if the two stocks have equal market betas; otherwise the pair keeps a net market exposure of the beta difference times the dollar size, so the hedge is approximate, and exact neutrality needs beta-weighted sizes under a stated risk model. Exposure limits, such as a ±5 percent net market exposure, apply to risks the alpha model does not forecast (p. 75). He prefers, "to the extent possible", to build exposure decisions into the alpha model's bet structure, for example forecasting stocks relative to their sectors, rather than relying on one risk model in an optimiser (pp. 76–77 and 80–81).

Boyd et al. express neutrality to factor $i$ as zero covariance of the portfolio with that factor, $(\Sigma_f F^\top w)_i = 0$, a linear constraint (sec. 3.3, eq. (15), p. 21). Interpretation: with correlated factors this differs from zero exposure, $(F^\top w)_i = 0$. For three assets with loadings $(1, 0.5)$, $(1, -0.5)$, $(0.8, 1)$, factor variances 0.04 and 0.01 and factor covariance 0.012, the weights $(0.5, -0.5, 0)$ have exposures $(0, 0.5)$ but a covariance of 0.006 with the first factor.

*Choosing a model.* Most quants use theory-driven models for their interpretability; empirical ones promise adaptiveness but lag a regime change and trade adaptiveness against estimation error. With two years of daily data each new day changes two of about 520 observations, while shorter windows lose statistical power. Intraday data is one partial remedy. The two kinds can be combined, and commercial models (BARRA, Northfield, Qontigo/Axioma, Wolfe) are mostly theory-driven and for equities (pp. 78–80). Boyd et al. add that a factor model can estimate a large covariance better than direct estimation and cuts the solve cost from $O(n^3)$ to $O(nk^2)$ (p. 21).

**Pitfalls.** *Risk model against decision.* Narang assigns the intentional selection and sizing of exposures to risk *management*, not to the risk model; in his black-box schematic (Exhibit 4.1, p. 83) the risk model is an input to the portfolio construction model, which selects and sizes. *Factor, exposure, signal or premium?* Narang's risk chapter names market, sector and market-capitalisation risk, not market, size, value and momentum as "the most common factors"; value is his example of an alpha (pp. 71–72). Whether such a factor is a risk to limit, a signal to forecast or a premium expected to be paid depends on the strategy's forecasts, not on the factor.

The same factor can play several roles. *Unintended is not the same as uncompensated.* Narang's rule is that "there should be no expectation of being compensated sufficiently" for unintentional exposures (p. 76): not that they earn nothing, but that any expected reward is not enough for the risk. Whether a given exposure is compensated, and by how much, is an empirical question about risk premia, not settled by its being unintended (Interpretation).

*Empirical, not adaptive by default.* Updating an empirical model with new data is not evidence that its forecasts improve; Narang himself describes the lag and the loss of power (p. 79). His remark that empirical models are "most commonly found among" statistical arbitrage traders (p. 78) describes their users; it is not a definition.

**See also.** mean–variance optimisation (where exposure limits, penalties and the covariance enter); rule-based position sizing; sources of strategy risk (regime change and model risk); common-investor contagion (VaR-based leverage under rising volatility).

**Further reading.** Covariance shrinkage is in covariance estimation; estimating factor-model loadings requires a separate estimation procedure.

**Method 14.26 Estimating a large covariance matrix: why the sample matrix fails when assets are many relative to observations, and linear and nonlinear shrinkage**

**Statement.** Ledoit and Wolf's review starts from the case "when the number of variables is of the same magnitude as the number of observations", where "the sample covariance matrix certainly will not do" [wide-ledoit-wolf-review, UZH ECON WP 323, version of February 2020, abstract, p. 1]. *Linear shrinkage* to the identity solves $\min_{\rho_1,\rho_2} E\|\hat{\Sigma} - \Sigma\|_F^2$ over $\hat{\Sigma} = \rho_1 I + \rho_2 S$, with the scaled Frobenius norm $\|A\|_F^2 = \mathrm{Tr}(A'A)/N$ (eqs. (2.1)–(2.5), pp. 4–5). The solution is

$$\Sigma^* = \gamma^* \mu I + (1 - \gamma^*) S, \qquad \gamma^* = \frac{\beta^2}{\delta^2},$$

with $\mu = \langle \Sigma, I \rangle$ the average variance, $\alpha^2 = \|\Sigma - \mu I\|_F^2$, $\beta^2 = E\|S - \Sigma\|_F^2$ and $\delta^2 = E\|S - \mu I\|_F^2 = \alpha^2 + \beta^2$ (eqs. (2.6)–(2.7), p. 5). It is a convex combination, positive definite and invertible "even when $N > T$". The optimal intensity depends on unknown population quantities, so a feasible estimator is derived under large-dimensional asymptotics, with $N/T \to c$ (sec. 2.2, p. 6).

- *Other targets.* Linear shrinkage can also target the single-factor matrix or the constant-correlation matrix.
- *Their own comparison.* In their 2004 back-test of a tracking-error-constrained manager, the single-factor target, the constant-correlation target and a five-principal-component factor model all "dominated the sample covariance matrix, but there was no clear winner": constant correlation was best for $N \leq 100$ and the single factor for $N \geq 225$ (sec. 3.3, p. 14).
- *Nonlinear shrinkage* keeps the sample eigenvectors and shrinks each eigenvalue differently. It "can deliver another level of performance improvement, especially if overlaid with" time-varying co-volatility or factor models (abstract; secs. 4–6).

**Notation.** $N$ assets, $T$ observations, $S$ sample covariance, $c = N/T$ the concentration; the norm divides by $N$, unlike the Frobenius norm.

**Conditions.** Section 2 assumes i.i.d., mean-zero observations; demeaned data and dynamic (DCC) and factor extensions are treated later (Remark 2.1; secs. 5–6). Optimality is in Frobenius loss, which is not the same as the out-of-sample variance of an optimised portfolio, though the review's applications are mostly Markowitz problems.

**Practice.** Own simulation with a known one-factor population of 100 assets and 120 observations:

- $\alpha^2 + \beta^2 = \delta^2$ holds (0.2035 against 0.2031);
- the oracle intensity is 0.045, and the Frobenius loss falls from 0.00910 to 0.00871;
- the true variance of the global minimum-variance portfolio falls from 0.0252 using $S$ to 0.0089 using $\Sigma^*$, against an optimum of 0.0044.

The small loss gain and the large portfolio gain illustrate that the two criteria differ. Narang's grouping of similar instruments (factor risk model) is a structured target of this kind, without a rule for the intensity.

- *The constant-correlation target and its feasible intensity* (s6-honey, 2003 manuscript). The target $F$ keeps the sample variances and sets every covariance to $\bar{r}\sqrt{s_{ii}s_{jj}}$, with $\bar{r}$ the average sample correlation (App. A, eq. (3), p. 12). Under Frobenius loss the optimal weight on $F$ behaves like $\kappa/T$ with $\kappa = (\pi - \rho)/\gamma$: $\pi$ sums the asymptotic variances of the $\sqrt{T}s_{ij}$, $\rho$ the asymptotic covariances of $\sqrt{T}f_{ij}$ with $\sqrt{T}s_{ij}$, and $\gamma = \sum_{ij}(\phi_{ij} - \sigma_{ij})^2$ measures the target's misspecification. Each is replaced by a consistent sample estimate, and the intensity is $\hat{\delta}^* = \max\{0, \min\{\hat{\kappa}/T, 1\}\}$ (App. B, eqs. (4)–(5), pp. 12–15) [s6-honey, App. A–B, pp. 12–15].
  Numerically, with 50 assets and 60 observations the shrunk matrix has the lower Frobenius loss; the intensity goes to one when the population really has a constant correlation (the target is then right) and is about 0.3 when correlations come in blocks; with more assets than observations the shrunk matrix is invertible and the sample matrix is not.
- *POET* (wide-fan-liao-mincheva-poet, arXiv 1201.0175). Keep the first $K$ principal components of the sample covariance and threshold the rest, the "principal orthogonal complement": off-diagonal entries $\hat{r}_{ij}$ of the remainder are set to $s_{ij}(\hat{r}_{ij})I(|\hat{r}_{ij}| \geq \tau_{ij})$ with an adaptive threshold $\tau_{ij} = \tau(\hat{r}_{ii}\hat{r}_{jj})^{1/2}$, so $\tau = 0$ gives the sample covariance and $\tau = 1$ a strict factor model (sec. 2, eqs. (2.4)–(2.7), printed pp. 9–10). It assumes *pervasive* factors (the eigenvalues of $p^{-1}B'B$ bounded away from zero, so the first $K$ eigenvalues grow with the number of assets $p$) and a sparse residual covariance [wide-fan-liao-mincheva-poet, sec. 2, printed pp. 6–10].
- *Dynamic correlation* (Engle and Sheppard's account of DCC, NBER WP 8554, 2001; Engle's 2002 *JBES* paper is wide-engle-dcc). Each series gets its own univariate GARCH; the standardised residuals $\epsilon_t$ drive $Q_t = (1 - \sum\alpha_m - \sum\beta_n)\bar{Q} + \sum\alpha_m\epsilon_{t-m}\epsilon'_{t-m} + \sum\beta_n Q_{t-n}$, with $\bar{Q}$ the unconditional covariance of the standardised residuals, and the correlation matrix is $R_t = Q_t^{*-1}Q_tQ_t^{*-1}$ with $Q_t^*$ the square roots of $Q_t$'s diagonal; $\alpha_m, \beta_n \geq 0$ with $\sum\alpha_m + \sum\beta_n < 1$ and a positive-definite $\bar{R}$ keep $R_t$ a correlation matrix (eqs. (1)–(2) and Proposition 2, printed pp. 5–6) [engle-sheppard-2001-dcc, secs. 2–3, printed pp. 5–6]. Estimation is in two steps, the variances first and then the correlation parameters.
- *The published review* (*Journal of Financial Econometrics* 20(1), 2022). As a rule of thumb, "one generally needs both $N$ and $T$ to be above 50 for nonlinear shrinkage to outperform linear shrinkage" when the optimal transformation is markedly nonlinear, and then "there is basically nothing to lose but potentially a lot to gain by upgrading" (sec. 3.3, pp. 205–206). For Markowitz problems with time-varying co-volatility they propose DCC-NL: univariate GARCH(1,1) to devolatilise, nonlinear shrinkage of the unconditional correlation matrix, and composite likelihood for the dynamics; RiskMetrics 2006 "performed poorly and cannot be recommended for the purpose of Markowitz's portfolio selection" (sec. 4, pp. 208–210, n. 13) [wide-ledoit-wolf-review, 2022 journal version, secs. 3.3–4, pp. 205–210]. The working paper read earlier and the journal version agree on the passages compared (abstract and the linear-shrinkage solution).

**Pitfalls.** The nonlinear estimator itself was not transcribed. For how the loadings, factor covariance and residual variances of a factor model are estimated, see Connor, Goldberg and Korajczyk (wide-connor-goldberg-korajczyk). *Time-varying estimators*. Boyd et al. list EWMA, Engle's DCC and iterated EWMA as approaches that "work well in practice", with or without a factor model. They give no comparison and refer to Johansson et al. (2023) for detail [wide-boyd-markowitz-seventy, arXiv v1, sec. 3.4, pp. 21–22]. In their backtest the estimator is $\hat{\Sigma}_t = \alpha_t \sum_{\tau=1}^{t} \beta^{t-\tau} r_\tau r_\tau^\top$ with $\alpha_t = (1 - \beta)/(1 - \beta^t)$. It uses the second moment, the mean being negligible, with a half-life of 125 trading days, $\beta \approx 0.994$ ($0.5^{1/125} = 0.9945$). The authors note that the half-life does not change their results qualitatively (pp. 31–32). Interpretation: the normalisation makes the weights sum to one in short samples, which a fixed $1 - \beta$ does not. The univariate GARCH and EWMA comparison is in GARCH volatility forecasts.

**See also.** factor risk model; mean–variance optimisation.

**Further reading.** wide-connor-goldberg-korajczyk.

**Evidence 14.27 How much of an empirical correlation matrix is noise: the Marchenko–Pastur band against 406 S&P 500 stocks**

**Statement.** For a $T \times N$ matrix of independent entries of variance $\sigma^2$, with $Q = T/N \geq 1$ fixed as both grow, the eigenvalue density of the correlation matrix is $\rho(\lambda) = \frac{Q}{2\pi\sigma^2}\frac{\sqrt{(\lambda_{\max} - \lambda)(\lambda - \lambda_{\min})}}{\lambda}$ on $[\lambda_{\min}, \lambda_{\max}]$, with $\lambda_{\max,\min} = \sigma^2(1 + 1/Q \pm 2\sqrt{1/Q})$ [wide-laloux-noise-dressing, arXiv cond-mat/9810255v1, eq. (3), p. 1]. Laloux, Cizeau, Bouchaud and Potters compare it with the correlation matrix of $N = 406$ S&P 500 stocks over $T = 1309$ days, 1991–96 ($Q = 3.22$). The largest eigenvalue, the market mode, is "25 times larger than the predicted $\lambda_{\max}$". Taking the rest as noise and fitting $\sigma^2$, the best fit ($\sigma^2 = 0.74$) "accounts quite satisfactorily for 94% of the spectrum, while the 6% highest eigenvalues still exceed the theoretical upper edge" (pp. 1–2).

**Notation.** $N$ assets, $T$ observations, $Q = T/N$; returns demeaned and scaled to unit volatility before the correlation matrix is formed.

**Conditions.** Evidence on 1991–1996 data. The band is an $N \to \infty$ limit for independent, identically distributed noise; for finite $N$ the edges blur. Heavy tails and volatility clustering in returns also move the edges, which the authors say a smaller effective $Q$ could absorb (p. 2). Eigenvalues inside the band are not proved to be noise, only indistinguishable from it on this test.

**Practice.** With pure noise at the paper's $N$ and $T$, the simulated eigenvalues fall within the band $[0.196, 2.424]$ up to the finite-sample blur. *A slip in the source* (own reading): the text derives the residual variance as "$\sigma^2 = 1 - \lambda_{\max}/N = 0.85$". With $\lambda_{\max} = 2.43$ that would give 0.994; with the market eigenvalue $\lambda_1 = 25\lambda_{\max} \approx 60.6$, $1 - \lambda_1/406 = 0.851$, so the formula must mean $\lambda_1$. The practical use is in covariance estimation: clipping or shrinking the eigenvalues inside the band, which Bun, Bouchaud and Potters review (`wide-bun-bouchaud-potters-cleaning`).

**Pitfalls.** The sample is 1991–96 daily data for one index; the 94/6 split is a fit with $\sigma^2$ chosen to match, not a test. Locators refer to the arXiv preprint of the letter (*Physical Review Letters* 83, 1999).

**See also.** covariance estimation; PCA risk-distribution weights.

**Further reading.** `wide-bun-bouchaud-potters-cleaning`; `wide-vershynin-hdp`.

### Model 14.28 GARCH(1,1) and EWMA (RiskMetrics) variance forecasts: mean reversion against a flat forecast, and the evidence on forecast accuracy

**Statement.** The GARCH(1,1) daily variance is $\sigma^2_{t+1} = \omega + \alpha R^2_t + \beta\sigma^2_t$ with $\alpha + \beta < 1$. Its long-run variance is $\sigma^2 = \omega/(1 - \alpha - \beta)$, so $\sigma^2_{t+1} = \sigma^2 + \alpha(R^2_t - \sigma^2) + \beta(\sigma^2_t - \sigma^2)$, and the $k$-day-ahead forecast reverts at the rate of the *persistence* $\alpha + \beta$ [`wide-christoffersen-elements`, chap. 4, secs. 2–3, printed pp. 70–71 of an uncorrected proof]:

$$E_t[\sigma^2_{t+k}] - \sigma^2 = (\alpha + \beta)^{k-1}(\sigma^2_{t+1} - \sigma^2).$$

RiskMetrics' exponential smoothing, $\sigma^2_{t+1} = \lambda\sigma^2_t + (1 - \lambda)R^2_t$ with $\lambda = 0.94$ set for every asset, is the case $\omega = 0$, $\alpha + \beta = 1$. It "ignores the fact that the long-run average variance tends to be relatively stable over time", so its forecast is flat in $k$ and "counterfactual" at longer horizons. Its advantages are a single parameter, no estimation and little stored data (pp. 70–71).

*Evidence on accuracy.* Hansen and Lunde compare 330 volatility models out of sample against intraday realised volatility: the DM–dollar rate (1,254 daily returns, October 1987 to September 1992) and IBM (2,378, January 1990 to May 1999). They use seven loss functions and White's and Hansen's tests for superior predictive ability. They find "the best models do not provide a significantly better forecast than the GARCH(1,1) model", while "if an ARCH(1) model is selected as the benchmark, it is clearly outperformed" [`hansen-lunde-2001`, abstract; sec. 2.4, pp. 7–8; sec. 6]. Their model set includes IGARCH, which they find "surprisingly poor" on most criteria (p. 25), but not RiskMetrics' fixed-$\lambda$ smoother as such.

*Two extensions.*

- *Asymmetry (GJR).* Glosten, Jagannathan and Runkle let the squared innovation enter the variance with a different coefficient according to its sign: $v_{t-1} = b_0 + b_1 v_{t-2} + g_1\epsilon^2_{t-1} + g_2\epsilon^2_{t-1}I_{t-1}$ (Model 2, eq. (9), printed p. 10) [`wide-gjr-garch`, Minneapolis Fed Working Paper 505, November 1992, printed p. 10]. In this working paper $I_{t-1} = 1$ when $\epsilon_{t-1}$ is *positive*; the common textbook form puts the indicator on negative shocks, so the sign of the asymmetry coefficient depends on the convention used.
  On monthly U.S. excess returns they find that "positive unanticipated returns result in a downward revision of the conditional volatility whereas negative unanticipated returns result in an upward revision … of a similar magnitude", that volatility is less persistent than the standard model suggests, and a negative relation between conditional mean and variance once the model allows for asymmetry, seasonality and interest rates (abstract). The paper was published in the *Journal of Finance* (1993).
- *Realised-volatility cascade (HAR-RV).* Corsi models next-day realised volatility as $RV^{(d)}_{t+1} = c + \beta^{(d)}RV^{(d)}_t + \beta^{(w)}RV^{(w)}_t + \beta^{(m)}RV^{(m)}_t + \omega_{t+1}$, where $RV^{(w)}$ and $RV^{(m)}$ average the daily values over 5 and 22 trading days (eqs. (4) and (6)–(8), pp. 177–181) [`wide-corsi-har`, *Journal of Financial Econometrics* 7(2), 2009, pp. 174–196]. It is a restricted AR(22) estimable by least squares, which reproduces long memory approximately without being a long-memory process; Corsi's $RV$ is realised *volatility*, the square root of summed squared intraday returns, and his averages are of volatilities, which by Jensen's inequality are not exactly the volatility of the longer period (n. 4). The HAR forecast equals the AR(22) with coefficients $\beta^{(d)} + \beta^{(w)}/5 + \beta^{(m)}/22$ at lag 1, $\beta^{(w)}/5 + \beta^{(m)}/22$ at lags 2–5 and $\beta^{(m)}/22$ at lags 6–22.

**Notation.** $R_t$ daily return (mean taken as zero); $\sigma^2_{t+1}$ the variance for day $t + 1$ known at the end of day $t$; $\lambda$ the smoothing weight.

**Conditions.** The GARCH forecast formula needs $\alpha + \beta < 1$ and parameters estimated by quasi-maximum likelihood. Comparisons depend on the volatility proxy: Hansen and Lunde argue that noisy proxies such as squared daily returns obscure the differences between models (sec. 1). They also depend on the loss function and the sample, here one currency and one stock in the 1990s.

**Practice.** Synthetic check. The $k$-step formula matches simulation. Where GARCH(1,1) is true (persistence 0.98), the EWMA forecast's mean squared error for cumulative variance is 0.9 percent above GARCH's at one day and 15 percent above at 20 days. That shows how the flat forecast costs more as the horizon grows; it is not evidence on market data. This bears on Narang's statements (mean–variance optimisation). That GARCH is "the most widely used such technique (particularly in equity trading)" is unsupported, since no source here measures usage. That a simpler EWMA "may perform just about as well" is consistent with the check at short horizons, but Hansen and Lunde did not test the fixed-$\lambda$ smoother.

**Pitfalls.** The GARCH formulation follows secondary sources, not Bollerslev's 1986 paper (`s7-bollerslev`). The Christoffersen file is an uncorrected proof of the second edition, so its page numbers may differ from print. The Hansen–Lunde text is their March 2001 working paper (CAF Working Paper Series, No. 84); the published version (*Journal of Applied Econometrics* 20(7), 2005) may summarise the IBM results differently.

**See also.** VaR and ES estimation (filtered historical simulation scales by a GARCH variance); covariance estimation; multiple testing (White's Reality Check, which Hansen and Lunde use).

**Further reading.** `s7-bollerslev`; `s7-realized` (realised volatility); loss functions for comparing volatility forecasts.

### Method 14.29 Comparing volatility forecasts with a noisy proxy: which loss functions keep the ranking, and why MSE and QLIKE do while MAE does not

**Statement.** True conditional variance is unobserved, so forecasts are compared against a proxy $\hat\sigma^2_t$ (a squared return, a range, realised variance) that is conditionally unbiased but noisy. Patton calls a loss function *robust* if ranking two forecasts by expected loss gives the same order with the proxy as with the true variance (Definition 1, printed p. 4). A loss is robust if and only if it can be written $L(\hat\sigma^2, h) = \tilde C(h) + B(\hat\sigma^2) + C(h)(\hat\sigma^2 - h)$ with $C$ strictly decreasing and $\tilde C$ its antiderivative (Proposition 2, printed p. 13).

MSE, $(\hat\sigma^2 - h)^2$, is the only robust loss that depends on the forecast error alone, and QLIKE, $\log h + \hat\sigma^2/h$, the only one that depends on the standardised error alone (Proposition 3, printed p. 14). The homogeneous robust losses form a one-parameter family that contains MSE ($b = 0$) and QLIKE ($b = -2$) (Proposition 4, printed p. 14) [`wide-patton-robust-loss`, UTS QFRC Research Paper 175, April 2006, secs. 2–3, printed pp. 4–14].

**Notation.** $h_t$ the forecast of the conditional variance $\sigma^2_t$; $\hat\sigma^2_t$ the proxy with $E_{t-1}[\hat\sigma^2_t] = \sigma^2_t$.

**Conditions.** The result needs a conditionally unbiased proxy; it concerns rankings, not the level of expected loss, which noise raises for every forecast (printed p. 4). "Robust" here does not mean robust to outliers (n. 6).

**Practice.** With a squared-return proxy, $\hat{\sigma}^2 = \sigma^2\chi_1^2$, MSE and QLIKE both rank the true variance first among forecasts scaled by 0.455, 0.8, 1 and 1.25, while MAE prefers $0.455\sigma^2$, the proxy's conditional median, so MAE (and MAE on logs or standard deviations, by the same argument) favours a forecast biased downward. This argues for reporting MSE and QLIKE when ranking GARCH, GJR and HAR forecasts against realised variance.

**Pitfalls.** Locators refer to the 2006 working paper (*Journal of Econometrics*, 2011). Hansen and Lunde's 2005 comparison, cited in GARCH(1,1) and EWMA, used seven loss functions, some of them non-robust in Patton's sense.

**See also.** GARCH(1,1) and EWMA; multiple testing (comparing many forecasts).

**Further reading.** `wide-hansen-lunde-garch11`; `s7-realized`.

#### Method 14.30 Testing for several structural breaks at unknown dates: least-squares break estimation, trimmed sup-F tests and the sequential choice of the number of breaks (Bai and Perron)

**Statement.** In a linear regression whose coefficients (all or some) shift at $m$ unknown dates, Bai and Perron estimate the break dates by least squares, the break fractions converging at rate $T$. To test no break against $k$ breaks they use $\sup F_T(k;q)$, the largest $F$ statistic over all break partitions in which breaks are at least a fraction $\epsilon$ apart and from the sample ends ("we need to restrict each break date to be asymptotically distinct and bounded from the boundaries of the sample"), generalising Andrews's single-break test; its limit depends on the regressors and on serial correlation and heterogeneity in the errors (sec. 4, printed pp. 16–17).

To choose the number of breaks they propose a sequential procedure: test $\ell$ against $\ell+1$ breaks with $F_T(\ell+1 \mid \ell)$ in each segment and add a break wherever the test rejects, stopping at the first non-rejection; information criteria are an alternative, with BIC consistent and AIC tending "to overestimate the dimension" (sec. 5, printed pp. 26–27) [`wide-bai-perron`, MIT Department of Economics Working Paper 95-17, September 1995, abstract and secs. 4–5].

**Notation.** $T$ observations; $m$ breaks with fractions $\lambda_i = T_i/T$; $q$ the number of coefficients allowed to shift; $\epsilon$ the trimming fraction.

**Conditions.** The critical values depend on $\epsilon$, $q$ and $k$; the asymptotics need breaks bounded away from each other and from the ends. Whether the sequential procedure uses breaks estimated jointly or one at a time matters for the null distribution (sec. 5).

**Practice.** To report break dates that change a forecast ranking, fix $\epsilon$ and the maximum number of breaks before looking, report the sequential test's choice alongside BIC, and state whether rankings of forecasts (loss functions) change across the estimated regimes.

**Pitfalls.** The source is the 1995 working paper behind the 1998 *Econometrica* article, not the 2003 *Journal of Applied Econometrics* paper with the dynamic-programming algorithm; formulas are not transcribed.

**See also.** volatility forecast comparison; GARCH and HAR.

**Further reading.** `wide-bai-perron` (the 2003 computation paper); `wide-hamilton-1989`.

#### Method 14.31 Estimating VaR and ES: historical simulation, filtered historical simulation, parametric dynamic models and Monte Carlo as overlapping choices, and the coverage backtest

**Statement.** Christoffersen's textbook [`wide-christoffersen-elements`, 2nd ed., 2012] treats the methods as layers, not exclusive categories:

- *Historical simulation* (HS) applies today's holdings to past prices. It assumes "the distribution of tomorrow's portfolio returns … is well approximated by the empirical distribution of the past $m$ observations" (chap. 2, sec. 2.1, PDF pp. 34–35). His conclusion is that "HS is problematic for computing VaR", which motivates dynamic models (chap. 2, overview). Weighted HS, which weights recent observations more, is an extension.
- *Filtered historical simulation* (FHS) "combines model-based methods of dynamic variance, such as GARCH, with model-free methods of distribution". It scales the empirical distribution of standardised returns by the model's current volatility (chap. 6, PDF p. 134).
- *Monte Carlo* and FHS are both used to simulate a dynamic risk model forward, for VaR and ES at horizons longer than a day, where closed forms are lacking. Monte Carlo "relies on artificial random numbers" from the model's assumed distribution, FHS "uses historical random shocks" (chap. 8, overview, PDF p. 179).

*Backtesting.* With $I_{t+1}$ the indicator of a VaR violation at coverage rate $p$, $T_0$ and $T_1$ the numbers of non-violations and violations, the unconditional coverage test is

$$LR_{uc} = -2\ln\left[(1-p)^{T_0}p^{T_1}\Big/\left((1-T_1/T)^{T_0}(T_1/T)^{T_1}\right)\right] \sim \chi_1^2,$$

rejected at 10 percent above 2.7055. An independence test and a combined conditional-coverage test follow (chap. 13, PDF pp. 303–304).

**Notation.** Losses $L$ (positive numbers are losses). The *confidence level* $c$ (for example 99 percent) and the *exceedance* or *coverage* probability $p = 1-c$ (1 percent) name the same tail from opposite ends; Christoffersen's $p$ is the latter. $\mathrm{VaR}_c = \inf\{\ell : P(L \le \ell) \ge c\}$, the lower $c$-quantile of the loss. Expected shortfall is the average of VaR over the tail confidence levels,

$$\mathrm{ES}_c = \frac{1}{p}\int_c^1 \mathrm{VaR}_u\,du = \frac{1}{p}\Big(E\left[L\,1\{L > \mathrm{VaR}_c\}\right] + \mathrm{VaR}_c\big(P(L \le \mathrm{VaR}_c) - c\big)\Big),$$

the loss-side form of Acerbi and Tasche's tail mean and expected shortfall [`acerbi-tasche-2001`, Definition 2.6, p. 5; Proposition 3.2, p. 6; eq. (3.3), p. 7, stated for a profit $X$ and tail level $\alpha = p$]. The second term gives the atom at VaR exactly the weight needed for the tail to carry probability $p$. $m$ is the HS window.

*Why not "the expected loss given that VaR is exceeded".* That phrase equals ES only when $L$ has no atom at $\mathrm{VaR}_c$, for example a continuous distribution; Acerbi and Tasche show that the tail conditional expectations can then differ from ES and fail sub-additivity (pp. 3–5 and Example 5.4). Example: a loss of 100 with probability 0.005, otherwise 0. At $c = 99$ percent ($p = 1$ percent), VaR $= 0$ because $P(L \le 0) = 0.995 \ge 0.99$. Then ES $= (0.005 \times 100 + 0 \times (0.995 - 0.99))/0.01 = 50$, while $E[L \mid L > \mathrm{VaR}] = 100$ and $E[L \mid L \ge \mathrm{VaR}] = 0.5$.

In *historical simulation* the same definition is the average of the worst $np$ losses of $n$, with a fractional weight on the last one when $np$ is not an integer: with $n = 250$ at 99 percent, $np = 2.5$ and ES $= (\ell_{(1)} + \ell_{(2)} + 0.5\,\ell_{(3)})/2.5$, not the mean of the worst two or three.

**Conditions.** HS needs no distributional assumption but assumes the window's returns are draws from tomorrow's distribution: a stationary, independent sample over $m$ days. FHS needs a correct variance model and i.i.d. standardised returns. Monte Carlo needs a fully specified distribution. The $\chi^2$ backtest is asymptotic and assumes independent violations under the null.

**Practice.** Synthetic check. Returns are simulated from a GARCH(1,1) with Student-$t$ shocks over 20,000 days, and 1 percent HS and FHS VaR are backtested, both on a 250-day window.

- HS violations occur on 1.62 percent of days, with $LR_{uc} = 66$. Their probability the day after a violation is 5.8 percent: they cluster.
- FHS violations occur on 1.38 percent of days, with $LR_{uc} = 25$, the 250-day empirical 1-percent quantile being imprecise. They do not cluster after a violation.

A second check verifies the discrete example above, the equality of the integral and fractional-mass forms, historical simulation with 1,000 and 250 observations, and agreement with the normal closed form $\varphi(z_{0.99})/0.01 = 2.665$ for a continuous law. Whether the methods are useful for dynamically updating risk assessments therefore needs this kind of evidence: forecasts with better coverage and independence. That a method can be updated is not enough. Here the evidence is only synthetic, in a world where the GARCH model is true.

**Pitfalls.** The book's EVT, ES backtesting and stress-testing chapters, the regulatory ES standard and Kupiec's 1995 paper (`wide-kupiec-backtest`) are not covered here. Page numbers are PDF pages; the printed numbering is offset by a varying amount.

**See also.** factor risk model; common-investor contagion (VaR-based leverage in August 2007).

**Further reading.** `s7-qrm`; `wide-kupiec-backtest`.

## 14.7 Transaction cost models as inputs to portfolio construction

**Model 14.32 Transaction cost functions for portfolio construction: flat, linear, piecewise-linear and quadratic forms, with units and size conventions**

**Statement.** Narang's transaction cost model quantifies "the cost of making a trade of a given size" so that the portfolio construction model can weigh it against the alpha and risk models when choosing the portfolio to hold [`s7-narang`, chap. 5, "Transaction Cost Models", p. 85]. It is "not designed to minimize the cost of trading, only to inform the portfolio construction engine"; minimising cost is the execution algorithm's job (p. 86), and cost is reduced in two phases, first by the portfolio construction model using the cost estimate and then by execution (sec. 5.3, p. 96). In his framing there are "only two reasons to make a trade", a better reward from the alpha model or a lower risk from the risk model, and a trade's benefit must "clear the hurdle of the cost of transacting" (p. 85).

He names four forms, each giving total cost against trade size, with commissions and fees as the baseline (sec. 5.2, pp. 91–96, Exhibits 5.1–5.4):

- *flat*: the same cost whatever the size; reasonable when the size traded is nearly always about the same and liquidity is fairly constant, "rarely correct" otherwise (pp. 92–93);
- *linear*: cost rising with a constant slope in size, overestimating at small sizes and underestimating at large ones in his exhibit (pp. 93–94);
- *piecewise-linear*: linear in ranges, with a steeper line beyond each breakpoint; "rather popular among quants as a happy medium between simplicity and accuracy" (p. 94);
- *quadratic*: cost rising ever faster with size; in his words the most "accurate" of the four in the exhibits and the most computationally intensive (p. 95).

**Notation.** Interpretation, for a single equity traded in shares: let $q$ be the signed trade size in shares ($q>0$ a buy, $q<0$ a sale), $V>0$ a volume scale in shares such as average daily volume, $P>0$ the price in currency per share, and $C(q)$ the *total* expected cost of the trade in currency. For $q\neq 0$ the *per-share* cost is $c(q)=C(q)/|q|$ in currency per share, and $c(q)/P$ is the same cost as a fraction of price (times $10^4$ for basis points); $c(0)$ is not defined. With coefficients $a\geq 0$ in currency, $b, b_0, h_i \geq 0$ in currency per share, $0<\kappa_1<\cdots<\kappa_m$ in shares, $k\geq 0$ in currency per share squared and $\eta\geq 0$ dimensionless, the forms are

$$C_{\text{flat}}(q)=\begin{cases} a, & q\neq 0,\\ 0, & q=0,\end{cases} \qquad C_{\text{lin}}(q)=b|q|, \qquad C_{\text{quad}}(q)=b|q|+kq^2,$$

$$C_{\text{pwl}}(q)=b_0|q|+\sum_{i=1}^{m} h_i\big(|q|-\kappa_i\big)_+, \qquad C_{\text{pow}}(q)=\eta P|q|\Big(\frac{|q|}{V}\Big)^{1/2},$$

with $(x)_+=\max(x,0)$. Every form has $C(0)=0$ and $C\geq 0$, and depends on $q$ only through $|q|$. All but the flat form are continuous; the piecewise-linear form has slope $b_0+\sum_{i:\kappa_i<|q|} h_i$, rising at each breakpoint $\kappa_i$, and equals $\max_j(s_j|q|-d_j)$ with $s_j=b_0+\sum_{i\leq j}h_i$ and $d_j=\sum_{i\leq j}h_i\kappa_i$ ($s_0=b_0$, $d_0=0$). A maximum of lines with increasing slopes alone does not give $C(0)=0$: its intercepts must take this form.

For $b_0=0.01$, $h=(0.02, 0.05)$ and $\kappa=(10{,}000, 50{,}000)$: $C=0$ at $q=0$; 50, 100, 700, 1,300 and 2,100 at $|q|=5{,}000$, 10,000, 30,000, 50,000 and 60,000, the same for buys and sales; continuous at both breakpoints; equal to the maximum of lines with slopes 0.01, 0.03, 0.08 and intercepts 0, 200, 2,700.

The forms assume a cost symmetric between buys and sales, never negative (no rebates or favourable price drift), expected rather than realised, for one instrument and one period, independent of the trade's duration and of other trades. Contracts or currency notional need their own size and price units, such as a contract multiplier; the per-share conversions above do not carry over unchanged. A total cost linear in size is a constant per-share cost; a quadratic total cost is a per-share cost linear in size, $c(q)=b+k|q|$ for $q\neq 0$. A per-share commission of, say, \$0.001 (Narang's example, p. 87) is therefore a linear, not a flat, cost; only a charge per ticket is flat. A per-share cost proportional to $(|q|/V)^{1/2}$, as in the square-root law, is a total cost proportional to $|q|^{3/2}$.

Numerical illustration: with $C_{\text{pow}}$, $\eta=0.5\times 2\%=0.01$, $P=\$50$ and $V=10^6$ shares, 10,000 shares bought or sold cost 10 bp per share, \$500 in total; 20,000 shares cost 14.1 bp, \$1,414; doubling the size multiplies the per-share cost by $2^{1/2}$ and the total by $2^{3/2}\approx 2.83$. Using the peak-impact law as a cost scale is a simplification: the evidence entry gives an execution shortfall per share of about two-thirds of peak impact.

**Conditions.** The ranking in the schematic comparison is not an established result. Narang's exhibits are schematic, with unlabelled axes and no data; the "true" cost curve in each is drawn convex and starting at zero (Exhibits 5.1–5.4, pp. 93–95), although his text puts fixed charges under it as a floor (p. 91). Against an assumed convex curve, the closeness of each form follows from the shape drawn, not from estimation. His own summary is conditional: simpler models suit traders whose sizes are roughly constant or small, more complex ones those who may trade significant or variable quantities quickly, and "any of the four models described here can be valid in the right set of circumstances" (sec. 5.3, p. 96).

His statement that a quadratic shape "is generally accepted by the quant community" (p. 92) is unsourced, and his note 2 qualifies it: "there is good evidence that, at least for U.S. equities, cost per share scales with an exponent of 1.5, or in other words, as the square root of trade size" (n. 2, p. 97). That sentence conflates two conventions: a per-share cost scaling as the square root of size is a total cost with exponent 1.5 (Notation).

Both a quadratic and a 3/2-power total cost are convex, so "the cost gets higher ever more quickly" as size grows (p. 92) does not distinguish them; which exponent fits is an empirical question (square-root impact, direct estimation, whose fitted temporary-impact exponent of 0.6 per share implies, at a fixed duration, a total exponent of about 1.6).

**Practice.** Narang describes models built separately for each instrument, "highly empirical", evolving with the quant's own recorded executions and refreshed to keep up with market conditions; a less liquid small-cap stock costs more to trade than a mega-cap (pp. 91–92 and 96). His Exhibit 5.4 contrasts the expected cost, specified before trading, with what is "observed empirically after trading"; the differences come from changes in liquidity, volatility and the mix of traders (p. 95). Interpretation: the realised cost observed after trading is measured against a benchmark price, and what is observed is a cost, not the impact component; how ex-ante cost forecasts are tested against realised costs is in transaction cost analysis.

Interpretation: inside an optimiser the cost enters the objective jointly with the alpha forecast and the risk penalty, so a trade is not tested against a separate hurdle one by one; and trades can also be forced by flows, margin or constraints, which Narang's two reasons leave out. The cost of underestimating or overestimating costs (too much turnover, or positions held too long, p. 86) is the subject of backtest cost and short assumptions.

**Pitfalls.** *No universal ranking of accuracy or computation.* Narang's comparisons of accuracy and computation rest on his drawn curve and do not hold generally. Which form is most accurate depends on the instrument, the size range traded and the true shape, which Narang's own note 2 says may not be quadratic. Computational cost depends on how the form enters the optimiser, not on how many terms it has.

What the solver needs: Boyd et al.'s generic separable cost per asset is $x \mapsto a\,|x| + b\sigma\,|x|^{3/2}/V^{1/2} + cx$ for a dollar trade $x$, with $a$ half the spread plus linear fees, $b$ a constant fitted to realised costs ("around one" by the rule of thumb that a day's volume moves the price by a day's volatility), $V$ the period's dollar volume and $c$ an asymmetry term, which makes the cost negative when $|c| > |a|$; they know of no empirical test of this exact form, though similar models with cost growing about as the 3/2 power of size are validated in the work they cite [exp-boyd-multiperiod-trading, sec. 2.3, eq. (2.2), pp. 9–10]. Piecewise-linear and added quadratic terms are common variants and "almost all of these are convex functions"; a fixed fee for any nonzero trade is not (p. 11).

The optimisation is fast and reliable so long as costs, risk and constraint sets are convex (the self-financing equality relaxed to an inequality, tight at the optimum); a single-period problem with a factor risk model costs $O(nk^2)$ flops with a leading coefficient of order 100 by interior-point methods, "even for … the 3/2 power transaction costs", and a purely quadratic objective with only linear equality constraints has a coefficient closer to 2, about 50 times faster (sec. 4.7, pp. 38–39). A multi-period problem over $H$ periods has $Hn$ variables, but its structure lets the effort grow linearly in $H$ (sec. 5.3, p. 51). So the computational difference that matters is convex against nonconvex (flat per-trade fees, integer and minimum-size constraints) and quadratic against general convex, not the number of terms.

Interpretation, not stated by Boyd et al.: a convex piecewise-linear cost can be written with linear constraints and auxiliary variables. Narang's own wording is hedged ("this perhaps more accurate one is more computationally intensive", n. 2, p. 97). For example, Boyd et al. use a spread term plus a $|z|^{3/2}$ impact term inside a convex problem [wide-boyd-markowitz-seventy, author version of 5 Jan. 2024, sec. 2.4, eq. (10), p. 17], and find integer, cardinality and minimum-size constraints nonconvex and much slower to solve (sec. 4.3, pp. 27–28); see mean–variance optimisation.

*Units.* A cost function stated without its convention (total or per unit; shares, currency or fraction of volume) cannot be compared with an impact law; say which. *Benefit and cost.* The trade-off is between expected benefit and expected cost over the same horizon; comparing a forecast return with a cost in different units or horizons misstates the hurdle.

*Single- against multi-period optimisation*. Boyd et al. plan trades over a horizon $H$ and execute only the first period's. They argue that this helps when forecasts decay, when risk, constraints or liquidity are known to change, and when a portfolio is being set up or shut down: "trading towards a lower risk position starts well before the increase in risk, trading it off with the transaction costs". Their example with $H = 2$ shows a "substantial advantage" over single-period optimisation, but they attribute it "mostly" to the extra forecast of tomorrow's return, and their return forecasts are built from realised returns plus noise, which they call "clearly not a practical forecast" [exp-boyd-multiperiod-trading, secs. 5.2 and 7.4, pp. 46–47 and 66–69]. The example therefore shows the mechanics, not a measured gain.

**See also.** transaction costs (the components, including Narang's commissions and fees, slippage and impact); square-root impact and direct estimation of impact (the empirical shape); mean–variance execution (linear temporary impact, a quadratic total cost for a given schedule); transaction cost analysis (forecasts against realised costs); backtest cost and short assumptions.

**Further reading.** Guasoni and Muhle-Karbe's survey summarises the literature [guasoni-muhle-karbe-2012, arXiv v1, sec. 1, PDF p. 3]. With proportional costs, "optimal portfolios entail a no-trade region, in which it is optimal to keep existing holdings", trading happens only at its boundaries, and the region "is wide, even for small transaction costs" while its welfare cost is small (Constantinides, 1986). Davis and Norman solve the free-boundary problem for power utility from consumption over an infinite horizon. The width of the no-trade interval and the utility loss are of order $\varepsilon^{1/3}$ and $\varepsilon^{2/3}$ in the proportional cost $\varepsilon$ (Shreve and Soner; Rogers). The survey derives the leading constant of the width for its long-run model (Lemma 3.2 and the expansion that follows it, PDF p. 9).

*Davis and Norman through Shreve and Soner*. Shreve and Soner treat the same problem, one money market and one stock, and describe Davis and Norman's paper as providing "a precise formulation and analysis, including an algorithm and numerical computations of the optimal policy", but "under restrictive and not fully verifiable assumptions", which they replace "by the sole assumption of a finite value function", confirming some of Davis and Norman's conjectures and refuting others. Their result: "regardless of model parameters, the proportion of wealth held in the stock by the optimal portfolio remains in an interval whose endpoints depend on all the model parameters, but not on the wealth"; when the Merton proportion exceeds one (leverage is optimal) it can lie outside the interval, so "the presence of transaction costs reduces the agent's desire for leverage".

They also show in their appendix that the liquidity premium behaves like the transaction cost to the 2/3 power in this model [shreve-soner-1994, *Ann. Appl. Probab.* 4(3), 1994, sec. 1, pp. 610–612]. That supports the wedge description above with the proof's conditions stated, from a source that read Davis and Norman, not from their paper. These are restated results, not Davis and Norman's own theorem. Interpretation: the rebalancing rule this implies, trading to the nearest boundary only when the position leaves the region, differs from penalising turnover in a single-period optimiser, which is the form in Boyd et al.

## 14.8 Performance evaluation

#### Method 14.33 Relative performance evaluation: total returns, market-adjusted and risk-adjusted returns, market timing, and smoothed returns

**Statement.** Portfolio performance depends on management and on every factor that moves the prices of its holdings, most of which nobody can foresee, so good results may reflect skill or luck [alpha-harris, chap. "Performance Evaluation and Prediction", sec. 22.1, pp. 443–444]. Relative evaluations are therefore more valuable than absolute ones: an absolute evaluation computes the total return; a relative one compares it with a benchmark chosen to show how the portfolio would have done without active management (sec. 22.2, p. 445). Harris's measures (secs. 22.2.1–22.2.2.3, pp. 445–450):

- *total return* over an interval, including income and paying out commissions, fees and margin flows; with capital added or withdrawn, either the *internal rate of return*, approximately a time- and value-weighted average of the returns between flows, or the *holding-period return* of one unit invested at the start with distributions reinvested, which differ when flows are not pro rata;
- *market-adjusted return*, the portfolio return minus a market index return;
- *risk-adjusted excess return* or realised alpha, the portfolio return minus the portfolio's beta times the market return, with the beta measured concurrently because it changes when the manager trades;
- *market-timing return*, beta times the market return minus the market return; so that

$$r^p = \underbrace{(r^p - \beta r^m)}_{\text{excess return}} + \underbrace{(\beta r^m - r^m)}_{\text{market timing}} + \underbrace{r^m}_{\text{market}}$$

(p. 450). Harris notes that the decomposition generalises to other risk factors, such as size and growth.

**Conditions.** Timeless (methods); presentation standards named by Harris (AIMR-PPS) as of 2002. Returns of illiquid assets depend on estimated values; managers who adjust them slowly, especially downwards, report smooth returns with positive serial correlation; the low volatility inflates $t$-ratios, and forced revaluation later produces large losses, an artificial peso problem [alpha-harris, same chap., box "Smooth Sailing, Rapids Ahead", p. 448, and sec. 22.5.2.1, pp. 469–470]. Holding-period returns compound: +100 percent then −50 percent is zero, not +50 percent (box, p. 447).

**Practice.** Interpretation: transaction cost analysis applies the same idea over an order's horizon, subtracting beta times the index move from the arrival cost (transaction cost analysis); in both, the residual is not proof of skill.

**Pitfalls.** Interpretation: Harris's realised alpha $r^p - \beta r^m$ omits the risk-free rate; the regression form $r_t^p - r_t^f = \alpha + \beta(r_t^m - r_t^f) + u_t$ (Jensen's alpha) differs from it by $(1-\beta)r^f$, for example −0.9 percentage points at $\beta = 1.3$ and $r^f = 3$ percent; the regression also estimates a constant beta, whereas Harris uses concurrent betas. The decomposition $r^p = r^m + (r^p - \beta r^m) + (\beta - 1)r^m$ is Harris's identity. Harris calls the IRR approximately time- and value-weighted, not money-weighted, and does not discuss who controls flows; fill quality belongs to transaction cost analysis. Return smoothing is in Harris a problem of fraudulent or careless valuation; for a formal treatment see `wide-getmansky-lo-makarov`.

**See also.** transaction cost analysis (the execution analogue).

**Definition 14.34 Return measures: holding-period, price and total returns; time-weighted and money-weighted returns; cumulative, arithmetic-mean, geometric-mean and annualised returns; log returns; gross and net, real and leveraged returns**

**Statement.** GIPS 2020 defines a *time-weighted return* (TWR) as "a method of calculating period-by-period returns that reflects the change in value and negates the effects of external cash flows", and a *money-weighted return* (MWR) as "the return for a period that reflects the change in value and the timing and size of external cash flows" [wide-gips-2020, Glossary, pp. 83 and 78]. An *external cash flow* is capital entering or leaving a portfolio; dividend and interest income are not external cash flows (p. 75). For portfolios in composites, firms must value at least monthly and on the date of all large cash flows, calculate sub-period returns at large cash flows if daily returns are not calculated, with exceptions for private-market investment portfolios, and "geometrically link periodic and sub-period returns" (provisions 2.A.23–2.A.24, p. 11).

When calculating money-weighted returns, firms must "calculate annualized since-inception money-weighted returns" using daily external cash flows (2.A.29, p. 12), but "returns for periods of less than one year must not be annualized" (2.A.12, p. 9); accordingly a composite money-weighted report presents the annualised since-inception return through the most recent annual period end, and, for a track record shorter than a full year, the *non-annualised* since-inception return through the initial annual period end (5.A.1.a–b, p. 33). A *gross-of-fees* return is "reduced by any transaction costs", and a *net-of-fees* return is "the gross-of-fees return reduced by investment management fees" (Glossary, pp. 76 and 78).

A share's *holding-period return* adds income to the price change: total dollar return is dividend income plus capital gain, and total percent return is dividend yield plus capital-gain yield, each divided by the purchase price; the *price return* is the capital-gain part alone [principles-of-finance-2026, sec. 15.1]. Indexes are published in both forms: in the FT Wilshire 5000 series "the reinvestment of all dividend payments is reflected in the total return indexes at the close on the ex-dividend date but is not reflected in the price indexes" [ft-wilshire-5000-2026, sec. 1.1, p. 3]. Over several periods the *arithmetic* mean adds the returns and divides by their number; the *geometric* mean multiplies the growth factors $1 + r_i$, takes the $n$th root and subtracts one. The geometric average "tells you what you actually earned per year on average, compounded annually", the arithmetic "what you earned in a typical year"; the geometric is smaller unless all the returns are equal [principles-of-finance-2026, sec. 15.1].

**Notation.** One period from price $P_0$ to $P_1$ with income $D_1$ received at the end: price return $P_1/P_0 - 1$, income return $D_1/P_0$, total return $(P_1 + D_1)/P_0 - 1$, their sum. A total-return index multiplies its level each period by $(P_t + D_t)/P_{t-1}$, which is the return of holding the share and reinvesting the dividend at the ex-dividend price. Arithmetic mean $\bar{r} = \frac{1}{n}\sum r_i$; geometric mean $g = \left(\prod(1+r_i)\right)^{1/n} - 1$; $g \le \bar{r}$, with equality only when all $r_i$ are equal (the inequality of arithmetic and geometric means). For small returns $g \approx \bar{r} - \sigma^2/2$, with $\sigma^2$ the population variance of the $r_i$; this is an approximation, not an identity.

Interpretation, derived here: with sub-period returns $r_1, \ldots, r_n$, each measured between external cash flows, the cumulative TWR is $R = \prod_{i=1}^{n}(1 + r_i) - 1$; the geometric mean per sub-period, for equal sub-periods, is $(1+R)^{1/n} - 1$; and a return $R$ over $d$ days annualises mathematically to $(1+R)^{365/d} - 1$. That formula is arithmetic; whether an annualised figure may be *presented* is a separate rule, and under GIPS it may not be for periods shorter than a year (2.A.12). The MWR is the internal rate of return $x$ solving $\sum_t CF_t(1+x)^{-t} = 0$, with contributions negative and withdrawals and the final value positive, from the investor's side. A continuously compounded (log) return is $\ln(1+R)$. A real return is $(1+r_{\text{nom}})/(1+\pi) - 1$ for inflation $\pi$.

With own capital $V_0$ and borrowing $V_B$ at rate $r_B$ invested at return $r$ for one period, the return on own capital is

$$r_L = \frac{r(V_0 + V_B) - r_B V_B}{V_0} = r + \frac{V_B}{V_0}(r - r_B).$$

**Conditions.** The TWR removes the effect of the investor's timing of contributions and withdrawals, which suits judging a manager who does not control them; the MWR includes it, and GIPS allows firms to present it only when they control the external cash flows and the composite or pooled fund is closed-end, fixed-life, fixed-commitment or significantly illiquid (1.A.35, pp. 5–6). The IRR need not exist or be unique: the polynomial in $(1+x)^{-1}$ can have several positive roots when cash flows change sign more than once, or none. Flows of −1, +2.5, −1.54 have IRRs of 10 and 40 percent; flows of −1, +1, −1 have none, since their net present value is at most −0.75.

Annualising a short-period return assumes it would recur; GIPS forbids annualising returns for periods of less than one year (2.A.12, a calculation-methodology provision that also governs what is presented), and gives no reason; interpretation, that assumption is the obvious one. For periods of a year or more, annualising is a change of units. The leveraged formula assumes one period, borrowing at a fixed rate $r_B$ with no margin calls, fees or forced sales, and $r$ and $r_B$ measured over the same period.

**Practice.** Examples, original. A share bought at 50.00 pays a dividend of 1.20 and is sold at 54.00 a year later: price return 8.0 percent, income return 2.4 percent, total return 10.4 percent. A share at 100 goes ex-dividend with a dividend of 2 at a price of 102 and ends at 105: the price index rises 5.0 percent, the total-return index 7.0588 percent, the return on one share whose dividend buys 2/102 more shares. Returns of +20, −10 and +5 percent have an arithmetic mean of 5.0 percent and a geometric mean of 4.2808 percent, which compounds to the realised growth factor 1.134; the approximation $\bar{r} - \sigma^2/2$ gives 4.25 percent. A portfolio of 1,000 that earns 8.0 percent after transaction costs (gross of fees) and pays a management fee of 1.0 percent of opening assets at the period end returns $(1{,}080 - 10)/1{,}000 - 1 = 7.0$ percent net of fees.

Example: \$100 grows 10 percent to \$110, \$100 is added, and the \$210 falls 10 percent to \$189. The TWR is $1.10 \times 0.90 - 1 = -1.0$ percent *cumulative over both periods*, unaffected by the addition, a geometric mean of −0.50 percent per sub-period. The MWR solves $-100 - 100(1+x)^{-1} + 189(1+x)^{-2} = 0$, giving $x = -3.71$ percent *per sub-period*, or −7.29 percent compounded over both. On matching horizons the MWR is lower, −7.29 against −1.0 percent over the two periods and −3.71 against −0.50 percent per sub-period, because more money was invested in the losing period. Sub-period returns of +10, −10 and +5 percent compound to 3.95 percent, a geometric mean of 1.30 percent per period.

Four percent over 200 days annualises mathematically to 7.42 percent, a figure GIPS would not allow a firm to present (2.A.12). With 8 percent nominal and 3 percent inflation the real return is 4.85 percent, not 5. With $V_0 = 100$, $V_B = 50$, $r = 10$ and $r_B = 4$ percent, $r_L = 13$ percent. Harris's IRR and holding-period return, and the effect of non-pro-rata flows, are in relative performance evaluation.

**Pitfalls.** The cumulative TWR and the geometric mean per sub-period are different quantities (Notation). Use the geometric mean to describe realised compound growth and the arithmetic mean as the estimate of a single period's expected return; quoting an arithmetic mean as an annual compound rate overstates growth. A price index omits dividends, so its return understates the holder's return; a total-return index assumes the dividend is reinvested at the ex-dividend date. Sub-periods must be split at external cash flows. A real return adjusts for inflation alone. After-tax returns depend on jurisdiction, investor and realisation timing. The return identities are algebraic derivations.

**See also.** relative performance evaluation; interest-rate measures (compounding and annualising rates).

**Further reading.** Composite construction and disclosures under GIPS go beyond these definitions.

**Method 14.35 Testing skill against luck: the $t$-test of mean excess return, its power and data needs, sample length against sampling frequency, distributional shape, smoothed and fraudulent returns, sample selection and regression to the mean**

**Statement.** To judge whether a manager or strategy beats the market systematically, Harris compares the average market-adjusted return with "the standard error of the mean", the size of average return luck alone would produce; if their ratio, Student's $t$, is large, "luck alone probably cannot explain the returns"; for an unskilled manager the one-sided $t$ exceeds 1.64 only 5 percent of the time [alpha-harris, chap. 22, sec. 22.4.1, p. 453]. With excess returns $x_1, \ldots, x_n$, $t = \bar{x}/(s_x/\sqrt{n})$. The test's *power*, the probability of calling a skilled manager skilled, rises as the confidence level falls, with skill, with more years of data, and falls as luck matters more (sec. 22.4.2, p. 454).

Harris's calibration: skill of 2 percent a year, and a market-adjusted standard deviation of 7 percent from $\sqrt{16^2 + 14.5^2 - 2 \times 0.9 \times 16 \times 14.5} = 6.97$ (p. 455); a 95 percent test then identifies the skilled manager with probability 15 percent from five years of monthly data and 23 percent from ten (p. 455; Table 22-2, p. 456), and a test with 75 percent confidence and 75 percent power needs 22 years of data, one with 95 and 95 percent 132 years (Table 22-3, p. 457). "In practice, more than 20 years of returns data are typically required to obtain useful results for a given investment manager" (p. 453). The cause is "a low signal to noise ratio" (box, p. 463).

**Notation.** $x_i$ excess (market-adjusted) return in period $i$; $\mu$ and $\sigma$ its true mean and standard deviation; $\bar{x}$, $s_x$ their sample estimates; $n$ the number of periods; $\alpha$ the significance level (one minus Harris's confidence level) and $1 - \beta$ the power; $z_q$ the standard normal $q$-quantile.

**Conditions.** *Power and sample size* (derived). With independent normal returns the $t$-statistic is centred near $\mu\sqrt{n}/\sigma$, so a one-sided test at level $\alpha$ with power $1 - \beta$ needs about $n = \left((z_{1-\alpha} + z_{1-\beta})\,\sigma/\mu\right)^2$ periods: of order $(\sigma/\mu)^2$, as the proposition says, with a factor set by the error rates. In years, this reproduces Harris's tables: power 0.157 and 0.229 for five and ten years; 22.3, 132.6 and 33.1 years for (75%, 75%), (95%, 95%) and (50%, 95%) at 2 percent skill, and 33.1 years for (95%, 95%) at 4 percent. The proof's "expected $t$-statistic is approximately $\mu\sqrt{n}/\sigma$" is a large-sample statement: with ten annual observations the mean simulated $t$ is 1.00 against 0.90.

*Sample length, not frequency.* "statistical power depends primarily on the length of the sample period and not on the frequency of sampling within that period", because total performance over a period does not depend on how often it is observed; more frequent sampling can estimate the standard error better and helps mainly when there would otherwise be fewer than 20 observations (sec. 22.4.6.1, p. 466). Simulated over ten years, the mean $t$ is 0.91 from monthly and 0.90 from daily data. With serially correlated returns the naive standard error is too small: for AR(1) returns with coefficient 0.3 the naive $t$ of a zero-mean series has standard deviation 1.36 instead of 1 (Interpretation). More observations within the same period do not add history.

*Distributional shape.* The test assumes normal returns; fat tails make it less powerful, asymmetric strategies such as selling volatility (the peso problem) look skilled until the rare loss, and smoothed valuations of illiquid assets lower measured volatility, giving "artificially high $t$-ratios, and therefore overconfident conclusions"; pyramid schemes look skilled until they collapse (sec. 22.5, pp. 467–470). A stream earning 1 percent a month with a 2 percent monthly chance of losing 40 percent has a true mean of 0.18 percent a month and shows no loss in 60 months with probability 0.30.

**Practice.** *Selection.* "The sample selection bias arises when some process selects the information that you see"; fund families advertise their best funds and merge away their losers (survivorship bias), and newsletters with poor records disappear (sec. 22.6, pp. 470–473). Significance must be judged against all the managers who might have been lucky: a $t$ of 4.9 for Buffett's 36 years is extreme for one manager, but Harris estimates that the best of 10,000 unskilled managers would beat the market by 11.8 percent a year with probability 0.5 percent, and the best of 100,000 with 5 percent (box, p. 463). With normal 36-year averages, these are 0.37 and 3.6 percent; Harris's method is not stated.

*Regression to the mean*: selection can raise an average of past returns but "cannot affect subsequent returns", so "When selection has caused the past return average to be high, the subsequent return average is invariably lower" (sec. 22.6.3, p. 474). In a simulation where skill has variance $\tau^2 = 0.25$ and luck $\sigma^2 = 1$, the top decile's next-period mean is 0.20 of its selection-period mean, $\tau^2/(\tau^2 + \sigma^2)$ (see zero-sum trading, skill and luck).

**Pitfalls.** Additional sources of bias include *look-ahead* (timestamps, corporate actions or benchmark constituents that use future information), which belongs to point-in-time reference data and time-validity of labels, and *fillability* (simulated fills assume unreachable liquidity), which is backtest cost assumptions; multiple testing is Harris's best-of-many problem, whose corrections (deflated Sharpe ratios, stepdown tests) are in deflated Sharpe ratio and multiple-testing control. That most backtest errors are sampling errors before they are modelling errors is a generalisation; Harris says sample selection biases "may be responsible for more trading losses than any other cause" (p. 472). The calibration (2 percent skill, 7 percent noise) is Harris's assumption for diversified U.S. equity managers as of 2002, and the test is one-sided on normal monthly returns. See `wide-fama-french-luck`, `wide-barras-scaillet-wermers` and `wide-getmansky-lo-makarov` on luck in fund returns, smoothing and hedge-fund returns.

**See also.** relative performance evaluation (market-adjusted returns and smoothing); zero-sum trading, skill and luck; parameter overfitting; backtest evaluation diagnostics; venue market share (proportion intervals).

**Further reading.** Harris, chap. 22, secs. 22.4–22.6.

**Method 14.36 A GIPS composite report: which portfolios form a composite, the statistics it must present and its first required disclosures (GIPS for Firms, 2020 edition)**

**Statement.** The 2020 edition of the GIPS standards for firms, effective 1 January 2020 [wide-gips-2020, Introduction, PDF p. 17], sets composite construction in section 3 [wide-gips-2020, provisions 3.A.1–3.A.11, pp. 18–19]:

- a firm "must create composites" for strategies managed or offered as segregated accounts, and "all actual, fee-paying, discretionary segregated accounts must be included in at least one composite"; non-discretionary portfolios must not be;
- composites are "defined according to investment mandate, objective, or strategy" and include every portfolio meeting the definition, whatever its legal structure, and a change of definition "must not be applied retroactively";
- new portfolios enter "on a timely and consistent composite-specific basis"; only portfolios managed for the full measurement period count; terminated portfolios stay in history "up to the last full measurement period"; portfolios move between composites only on documented client-directed changes or a redefinition, and their history stays with the original composite.

A composite time-weighted return report must present (4.A.1, pp. 21–22):

- at least five years of compliant performance, or since inception, building up to ten;
- annual composite returns and "the total return for the benchmark for each annual period and for all other periods for which composite returns are presented, unless the firm determines there is no appropriate benchmark" (4.A.1(b) and (e), PDF p. 39, printed p. 21);
- the number of portfolios, if more than five, composite assets and total firm assets at each annual period end;
- a measure of internal dispersion of portfolio returns, if more than five portfolios were held for the full year;
- where monthly returns are available, the three-year annualised ex post standard deviation of composite and benchmark, from monthly returns.

It must label the periods and whether returns are gross or net of fees (4.A.3). Its disclosures begin with one of three prescribed compliance statements: verified, verified with a performance examination, or not verified. Then come the trademark statement, the definition of the firm, the composite description and the benchmark description (4.C.1–4.C.5, pp. 25–27).

*No appropriate benchmark.* The benchmark column has its own exception, 4.A.1(e) above. When it is used, "If the firm determines no appropriate benchmark for the composite exists, the firm must disclose why no benchmark is presented" (4.C.31, PDF p. 47, printed p. 29). Money-weighted composite reports and pooled-fund reports carry the same disclosure (5.C.30, 6.C.26, 7.C.26), and the time-weighted pooled-fund report the same

exception (6.A.1(e)). In GIPS advertisements the explanation is only recommended (8.D.4, 8.F.4, 8.H.1: “should disclose”). Money-weighted composite reports, for example for closed-end funds, present since-inception money-weighted returns instead (5.A.1, p. 33).

**Notation.** Composite: an aggregation of portfolios managed to a similar mandate; internal dispersion: the spread of annual returns across the composite’s portfolios (the firm discloses which measure); ex post standard deviation: of 36 monthly returns, annualised.

**Conditions.** The provisions are requirements of a voluntary standard: they bind a firm that claims compliance, and the firm must disclose when the three-year standard deviation is not presented because 36 monthly returns are not available (4.C, p. 30). Footnotes date some requirements: total firm assets from periods ending 31 December 2020 (n. 31), part-period returns for composites starting from 1 January 2011 (nn. 29–30).

**Practice.** Interpretation: the composite rules are survivorship and cherry-picking controls. Every qualifying account must be included, terminated accounts stay in the history, nothing is applied retroactively and portfolios move only for documented reasons. The required statistics let a reader see scale, dispersion across clients and risk next to return. A toy report needs the five-year table with these columns and the compliance statement wording, not only returns.

**Pitfalls.** Edge cases in computing internal dispersion or the 36-month figure are covered by the GIPS Guidance Statements and Q&A. Pooled-fund reports (section 6 onwards) differ and were only searched for the matching provisions.

**See also.** return measures (time- and money-weighted returns); relative performance evaluation; tests of skill against luck.

**Further reading.** GIPS Guidance Statements and the Q&A database.

# 15 Funding, collateral and liquidity

## 15.1 Crowding and common-investor contagion

**Definition 15.1 Common-investor (contagion) risk: crowded positions and forced exits through what else their holders own, with Narang's account of August 2007**

**Statement.** Contagion, or common-investor, risk is the risk a strategy bears "not because of the strategy itself but because other investors hold the same strategies" [s7-narang, chap. "Risks Inherent to Quant Strategies", sec. 10.4, p. 202]. Narang splits it in two:

- *crowding*: how many others hold the same positions;
- the *ATM effect*: what else those holders own that could force them out. When highly levered investors take losses on an illiquid holding and face margin calls, they sell what they can, the liquid strategy, to raise cash. Losses in one strategy thus cause the liquidation of an unrelated one (p. 202).

His examples are August 1998, in which LTCM's losses on Russian debt forced it to sell positions such as Royal Dutch against Shell, whose spread widened from 8–10 percent to over 20 percent (pp. 202–203), and August 2007 (sec. 10.4.1, pp. 203–209).

**Conditions.** Practitioner account as of 2024 of events in 1998 and 2007. *Narang's account of August 2007, a practitioner's explanation.* He names four drivers (pp. 203–205):

1. the size and popularity of quant long/short equity strategies from 2004 to 2007. He estimates "likely" hundreds of billions of capital and about $1 trillion of gross positions, mostly in large-cap U.S. stocks; despite diverse models, the overlap was enough to crowd individual trades;
2. weak returns before August, with value lagging growth from about the end of May 2007, and multi-strategy and proprietary desks that cut whatever had done poorly;
3. the practice, "in my view, a critically important one", of banks' proprietary desks and multi-strategy funds cross-collateralising strategies explicitly or implicitly. When credit strategies lost heavily in July 2007, the liquid quant books were sold to raise cash;
4. risk targeting: leverage set inversely to measured risk, usually VaR. Falling volatility from 2003 to 2006 called for one and a half to two times more leverage, so a four-times gross strategy became six or eight times. Higher returns were also sought by levering up. When volatility and correlations jumped together in late July 2007, VaR rose and many players had to cut leverage at once.

In his sequence, multi-strategy funds and desks began deleveraging the most liquid book, U.S. quant long/short. Statistical arbitrageurs first supplied liquidity, then took losses and joined the selling. By 8–9 August the losses had spread across strategies and to Japan, while U.S. indices were roughly flat for the first ten days of August (pp. 205–209). He calls the losses "solely the result of market impact" (p. 208), reports losses of $-5$ to $-45$ percent at funds (p. 208), and concludes that "for the first time, crowding became a risk of quant trading strategies" (p. 209). Phrases such as "it is likely", "it appears" and "in my view" mark these as his inferences: the chapter gives no data on positions, leverage or flows.

**Practice.** Interpretation: the entry separates two channels. In *common positions*, crowding, the same trades are held by many, so any one holder's exit moves the others' prices. In *common funding*, the ATM effect, an unrelated loss forces a holder to sell. Cross-collateralisation is what joins the second channel to a liquid strategy, and volatility targeting can make many holders sell at the same moment. The same rule also has a return case. Moreira and Muir "scale monthly returns by the inverse of their previous month's realized variance". They report that such volatility-managed portfolios "produce large alphas" and raise factor Sharpe ratios, "because changes in factor volatilities are not offset by proportional changes in expected returns" [s16-r25, NBER WP 22208, abstract and p. 1]. Their result concerns one investor's scaling.

It does not address what happens when many investors with the same rule deleverage into the same volatility jump; that is Narang's and Pedersen's concern. *Size and a stylised deleveraging*. The ECB's May 2020 Financial Stability Review gives "varying estimates" of the capital involved: globally "up to USD 2 trillion" in some form of volatility strategy, and "USD 300 billion invested in some 100 risk parity funds" [ecb-fsr-2020-05-volbox, box "Volatility-targeting strategies and the market sell-off", May 2020]. Its stylised risk-parity fund holds euro-area equity, high-yield, investment-grade and sovereign bond ETFs with an 8 percent volatility target. Before mid-March 2020 low volatilities and correlations let it leverage bonds and equities "by up to twice the amount of assets under management".

In March, rising volatility and correlation meant "assets worth nearly 225% of the portfolio's capital had to be sold", leaving a cash share near 25 percent, with sales in every asset class. That is a model rule applied to ETF prices, not observed fund flows. It makes a leverage of about two plausible for such strategies, without measuring Narang's 1.5–2 times or how much capital actually deleveraged together.

*Crowding as measured in the literature*. Chincarini, Lazo-Paz and Moneta define crowded trades as those "in which many investors hold the same stocks possibly exhausting their liquidity provision". They find anomaly risk-adjusted returns generated mainly by the most crowded stocks in the long leg and the least crowded in the short leg, persisting after publication dates, and read crowding as exposure to crash risk and a limit to arbitrage [w4-chincarini-crowded-anomalies, abstract].

Kyle, Obizhaeva and Wang model traders who misjudge how crowded a market is: those who underestimate it trade too aggressively and earn less than expected, crowded markets are prone to abrupt crashes, and the size of price dislocations and the speed of recovery in fire sales can reveal traders' beliefs about crowdedness [w3-kyle-crowded-markets, abstract]. Neither abstract uses a delayed-signal decay as a crowding measure. The same mechanics in arbitrage, where staying power fails under leverage, are in arbitrage risks. Leverage and margin terms are in futures margin and settlement and lending fees and rebates.

**Pitfalls.** This practitioner's reconstruction is not causal evidence that the four drivers caused the losses, nor a measure of their weights.

*Measured evidence*. Khandani and Lo simulate long/short portfolios on five valuation factors and a market-making (contrarian) strategy [exp-khandani-lo-quant-meltdown, NBER WP 14465, November 2008, abstract and sec. 4.3, PDF pp. 2 and 19]. They find:

- evidence "that the unwinding of these portfolios began in July 2007 and continued until the end of 2007";
- market-making returns "significantly negative during the week of August 6, 2007, but positive before and after";
- two unwinds, a mini-unwind on 1 August, 10:45 to 11:30, and a sustained one from the open on 6 August to 1 p.m., beginning with financials and with long book-to-market and short earnings momentum;
- book-to-market, cash-flow and earnings-to-price portfolios unwound on 2–3 August, momentum portfolios hit on 8–9 August, and on 10 August "sharp reversals in all five strategies erased nearly all of the losses of the previous four days".

An 8:1 levered book-to-market portfolio would have lost 24 percent by the close of 7 August, "likely to have triggered a reduction in leverage". Weekly NYSE volume in the weeks of 23 July to 13 August reached 2.9, 3.0, 3.6 and 3.1 billion shares (PDF p. 13). They read the episode as portfolio deleveraging plus "a temporary withdrawal of marketmaking risk capital starting August 8th". *The market-making path*: an intraday mean-reversion strategy rebalanced every $m$ minutes earns more the shorter the interval, and for every $m$ shows "the same dip in profits during the second week of August"; at five minutes the cumulative profit levels off on 6 August, falls from 7 to 13 August and then resumes "at nearly the same rate". The paper gives these paths as figures, not as daily percentages, so no daily loss is quoted here.

This supports Narang's sequence, a liquid-book unwind followed by liquidity providers' losses, with dated, simulated returns. It does not observe positions, leverage or cross-collateralisation, so his first, third and fourth drivers remain inferences: the evidence is consistent with them but does not identify them. Pedersen gives the mechanism of the fourth: as volatility rises over a lengthening liquidation horizon, "one trader's prudent risk management can be another trader's vanishing market liquidity and funding", a risk-management spiral; and sophisticated traders' portfolios overlap because they pursue the same goal on common data [wide-pedersen-exit, NBER WP 15297, August 2009, secs. 2.2 and 3.2, PDF pp. 8 and 15]. Neither paper gives a measure of overlap. Narang's Pulte Homes illustration (Exhibit 10.6, p. 206) does not support all its words.

By the table, the stock fell 22.0 and then 12.5 percent (31.75 percent together) and then rose 15.6 percent, which recovers about a third of the lost price, not "about half". Over his 44-day and four-day periods that is roughly four to five times the earlier pace, not "about 20 times". He also calls the episode "an interesting illustration of the quadratic nature of market impact" (p. 206). Interpretation: the table cannot test that. Aggregate returns and average daily volume over a crisis period do not identify a metaorder impact exponent: they give no metaorder sizes or execution schedules, the volume mixes many traders' buying and selling, and the price that would have prevailed without the liquidation is not observed.

The illustration is therefore evidence neither for a quadratic nor against the square-root law for metaorders (square-root impact), which is estimated from identified metaorders. His "as much as 15 percent in market impact" on 9 August is likewise unsourced in the chapter. Whether faster decay of a delayed signal identifies crowding is a separate question. Common positions and forced liquidation, the subject here, do not answer it.

**See also.** arbitrage risks (staying power under leverage); sources of strategy risk (the other three quant-specific risks); square-root impact (impact of large liquidations); futures margin and settlement (margin calls).

#### Model 15.2 Market liquidity and funding liquidity: the margin constraint, destabilising margins, and the margin and loss spirals of Brunnermeier and Pedersen

**Statement.** Speculators who supply immediacy to customers arriving at different times must finance positions by collateralised borrowing, and "the total margin on all positions cannot exceed a trader's capital" [exp-brunnermeier-pedersen-liquidity, RFS Advance Access version of 10 December 2008, p. 1]:

$$\sum_j \left(x_t^{j+} m_t^{j+} + x_t^{j-} m_t^{j-}\right) \le W_t, \qquad W_t = W_{t-1} + (p_t - p_{t-1})' x_{t-1} + \eta_t \quad \text{(eqs. (4)–(5), p. 9)},$$

with financiers setting each dollar margin so that an adverse price move larger than it has probability $\pi$, a value-at-risk rule (eqs. (6)–(7), p. 9). Market illiquidity is the gap between price and fundamental value; funding liquidity is the speculators' shadow cost of capital (p. 2). Margins are *destabilising* if they can rise with illiquidity, which happens when financiers cannot tell fundamental news from liquidity shocks and volatility is time-varying (p. 2; Proposition 3, p. 13). Then two *liquidity spirals* amplify a shock to speculator capital (sec. 3.2, pp. 18–19): a *margin spiral*, in which lower liquidity raises margins, tightening funding and lowering liquidity further; and a *loss spiral*, in which speculators holding a position opposite to the customers' demand pressure lose as prices move away from fundamentals and must sell more. In a stable illiquid equilibrium with customer selling pressure the price sensitivity to a wealth shock is

$$\frac{\partial p_1}{\partial \eta_1} = \frac{1}{\frac{2}{\gamma(\sigma^2)^2} m_1^+ + \frac{\partial m_1^+}{\partial p_1} x_1 - x_0} \quad \text{(eq. (24), Proposition 5(ii), p. 18)};$$

a margin spiral arises if $\partial m_1^+ / \partial p_1 < 0$ (margins rise as prices fall), and a loss spiral if the speculators' previous position is opposite to the demand pressure, $x_0 Z_1 > 0$; together the spirals' effect "is greater than the sum of their separate effects" (p. 19).

**Notation.** $x_t^{j\pm}$ long and short holdings of asset $j$; $m_t^{j\pm}$ dollar margins per share; $W_t$ speculator capital, $\eta_t$ an independent wealth shock; $Z_1$ the demand shock of customers present at time 1; $\gamma$ customers' risk aversion and $\sigma^2$ the fundamental volatility at time 2 in the model's notation; $p_1$ the time-1 price. Signs: with $Z_1 > 0$ customers are selling.

**Conditions.** Theory, published 2009. Three trading dates before a time-3 payoff; risk-neutral speculators, CARA customers who arrive sequentially with probability $a$; fundamentals follow $v_{t+1} = v_t + \sigma_{t+1}\varepsilon_{t+1}$ with ARCH-type volatility $\sigma_{t+1} = \bar{\sigma} + \theta\,|\Delta v_t|$ (eqs. (1)–(2), p. 8); speculators cannot raise new capital before time 2, "slow moving capital" (p. 9). The mechanism of eq. (24): with constant margins and no inventory a $1 wealth loss cuts the position by $1/m_1$ and moves the price by $1/(2m_1/(\gamma(\sigma^2)^2))$; writing the denominator as $k - l$, the expansion $1/(k-l) = 1/k + l/k^2 + l^2/k^3 + \cdots$ has one term per loop around the spiral, and the effect is amplified when $l > 0$ (p. 19).

The expansion and the super-additivity hold on illustrative numbers (an illustration, not a calibration). Spirals can also start from shocks to liquidity demand, fundamentals or volatility (p. 19). With destabilising margins or large existing positions there can be multiple equilibria, so liquidity is *fragile* and can dry up discontinuously (p. 3; Proposition 4, p. 14).

**Practice.** Implications the authors draw (pp. 2 and 5): speculators' mark-to-market capital and volatility are state variables for market liquidity and risk premiums; a loss of capital reduces liquidity more when capital is already low and for high-margin securities; across assets the ratio of illiquidity to margin is common, which gives commonality and flight to quality. They show S&P 500 futures margins rising in the liquidity crises of 1987, 1990, 1998 and 2007 (Figure 1, p. 3). Interpretation for a funding ladder: margin and haircut schedules belong in the stress scenario as functions of the shocked prices and volatility, not as constants, and a portfolio's existing positions relative to the crowd's flow decide whether a loss spiral applies (common-investor risk).

**Pitfalls.** The spirals are conditional: with slack capital the price equals fundamental value and does not respond to small wealth shocks (Proposition 5(i), p. 18), and with informed financiers margins can be stabilising (Proposition 2, p. 12); the mechanism is a model result, not a description of every sell-off. Locators are to the journal's Advance Access pages 1–38, not the issue pagination (*RFS* 22(6), 2201–2238).

**See also.** common-investor (contagion) risk; Grossman–Miller (the liquidity-provision model it extends with a funding constraint, p. 2); futures margin and settlement; arbitrage risks.

## 15.2 Securities lending and short selling

#### Definition 15.3 Street name and beneficial ownership: holder of record, beneficial owner, custody, permission to lend, and who votes lent shares

**Statement.** Clients hold securities in *street name* when their broker or a depository holds them on their behalf. The issuer then registers the broker or depository as the *holder of record*, and the client becomes a *beneficial owner* holding a claim on the broker rather than legal title [alpha-harris, chap. "Brokers", secs. 7.2.2.2 and 7.5.6.1, pp. 148 and 169–170]. Four things are kept apart:

- *registration*: whose name is on the issuer's register (Harris notes that most securities held by the Depository Trust Company are registered in the name of its nominee, Cede & Co., p. 169);
- *beneficial ownership*: the client's economic claim, recorded in the broker's books;
- *custody*: the broker's or depository's safekeeping, with duties to collect and allocate dividends and interest and to process corporate actions (p. 148);

- *permission to lend*: in Harris's U.S. account, a margin agreement lets the broker hypothecate the client's securities as loan collateral and lend them to short sellers; clients generally do not know whether their securities were lent and generally receive no lending fee, which is why some large institutions hold securities in their own name (pp. 169–170).

Traders hold in street name to avoid losing certificates, to use securities as margin collateral and to have them available for settlement (p. 148).

**Conditions.** U.S. practice as described in 2001–2007; ICMA as of October 2026. Voting. Only holders of record have the legal right to vote; a beneficial owner can only instruct its broker. A securities loan transfers legal title, and with it the voting rights attached to equities [al-icma, Repo FAQ 13]. Several parties may hold economic claims; only record shares carry legal votes.

When a lent share is sold short, its buyer may register it and acquire the vote, so the beneficial owners in a broker's books can together believe they own more shares than the broker holds of record. Harris's example: the broker then votes its record shares in proportion to its clients' instructions, and since many clients do not instruct, it usually has enough shares unless it has lent too many [alpha-harris, sidebar "Can Investors Vote More Than 100 Percent of the Shares Outstanding?", p. 169].

Because a loan transfers votes, lenders keep an express right of recall so that they can recover securities to vote or respond to corporate actions [al-icma, Repo FAQ 13]; the borrower pays the lender amounts equal to dividends ("cash-in-lieu") [al-duffie, sec. 2, p. 8].

**Practice.** Record dates decide who may vote. Hu and Black call borrowing shares shortly before a meeting's record date and returning them soon after "record date capture": the borrower holds votes without economic ownership and the lender economic ownership without votes [cfd-hu-black-2007, working-paper version, sec. 2.2.3, p. 5]. They report that in the United States Federal Reserve Board Regulation T bars lending shares for voting purposes "in many circumstances". Decoupling through equity swaps and options, and beneficial-ownership disclosure, are a separate matter (Pitfalls).

**Pitfalls.** Stock lending can separate economic exposure from voting rights, and brokers need entitlement records because more investors may believe they own voting shares than there are record shares available. Both match Harris's sidebar, which describes how brokers vote, not a legal requirement. Do not infer duplicate legal votes from several economic claims. Empty voting, hidden ownership through equity swaps and the 2023 U.S. beneficial-ownership reporting rules are not covered here. Harris describes U.S. practice as of 2002.

**See also.** securities lending for short sales; lending fees and rebates; broker conflicts.

### Mechanism 15.4 Securities lending for short sales: locate, borrow, delivery, recall, replacement borrow and buy-in, with broker and institutional layers kept apart

**Statement.** A short seller sells securities it does not own and must deliver them at settlement; the standard route is to borrow them for delivery through a stock loan. In Duffie, Gârleanu and Pedersen's description of U.S. equity lending in 2001 [al-duffie, sec. 2, pp. 5–8]:

1. *locate*: the would-be short seller asks its broker for a "locate"; the broker looks in its own inventory, in customer accounts that permit lending, then to custodian banks or other lenders, such as insurers, index funds and pension funds. Hard-to-locate stocks may get partial fills or delays (pp. 5–6);
2. *borrow*: given a locate, the loan can be done the same day; through a broker, the broker borrows from the outside lender and lends to the outside borrower (p. 7). A broker may pay a lender to hold located shares until the short sale ("pay-for-hold", p. 6);
3. *delivery*: the borrowed shares are delivered to the buyer at settlement;
4. *recall*: loans are normally open, renewed daily; the lender may recall, and the borrower must return the shares, typically by borrowing them from another lender (*replacement borrow*) or by its broker recalling them from another borrower (pp. 7–8);
5. *buy-in*: if no lendable shares can be found the borrower must buy the stock; if the borrower fails to deliver within standard settlement time, the lender may buy it with the cash collateral, the borrower remaining liable for extra costs. A buy-in interrupts the short sale (p. 8).

A securities loan transfers legal title in exchange for collateral, and lenders keep an express right of recall [al-icma, Repo FAQ 13].

**Conditions.** *U.S. rules (Regulation SHO, 17 CFR part 242, as published on eCFR for 13 October 2026).* A broker-dealer may not accept a short-sale order in an equity security, or short for its own account, unless it has borrowed the security or arranged in good faith to borrow it, or has reasonable grounds to believe it can be borrowed for delivery when due, and has documented compliance, with stated exceptions including bona-fide market making [cfr-242-203, § 242.203(b)]. A clearing-agency participant with a fail to deliver on a short sale must close it out by borrowing or buying securities of like kind and quantity by the beginning of regular trading hours on the settlement day after settlement date, with longer periods for some cases [cfr-242-204, § 242.204(a)].

A locate is therefore a belief or arrangement before the sale, not a completed borrow; delivery and close-out are later obligations. Harris's 2002 "affirmative determination" (p. 155) predates Regulation SHO. Other jurisdictions' rules are not covered.

*U.S. margin and price test, current.*

- *Initial margin.* Regulation T sets the initial margin on a short sale of a nonexempted equity security at "150 percent of the current market value of the security", or 100 percent when a security exchangeable or convertible within 90 days is held [ecfr-12cfr220, 12 CFR 220.12(c), eCFR text for 15 October 2026]. The 150 percent is the sale proceeds plus a further 50 percent.
- *Maintenance margin.* FINRA's rule requires, for each stock short, the greater of $2.50 a share or 100 percent of market value below $5.00 a share, and the greater of $5.00 a share or 30 percent of market value at $5.00 or above [finra-rules-2026, Rule 4210(c)(2)–(3), version from 4 June 2026]. Brokers may require more.
- *Price test.* Rule 201 of Regulation SHO is triggered when a covered (NMS) stock falls 10 percent or more from its prior closing price on the listing market. Trading centers must then prevent the execution or display of a short sale "at a price that is less than or equal to the current national best bid" for the rest of that day and the following day. A short sale order displayed above the national best bid when first displayed may still execute, and brokers may mark qualifying orders "short exempt" [ecfr-17cfr242-201, 17 CFR 242.201(b)–(c), adopted 75 FR 11323, 10 March 2010, last amended 2024].

DGP's 2001 description of a price test is therefore historical. The present test is this circuit-breaker form, the "alternative uptick rule", not a tick test on every sale; the name is common usage, not the rule's heading.

*Market practice, not law.* ISLA's best-practice handbook, which yields to the Global Master Securities Lending Agreement and to local codes, recommends that a recall notice give the ISIN, quantity and settlement date, reach the borrower at the latest one hour before the relevant market close minus two business days or, if greater, the standard settlement cycle; that the borrower acknowledge; and that, if the return fails to settle or the borrower does not respond, the lender calculate and claim the costs under GMSLA 9.3, after which the loan is closed and collateral returned [al-isla, "Loan Recall to buy-in/close out" and "Legal", pp. 2–3 and 24–25; sections last updated 28 January 2022].

**Practice.** Keep apart, Interpretation using the sources above:

- *broker–customer arrangements*: the customer's margin agreement, its short-sale margin and what the broker pays it, and the broker's permission to lend customer securities;
- *the institutional loan*: lender (or agent) and borrower under a master agreement, with collateral, fee or rebate, mark-to-market, recall and claims;
- the *availability* of a security to borrow (general collateral or special, lending fees and rebates) from *operational* risks: a locate that cannot be turned into a borrow, a delivery failure, a recall, a failed return, a buy-in.

Factors said to make shares hard to locate include small capitalisation, low float, exclusion from indices, illiquidity, concentrated ownership and events such as IPOs, mergers, spinoffs and acquisitions; the authors had no data on locate delays (p. 6).

**Pitfalls.** Brokers source inventory from street-name holdings, stock-loan desks, other brokers or institutions, earning lending fees or rebate spreads, which matches Harris (pp. 155 and 158) and Duffie, Gârleanu and Pedersen. Recall risk and settlement risk are operational risks, not availability classes beside general collateral and specials. A failed locate does not by itself break a short sale: under Regulation SHO the locate precedes the order, and it is a delivery failure that triggers close-out. DGP's description of retail short-sale margin and of a price test in 2001 is historical; the current rules are stated under Conditions.

**See also.** lending fees and rebates; street name and beneficial ownership; synthetic short through options; arbitrage risks.

#### Definition 15.5 Lending fees and collateral rebates: cash and non-cash collateral, general collateral and specials, reinvestment returns, broker spreads and customer margin

**Statement.** A securities borrower pays the lender a fee for the use of the security. Against non-cash collateral (other securities) the fee is paid outright. Against cash collateral, the lender reinvests the cash and *rebates* an agreed part of the reinvestment return to the borrower, usually deducting the fee, so that the fee is implicit in the rebate rate [al-icma, Repo FAQ 13]. ISLA describes the rebate as a negotiated spread in basis points over or under a benchmark rate [al-isla, "Cash Rebate", pp. 7–8].

In Duffie, Gârleanu and Pedersen's U.S. description, cash collateral is normally 102 percent of the market value of the shares for domestic securities (105 percent for international). An overnight rebate rate $r$ pays $r/360$ times the cash collateral per day, and the shortfall of the rebate below a market rate such as the federal funds rate is the lender's benefit [al-duffie, sec. 2, p. 7].

With $r_m$ a market overnight rate and $r_b$ the rebate rate on cash collateral, the implicit annualised lending fee is approximately $r_m - r_b$ (Interpretation); the definitions above do not fix the benchmark.

**Conditions.** Market practice as described in 2001–2002 and by ICMA and ISLA in 2021–2026. *Availability classes.* Easy-to-borrow securities carry small fees; in Harris's 2002 U.S. account the fee is "only about 10 basis points per year", and hard-to-borrow securities are said to be *on special*, with fees that depend on scarcity [alpha-harris, sec. 7.3.1.4, pp. 156–157]. Fees are highest for closely held securities that interest short sellers (sec. 7.3.1.7, p. 158). Duffie, Gârleanu and Pedersen model lending fees as the outcome of search for lenders and bargaining: if lendable securities are hard to find, the price is initially elevated and expected to decline, and may exceed even the most optimistic buyer's valuation because owners expect lending fees (abstract; a model result).

*Measured fees.* Sikorskaya uses S&P Global (Markit) buy-side data, whose fees are "typically derived from the quoted rebate rates", the difference between a short-term market rate and the rebate on cash collateral (n. 7, p. 4). She calls a stock on special when its annualised borrowing fee is above 1 percent, following the literature, and studies Russell 3000 stocks near the Russell 1000/2000 cutoff, 2006–2018 [catalogue-0897, author version of 27 April 2026, secs. 4.1–4.2, pp. 16–17].

- *General collateral* (13,047 stock-years): the May borrowing fee averages 0.39 percent a year (median 0.38, 99th percentile 0.79).
- *Special* (638 stock-years, about 5 percent of stocks near the cutoff): it averages 5.61 percent (median 3.12, 99th percentile 31.82), with about 17 percent of shares on loan and utilisation of lendable inventory near 76 percent (Table 1, p. 18).
- Over 80 percent of the U.S. sample's stocks are special at some point (p. 6).

These figures support the ordering in the sources above, small fees for general collateral and much larger, dispersed fees for specials, within that sample. They do not give a negative rebate, and they describe stocks near one index cutoff, not the whole market. The 2001 "about 10 basis points" is not comparable: it was Harris's illustrative U.S. figure, and the 0.39 percent here is a borrower-side measure. *General collateral* is used here for the easy-to-borrow class; the term's meaning in repo, collateral not specific to an issue, is related but not identical (Interpretation).

**Practice.** Who receives what, in the sources' U.S. descriptions:

- the *lender* earns the fee, or the reinvestment return minus the rebate, and bears reinvestment risk on cash collateral (Interpretation);
- the *broker* intermediating the loan normally offers its outside borrower a lower rebate than it receives (DGP, p. 7), a broker spread;
- *retail short sellers* typically receive no interest on their short proceeds, and retail customers whose stocks are lent typically receive no fee (DGP, p. 7; Harris, pp. 157 and 170); large clients demand a rebate (Harris, p. 157);
- *customer margin* is a separate obligation of the customer to its broker, not collateral in the institutional loan; DGP give the Regulation T requirement of 2001 as an additional 50 percent of market value (p. 7, footnote 9). The current text agrees: 150 percent of market value including the proceeds (12 CFR 220.12(c)(1); securities lending for short sales).

**Pitfalls.** The short seller does not post the cash collateral or receive the rebate directly: cash collateral passes from the borrower (often the broker) to the lender, and the retail short seller's obligation is margin to its broker. The harder the borrow, the lower the rebate and the higher the effective shorting cost, as the sources show for cash-collateral loans; rebates can in principle be negative when the fee exceeds the reinvestment rate, though the sources give no example. Harris's rebate benchmarks (federal funds or LIBOR) are of 2002 and are not current.

**See also.** securities lending for short sales; synthetic short through options; futures margin.

## 15.3 Money markets and reference rates

#### Definition 15.6 Money-market instruments and reference rates: bills, commercial paper, repo, CDs and bankers' acceptances; fed funds, SOFR, LIBOR's end, EONIA and €STR, SONIA; and the LIBOR–OIS spread as a historical stress gauge

**Statement.** Money markets are short-term, liquid credit instruments: commercial paper, Treasury bills, agency and municipal paper, bankers' acceptances, time deposits, interbank loans and repurchase agreements [s7-gliner, chap. 9, sec. "Funding/Money Markets", PDF pp. 246–253]. In Gliner's account commercial paper is unsecured, "Most has a maturity of between 90 and 270 days, since it can avoid registering with the Securities and Exchange Commission (SEC) if the security has a maturity of less than 270 days", and it is cheaper for issuers than bank borrowing (p. 246).

Time deposits (CDs) run from 30 days to five years, those of a year or less counting as money-market instruments (p. 252). Bankers' acceptances are short-term obligations guaranteed by a bank, mainly for trade (pp. 251–252). A repo is "a sale of securities with a simultaneous agreement by the seller to repurchase them at a later date, which is usually done overnight", mostly against Treasuries or GSE debt (p. 252).

*Fed funds* are unsecured overnight loans of reserve balances, and the effective rate serves as the reference for OIS (p. 257). *LIBOR*, in the BBA's definition, was the rate at which a panel bank could borrow "by asking for and then accepting interbank offers in reason [sic] market size, just prior to 11:00 A.M. London time", computed as a trimmed mean of 16 banks' submissions (p. 254). The LIBOR–OIS spread compares unsecured term bank borrowing with the compounded overnight rate; it "skyrocketed" in 2008, rose with European banks' dollar funding needs in the euro crisis, and "does not automatically indicate a crisis" but is "a useful barometer for credit risk" (pp. 261–262).

The SEC staff's interpretations add that commercial paper "payable on demand but in any event no later than 9 months from issuance" meets the maturity requirement, and that tracing proceeds to actual current transactions is not necessary where they are commingled with general funds and an equal amount is applied to current transactions (Corporation Finance interpretations 219.01 and 219.02, both dated 26 November 2008) [sec-cdi-securities-act, Securities Act Sections, 219.01–219.02, page updated 23 June 2026].

The Richmond Fed's money-market handbook (chapter last updated 1993) lists further historical conditions: notes "of a type not ordinarily purchased by the general public" (in practice minimum denominations usually $100,000), current-transaction use, and, under SEC Release 33-4412 (1961), "prime quality" and discountability at a Federal Reserve Bank, the discounting requirement having been "dropped in 1980"; it puts typical maturities at 5 to 45 days, averaging 30–35, against Gliner's 90 to 270, and notes that rollover is allowed if not automatic [richmondfed-cp-1993, chap. 9, pp. 105–107 and n. 4]. Private placements (Section 4(a)(2)) and bank-guaranteed paper (Section 3(a)(2)) are other exemptions.

**Notation.** SOFR the Secured Overnight Financing Rate; EFFR the effective federal funds rate; €STR the euro short-term rate; SONIA the Sterling Overnight Index Average.

**Conditions.** *Current benchmarks.* SOFR measures the cost of overnight borrowing collateralised by Treasuries, and the EFFR is a volume-weighted median of overnight federal funds transactions, both published by the New York Fed [exp-sofr-data, SOFR description; EFFR and OBFR methodology]. The U.S. dollar LIBOR panel ended after 30 June 2023 [arrc-sofr-transition, web page], with synthetic settings to 30 September 2024 [cbot-23-214, sec. 2, n. 11]. EONIA was discontinued on 3 January 2022 [emmi-eonia, web page]; the ECB first published €STR on 2 October 2019 [ecb-estr, web page]. A LIBOR–OIS spread therefore exists only historically.

*One documented replacement.* The Office of Financial Research's Financial Stress Index replaced all seven variables in its funding category from 1 January 2022 (published 27 June 2023): 3-month U.S. LIBOR–OIS by "3-Month Commercial Paper - SOFR OIS", the TED spread by "3-Month Commercial Paper - 3-Month Treasury", the 2-year LIBOR swap spread by the 2-year SOFR swap spread, 3-month Euribor–EONIA OIS by Euribor–€STR OIS, and the yen and euro cross-currency spreads by SOFR, €STR and TONAR versions, with the commercial-paper rate the Federal Reserve Board's 3-month AA financial rate; the index's methodology was unchanged [ofr-fsi-factsheet-2023, pp. 1–2, Tables 1–2]. In the index's daily data, the funding category's contribution was 9.58 on 10 October 2008 and 0.96 on 30 March 2020 (both with the old variables), and −0.16 on 13 October 2026, below its long-run average of zero [ofr-fsi-data, `fsi.csv`].

That is one official monitor's choice, not a survey of what market studies use; the Federal Reserve's May 2026 *Financial Stability Report* judges funding risk from quantities (runnable liabilities, uninsured deposits, liquid assets against short-term debt), not from a spread [fed-fsr-2026-05, sec. 4].

**Practice.** Treasury sells bills "for terms ranging from four weeks to 52 weeks", auctioning 4-, 6-, 8-, 13-, 17- and 26-week bills weekly and 52-week bills every four weeks, plus cash management bills on no regular schedule [treasurydirect-bills, Treasury Bills page] (Gliner gives the history of bill auctions, p. 247). That money-market stress often precedes broader financial stress is a plausible but unsourced generalisation.

- *Commercial-paper maturities.* The BIS primer gives the legal limit as "270 days in the United States and one year in Europe and Japan", but "In practice, the maturity of CP tends to be much shorter. In the US, the average maturity is around 30 days, while in Europe it is approximately four months" [gap-money-cp-cd-primer, sec. "Main funding instruments and their characteristics"]. Gliner's "Most has a maturity of between 90 and 270 days" is therefore not a description of the U.S. market as the BIS reports it; the 270-day figure is the registration exemption, not typical practice.
- *How SOFR is calculated and revised.* SOFR covers the trades in the BGCR (tri-party and GCF) plus cleared bilateral (DVP) repo; to limit the influence of specials, "20 percent of the lowest-rate transaction volume from the DVP segment is removed each day"; the rate is the volume-weighted median, rounded to the basis point, published at about 8:00 a.m. ET with the 1st, 25th, 75th and 99th volume-weighted percentiles and the volume; a revision can follow at about 2:30 p.m. ET the same day, and "Rate revisions will only occur if the change in the rate exceeds one basis point and only on the same day as initial publication" [exp-sofr-data, sections on SOFR, calculation methodology and publication and revisions]. A backtest that uses SOFR as of the morning should record whether the rate was later revised that day.
- *Why overnight RFRs.* Schrimpf and Sushko trace the switch to the collapse of unsecured interbank term lending: few transactions underpinned LIBOR even before 2008, and interbank trading "plummeted" after it. RFR-linked OIS give a risk-free term structure for discounting, but banks lose a benchmark for their marginal term funding cost. Backward-looking (compounded in arrears) term rates are known only at the end of the period and lag rising rates, while forward-looking term rates from futures or OIS are known in advance [rt-schrimpf-sushko-rfr-2019, BIS Quarterly Review, March 2019, sections on the case for reform and on term rates].
- *A liquidity premium inside repo rates.* Copeland and Engbretson measure the premium dealers charge for holding liquid buffers: at 2024's average rates and reserves, "a 100-basis-point increase in the interest rate on reserve balances results in a 0.9 basis point increase in the liquidity risk premium", about 10 percent of the dealer spread, the same effect as a $429 billion fall in reserves, and the relation is nonlinear [latest-copeland-repo-premium-2026, Staff Report 1189, March 2026, abstract].

**Pitfalls.** LIBOR did not end simply because of the manipulation scandals: the cases undermined confidence, but the cessation followed the lack of underlying unsecured transactions and the FCA's 2021 announcements (policy path); U.S. dollar panel settings continued to 30 June 2023. €STR began in 2019 and EONIA ended on 3 January 2022. SONIA was reformed in 2018 to cover more transactions. Gliner describes a pre-2013 LIBOR calculation by the BBA; administration passed to ICE Benchmark Administration later.

**See also.** interest-rate measures; policy path from OIS and futures; securities lending; fixed-income universe.

**Further reading.** On funding liquidity, [exp-brunnermeier-pedersen-liquidity].

#### Evidence 15.7 Runs on prime money market funds in March 2020: institutional investors ran on fundamentals, retail investors followed within-family outflows

**Statement.** At the height of the March 2020 run, cumulative redemptions from U.S.-dollar prime money market funds, onshore and offshore, were 22 percent of the industry's total net assets at end-2019: 32 percent for institutional and 11 percent for retail investors [gap-money-prime-fund-runs, Staff Report 956, December 2020, revised April 2024, sec. 1, PDF p. 3]. Cipriani and La Spada find that "sophisticated institutional investors ran preemptively based on fundamentals" while "unsophisticated retail investors ran based on herd-like informational spillovers", leaving funds of families with large institutional outflows; institutional investors looked up fund portfolios more (website traffic), and both types ran more when switching to a safer fund was cheaper, "suggesting that safe-haven availability exacerbates runs" (abstract, PDF p. 2).

**Conditions.** One episode and market; sophistication is split by share class, and website traffic proxies for information acquisition.

**Practice.** For a run scenario, model institutional and retail outflows separately, and allow outflows to depend on a cheap switch into government funds. The 2023 SEC reforms changed liquidity fees and gates (`gap-money-sec-2023-reform`).

**Pitfalls.** Figures are from the working paper's abstract and introduction.

**See also.** money-market instruments and reference rates; stablecoin runs and flights to safety.

**Further reading.** `gap-money-sec-2023-reform`; `gap-money-nmfp-data`.

## 15.4 Synthetic short positions

#### Result 15.8 Synthetic short through options: long put plus short call is a short forward, not short stock, and does not remove the cost of scarce borrow

**Statement.** Take a European put and a European call on the same underlying, with the same strike $K$, maturity $T$ and contract multiplier, both settled the same way. Long the put and short the call pays at $T$

$$\max(K - S_T, 0) - \max(S_T - K, 0) = K - S_T$$

for every terminal price $S_T$: a short forward position at delivery price $K$. With no arbitrage, a constant continuously compounded rate $r$ and a continuous dividend yield $q$, put–call parity gives the initial value

$$P - C = Ke^{-rT} - S_0e^{-qT} = e^{-rT}(K - F), \qquad F = S_0e^{(r-q)T},$$

so the package costs $e^{-rT}(K - F)$ upfront, positive or negative, and is a zero-cost short forward only when $K = F$.

Adding a *short* zero-coupon bond that repays $K$ at $T$ (borrowing $Ke^{-rT}$ now) turns the package's payoff into $K - S_T - K = -S_T$. For a stock paying no dividends ($q = 0$) that is the terminal value of one share sold short, and the initial values agree: $P - C - Ke^{-rT} = -S_0$. So long put, short call and short bond are equivalent to short stock; equivalently, long put and short call are equivalent to short stock plus a long bond paying $K$, a short sale whose proceeds are invested. With dividends the package less the bond is a terminal claim $-S_T$, worth $-S_0e^{-qT}$ now, while a share sold short is worth $-S_0$ and owes the dividends paid before $T$; the two differ by those dividend liabilities.

Harris's sidebar says that a long put, a short call with the same strike and maturity, "plus a long position in a bond with the same maturity", is economically equivalent to a short position in the underlying [alpha-harris, chap. "Brokers", sidebar "Synthetic Short Positions", p. 157]. As worded, with a long bond paying $K$ the payoff is $2K - S_T$; the bond belongs on the other side, as above. The discrepancy is recorded here and Harris's wording is quoted as printed.

**Notation.** $S_0$, $S_T$: underlying price now and at maturity; $P$, $C$: European put and call premia per unit of underlying (multiply by the contract multiplier for one contract); $F$: forward price; $r$: financing rate; $q$: dividend yield. With discrete dividends of present value $D$, replace $S_0e^{-qT}$ by $S_0 - D$.

**Conditions.** The identity needs matching strike, maturity, underlying and multiplier, the same settlement (both cash or both physical), European exercise, and frictionless borrowing and lending at $r$. Parity example (own computation, Black–Scholes prices with $S_0 = 100$, $K = 95$, $r = 5\%$, $q = 2\%$, $\sigma = 25\%$, $T = 1$): $C = 13.6847$, $P = 6.0317$, $P - C = -7.6531 = 95e^{-0.05} - 100e^{-0.02}$; $F = 103.0455$ and $e^{-rT}(K - F) = -7.6531$. Repricing both options by numerical integration over the risk-neutral density confirms these values, and the payoffs at $S_T = 60, 95, 100, 140$ are $K - S_T$ for the package, $-S_T$ with a short bond and $2K - S_T$ with a long bond (Interpretation).

**Practice.** Against a short sale, Interpretation using the numbers above:

- *package against short sale, no dividends*: take $q = 0$, European options priced at frictionless parity, and borrowing or lending at $r$. Financing the premium $P - C = Ke^{-rT} - S_0$ at $r$, the package ends with $K - S_T - (P - C)e^{rT} = S_0e^{rT} - S_T$. Sell one share short, hold the whole proceeds $S_0$ as cash collateral credited at a constant rebate $r_b$ (the lending fee implicit in $r_b$, no other fee, no extra margin, no recall), and buy back at $T$: $S_0e^{r_bT} - S_T$. The two agree only when $r_b = r$, and the package is ahead by $S_0(e^{rT} - e^{r_bT})$ otherwise.
  With $S_0 = 100$, $K = 95$, $r = 5\%$, $\sigma = 25\%$, $T = 1$: $C = 15.0471$, $P = 5.4138$, $P - C = -9.6332$; at $S_T = 100$ the package ends with 5.1271 and the short sale with 5.1271, 3.0455 or −0.9950 for $r_b = 5\%$, 3% or −1%, gaps of 0, 2.0817 and 6.1221 at every $S_T$ (day-by-day accounts). The synthetic therefore replicates a short sale that earns the full market rebate, which is Harris's point: when options dealers receive the rebate and competition makes them price it in, the synthetic short "in effect" obtains short interest rebate (p. 157);
- *borrow cost*: a market maker who sells the put or buys the call is left long the stock's delta and hedges by selling the stock, short if it holds none (Interpretation), so a scarce borrow can be priced into the options and parity with a frictionless $r$ need not hold. It does not guarantee access to scarce borrow or remove its cost.
  *Evidence*. Ofek, Richardson and Whitelaw test put–call parity on individual U.S. stock options from OptionMetrics end-of-day quotes, on 118 dates about five business days apart from July 1999 to November 2001. They measure the cost of shorting by the *rebate rate spread*, the gap between the rate a short seller earns on sale proceeds and the standard rate [orw-2002-nber, abstract; sec. II.C, pp. 8–9]. Violations are "asymmetric in the direction of short sales constraints" and strongly related to the rebate spread, and they persist after option and equity-lending transaction costs. Violations and rebate spreads also predict future stock returns (abstract). So scarce borrow is priced into options, in that sample and period: the synthetic short does not escape the borrow cost but pays it through the option prices;
- *upfront and dividends*: the premium $P - C$ is paid or received at the start; dividends enter through $F$, whereas a short seller pays them to the lender as they fall due;
- *costs*: option spreads (option spread measurement) and the absence of listed options on many securities limit use; Harris notes that transaction costs "can significantly reduce the economic benefits" and that the strategy works only where options trade (p. 157).

**Pitfalls.** Listed U.S. equity options are typically American and physically settled, a general description unverified against exchange or OCC rules. With American options the payoff identity holds only to maturity: the short call can be assigned early, notably before an ex-dividend date, which delivers a real short stock position that must then be borrowed or covered; the long put may be worth exercising early. Parity for American options is an inequality, not an equality. The package replicates a short forward exactly only for European options, and replacing scarce borrow with it is subject to funding, dividends, exercise and option-liquidity costs, as qualified above. The put–call parity derivation is standard. Hu and Black note that a put and call with the same exercise price and expiration are economically equivalent to selling shares [cfd-hu-black-2007, working-paper version, p. 5, footnote 6].

**See also.** securities lending for short sales; lending fees and rebates; option market-maker spreads; arbitrage; arbitrage risks.

#### Evidence 15.9 Decoupling votes from economic ownership through equity derivatives: empty voting and hidden (morphable) ownership, two cases, and the SEC's 2023 treatment of cash-settled derivatives

**Statement.** Hu and Black define *empty voting* as holding more votes than economic ownership, and *hidden (morphable) ownership* as undisclosed economic ownership "coupled with likely possession of informal voting rights" [cfd-hu-black-2007, abstract and sec. 2.1, pp. 1 and 3]. Derivatives do both.

- *Empty voting.* Holding shares while hedging their return through "(i) a short equity swap position; (ii) a short position in a single stock future; and (iii) a (short call, long put) position" (sec. 2.2.1, p. 3). Perry Corp., holding seven million King Pharmaceuticals shares when Mylan agreed in late 2004 to buy King for stock, bought 9.9 percent of Mylan and fully hedged it, "ending up with 9.9% voting ownership but zero economic ownership". Its overall interest in the merger's completion was negative for Mylan's shareholders (p. 3). The derivatives dealers' own hedges, short sales of Mylan, offset the market impact of the purchases.
- *Hidden (morphable) ownership.* In May 2001 Perry sold 31 million Rubicon shares to two dealers and took an equivalent long equity swap, which it treated as needing no disclosure. In July 2002, just before Rubicon's meeting, it terminated the swaps, reacquired the shares and disclosed 16 percent. New Zealand's courts upheld the non-disclosure; the Court of Appeal found it "almost certain" that the dealers, holding the shares as a hedge, would sell them back (sec. 2.3, p. 6).

*U.S. disclosure since 2024.* The SEC did not adopt its proposed Rule 13d-3(e), which would have deemed holders of non-SBS cash-settled derivatives held to change or influence control to be beneficial owners of the reference class. Instead it discussed when such a holder "may be deemed the beneficial owner" under the existing Rule 13d-3 [fr-2023-22678, 88 FR 76896, 7 November 2023, summary, p. 76898, and sec. II.B.3, "Commission Guidance", pp. 76928–76929]. It amended Item 6 of Schedule 13D to "remove any implication that a person is not required to disclose interests in all derivative securities that use a covered class as a reference security", including a derivative not issued by the issuer, "such as a cash-settled option or SBS" (p. 76899). The amendments took effect on 5 February 2024.

**Conditions.** Hu and Black's cases are illustrations, not a sample: they show the mechanism, not its frequency. Locators refer to the May 2005 ECGI working paper, not the 2007 *Journal of Corporate Finance* article. The SEC text binds Schedule 13D filers (more than 5 percent of a covered class); a holder below the threshold, or one whose derivative confers no beneficial ownership under Rule 13d-3, has no Schedule 13D to amend.

**Practice.** Interpretation: a synthetic long held through a total return swap or a CFD carries no vote. Whether the dealer's hedge shares can later be voted or acquired depends on the dealer's practice, not on the contract. Since 2024 that position must still be disclosed in Item 6 by anyone who files a Schedule 13D. Vote transfer through securities lending is a separate route (street name and beneficial ownership).

**Pitfalls.** Section II.B.3 is used only for its framing.

**See also.** synthetic short through options (route (iii)); street name and beneficial ownership; Schedules 13D and 13G.

**Further reading.** Hu and Black's companion articles (2006a, 2006b).

# 16 Market microstructure and market making

## 16.1 Markets, traders and venues

**Definition 16.1 What markets do: liquidity, price discovery, risk transfer and capital allocation; private and public benefits; why the mechanism matters**

**Statement.** Harris organises the study of trading around five characteristics of market quality: liquidity, transaction costs, informative prices, volatility and trading profits, and the secondary question of how market structure, "trading rules and information systems", affects each [alpha-harris, chap. 1, sec. 1.2, p. 4]. What markets produce falls into private benefits, which accrue to traders when they trade (they trade voluntarily, so they expect to gain), and public benefits, positive externalities that reach people who do not trade, in two classes: those from informative prices and those from liquid markets [alpha-harris, chap. 9, secs. 9.1.1–9.3, pp. 205–206]. These functions can be organised as follows:

- *liquidity*: the buy side buys exchange services, of which liquidity, "the ability to trade when you want to trade", is the most important, and the sell side sells it [alpha-harris, chap. 3, sec. 3.1, p. 32]; patient traders get better prices, and impatient traders "pay for the privilege of trading when they want to trade" (chap. 1, p. 5); the dimensions are in liquidity and market quality;
- *price discovery*: markets "aggregate information from many different sources to produce prices that are closer to fundamental values than anyone could consistently estimate by himself or herself" (chap. 9, sec. 9.4, p. 206); how informed traders do this is informed trading and aggregation;
- *risk transfer*: hedgers trade to exchange risks (chap. 8, sec. 8.1.3, pp. 182–188); liquid hedging and insurance markets let producers specialise, and risk sharing through public corporations lets large projects be financed, which lowers costs of capital and prices even for those who never trade (chap. 9, secs. 9.5.2–9.5.3, pp. 214–215);
- *capital allocation*: informative primary-market prices decide which new projects receive capital, and informative secondary-market prices help investors choose, pay and replace managers of existing capital (chap. 9, secs. 9.4.2–9.4.3, pp. 211–212); when prices do not reflect how well managers perform, good managers may be dismissed and poor ones kept (p. 211).

*The mechanism matters.* Hasbrouck: "The market-clearing price, at least at it arises in usual Walrasian tatonnement, rarely appears in microstructure analyses"; at one instant there are many prices, by direction, urgency, identity and relationship to the counterparty [s6-hasbrouck, chap. 1, sec. 1.1.3, p. 4]. Harris: market structure "determines what people can know and do in a market" and so the power relations among traders [alpha-harris, chap. 5, p. 89].

**Conditions.** The list of functions is a classification, not a theorem; Harris adds that markets work well only when negotiation costs are small relative to the value traded, so economies mix markets with firms, which are small command economies (chap. 9, p. 209). Which benefit dominates is an empirical matter for each market. The welfare ranking of these functions is Harris's opinion, which he labels as such: public policy should first serve utilitarian traders, then the public benefits of liquid, informative markets, and support profit-motivated traders only as far as they serve those (sec. 9.6, pp. 216–217); see regulation as market design.

**Practice.** *Trade-offs.* Objectives conflict: rules that restrict insider trading "usually increase liquidity while making prices less informative" (chap. 9, p. 218); and the transparency that improves price formation is what each trader wants to avoid revealing (Lehalle and Laruelle, quoted in market structures and venues). *Visible activity against economic importance.* Harris's own estimates for the United States (Table 3-4, "Author's estimates", c. 2000) put common stocks at about 20 percent of capital wealth and real estate at about 50 percent; derivatives represent no wealth because they are in zero net supply; "Stocks represent less wealth than the widespread attention given to them by the media would suggest" [alpha-harris, chap. 3, sec. 3.4.1, p. 45]. The figures are dated rough estimates; the point is that trading volume and attention are not measures of a market's economic weight.

**Pitfalls.** Immediacy has a price: the spread compensates traders who absorb inventory, processing and adverse-selection risk; the components and their identification are spread determinants and spread components, and funding risk is an addition to Harris's list. That liquid markets "lower financing costs" is Harris's argument through investors' required returns (p. 215), not a measured effect. Harris's claim that liquidity benefits even non-traders through cheaper goods is an argument about externalities, stated without evidence in these pages.

**See also.** zero-sum trading, skill and luck; trader motives; liquidity and market quality; market efficiency; regulation as market design.

**Further reading.** Harris, chaps. 1 and 9; Hasbrouck, chap. 1.

**Definition 16.2 Trader motives: utilitarian, profit-motivated and futile traders; patience and the liquidity side; instruments for hedging; sources of a trading edge**

**Statement.** Harris classifies traders by why they trade, using stylised traders who each trade for one reason [alpha-harris, chap. 8, p. 177]:

- *utilitarian traders* trade for a benefit other than trading profit: investors and borrowers move money forward or backward through time, asset exchangers trade one asset for another of greater immediate use, hedgers trade risks, gamblers trade for entertainment, and also fledglings (to learn), cross-subsidisers (to pay for broker services through commissions) and tax avoiders (sec. 8.1, p. 178; Table 3-1, p. 33);
- *profit-motivated traders* trade only because they rationally expect to profit: speculators, who are informed traders or parasitic traders (order anticipators and bluffers), and dealers, who sell liquidity (sec. 8.2, pp. 194–196, Table 8-4);
- *futile traders* "believe that they are profit-motivated traders. Although they expect to trade profitably, their expectations are not rational" (p. 177): inefficient traders, of whom the commonest are pseudo-informed traders trading on old news, victimised traders whose agents fail their fiduciary duties, and rogue traders who hide losses from employers or clients (sec. 8.3, pp. 197–198).

A trader with a *long* position owns the instrument and profits when prices rise; one with a *short* position "ha[s] sold something that they do not own" and repurchases, or *covers*, later (chap. 3, p. 32). Equivalent negative exposure through derivatives is an extension beyond Harris's definition.

**Conditions.** Viewing trading as a game, with players submitting orders under venue rules, follows Harris's point that market structure "determines what people can know and do" and so "the power relationships among different types of traders" (chap. 1, p. 7; chap. 5, p. 89). Motives are a modelling device: "In practice, traders often trade for many reasons", and Harris's sidebar *Multiple Identities* says many traders "simultaneously invest, speculate, and gamble" (p. 177).

Motive is therefore inferred from context and exposure, not from the order: Harris notes that "Order type therefore is not always the best indicator of whether a trader is offering liquidity or taking it" [alpha-harris, chap. 19, sec. 19.3.1, p. 400]. *Patience and the liquidity side*: impatient traders search actively and demand liquidity; patient traders search passively and supply it (sec. 19.1.2, p. 396); Hasbrouck adds that with electronic access the role "is a strategic choice that can be quickly reversed" [s6-hasbrouck, chap. 1, sec. 1.2, p. 5].

**Practice.** *Instruments for a problem.* Investors favour high expected return, low risk and low transaction cost; borrowers the cheapest instruments to create, service and repay (chap. 8, p. 180). Hedgers like liquid markets and instruments closely correlated with their risks (sec. 8.1.3.4, p. 188).

Harris's hedging examples give the vehicle trade-offs: forwards need the parties to find and trust each other and agree delivery terms, so they suit traders who deal with each other in the normal course of business and "forward markets generally are not liquid" (pp. 184–185); futures are

standardised forwards guaranteed by a clearinghouse, liquid, and a *linear* hedge that removes upside as well as downside (p. 185), with the delivery location differing from the hedger's (pp. 185–186); options are a *nonlinear* hedge that limits downside for a premium and keeps upside (pp. 186–187); swaps exchange fixed for floating cash flows and leave basis risk, since a bank's deposit rates are not the swap's index rate (pp. 187–188).

Basis risk is developed in futures hedging and basis. *Comparative advantage.* "Successful profit-motivated traders must have some advantage that allows them to trade profitably. Their various advantages usually involve information" (p. 196), and free entry drives the net profits of alternative strategies towards equality after all costs (chap. 1, pp. 7–8). Harris advises speculators to expose themselves "only to the risks they understand best" (p. 186).

Applied to signals, a feature is useful only if it captures information or execution capacity that competitors do not fully exploit; a high in-sample score without an economic reason is usually a selection result, not an edge. The first condition is Harris's advantage requirement; the second concerns selection bias, treated in tests of skill against luck and parameter overfitting, and is untested here. Practitioners list information, models, execution, risk bearing and discipline as edges; only information and the cost of trading are Harris's in these pages.

**Pitfalls.** Arbitrageurs are not simply informed traders who supply or demand liquidity: Harris counts them as informed traders about relative values (Table 8-4) and, as liquidity suppliers, calls them "porters of liquidity" (chap. 19, p. 404); see arbitrage. Treating noise traders as rebalancing, cash needs, index changes, behavioural demand or errors mixes utilitarian traders with futile ones; Harris's point is that utilitarian and futile traders lose on average to profit-motivated traders "because trading is a zero-sum game" (p. 177). Pseudo-informed traders are a kind of inefficient trader, not a separate type. Gamblers are utilitarian in Harris's scheme because they get entertainment, not because of payoff convexity.

**See also.** zero-sum trading; informed trader types; order anticipation; bluffing and manipulation; market intermediaries; what markets do.

**Further reading.** Harris's chapter 8 is the source of the classification.

#### Result 16.3 Zero-sum trading accounting, its two corollaries, and what past profit says about skill

**Statement.** *Accounting.* Harris: trading is "a zero-sum game in an important accounting sense": "the combined gains and losses of buyers and sellers always sum to zero. If a buyer profits from a trade, the seller loses the opportunity to profit by the same amount" [alpha-harris, chap. 8, p. 176]; elsewhere, "when gains and losses are measured relative to the market average" (chap. 1, p. 6). For a trade of $q$ at price $P$ and any common terminal value $V$, the buyer's gain $q(V - P)$ and the seller's $q(P - V)$ sum to zero; with costs $C_i \geq 0$ the sum of net gains is $-\sum_i C_i$.

*Corollary 1*: "Since trading is a zero-sum game, speculators and dealers cannot profit on average if they trade only among themselves. They can profit only if utilitarian traders are willing to trade" (chap. 9, p. 206). *Corollary 2* (the arithmetic of active management, derived here, not from Harris): if active managers' holdings aggregate, value-weighted, to the market portfolio, their aggregate gross active return is zero and their aggregate net active return is $-\sum_i w_i c_i \leq 0$; a single manager beats a passive alternative with drag $c_P$ only if $a_i > c_i - c_P$. *Skill and luck*: write profit as $\Pi_{t+1} = S_{t+1} + L_{t+1}$, skill plus luck, and let $\mathcal{F}_t$ be the evaluator's information. Then

$$E[\Pi_{t+1} \mid \mathcal{F}_t, \Pi_t] = E[\Pi_{t+1} \mid \mathcal{F}_t] \quad \text{if} \quad E[S_{t+1} \mid \mathcal{F}_t, \Pi_t] = E[S_{t+1} \mid \mathcal{F}_t] \text{ and } E[L_{t+1} \mid \mathcal{F}_t, \Pi_t] = E[L_{t+1} \mid \mathcal{F}_t].$$

Both conditions are needed. Unconditional zero-mean luck, $E[L] = 0$, is not enough: persistent luck makes past profit predictive without skill.

**Conditions.** *Accounting.* The identity needs a common valuation of the asset for both sides (a common mark or terminal value); it says nothing about utility, so a hedger who pays to shed risk or an asset exchanger can gain in welfare while losing in money (chap. 8, pp. 176–177). Corollary 1 needs a closed set of traders and costs that are not income of the group (fees paid to members are transfers within it). Corollary 2 needs the actives to hold the market in aggregate, which fails if other holders (for example, non-market holdings by passive owners or issuers) take the other side.

*Skill and luck.* The result assumes that $S_{t+1}$ is independent of past profit given current information and that luck satisfies the second condition above. If skill is *not* in $\mathcal{F}_t$, past profit is informative about it: with persistent skill $S \sim N(\mu, \tau^2)$ and i.i.d. luck of variance $\sigma^2$, the regression slope of $\Pi_{t+1}$ on $\Pi_t$ is $\tau^2/(\tau^2 + \sigma^2)$, which is the shrinkage a performance evaluator applies.

Harris states the evaluator's version: predicting from past performance needs past performance to reflect skill, the skill to keep working in future market conditions, and the manager to keep it [alpha-harris, chap. 22, sec. 22.3, p. 451]; a statistical test works because luck "is random and completely unpredictable" and its net effect on average returns shrinks with time, while skill has a systematic effect (sec. 22.4.1, p. 454).

**Practice.** Simulation: 200,000 simulated managers with $\tau = 0.5$, $\sigma = 1$ give a slope of 0.200 against $\tau^2/(\tau^2 + \sigma^2) = 0.200$; with no skill and i.i.d. luck the slope is −0.006; with no skill and AR(1) luck of persistence 0.5 and unconditional mean zero the slope is 0.501, so past profit predicts luck; with skill in the information set and conditionally mean-zero luck the incremental slope is 0.0005. Random bilateral trades among ten traders marked at a common value sum to zero gross ($-7 \times 10^{-13}$) and to minus total costs net; value-weighted active returns of managers aggregating to the market sum to zero, and do not when they do not.

In practice: test skill claims against a benchmark, over long samples (Harris notes more than 20 years of returns are typically needed for a reliable test, p. 453), and correct for survivorship (sec. 22.6, pp. 470–471).

**Pitfalls.** "Zero-sum" is an accounting statement about money, not a claim that trade creates no value. Harris's two wordings differ: relative to a common terminal value (p. 176) and "relative to the market average" (p. 6); the second makes it a statement about excess returns. A skill claim must explain why the decision was right before the outcome was known, why the edge should persist, and why it is not already competed away, which matches Harris's three conditions and his free-entry argument (chap. 1, pp. 7–8). A proposition that conditions on skill being known cannot be used to say that past profit is useless for an evaluator who does not know it.

**See also.** trader motives; relative performance evaluation; backtest evaluation diagnostics; what markets do.

**Further reading.** Harris, chaps. 8, 9 and 22.

#### Definition 16.4 Market intermediaries: dealers and market makers, brokers and their services, broker-dealers, and the suppliers of liquidity

**Statement.** *Dealers* trade with their clients when the clients want to trade and profit by buying low and selling high; *brokers* arrange trades as agents and profit from commissions; firms that do both are *broker-dealers* or dual traders [alpha-harris, chap. 3, sec. 3.1.2, pp. 33–34]. "The sell side exists only because the buy side will pay for its services" (p. 34). Dealers sell liquidity at their bid and ask; to profit they must buy and sell equal volumes and so must find the prices that balance supply and demand; Harris classes them as uninformed about fundamental values. Market makers provide liquidity on demand in small quantities and trade in and out many times a day; block facilitators provide liquidity to large traders, may take days or weeks to unwind, and generally choose their clients [alpha-harris, chap. 8, sec. 8.2.2, pp. 195–196].

*Why brokers are cheaper than trading directly*: they solve clearing and settlement problems, give access to exchanges and dealers their clients cannot reach, know better who might trade, negotiate better, and represent orders when clients are absent [alpha-harris, chap. 7, sec. 7.1, pp. 140–141]; "The most important, but perhaps least appreciated" of these is clearing and settlement, because order-driven markets arrange trades among strangers (p. 141).

**Conditions.** Concepts timeless; institutional examples as of 2002. *Who supplies liquidity.* Harris's five suppliers, by what they know and the dimension they supply [alpha-harris, chap. 19, secs. 19.3.2–19.3.7, pp. 401–405]: market makers supply immediacy to small traders and avoid informed traders; block dealers supply depth to clients they know to be uninformed; value traders, "the ultimate suppliers of liquidity", supply depth and make markets resilient, and lose only to better news or value traders (p. 403); precommitted traders post aggressive limit orders to lower the

cost of trades they intend anyway and supply immediacy at narrow spreads, which can drive dealers out (p. 404); arbitrageurs are "not suppliers of liquidity but porters of liquidity" between markets (p. 404).

Harris's short list: "Impatient traders take liquidity. Dealers, limit order traders, and some speculators offer liquidity. Brokers and exchanges organize liquidity" (p. 394). Capital standards for dealers only ensure that they *can* supply liquidity; "no amount of capital will make them willing" under heavy adverse selection (p. 402).

**Practice.** *Brokered transactions.* Harris's Table 7-1 distinguishes order-flow, block, new and seasoned offering, and merger-and-acquisition business (p. 140). Order-driven exchanges "are essentially brokerages" (chap. 3, p. 35), and automated order-driven systems are "essentially electronic brokers" (chap. 7, p. 139); regulators classify similar systems sometimes as exchanges and sometimes as brokers (p. 139). Introducing brokers pass orders to clearing members, and sponsored subscribers of electronic systems trade under their broker's real-time credit limits (pp. 142–143). Inventory risk and quote management are dealer inventory risk; agency problems are broker agency conflicts; block roles are block trades; fast intermediaries are high-frequency trading strategies.

**Pitfalls.** Harris's five types of liquidity supplier each occupy a niche (p. 405), so calling all passive flow "market making" hides the source of depth. Confirmation, custody, margin and records extend Harris's post-trade clearing and settlement, and information control belongs to his block-trading material (block trades); in Harris custody is a service of depositories and custodians (chap. 3, sec. 3.2.2.4, p. 38). Dealer compensation from rebates and information is an additional interpretation.

**See also.** trader motives; market structures and venues; liquidity and market quality; the outside spread; inventory-based quotes.

**Further reading.** Harris, chaps. 7, 13 and 19.

**Definition 16.5 Dealers and their quotes: the four problems of dealing, two-sided and one-sided, firm and soft quotes, trading with dealers, and how dealers attract order flow (as of 2002)**

**Statement.** "Dealers are profit-motivated traders who allow other traders to trade when they want to trade. The liquidity service they sell—immediacy—is valuable to impatient traders" [alpha-harris, chap. 13, sec. 13.1, p. 279]. They are passive traders who buy when clients want to sell and sell when they want to buy, and lose when market conditions force them to sell low or buy high, often after trading with informed traders (p. 278). "All dealers face the same problems regardless of what they trade. They must set prices, they must market their services to acquire clients, they must manage their inventories, and they must be careful that they do not trade with better-informed traders" (p. 278). Their quotes [alpha-harris, same chap., sec. 13.2, pp. 280–281]:

- *two-sided* or *one-sided*: most dealers quote both sides but "usually aggressively price only the side on which they would prefer to trade", a dealer who wants to buy quoting a high bid and an uncompetitive ask (p. 280);
- *firm* or *soft*: firm quotes must be honoured "only up to some maximum quantity", the bid and ask sizes; a soft quote "is simply an indication of interest" (pp. 280–281);
- *continuous* or *on request*: dealers in most corporate bond and some foreign exchange markets quote only on request, and such quotes are good for a limited time, shorter in active markets with volatile prices or narrow spreads, while organised quote-driven stock markets such as Nasdaq required continuous firm two-sided quotes (p. 281);
- the *inside spread*, best bid against best offer across dealers, "can be no wider than the narrowest individual dealer spread": five dealers each quoting a 0.50 spread make a 0.10 inside spread in Harris's Table 13-1 (pp. 280–281).

Dealers attract order flow "by quoting aggressive prices and large sizes, providing high-quality service at low prices, advertising, creating marketing relationships, and purchasing it" [alpha-harris, same chap., sec. 13.4, pp. 282–283].

**Conditions.** *Trading with dealers* [alpha-harris, same chap., sec. 13.3, pp. 281–282]. Dealers often trade at better prices than they quote, especially with small, active and apparently uninformed customers, so traders negotiate; institutions dealing directly pay no commission but a net price. Brokers route small orders to dealers they expect to give the best combination of price and service and insist on the best quoted price; "When experienced traders negotiate prices with dealers, they usually ask for a two-sided quote before they say whether they want to buy or sell", which disciplines dealers who would otherwise quote high to known buyers and low to known sellers, most where dealers are few (p. 282).

*Order flow* [alpha-harris, same chap., sec. 13.4, pp. 282–283]: where quotes are public, aggressive prices win orders; where they are not, or price improvement is common, reputation, service and published statistics on execution quality do; relationships with brokers and institutions are cultivated with information, research, routing and accounting systems and entertainment; and some dealers buy order flow, typically market orders only, at payments that then averaged "less than 1 cent per share" in U.S. markets (up to 3 cents in the past), or exchange it reciprocally (p. 283).

**Practice.** *Preferencing.* A broker who sends an order to a specific dealer *preferences* it to that dealer; wholesalers trade mainly with orders introduced by retail brokers, and "By forming stable relationships, brokers and dealers often can lower the total costs of trading" (p. 282); the regulatory problems are in broker agency conflicts and best execution. *Position and spread traders.* Where competition among dealers is intense, dealers must also speculate successfully to stay in business (position traders); spread traders profit only from buying at the bid and selling at the ask (sec. 13.1, p. 279).

*Reading a wide quote.* A dealer who suspects informed flow, carries an unwanted inventory or expects a costly unwind widens or skews the quote, so a wide two-sided quote signals one of these; Harris's tactics for inventory and suspected informed traders are in dealer quote adjustment (Tables 13-2 and 13-3). The source tells the client only to seek another dealer or refuse to trade when the quoted spread is too wide (p. 282).

**Pitfalls.** Dealers are not necessarily compensated directly for relationships or order-flow value; Harris names the spread (buying low, selling high) and, for some dealers, speculation, and treats relationships and purchased order flow as ways of acquiring flow, not as compensation (pp. 279, 282–283). Inferring value from trades, quotes, news and client behaviour goes beyond Harris's fourth problem, avoiding better-informed traders, but matches his account of dealers inferring values from order flow and quote adjustment (secs. 13.6–13.8). Preferencing can route orders toward economic arrangements rather than best all-in execution, the agency conflict of broker agency conflicts; it is a possibility, not a finding. Harris's U.S. figures (payments under a cent, Nasdaq's continuous-quote obligation) are as of 2002; the EU ban on receiving payment for order flow is in EU trading venues.

**See also.** market intermediaries; specialists and designated market makers; dealer inventory risk; dealer quote adjustment; the outside spread; FX market structure and fixed-income market structure (quotes on request).

**Further reading.** Harris, chap. 13.

**Definition 16.6 Specialists and designated market makers: affirmative and negative obligations, privileges, the participation option, tick size and participation, and the regulator's question (as of 2002)**

**Statement.** Exchanges register some dealers as specialists or designated market makers and regulate what they must and may not do: "Affirmative obligations obligate specialists to offer liquidity in various circumstances. Negative obligations prevent them from trading in other circumstances. Specialists accept these obligations because they enjoy the various privileges that come with them" [alpha-harris, chap. 24, sec. 24.2, p. 496]. As of 2002:

- *affirmative*: ensure a reasonable market, quoting meaningful two-sided markets when no one else will, as "traders of last resort", without having to quote block sizes or support falling prices; smooth prices against large reversals to create *price continuity*, judged by quoted spreads and depths and the number and size of reversals (sec. 24.2.1, pp. 496–498);

- *negative*: the public order precedence rule (no trading at a price at which a public trader is willing to trade) and price priority, and the *public liquidity preservation principle*, which discourages specialists from trading with standing limit orders, so that they can only offer liquidity to incoming marketable orders or step ahead with a better price (sec. 24.2.2, pp. 498–499);
- *auction*: conducting the opening single-price auction and orderly markets, without direct compensation (sec. 24.4, p. 501);
- *capital*: exchanges' capital adequacy standards "also ensure that specialists will have adequate capital to provide liquidity if no one else will" (sec. 24.7, p. 511).

The privileges are "access to information about order flows, the right to make decisions after others make their decisions, the ability to create the market quote, the ability to create and exercise certain look-back timing options, and the right to collect brokerage commissions from executing system order flow" (sec. 24.5, p. 501), with dealing profits in thin markets where they are always present (sec. 24.5.9, p. 508).

**Conditions.** *The participation option* (Interpretation: a formalisation of Harris's argument). With $V_{\text{step}}$ the expected value of stepping ahead of or trading with the book, $C_{\text{obligation}}$ the expected loss from required liquidity provision and $K$ compliance and capital costs, an assignment is privately attractive when $V_{\text{step}} > C_{\text{obligation}} + K$. Harris states the qualitative version: the negative obligations "must be costly", and specialists accept them "because the value of their special privileges more than compensates" (p. 499). The public version is his: "Regulators must consider whether the value specialists obtain from their privileges is commensurate with the value of the additional liquidity they provide", which is hard because liquidity services are hard to measure and value (sec. 24.7, p. 510).

*Tick size and participation.* With a one-tick spread, the precedence rule and the preservation principle leave a specialist who wants to buy ahead of the book only the option of buying at the ask from an incoming marketable sell order; a smaller tick relaxes the public order precedence rule "by decreasing the cost of stepping in front of booked orders", since the specialist can then step inside the public quotes, and when U.S. ticks fell to one-sixteenth in 1997 and to a penny in 2000 "specialist participation rates and specialist profitability both increased substantially" (box, p. 500).

**Practice.** *Who pays and who gains.* The beneficiaries are mainly small uninformed market-order traders who get price improvement and momentum traders who get liquidity in volatile markets; the payers are the limit-order traders in front of whose orders specialists step, and the system transfers wealth between them (p. 510). The opening option: Harris's specialist who wants to sell 16 lots at the open chooses the clearing price after seeing the book, selling 15 at 35.2 rather than depress the price (box, p. 509).

The liquidity a specialist supplies in stress helps all markets, while the profits come only from trading at its own exchange. Exchanges with small market shares therefore "cannot expect that their traders will provide much price continuity". With many competing dealers, free riding makes price continuity scarce, which Nasdaq's 20-day rule for dealers who stop making a market only partly addressed (pp. 510–511).

**Pitfalls.** Capital standards do not mean price support: specialists "are not required to support prices when values are falling" (p. 497), so the obligation is a meaningful market, not a price floor. Larger ticks protect queue rents but can widen quoted spreads (tick size), and the value of queue position in large-tick markets is queue-position value. Everything here describes 2002 U.S. exchange specialists (NYSE, AMEX, regional exchanges, CBOE's designated market makers, pp. 495–496).

*One current venue.* NYSE Rule 104, in the rulebook compilation NYSE publishes (file last modified 9 May 2019), requires a DMM to maintain "a continuous two-sided quote with a displayed size of at least one round lot" and DMM units to be at the national best bid or offer at least 15 percent of the trading day in securities with consolidated average daily volume below one million shares and 10 percent above, executions crediting and hidden orders not counting.

For NMS stocks its quotes must be entered within a *Designated Percentage* of the NBBO (8 percent for LULD Tier 1 stocks, 28 for Tier 2 at $1 or more, 30 below $1; 20, 28 and 30 near the open and close) and refreshed if the NBBO moves beyond a *Defined Limit* (9.5, 29.5 and 31.5 percent; 21.5 for Tier 1 near the open and close); DMMs facilitate openings, reopenings, the Midday Auction and the close, and their algorithms see aggregate order information for that purpose [nyse-rules-2019, Rule 104(a)(1)–(3) and (b)].

Supplemental liquidity providers, by contrast, must average at least 10 percent of the day at the NBB or NBO in assigned securities in return for rebates (Rule 107B(a)). The Federal Register shows a 2023 proposal to restructure the DMM programme that NYSE withdrew in July 2024 [fr-nyse-dmm-2023-withdrawn, 88 FR 77625; 89 FR 56456; 84 FR 51205]. *Current text* (NYSE online rulebook, 16 October 2026). The current Rule 104 keeps the 15 and 10 percent inside-quote requirements for securities that are not ETPs and adds at least 25 percent of the trading day for ETPs; the Designated Percentages (8, 28 and 30; 20, 28 and 30 between 9:30 and 9:45 a.m. and from 3:35 p.m. to the close) and Defined Limits (9.5, 29.5 and 31.5; 21.5 for Tier 1 in those periods) are as in the 2019 compilation [nyse-rule104-2026, Rule 104(a)(1) and (a)(1)(A), export pp. 1–2].

Its amendment history runs to 22 November 2022 (SR-NYSE-2022-52), with amendments in January 2020, June 2021 and four in 2022 after those of 2019 (export p. 7). Rule 107B's 10 percent average for SLPs is unchanged, last amended 25 January 2023 (SR-NYSE-2023-05) [nyse-rule107b-2026, Rule 107B(a), export pp. 1 and 9]. The rule texts are rendered pages, not filings.

**See also.** dealers and their quotes; trading sessions and execution systems (specialist roles); order matching and trade pricing (public order precedence); tick size; order anticipation (quote matching); circuit breakers.

**Further reading.** Harris, chap. 24.

### Definition 16.7 Market structures and venues: forums and access, exchanges, OTC and alternative systems, market information systems and transparency, clearing and settlement, and brokered credit

**Statement.** "The trading rules and the trading systems used by a market define its market structure. They determine who can trade; what they can trade; and when, where, and how they can trade. They also determine what information traders can see about orders, quotations, and trades; when they can see it; and who can see it" [alpha-harris, chap. 5, p. 89]. Harris classifies a structure by its trading sessions, its execution system and its market information systems (p. 89; sessions and execution systems are trading sessions and execution systems), with a minimum price increment (p. 90; tick size). Forums are *physically convened* (pits, posts) or *distributed access* (telephone or screen); some jurisdictions require all trading in an instrument at one exchange (p. 90).

*Exchanges* provide forums where members trade; some only host negotiation, others match orders by rules; corporate ownership replaced member ownership at many exchanges around 2000 (chap. 3, sec. 3.2.1, pp. 34–35). In *over-the-counter* markets dealers and brokers arrange trades away from exchanges (p. 35). Harris's 2002 terms for U.S. equities: ECNs are order-driven systems not regulated as exchanges; the *third market* is dealers trading listed stocks away from the exchange; alternative trading systems, ECNs among them, trade in the *fourth market* (pp. 35 and 49). Exchanges are search engines: they "collect information about who wants to trade" and organise it so that traders find the best opportunities [alpha-harris, chap. 19, box, p. 396].

**Notation.** *Pre-trade* (ex ante) and *post-trade* (ex post) transparency: quick reporting of quotes and orders, and of trades; *top of book* against *market by price* [alpha-harris, chap. 5, sec. 5.4.4, p. 101]. Open-book and closed-book markets (sec. 5.4.7, p. 109).

**Conditions.** *Market information systems* (sec. 5.4, pp. 97–109): collection (electronic systems collect data cheaply; floors need reporters; data were over 20 percent of revenues at many exchanges, p. 97), distribution (members often see more and sooner; real-time and delayed services price-discriminate; U.S. securities information processors consolidate quotes and trades, p. 98), order routing, order presentation and order books.

*Transparency*: "Traders are often ambivalent about transparency … those who know the least about market conditions most favor transparency. Those who know the most oppose transparency" (p. 101); Lehalle and Laruelle call the same tension a paradox: more shared information should improve price formation, but "each trader fears 'information leakage' " [alpha-laruelle, Introduction, pp. 19–20]. *Post-trade infrastructure*:

a clearing agent matches buyer and seller records (failures are out-trades or DKs); settlement agents exchange cash and securities, usually net; clearinghouses act "as buyer for every seller and as seller for every buyer", require margins and position limits, and are mutualised among clearing members; depositories and custodians hold assets; systems that automate clearing and settlement provide straight-through processing [alpha-harris, chap. 3, sec. 3.2.2, pp. 35–38].

**Practice.** *Brokered credit.* Harris's sidebar *Multiplying Credit Checks*: with $10^6$ traders and no brokers, each trader must be ready to check every other, 999,999,000,000 potential relationships; with 100 brokers of 10,000 clients each who guarantee their clients' trades, brokers check clients, clients check brokers and brokers check each other: 2,009,900, "500,000 times smaller" [alpha-harris, chap. 7, p. 142]. These are ordered counts, $n(n-1)$ and $2n + B(B-1)$. Counted as unordered pairs they are $n(n-1)/2$ against $n + B(B-1)/2$ (499,999,500,000 against 1,004,950, the same ratio of about 497,500). Either form holds when each client uses exactly one broker that guarantees its trades, clients do not check one another or other brokers, every pair of brokers may trade, and $n$ counts clients only.

*Reported volume* depends on architecture: a 100-share trade between public traders reports 100 shares at an order-matching market, 200 with one dealer in between and 300 with two, and the World Federation of Exchanges distinguishes trading-system-view from regulated-environment-view counts (chap. 3, box, p. 48). *RFQ against CLOB.* In fixed income the dealing desk requests quotes from several market makers and takes the best, while the most liquid products trade on electronic books close to equity ones [alpha-laruelle, chap. 2, sec. 2.6, pp. 190–191]; the protocols are fixed-income market structure. Electronification has not produced one architecture: futures are almost unfragmented, which Lehalle and Laruelle attribute to post-trade complexity, and U.S. options are fragmented while European ones are not, because of fragmented clearing (pp. 189–190).

*Which venue for which order*: small traders like wide exposure, while large traders want exposure only to traders who will commit to trade with them, through crossing networks, negotiated size or brokers (Harris, chap. 26, sec. 26.3.1, pp. 530–531).

**Pitfalls.** Harris's venue names, regulatory classes and settlement cycle (T+3 in 2002, with T+1 planned for 2005, p. 36) are historical; current U.S. and EU venue types are in U.S. equity market structure rules and EU trading venues and transparency. Consolidated, fragmented and segmented market centres are Harris's chap. 26 vocabulary, treated in venue competition and fragmentation; whether dark pools, internalisers and crossing networks weaken public price discovery is in dark pools and crossing networks. Designated market makers follow Harris's chap. 24 [alpha-harris, sec. 24.1, pp. 495–496]: specialist systems impose obligations to supply liquidity, usually one specialist per stock; designated multiple-market-maker systems oblige several, obligations being "best enforced in electronic markets"; third-market dealers' obligations are voluntary; a pure CLOB has none.

**See also.** floor and automated systems; OTC clearing and collateral; market data levels; market intermediaries; trading sessions.

**Further reading.** Harris, chaps. 3, 5, 7 and 24.

### Definition 16.8 Trading sessions and execution systems: continuous and call markets, quote-driven, order-driven, brokered and hybrid markets, auctions and fixings, and specialists

**Statement.** *Sessions.* "In continuous trading, traders can attempt to arrange their trades whenever the market is open. In call markets, all trades take place only when the market is called" [alpha-harris, chap. 5, sec. 5.1, p. 89]. A call may cover all instruments at once or call them in *rotation*, by 2002 rare; many continuous markets open with a call and restart with one after halts; governments sell debt in calls; some markets (Deutsche Börse, Euronext Paris in 2002) trade their least active stocks only in calls (sec. 5.2.2, pp. 90–91). Calls focus attention "at the same time and place"; continuous markets serve impatient traders (sec. 5.2.3, p. 91). Some markets run after-hours sessions (sec. 5.2.4, p. 92). *Execution systems*, "the defining characteristic of a market" (sec. 5.3, p. 92):

- *quote-driven* (dealer) markets: in pure form dealers take part in every trade and choose their clients; interdealer brokers let dealers trade anonymously (sec. 5.3.1, pp. 92–93);
- *order-driven* markets: order precedence rules decide who trades with whom and pricing rules the price; most are auction markets, by single-price auction, continuous two-sided auction or crossing network; auctions are oral (open outcry) or rule-based; traders cannot choose counterparties, so these markets need strong credit arrangements (sec. 5.3.2, pp. 94–95);
- *brokered* markets: brokers search for *concealed* traders, who will not expose orders, and *latent* traders, who do not know they want to trade until shown an opportunity; typical for blocks, real estate and going concerns (sec. 5.3.3, pp. 95–96);
- *hybrid* markets mix these, as the 2002 NYSE (order-driven, with specialists obliged to supply liquidity) and Nasdaq (quote-driven, displaying public limit orders) did (sec. 5.3.4, p. 96).

**Conditions.** *Auctions and fixings.* A single-price double-sided auction ranks bids and offers and usually maximises feasible volume; securities with little natural interest trade mainly in periodic auctions, also called fixings; opening and closing auctions set reference prices used for margins, fund valuations, cash-settled derivatives and some mergers, which invites attempts to "mark the close" [s6-hasbrouck, chap. 2, sec. 2.4, pp. 17–18]. Design details matter: random stopping times discourage last-instant orders (Euronext), deadlines may be extended after large indicated moves, and large or destabilising orders and the cancellation of stabilising orders may face earlier deadlines (p. 18).

Clearing rules and incidence are order matching and trade pricing; primary markets' hold on auctions and its effect on venue shares is venue market share. *Specialists* play three roles, as dealers trading for their own account, brokers representing other brokers' orders, and exchange officials responsible for orderly markets, with the auctioneer's role under the third [alpha-harris, chap. 24, sec. 24.1, p. 496; secs. 24.2–24.4, pp. 496–501].

**Practice.** *The London gold price as a sequential call.* From 12 September 1919 a daily gold price was set by five founding members, chaired by N M Rothschild & Sons, by telephone for the first days and then at a daily meeting [lbma-gold-centenary, web page]; LBMA describes that arrangement, "an exclusive membership of five trading houses agreeing the daily price", as no longer valid, and on 20 March 2015 announced ICE Benchmark Administration as the new independent administrator of the auction (same page).

As of October 2026, IBA runs electronic auctions for gold at 10:30 and 15:00 London time in rounds of fixed duration (for example 30 seconds): it publishes a price, participants enter, change or cancel quantities, and the round ends the auction if the buy–sell imbalance is within a preset threshold; otherwise the price is adjusted and a new round starts; any residual imbalance is shared equally among direct participants [iba-precious-metals-auctions, web page]. The auction produces one price, not continuous immediacy.

*Why calls suit large orders*: a call concentrates opposite interest at one price, and Harris notes that trading halts which reopen with a call switch the pricing rule from discriminatory to uniform, protecting standing limit orders [alpha-harris, chap. 28, pp. 572–573].

**Pitfalls.** Session hours, call rotations and the hybrid examples are as of 2002 (Harris) and 2007 (Hasbrouck). Opening, intraday, closing and volatility calls are a classification; venue rules differ. More transparency improves price discovery only when participants cannot cheaply game the signal (Interpretation); Hasbrouck's deadline rules (p. 18) are the documented countermeasures. How floor and brokered systems weaken equal access is in floor and automated trading systems. The gold price's 1919–2015 procedure is from LBMA's history page; the IBA methodology and participant list change over time.

**See also.** market structures and venues; order matching and trade pricing; floor and automated trading systems; circuit breakers and trading halts; dealer inventory risk.

**Further reading.** Harris, chaps. 5, 6 and 24; Hasbrouck, chap. 2.

**Definition 16.9 Liquidity and market quality: liquidity as bilateral search, immediacy, width, depth and resiliency, displayed and latent liquidity, the four observable variables, and liquidity risk**

**Statement.** Harris: "Liquidity is the ability to trade large size quickly, at low cost, when you want to trade" [alpha-harris, chap. 19, p. 394]. It "is the object of a bilateral search" in which buyers look for sellers and sellers for buyers, actively or passively (sec. 19.1, p. 395). Searching has inputs (time) and outputs (price and size), whose trade-offs define three dimensions (sec. 19.2, p. 398):

- *immediacy*: "how quickly trades of a given size can be arranged at a given cost";
- *width*: "the cost of doing a trade of a given size", the spread for small trades plus commissions, the cost per unit of liquidity, also called breadth;
- *depth*: "the size of a trade that can be arranged at a given cost";

and a fourth, *resiliency*: "how quickly prices revert to former levels after they change in response to large order flow imbalances initiated by uninformed traders" (p. 400). Both width and depth are stated per unit: width "is the cost per unit of liquidity", and depth "is measured in units available at a given price of liquidity" (p. 398). Width and depth are duals: minimising the cost of trading a given size and maximising the size traded at a given cost are "essentially identical" problems, so depth and breadth "summarize essentially the same information" (p. 399).

Formalised: on a size domain $Q$ of positive sizes, let $w(q)$ be the per-unit cost of trading $q$ with a given search effort, measured from the midpoint with commissions included; depth at a per-unit cost $\bar{w}$ is $D(\bar{w}) = \max\{q \in Q : w(q) \leq \bar{w}\}$, and if $w$ is nondecreasing on $Q$ and the maximum is attained, then for $q \in Q$, $q \leq D(\bar{w})$ if and only if $w(q) \leq \bar{w}$. A total cash budget is a different question: with total cost $C(q) = q\,w(q)$ for $w$ in currency per share (or $C(q) = q\,m\,w_{\text{bp}}(q)/10{,}000$ for $w$ in basis points of the midpoint $m$), the size affordable within a budget $K$ is $\max\{q : C(q) \leq K\}$, which is not Harris's depth. Lehalle and Laruelle's four observable liquidity variables are traded volume, the quantity at the best limits (the *book size*), the bid–ask spread and volatility [alpha-laruelle, chap. 2, sec. 2.2, pp. 143–148].

**Notation.** $q$ size in shares (or lots), on a stated domain $Q$: a finite set of lot multiples $\{q_{\min}, 2q_{\min}, \ldots, \bar{q}\}$ or an interval $[q_{\min}, \bar{q}]$ with $0 < q_{\min} \leq \bar{q} < \infty$ (a per-unit cost is undefined at zero size); $w(q)$ per-unit cost of size $q$ in currency per share, *average* unless stated, or $w_{\text{bp}}(q) = 10{,}000\,w(q)/m$ in basis points of the midpoint $m$; $\bar{w}$ a per-unit cost level in the same units; $D(\bar{w})$ depth at that level, in shares; $C(q) = q\,w(q) = q\,m\,w_{\text{bp}}(q)/10{,}000$ total cost in currency; $K$ a cash budget in currency; $B$, $c$ and $L$ in the search rule below: expected improvement from one more search, its direct cost, and the expected loss if the current best match disappears.

**Conditions.** Concepts timeless; Lehalle and Laruelle's data 2008–2016. *Per unit against total.* With the midpoint at 99 and 100 shares offered at 100, buying $q \leq 100$ shares costs $w(q) = 1$ per share, so depth at a per-unit cost of 1 is 100 shares; inverting the total cost $C(q) = q$ at a budget of 1 gives one share. The alternative definition $D(c) = \max\{q : C(q) \leq c\}$ with $C$ an all-in cost and $c$ a budget answers the second question.

*Average or marginal.* With 100 shares more offered at 101, the average per-unit cost of 200 shares is $(100 \times 1 + 100 \times 2)/200 = 1.5$, so depth at an average cost of 1.5 is 200 shares, while at a marginal cost (the cost of the last share) of 1.5 it is 100; say which is meant.

*Monotonicity.* Sweeping a book makes the average per-unit cost nondecreasing in size; a fixed commission per order makes it fall with size ($F/q$ added to the price cost), so $w$ need not be monotone, the set $\{q : w(q) \leq \bar{w}\}$ need not be an interval, and the equivalence can fail (with $F = 50$ on the book above, $w(1) = 51$ while $w(100) = 1.5$).

*Attainment.* On a finite domain the maximum exists whenever some size is feasible. On an interval domain it exists if and only if the feasible set is nonempty, bounded above and contains its supremum $q^\star$, that is, $w(q^\star) \leq \bar{w}$. A compact domain with $w$ lower semicontinuous is sufficient, because the feasible set is then closed and bounded. Lower semicontinuity is not necessary: $w(q) = q$ on $[q_{\min}, 1]$, 2 on $(1, 2)$ and 5 on $[2, 3]$, which jumps up at 2 and is not lower semicontinuous there, attains the maximum 1 at $\bar{w} = 1.5$. It is not enough without boundedness: on $[q_{\min}, \infty)$ a constant $w$ below $\bar{w}$ is continuous and the feasible set has no maximum.

And with $w(q) = q$ for $q < 1$ and $w(q) = 2$ for $q \geq 1$ at $\bar{w} = 1.5$ the feasible sizes are $[q_{\min}, 1)$, the supremum 1 is infeasible, and the equivalence fails at $q = 1$. *Definitions differ.* Hasbrouck's "depth, breadth, and resiliency" uses breadth for a market with many participants, none with market power [s6-hasbrouck, chap. 1, sec. 1.2, pp. 4–5], not Harris's cost per unit; precise definitions "only exist in the contexts of particular models" (p. 4).

*Search.* Continue an active search while the expected benefit of another inquiry exceeds its cost; in a bilateral search "the cost of continuing to search must include the potential loss of your previous best match" [alpha-harris, chap. 19, secs. 19.1.1–19.1.2, pp. 395–396], hence the rule, search again if $B > c + L$. It is a one-step comparison; with recall and independent offers it coincides with the reservation-value rule (for uniform offers with cost 0.02: reservation 0.80, net value 0.8005 against 0.784 and 0.792 for earlier and later stopping), but it is not a general optimal-stopping theorem. Harris's camera example: one more search has expected gain below 2.50 (a gain of at most 10 with probability below 0.25) and costs more than 2.50 at 30 an hour for five minutes, so the buyer stops (box, p. 395).

**Practice.** *Search strategies*: active searchers buy speed with exposure, passive searchers wait and risk not trading, and large traders search actively only among traders who post firm quotes or become passive and undisplayed (Harris, pp. 396–397); the "selective search" (trusted counterparties first, then wider) is how brokers use what they know about who wants to trade (pp. 397–398).

*Displayed and undisclosed liquidity.* Harris's Table 19-1, after a classification by George Sofianos, lists displayed quotes and orders, liquidity not displayed at a venue, displayed indications, liquidity not displayed at broker-dealers or at the buy side, and latent liquidity, each with its search strategy ("take it", "offer to trade", "ask", "use Liquidnet") (p. 397). *Book size* averages the quantities at the best bid and ask, summing several limits for small-tick instruments; it is neither an upper nor a lower bound on traded volume, because books replenish (a best bid of 20 throughout and 100 trades of 19 in five minutes trade 1,900) and a large order sweeps several limits [alpha-laruelle, chap. 2, sec. 2.2, pp. 143–144].

*What drives liquidity for an order*: why the trader trades (informed traders find less), what is traded (interest), how well values are known, when (closed markets, suspected news, holidays), what others are doing, displayed imbalances, and who is trading (Harris, sec. 19.2, p. 399; sec. 19.1.2, p. 398). *Resiliency* comes from value traders when they are "well capitalized, well informed, and willing to trade" (sec. 19.3.4, p. 403).

*Time of day*: volume, volatility, spread and book size share an intraday shape; high volume, volatility and spread after the open, a calm middle of the day, rising volume into the close with the spread at its minimum and the book larger (Royal Dutch Shell, 2011–2016, Lehalle and Laruelle, Figure 2.12, pp. 145–148). *Flow reactivity*, Lehalle and Laruelle's "correlation between the number of trades and high frequency volatility", measures whether a venue keeps trading when volatility rises (sec. 2.3.3, pp. 156–158).

**Pitfalls.** Measures see one face each: "liquidity cannot be reduced to past traded volumes" (Lehalle and Laruelle, p. 143), and the spread misses depth. Book size is not a lower bound on executable size: hidden, reserve and replenishing orders can make it exceed the display, and cancellations before an order arrives can make it smaller (order exposure decisions). Liquidity risk, uncertainty about the cost of turning a desired position into a trade, and price continuity, the absence of jumps and reversals for ordinary order flow, are working definitions, not Harris's. Forcing continuity against informed flow only transfers losses to liquidity suppliers, an interpretation consistent with adverse selection (adverse-selection component) and with Harris's analysis of halts (circuit breakers).

Market quality is often listed as liquidity, tightness, depth, resiliency, transparency, volatility control and informational efficiency, which adds transparency and volatility control to Harris's five characteristics (what markets do); that these criteria conflict is Harris's point (chap. 9, p. 218).

**See also.** spread determinants (spread and book size when the tick binds; the bounce statistic); spread–volatility regression; price impact; intraday deseasonalisation; market intermediaries; spread measures.

**Further reading.** Harris, chap. 19; Hasbrouck, chap. 1; Lehalle and Laruelle, secs. 2.2–2.3.

## 16.2 Orders, limit order books and queues

**Mechanism 16.10 Order lifecycle and messages: order states, insert, modify, cancel and execute, private and public channels, and the instructions that change priority**

**Statement.** Lehalle and Laruelle's account starts from a book that is "most often empty" at the start of the day; orders are stored while they do not match a resting order, and two orders match when their prices are compatible, each match producing one public trade message and two private messages, one to the owner of each order [alpha-laruelle, chap. 3, sec. 3.3.1, p. 217]. Members usually reach the matching engine through two channels: a private one, on which they send their orders and receive the messages that concern them, and a public one, on which everyone connected sees the transactions and the state of the book; the private channel is generally the faster (p. 218).

At any time a trader can insert, modify or cancel an order and attach tags to it, such as a time in force (immediate or cancel, good till day, good till time) or a displayed quantity, which creates an iceberg order (p. 218). An order then leaves the book by execution against an incoming order that crosses it, by cancellation, or by expiry; an iceberg's hidden part usually has lower time priority than displayed orders at the same price submitted later, and an immediate-or-cancel order that matches nothing leaves no visible trace [catalogue-0528, chap. 3, sec. 3.2.2, p. 55].

The states an order passes through are those a broker or venue reports for it. Harris calls an order *pending* after submission and before the broker accepts it (for example while an account or a borrow is confirmed) and *working* once accepted and until filled or cancelled [alpha-harris, chap. 4, sec. 4.2, p. 71].

In the FIX 4.4 message standard, OrdStatus (tag 39) "identifies current status of order" with the values new, partially filled, filled, done for day, cancelled, pending cancel, stopped, rejected, suspended, pending new, calculated, expired, accepted for bidding and pending replace; there is no value for *replaced*. ExecType (tag 150) "describes the specific ExecutionRpt … while OrdStatus (39) will always identify the current order status": a confirmed replacement is reported with ExecType 5 (Replace) and an OrdStatus such as new or partially filled, and a fill with ExecType F (Trade) [fix44-orchestra, OrdStatus and ExecType code sets, FIX.4.4_EP311].

**Notation.** *Insert* adds displayed or hidden quantity at a price; *modify* changes price, quantity or instructions; *cancel* removes resting quantity; *execute* is a match against an incoming marketable order. A replacement (cancel-replace) is reported by its execution type, not as an order state. *Working* (resting) and *part-filled* orders have quantity still governed by their instructions; filled, cancelled, expired and rejected orders are terminal.

**Conditions.** FIX 4.4 (2003) code sets; Harris's broker practice as of 2002; CME pages of 2026; MIAX PEARL as filed in 2021; Nasdaq's rules as of 16 October 2026. *Sessions.* Whether a book starts empty depends on the venue and on the orders' durations. A day order is valid for the trading day on which it is submitted and expires at the close if unfilled, and most brokers assume a day order when no instruction is given; a good-till-cancel order stays valid until cancelled (some brokers cancel such orders after a month or two), and good-until orders until a stated date [alpha-harris, chap. 4, sec. 4.9, pp. 82–84, Table 4-3].

FIX 4.4's TimeInForce treats an absent value as "Day (or session)" and also defines good till cancel, good till date, at the opening, immediate or cancel, fill or kill, good till crossing and at the close [fix44-orchestra, TimeInForce code set]. Orders carried across sessions therefore rest in the book when it reopens, which a replay must load.

On CME Globex, day orders "automatically expire at the end of the day and do not carry over to the next trade date", and orders are day orders unless otherwise specified; good-till-cancel orders remain until executed, cancelled or the instrument expires, and good-till-date orders until the date, cancellation or expiry [cme-wiki-2026, page "Order Qualifiers", version 3, 11 March 2026]; under FIFO a GTC order "will retain its timestamp over subsequent trading sessions from initial entry" unless modified in a way that loses priority, and GTC orders "do not maintain TOP over a weekend" (page "CME Globex Matching Algorithm Steps", version 5).

Interpretation: a replay of a CME product must therefore start each session with the carried GTC and GTD orders at their original time priority, and without carried TOP status after a weekend. How those orders appear in CME's market-data recovery snapshots, and other venues' carry-over rules, are unverified.

Priority after a modification is a venue rule, not a general one. On CME Globex under FIFO matching an order loses its priority and is requeued if its quantity is increased, its price changed or its account changed [ex-cme-matching, page "Supported Matching Algorithms", updated 11 February 2025]; a reduction in quantity is not on that list. Under pro-rata matching, which Bouchaud et al. describe as common in short-rate futures, time does not break ties at a price, and each resting order receives a share of an incoming order in proportion to its size [catalogue-0528, chap. 3, sec. 3.2.1, p. 54].

A rule that applies to one venue, order type or matching algorithm must be stated with that venue and its rule date. Lehalle and Laruelle note in passing that "inserting a new order will have a better rank than a modified order" on reaching a matching engine [alpha-laruelle, chap. 1, sec. 1.2.1, p. 70], without naming a venue.

**Practice.** Limit and stop instructions let a trader who is absent take part: many traders use limit orders to tell their brokers the conditions under which they would trade, and brokers then watch the market on their behalf [alpha-harris, chap. 4, sec. 4.4.5, pp. 77–78]. A stop order becomes an ordinary order once triggered and "remains valid even if price crosses back over the stop price", so it can fill well beyond its stop price in a fast market (Harris's cotton example: a stop at 70 cents sold at 66.90; sec. 4.5, p. 78). For a replay, an execution record therefore needs the acknowledgement, every state change and the channel timestamps of each order, not only its fills; the public feed alone cannot show which resting order was the replayer's own.

**Pitfalls.** "Replaced" is an execution type, not an order state in FIX 4.4. Resting orders can carry across sessions according to their time-in-force. Priority loss is not limited to worsening the price or increasing size. On CME's FIFO algorithm any price change loses priority, an improvement included. Their five "order states" (pending, accepted, working, part-filled, terminal) are a simplification of the reported states above.

*Minimum quantity, cancel on disconnect and modification at named venues.* On CME Globex a minimum quantity is an optional attribute of an order qualifier: at least that quantity must fill immediately on entry or the whole order is cancelled, the entire quantity is exposed to the market, and any remainder rests under the order's time in force; a fill-or-kill order is a fill-and-kill order whose minimum equals its size, and both are accepted only in the Open market state [cme-wiki-2026, page "Order Qualifiers", version 3, 11 March 2026].

Cancel on disconnect cancels all resting futures and options orders, except good-till-cancel and good-till-date orders, for the session and trader ID pairs of an iLink session that disconnects involuntarily, detected by an unanswered test request within the heartbeat interval; it is on by default and mandatory except on iLink CGW sessions, it may fail to cancel during no-cancel states or emergencies, and an order already filled at the host returns "Too Late to Cancel" (page "Cancel on Disconnect", version 7, 3 September 2026).

Under CME's FIFO algorithm a modification loses timestamp priority on an increase of working quantity, a change of price or a change of account. A good-till-cancel order keeps its timestamp across sessions unless so modified, and a display-quantity order goes to the back of the queue each time its display is refreshed (page "CME Globex Matching Algorithm Steps", version 5, 2 June 2026), consistent with the page cited above.

On MIAX PEARL Equities, Rule 2616(a)(5) as described in a July 2021 filing keeps an order's timestamp only for a size decrease, a change to the Max Floor of a reserve order or certain sell/short-sale marking changes. Any other modification, including a size increase or a price change, loses time priority, and the filing proposed that marking changes during a Short Sale Period also lose it [fr-2021-16789, 86 FR 43279–43281]; whether that proposal was approved is unverified.

On Nasdaq (Equity 4, current text as of October 2026), Minimum Quantity is an order attribute under which an order "will not execute unless a specified minimum quantity of shares can be obtained". One of two modes is chosen at entry: satisfied by several resting orders in aggregate, or only by orders each at least the minimum, in which case execution stops at the first order that is too small and the remainder is cancelled. An order that would lock or cross only smaller orders is repriced one increment away.

Once posted, the order is not displayed and executes only against incoming orders of at least the minimum. It must be at least one round lot, and after a partial fill the minimum falls to the shares remaining. With a Display attribute it is accepted as IOC, with Routing it is rejected, and it is excluded from the opening, halt and closing crosses [nasdaq-equity4-2026, Rule 4703(e), export pp. 19–20].

A modification keeps priority only when it re-marks a sell order as long, short or short exempt (through OUCH or FLITE only) or is a partial cancellation reducing size; "all other modifications of orders will result in the replacement of the original order with a new order with a new time stamp" (Rule 4756(a)(3), export p. 29), and the book ranks by price, then displayed before non-displayed orders, each in time priority (Rule 4757(a)(1), export p. 30). So Nasdaq, CME and MIAX PEARL agree that a size reduction keeps priority and a price change loses it, and a CME minimum quantity applies to the immediate fill on entry while Nasdaq's persists on the resting order.

**See also.** limit order book; order instructions; matching and trade pricing; CME Globex trading conventions (matching algorithms); order exposure decisions (hidden and iceberg orders); operational controls (unknown order state); market data levels (Level III order messages).

**Further reading.** On replaying order books: Huang, Lehalle and Rosenbaum; Noble, Rosenbaum and Souilmi.

### Definition 16.11 Order instructions: what an order specifies, the standard order types, validity, quantity, display and handling instructions, and how they interact

**Statement.** Orders are trade instructions: they always specify the instrument, quantity and side, and may add conditions, most often limits on price, and also how long the order is valid, when it can be executed, whether partial fills are acceptable, where to present it, how to search for the other side and even with whom to trade [alpha-harris, chap. 4, sec. 4.1, pp. 68–69]. Traders who cannot watch the market must write contingencies into their orders, and lose when old orders trade before they can be cancelled (p. 69). Harris's standard instructions are:

- *market*: trade at the best price currently available; small market orders pay the spread, may receive price improvement from a dealer, large ones have market impact, and all bear execution-price uncertainty from quote changes between submission and execution (sec. 4.3, pp. 71–73);
- *limit*, including the *marketable* limit order: trade only at the limit price or better (sec. 4.4, pp. 73–78; limit order book);
- *stop* and *stop limit*: a stop price activates the order when the market reaches or passes it, after which a stop order is an ordinary market order and a stop limit order an ordinary limit order; "a stop instruction provides for the activation of an order", whereas a limit price sets the terms of trade (sec. 4.5.1, p. 79); stop orders demand liquidity when it is least available and add momentum (sec. 4.5.2, p. 79);
- *market-if-touched*: a market order activated when price falls to the touch price for a buy (rises, for a sell), contrarian, and very rare (p. 80, Table 4-2);
- *tick-sensitive*: a buy downtick order can fill only on a downtick or zero downtick, a sell uptick order only on an uptick or zero uptick; they cannot have market impact, act as limit orders with a limit that follows the last different price, and are most attractive when the tick is large (sec. 4.7, p. 81);
- *market-not-held*: the broker may use discretion over timing and price and is not accountable for failing to trade, used mostly for large orders (sec. 4.8, p. 82);
- *validity*: day (the most common, and the default at most brokers), good-till-cancel, good-until a date (good this week, good this month), immediate-or-cancel, good-after, market-on-open and market-on-close (sec. 4.9, pp. 82–84, Table 4-3);
- *quantity*: all-or-none, which must fill at once, and minimum-or-none (minimum acceptable quantity), each of whose fills must exceed a stated size (sec. 4.10, pp. 83–84);
- *other*: spread orders, with a limit on the price difference stated as a premium to the buy or sell side; display instructions that cap the displayed size (undisclosed, hidden, reserve or iceberg orders); substitution orders that let the broker choose among securities; and special settlement instructions (sec. 4.11, pp. 84–86).

**Notation.** FIX 4.4 carries these in separate fields: OrdType (40) for market, limit, stop, stop limit, market-if-touched, pegged and others; TimeInForce (59) for day, good till cancel, at the opening, immediate or cancel, fill or kill, good till crossing, good till date and at the close; and ExecInst (18) for handling instructions such as not held, all or none, do not increase or reduce, and the pegs (market, primary, mid-price, last and others) [fix44-orchestra, OrdType, TimeInForce and ExecInst code sets, FIX.4.4_EP311]. A label such as "hidden pegged IOC" is shorthand for values in several fields.

**Conditions.** Harris's instructions and settlement as of 2002; FIX 4.4 (2003). *Interactions.* An order type and its instructions combine, and the combinations are not all equivalent across sources. Harris lists immediate-or-cancel, good-on-sight and fill-or-kill together as orders valid only on submission, "in some markets" called by the other names, and in his Euronext example a fill-or-kill limit order for 60,000 shares, sent against a 10,000-share display, fills in full from the hidden size behind it, while he says that any unfilled remainder of that order would have been cancelled, which is immediate-or-cancel behaviour in FIX's terms (pp. 83–85); Bouchaud et al. and FIX distinguish them: an immediate-or-cancel order executes what it can and cancels the rest, while fill-or-kill (all-or-nothing) executes in full or not at all [catalogue-0528, chap. 3, sec. 3.2.2, p. 55].

A marketable limit order without such an instruction leaves its unfilled remainder resting at its limit, and "traders who do not want their unfilled orders to stand in the book must attach a fill-or-kill or an immediate-or-cancel instruction" [alpha-harris, chap. 6, sec. 6.4, p. 126]. Orders with size restrictions usually rank behind unrestricted orders at the same price (p. 117). A stop limit order activated by a gap through both prices does not trade, and a stop market order that does trade may fill far beyond its stop (pp. 78–80; order lifecycle). Market-on-open and market-on-close orders participate in the opening or closing auction where there is one (p. 83; matching and trade pricing).

**Practice.** The order type is the trader's choice of how to trade off price, certainty and exposure; which orders a venue accepts, and how each behaves at that venue, are its rules, to be read from its rulebook with a date. Interpretation: in a replay, each order's type, time in force and quantity and display instructions are needed to know whether its remainder rests, where it ranks and whether it could have traded.

**Pitfalls.** IOC and FOK are not one category: Harris uses the names interchangeably and FIX does not, so the source of a definition must be named. Pegged orders are not in Harris's 2002 list; they are FIX 4.4 ExecInst values, and pegging references are venue rules. Harris's special-settlement example assumes U.S. regular-way settlement of T+3 (p. 85), as of 2002; it is not current. The rule on tick-sensitive orders is a description of an order instruction, not a market rule; short-sale tick rules are a different matter, not covered here.

**See also.** order lifecycle; limit order book; limit orders as options; matching and trade pricing; order exposure decisions (display and hidden orders); CME Globex trading conventions.

### Definition 16.12 Limit order book: best quotes, limit-price placement, visibility, the three event types and how prices change, and quote-driven requests for quotes

**Statement.** *Order type and aggressiveness are different things.* Harris's market order is "an instruction to trade at the best price currently available" [alpha-harris, chap. 4, sec. 4.3, p. 71]; a limit order trades only at its limit price or better, and a *marketable* limit order is one that can execute on arrival because its price reaches the opposite best quote, "like market orders, except that they limit the price concessions" (sec. 4.4.1, p. 73). Bouchaud et al. use a convention for event categories instead: "orders that match upon arrival are called market orders" and those that do not are limit orders, so in their counts a marketable limit order is a market order [catalogue-0528, chap. 3, sec. 3.1.3, p. 45].

A crossing limit order can do both: the part that matches executes at once, and "any portion of $x$ that does not match instead becomes active at the price $p_x$" (sec. 3.1.2, p. 45), unless an instruction such as immediate or cancel removes the remainder (p. 55). In this entry "market order" follows Bouchaud et al.'s convention only where stated. A *limit order book* is the set of all limit orders for an asset on one platform at a given time, its best buy price is the bid $b(t)$, its best sell price the ask $a(t) > b(t)$, with mid-price $m(t) = \frac{1}{2}[a(t) + b(t)]$ and spread $s(t) = a(t) - b(t)$, and the book is a set of queues, one per price level [catalogue-0528, chap. 3, secs. 3.1.3–3.1.4, pp. 45–47].

In that convention, limit orders supply liquidity and market orders demand it: patient traders meet impatient ones (p. 43). Harris classifies a limit order by its price relative to the market [alpha-harris, chap. 4, sec. 4.4.1, Table 4-1, pp. 73–74]: *marketable* (a buy at or above the best offer), *making a new market* (between the best bid and offer), *at the market* (at the best bid or offer), and *behind* or, further away, *far from* the market; aggressiveness falls in that order. Open-book markets display their books to all traders, closed-book markets do not, and some let traders hide their orders [alpha-harris, chap. 5, p. 109].

**Conditions.** Book dynamics are driven by three kinds of event: insertions add liquidity, while cancellations and transactions (by market or marketable limit orders) remove it, and trades can occur only at the best limits [alpha-laruelle, chap. 3, sec. 3.3.1, p. 219]. The bid rises when a buy limit order arrives inside the spread and falls when sells exhaust the volume at the bid; a sell market order larger than the bid queue walks down the book and trades at several prices [catalogue-0528, chap. 3, sec. 3.1.8, p. 51; Table 3.1, p. 53].

Request-for-quote systems differ: a trader asks several market makers, each free to answer, and modern systems aggregate the answers into a book-like view; Lehalle and Laruelle note fixed-income markets' move towards all-to-all RFQs that look like central limit order books [alpha-laruelle, chap. 3, p. 216]. Interpretation: an aggregated RFQ screen shows conditional, request-specific prices, not firm resting orders, so its depth is not comparable with a book's.

**Practice.** Bouchaud et al. measure prices relative to the same-side best quote, $d = b(t) - p_x$ for a buy and $p_x - a(t)$ for a sell, to aggregate order flow across traders (sec. 3.1.6, p. 50). The spread and mid-price are measured as in spread measures, which this entry does not repeat.

**Pitfalls.** An order that matches on arrival is not necessarily a market order: Bouchaud et al. use "market order" as an event convention, which must be distinguished from the submitted order type. Orders far from the market do not have a low immediacy cost: they give a better price if they fill but rarely fill. Top-of-book and depth-book visibility are data products, defined in market data levels. A displayed book is not all liquidity: in Bouchaud et al.'s 2015 NASDAQ sample, 3 to 5 percent of market orders matched a hidden order inside the spread (Table 4.1, p. 60).

**See also.** order lifecycle; limit orders as trading options; queue-reactive intensities; spread measures; tick size; floor and automated trading systems; fixed-income market structure (RFQ protocols).

### Mechanism 16.13 Matching and trade pricing: order precedence, the matching procedure, pro-rata allocation and its remainders, uniform, discriminatory and derivative pricing, and call-auction uncrossing

**Statement.** Rule-based order-matching systems arrange trades from submitted orders in the same sequence, in calls and in continuous trading: they rank orders by their *order precedence rules*, match the highest-ranked buy with the highest-ranked sell, trade if the buyer will pay at least what the seller asks, carry the remainder of the larger order to the next match, and price each trade by their *trade pricing rules*; in a call the system makes one attempt after the call, in continuous trading one whenever an order arrives [alpha-harris, chap. 6, secs. 6.2–6.2.2, pp. 116–119].

*Precedence* is hierarchical: price priority first everywhere, market orders ranking highest; then at least one secondary rule: time precedence (floor time, giving only the first order at a price precedence, or strict time), display precedence (displayed before undisclosed orders at a price, the parts of a partly displayed order treated separately), size precedence (large first in some markets, small first in others) or pro-rata allocation among orders at parity; orders with size restrictions usually rank lower (sec. 6.2.1, p. 117). Some equity exchanges also apply *public order precedence*, forbidding members to trade ahead of a public order at the same price (sec. 6.1.1.3, p. 115).

*Pricing*: single-price auctions use the *uniform* rule, one clearing price for all trades; continuous two-sided auctions use the *discriminatory* rule, each trade at the standing order's limit price; crossing networks use the *derivative* rule, a price taken from another market (sec. 6.2.3, p. 120; secs. 6.3–6.5, pp. 120–136).

**Notation.** For a candidate price $p$, supply $S(p) = \sum_i q_i^s 1\{s_i \le p\}$, demand $D(p) = \sum_j q_j^b 1\{b_j \ge p\}$ (market orders counted at every price), matched volume $\min\{S(p), D(p)\}$ and excess demand $D(p) - S(p)$; the single-price auction clears where the volume is greatest (Harris, sec. 6.3.2, pp. 121–122). Bouchaud et al. write the auction price as $\arg\max_p Q(p)$ [catalogue-0528, chap. 3, sec. 3.2.3, eq. (3.8), p. 56]. The *surplus* at an auction price is the quantity left unmatched there [alpha-laruelle, chap. 2, sec. 2.1.1, p. 122].

**Conditions.** Mechanisms timeless; venue settings (random ends, public order precedence) as of 2002–2012; the batch-auction design as published in 2015. *Call auctions in European markets.* Lehalle and Laruelle describe four successive rules (pp. 121–122): maximum executable volume; then minimum surplus; then market pressure (all surplus on the buy side: the highest candidate price; all on the sell side: the lowest); then the price nearest a reference price, generally the last traded price; if orders do not cross, no price is formed.

Some venues end the call at a random time after its theoretical end; their Table 2.2 gives maxima of 0 seconds for the LSE and Euronext Paris, 30 for Deutsche Börse, 60 for Borsa Italiana and 300 for U.S. markets, with variation by day and stock (pp. 122–123); those are as of their study, not current rules.

*Surplus and price priority.* The total trader surplus of a trade is the buyer's valuation minus the seller's whatever the price; the price only divides it. If traders bid and offer their valuations, matching by price priority puts every buyer who values the asset above the clearing price with every seller who values it below, so no other arrangement of the same orders yields more total surplus (sec. 6.3.3, pp. 122–125).

*Incidence.* For a given set of orders, large impatient traders prefer discriminatory pricing, which fills the first parts of their orders at better prices, and standing limit-order traders prefer uniform pricing; but traders bid more aggressively under uniform pricing, so the comparison overstates the effect, and continuous markets cannot enforce uniform pricing because large traders split their orders (sec. 6.4.2, pp. 129–130). Continuous trading lets traders trade when they want to, a benefit the surplus does not measure (p. 131).

**Practice.** On Harris's nine-order example (Table 6-1): demand and supply give matched volumes 1, 7, 6 and 4 at 19.8, 20.0, 20.1 and 20.2, so the call clears 7 at 20.0 by the first rule alone; total surplus is 1.6 in the call against 1.0 for the same orders traded continuously (11 units), Harris's Tables 6-7 and 6-9; Bif's market buy averages 20.15 under discriminatory pricing (p. 128).

*Pro-rata allocation* fills orders at parity in proportion to size: 18 against resting 10 and 20 gives 6 and 12 (p. 117), and POSIT's 3,000 to sell against 5,000 and 10,000 to buy gives 1,000 and 2,000 (p. 134). The executable quantity is first capped by the eligible resting liquidity; what an incoming order cannot fill is its own residual, which rests or cancels by its instructions, not a rounding remainder (10 against three resting orders of 3 fills 3, 3 and 3 and leaves 1 of the incoming order unexecuted).

When proportional shares of the executable quantity are not whole lots, a rounding remainder must be allocated by a further rule: 10 against three resting orders of 4 gives $\frac{10}{3}$ each, floors of 3 and one lot to allocate, and 7 against 1, 1 and 100 floors the small orders to nothing; which rule applies (time, size, rounding, minimum allocation) is venue-specific.

*CME Globex* publishes its allocation steps [cme-wiki-2026, page "CME Globex Matching Algorithm Steps", version 5, 2 June 2026]. Each product is assigned an algorithm built from steps (TOP, LMM, Split, FIFO, Pro Rata, Leveling): for example A is TOP, Pro Rata, FIFO; C is Pro Rata, FIFO; F is FIFO; K is TOP, LMM, Split, FIFO, Pro Rata, optional Leveling, FIFO. The Pro Rata step allocates $\lfloor$(displayed working quantity/total working lots) $\times$ match quantity$\rfloor$, but only to orders whose allocation would reach the product's Pro Rata Minimum; "fractional lots … are rounded down", so Pro Rata "is never the last step" and is followed by FIFO, or by Leveling and FIFO.

In CME's example a 50-lot sell against 5, 9, 57, 4, 28 and 300 lots with a minimum of 2 allocates 0, 0, 7, 0, 3 and 37, leaving 3 lots; Leveling, in algorithm K only, gives at most one lot to each order that took part in Pro Rata and received nothing, ranked by working quantity and then time (orders of 9, 5 and 4 lots), and the final FIFO step takes what is left. Split divides a match between FIFO and Pro Rata with FIFO rounded up (49 lots at 40/60 give 20 and 29). TOP gives one order that improved the market (with displayed quantity at least TOP Min) all fills up to TOP Max; LMM entitlements, rounded down to no less than one lot, exist only for options, not futures.

When an aggressor's quantity is at least the displayed quantity at a price, FIFO applies instead of the product's algorithm, including hidden quantity. *Display (iceberg) orders*: under FIFO a refreshed slice goes to the end of the queue at its price; under Pro Rata only displayed quantity counts; only the initial display can be TOP (same page). ICE and Eurex rules may differ.

*Matching curves.* In Lehalle and Laruelle's study of auction feeds for FTSE 100, FTSE MIB, SBF 120, DAX and OMXS 30 stocks from 1 January to 30 March 2011, the indicative volume relative to the final volume rises late and convexly in opening calls and early and concavely in closing calls; in the German closing auction, whose book is not shown, the indicative volume exceeds the final one in about a quarter of cases, which they read as orders placed to test the book and then cancelled (pp. 123–127). Indicative prices and volumes are therefore signals, not commitments; CME publishes an indicative opening price in its pre-open (CME Globex).

*The open-outcry rule*. In an oral auction traders "must publicly express all bids and offers so that all traders can act on them" and must announce acceptances publicly, which makes participation fair, informs traders of market conditions and protects customers from brokers arranging private trades for friends [alpha-harris, chap. 6, sec. 6.1, p. 113]. Price priority is self-enforcing; time precedence is not, and "is meaningful only when the minimum price increment is not trivially small" (pp. 113–114). Harris's order precedence and trade pricing rules (p. 112), with access rules added, are why the same latent supply and demand can produce different prices under different rules.

**Pitfalls.** Frequent batch auctions come from Budish, Cramton and Shim's proposal. Within a batch, time priority is removed, not merely reduced: in the published design, rationing at the clearing price gives priority to orders outstanding for more whole batch intervals, while "if two orders were submitted in the same batch interval they have the same priority irrespective of the precise time they were submitted within that batch interval", with ties broken at random (pro rata) [wide-budish-cramton-shim, sec. VII.A, p. 1596]; the September 2013 working paper had proposed earlier batches first and pro rata within the last batch filled, saying "Time priority is only relevant across batch intervals, not within a batch interval" [bcs-nber-2013, n. 33, printed p. 42]. So time priority survives in discrete form across batches and is removed within one.

Latency value is reduced, not abolished: a fast firm's advantage matters only for jumps in the window $(\tau - \delta_{\text{slow}}, \tau - \delta_{\text{fast}}]$, a fraction $\delta/\tau$ of the time with $\delta = \delta_{\text{slow}} - \delta_{\text{fast}}$ (a 100 ms batch and a 100 $\mu$s speed difference give 1/1000), and a symmetrically informed liquidity provider is never sniped in the stylised model, against probability $(N-1)/N$ per large jump in the continuous market (sec. VII.B, pp. 1597–1598); with random latency draws on $[0, \epsilon]$ the sniping exposure becomes $\epsilon/(2\tau)$ (n. 34). Rents remain for information that only one trader has by the end of the batch.

The authors' rough calibration puts the batch length needed to stop the race at 10–100 ms or 100 ms–1 s depending on the meaning of $\delta$ (p. 1606, from an online appendix), which they call "at best" a sense of magnitudes; random message delays, they argue, do not address sniping (sec. VIII.C, pp. 1610–1611). Their surplus proposition, "trade the largest $Q$ with $b_Q \geq s_Q$", is Harris's result for truthful limit prices (an exchange argument), not a property of any auction whatever traders submit. Public order precedence is described for U.S. equity exchanges as of 2002, when Harris notes that decimalisation had greatly weakened it (p. 115); it is not stated as a current rule. That discriminatory pricing favours aggressive size holds only for a given set of orders.

**See also.** limit order book; order instructions; order lifecycle (priority after modification); CME Globex trading conventions (matching algorithms; pre-open); dark pools and crossing networks (derivative pricing); latency arbitrage; passive order split (queue position).

### Result 16.14 Standing limit orders are trading options: what sets their value, the two risks of using them, and why fills are adversely selected

**Statement.** Standing limit orders are options to trade: a sell limit order is a call that lets others buy when they want to, a buy limit order a put that lets them sell, and the strike is the limit price [alpha-harris, chap. 4, sec. 4.4.2, p. 75]. They are options but not option contracts: their writers grant them free to the market and can cancel them, and anyone can exercise one with a market or marketable limit order (p. 75). Their value to others rises with how close the limit price is to the market, how long the order will stand, and volatility; to limit that value, traders place orders further from the market when it is volatile, so spreads are wide in volatile markets (pp. 75–76).

Bouchaud et al. make the same point, adding that limit orders "typically earn profits when prices mean-revert but suffer losses when prices trend" [catalogue-0528, chap. 3, sec. 3.1.3, p. 46]. The formalisation, derived here: for an order of size $q$ at limit $L$, a contra trader with value $V_t$ exercises only when that is worth while, so the exercise value is $q(V_t - L)^+$ for a sell order and $q(L - V_t)^+$ for a buy order, while the order is live.

**Conditions.** The option view needs a contra trader whose value differs from the limit price and who can act before the writer cancels; the payoff is the exerciser's surplus, not the writer's loss on a market that later moves. Two risks follow [alpha-harris, chap. 4, sec. 4.4.4, p. 77]: *execution uncertainty*, when prices move away and the order fails to trade (Harris's example: a buyer who chased the price paid 9 cents more than a market order would have, p. 77), and *ex post regret*, when prices move through the limit and the order fills just before the market moves against the new position. Adverse selection, trading with better-informed traders, is "the most important cause of ex post regret". Placing orders further away lowers regret and the fill probability; placing them closer does the reverse.

**Practice.** Interpretation, from the passages above: fills are not a random sample of the times an order was available, because informed or faster traders exercise it when it has become cheap, so an evaluation of a passive strategy must measure the price move after its own fills (as in the realised spread of spread measures), not only its fill rate. In a continuous order-driven market the equilibrium spread makes the marginal trader indifferent between limit and market orders; that result, with its conditions, is spread determinants, which this entry does not restate.

*Order choice.* Market orders trade at once and pay the spread; limit orders get good prices if they execute and risk failing to trade, after which a trader may pay more than a market order would have cost; "if the spread is wide, market orders will be expensive … if the spread is narrow, market orders may be more attractive" [alpha-harris, chap. 14, sec. 14.4, p. 304]. Information changes the comparison: Harris's Table 14-1 shows uninformed traders losing to informed ones whichever order they use, a limit order filling when informed traders want the other side and failing when they want the same side (p. 304).

Interpretation, derived from explicit cash flows: to buy one unit by a horizon $T$, crossing pays the ask $a_0$ now; posting pays the bid $b_0$ if it fills and otherwise buys at a chase price $c_T$ at the horizon; both strategies then hold the same unit, marked at the same terminal price $P_T$. Posting's gain over crossing is therefore $a_0 - b_0 = S$ on a fill and $a_0 - c_T$ on a miss, and its expectation is

$$pS - (1-p)M, \qquad M = E[c_T - a_0 \mid \text{miss}],$$

with $p$ the fill probability. A price fall after a fill (buying at 99 instead of 100, marked at 97) costs both strategies the same 2 or 3, so it is not a cost of posting: posting still gains 1. Adverse selection enters through $p$ and $M$: fills are likelier when the price is about to fall, and misses when it is about to rise, which makes the chase dear. Pathwise, on 200,000 random-walk paths the mean gain equals $pS - (1-p)M$ exactly ($-0.046$ with $p = 0.43$), while the alternative formula $p(S - A) - (1-p)M$, with $A$ the post-fill fall, gives $-0.313$ because it charges that fall to posting alone.

A separate $A$ would be justified only as a cost that posting incurs and crossing does not, for example if the strategies end with different quantities. A rule based on spread width alone ignores the trader's own information about whether the quote is stale. Aggressive orders buy certainty and reveal urgency; passive and hidden orders reveal less and bear queue, non-execution and adverse-selection risk (order exposure decisions).

**Pitfalls.** The option a limit order gives "to the market" is exercised at the other trader's discretion, and the writer may cancel it. Its timing value depends on cancellation and monitoring speed, which match Harris's argument about volatility and time to cancel (spread determinants) but are not stated as such by Harris.

**See also.** Glosten–Milgrom; latency arbitrage; spread determinants; trading aggressiveness; static limit-order depth; order exposure decisions; order instructions; limit order book.

**Definition 16.15 Queue imbalance at the best quotes: two normalisations, its predictive power for the next price move, and its limits**

**Statement.** With $V_b$ and $V_a$ the volumes at the best bid and ask, Bouchaud et al. define the *queue imbalance*

$$I = \frac{V_b}{V_b + V_a} \in (0,1), \qquad \text{equivalently} \qquad QI = \frac{V_b - V_a}{V_b + V_a} = 2I - 1 \in (-1,1),$$

invariant to rescaling both volumes; $I$ near 1 (a much larger bid queue) suggests net buying pressure and an upward move [catalogue-0528, chap. 7, sec. 7.2, eq. (7.1), p. 119]. For a large-tick stock whose spread is one tick, the price changes only when one best queue empties, so the next move is a race between the two queues (sec. 7.1, p. 118). Estimating from random times the share of races that end with an up move, they find "a strong, monotonic correlation between the queue imbalance and the direction of the next price move" (p. 120). Lehalle and Laruelle state the same fact in queue-reactive terms: "when the first queue is smaller than its opposite queue, it will most probably be consumed" [alpha-laruelle, chap. 3, sec. 3.3.2, pp. 223–224].

**Conditions.** *Sample.* Bouchaud et al.'s figure uses two large-tick NASDAQ stocks, INTC and CSCO, in 2015 (chap. 4, p. 58; Table 4.1, p. 60); Lehalle and Laruelle's statistics are for a large-tick stock with an average spread of about 1.3 ticks (p. 227). *Horizon.* The prediction is for the next mid-price move, and Lehalle and Laruelle add that "the amplitude of this future price move is far lower than a tick" in most cases, so that no rewarding arbitrage can be built on it alone (p. 224).

*Large and small ticks.* The race argument needs a spread pinned at one tick; for small-tick stocks the spread can be filled inside and the argument does not apply as stated. *Placement and latency* (Lehalle and Mounjid, arXiv version 4 of 15 March 2018). Their evidence is one stock, AstraZeneca, on its primary market NASDAQ-OMX (72 percent of continuous-trading value, against Chi-X, BATS and Turquoise), from 2 January to 30 September 2013 (376,672 trades), using $\mathrm{Imb} = (Q^{\mathrm{Bid}} - Q^{\mathrm{Ask}})/(Q^{\mathrm{Bid}} + Q^{\mathrm{Ask}})$ at the first limits only; the imbalance just before a trade is "highly linearly positively correlated" with the mid-price move over the next 50 trades, shown only as a figure whose units are not stated in the text [wide-lehalle-mounjid, secs. 2.1–2.2, Table 1 and Figure 1, pp. 3–5].

They add that such predictions "usually" cannot pay for a round trip, "does not beat transaction costs (bid-ask spread and fees)", and are worth having for a trader who has already decided to buy or sell (p. 4). Members' identities on NASDAQ Nordic allow price profiles by class: high-frequency participants' passive buys follow a price rise, consistent with inserting shortly before the trade (sec. 2.3.2, pp. 7–8). Latency enters only in their stochastic-control model, as a factor $\tau \in \mathbb{N}^*$ (the trader acts every $\tau$ book changes), whose cost $V_{U_0,f} - V_{U_0,f,\tau}$ rises with $\tau$ and with the adverse-selection parameter $\alpha$ (eq. (5) and Figure 11, pp. 18–19); these are model values, not a calibrated price of latency in money or time. So the paper supplies a sample and a mechanism, not an out-of-sample effect size.

*An out-of-sample size and a calibrated latency cost*. Gould and Bonart fit a logistic regression of the direction of the next mid-price change on the queue imbalance, sampled at a random time between changes, for ten liquid Nasdaq stocks over 2014 (LOBSTER, 10:00–15:30, 252 days; 100 changes drawn per stock-day), with a random 80/20 split of the pooled observations into training and testing sets [gould-bonart-2015, arXiv v1, 11 December 2015, secs. 4–5.2, pp. 9–12]; out-of-sample areas under the ROC curve are 0.752–0.805 for the five large-tick stocks (MSFT, INTC, MU, CSCO, ORCL) and 0.581–0.642 for the five small-tick ones (GOOG, AMZN, TSLA, PCLN, NFLX), against 0.5 for the null, with in-sample values almost the same, and mean squared residuals of 0.180–0.202 and 0.235–0.246 against 0.25 (Tables 4–5, pp. 20–21).

Cautions (interpretation): the split is random across days of one year, not forward in time; the stocks were chosen by their 2014 dollar volume and price range; and the target is the direction of the next change given that one occurs, not its timing or a fill, so these are classification scores, not trading results. Moallemi and Sağlam price latency, the delay $\Delta t$ between a trading decision and its execution, in a model of a trader who must sell within a short horizon with limit orders, using a market order at the deadline if unfilled: as a fraction of the cost of immediacy, latency cost is asymptotically $\frac{\sigma\sqrt{\Delta t}}{\delta}\sqrt{\log\frac{\delta^2}{2\pi\sigma^2\Delta t}}$, with $\sigma$ the price volatility and $\delta$ the bid–offer spread (eq. (1), p. 5; Corollary 1, p. 19), and the optimal limit premium falls by about $\sigma\sqrt{\Delta t\,\log(\delta^2/(2\pi\sigma^2\Delta t))}$ (eq. (13), p. 23) [moallemi-saglam-2013, revision of 5 February 2013].

Calibrated to Goldman Sachs on 4 January 2010 (spread \$0.058, daily volatility \$1.92), the reduction is about 1.4, 1.6 and 2.1 standard deviations of the price move over 500, 250 and 50 ms (p. 23), values reproduced here with a 6.5-hour trading day; across NYSE stocks the median latency cost of a 500 ms "human" trader rose from about 5 to about 14 percent over 1995–2005 (Figure 8, pp. 26–27). Neither paper measures the after-cost performance of an imbalance-based placement rule.

**Practice.** $QI = 2I - 1$ to machine precision, and both are unchanged when both volumes are scaled. Interpretation: as a model feature it must be bound to a stated forecast horizon and validated on later days and other stocks, since hidden orders, replenishment after depletion and the next queue's size all enter after the next move.

**Pitfalls.** Queue imbalance is $(B_t - A_t)/(B_t + A_t)$. Queue imbalance is a state of displayed volume, distinct from the event-based order-flow imbalance and from signed trade volume.

**See also.** queue-reactive intensities; order-flow imbalance; tick size (large and small ticks).

**Further reading.** Pulido, Rosenbaum and Sfendourakis on why imbalance predicts prices; Bouchaud et al.'s models of the race (secs. 7.3–7.5).

**Model 16.16 Queue-reactive order-book intensities: the Poisson baseline and why it fails, intensities conditioned on queue sizes, the arrival–departure ratio, and the reset that limits mean reversion**

**Statement.** Conditioning a variable means specifying the contexts in which it is "less random"; for an order book, the probability of an insertion at the first limit when it is large, rather than of any insertion anywhere [alpha-laruelle, chap. 3, sec. 3.3.2, pp. 219–220]. The baseline models three intensities, of insertions $\lambda^+$, cancellations $\lambda^c$ and transactions $\lambda^x$, as homogeneous Poisson processes independent of the book; each is estimated by starting at many times $t$, recording the waiting time $\tau - t$ to the next event of that type and averaging, which estimates $1/\lambda$ (p. 221). Simulated with those constant intensities, the book's queue sizes are "very far away from the reality" (Figure 3.11, p. 222).

Huang, Lehalle and Rosenbaum's *Model I* keeps one intensity per event type and per size of the queue concerned, measured in average event sizes, by filing each waiting time under the queue size at the start time; Models II and III also condition on the sizes of the neighbouring queues, for example $\lambda^c(4,2,1)$ for a first bid of 4 with a second bid of 2 and a best ask of 1 (pp. 222–223).

**Notation.** The *arrival–departure* (A/R) ratio of a queue is its insertion intensity over the sum of its cancellation and transaction intensities; Lehalle and Laruelle read a ratio above one as a queue that grows on average and below one as one that depletes (p. 224). In the original paper the ratio is shifted by one size, $\rho_i(n) = \lambda_i^L(n)/(\lambda_i^C(n+1) + \lambda_i^M(n+1))$, because it is the birth–death ratio whose products give Model I's invariant law, $\pi_i(n) = \pi_i(0)\prod_{j=1}^{n}\rho_i(j-1)$ [exp-huang-lehalle-rosenbaum-queue-reactive, arXiv v2, sec. 2.3.3, p. 10]; with size-dependent intensities the shifted and same-size ratios differ (the product formula matches a direct solution of the generator to $10^{-10}$). AES: average event size, the average size of an insertion, cancellation or trade; queue sizes are rounded up to whole AES of the queue concerned (p. 7).

**Conditions.** In their figures the first queue's ratio is below one at every size and the second's above one, and the book is stable only because, when the first queue empties, the second is promoted and starts to deplete (pp. 224–225; the book is ergodic, n. 13). *The original paper* (arXiv version 2, 3 September 2014). The book is a $2K$-dimensional Markov jump process of queue sizes around a reference price, with unit jumps; Theorem 2.1 proves ergodicity under two sufficient conditions: a negative individual drift, $f_i(q) - g_i(q) < -\delta$ whenever $q_i > C_{\text{bound}}$, and a bound $H$ on the total insertion rate (sec. 2.1, pp. 4–5).

Model I treats the $2K$ queues as independent birth–death processes whose insertion, cancellation and market-order intensities depend only on the queue's own size, symmetric between bid and ask; Model IIa lets the second queue's insertion and cancellation rates depend also on whether the first queue is empty, with market orders hitting the first non-empty queue; Model IIb lets the first queue's rates depend on the opposite first queue's size in four ranges (empty, below the lower tercile, middle, above the upper tercile) (secs. 2.3–2.4, pp. 7, 10–11 and 13).

Intensities are estimated by maximum likelihood: for each size $n$, $\hat{\Lambda}_i(n)$ is the inverse of the mean waiting time since the previous event at that queue, and each event type's intensity is $\hat{\Lambda}_i(n)$ times that type's share of events at size $n$, restarting whenever the reference price changes (pp. 7–8; recovered within 0.01 on a simulated queue). Sample: Cheuvreux's order-book records, Euronext Paris, January 2010 to March 2012, five levels, first and last hour of trading removed; France Telecom (159,250 orders and 7,282 trades a day, average spread 1.43 ticks) and Alcatel-Lucent (1.99 ticks), with $K = 3$ (sec. 2.2, Table 1, pp. 5–7).

Two regularities explain why conditioning helps: a first queue smaller than the opposite one is usually consumed, and a large first queue brings more cancellations to the second (pp. 223–224). The statistics are for a large-tick stock with a spread of about 1.3 ticks; with small ticks a queue carries less meaning (p. 227). Interpretation, on a birth–death toy: with intensities that do not depend on size, a ratio of 1.2 gives an upward drift of +0.09 per event and 0.8 a drift of −0.11; with size-dependent intensities the ratio has to be read at each size.

**Practice.** Liquidity dynamics fitted this way give simulated prices whose volatility is "far too low": a newly promoted first queue is large, which pulls the price back, so mean reversion is too strong (pp. 225–226). Lehalle and Laruelle suggest Hawkes processes to add memory, or a reset: with some probability $\theta$, when a first queue depletes, the book is redrawn from the database as if participants accepted the new price (p. 226). In the paper (Model III, sec. 3.1, pp. 17–18) the reference price moves by a tick with probability $\theta$ when an event changes the mid-price while the relevant first queue is empty, and with probability $\theta^{\text{reinit}}$ the book is then redrawn from its invariant distribution, read as the share of price changes due to exogenous information.

With $\theta^{\text{reinit}} = 0$ (a purely order-book-driven price), the largest attainable ten-minute volatility, at $\theta = 1$, was 5 bp against 14 bp observed for France Telecom, with a mean-reversion ratio $\eta = N_c/(2N_a)$ of 0.08 against 0.39 observed; the authors attribute the gap to the reversed imbalance just after a price change. They calibrate $\theta = 0.7$ and $\theta^{\text{reinit}} = 0.85$ to the ten-minute volatility and $\eta$ (p. 17 and sec. 3.2.1, p. 20). The *starting time* of the waiting-time estimator matters: from every second, from just after any event, or just after a trade gives different estimates (p. 221, n. 12). For a clustered (Hawkes) sample the mean wait from uniform start times is 0.594 against a mean inter-event time of 0.405.

Interpretation, derived: for a stationary renewal process with finite second moment the mean wait from an arbitrary time is $E[X^2]/(2E[X])$, which equals $E[X]$ exactly when the coefficient of variation is 1; that does not single out exponential gaps (gaps of 0.5 with probability 0.8 and 3 with probability 0.2 have $E[X] = 1$, $E[X^2] = 2$, and a simulated mean wait of 1.0006). The Hawkes sample's gaps have a coefficient of variation of 1.37 and $E[X^2]/(2E[X]) = 0.579$, close to its measured wait.

**Pitfalls.** The queue-reactive model is a conditional Markov model with exponential waiting times; the guide's catalogue notes that it misses clustered latency races and a trader's longer-lived impact, so a simulator built on it must reproduce both book statistics and policy costs before its fills are trusted. Interpretation: latency matters most when a trader can change queue position, less with wide ticks, deep queues or orders not competing for priority; Lehalle and Laruelle say a small tick shifts attention "from queuing issues to limit price issues" (chap. 1, p. 89), and Bouchaud et al. that activity at the best quotes is faster for large-tick stocks, reflecting the importance of queue position (chap. 4, p. 59).

**See also.** queue imbalance; Hawkes processes for order flow; point processes and intensities; static limit-order depth; passive order split (queue position across venues).

**Further reading.** Cont, Stoikov and Talreja (2010) on queue dynamics; equation numbers above follow Huang, Lehalle and Rosenbaum's arXiv version 2 (2014), not the journal version.

### Definition 16.17 Order-flow imbalance: Cont, Kukanov and Stoikov's event-based measure at the best quotes, against signed trade volume, and the family of imbalance features

**Statement.** For consecutive best-quote states $(P_n^B, q_n^B, P_n^A, q_n^A)$, Cont, Kukanov and Stoikov define the contribution of the $n$th event to supply and demand at the best quotes,

$$e_n = 1\{P_n^B \geq P_{n-1}^B\} q_n^B - 1\{P_n^B \leq P_{n-1}^B\} q_{n-1}^B - 1\{P_n^A \leq P_{n-1}^A\} q_n^A + 1\{P_n^A \geq P_{n-1}^A\} q_{n-1}^A,$$

so that at an unchanged bid $e_n$ is the change in bid size, a higher bid contributes the new bid size and a lower bid minus the old one, and the ask side enters with opposite signs; the *order-flow imbalance* over $[t_{k-1}, t_k]$ is $\text{OFI}_k = \sum_{n=N(t_{k-1})+1}^{N(t_k)} e_n$, which "encompasses trades, limit orders and cancelations" [catalogue-0503, arXiv v3, sec. 2.1, p. 4]. The *trade imbalance* is $\text{TI}_k = \sum(b_n - s_n)$, buyer- minus seller-initiated volume (sec. 4.1, p. 16).

**Conditions.** *Evidence.* On one month (April 2010) of consolidated NYSE TAQ quotes and trades for 50 S&P 500 stocks drawn at random, with 10-second intervals and regressions per half-hour, $\Delta P_k = \beta\, \text{OFI}_k + \epsilon_k$ is linear with an average $R^2$ of 65 percent, against 32 percent for $\text{TI}_k$; with both, the trade imbalance is significant in only 31 percent of subsamples (pp. 8 and 16), and the slope is inversely proportional to depth (abstract, p. 1). Timestamps were rounded to the second and quotes aggregated into an NBBO; Level 1 data only (p. 8). *Names.* Bouchaud et al. call signed market-order volume $\sum \varepsilon_n v_n$ the "order-flow imbalance" [catalogue-0528, chap. 11, sec. 11.4, p. 222]; that is Cont et al.'s trade imbalance. A paper's "OFI" must be read from its formula.

**Practice.** On a toy sequence, a market sell of 200 that is replenished, a cancellation of 300 at the ask and a market buy of 100 that empties the ask give OFI = +400 and TI = −100; the two can differ in sign. Trade imbalance, best-quote imbalance as in queue imbalance, multi-level depth imbalance, event imbalance by queue, and pressure aggregated across venues are a menu of constructions, not results.

*Multi-level and cross-asset OFI.* Cont, Cucuringu and Zhang compute, for each of the top $M = 10$ levels $m$, a per-event flow $\text{OF}_n^{m,b}$ equal to $q_n^{m,b}$ if the level-$m$ bid price rose, $q_n^{m,b} - q_{n-1}^{m,b}$ if unchanged and $-q_n^{m,b}$ if it fell (and symmetrically on the ask), sum $\text{OF}^{m,b} - \text{OF}^{m,a}$ over each minute, and scale by the average depth of the first ten levels; the *integrated* OFI is the first principal component of the ten scaled OFIs, with weights normalised to sum to one, which explains on average 89.06 percent of their variance (secs. 2.1–2.2, pp. 5–8, Table 2) [wide-cont-cross-impact-ofi, arXiv v4, 13 June 2023].

Note that when a price moves away the printed per-level formula uses the *new* size where Cont, Kukanov and Stoikov use the old one. Sample: Nasdaq ITCH data from LOBSTER for the 100 largest S&P 500 stocks, 2017–2019, one-minute bins, 10:00 to 15:30, regressions per non-overlapping 30-minute window and out-of-sample $R^2$ on the next 30 minutes (p. 11). The universe is "the top 100 components of S&P 500 index, existing from 2017-01-01 to 2019-12-31", selected "based on their market capitalization as of the most recent market close at the time of our analysis, i.e. 2019-12-31" (sec. 2.1 and n. 3, p. 5).

Contemporaneously, the best-level OFI explains 71.16 percent of returns in sample and 64.64 out of sample, the integrated OFI 87.14 and 83.83, and adding other stocks' OFIs (by LASSO) adds little: 73.87 and 66.03 for best-level, 87.85 and 83.62 for integrated (Tables 3 and 5, pp. 12 and 16).

For forecasting one-minute-ahead returns, re-estimated on a rolling 30 minutes, every model has a *negative* out-of-sample $R^2$ ($-0.37$ for own best-level OFIs, $-0.10$ with cross-asset terms), though forecast-implied portfolios, before trading costs, earn more with cross-asset terms, and the advantage decays within minutes (Table 8, p. 23; Table 11 and Fig. 10, pp. 26–27); integrated OFIs do not beat best-level ones for forecasting (p. 28). So "multi-level" OFI improves contemporaneous fit, not one-minute forecasts.

*Universe selection*. Selecting by market capitalisation at the *end* of the sample means that a stock's membership in 2017 and 2018 is decided by information from December 2019: the universe contains only firms that were large, listed and in the index at the end of 2019, and leaves out those that shrank, were acquired or left the index earlier. That is a look-ahead in the choice of stocks, separate from the temporal design, in which each regression is fitted on past minutes and evaluated on later ones (the next 30 minutes contemporaneously; rolling one-minute-ahead forecasts from the previous 30 minutes, pp. 11 and 22). The rolling design keeps parameters out of sample; it does not make the universe point-in-time.

The reported calculations are not thereby invalid, but the forecasting and portfolio results (most of all for 2017–2018, and the cross-sectional breadth of the portfolios) describe a survivor-selected set of large stocks, and their transfer to a universe formed with information available at each date is untested here; the paper reports no robustness check with a point-in-time universe. Normalising by volume, average daily volume or depth before comparing stocks is interpretation; Cont et al.'s depth result suggests depth.

**Pitfalls.** Defining order-flow imbalance solely as signed trade volume conflates two measures: that is trade imbalance, not the event-based OFI. Cont et al. report contemporaneous fits, not forecasts, so they show neither that imbalance predicts the next quote move nor that it fails over longer horizons; on a subsample of stocks the results held for intervals from 10 quote updates to 10 minutes, the fit improving with the interval (p. 8). Any OFI feature used to forecast must be computed only from events before the forecast time.

*Horizon and transfer in deep order-book models*. The LOBCAST benchmark re-ran 15 deep models on FI-2010 and on two LOBSTER sets it built, LOB-2021 (1–15 July 2021) and LOB-2022 (1–15 February 2022), each ten trading days of six NASDAQ stocks (SOFI, NFLX, CSCO, WING, SHLS, LSTR) chosen from the more liquid half of NASDAQ, ten levels sampled every ten events, split 6–2–2 days into training, validation and test, with up/stable/down labels from the mean mid-price over the next $k \in \{1, 2, 3, 5, 10\}$ observations against a fixed threshold [`w2-lobcast`, arXiv v2, secs. 3–4.1, pp. 3–5].

F1 scores on the LOBSTER sets were 48–61 percent against up to 82.6 (BINCTABL) on FI-2010, BINCTABL losing about 19.6 percent on average across horizons; about half of hyperparameter runs diverged; and the authors' trading simulation (in a supplement) found profitability "far from guaranteed" (sec. 4.4, pp. 6–9). The horizon effect is not monotone: BINCTABL's correct-class rates on LOB-2021 were 54.5/65.5/57.0 percent (down/stable/up) at $k = 1$ and 48.2/83.9/53.2 at $k = 10$, the stable class gaining as the label smooths (Figure 2, p. 8).

Briola, Bartolucci and Aste label the mid-price change over 10, 50 and 100 book updates against one tick, on 15 large- and mega-cap NASDAQ stocks with LOBSTER data for 2017–2019, training on 45 consecutive days and testing on the next 10 each year, with five validation days drawn at random *from the training period* [`w2-briola-lob-guide`, arXiv v4, secs. 4–5, Table 2 and eq. (2), pp. 7–8]; they group stocks by mean spread (small tick $\gtrsim 3$ ticks, large tick $\lesssim 1.5$) and find predictability rising from small- to large-tick stocks (sec. 6). Two cautions follow (interpretation). Validation days inside the training window make model selection in-period rather than forward, and choosing stocks that stayed large- or mega-cap through 2019 conditions on the period's outcome, as with Cont, Cucuringu and Zhang's universe above.

Their transaction metric $p_T = CT/(PT + TT - CT)$, the overlap between round trips implied by predicted and by realised labels, is described by the authors as "assumption-free": it assumes no costs, fills or latency (sec. 7, eq. (3), pp. 22–23), so it measures whether entries and exits are placed correctly, not "profitable round trips after spread"; small-tick stocks at the 10-update horizon gave $p_T$ of about 0.06 (CHTR, GOOG, GS) and 0.12 (IBM, MCD, NVDA) without probability thresholds (p. 24).

*Order-flow inputs across horizons* (SSRN version of 6 August 2021, not the 2023 journal version). Kolm, Turiel and Westray train one network per stock on LOBSTER data for 115 Nasdaq stocks, January 2019 to January 2020, chosen as Nasdaq-100 members at some time in the period with at least 260 trading days and no corporate actions over it (so, again, a selection that uses the whole period), defining ten-level book states (LOB), order flows (OF) and OFI and comparing six models trained on LOB or on OF inputs [`wide-kolm-deep-ofi`, SSRN version, secs. 2.1 and 3.1, PDF pp. 5–7 and 12].

Horizons are stock-specific: $\Delta t = 2.34 \cdot 10^7 / N$ milliseconds, with $N$ the average daily number of non-zero mid-price changes, $h_k = k\Delta t/5$ for $k = 1, \ldots, 10$, and returns measured after a ten-millisecond latency buffer; windows of one validation week, four training weeks and one test week roll forward by three weeks, 18 windows in all (sec. 3.2, PDF pp. 14–15). In pairwise comparisons of each model trained on the two inputs, a Table 3 entry is, for one model and horizon, the number of stocks with a significantly positive $t$-statistic (1 percent level) minus the number with a significantly negative one (sec. 4.1.2, PDF p. 19; caption, PDF p. 21).

The CNN-LSTM row runs from 70 at 0.2 to 93 at 1.8 average price changes (89 on the horizon average), the LSTM rows 114–115. A net count is not a count of winning stocks, and 115 minus it is not a count of losing ones: stocks with no significant difference count on neither side, and the raw counts are not reported. The source is ambiguous here, and no exact count is stated in this entry: the caption and text also read an entry $X$ as OF outperforming "for $X$ stocks", the caption calls the tested quantity a "loss difference" where the text says a difference in $R^2_{OS}$, and the text's "about 40 stocks for which the CNN-LSTM with LOB inputs perform better" at the shorter horizons (PDF p. 20) cannot be derived from the net counts.

Out-of-sample $R^2$ rises up to about two average price changes (Fig. 2, PDF p. 18, whose horizons stop at two, so it cannot show what follows); the decline beyond two is shown by Fig. 8 (PDF p. 28), on horizons to ten: read from the plot, CNN-LSTM peaks near 1.5 percent at about two and falls to roughly 0.64 percent at ten, although the text describing it says $R^2_{OS}$ "plateaus at around two average price changes" (sec. 4.3, PDF p. 27). These are statistical forecasts with no costs; the authors say that for trading strategies "proper modeling of transaction costs and application of an appropriate execution strategy is critical to their profitability" (PDF p. 31). None of this establishes that any signal here pays after costs.

*Judging a signal by what it is worth after trading*. Briola, Bartolucci and Aste list the assumptions that a history-only backtest of a deep-learning forecast must make: (i) the infrastructure to record and process live data, forecast and execute in time; (ii) always being executed; (iii) zero market impact; (iv) zero transaction costs, whose combination "ruins any attempt to produce a reliable backtest" [`w2-briola-lob-guide`, arXiv v4, sec. 7, pp. 22–23]; they also report how often their tick-time horizons (10, 50, 100 updates) take under one second, one to ten seconds or longer in clock time, because practitioners care about "actual practicability in real-world scenarios" (Table 7, pp. 16–17).

LOBCAST's own trading simulation found profitability "far from guaranteed" (above), and Kolm et al. make costs and execution "critical to their profitability" (above); Gould and Bonart's classification scores (queue imbalance) and Moallemi and Sağlam's latency cost give the predictability and the delay penalty but no net result. So the checklist's items on tradability (spread, fees, fill and queue loss), latency and tick size now have sources; its capacity and crowding items do not.

*A costed test, specified but not run.* Data: message and book files (for example LOBSTER, ten levels) for named stocks and days, under the provider's licence; LOBSTER's free samples require a request and acceptance of terms. Signal: queue imbalance or event-based OFI computed only from events before the decision time. Decision: trade when the signal passes a threshold chosen on training days; order: a market order at the touch after a stated latency (in events and in milliseconds), filled up to the displayed size, the rest unfilled; exit: at the next mid-price change or a fixed horizon, again crossing the spread. Costs: the spread paid on entry and exit and the venue's taker fee at the data date.

Validation: forward by day, with each threshold and every variant logged, reporting net profit per trade and per day, the Sharpe ratio with Lo's standard error and the deflated Sharpe ratio over the variants tried, and the share of displayed depth used. Until such a test is run on licensed data, no signal here is shown to pay after costs.

**See also.** trade-sign classification; queue imbalance; information-driven bars (imbalance bars, a different use); Kyle (impact linear in net order flow); order-flow impact regression.

**Further reading.** Kolm, Turiel and Westray, SSRN working-paper version of 6 August 2021 (Pitfalls); later published in *Mathematical Finance*.

## 16.3 Order flow in time: stylised facts, clocks and point processes

**Evidence 16.18 Stylised facts of high-frequency prices and order flow: heavy tails, volatility clustering, long memory of order signs, activity clustering and intraday patterns, with their samples**

**Statement.** Bouchaud et al. report, each with its own sample:

- *heavy tails*: unconditional return distributions decay as a power law; between about a minute and a few hours a Student-$t$ with tail parameter $\mu$ near 3 fits "reasonably well", and their figure of normalised daily returns of U.S. stocks, CDS spreads and implied volatilities for 2011–2013 shows tails with exponent about 3.5 [catalogue-0528, chap. 2, sec. 2.2.1, pp. 28–30];
- *volatility clustering*: returns are nearly uncorrelated but far from IID; writing $r_t = \sigma_t \xi_t$, log-volatility has a variogram growing like $\ln(1+\tau)$ up to a cut-off of years, so bursts occur on all time scales (sec. 2.2.2, pp. 30–31);
- *long memory of order signs*: the autocorrelation of market-order signs in event time decays roughly as $\ell^{-\gamma}$ with $\gamma < 1$, close to $\ell^{-0.5}$ for four NASDAQ stocks in 2015; with $C(\ell) \approx 0.5\,\ell^{-1/2}$, $C(10{,}000) \approx 0.005$ (chap. 10, sec. 10.1, pp. 187–189);
- *activity clustering*: event arrivals cluster beyond what intraday patterns explain; counts in windows have variance well above their mean (chap. 9, secs. 9.1.2 and 9.3.3, pp. 165 and 174–175);
- *intraday patterns*: market-order and limit-order volume follow a J-shaped profile over the day, the busiest periods about four times the quietest, for PCLN, TSLA, CSCO and INTC on NASDAQ in 2015 (chap. 4, sec. 4.2.1, p. 61); volume at the best quotes behaves differently for large- and small-tick stocks (same page).

**Conditions.** Long memory of signs and clustering of activity are different facts: Poisson arrivals can carry persistent signs, and uncorrelated signs can arrive in clusters (chap. 10, pp. 187–188). Each fact is reported for a sample and a method; "universality" of the tail exponent is Bouchaud et al.'s reading of many studies, whose estimates vary (p. 29). Hasbrouck notes the same intraday seasonality, volume and volatility elevated at the start and end of sessions [s6-hasbrouck, note 1 to chap. 4, p. 180].

**Practice.** For a simulator, these are the validation targets: a reactive simulation is checked against them, not tuned until it matches one of them by construction. They describe data; none is by itself a trading signal. *What simulators are scored on*. Vyetrenko et al. catalogue realism checks in five groups: return distributions (absence of linear autocorrelation beyond about 20 minutes, heavy tails with aggregational normality, volatility clustering, absolute-return autocorrelation decaying as $\tau^{-\beta}$ with $\beta \in [0.2, 0.4]$, gain/loss asymmetry, volume–volatility and return–volatility correlations); volumes and order flow (best-queue volumes, order sizes, counts per window, inter-arrival times, placement distance and order lifetimes); non-stationary patterns (intraday and seasonal volume, event days); impact (average mid-price move against binned participation $P_\tau = |V_{\text{buy}} - V_{\text{sell}}|/(V_{\text{buy}} + V_{\text{sell}})$, fitted as $M_i \sim \alpha P_i^{\beta}$); and cross-asset correlation [w2-vyetrenko-realism, arXiv v1, sec. 2, pp. 4–6].

Their comparisons are visual and distributional, without a single score: two ABIDES configurations against minute returns of 50 U.S. stocks resampled daily over 2011 and JPM's NASDAQ book in June 2019, where neither configuration reproduced the spike of near-zero return autocorrelations, the volume–volatility correlation or the U-shaped intraday volume (the richer one reversed it) (secs. 3–4, pp. 6–8). Noble, Rosenbaum and Souilmi validate a queue-reactive simulator on the event-type mix, the imbalance before trades, hourly volume, five-minute realised volatility and the five-minute return distribution, calibrated and compared on the same Databento MBP-10 data for PFE (also INTC, VZ and T), December 2023 to December 2025, 10:00–15:30, with parameter stability checked on disjoint six-month windows; the baseline's return tails are too light, since an ergodic queue-reactive mid-price is asymptotically Brownian [exp-noble-rosenbaum-souilmi-reality-gap, arXiv v1 of 25 March 2026, secs. 2.7–2.8, pp. 8–9].

They then fit inter-event times to the data, finding a mode at about 29 $\mu$s ($\log_{10} \Delta t \approx 4.47$ in nanoseconds) with roughly 25 percent of events faster than that, read as latency races, and add an impact state $\phi_t = \sum_{t_k < t} G(t - t_k)\varepsilon_k V_k$ with a power-law kernel that tilts trade probabilities toward reversion (secs. 3.1 and 4.1, pp. 11–13). Their validation is in-sample except for that stability check, and the paper is an unrefereed preprint.

**Pitfalls.** Long memory of order signs, durations and volatility is not one property: the sources support long memory of order signs and of volatility, and clustering of activity, each with a different measure.

*Cont's eleven facts retested*. Ratliff-Crain, Van Oort, Bagrow, Koehler and Tivnan test each of its 11 facts on intraday returns of the 30 Dow stocks from 18 October 2018 to 19 March 2019, using the data the U.S. regulator used, and "find conclusive evidence for eight of Cont's original facts and no support for the remaining three" [ratliff-crain-2023-stylized, arXiv 2311.07738v2, abstract and sec. 1, pp. 1–2].

The three are gain/loss asymmetry (fact 3), the leverage effect (fact 9) and asymmetry in timescales (fact 11), whose measures "vary in sign and strength depending on the symbol and timescale"; the volume–volatility correlation holds in clock time but not at the trade level, and aggregational Gaussianity shows more clearly in event time (sec. 6, PDF p. 15). They note that daily data or pooled stocks have shown the leverage effect and timescale asymmetry elsewhere. So these three are not safe defaults for an intraday single-stock simulator.

**See also.** realised variance and the signature plot; intraday deseasonalisation; Hawkes processes for order flow; propagator model (sign persistence and impact); activity-sampled bars (heavy tails under different clocks); intraday and overnight returns.

**Further reading.** Vyetrenko et al.'s realism metrics and Noble, Rosenbaum and Souilmi's reality-gap tests, in their arXiv versions (Practice).

**Method 16.19 Realised variance and covariance at high frequency: the signature plot, microstructure noise and the Epps effect**

**Statement.** For a log price $X = \log S$ observed over $[0, T]$ with step $\Delta$ and returns $r_\Delta(j) = X_{j\Delta} - X_{(j-1)\Delta}$, Lehalle and Laruelle state that "in the Itô semi-martingale framework" the realised variance $\hat{V}_R(\Delta) = \sum_{j=1}^{T/\Delta} r_\Delta(j)^2$ converges in probability to the integrated variance $\int_0^T \sigma_s^2\, ds$ as $\Delta \to 0$, and the realised covariance $\hat{C}_R(\Delta) = \sum_j r_\Delta^1(j) r_\Delta^2(j)$ to $\int_0^T \rho_s \sigma_s^1 \sigma_s^2\, ds$ [alpha-laruelle, app. A, secs. A.13.1–A.13.2, pp. 316–318]. Interpretation (standard): those limits hold for continuous Itô processes; for a semimartingale with jumps $\hat{V}_R$ converges to the quadratic variation $\int_0^T \sigma_s^2\, ds + \sum_{s \le T} (\Delta X_s)^2$ and $\hat{C}_R$ to $\int_0^T \rho_s \sigma_s^1 \sigma_s^2\, ds + \sum_{s \le T} \Delta X_s^1 \Delta X_s^2$, the last term from co-jumps.

If observed prices are $X = M + \varepsilon$ with $M$ a semimartingale and $\varepsilon$ noise, $\hat{V}_R$ adds the noise terms $\sum \eta_\Delta(j)^2$ and a cross term, and on Rio Tinto data (1 January to 19 June 2012) the estimator diverges as $\Delta \to 0$ (pp. 316–318). The covariance estimator needs synchronous data, which quotes and trades are not; the *Epps effect* is Epps's (1979) finding that "correlations among price changes … are found to decrease with the length of the interval for which the price changes are measured", shown for Total and France Telecom on Euronext Paris, 23–27 July 2012 (pp. 318–319, Figure A.26).

**Notation.** Bouchaud et al.'s *volatility signature plot* is $\sigma^2(\tau) = V(\tau)/\tau$ against the lag $\tau$, with $V(\tau)$ the price variogram: flat for uncorrelated returns, rising with trends and falling with mean reversion or with mean-reverting high-frequency noise [catalogue-0528, chap. 2, secs. 2.1.2–2.1.3, pp. 25–27]. Lehalle and Laruelle plot $\hat{V}_R(\Delta)$ against $\Delta$; the two plots carry the same information in different units.

**Conditions.** *Jumps.* A unit-jump Poisson path with five jumps on $[0, 1]$ has zero diffusion variance, yet its realised variance is 9 on a 10-step grid (two jumps in one interval) and 5, the jump count, from $10^4$ steps on; three common jumps of $+1$ and $-0.5$ give a realised covariance of $-1.5$.

*Noise.* The finite-grid expectation below is exact for a driftless Brownian $M$ with constant volatility and observation noise i.i.d., centred and independent of $M$; with drift it adds the squared drift increments, with autocorrelated noise (as in Bouchaud et al.'s mean-reverting noise) the noise

term is no longer $2no_\eta^2$, and for a general semimartingale only the limits above are claimed. Explanations offered for the Epps effect are lead–lag between related assets, asynchronous transactions, and tick and other microstructure effects (Lehalle and Laruelle, p. 318).

In the simulated model, $X = M + \varepsilon$ with $M$ a driftless Brownian motion of volatility $\sigma$ and $\varepsilon$ i.i.d. centred noise of standard deviation $\sigma_\eta$ independent of $M$, $E[\hat{V}_R] = \sigma^2 T + 2no_\eta^2$ exactly for $n = T/\Delta$ returns (each noise difference has variance $2\sigma_\eta^2$ and the cross terms have mean zero), and simulated values track it from 13 to 23,400 returns a day; two Brownian motions with correlation 0.6, observed at independent random times and sampled at the previous tick, show a mean correlation over 40 paths of 0.12 at a step of 5 time units against 0.63 at 600, from asynchrony alone.

**Practice.** Bouchaud et al.'s E-mini S&P 500 signature plot is almost flat from one second to about 20 trading days, decreasing by about 20 percent, which indicates weak mean reversion (sec. 2.1.4, pp. 27–28): for that contract, noise inflates short-horizon volatility only modestly. Whether the plot rises at fine steps depends on the price used (trades bounce between bid and ask; mid-quotes much less) and on the instrument. Lehalle and Laruelle list range-based, multi-scale and bid–ask-model estimators as alternatives, and Barndorff-Nielsen–Shephard, Hayashi–Yoshida and Zhang for covariance (pp. 317–318).

**Pitfalls.** Realised variance can diverge as sampling becomes too fine, and the Epps effect can come from tick effects, but both hold under stated noise and asynchrony, not for every series. Previous-tick sampling is not a remedy for covariance, since it itself produces the Epps effect in the check above; refresh-time sampling is discussed above.

*Refresh time*. Barndorff-Nielsen, Hansen, Lunde and Shephard define it for $d$ assets with trade times $t_j^{(i)}$ and counts $N^{(i)}(t)$ as $\tau_1 = \max_i t_1^{(i)}$ and $\tau_{j+1} = \max_i t^{(i)}_{N^{(i)}(\tau_j)+1}$, the first time after $\tau_j$ at which every price has been updated; they retain $p = dN/\sum_i n^{(i)}$ of the data (21/27 in their three-asset figure), the sample falls with log $d$ when trades arrive as Poisson processes, and they treat the vector at $\tau_j$ as if fully updated although "we only see a single new price and $d-1$ stale prices", showing that this staleness does not affect the asymptotic distribution of their *realised kernel*, which includes lagged cross-products [bnhls-2008, sec. 2.1.1, Definition 1 with nn. 2–3 and Figure 1, pp. 4–5].

Their limit theory assumes noise that is covariance stationary given the efficient price, with summable autocovariances (Assumption 1, not white noise), and refresh times $\tau_j = T(j/n)$ from a strictly positive càdlàg time change adapted to the filtration (Assumption 2, pp. 7–8). Refresh time is therefore a synchronisation step, not a covariance estimator on its own. Two Brownian motions with correlation 0.6 observed at independent Poisson times (400 per unit time, no noise) give a mean realised covariance over 60 paths of 0.39 at refresh times, 0.60 once one lead and one lag are added, 0.59 for the Hayashi–Yoshida sum of products of overlapping returns, and 0.02 at a previous-tick grid of 1/5000.

The Hayashi–Yoshida estimator is the paper's sum $U_n = \sum_{i,j} \Delta P^1(I^i)\Delta P^2(J^j)1_{\{I^i \cap J^j \neq \varnothing\}}$ of products of returns over overlapping observation intervals [hayashi-yoshida-2005, Definition 3.1, eq. (3.1), p. 368]. Its consistency (Theorem 3.1, p. 369: in $L^2$ and in probability) holds for diffusions with deterministic, bounded volatilities and deterministic correlation (p. 367) under Condition C: sampling intervals independent of the prices, and the expected longest interval going to zero (p. 368); without drift it is unbiased for every $n$ (Remark 3.3, p. 369), and realised covariance at a fixed interval is not consistent (Remark 3.4, p. 375). Microstructure noise is not in the model; the authors mention it only as a possible extension (p. 367).

The simulation above meets these conditions (independent Poisson times, no noise), so its 0.59 is what the theorem predicts; the observation times of real trades depend on prices, and real prices carry noise. In Barndorff-Nielsen et al.'s data (ten NYSE stocks, TAQ, 3 January 2005 to 29 June 2007, 626 days, trades and mid-quotes), the Hayashi–Yoshida estimator was "even more distorted" than the 15-second realised covariance relative to open-to-close covariances, which noise-free simulations cannot show (sec. 5.3, p. 18). In pairs, refresh time "rarely" cost more than half the observations of the most frequently traded stock and typically 30 to 40 percent of the least active one; for all ten together more than 20 percent remained in the worst case (Table 3, pp. 17–18; page numbers are the paper's printed ones).

**See also.** fundamental and transitory volatility; Roll's estimator (bounce as negative autocovariance); Garman–Klass volatility; stylised facts.

**Further reading.** [bnhls-2008] for refresh time and realised kernels; [hayashi-yoshida-2005] for the overlapping-interval estimator.

### Method 16.20 Intraday deseasonalisation: additive and multiplicative decompositions, centred moving averages, seasonal coefficients and a factorised volume model

**Statement.** A series is decomposed into a trend $C_t$, a seasonal component $S_t$ and an irregular part $\varepsilon_t$, additively $X_t = C_t + S_t + \varepsilon_t$ or multiplicatively $X_t = C_t S_t + \varepsilon_t$ [alpha-laruelle, app. A, sec. A.11.1, p. 299]. A centred moving average of odd length $2k+1$ is $MA_t = (X_{t-k} + \cdots + X_{t+k})/(2k+1)$; of even length $2k$ it weights the end points by $\frac{1}{2}$, $MA_t = (\frac{1}{2}X_{t-k} + X_{t-k+1} + \cdots + X_{t+k-1} + \frac{1}{2}X_{t+k})/(2k)$, and it cannot be computed for the first and last $k$ observations (pp. 299–300).

For period $p$ and $t = (i-1)p + j$, with $MA$ of length $p$ for the trend: additively, $\hat{S}_j = \frac{1}{n}\sum_i (X_{ij} - MA_{ij})$, centred as $S_j = \hat{S}_j - \frac{1}{p}\sum_k \hat{S}_k$; multiplicatively, $\hat{S}_j = \frac{1}{n}\sum_i X_{ij}/MA_{ij}$, normalised as $S_j = \hat{S}_j / (\frac{1}{p}\sum_k \hat{S}_k)$, so that the coefficients sum to $p$; medians may replace means (pp. 300–302).

**Conditions.** The moving average's length must equal the period of the seasonal variation or a multiple of it, so that it carries no seasonality (p. 300). The multiplicative estimate assumes $\varepsilon_{ij}$ small relative to the trend (p. 301). Interpretation: a centred average uses $k$ later observations, so a shock at $t$ changes the estimates for the preceding $k$ times; seasonal coefficients estimated this way on a sample cannot be used at a decision time inside that sample without look-ahead.

**Practice.** Lehalle and Laruelle's volume model for slot $n$, day $d$, month $m$ and year $y$ is $V(y,m,d,n,\omega) = (f(n) + g(d,n))F(m)G(y)v(n,\omega)$, with $f$ the mean slot profile, $g$ special-day effects such as expiries or macroeconomic news, $F$ and $G$ monthly and yearly effects and $v$ random; with $E[H] = 1$ and $\sum_m F(m) = 1$, $E[V_M(y,m,\cdot)] = G(y)F(m)$ and $E[V_Y(y,\cdot)] = G(y)$ (sec. A.11.2, pp. 302–303). Their Total SA example (2 January 2007 to 1 June 2012) shows a bump at noon on Fridays from derivative expiries (pp. 303–304). Bouchaud et al.'s alternative for point processes is a *business time* $\hat{t}(t) = \int^t \bar{\phi}(u)\,du$ built from the average activity profile, which flattens it [catalogue-0528, chap. 9, sec. 9.3.1, p. 173]; see market clocks. On a synthetic series with 13 slots and 60 days: multiplicative coefficients recovered to within 0.005.

**Pitfalls.** For an even-length moving average, the window must match the period precisely; Lehalle and Laruelle require it to equal the period or a multiple. Removing predictable clock-time effects before comparing venues, estimating impact or training intraday features keeps a noon lull, a close auction, an expiry day or a news window from looking like a signal, but the removal must use profiles estimated from earlier data (interpretation).

*Mean against path*. "To show that the average profile of a quantity does not mean that this profile exists in the real world", Lehalle and Laruelle compare Adidas's mean intraday volatility from 27 April to 27 July 2012 with its path on 25 May 2012, which "does not have the same behavior as the mean" [alpha-laruelle, app. A, sec. A.14.1, pp. 318–320, Figure A.27]. A seasonal profile is an expectation, not a typical day: in a simulation of 500 calm days each with one short spike at a random time, the average profile peaks at 1.36 while every day peaks at 7. Averaging across days therefore hides event days, closing auctions and isolated liquidity shocks, and a simulator validated only on average profiles can miss the day-level behaviour.

**See also.** stylised facts (intraday patterns); market clocks; time-validity of labels and fitted transforms; execution benchmarks (volume curves).

### Definition 16.21 Market clocks and durations: clock, event, trade, quote, volume and business time, and when each is the right index

**Statement.** For event times $0 < \tau_1 < \tau_2 < \cdots$ the *duration* is $\Delta\tau_i = \tau_i - \tau_{i-1}$ [alpha-laruelle, app. A, sec. A.12.1, Definition 5, p. 306]. An analysis can be indexed by an event counter (trades, quote revisions, or both) or by wall-clock time at fixed intervals; Hasbrouck's rule is that for a single precisely time-stamped security an event index is "probably the first choice", because the process is more likely to be covariance stationary in event time, while comparisons across securities may favour wall-clock time, volatility per minute being more comparable than volatility per trade [s6-hasbrouck, sec. 9.7.3, p. 90]. *Business time* rescales clock time by the average activity profile $\bar{\phi}$, $\hat{t}(t) = \int^t \bar{\phi}(u)\,du$, so that intraday

activity is flat [catalogue-0528, chap. 9, sec. 9.3.1, p. 173]. Volume and dollar clocks, one step per fixed quantity or value traded, are the sampling rules of activity-sampled bars.

**Conditions.** With a wall-clock index, several events in one interval must be aggregated or transformed (Hasbrouck, p. 90). Propagator models are almost always written in trade time and Hawkes processes in physical time; hybrids exist [alpha-laruelle, app. A, sec. A.12.3, p. 311]. For a homogeneous Poisson process the durations are i.i.d. exponential with mean $1/\lambda$ (p. 306), so high intensity means short expected durations; for clustered processes the expected duration from a given time depends on the recent history.

**Practice.** Lehalle and Laruelle note that replacing physical time by event time "would change a lot the meaning of the obtained models" (chap. 3, p. 227). Interpretation: latency, risk limits, news and benchmark windows are defined in clock time, and an event-time or business-time model must be mapped back before it is used for them; a transformation that flattens activity also hides the periods of intense activity a risk manager cares about, so diagnostics are kept in both clocks.

**Pitfalls.** A duration is not only the gap between trades: Hasbrouck uses duration analysis for other waiting times too, such as a limit order's time to execution, whose observations are censored when the order is cancelled first [s6-hasbrouck, sec. 15.2.3, pp. 163–164]. Quote time and business time are covered above; business time as activity-adjusted time after removing intraday seasonality matches Bouchaud et al.'s construction only when $\bar{\phi}$ is estimated from data before the period it rescales.

**See also.** point processes and intensities; activity-sampled bars; intraday deseasonalisation; propagator model (trade time).

**Further reading.** Engle and Russell's autoregressive conditional duration model is named by Bacry et al. (wide-bacry-hawkes, p. 1).

#### Definition 16.22 Point processes for order-book events: counting process, conditional intensity, the homogeneous Poisson baseline and the time-change theorem

**Statement.** A simple point process is an increasing sequence of event times $(\tau_i)_{i\geq 1}$, for example trades or limit-order arrivals; its *counting process* is $N_t = \sum_{i\geq 1} 1\{\tau_i \leq t\}$ [alpha-laruelle, app. A, sec. A.12.1, Definitions 3–4, pp. 305–306]. For $N$ adapted to a filtration $\mathcal{F}_t$, the left-continuous *intensity* is $\lambda(t \mid \mathcal{F}_t) = \lim_{h\downarrow 0} h^{-1} E[N_{t+h} - N_t \mid \mathcal{F}_t]$, or equivalently $\lim_{h\downarrow 0} h^{-1} P(N_{t+h} - N_t > 0 \mid \mathcal{F}_t)$; it depends on the filtration, taken to be the process's own history (Definition 6, p. 306). A homogeneous Poisson process has $P(N_{t+h} - N_t = 1 \mid \mathcal{F}_t) = \lambda h + o(h)$, an intensity that ignores the history, and i.i.d. exponential durations (p. 306). With the integrated intensity $\Lambda(\tau_{i-1}, \tau_i) = \int_{\tau_{i-1}}^{\tau_i} \lambda(s)\, ds$, the *time-change theorem* states that $\tilde{N}_\tau = N_{t_\tau}$, with $\int_0^{t_\tau} \lambda(s)\, ds = \tau$, is a unit-rate Poisson process (Theorem 1, pp. 307–308).

**Notation.** Lehalle and Laruelle model order-book events with three intensities, $\lambda^+$ for limit-order insertions, $\lambda^c$ for cancellations and $\lambda^x$ for transactions [alpha-laruelle, chap. 3, p. 221]. Bouchaud et al. write $\phi(t)$ for the intensity and define the clustering ratio $r(\tau) = V[N(t+\tau) - N(t)]/E[N(t+\tau) - N(t)]$, equal to 1 for a Poisson process, above 1 for clustering and below 1 for inhibition [catalogue-0528, chap. 9, secs. 9.1–9.1.2, pp. 164–165].

**Conditions.** The intensity is conditioned on information strictly before $t$: interpretation, the intensity must be predictable (left-continuous), which is what makes it usable for prediction and simulation. Lehalle and Laruelle print the theorem's condition as $\int_0^\infty \lambda(s)\, ds = 1$ (p. 308); the integrated intensity must instead diverge, $\int_0^\infty \lambda(s)\, ds = \infty$, for the rescaled times to cover the half-line (a statement attributed to Meyer (1969) and Papangelou (1972)).

*Half-line statement.* Laub, Taimre and Pollett restate Daley and Vere-Jones's Proposition 7.4.IV: for "an unbounded, increasing sequence of time points $\{t_1, t_2, \dots\}$ in the half-line $(0, \infty)$, and a monotonic, continuous compensator $\Lambda(\cdot)$ such that $\lim_{t\to\infty} \Lambda(t) = \infty$ a.s.", the transformed sequence $\{\Lambda(t_1), \Lambda(t_2), \dots\}$ is a realisation of a unit-rate Poisson process if and only if the original is a realisation of the point process defined by $\Lambda$ [laub-taimre-pollett-2015, arXiv v1, July 2015, sec. 5.1, Theorem 5, p. 18]; their finite-window version (Theorem 4, after Brown et al.) needs a positive intensity and $\Lambda(T) < \infty$. This confirms the divergence condition stated above and that Lehalle and Laruelle's "= 1" is a misprint; the attribution to Daley and Vere-Jones is through this survey.

*Finite-window statement.* Brown, Barbieri, Ventura, Kass and Frank state the theorem for event times $0 < u_1 < \cdots < u_n < T$ of a point process whose conditional intensity satisfies $0 < \lambda(t \mid H_t)$ for all $t \in (0, T]$, with $\Lambda(u_k) = \int_0^{u_k} \lambda(u \mid H_u)\, du$ and $\Lambda(t) < \infty$ with probability one for all $t \in (0, T]$: then the $\Lambda(u_k)$ "are a Poisson process with unit rate", proved by showing that $\tau_k = \Lambda(u_k) - \Lambda(u_{k-1})$ are i.i.d. unit exponential, including the censored interval after $u_n$ [brown-time-rescaling-2002, sec. 2.2, eqs. (2.6)–(2.11), p. 329]. Existence of a conditional intensity makes the compensator absolutely continuous, so the continuity condition is built in; on a finite window no divergence is needed, since only $(0, \Lambda(T)]$ is covered.

Applied as a goodness-of-fit test, a correctly specified model's rescaled durations are i.i.d. unit exponential, and $z_k = 1 - e^{-\tau_k}$ are i.i.d. uniform; Brown et al. plot the ordered $z_{(k)}$ against $b_k = (k - \frac{1}{2})/n$ with 95 percent Kolmogorov–Smirnov bounds $b_k \pm 1.36/\sqrt{n}$ for moderate to large $n$ (eqs. (2.16)–(2.17), p. 331). Those bounds are for a fully specified intensity; with parameters estimated from the same data they are only approximate (interpretation, not a statement of the paper).

**Practice.** Durations of a simulated exponential-kernel Hawkes process are far from exponential (Kolmogorov–Smirnov $p < 10^{-260}$), while the same durations rescaled by the true compensator have mean 1.005 and are not rejected as unit exponential ($p = 0.35$). The Poisson baseline is diagnostic: as queue-reactive intensities shows, context-free intensities do not reproduce queue sizes.

**Pitfalls.** $\mathcal{F}_t$ is the information available before $t$ only with left-continuity, and the time change needs its stated conditions. That a Poisson baseline "cannot reproduce realistic queue-size distributions because event rates are strongly state-dependent" is Lehalle and Laruelle's finding for one large-tick stock (Figure 3.11), not a general theorem.

**See also.** Hawkes processes for order flow; queue-reactive intensities; market clocks and durations.

**Further reading.** Daley and Vere-Jones on point processes (its Proposition 7.4.IV is cited through Laub, Taimre and Pollett's survey); Brown et al. (2002) for the finite-window theorem and its goodness-of-fit use.

#### Model 16.23 Hawkes processes for order flow: the linear self-exciting intensity, its conditions, branching ratio and stationarity, multivariate and marked forms, inhibition through a non-linear link, and what calibrations show

**Statement.** A linear Hawkes process has intensity

$$\lambda(t) = \lambda_0(t) + \int_{-\infty}^{t} \nu(t-s)\, dN_s = \lambda_0(t) + \sum_{\tau_i < t} \nu(t - \tau_i),$$

with a deterministic baseline $\lambda_0 \geq 0$ and a kernel $\nu \geq 0$ that "expresses the positive influence of the past events" [alpha-laruelle, app. A, sec. A.12.2, Definition 7, p. 308]. Bacry, Mastromatteo and Muzy's multivariate definition, $\lambda_t^i = \mu^i + \sum_j \int \varphi_{ij}(t - t')\, dN_{t'}^j$, requires each kernel component to be non-negative, causal ($\varphi_{ij}(t) = 0$ for $t < 0$) and integrable, and the process has asymptotically stationary increments if the matrix of kernel norms $\|\Phi\| = \{\int |\varphi_{ij}|\}$ has spectral radius below 1 [wide-bacry-hawkes, arXiv v2, sec. 2.1, Definition 1 and Proposition 1, pp. 3–4]. In one dimension, with norm $g = \int_0^\infty \nu(u)\, du < 1$, the mean intensity is $\bar{\lambda} = \lambda_0/(1-g) \geq \lambda_0$; $g$ is the *branching ratio*, the expected number of events directly triggered by one event, and a family of events has mean size $1/(1-g)$ [catalogue-0528, chap. 9, secs. 9.2.2 and 9.2.6, pp. 167–168 and 172]. For $\nu(u) = \alpha e^{-\beta u} 1\{u \geq 0\}$ and constant $\lambda_0$, $g = \alpha/\beta$ and $E[\lambda(t)] = \lambda_0/(1 - \alpha/\beta)$ (Lehalle and Laruelle, pp. 309 and 311).

**Notation.** Parameterisations differ: Lehalle and Laruelle's $\alpha e^{-\beta t}$ has norm $\alpha/\beta$; Bacry et al.'s $\alpha\beta e^{-\beta t}$ (p. 5) and Bouchaud et al.'s $g\omega e^{-\omega t}$ (eq. (9.21), p. 170) have norms $\alpha$ and $g$. A fitted "$\alpha$" must be read with its kernel.

**Conditions.** *Excitation only.* A non-negative kernel is necessary and sufficient for the intensity to stay non-negative for every realisation, and always gives self-excitation and clustering [catalogue-0528, chap. 9, sec. 9.5, p. 181]. *Inhibition* needs a non-linear form, $\lambda(t) = F\big(\lambda_0 + \int \nu(t-u)\, dN_u\big)$ with $F \geq 0$, such as the positive part or $\ln(1+e^x)$; if $F$'s slope never exceeds 1 the same stability condition holds (Brémaud and Massoulié, 1996, as reported, p. 181). Zhu's thesis restates their theorems, for $\lambda_t = \lambda\big(\int_{-\infty}^{t} h(t-s)N(ds)\big)$, in two parts [zhu-nonlinear-hawkes-2013, arXiv v3, sec. 1.5, Theorems 4–5, pp. 33–34]:

- *Existence* of a stationary process needs $\lambda(\cdot)$ non-negative, non-decreasing and left-continuous with $\lambda(z) \leq C + \alpha z$ and $\alpha \int_0^\infty h < 1$, without Lipschitz continuity (Theorem 4).
- *Uniqueness* of the stationary distribution with finite intensity, and stability in distribution, need $\lambda(\cdot)$ $\alpha$-Lipschitz with $\alpha\|h\|_{L^1} < 1$. Stability in variation needs in addition $\int_0^\infty t\,h(t)\,dt < \infty$ (Theorem 5).

Zhu notes that the original allows $h$ to take negative values, while his own work keeps $h \geq 0$. *Nonlinear Hawkes processes*. Brémaud and Massoulié write $\lambda(t) = \phi\big(\int_{(-\infty,t)} h(t-s)N(ds)\big)$ with $\phi : \mathbb{R} \to \mathbb{R}^+$ and $h : \mathbb{R}^+ \to \mathbb{R}$, so a signed kernel is allowed; the linear Hawkes process is the case $h \geq 0$, $\phi(x) = \nu + x$ [bremaud-massoulie-1996, eq. (1), p. 1564]. For a signed kernel the norm is that of $|h|$: Theorem 1 requires $\phi$ $\alpha$-Lipschitz and $\alpha \int_{\mathbb{R}^+} |h(t)|\, dt < 1$ (eq. (3)), and then gives a unique stationary distribution with finite intensity and stability in distribution; stability in variation needs in addition $\int t|h(t)|\, dt < \infty$ (eq. (4), p. 1567). Theorem 2 drops the bound on the $L^1$ norm when $\phi$ is also bounded: $h \in L^1$ and (4) then suffice (p. 1569).

So "slope never exceeds 1" is the $\alpha = 1$ case and needs $\int |h| < 1$, not $\int h < 1$; the "same stability condition" as reported holds only when the inhibiting part is counted in absolute value. Such models lose most analytical results, while simulation and calibration remain possible (Bacry et al., sec. 2.2.4, p. 8). With $\lambda_0 = 0.5$ and a kernel $-0.8e^{-2u}$, the linear intensity just after an event is $-0.3$, not an intensity; with the positive-part link the simulated rate is 0.37 and the clustering ratio 0.62, below 1 as for repelling events.

*Stationarity.* With $g \geq 1$ the process explodes or, at $g = 1$, is critical; the stationary formulas then fail (Bouchaud et al., p. 168). *Multivariate and marked flow.* Market orders can trigger market orders, limit orders and cancellations, and a $K$-type process has $\bar{\phi} = (I - g)^{-1}\phi_0$ with $g_{ij} = \int \Phi_{ij}$ (sec. 9.5, pp. 182–183); Lehalle and Laruelle sketch classes of agents (banks, brokers, high-frequency traders) with buy and sell intensities excited across classes under $\|\varphi\|_1 < 1$ [alpha-laruelle, app. A, sec. A.12.3, pp. 315–316].

**Practice.** *Calibrations.* Lehalle and Laruelle fit a one-dimensional exponential Hawkes process to trades in France Telecom on the Chi-Delta dark pool on 23 July 2012, with $\lambda_0 = 140$, $\alpha = 250$, $\beta = 300$, a branching ratio of 0.83 (Figure A.24, p. 310). Bouchaud et al. report maximum-likelihood fits across many contracts with $g$ close to 1 and power-law kernels, the short time scale falling from seconds in 1998 to milliseconds in 2012, and TSLA mid-price changes whose effective feedback approaches 1 as the window grows (pp. 174–175 and 178). Bacry et al. report, from Bacry and Muzy's model of EuroStoxx and Euro-Bund futures over 800 trading days in 2009–2012, slightly negative self-kernels in the feedback of price moves on market-order flow: an up move raises sell intensity and lowers buy intensity, an inhibitory effect [wide-bacry-hawkes, arXiv v2, sec. 5.1, p. 27].

By simulation, an exponential-kernel process with branching ratio 0.6 has mean rate 2.47 against the theoretical 2.50 and a clustering ratio of 6.4 in windows of 50 against a long-window limit of 6.25.

**Pitfalls.** In a linear marked Hawkes model one class of event can excite another but not inhibit it: kernels are non-negative. Calibration "does not prove the existence of a genuine causal self-excitation mechanism": two-point correlations do not imply causality, and near-critical fits may mean either fragile markets or a framework too restrictive (Bouchaud et al., pp. 169, n. 4, and 184). Linear Hawkes processes do not reproduce strong volatility clustering and fat tails without extensions such as price feedback (p. 185). Time-stamp errors distort maximum-likelihood fits of power-law kernels (p. 178).

**See also.** point processes and intensities; queue-reactive intensities (Hawkes memory); stylised facts; propagator model (event-time alternative).

**Further reading.** wide-bacry-hawkes, arXiv version 2 (17 May 2015). Hawkes (1971), Brémaud and Massoulié (2001) and Bacry and Muzy; Brémaud and Massoulié (1996) for eq. (1) and Theorems 1–2.

## 16.4 Inventory risk and the price of immediacy

#### Model 16.24 Grossman–Miller market making: the price of immediacy

**Statement.** Competitive, risk-averse market makers supply immediacy to a liquidity trader who arrives before the trader with the offsetting need, and the price concession they charge compensates exactly for carrying the inventory until that trader arrives [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 21]. In the simplified three-date version, $n$ identical market makers, liquidity trader LT1 holding $i$ units and liquidity trader LT2 holding $-i$ units (arriving at date 2) trade an asset worth $S_3 = \mu + \epsilon_2 + \epsilon_3$ at date 3, and the date-1 equilibrium is [crypto-cartea, same chap., pp. 23–24]

$$S_1 = \mu - \gamma\sigma^2 \frac{i}{n+1}, \qquad q_1^{MM} = q_1^{LT1} = \frac{i}{n+1},$$

so LT1 sells $\frac{n}{n+1}\, i$ at once and each market maker holds $\frac{i}{n+1}$ until date 2, when LT2 arrives, the price is efficient, $S_2 = \mathbb{E}[S_3 \mid \epsilon_2] = \mu + \epsilon_2$, and every agent exits with zero inventory [crypto-cartea, same chap., pp. 22–24]. The liquidity discount is $|S_1 - \mu| = \gamma\sigma^2\, |i|\, /(n+1)$; when LT1 buys ($i < 0$) the market makers earn the same amount as a premium [crypto-cartea, same chap., p. 24]. Figure 16.1 shows the holdings and trades.

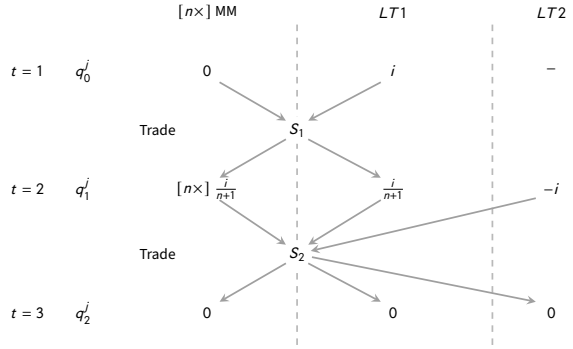



Figure 16.1: Holdings $q_t^j$ and trading in the Grossman–Miller model, after [crypto-cartea, Figure 2.1, p. 23].

**Notation.** Local override for this entry, the fee result and the liquidity measures: $\sigma^2$ is the variance of one period's value shock, in squared price units and not annualised; $\gamma$ is the coefficient of absolute risk aversion; $q_t^j$ is agent $j$'s holding on leaving date $t$, and $S_t$ the date-$t$ trade price.

**Conditions.** Dates $t \in \{1,2,3\}$; shocks $\epsilon_2, \epsilon_3$ independent $N(0,\sigma^2)$, each announced between dates $t-1$ and $t$; every agent has CARA utility $U(X) = -\exp(-\gamma X)$; market makers start with no inventory; the arrival of LT2 at date 2 is certain; no trading or holding costs [crypto-cartea, same chap., p. 21]. Interpretation: the concession charges for risk, not for information; no agent is informed, which separates this model from Kyle's. Idea of the proof (full working in `notes/artefacts/market-making-lob/model-derivations.tex`): solve backwards; with CARA utility and normal shocks each agent's demand is $(\mathbb{E}[\text{next price}] - \text{price})/(\gamma\sigma^2)$, and market clearing at each date gives the prices [crypto-cartea, same chap., pp. 22–23].

**Practice.** The discount rises with the size of the liquidity need $|i|$, risk aversion $\gamma$ and volatility $\sigma^2$, and falls with competition $n$; as $n$ grows the premium goes to zero, $S_1$ tends to $\mu$, and LT1's immediate trade tends to $i$ [crypto-cartea, same chap., p. 24]. Competition is itself set by participation costs: in Grossman and Miller's earlier entry stage, the number of market makers falls monotonically as the participation cost $c$ rises, which raises the discount [crypto-cartea, same chap., p. 25].

*Free entry in the original* (NBER working paper, July 1988). A market maker present on the floor forgoes opportunities worth $c$, and "free entry of market makers will occur until" $EU(W_0 - c + (\tilde{P}_2 - P_1)x_1^m) = EU(W_0)$ (eq. (16)); with exponential utility and the market makers' demand (12) this becomes $e^{ac}\,E\exp\left(-\frac{t}{2}z^2\right) = 1$ with $t = a^2\,\mathrm{Var}_1(\tilde{P}_2)\,\mathrm{Var}(i)/(1+M)^2$ and $z^2 = i^2/\mathrm{Var}(i)$ (eq. (17b)), and, when the expected imbalance is zero, $1/\sqrt{1+t} = e^{-ac}$ (eq. (19)) [grossman-miller-1988-nber, eqs. (16)–(19), printed p. 18]; here $a$ is absolute risk aversion, $M$ the number of market makers and $i$ the order imbalance.

The paper says that $t$ "is determined solely by $ac$ and is an increasing function of $ac$", that the number of market makers is greater the lower $c$ and the smaller $a$, and that it rises with either variance, $\mathrm{Var}(i)$ measuring the average size of hedging demand and $\mathrm{Var}_1(\tilde{P}_2)$ the predictability of the price change [grossman-miller-1988-nber, printed p. 19]. From (19): $1 + M = a\sqrt{\mathrm{Var}_1(\tilde{P}_2)\,\mathrm{Var}(i)}\,/\sqrt{e^{2ac}-1}$, so $M$ falls with $c$ and rises with each variance, as stated, but its response to $a$ at fixed $c$ is not monotone: it rises with $a$ while $2ac < 1.594$ and falls beyond, so the paper's "That number would also be larger, of course, the smaller the risk-aversion parameter $a$" holds only for large enough $ac$.

The free-entry condition equates expected utilities, not expected spread capture and cost. Per-share fees and two empirical liquidity measures follow from the same equilibrium: see trading fees and price impact and return reversal. Interpretation: the quoted inside spread implied by the model, a bid at $\mu - p$ and an ask at $\mu + p$ with $p = \gamma\sigma^2\,|i|\,/(n+1)$, is $2p$.

**Pitfalls.** The model solves a single Walrasian clearing rather than a trading mechanism [crypto-cartea, same chap., pp. 26–27], and the offsetting trader arrives for sure; the arrival time and its uncertainty, which drive inventory risk in continuous-time quoting models, are absent. Interpretation: reading every concession as risk compensation overstates inventory costs where order flow is informed.

The market-making equilibrium is one of several structures in the original paper. On the NYSE of 1988 an order "larger than can be absorbed by the specialist or by any previously entered 'limit orders' " then resting in the book means "the market can switch to search mode". The specialist, with the Exchange's permission, can suspend trading and search for counterparties, more likely at the upstairs block desks, which "until recently at least … functioned primarily as a search market" [grossman-miller-1988-nber, printed p. 9], though their positioning had grown as index futures and options let them hedge inventories (printed p. 10). Those are 1988 descriptions of NYSE practice, not current rules; current block handling is in block trades.

**See also.** Kyle single-auction model (contrast: risk-neutral market makers, price impact from information); Avellaneda–Stoikov quoting (generalisation: inventory risk in continuous time with random order arrival); static limit-order depth (the same immediacy supplied through posted orders).

**Further reading.** [grossman-miller-1988], the original model, including participation costs; equation numbers above follow its NBER working-paper version (`grossman-miller-1988-nber`, July 1988) and may differ in the *Journal of Finance* version.

#### Result 16.25 Grossman–Miller with per-share trading fees

**Statement.** If every trader pays a fee $\eta$ per share bought or sold, the date-2 price is unchanged, $S_2 = \mu + \epsilon_2$, but for a seller ($i > 0$) the date-1 price falls to [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", pp. 25–26]

$$S_1 = \mu - \gamma\sigma^2\,\frac{i}{n+1} - 2\Big(\frac{n}{n+1}\Big)\eta, \qquad q_1^{LT1,*} = \frac{i}{n+1} + 2\Big(\frac{n}{n+1}\Big)\frac{\eta}{\gamma\sigma^2}.$$

The initiating liquidity trader pays his own fee and, through the lower price, the fraction $n/(n+1)$ of the market makers' round-trip fees $2\eta$, and keeps more inventory after date 1 than without fees [crypto-cartea, same chap., p. 26].

**Notation.** As in the Grossman–Miller model.

**Conditions.** The model of the Grossman–Miller entry, with fees proportional to shares rather than to dollar volume, positions remaining after date 2 liquidated at date 3, and desired trades large enough that every participant prefers trading to not trading [crypto-cartea, same chap., p. 25]. Idea of the proof: fees cancel at date 2 because remaining positions are sold either then or at date 3; at date 1 they reduce the market makers' demand by the round-trip fee, $q_1^{MM} = (\mu - S_1 - 2\eta)/(\gamma\sigma^2)$, and clearing $i = nq_1^{MM} + q_1^{LT1}$ gives the price [crypto-cartea, same chap., pp. 25–26].

**Practice.** Liquidity traders whose desired trade is small relative to the fee do not trade at all [crypto-cartea, same chap., p. 25]. Participation costs and fees act differently: participation costs work through the number of market makers, while fees work mainly through the discount and the immediacy obtained, since each market maker pays $2\eta$ per trade but recovers $2n\eta/(n+1)$ through the discount [crypto-cartea, same chap., p. 26].

**Pitfalls.** Real exchange fees are often proportional to dollar volume and differ by side and by maker or taker [crypto-cartea, same chap., p. 25]; the closed form assumes the same per-share fee for everyone.

**See also.** Grossman–Miller model (the base case, $\eta = 0$).

#### Result 16.26 Grossman–Miller liquidity measures: linear price impact and negative autocovariance

**Statement.** Read as trading in a limit-order book, where LT1's market orders walk the book posted by the market makers, the Grossman–Miller date-1 price is linear in the quantity LT1 trades [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", pp. 26–27]:

$$S_1 = \mu + \lambda\,q^{LT1}, \qquad \lambda = -\frac{1}{n}\,\gamma\sigma^2, \qquad q^{LT1} = i\,\frac{n}{n+1}.$$

With public news $\epsilon_1$ before LT1 trades and efficient price $\mu_t = \mathbb{E}[S_3 \mid \epsilon_1, \dots, \epsilon_t]$, the price changes $\Delta_1 = S_1 - S_0$ and $\Delta_2 = S_2 - S_1$ satisfy [crypto-cartea, same chap., pp. 27–28]

$$\mathrm{Cov}[\Delta_1, \Delta_2] = -\lambda^2\,\mathrm{Var}[q^{LT1}] < 0.$$

**Notation.** As in the Grossman–Miller model; here $q^{LT1}$ is the quantity LT1 trades at date 1, not a holding, and $\lambda$ is the slope of price on that quantity.

**Conditions.** The model of the Grossman–Miller entry, with shocks $\epsilon_1, \epsilon_2, \epsilon_3$ i.i.d. normal with variance $\sigma^2$, and liquidity demand independent of news [crypto-cartea, same chap., pp. 27–28]. Idea of the proof: $S_1 = \mu_1 + \lambda q^{LT1}$ and $S_2 = \mu_2$, so the order-pressure term enters $\Delta_1$ and reverses in $\Delta_2$ [crypto-cartea, same chap., p. 28].

**Practice.** A more liquid market has a smaller $|\lambda|$, from more competition, less risk aversion or lower volatility [crypto-cartea, same chap., p. 27]. As $\lambda$ goes to zero, so does the autocovariance, and the price process converges to the efficient-price martingale [crypto-cartea, same chap., p. 28]. The two measures coincide here only because the model is essentially static; in dynamic models they capture different things, permanent effects of order flow on the efficient price as against the temporary effect in each trader's execution price [crypto-cartea, same chap., p. 28].

**Pitfalls.** Interpretation: an empirical regression of price changes on signed order flow, or a negative return autocovariance, mixes this inventory effect with information and bid–ask bounce; this model alone cannot attribute it. Compare the information slope of Kyle's model.

**See also.** Grossman–Miller model (source of the equilibrium); Kyle single-auction model (contrast: a price-impact slope from information).

## 16.5 Quoting with limit orders

### Model 16.27 Static limit-order depth with exponential fill probability

**Statement.** A small, risk-neutral market maker with costless inventory and infinite patience chooses depths $\delta^{\pm}$ from the midprice $S_t$ for a sell order at $S_t + \delta^+$ and a buy order at $S_t - \delta^-$; a buy (sell) market order arrives with probability $p^+$ ($p^-$) and reaches depth $\delta$ with probability $e^{-\kappa^{\pm}\delta}$, and expected profit per trade is maximised by [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 29, eq. (2.3)]

$$\max_{\delta^+,\delta^-} \left\{ p^+ e^{-\kappa^+\delta^+}\delta^+ + p^- e^{-\kappa^-\delta^-}\delta^- \right\}, \qquad \delta^{\pm,*} = \frac{1}{\kappa^{\pm}}.$$

The optimal depth equals the mean depth of the book on that side [crypto-cartea, same chap., p. 30]. This is a static simplification set up in place of the dynamic Ho–Stoll dealer model, which the authors give as the usual first reference [crypto-cartea, same chap., p. 28]. Figure 16.2 shows the depths and fill probabilities.

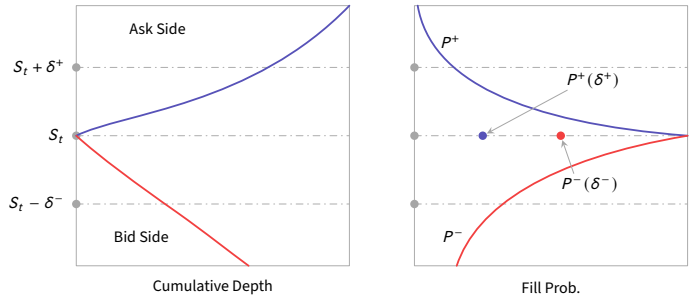



Figure 16.2: Limit-order depths and probability of execution, after [crypto-cartea, Figure 2.2, p. 30].

**Notation.** Local override for this entry: $p^{\pm}$ are probabilities of market-order arrival, not prices, and $P^{\pm}(\delta)$ is the probability that an arriving market order reaches depth $\delta$, not a bond or put price.

**Conditions.** Everyone agrees on the value $S_t$ (no information asymmetry); other market makers' orders form a fixed book that does not react; the market maker has no inventory or trading costs and liquidates at the midprice [crypto-cartea, same chap., pp. 28–29]. Idea of the proof: the problem separates by side, and the first-order condition $p^{\pm}e^{-\kappa^{\pm}\delta^{\pm}}(1-\kappa^{\pm}\delta^{\pm}) = 0$ gives the depth; the arrival probabilities scale profit but not the optimal depth (Interpretation, own derivation).

**Practice.** The model isolates the trade-off between margin per trade and probability of execution [crypto-cartea, same chap., p. 30]; quoting deeper earns more per fill and fills less often.

**Pitfalls.** The authors list what makes it unrealistic: the arrival and fill probabilities are special, constant and exogenous; the market maker's decision does not affect other traders; and the objective is static [crypto-cartea, same chap., p. 30]. Interpretation: there is no inventory penalty and no adverse selection, so the depth does not respond to position or to informed flow; Avellaneda–Stoikov adds inventory.

**See also.** Avellaneda–Stoikov quoting (generalisation: dynamic, with inventory and the same exponential intensity); Grossman–Miller model (the immediacy this quoting supplies).

**Further reading.** [ho-stoll-1981], the dynamic dealer model this simplifies.

### Model 16.28 Avellaneda–Stoikov inventory-aware quotes

**Statement.** A dealer with CARA utility, inventory $q$ and horizon $T$ quotes around a reservation (indifference) price rather than the midprice $s$ [s6-stoikov, sec. 2.2, eq. (8), p. 219]:

$$r(s,q,t) = s - q\gamma\sigma^2(T-t).$$

With symmetric exponential execution intensities $\lambda^a(\delta) = \lambda^b(\delta) = Ae^{-k\delta}$, an expansion of the value function in $q$ with a first-order approximation of the order-arrival terms gives the same reservation price and a bid–ask spread, independent of inventory, of [s6-stoikov, sec. 3.2, eqs. (29)–(30), p. 221]

$$\delta^a + \delta^b = \gamma\sigma^2(T-t) + \frac{2}{\gamma}\ln\left(1+\frac{\gamma}{k}\right).$$

**Notation.** Local override for this entry: the midprice follows $dS_u = \sigma\, dW_u$, so $\sigma$ is an absolute volatility in price units per square root of the model's time unit, not the annualised volatility of returns; $s$ is the current midprice; $\delta^a = p^a - s$ and $\delta^b = s - p^b$ are the quote distances; $\lambda^a$, $\lambda^b$ are Poisson execution intensities, not price-impact slopes [s6-stoikov, secs. 2.1 and 2.4, pp. 218–219].

**Conditions.** As published in 2008; review annually with the catalogue record. Arithmetic Brownian midprice with no drift and a zero interest rate; exponential utility of terminal wealth plus inventory marked at the midprice; one-share limit orders, updated continuously at no cost, filled at Poisson rates that decrease with distance from the midprice [s6-stoikov, secs. 2.1–2.4, pp. 218–219]. The exponential intensity follows from a constant market-order frequency, a power-law order-size distribution and a logarithmic temporary impact, with $A = \Lambda/\alpha$ and $k = \alpha K$; a power-law impact gives a power-law intensity instead [s6-stoikov, sec. 2.5, eq. (12), p. 220]. Exact quotes are found in two steps: solve the HJB equation (15) for $\theta(s,q,t)$, whose differences give the reservation bid and ask, $r^b = \theta(s,q+1,t) - \theta(s,q,t)$ and $r^a = \theta(s,q,t) - \theta(s,q-1,t)$ (eqs. (16)–(17)); then solve the implicit first-order conditions [s6-stoikov, sec. 3.1, eqs. (15)–(19), pp. 220–221]

$$s - r^b = \delta^b - \frac{1}{\gamma}\ln\left(1-\gamma\frac{\lambda^b(\delta^b)}{\partial_\delta\lambda^b(\delta^b)}\right), \qquad r^a - s = \delta^a - \frac{1}{\gamma}\ln\left(1-\gamma\frac{\lambda^a(\delta^a)}{\partial_\delta\lambda^a(\delta^a)}\right),$$

which the authors read as calibrating the indifference prices "to the current market supply $\lambda^b$ and demand $\lambda^a$" (p. 221). With exponential intensities $\lambda/\partial_\delta\lambda = -1/k$, so each distance is the gap to the reservation price plus $\gamma^{-1}\ln(1+\gamma/k)$, which summed gives the spread term above; the closed form above is an approximation because the reservation prices come from an expansion of $\theta$ in $q$ with linearised order-arrival terms, under which, with symmetric exponential intensities, they coincide with their frozen-inventory values [s6-stoikov, sec. 3.2, p. 221]. A long position ($q > 0$) puts the reservation price below the midprice, so both quotes shift down to sell [s6-stoikov, sec. 2.2, p. 219].

**Practice.** In 1000 simulations with $s = 100$, $T = 1$, $\sigma = 2$, $\gamma = 0.1$, $k = 1.5$, $A = 140$ and $dt = 0.005$, the inventory strategy earned a mean profit of 65.0 with standard deviation 6.6, against 68.4 and 12.7 for a strategy quoting the same average spread of 1.49 symmetrically around the midprice; final inventory had standard deviation 2.9 against 8.4 [s6-stoikov, sec. 3.3, Table 1, p. 222]. The symmetric strategy earns slightly more because it trades more, but its P&L is much more dispersed [s6-stoikov, sec. 3.3, p. 222].

**Pitfalls.** Interpretation: the closed form is a control template, not a production model; real books add queue position, tick size, discrete inventory limits, latency, rebates, hedging instruments and fill probabilities that change with regime. The approximate spread does not depend on inventory only because the intensities are exponential [s6-stoikov, sec. 3.2, p. 221]. Interpretation: the formula has no inventory limit; a hard limit needs the bounded-inventory solution of Guéant, Lehalle and Fernandez-Tapia (`wide-gueant-lehalle-tapia`).

*Infinite horizon.* The reservation price depends on $T - t$; for a stationary version the paper takes, for an agent who does not trade, $\bar{v}(x,s,q) = \mathbb{E}\left[\int_0^\infty -e^{-\omega t}e^{-\gamma(x+qS_t)}\,dt\right]$ and prints stationary reservation prices $\bar{r}^{a,b}(s,q) = s + \gamma^{-1}\ln\left(1 + (\pm 1 - 2q)\gamma^2\sigma^2/(2\omega - \gamma^2q^2\sigma^2)\right)$, "where $\omega > (1/2)\gamma^2\sigma^2q^2$"; $\omega$ "may therefore be interpreted as an upper bound on the inventory position", and the "natural choice of $\omega = (1/2)\gamma^2\sigma^2(q_{\max}+1)^2$ would ensure that the prices defined above are bounded" [s6-stoikov, sec. 2.3, p. 219].

Derivation: $\bar{v} = -e^{-\gamma(x+qs)}/D(q)$ with $D(q) = \omega - \frac{1}{2}\gamma^2\sigma^2q^2$, finite only when $D(q) > 0$, which is the stated condition; the indifference definitions of eqs. (4)–(5) then give $\bar{r}^b = s + \gamma^{-1}\ln(D(q+1)/D(q))$, equal to the printed bid, and $\bar{r}^a = s + \gamma^{-1}\ln(D(q)/D(q-1))$, which differs from the printed ask beyond first order in $\gamma^2\sigma^2/\omega$ (at $q = 0$ the derived quotes are symmetric about $s$, as the problem is, and the printed ones are not), so the printed ask looks like a misprint, unconfirmed against the journal's errata. A later source does not settle it.

Balakaeva and Veretennikov's preprint, which says it corrects "seemingly serious gaps" in the paper, prints the same ask as a "trivial computation" [balakaeva-veretennikov-2025, arXiv v1, 13 October 2025, Lemma 3, p. 4]. Its own value function [balakaeva-veretennikov-2025, Lemma 2, p. 3] and indifference definitions [balakaeva-veretennikov-2025, Definition 3, p. 3] yield the ask derived here, with $D(q-1)$ in the denominator, so it repeats the candidate misprint rather than confirming the printed form. The printed formula is quoted unchanged above. Under the natural choice $D(q) > 0$ for $|q| \le q_{\max}$ and $D(q_{\max}+1) = 0$, so the bid at $q_{\max}$ (and the ask at $-q_{\max}$) is infinite: "bounded" holds strictly inside the limit, and the limit acts by making further accumulation unacceptable. Section 2.3 concerns the frozen-inventory agent; the paper does not solve the infinite-horizon quoting problem.

**See also.** Static limit-order depth (special case without inventory risk); Grossman–Miller model (same premium, three-date equilibrium).

## 16.6 Informed trading and informative prices

#### Definition 16.29 Fundamental value, market value and noise: what informative prices are informative about

**Statement.** The *market value* of an instrument is the price at which traders can buy or sell it. Its *fundamental value* (intrinsic value) is the expected present value of all present and future benefits and costs of holding it: the value everyone would agree on if all knew everything known about the instrument, analysed it properly and perceived its benefits and costs alike [alpha-harris, chap. "Informed Traders and Market Efficiency", sec. 10.1, pp. 222–223]. The difference between the two is *noise*; prices are informative when they are near fundamental values and completely informative when they equal them [alpha-harris, same sec., p. 223]. Writing $V_t$ for fundamental value and $P_t$ for the price, noise is $\epsilon_t = P_t - V_t$.

Fundamental values use only information available now, so they are the best estimates of perfect-foresight values, not those values; and their changes are unpredictable, since a predictable change would mean they did not yet reflect the information behind the prediction [alpha-harris, same sec., p. 223]. Relative to a price, information is of three kinds:

- *public*: available to any trader, such as public news and past prices and volumes [alpha-harris, same chap., pp. 223 and 240];
- *private*: held only by some traders, whether learnt first or produced by their own research [alpha-harris, same chap., pp. 223 and 228];
- *stale*: already in the price, meaning it cannot be used to forecast future price changes; traders who act on it are *pseudo-informed*, and in fact uninformed [alpha-harris, same chap., sec. 10.5.2.1, p. 229].

**Notation.** $V_t$ fundamental value, $P_t$ market price, $\epsilon_t = P_t - V_t$ noise; the sign is a convention, since Harris defines noise only as the difference.

**Conditions.** Fundamental values are not observed, so whether a price is informative or noisy cannot easily be determined [alpha-harris, same chap., p. 223]. Prices become informative in two ways: when values are common knowledge no one trades at other prices; otherwise informed traders' trading moves prices towards their estimates, which is the more important route because values are rarely common knowledge [alpha-harris, same chap., sec. 10.6.1, pp. 236–237]. Information gets into the price when all traders understand its significance or when informed traders push the price towards their estimates [alpha-harris, same chap., p. 229].

**Practice.** Whether information is stale can rarely be checked directly. Harris's guide: information that is unique and that others could not have anticipated is probably not stale; information acted on slowly, widely known, cheap to obtain, or reasonably anticipated probably is. His example: Occidental Petroleum rose 10 percent the day after its long-ill chairman died and fell back the next day, because his death was expected and its date was not material [alpha-harris, same chap., p. 229 and box, p. 230]. Fischer Black would call stock prices informative if they lay between half and twice their fundamental values [alpha-harris, same chap., box, p. 223].

**Pitfalls.** Two conditioning sets are in use. Harris's fundamental value conditions on everything known to anyone, private information included. Hasbrouck calls the expectation of the payoff conditional on *all public information* the fundamental value or efficient price [s6-hasbrouck, sec. 3.2, p. 26]: the $m_t$ of the random-walk decomposition. Harris's noise is therefore not Hasbrouck's pricing error $s_t = p_t - m_t$. A price equal to the public-information expectation has a zero pricing error but nonzero noise whenever private information has not yet reached it (Interpretation). Fundamental value is also not the perfect-foresight value implied by all payoff-relevant information, which Harris distinguishes from it. Private information need not be restricted: research can produce it lawfully from public data [alpha-harris, chap. "Insider Trading", p. 585].

**See also.** fundamental and transitory volatility (the statistical version, conditioned on public information); informed trading and aggregation (how prices become informative); market efficiency; inside information; outside spread (value traders' estimates in use).

#### Result 16.30 Informed trading makes prices informative: the price as an aggregate of value estimates

**Statement.** Informed traders buy when the price is below their estimate of fundamental value and sell when it is above. Their buying pushes prices up and their selling pushes them down, so their trading moves prices towards their estimates [alpha-harris, chap. "Informed Traders and Market Efficiency", secs. 10.2–10.3, pp. 223–224]. Harris's algebraic illustration [alpha-harris, same chap., box, p. 225]: $N$ traders hold unbiased forecasts $f_i = V + e_i$ of the value $V$, each wants the position $D_i = a(f_i - P)$, and the security is in zero net supply. Market clearing, $\sum_i a(f_i - P) = 0$, gives

$$P = \frac{1}{N}\sum_{i=1}^{N} f_i = V + e_M, \qquad e_M = \frac{1}{N}\sum_{i=1}^{N} e_i.$$

With independent errors, $e_M \to 0$ as $N$ grows, by the law of large numbers: prices are most informative when many informed traders research independently. Interpretation: if the errors are also independent of $V$ with common variance $\sigma^2$, then $\mathbb{E}[P \mid V] = V$ and $\mathrm{Var}(P - V) = \sigma^2/N$.

**Conditions.** When forecasts differ in accuracy, the best combination is a weighted average that gives more weight to the more accurate. Well-informed traders tend to be wealthier and take larger positions, so prices approximate such an optimally weighted average: the market is a statistical calculator [alpha-harris, same chap., p. 224 and box, p. 226]. Interpretation: with trader-specific slopes $a_i$ the same clearing condition gives $P = \sum_i a_i f_i / \sum_i a_i$, still unbiased when each $f_i$ is; with independent errors of variances $\sigma_i^2$, $\mathrm{Var}(P - V) = \sum_i a_i^2 \sigma_i^2 / (\sum_i a_i)^2$, minimised by $a_i \propto 1/\sigma_i^2$ at $\left(\sum_i \sigma_i^{-2}\right)^{-1}$, below every $\sigma_i^2$. Many mistaken traders, or one large one, make prices less informative, until their losses drive them out [alpha-harris, same chap., p. 224].

Informed traders do not trade to make prices informative: price impact is their cost, and they want prices to adjust only after they have built their positions [alpha-harris, same chap., p. 224]. They profit only from uninformed traders; competition among them pushes prices closer to values and lowers their profits until the least successful just cover their costs [alpha-harris, same chap., secs. 10.6.3 and 10.6.5, pp. 237–239].

**Practice.** Trade aggressively when the information will soon be common knowledge or many others will act on it, since the first to trade profit most and a flood of orders makes liquidity suppliers wary; trade slowly (stealth trading) when the advantage will last, so that others cannot infer that the trader is well informed [alpha-harris, same chap., sec. 10.4, pp. 225–226]. Information already incorporated in the price is stale, and trading on it does not pay [alpha-harris, same chap., p. 229].

The most profitable traders have estimates that are both precise and *orthogonal*, uncorrelated with other traders'; they face little competition for liquidity when they trade. Precision and orthogonality trade off: a precise but crowded estimate and an imprecise but orthogonal one can be equally profitable [alpha-harris, same chap., sec. 10.6.2.1, p. 237].

**Pitfalls.** Aggregation removes independent errors, not shared ones. Interpretation: if the errors have common variance $\sigma^2$ and pairwise correlation $\rho$ (which requires $\rho \geq -1/(N-1)$), then $\mathrm{Var}(e_M) = \frac{1}{N^2}[N\sigma^2 + N(N-1)\rho\sigma^2] = \sigma^2[\rho + (1-\rho)/N] \to \rho\sigma^2$ for $\rho \geq 0$, so a shared data source, model or stale signal leaves an error that no number of traders removes; Harris notes that traders who use the same information and methods have highly correlated estimates [alpha-harris, same chap., p. 237]. Harris also says the market's forecast error is smaller than the average individual error whenever the individual errors are not identical, and that the price estimates value better than any individual trader can [alpha-harris, same chap., box, p. 225]. Interpretation, by error measure: the signed errors average exactly, $e_M = \frac{1}{N}\sum_i e_i$, so "smaller" needs a loss.

For one realisation, squared error gives $e_M^2 \leq \frac{1}{N}\sum_i e_i^2$, since the gap is $\frac{1}{N}\sum_i (e_i - e_M)^2$, with equality only when all $e_i$ are equal, as Harris's condition says. Absolute error gives $|e_M| \leq \frac{1}{N}\sum_i |e_i|$ by the triangle inequality, with equality whenever the errors share a sign, so $e = (1, 2)$ is not identical yet gives 1.5 on both sides. In expectation, with independent, zero-mean, nondegenerate errors, both inequalities are strict, since errors of opposite sign then occur with positive probability. With independent, unbiased errors, equal accuracy is sufficient for the equal-weighted average to beat every individual in expected squared error, but not necessary: with variances 1 and 2 the average has variance 0.75, below both. It can also lose to the best individual: with variances 0.01 and 1 the average has variance 0.2525, worse than the better trader's, while inverse-variance weights give 0.0099.

The assumption is that the price is the average forecast; in the box this follows from linear demands with a common slope and zero net supply, an assumption Harris says simplifies the arithmetic without changing the qualitative results. Informed traders act on estimates of value, not on value itself, and lose when their estimates are poor [alpha-harris, same chap., p. 223].

**See also.** fundamental value; market efficiency; Kyle single-auction model and Glosten–Milgrom model (equilibrium models of informed trading); price formation process; informed-trader types; order anticipation (front runners accelerate informed traders' impact).

#### Definition 16.31 Informed-trader types: value traders, news traders, information-oriented technical traders and arbitrageurs

**Statement.** Harris classifies informed traders by what they estimate [alpha-harris, chap. "Informed Traders and Market Efficiency", sec. 10.5, pp. 226–235, Table 10-3, p. 235]:

- *value traders* estimate total fundamental value from all available information; slow, because careful review guards against biased estimates;
- *news traders* estimate changes in value from new information, assuming the price already reflects everything else; fast on public news, slower on information they produce by their own research;
- *information-oriented technical traders* identify price patterns left by informed traders' systematic mistakes, such as under- or overreaction; fast;
- *arbitrageurs* estimate relative values of instruments that share valuation factors; fast.

**Conditions.** Value and news traders profit in different circumstances. Price departs from value either because value changes and price lags, where news traders, first to learn the news, profit; or because uninformed traders push price away from value, where value traders profit [alpha-harris, chap. "Value Traders", sec. 16.1, pp. 338–339]. Their mistakes put them on opposite sides. News traders who misjudge the size of a change leave underreaction for value traders to trade on, or overreaction for them to trade against (Table 10-1, p. 230); pseudo-informed traders acting on stale news lose to value traders (box, p. 229). Value traders who mistake a price move driven by news for uninformed pressure lose to news traders [alpha-harris, chap. "Informed Traders and Market Efficiency", p. 239].

**Practice.** Organisation follows speed: large value traders have pyramid-shaped organisations with layers of review, news traders flat organisations that decide quickly [alpha-harris, same chap., pp. 227–229]. Technical traders who trade on the predictable impact of *uninformed* traders are not informed traders: they act as dealers if they supply liquidity, or as order anticipators if they front-run [alpha-harris, same chap., p. 231]. Information-oriented technical strategies decay as informed traders learn from their mistakes [alpha-harris, same chap., p. 231].

*Liquidity roles.* Harris's Table 8-4 lists value traders, news traders, information-oriented technical traders and arbitrageurs as the informed traders [alpha-harris, chap. "Why People Trade", sec. 8.2.1, pp. 194–196]. As liquidity suppliers, value traders trade when uninformed demand has moved prices away from value and are "the ultimate suppliers of liquidity", supplying depth, while news traders trade on changes in value before prices adjust [alpha-harris, chap. "Liquidity", sec. 19.3.4, pp. 402–403]; arbitrageurs move liquidity between markets (arbitrage). The distinction, whether an order corrects stale prices, absorbs non-value flow or links related prices, maps onto news traders, value traders and arbitrageurs.

**Pitfalls.** Technical traders do not in general exploit predictable errors by other traders. In Harris that holds for errors by informed traders; errors by uninformed traders belong to sentiment-oriented technical traders, who are order anticipators (order anticipation). Dealer inventory pressure is not a separate source of value-trader profit: in Harris it is a channel of uninformed demand (see outside spread).

**See also.** informed trading and aggregation; outside spread; arbitrage; order anticipation; fundamental value (stale information).

#### Definition 16.32 Winner's curse: what winning a common-value trade reveals, and how bids adjust

**Statement.** Buyers who compete for an item of common but unknown value suffer the *winner's curse* when they pay more than it is worth. The highest bidder tends to be the one who most overestimated the value, so bids near one's own estimate lose on average when they win. Buyers avoid it by lowering their bids to reflect what winning would tell them about their estimate; sellers whose offers are taken symmetrically raise their offers [alpha-harris, chap. "Value Traders", sec. 16.2.1.2, pp. 341–342]. Interpretation: let $\widehat{V}$ be the buyer's estimate, $c$ an all-in cost, and $\omega(b) = \mathbb{E}[\widehat{V} - V \mid \widehat{V},\ \text{win at bid } b]$ the expected overestimate given one's own estimate and winning at bid $b$. The expected profit of winning is $\widehat{V} - \omega(b) - b - c$, so the highest bid that breaks even solves $b = \widehat{V} - \omega(b) - c$.

**Conditions.** The required adjustment is larger [alpha-harris, same sec., p. 342]:

- the more bidders there are, because outbidding many implies a larger overestimate;
- the harder the value is to estimate, because extreme estimates are then further from the truth.

Against market power: with only one or very few competitors a bidder may underbid in the hope of winning cheaply, so bids are more aggressive against several competitors than against one or two; this matters only with very few [alpha-harris, same sec., p. 342]. In a two-sided single-price auction the curse matters little, because traders do not trade at their own bids and offers and the clearing price is near the median estimate. In continuous markets they do, and aggregation is sequential, so traders who misestimate lose when they push prices [alpha-harris, same sec., pp. 342–343].

**Practice.** Oil companies overpaid for drilling leases until they allowed for the curse; their geologists were right on average over all tracts but optimistic about the tracts they won [alpha-harris, same chap., box, p. 342]. Value traders face the curse whenever they trade on a large apparent mispricing and widen their outside spreads in response; Harris attributes part of late-1990s Internet-stock volatility to prices bouncing within wide outside spreads [alpha-harris, same chap., p. 343 and box]. Do not compete against foolish bidders: one cannot profit bidding against strategies that ensure losses [alpha-harris, same chap., p. 342].

**Pitfalls.** The break-even bid $b = \widehat{V} - \omega(b) - c$ conditions on one's own estimate and on winning, which depends on the bid itself and, in equilibrium, on rivals' strategies, so it is a fixed-point condition, not a closed form. It also gives the highest acceptable bid, not the profit-maximising one, which shades further when competitors are few (Interpretation). Shared data make estimates move together, which reduces how much winning reveals, while independent noisy estimates increase selection risk (Interpretation; not in the source).

**See also.** outside spread (the curse widens value traders' spreads); informed trading and aggregation (correlated and independent estimates); Glosten–Milgrom model (adverse selection, the analogous inference a dealer makes from the side of a trade).

#### Definition 16.33 Market efficiency: weak, semistrong and strong forms, and the definition with information costs

**Statement.** In an *efficient market*, prices reflect all information that traders can acquire and profitably trade upon. The definition builds in the costs of acquiring information and of acting on it, and the impact of informed trading on prices [alpha-harris, chap. "Informed Traders and Market Efficiency", sec. 10.6.6, p. 240]. Harris sets it beside the three traditional forms [alpha-harris, same sec., p. 240]:

- *weak form*: prices reflect all information in past prices, so no one can predict price changes from past prices alone;
- *semistrong form*: prices reflect all public information, which includes all public news and past prices and volumes in all securities and contracts;
- *strong form*: prices reflect all public and private information as soon as it is known; since informed traders could never profit, only markets in instruments whose values are common knowledge are strong-form efficient.

**Conditions.** Timeless (concepts); the empirical summary as of 2002. Prices never fully reflect all the information traders could collect: information that is too expensive or too inconsequential is not acquired [alpha-harris, same sec., p. 240]. Harris's market paradox: if prices were fully informative, informed trading would not pay, so no one would make them informative. Prices and values change constantly, so gaps open and informed trading closes them: prices are informative but not always [alpha-harris, same chap., sec. 10.6.4, pp. 238–239].

Lehalle and Laruelle make the same point with Grossman and Stiglitz (1980): market dynamics must contain enough inefficiency to reward those who improve the information in prices [alpha-laruelle, Introduction, p. 4]. How informative prices are depends on the costs of information and of trading, and on there being uninformed traders to lose to informed ones [alpha-harris, same chap., secs. 10.6.3 and 10.7, pp. 238 and 243–244].

**Practice.** Informative prices are not free. What uninformed traders lose to informed traders pays for much of the information in prices, so policies that frustrate informed traders can make prices less informative and markets more liquid; restrictions on insider trading are an example [alpha-harris, same chap., sec. 10.6.7, p. 241].

The trade-off is not universal: publishing material fundamental information raises both liquidity and informativeness [alpha-harris, chap. "Informed Traders and Market Efficiency", sec. 10.6.8, pp. 241–242]. Trading halts before material releases protect uninformed traders, especially those who offer liquidity, at the cost of a few minutes' delay in price formation, a trade Harris judges worth making [alpha-harris, same chap., box, p. 241]. Most published studies find markets weak-form efficient, and semistrong-form efficient for public information that is easy to obtain and interpret [alpha-harris, same chap., p. 240].

*Liquidity and informative prices*. Liquid markets make informed trading cheaper, so informed traders can afford more research and trade on smaller pieces of information, and "Prices therefore should be more informative in liquid markets than in illiquid markets"; informed traders profit only if uninformed traders are willing to lose to them [alpha-harris, chap. "Good Markets", sec. 9.5.4, pp. 215–216]. Visibility, update speed and uninformed flow therefore decide how much information enters prices, through this mechanism and that of informed trading and aggregation.

**Pitfalls.** Comparing market structures needs the change in benefits, not their totals, and the benefits of informative prices and of liquidity are both hard to measure [alpha-harris, same chap., sec. 10.6.7, p. 241].

**See also.** fundamental value; informed trading and aggregation; inside information (the main example of the trade-off); spread determinants (asymmetric information widens spreads); order anticipation (front running of informed traders lowers informativeness in the long run).

#### Definition 16.34 Inside information: material nonpublic information and who may not trade on it (United States, as summarised in 2002)

**Statement.** For economic analysis, Harris defines *inside information* as material information about the value of a security that is not available to public traders, and *material* information as information that would cause prices to change if it were widely known; *insider trading* is trading on it [alpha-harris, chap. "Insider Trading", sec. 29.1, p. 584]. Inside information keeps its status however many people pass it on, and loses it only when it is publicly available, through a broadly distributed press release or a public filing [alpha-harris, same sec., p. 585].

The legal definitions come from statutes, regulations and case law, are complex, and call for an attorney [alpha-harris, same sec., p. 584]. The SEC's summary, as quoted by Harris from its web page of January 2002, covers trading in breach of a fiduciary duty or other relationship of trust and confidence while in possession of material nonpublic information, and tipping, trading by those tipped, and misappropriation [alpha-harris, same sec., box, p. 585]. Those who may not trade on such information include:

- *corporate insiders*: senior managers, directors, large shareholders, key employees, and anyone involved in corporate activities that will materially affect the firm's value [alpha-harris, sec. 29.1.1, p. 587];
- employees of law, banking, brokerage and printing firms given the information to serve the corporation, and government employees who learn it at work [alpha-harris, sec. 29.1, box, p. 585];
- *tippees*: friends, family and business associates of insiders, and the friends of their friends [alpha-harris, same sec., pp. 584–585];
- *misappropriators*, who take confidential information from an employer or other source. Chiarella's conviction was reversed in 1980 because he owed no fiduciary duty to the firms whose stock he traded; Harris says that under the later law he would be convicted for misappropriation [alpha-harris, same sec., box, p. 586]. Winans was convicted under the misappropriation doctrine [alpha-harris, same sec., box, p. 587].

Analysts whose private information comes only from analysing public information are not trading on inside information [alpha-harris, same sec., p. 585]. Harris's box "Inside information, skilled speculation, or good advice?" shows why the line is hard to draw from the trade alone: the same well-timed purchase before a takeover may rest on a tip, on independent analysis of public information, or on analysis that the trader then sold to the target's management; and a trader who had a tip will claim the second [alpha-harris, sec. 29.2, box, p. 590].

**Conditions.** Jurisdiction and date: United States law as Harris summarises it in 2002. Most countries prohibit insider trading, but only a few, mainly the United States, Canada and Great Britain, regularly and seriously enforce their laws [alpha-harris, chap. "Insider Trading", p. 584]. Regulation FD (2000) requires an issuer that discloses material nonpublic information to anyone to disclose it publicly [alpha-harris, same chap., box, p. 586].

**Practice.** Corporations restrict their employees' trading: many require corporate approval before a trade, may refuse it before a release, or allow trading only in a window after quarterly earnings [alpha-harris, same chap., sec. 29.1.1, p. 588]. Exchange surveillance looks for large price changes on large volume before significant news [alpha-harris, same chap., sec. 29.2, p. 588]. Insider trading is hard to prove: a sophisticated speculator can claim independent analysis of public data, and trading on information that is not revealed in a discrete event is essentially undetectable from prices and volumes [alpha-harris, same chap., pp. 589–590]. Like all informed trading, insider trading hurts liquidity suppliers and raises spreads for uninformed traders [alpha-harris, same chap., sec. 29.3.1.2, pp. 591–592].

*Controls on insiders' own trading*. Harris lists, for the United States in 2002, reporting of insider trades to the SEC, prearranged trading programmes that let insiders trade while knowing material information, return of short-swing profits on positions opened and closed within six months, restrictions on short positions in the employer's stock, corporate pre-approval of trades, and trading windows after quarterly earnings (sec. 29.1.1, pp. 587–588).

Current U.S. rules, as of the eCFR text for 14 October 2026: Rule 10b5-1 gives an affirmative defence for trades under a plan adopted before the trader was aware of the information; since its 2022 amendments (effective 27 February 2023) directors and officers must wait out a cooling-off period of the later of 90 days or two business days after the issuer's results for the quarter in which the plan was adopted are disclosed, up to 120 days, and other persons except the issuer 30 days [cfr-240-10b5-1, paras. (b) and (c)(1)(ii)(B)] [sec-34-96492, summary and dates, p. 80362]; the same release added disclosure of plan adoptions and terminations and a Form 4 flag for plan trades (summary, p. 80362).

Form 4 must be filed "before the end of the second business day following the day on which the subject transaction has been executed" [cfr-240-16a-3, para. (g)(1)]. Regulation FD requires public disclosure "simultaneously" for intentional and "promptly" for non-intentional selective disclosures [cfr-243-100, para. (a)].

**Pitfalls.** Harris's categories are legal categories summarised for economists, not a legal test; check the law in force. The reporting deadlines, prearranged-plan rules and penalties he gives [alpha-harris, same chap., pp. 587–591] are as of 2002; the reporting deadline and the plan rules here reflect the October 2026 rules above, while the penalties remain those of the 2002 account. The terms "temporary insider" and a "relationship, benefit or confidence" test for tipping do not appear in the source and need separate legal sources, as does the status of auditors, consultants and exchange staff, whom the SEC summary does not list. Whether a predictive signal may be used depends on data provenance, confidentiality duties, firm restricted lists, compliance approval and execution capacity.

The source supports confidentiality duties (misappropriation), employer approval of employees' own trades (p. 588), and the dependence of informed trading profits on liquidity [alpha-harris, chap. "Informed Traders and Market Efficiency", box, p. 236]. Firm restricted lists, compliance approval of signals, the provenance of purchased data, and the rights status of alternative data and channel checks are not discussed there. Two of Harris's statements are superseded: the ten-day reporting deadlines (p. 587) have been replaced by the two-business-day Form 4 deadline, and "insiders do not have to disclose these plans to the public" (p. 588) no longer holds after the 2022 amendments. Disclosure controls, ownership reports, prearranged plans, blackout windows, short-swing rules, surveillance and audit trails summarise the controls; blackout windows are Harris's trading windows seen from the other side.

**See also.** fundamental value (private and public information); market efficiency (insider-trading restrictions in the information–liquidity trade-off); spread determinants (enforced insider-trading rules as a secondary determinant); insider-trading profitability, detection and deterrence.

#### Result 16.35 Insider-trading profitability, detection and deterrence, and the policy debate (United States, as described in 2002)

**Statement.** Harris's account of when trading on inside information pays and when it is caught [alpha-harris, secs. 29.2–29.3, pp. 588–595]:

- *competition among insiders*: one insider with information that will not soon be released can trade slowly and extract the most; many insiders who learn it at once must race, which quickly reveals the information and lowers their total profit [alpha-harris, sec. 29.3.1.2, p. 592];
- *discreteness of release*: trading on information revealed in a discrete event can be traced back from it, whereas trading on information that firms do not reveal for months or years is essentially undetectable from prices and volumes [alpha-harris, sec. 29.2, p. 590]; restrictions are therefore effective mainly against information that will soon be public [alpha-harris, sec. 29.3.2.1, p. 595];
- *deterrence*: a rational criminal trades when the expected benefit exceeds the expected cost, the probability of being caught times the penalty, so a low probability of detection requires a large penalty [alpha-harris, sec. 29.2, p. 590].

Interpretation: with expected gross gain $G$, execution and leakage costs $K$, a risk charge $R$, detection probability $q$ and penalty $F$, the trade is deterred when $G \leq K + R + qF$. If the penalty were only disgorgement of the gain, $F = G$, the condition becomes $(1 - q)G \leq K + R$, which a small $q$ rarely satisfies; hence fines on top of disgorgement.

**Conditions.** United States as Harris describes it in 2002. Penalties then included disgorgement, fines and, rarely, prison; they were heavier for traders closer to the source. He gives maximum criminal penalties of ten years and a million dollars, a maximum civil penalty of three times the profit gained or loss avoided, and an SEC bounty of up to 10 percent of civil penalties collected [alpha-harris, sec. 29.2, pp. 590–591 and box]. These figures are dated and unverified against current law. $R$ is a cost only for a risk-averse trader; a risk-neutral trader would not subtract it (Interpretation).

**Practice.** Exchange surveillance identifies most cases. It looks for large price changes on large volume and asks the listed firm whether news explains them; trading before releases of significant unexpected news often indicates insiders or confederates [alpha-harris, sec. 29.2, p. 588]. Investigators then match a list of those who knew with a list of those who profited, in stocks, options or bonds, through links such as family, neighbours or classmates [alpha-harris, same sec., pp. 588–589 and box]. In the Colt Industries case, deep out-of-the-money options bought by friends and relatives of a law-firm employee were traced through the buyers' home addresses [alpha-harris, same sec., box, p. 589].

*The policy debate*. Harris's arguments for restricting insider trading: fairness and investor confidence (rebutted by the claim that firms can restrict it by contract, and answered by the difficulty of private enforcement), liquidity (insiders, like all informed traders, widen spreads and hurt limit-order traders), and corporate control (insiders who may trade share less information, may choose projects for their trading advantage, and may front-run their firms' own trades); against: informative prices (answered by the fact that enforcement works mainly against information soon to be public, so the efficiency gain is small), costs of enforcement and selective prosecution, and Manne's entrepreneurial incentives (answered by the need for positions to be taken before outcomes are known, and the general ban on insiders' short positions) [alpha-harris, sec. 29.3, pp. 591–597].

His summary: the laws "narrow bid/ask spreads and make prices less informative in the short run" (pp. 597–598). Saying that a restriction improves welfare if and only if its liquidity and control gains exceed enforcement and delay costs, $L + C > E + D$, is an accounting identity that lists these terms; its content lies in their sizes, which Harris does not quantify. Enforcement detects abnormal trading before an event, lists who knew, matches traders to access, tests alternative explanations and then sanctions, as in the surveillance and investigation steps above.

**Pitfalls.** These footprints are leads, not proof. Relationships between the two lists do not prove insider trading, and a sophisticated speculator will claim independent analysis of public data [alpha-harris, same sec., pp. 589–590]. Widened spreads, quote fading and dark-pool activity before news are not among Harris's signals, although wider spreads in the presence of insiders follow from adverse selection [alpha-harris, sec. 29.3.1.2, p. 591]. Market liquidity as a factor in insider profits is Harris's point for informed trading generally (chap. "Informed Traders and Market Efficiency", box, p. 236).

**See also.** inside information (definition and who is covered); informed trading and aggregation (aggressive against stealth trading); market efficiency (restrictions in the information–liquidity trade-off); bluffing and manipulation (the same proof problem).

**Definition 16.36 Price formation process: immediate impact and its decay**

**Statement.** The *price formation process* (PFP) is how trades and information become prices. Lehalle and Laruelle report a consensus that it consists of the impact of each trade combined with the variation of the asset's "fair price", after Bouchaud et al. (2002) [alpha-laruelle, Introduction, p. 27]. They split it into two phases [alpha-laruelle, same, Figure 8, pp. 27–29]:

1. *immediate impact*: trades move the price through the balance of buyers and sellers, and of liquidity-providing and liquidity-consuming orders, together with moves of the spread;
2. *decay*, or the *resilience* of the book: liquidity returns, or does not, according to the distance between the best quotes and investors' views of the fair price.

Pressure inconsistent with participants' consensus generates only temporary impact; pressure consistent with it is not pushed back, and the impact is permanent [alpha-laruelle, same, pp. 3 and 28]. The PFP is mainly driven by information [alpha-laruelle, same, p. 7].

**Conditions.** Timeless (concept); the regulatory context is European, 2007–2018. The more investors with views of the fair price take part, the faster impact decays. Market makers with inventory constraints contribute to the decay without any view of the fair price, but only if their inventory exceeds the turnover of a price jump [alpha-laruelle, Introduction, p. 28]. Post-trade transparency speeds the decay by warning traders that the last price is out of line with their views [alpha-laruelle, same, p. 28]. A design tension runs through it: the more information traders have, the better the PFP should be, but each trader fears information leakage, front-running and excessive impact [alpha-laruelle, same, pp. 19–20].

**Practice.** Market makers buy from early sellers and sell to later buyers, so that most temporary oscillations shrink to a spread related to the asset's risk [alpha-laruelle, Introduction, pp. 3–4]. Volatility is the random, unexpected component of the PFP, and the more efficient the microstructure, the lower intraday volatility [alpha-laruelle, chap. 2, sec. 2.5.1, pp. 171–172]. In a fragmented market, a consolidated tape organises how information is shared [alpha-laruelle, Introduction, p. 7]. The authors see high-frequency traders' growing role as a possible threat to the decay, since they need not hold a view of the fair price and have small inventory limits [alpha-laruelle, same, p. 28]; that question belongs to `hft-market-design`.

**Pitfalls.** The impact of a single large order depends on what others are doing. If many others buy at the same time, the impact of a large buy is permanent; if most sell, it can be almost invisible. Impact is measured only by averaging over many market configurations [alpha-laruelle, Introduction, p. 3]. "Clean" and "noisy" PFP, these terms, are not defined by the authors, although they speak of a "less noisy PFP" [alpha-laruelle, chap. 2, sec. 2.5.1, p. 177]. Interpretation: the noise is the transitory component of price changes.

**See also.** informed trading and aggregation (the information side); component identification (permanent and transitory price effects of trades); Kyle single-auction model (permanent impact from information); Grossman–Miller model (temporary concession for immediacy); returns by time scale; price impact (reaction and prediction impact, and the two readings of impact).

**Further reading.** Bouchaud et al. (2002), cited for the decomposition.

## 16.7 Arbitrageurs, order anticipators and bluffers

**Definition 16.37 Arbitrage: hedge portfolio, basis, fair value, arbitrage spread and bounds; pure and speculative arbitrage**

**Statement.** Arbitrageurs trade instruments with correlated prices. An *arbitrage opportunity* arises when those prices diverge from their normal relation: the arbitrageur buys the relatively cheap instruments, sells the relatively expensive ones, and profits if the prices converge [alpha-harris, chap. "Arbitrageurs", sec. 17.1, p. 348]. The positions form a *hedge portfolio*, whose legs are scaled by hedge ratios against a reference instrument, the *arbitrage numerator*, chosen to minimise the portfolio's risk. Its *carrying costs* include financing, dividends paid on short positions, storage and spoilage, offset by dividends, lending fees and interest received [alpha-harris, same sec., pp. 348–349]. With $B_t$ the *basis*, the difference in prices between the instruments in the hedge portfolio, and $B_t^\star$ its *fair value*, the basis if all were correctly priced relative to each other, which depends on carrying costs and must be estimated, the *arbitrage spread* is

$$A_t = B_t - B_t^\star$$

[alpha-harris, same sec., p. 349]. Arbitrageurs trade only when the basis lies outside the *arbitrage bounds*, the values on either side of fair value at which they are just willing to trade [alpha-harris, same sec., p. 349]. Interpretation: if one unit of basis converges fully to fair value, the trade earns $|A_t|$ gross; with all-in round-trip cost $K_t$ and a charge $R_t$ for residual basis risk it is attractive only when $|A_t| > K_t + R_t$, so the bounds are $B_t^\star \pm (K_t + R_t)$.

**Conditions.** Timeless (concepts); examples to 2000. Harris distinguishes two types by the behaviour of the hedge portfolio's value [alpha-harris, same chap., sec. 17.3, pp. 350–363]:

- *pure arbitrage*: the value is strictly mean reverting, because some mechanism forces convergence. The mechanisms are *shipping* (the same instrument in two places, bounded by shipping and financing costs), *delivery* (contract and cash prices converging at maturity, the basis set by carrying costs before it) and *conversion* (the same risk in different forms, bounded by the costs of converting and carrying). Cyclic arbitrages such as currency cross-rates are a kind of conversion arbitrage.
  *Cyclic currency arbitrage with quotes.* Write $S_{X/Y}$ for units of $Y$ per unit of $X$. The exact equality $S_{A/B} S_{B/C} = S_{A/C}$ is the no-arbitrage condition only in a frictionless market with one price per pair; with quotes it need not hold for midpoints. For example, with EUR/USD 1.0999/1.1001, USD/JPY 109.99/110.01 and EUR/JPY 120.99/121.03, the synthetic midpoint is 121.00 and the direct one 121.01, yet the two cycles return 0.99957 (via USD) and 0.99974 (via JPY) per euro, so neither is profitable.
  With bid and ask, converting $X$ into $Y$ earns the bid of the $X/Y$ quote and converting $Y$ into $X$ pays its ask; starting from one unit of EUR, the two directions of the EUR–USD–JPY cycle return $b_{\text{EUR/USD}}\, b_{\text{USD/JPY}} / a_{\text{EUR/JPY}}$ and $b_{\text{EUR/JPY}} / (a_{\text{USD/JPY}}\, a_{\text{EUR/USD}})$, so there is no cyclic arbitrage if and only if the direct quote overlaps the synthetic band:

  $$b_{\text{EUR/USD}}\, b_{\text{USD/JPY}} \le a_{\text{EUR/JPY}} \quad \text{and} \quad b_{\text{EUR/JPY}} \le a_{\text{EUR/USD}}\, a_{\text{USD/JPY}}$$

  Arbitrage parity holds only within this band, widened by spreads, funding and settlement costs; it is not an equality.
- *speculative arbitrage*, also called risk arbitrage: the value is nonstationary, because instrument-specific factors make it wander, and arbitrageurs bet that short-run mean reversion dominates. Kinds: spreads (maturity, calendar, yield-curve and credit spreads), pairs trading, statistical arbitrage with factor models, and merger arbitrage, the usual sense of "risk arbitrage".

Arbitrage is understood two ways [alpha-harris, same chap., secs. 17.6 and 17.9, pp. 375–378]. In the *price characterisation*, arbitrageurs are informed traders in relative values who enforce the law of one price [alpha-harris, chap. "Informed Traders and Market Efficiency", sec. 10.5.4, pp. 232–235]. In the *quantity characterisation*, they discipline slow prices, ship common-factor risk between markets and repackage it, manufacturing derivative positions; the arbitrage spread is the payment for these services.

**Practice.** Arbitrageurs are dealers or value traders in the hedge portfolio: fast dealer-type arbitrageurs trade short-term reversals, and value-type arbitrageurs take large positions when sure of fair value [alpha-harris, chap. "Arbitrageurs", sec. 17.2, pp. 349–350]. Dealers span time, matching buyers and sellers who arrive at different times in one market; arbitrageurs span place, matching those who arrive together in different markets [alpha-harris, same chap., sec. 17.7, p. 376]. Free entry narrows arbitrage spreads until they just cover the costs of the most efficient

arbitrageurs. These are often traders with a natural interest in one leg, such as grain operators near a delivery point [alpha-harris, same chap., sec. 17.8, pp. 377–378].

*Economic function*. Arbitrageurs "buy in cheaper markets and sell in more expensive markets", connecting sellers in one market to buyers in another [alpha-harris, chap. 1, p. 6]; as liquidity suppliers they are "not suppliers of liquidity but porters of liquidity", demanding it where it is available and supplying it where it is demanded, and they compete with dealers, who connect buyers and sellers arriving at different times in one market [alpha-harris, chap. "Liquidity", sec. 19.3.6, pp. 404–405]. They enforce the law of one price only when costs, funding, hedge error and execution risk are small enough (arbitrage risks).

*Cross-asset and cross-venue arbitrage*. Dealers connect buyers and sellers across time, while arbitrageurs connect them across instruments, venues, locations and risk forms, often selling a transformed exposure: Harris's dealers span time and arbitrageurs span place, with conversion arbitrageurs as manufacturers: "the conversion arbitrageur is a manufacturer who produces financial instruments that interest traders" (sec. 17.3.1.3, p. 354).

Common cross-asset arbitrages map onto Harris's types: cash against futures is a delivery arbitrage; ETF shares against their basket, depositary receipts against local shares and one currency against a synthetic cross rate are conversion arbitrages, the last a cyclic one; option conversions manufacture option exposure from stock and cash (Harris's box "Manufacturing Stock Option Contracts", p. 354: arbitrageurs who sell options "convert equity risk from stock shares to option contracts" and must rebalance because option values are nonlinear in the stock, pp. 354–355); and merger spreads are speculative risk arbitrage (pp. 360–363).

Harris's arbitrage chapter does not discuss ETFs or depositary receipts; their placement is Interpretation, and ETF creation and redemption is in ETF creation and redemption. *Derivative conversion.* A market maker who buys an option and delta-hedges converts option demand into stock trading, funding and volatility exposure: "Such hedges are dynamic hedges. They arise whenever the value of one instrument is not an exact linear function of the value of another instrument", and volatility makes them perform poorly, so option dealers charge for expected hedging costs and manage the risk of their combined positions rather than each arbitrage separately (pp. 354–355). The changing sensitivities include delta, gamma, vega and funding costs; Harris names only delta (n. 1, p. 355).

**Pitfalls.** Full convergence is the assumption behind the $|A_t|$ in the condition above; with partial or slow convergence the gross gain is the expected narrowing net of extra carry (Interpretation; see arbitrage risks). Pure, speculative, risk and statistical arbitrage are not four parallel types: in Harris risk arbitrage is another name for speculative arbitrage, and statistical arbitrage is one of its kinds. Cross-rate arbitrage is not a fourth pure mechanism beside shipping, delivery and conversion; Harris treats it as a cyclic conversion arbitrage.

Harris's box "A Three-legged Currency Money Machine" (p. 357) is internally inconsistent: 110 yen per dollar and 1.10 dollars per euro imply 121 yen per euro, not the printed 100; and its cycle (buy $1.10 million for 100 million yen, buy 1 million euros for the dollars, buy 99.9 million yen for the euros) loses 0.1 million yen on its own prices, because it buys euros at the implied rate and sells them where they are cheap. A consistent version: with the euro quoted directly at 120.9 yen against the implied 121, buy 1 million euros for 120.9 million yen, sell them for $1.10 million and the dollars for 121 million yen, a gain of 0.1 million yen, about $909.

The point of the box, that price pressure from the arbitrage closes the gap and that dealers, the fastest traders, are the natural arbitrageurs, is unaffected. Even pure arbitrage can lose if convergence is slow or financing fails [alpha-harris, same chap., secs. 17.3.1 and 17.9, pp. 350 and 378].

**See also.** arbitrage risks; informed-trader types; outside spread (value traders' no-trade band, the analogue of arbitrage bounds); dealer inventory risk (dealers span time).

### Result 16.38 Arbitrage risks: implementation, basis, model and carrying-cost risk, and staying power

**Statement.** Harris names four arbitrage risks [alpha-harris, chap. "Arbitrageurs", sec. 17.4, pp. 363–373]:

- *implementation risk*: transaction costs larger than expected, from adverse prices on market orders or failed limit orders. It is worst when one leg fills and another does not, when markets move fast, and when the arbitrage spread is small relative to short-term volatility;
- *basis risk*: the basis moves against the position, because specific factors diverge or common factors are mispriced. Total basis risk scales with the size of the position, and adverse moves force liquidation, at the worst time in a pure arbitrage, when the trader cannot finance the losses;
- *model risk*: misunderstanding the value relation, so that the trader takes the wrong side, carries unrecognised risks or uses wrong hedge ratios. The commonest error is treating a move of the basis towards a changed fair value as an opportunity;
- *carrying-cost risk*: higher financing, borrowing, storage or dividend costs than expected; unexpected price rises whose losses on short legs must be funded at once; slow convergence; and forced buy-ins of borrowed securities, which tend to come when the basis has moved against the arbitrageur.

Risky arbitrage needs compensation, so arbitrage bounds are wide when arbitrage is risky [alpha-harris, same sec., p. 364].

**Conditions.** Timeless (concepts); examples to 2000. Arbitrage opportunities have two causes [alpha-harris, same chap., sec. 17.5, pp. 373–375]. In the *slow price adjustment* scenario a common factor changes and some prices lag. Arbitrage then moves prices to their new values, and the arbitrageur should trade quickly, demanding liquidity. In the *uninformed liquidity demand* scenario values are unchanged and uninformed traders push prices apart in different markets. The arbitrageur then supplies liquidity, competing with value traders and dealers, and need not trade aggressively. Opportunities do not arise when an instrument-specific value changes: then the fair value of the basis has moved, and a pairs trade against it loses [alpha-harris, same chap., pp. 358–360 and 368].

**Practice.** Never leverage to the limit of capital: those who do have no staying power when the basis moves against them [alpha-harris, same chap., secs. 17.4.2 and 17.4.4.2, pp. 368 and 372]. LTCM, highly leveraged in mean-reverting positions, failed in 1998 although many positions later paid [alpha-harris, same chap., box, p. 369]. In illiquid markets, trade the illiquid leg first, or interleave small orders in both legs, so as never to be far from hedged [alpha-harris, same chap., pp. 366–367]. To judge whether a move in the spread is an opportunity, ask which leg moved and why. In Harris's GE against S&P 500 futures example, a widening caused by the index falling while GE lags is more likely an opportunity than one caused by GE rising, which may be firm-specific news [alpha-harris, same chap., box and Table 17-2, pp. 361–362].

*Legging and execution*. Arbitrageurs "get into trouble when they are stuck in the middle with one leg on and one leg off" (box, p. 366); they "try very hard to trade all legs of their hedge portfolio at the same time", using only market orders when spreads are small relative to the execution uncertainty of limit orders, trading the illiquid leg first when one leg is illiquid so that they can cancel cheaply if the basis closes, and interleaving split orders when both are, often with limit orders (p. 366).

In execution, the arbitrageur estimates the fair basis, costs and hedge ratios, chooses the leg order, submits synchronised orders or trades the illiquid leg first, monitors partial fills, basis and residual exposure, and cancels, completes or rebalances. Trading the illiquid instrument first is Harris's rule; the rest is an illustrative procedure (Interpretation).

*Capital.* Forced liquidation by arbitrageurs who cannot finance losses, meet margin, borrow or carry inventory can widen the basis "precisely when the forward-looking trade is most attractive": this is Harris's staying power and carrying-cost risk. That the widest spreads appear when balance sheet is scarce is a generalisation, not a measured result.

*Relative-value features*. Useful features map onto this entry's risks: hedge quality from rolling betas, cointegration or factor residuals and hedge-ratio instability onto model risk; a convergence clock to delivery, expiry, fixing, rebalance or event, and carry from funding, borrow, dividends, storage and roll, onto carrying-cost risk; execution synchrony across legs onto implementation risk; and stress capacity from margin, drawdown to liquidation and recall risk onto staying power. Basis level, carry-adjusted spread, z-score and distance to the bounds measure the opportunity itself. That such features separate cheapness from tradability is an untested interpretation.

**Pitfalls.** Staying power and specific-factor risk are not separate risk categories: in Harris staying power is the consequence of basis and carrying-cost risk under leverage, and specific factors are a source of basis risk and model error. Separating common from specific factors tells whether a spread move is an opportunity or new information, as the passages support. Writing $r_i = \beta_i^\top f + \epsilon_i$ and $r_j = \beta_j^\top f + \epsilon_j$ and attributing spread moves to common factors, a hedge-ratio error $\beta_i - \beta_j$, or residuals $\epsilon_i - \epsilon_j$ holds only for a one-for-one spread.

For a hedge portfolio $r_i - h\,r_j$ with hedge ratio $h$, the spread is $(\beta_i - h\beta_j)^\top f + (\epsilon_i - h\epsilon_j)$, so the factor exposure is $\beta_i - h\beta_j$ and the residual $\epsilon_i - h\epsilon_j$; a hedge ratio chosen to neutralise one factor ($h = \beta_{i1}/\beta_{j1}$) leaves the others (with two factors, $h = 1.5$ zeroes the first exposure and leaves $-0.45$ on the second; for independent residuals the spread variance is $(\beta_i - h\beta_j)^\top \Sigma_f (\beta_i - h\beta_j) + \sigma_{\epsilon_i}^2 + h^2\sigma_{\epsilon_j}^2$).

**See also.** arbitrage; order anticipation (short squeezes hit arbitrageurs); bubbles and crashes (limits to arbitrage).

#### Definition 16.39 Order anticipation: front running, quote matching, sentiment-oriented technical trading, squeezes and stop gunning

**Statement.** *Order anticipators* profit from information about other traders' orders, not about fundamental values, by trading before them [alpha-harris, chap. "Order Anticipators", pp. 245 and 256–257]:

- *front runners* trade on orders others have decided to arrange. Ahead of aggressive traders they profit from the orders' price impact. Ahead of passive traders they are *quote matchers*, or penny jumpers, who step in front of standing limit orders and keep the limit order as an option to exit: long stock plus the implicit put of a large buy order is a call [alpha-harris, sec. 11.1, pp. 245–250];
- *sentiment-oriented technical traders* predict trades uninformed traders have not yet decided to make, from cash-flow timing, hedging needs, asset exchanges, gamblers' interests and speculators' mistakes [alpha-harris, sec. 11.2, pp. 251–254];
- *squeezers* corner one side of a market so that those who must close positions have to trade with them [alpha-harris, sec. 11.3, pp. 254–256].

*Gunning the market*, pushing prices to trigger clustered stop orders and trading with them, is a manipulation of stop orders that Harris treats alongside these strategies [alpha-harris, sec. 11.4, p. 256].

**Conditions.** Effects [alpha-harris, same chap., secs. 11.1.3–11.1.4, 11.2.1 and 11.3, pp. 250–256]. Front running accelerates the impact of the traders it front-runs. It makes prices more informative in the short run when it front-runs informed traders, but in the long run it lowers their profits and investment in information, and so informativeness. It makes prices less informative when it front-runs uninformed traders. It generally makes markets less liquid: front runners' profits are other traders' costs, they offer liquidity back at worse prices, and they drive away the orders they feed on.

Front runners benefit large traders only in narrow conditions: better traders who do not compete with each other and do not squeeze. Sentiment-oriented technical trading makes prices less informative, especially in bubbles, and loses when value traders push prices back, so it pays best in hard-to-value instruments. Squeezes make prices less informative and greatly raise others' costs.

**Practice.** Time precedence with a meaningful minimum price increment makes quote matchers pay to step ahead, to the traders on the other side rather than the passive traders they front-run; a small tick weakens this protection. Passive traders defend themselves by hiding or splitting orders or switching to market orders, which raises their costs and reduces displayed size [alpha-harris, same chap., pp. 250 and 257]. Traders protect themselves from front runners by guarding order information, from sentiment traders by trading quickly, and from squeezers by keeping several ways to close [alpha-harris, same chap., p. 257]. NYSE and AMEX specialists, as Harris describes them in 2002, had privileged access to the order flow and used it in the same ways [alpha-harris, chap. "Specialists", sec. 24.5, pp. 501–508]:

- speculating on short-term forecasts from the order flow;
- quote matching against a heavy side of their book;
- *cream skimming*, stepping in front of the book to fill only the more profitable, usually small retail, marketable orders;
- anticipating stop orders in their books;
- *stopping stock*, guaranteeing an execution price and then deciding when to fill, a look-back timing option that harms the limit-order traders who would have filled the order.

Public order precedence, the public liquidity preservation principle and audit trails limit these privileges [alpha-harris, sec. 24.2.2, pp. 498–499].

**Pitfalls.** Not all front running is illegal: it is illegal when the information is obtained improperly, through a confidential brokerage relationship or eavesdropping, but an observant floor trader may legally infer a broker's order [alpha-harris, chap. "Order Anticipators", sec. 11.1.1, pp. 246–247]. Squeezers and gunners defend themselves as speculators, which makes the laws against them hard to enforce [alpha-harris, same chap., pp. 256–257]. Quote matching and stop gunning are not categories parallel to front running: in Harris quote matching is front running of passive traders, and gunning a manipulation of stop orders. The specialist strategies describe a floor-based market structure of 2002 and are dated.

**See also.** bluffing and manipulation; informed-trader types (information-oriented against sentiment-oriented technical trading); arbitrage risks (short squeezes); bubbles and crashes (order anticipators accelerate the rise); spread determinants (tick size); HFT strategies (order anticipation as one of the SEC's 2010 strategy types, and pinging as a tool).

#### Definition 16.40 Bluffing and market manipulation: rumourmongers, price manipulators and the asymmetric-impact condition

**Statement.** *Bluffers* try to fool other traders into trading unwisely and then profit from them. Harris names two techniques [alpha-harris, chap. "Bluffers and Market Manipulation", p. 259 and Table 12-3, p. 274]:

- *rumourmongers* spread false information, or true information presented so that it will be misread;
- *price manipulators* arrange trades at prices, volumes and times meant to change others' opinions of value. The trades may be genuine arm's-length trades or *wash trades* with confederates that create artificial activity.

*Market manipulation* occurs when bluffers or their victims change prices from what they would otherwise be. Bluffing is profitable when the price impact of the bluffer's purchases differs from that of its sales: if buys move prices more, buy first and sell later, and conversely [alpha-harris, same chap., sec. 12.3, pp. 270–273]. In Harris's example, liquidity suppliers raise prices 10 cents per 100 contracts for 1,000-contract orders but only 5 cents for 500-contract orders. Buying 4,000 contracts in 1,000s and selling them in 500s then averages 12.50 bought against 12.875 sold, a profit of $1.5 million on a 1,000-unit contract, and leaves the price at 12 rather than 10 [alpha-harris, same sec., p. 270]. With the same impact per unit for all trades, the same round trip loses $3 million (p. 271; both figures rechecked, Interpretation).

**Conditions.** Timeless (concepts); legal remarks as of 2002. Bluffers imitate informed speculators: both promote their positions and trade to show confidence. The difference is that speculators trade on well-founded opinions about value, whereas bluffers manufacture the signals; bluffs make prices less informative and are more likely followed by reversals [alpha-harris, same chap., sec. 12.2.1, pp. 265–267]. Value traders foil bluffs by trading against prices pushed from value, unless bluffers have more capital and force them out. Bluffs therefore succeed most in securities value traders do not follow or cannot trade: illiquid, hard to value or hard to borrow. Long-side bluffs are probably commoner than short-side ones [alpha-harris, same chap., sec. 12.2.2, pp. 268–269].

**Practice.** Liquidity suppliers who set prices from net order flow must make buys and sells of every size, speed and timing move prices equally per unit traded; otherwise bluffers can exploit them [alpha-harris, same chap., sec. 12.3, pp. 271–273]. Momentum traders and liquidity suppliers, who react to the trades they see, are the most vulnerable [alpha-harris, same chap., sec. 12.4, p. 274]. Transaction costs limit bluffing, which trades intensively [alpha-harris, same chap., p. 273].

Symmetric impact disciplines bluffing in this sense. If impact depended on net pressure and risk, of which Harris names only net quantity, a zero-net sequence could not create a higher average sale price than purchase price. That conclusion holds in Harris's example, where each trade's impact is immediate and permanent. In impact models with decay and changing liquidity, equal treatment of buys and sells does not exclude manipulation (see absence of price manipulation).

**Pitfalls.** A price reversal does not prove manipulation, because good speculators also misjudge values. Conviction usually requires evidence that the bluffer spread information known to be false or arranged wash trades, or testimony on intent; careful bluffers therefore pass as speculators [alpha-harris, same chap., sec. 12.2.3, pp. 269–270]. Harris's legality column (Table 12-3): false rumours and wash trades are illegal in most jurisdictions and easy to prosecute if caught; misleadingly framed true information and genuine trades meant to mislead are of questionable legality and hard to prove. Wash traders are not a third type beside rumourmongers and price manipulators: in Harris wash trading is a technique of price manipulators.

**See also.** order anticipation (gunning stop orders); insider-trading detection (the same proof problem); informed-trader types; Kyle single-auction model (Interpretation: impact linear and symmetric in signed flow, as in Harris's no-profit case); absence of price manipulation (manipulation through one's own impact in a model, not through others' inferences).

## 16.8 Information and price impact

#### Model 16.41 Kyle single-auction model: informed trading and linear price impact

**Statement.** One risk-neutral insider who knows the asset value $v \sim N(\mu, \sigma_v^2)$, price-insensitive noise traders with net order $u \sim N(0, \sigma_u^2)$ independent of $v$, and competitive risk-neutral market makers who see only the net order flow $y = x + u$ and set $S(y) = \mathbb{E}[v \mid y]$ [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", pp. 31–32] have a linear equilibrium [catalogue-0496, chap. "Strategic Trader Models I: Informed Traders", p. 93, eqs. (4.3)–(4.5)]:

$$x(v) = \beta(v - \mu), \qquad S(y) = \mu + \lambda y, \qquad \beta = \frac{\sigma_u}{\sigma_v}, \qquad \lambda = \frac{\sigma_v}{2\sigma_u}.$$

The insider's ex ante expected profit is $\frac{1}{2}\sigma_v\sigma_u$ [catalogue-0496, same chap., p. 97, eq. (4.17)], and the posterior variance of $v$ after the auction is exactly half the prior, whatever the order flow: price reveals half of the insider's information [catalogue-0496, same chap., p. 98, eq. (4.20)].

**Notation.** Source notation: `catalogue-0496` writes $p_0$ for $\mu$, $\Sigma_0$ for $\sigma_v^2$, $\sigma_\mu$ for $\sigma_u$ and $P$ for the price; `crypto-cartea` writes $\sigma$ for $\sigma_v$. Here $\lambda$ is a price-impact slope, in price per share.

**Conditions.** A single auction; the insider does not observe $u$ before trading; normality of $v$ and $u$; market makers earn zero expected profit given the order flow, so price is semi-strong efficient [catalogue-0496, same chap., pp. 91–92]; [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 32]. There is no other linear equilibrium, but nonlinear equilibria are not excluded [catalogue-0496, chap. "Strategic Trader Models I: Informed Traders", p. 97]. Idea of the proof: given a linear price rule the insider's expected profit $x(v - \mu) - \lambda x^2$ is quadratic, so $\beta = 1/(2\lambda)$; given a linear strategy, $v$ and $y$ are jointly normal and the projection $\mathbb{E}[v \mid y]$ is linear in $y$; the two conditions fix $\beta$ and $\lambda$ [catalogue-0496, same chap., pp. 93–96].

**Practice.** Doubling the noise-trader standard deviation doubles the insider's order and expected profit and leaves the ex ante price unchanged [catalogue-0496, same chap., p. 97]. Market makers are risk-neutral, so any premium compensates their informational disadvantage and is borne by the price-insensitive liquidity traders [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 31]. Interpretation (own derivation): the insider trades only part of the mispricing, because a larger order reveals more through $y$; $\lambda$ rises with $\sigma_v$ and falls with $\sigma_u$.

**Pitfalls.** A linear price rule implies a negative price for a large enough sell imbalance, so it is at best an approximation of actual price behaviour [catalogue-0496, chap. "Strategic Trader Models I: Informed Traders", p. 96]. The joint-normality assumption drives the predictions on trade size and volume [catalogue-0496, same chap., p. 98]. The textbook account in `crypto-cartea` prints the equilibrium price as $\mu + 2(x+u)\sigma_u/\sigma$, so $\lambda = 2\sigma_u/\sigma$ [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 33]; this is the reciprocal of the value in `catalogue-0496`, eq. (4.5), and of the value implied by its own preceding step $\beta = (2\lambda)^{-1}$. The authors' errata correct the price to $\mu + (x+u)\sigma/(2\sigma_u)$, so $\lambda = \sigma/(2\sigma_u)$, in agreement with `catalogue-0496` [crypto-cartea-errata, entries for p. 33]. Do not copy $\lambda$ from the printed page.

**See also.** Grossman–Miller model (a premium for risk, not information); Grossman–Miller liquidity measures (an inventory price-impact slope).

**Further reading.** [s6-kyle], the original model and its dynamic version.

## 16.9 Adverse selection and the bid–ask spread

#### Model 16.42 Glosten–Milgrom sequential-trade model: spreads from adverse selection

**Statement.** In a pure dealership market, investors arrive one at a time, randomly and anonymously, see the specialist's bid $b_t$ and ask $a_t$, and buy one unit at the ask, sell one unit at the bid, or leave; some arrivals are informed about the asset's future value $V$, the rest trade for liquidity reasons [glosten-milgrom-1983, sec. II, pp. 3–4]. An arrival with information $\mathcal{F}_t$ has reservation price $Z_t = \rho_t\,\mathbb{E}[V \mid \mathcal{F}_t]$ and buys if $Z_t > a_t$, sells if $Z_t < b_t$; competitive, risk-neutral specialists with information $\mathcal{S}_t$ earn zero expected profit on each purchase and each sale, so the quotes solve [glosten-milgrom-1983, p. 6]

$$a_t = \mathbb{E}[V \mid \mathcal{S}_t,\ Z_t > a_t], \qquad b_t = \mathbb{E}[V \mid \mathcal{S}_t,\ Z_t < b_t].$$

Then $a_t \geq \mathbb{E}_t[V] \geq b_t$, with strict inequalities whenever informed buying above and informed selling below $\mathbb{E}_t[V]$ have positive probability: adverse selection alone produces a spread, with no inventory cost, risk aversion or monopoly power [glosten-milgrom-1983, Prop. 1, pp. 7–9]. Each transaction price is the specialist's updated expectation of $V$, so transaction prices form a martingale with respect to the specialist's and the public information [glosten-milgrom-1983, Prop. 2, p. 10].

**Notation.** $V > 0$ is the asset's value, realised at a future date, with finite variance; $\rho_t$ is the arriving investor's liquidity (time-preference) parameter, with $\rho = 1$ for the specialist; $\mathcal{F}_t$ and $\mathcal{S}_t$ are the information of the arrival and of the specialist; $\mathbb{E}_t$ is expectation given the information both share [glosten-milgrom-1983, pp. 4–7]. The paper writes $A_t, B_t$ for $a_t, b_t$, $F_t, S_t$ for $\mathcal{F}_t, \mathcal{S}_t$.

**Conditions.** Every participant is risk-neutral with utility $\rho x V + c$ over shares $x$ and consumption $c$; $\rho$ is independent across traders and carries no information about $V$ [glosten-milgrom-1983, pp. 4–5]. Uninformed participation is exogenous; trades are single units; short positions in cash or stock cost nothing; zero profit on each side is motivated by Bertrand competition between equally informed specialists, and existence of the quotes is not shown in general [glosten-milgrom-1983, pp. 4–6]. If the specialist sees only public information, the ask is the smallest $a$ with $a \geq \mathbb{E}[V \mid \mathcal{H}_t, Z_t > a]$ and the bid the largest $b$ with $b \leq \mathbb{E}[V \mid \mathcal{H}_t, Z_t < b]$ [glosten-milgrom-1983, p. 7]. Idea of the proof of Proposition 1: conditional on a buy, the ask equals the expectation of $V$ given the buy; since buying selects arrivals whose own expectation exceeds a threshold and $\rho$ is uninformative, that conditional expectation is at least $\mathbb{E}_t[V]$ [glosten-milgrom-1983, p. 8].

**Practice.** The spread is larger, for given information, when insiders' information is finer, when the ratio of informed to uninformed arrivals rises, or when the dispersion of the uninformed liquidity parameter around its median falls [glosten-milgrom-1983, Prop. 5, pp. 15–16]. If buy

and sell probabilities stay above $\theta_a, \theta_b > 0$, prices converge and the root-mean-square spread over the first $k$ trades is at most $\sqrt{\mathrm{Var}(V)/(\theta_a \theta_b k)}$ [glosten-milgrom-1983, Prop. 3, p. 11]; with Poisson arrivals the average spread is then of the order of one over the square root of volume, because trades reveal information [glosten-milgrom-1983, pp. 18–19].

The specialist and the arrivals come to agree on the expected value of $V$ [glosten-milgrom-1983, Prop. 4, p. 13]. The market maker need not know which traders are informed: since well-informed traders rarely reveal themselves, dealers treat every order as possibly informed and revise prices with the side of each trade [alpha-harris, chap. "Dealers", pp. 288–289]. The two-point version with closed-form half-spreads is a separate result.

**Pitfalls.** If the liquidity motive is weak relative to the insiders' advantage, the zero-profit quotes may preclude trade, and the market shuts down until enough public information neutralises the insiders' advantage [glosten-milgrom-1983, p. 9]; if the bounded-probability condition fails, the spread need not shrink: the market may close, or every trade may occur at the bid [glosten-milgrom-1983, p. 12]. Proposition 5 compares spreads for given information; it does not say that stocks with more insiders have wider spreads on average, since more insiders also reveal information faster [glosten-milgrom-1983, p. 17]. The authors list unit lots, the absence of risk aversion and inventory costs, and a fixed underlying value as limitations [glosten-milgrom-1983, sec. IV, pp. 24–25]. The locators are to the 1983 discussion paper, not the 1985 journal article that the catalogue names.

**See also.** Glosten–Milgrom half-spreads (special case: two values, closed form); market collapse (special case: price-sensitive liquidity traders); adverse-selection spread component (special case: a textbook example); Kyle single-auction model (contrast: batch order flow and a strategic insider); Grossman–Miller model (contrast: a spread for inventory risk, without information).

**Further reading.** [s6-glosten], the published version;

#### Result 16.43 Glosten–Milgrom half-spreads with two values

**Statement.** Let the asset's value be $V_H$ with probability $p$ and $V_L < V_H$ otherwise, let a fraction $\alpha$ of traders be informed, and let competitive risk-neutral market makers quote $a = \mu + \Delta_a$ and $b = \mu - \Delta_b$ around $\mu = \mathbb{E}[V \mid \mathcal{F}]$. Zero expected profit on each side gives the half-spreads [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 35]

$$\Delta_a = \frac{\alpha p}{\alpha p + (1-\alpha)/2}\,(V_H - \mu) = \frac{V_H - \mu}{1 + \frac{1-\alpha}{\alpha}\,\frac{1/2}{p}}, \qquad \Delta_b = \frac{\mu - V_L}{1 + \frac{1-\alpha}{\alpha}\,\frac{1/2}{1-p}}.$$

With trading over time, $p_t = \mathbb{P}(V = V_H \mid \mathcal{F}_t)$ and $\mu_t = \mathbb{E}[V \mid \mathcal{F}_t]$ are updated by Bayes' rule from the order flow, the same formulas hold with $p_t, \mu_t$, and each execution price equals the expected value given the history and the side of the trade, so the price at execution times is a martingale [crypto-cartea, same chap., p. 36].

**Notation.** Local override for this entry and market collapse: $p$ is the probability of the high value, not a price; $\alpha$ is the fraction of informed traders; $\Delta_a, \Delta_b$ are the ask and bid half-spreads; $\mathcal{F}$ is public information, including the order flow. Source notation: `crypto-cartea` writes $\nu$ for $V$.

**Conditions.** A simplified, essentially static version of the model: one-unit orders; informed traders know $V$ and buy one unit if $a < V_H$ and $V = V_H$, sell one unit if $b > V_L$ and $V = V_L$; liquidity traders are price-insensitive and buy or sell with probability $\frac{1}{2}$ each, whatever the value; market makers are risk-neutral, carry inventory at no cost and compete to zero expected profit, using only public information [crypto-cartea, same chap., pp. 34–35]. Idea of the proof: a buy comes from a liquidity trader with weight $(1-\alpha)/2$, who pays $\Delta_a$, and from an informed trader with weight $\alpha p$, who costs $V_H - \mu - \Delta_a$; setting the weighted profit to zero gives $\Delta_a$, and the bid is symmetric [crypto-cartea, same chap., p. 35].

**Practice.** Reading asymmetric information as toxicity, $\alpha$ is its prevalence, and $p$ with $V_H - \mu$, or $1-p$ with $\mu - V_L$, its magnitude on the buy or sell side; market makers widen each half-spread as the prevalence and magnitude of toxicity on that side rise [crypto-cartea, same chap., pp. 35–36]. Interpretation (own derivation): as $\alpha \to 0$ both half-spreads vanish, and as $\alpha \to 1$ the quotes reach $V_H$ and $V_L$, where only informed traders would trade.

**Pitfalls.** This is a textbook simplification: the paper allows a general value distribution and price-sensitive uninformed traders [glosten-milgrom-1983, sec. II, pp. 3–6]. With price-insensitive liquidity traders a monopolist would quote $a = V_H$, $b = V_L$; the formulas rely on competition between market makers [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 35].

**See also.** Glosten–Milgrom model (generalisation); market collapse (extension: price-sensitive liquidity traders); adverse-selection spread component (contrast: a parametrisation by the dealer's pricing error).

#### Result 16.44 Glosten–Milgrom with price-sensitive liquidity traders: market collapse

**Statement.** If liquidity trader $i$ trades only when the half-spread is at most his urgency $c_i$, with $c_i$ distributed with cumulative distribution function $F$ across liquidity traders, the zero-profit half-spreads are defined implicitly by [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 37]; [crypto-cartea-errata, entry for p. 37]

$$\Delta_a = \frac{V_H - \mu}{1 + \frac{1-\alpha}{\alpha}\,\frac{(1-F(\Delta_a))/2}{p}}, \qquad \Delta_b = \frac{\mu - V_L}{1 + \frac{1-\alpha}{\alpha}\,\frac{(1-F(\Delta_b))/2}{1-p}}.$$

Widening a half-spread drives out liquidity traders; if urgency is small, the only solutions may be the extreme ones, $\Delta_a = V_H - \mu$ and $\Delta_b = \mu - V_L$: the market collapses, only informed traders, who are then indifferent, might trade, and any trade reveals the value [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 37].

**Notation.** As in the two-value half-spreads; $c_i$ is the cash-equivalent gain to liquidity trader $i$ from completing the desired trade.

**Conditions.** The two-value model of the half-spreads entry, with only the fraction $1 - F(\Delta)$ of would-be liquidity buyers or sellers trading at half-spread $\Delta$ [crypto-cartea, same chap., p. 37]. "Small" urgency means that no liquidity trader's urgency exceeds the expected value of the asset: $F(\mu - \epsilon) = 1$ for some $\epsilon > 0$ [crypto-cartea, same chap., p. 37, n. 5].

**Practice.** The paper gives the same mechanism: when the proportion of insiders is high or the liquidity motive of the uninformed weak, the zero-profit quotes can preclude trade, a lemons problem as in Akerlof, and the market reopens when public information neutralises the insiders' advantage [glosten-milgrom-1983, pp. 9 and 17].

**Pitfalls.** The book prints $F(\Delta_a)$ in the bid formula; the authors' errata correct it to $F(\Delta_b)$, as given above [crypto-cartea-errata, entry for p. 37]. Do not copy the bid formula from the printed page.

**See also.** Glosten–Milgrom half-spreads (the price-insensitive case, $F \equiv 0$); Glosten–Milgrom model (generalisation, with market shutdown).

#### Result 16.45 Spread components: the adverse-selection component, its two views, and the total spread

**Statement.** The bid–ask spread is split, for analysis, into a *transaction-cost* (transitory) component $C$, which pays the dealer's normal costs of doing business, and an *adverse-selection* (permanent) component $A$, which recovers from uninformed traders what the dealer loses to informed ones [alpha-harris, chap. "Bid/Ask Spreads", pp. 298–299]. A dealer first estimates the value $V_0$ from all current information, then the values $V^B$ and $V^S$ she would estimate if the next trader were a buyer or a seller; $A = V^B - V^S$, and she quotes [alpha-harris, same chap., Figure 14-1, p. 301]

$$a = V^B + \tfrac{1}{2}C, \qquad b = V^S - \tfrac{1}{2}C, \qquad a - b = A + C.$$

If the dealer learns nothing more from the next trade than its side, the conditional estimate becomes her new value: after a buy both quotes rise by $A/2$ [alpha-harris, same chap., Figure 14-2, pp. 301–302]. In a simple example based on the Glosten–Milgrom model, a dealer believes that the security is worth $V$ unconditionally, that the next trader is equally likely to buy or sell and is informed with probability $\alpha$, and that it is worth $V + e$ if an informed trader buys and $V - e$ if an informed trader sells [alpha-harris, chap. "Bid/Ask Spreads", p. 320].

Information view: conditional on a buy the dealer's value estimate is $(1 - \alpha)V + \alpha(V + e) = V + \alpha e$, conditional on a sell $V - \alpha e$, so the adverse-selection component is $2\alpha e$ [alpha-harris, same chap., p. 320]. Accounting view: expected profit on the next trade is $\frac{1}{2}(a - b) - \alpha e$, so breaking even requires $a - b = 2\alpha e$ [alpha-harris, same chap., pp. 320–321]. Both views give the same component [alpha-harris, same chap., p. 300]; in the example $A = 2\alpha e$ and $C = 0$.

**Notation.** $A$ and $C$ are amounts of the full spread, in price units; $a$, $b$ are the ask and bid. Source notation: Harris names the components in words and writes $V_0^B$, $V_0^S$ for $V^B$, $V^S$ (Figure 14-1); in his appendix he writes $P$ for $\alpha$, $E$ for $e$ (the dealer's pricing error when the trader is informed), and $A$, $B$ for the ask and bid $a$, $b$, so his $A$ there is not the component.

**Conditions.** Normal costs of doing business are ignored; the example assumes that dealers know the value when informed traders trade, which the result does not need [alpha-harris, same chap., pp. 300 and 320]. The two views agree when the loss from trading with an informed trader equals the pricing error, as when the dealer cannot restore inventory before prices adjust; if the dealer can rebalance but does not recognise the informed trade, informed traders keep trading with him until the price adjusts, and his cumulative loss is still the original pricing error [alpha-harris, same chap., p. 302].

Low barriers to entry and competition from public limit orders, which undercut wide dealer quotes, limit monopoly spreads [alpha-harris, same chap., p. 298]; in a competitive dealer market free entry and exit drive dealers to normal profits, so the total spread is this component plus a transaction-cost component that covers operating costs, financing, any monopoly profit and any premium for inventory risk [alpha-harris, same chap., pp. 298–299]. $V^B$ and $V^S$ lie equally far from $V_0$, at $V_0 \pm A/2$, when informed buyers and sellers are equally likely and their expected pricing errors equal [alpha-harris, same chap., p. 301]. A three-way split (order processing, inventory, adverse selection) separates the inventory-risk premium out of $C$; Hasbrouck likewise notes that a spread impounds noninformational costs and inventory effects as well as asymmetric information [s6-hasbrouck, sec. 5.5, p. 53].

In the structural sequential-trade version with a cost $c$ per trade, $a = \mathbb{E}[V \mid \text{buy}] + c$ and $b = \mathbb{E}[V \mid \text{sell}] - c$; the bid and ask sequences are martingales but trade prices are not [s6-hasbrouck, sec. 5.4.2, p. 49]. There $c$ is a per-side amount, so Harris's $C$ corresponds to $2c$ (Interpretation).

**Practice.** The component rises with the probability that the next trader is informed, with the materiality of the information and with trade size, because dealers believe large traders are often well informed [alpha-harris, chap. "Dealers", pp. 289–291 and 295]. Price changes from the adverse-selection component are permanent, while the transaction-cost component causes bid–ask bounce [alpha-harris, chap. "Bid/Ask Spreads", pp. 299–300].

Volatile instruments have wider spreads, both through inventory risk for risk-averse dealers and as a proxy for asymmetric information [alpha-harris, same chap., pp. 312–313]; the other determinants are in spread determinants. Uninformed traders lose to informed ones whichever order they use: their limit orders fill quickly when informed traders want the other side and fail to fill when informed traders compete on the same side, and their market orders pay spreads widened by adverse selection, so that the component is in effect a fee dealers charge market-order traders for bearing adverse-selection risk [alpha-harris, same chap., sec. 14.3, pp. 303–304, Table 14-1].

**Pitfalls.** Consider the claim that $A$ is twice the dealer's mark-to-market loss against a correctly signed informed trade. In the example that holds only if $\alpha = 1$: against an informed buyer the dealer's loss relative to $V$ is the whole pricing error $e$, while $A/2 = \alpha e$ is the expected revision given a buy, that is, the expected loss per buy, informed and uninformed buyers together, of a dealer whose ask is at $V$ (Interpretation: own derivation from the example, pp. 300–302 and 320). The two views equate a probability-weighted loss with a conditional revision, not $A$ with the loss on one informed trade.

This is a textbook example, not the Glosten–Milgrom paper, whose quotes are expectations conditional on a trade at the quote (the model); the note that attributed $a = V + \alpha e$, $b = V - \alpha e$ to Glosten and Milgrom followed this example. The accounting view fixes only the width $a - b$; the level $V \pm \alpha e$ comes from the information view (Interpretation). Harris prints the dealer's profit against an informed buyer as $A - (V - E)$; his stated result requires $A - (V + E)$ [alpha-harris, chap. "Bid/Ask Spreads", p. 321].

**See also.** Glosten–Milgrom model (generalisation); Glosten–Milgrom half-spreads (contrast: a parametrisation by the value distribution); dealer quote adjustment (practice: how dealers set and revise quotes); spread measures (effective and realised spreads, which measure the components after the fact); component identification (estimating $A$ and $C$ from prices and trade signs).

## 16.10 Dealer inventory and quote adjustment

#### Definition 16.46 Dealer inventory imbalance and the two inventory risks

**Statement.** A dealer's *target inventory* is the position it wants to hold, and its *inventory imbalance* is actual inventory minus target [alpha-harris, chap. "Dealers", p. 283]. Dealers face two inventory risks, distinguished by whether future price changes are correlated with the imbalance [alpha-harris, same chap., p. 285]:

- *diversifiable inventory risk*: price changes independent of the imbalance; gains and losses are equally likely, they do not cause losses in the long run, and dealing in many instruments reduces their effect on total profit [alpha-harris, same chap., pp. 285–286];
- *adverse-selection risk*: price changes inversely correlated with the imbalance, because informed traders take the side their information favours, make order flow one-sided and push the dealer's inventory away from target before prices move [alpha-harris, same chap., p. 286].

**Conditions.** If long and short positions are equally costly, a dealer that does not also speculate, hedge or invest has a target of zero; where shorting costs more, targets are positive; speculating dealers set targets by their views [alpha-harris, same chap., pp. 283–284]. Large imbalances are costly to finance and expose the dealer to losses if prices move, and an imbalance beyond the dealer's capital forces liquidation [alpha-harris, same chap., p. 284]; a dealer can also hedge an imbalance with a correlated instrument [alpha-harris, same chap., p. 285, Table 13-2]. Interpretation: the costs of carrying inventory thus group into price risk, the financing of long and the borrowing of short positions, the use of capital, and the basis risk and costs of hedges; recall risk and risk-limit use are additions to Harris's list.

**Practice.** Dealers avoid both risks by keeping inventory under control, so most attend to *market values*, the prices that produce two-sided order flow, more than to fundamental values, which they discover as a by-product [alpha-harris, same chap., pp. 286–287]. Adverse selection is still the most important determinant of dealer profitability [alpha-harris, same chap., p. 286]. Inventory risk does not need informed traders: diversifiable inventory risk is present when no one is better informed, risk-averse dealers require a premium for it, mainly in highly volatile instruments, and that premium is part of the transaction-cost spread component [alpha-harris, chap. "Bid/Ask Spreads", pp. 299 and 313]; see spread components.

**Pitfalls.** "Diversifiable" here means unrelated to the dealer's imbalance; within one instrument it is still a risk that a risk-averse dealer is paid to bear, as in Grossman–Miller (Interpretation). A falling inventory after buys does not by itself say which risk is present; separating them needs the subsequent price path (Interpretation).

**See also.** dealer quote adjustment (how dealers control the imbalance); Grossman–Miller model (a model of the first risk); Glosten–Milgrom model (a model of the second); inventory-based quotes (models of quotes as functions of the imbalance).

**Method 16.47 Dealer quote adjustment for inventory and suspected informed trading**

**Statement.** Dealers manage inventory and adverse selection through quote levels, quote skew, spread width and displayed size.

1. *Inventory.* To reduce inventory, lower bid and ask, decrease the bid size and raise the ask size; to increase it, do the opposite; an impatient dealer can take another trader's offer or hit another trader's bid, or hedge with a correlated instrument [alpha-harris, chap. "Dealers", pp. 284–285, Table 13-2].
2. *After a suspected informed trade.* After selling to a suspected well-informed buyer, raise the ask and lower its size to discourage further informed buying, raise the bid and its size to rebuild inventory, or buy at another trader's ask or buy a correlated instrument; after buying from a suspected informed seller, do the reverse [alpha-harris, same chap., pp. 287–288, Table 13-3].
3. *Before the next trade.* Set the ask and bid at the values expected if the next trader buys or sells, adjusting more when informed trading is more likely and the information more material [alpha-harris, same chap., pp. 288–290]; see the adverse-selection component.
4. *Size.* Avoid displaying large size, so that traders must ask for it and reveal themselves [alpha-harris, same chap., p. 291].

**Conditions.** Timeless (practice description from 2002; venue examples dated). Dealers rarely know who is informed: informed traders hide behind brokers and claim to be uninformed, and rules of thumb (large, active, impatient traders are often informed; small retail traders rarely) are useful but unreliable, so every order is treated as possibly informed [alpha-harris, same chap., pp. 288–289]. Spreads widen with asymmetric information, volatility and weak utilitarian trading interest [alpha-harris, chap. "Bid/Ask Spreads", pp. 312–313] (spread determinants).

**Practice.** Searching for prices that bring two-sided order flow is price discovery: it finds market values [alpha-harris, chap. "Dealers", p. 284]. Demanding liquidity to rebalance gives a negative realised spread [alpha-harris, same chap., p. 284]. After a buy, raising quotes is right if the buyer was informed and a mistake if not, when value traders sell to the dealer and realised spreads fall; doing nothing is the reverse; buying at another trader's ask may limit losses against an informed buyer but is a big mistake against an uninformed one [alpha-harris, same chap., pp. 291–292, Table 13-4]. Poorly informed dealers more often mistake uninformed for informed clients, so they turn inventory quickly and earn smaller realised spreads; well-informed dealers can hold larger positions longer and earn larger realised spreads [alpha-harris, same chap., pp. 292–294].

Value traders trade when prices diverge from fundamental value, often after such mistakes or after risk-averse dealers obtain large concessions for taking big positions, and in doing so restore dealers' inventories [alpha-harris, same chap., pp. 291 and 294]. Some dealers avoid informed traders by trading only with retail or known clients, refusing anonymous size, or slicing large orders over time, which lets them offer more liquidity to the flow they accept [alpha-harris, same chap., pp. 290–291].

**Pitfalls.** Informed traders can bluff, sending small orders on one side to move a dealer before trading on the other; against traders they cannot identify, dealers should adjust prices at the same rate per quantity whatever the side, speed, size or sequence of orders [alpha-harris, same chap., pp. 292 and 294]. Interpretation: refusing flow protects the dealer but reduces the immediacy available to others; one-sided flow turns dealing into warehousing risk. These are heuristics without an objective function; Avellaneda–Stoikov derives the inventory skew from one.

**See also.** dealer inventory risk (prerequisite); adverse-selection spread component (the spread set before the next trade); Avellaneda–Stoikov quoting (a model of the inventory skew); Glosten–Milgrom model (a model of the revision after each trade); inventory-based quotes (textbook models of quotes as functions of inventory); spread determinants (why spreads differ across instruments).

**Model 16.48 Inventory-based dealer quotes: reservation prices linear in inventory (Stoll; Ho–Stoll as reported by O'Hara)**

**Statement.** In Hasbrouck's version of Stoll's one-period model, a dealer with CARA utility $U(W) = -e^{-\gamma W}$ holds $n$ shares of a security with payoff $X \sim N(\mu_X, \sigma_X^2)$, borrowing and lending at a zero rate, and posts a bid $B$ for one share that leaves him indifferent to its execution; equating certainty equivalents gives [s6-hasbrouck, sec. 11.2, eq. (11.2), p. 110]

$$B = \mu_X - (2n+1)\,\frac{\gamma\sigma_X^2}{2}.$$

If the dealer could first trade at a notional price $P$, his subjective valuation, his optimal holding would be $n^* = (\mu_X - P)/(\gamma\sigma_X^2)$, and at that optimum $B = P - \gamma\sigma_X^2/2$ [s6-hasbrouck, same sec., pp. 110–111]. The same indifference for a one-share sale gives $A = \mu_X - (2n-1)\gamma\sigma_X^2/2$ (Interpretation: own derivation; the source gives only the bid), so the quotes are

$$B = r - \tfrac{1}{2}\gamma\sigma_X^2, \qquad A = r + \tfrac{1}{2}\gamma\sigma_X^2, \qquad r = \mu_X - \gamma\sigma_X^2\, n = P - \gamma\sigma_X^2\,(n - n^*):$$

the midpoint of the quotes falls linearly as inventory rises above target, and the spread $\gamma\sigma_X^2$ does not depend on inventory.

**Notation.** $n$ is the dealer's inventory and $n^*$ his target; $r$ is the reservation midpoint; $A$, $B$ are the dealer's ask and bid. The source writes $\alpha$ for the risk aversion $\gamma$, and $\mu_x$, $\sigma_x^2$ in eq. (11.2). An alternative notation is $r_t = m_t - \kappa(q_t - q^\star)$ with spread $s_t$; here $\kappa = s = \gamma\sigma_X^2$, $q^\star = n^*$ and $m = P$.

**Conditions.** One period, a normally distributed payoff, exponential utility, one-share trades, and no information asymmetry: the quotes compensate only for being pulled off the optimal portfolio [s6-hasbrouck, same sec., p. 110]. Idea of the proof: with CARA utility and normal wealth the certainty equivalent is $\mu_W - \gamma\sigma_W^2/2$; set it equal with and without the trade, and solve for the price [s6-hasbrouck, same sec., p. 110]. With two correlated securities the bid for the first is $B_1 = \mu_1 - \frac{1}{2}\gamma\sigma_1\left[(1+2n_1)\sigma_1 + 2\rho n_2\sigma_2\right]$, lower when the securities are positively correlated, but at the optimum it is again $P_1 - \gamma\sigma_1^2/2$ and the correlation drops out, a consequence of CARA utility [s6-hasbrouck, same sec., eq. (11.3), p. 111].

**Practice.** O'Hara's account of the multiperiod model of Ho and Stoll (1981): a monopolistic, risk-averse dealer facing Poisson buy and sell arrivals whose rates are linear in his ask and bid adjustments around a fixed true price $p$, quoting $p + a$ and $p - b$ to maximise expected utility of terminal wealth over a finite horizon; the problem has no known closed form, and Ho and Stoll approximate it with a first-order expansion and quadratic utility [catalogue-0496, sec. 2.3, pp. 29–35].

As O'Hara reports them, three properties follow: the spread is the risk-neutral monopoly spread, set by the slopes of the arrival functions, plus a risk adjustment that rises with relative risk aversion, transaction size, the stock's variance and the time remaining; transaction uncertainty by itself does not affect the spread; and the spread does not depend on inventory, the dealer instead shifting both quotes (with true price 50, quotes 48 and 52 move to 47 and 51 after a purchase at the bid) by an amount set by risk aversion, the stock's risk and his wealth [catalogue-0496, same sec., pp. 35–36].

Garman's fixed-quote dealer is ruined with probability one, within days for realistic parameters, which is why quotes must depend on inventory; in Amihud and Mendelson's risk-neutral model with inventory bounds both quotes fall as inventory rises, but the spread *widens* with distance from the preferred position and the quotes are not symmetric about the true value [s6-hasbrouck, secs. 11.1.1–11.1.2, pp. 107–109].

**Pitfalls.** The model is Hasbrouck's version of Stoll (1978), and the Ho–Stoll properties are as O'Hara reports them, ho-stoll-1981. A spread independent of inventory is a feature of these models, not a general result (Amihud and Mendelson above). O'Hara notes that a fixed true price is fundamental to the Ho–Stoll analysis, and that its finite horizon, with liquidation certain at the end, implies spreads narrowing predictably through the day, which seems unrealistic [catalogue-0496, sec. 2.3, p. 36]. Empirically, quotes rising after a customer purchase fit inventory control and adverse selection alike; NYSE specialist quote changes are explained mostly by trades, with inventories adding little, and dealers manage positions more through nonpublic quotes, interdealer trading and discretion over participation than through posted quotes [s6-hasbrouck, secs. 11.4–11.5, pp. 115–117]; see component identification.

**See also.** Avellaneda–Stoikov quoting (generalisation: continuous time, with the spread also independent of inventory in its approximation); Grossman–Miller model (contrast: compensation set by market clearing); dealer inventory risk; dealer quote adjustment.

**Further reading.** [ho-stoll-1981], the multiperiod model;

## 16.11 Measuring and decomposing the spread

**Definition 16.49 Spread measures: quoted, inside, relative, effective and realised spreads**

**Statement.** With best bid $b_t$, best ask $a_t$ and midpoint $m_t = \frac{1}{2}(a_t + b_t)$:

- the *quoted spread* is $a_t - b_t$; the *inside spread* is the highest bid's distance from the lowest ask across all dealers, and is no wider than the narrowest dealer's spread [alpha-harris, chap. "Dealers", p. 280];
- the *relative spread* is the spread as a fraction of price, the only basis on which spreads of different instruments compare [alpha-harris, chap. "Bid/Ask Spreads", p. 312];
- for a trade at $p_t$ with sign $Q_t = +1$ if buyer-initiated and $-1$ if seller-initiated, the *effective spread* is twice the signed difference from the midpoint at the time of the trade, and the *realised spread* twice the signed difference from the midpoint a set interval $\Delta$ later, commonly 5, 10, 15 or 60 minutes [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", pp. 424–425]:

$$ES_t = 2Q_t(p_t - m_t), \qquad RS_t = 2Q_t(p_t - m_{t+\Delta}), \qquad ES_t = RS_t + 2Q_t(m_{t+\Delta} - m_t).$$

The last term, the signed midpoint revision after the trade, is sometimes used as an estimate of the trade's price impact [s6-hasbrouck, sec. 14.2.2, eq. (14.2), p. 148].

**Notation.** $Q_t$ is the trade sign. Harris's effective and realised spreads are doubled, full-spread amounts; Hasbrouck's *effective cost* $p_t - m_t$ and *realised cost* $p_t - m_{t+5}$ are per-side amounts written for a buy order, with the midpoint five minutes after the order's receipt, as in the SEC's Rule 605 [s6-hasbrouck, same sec., p. 148]. Harris calls the undoubled signed difference the *liquidity premium* [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 425]. Alternative symbols are $\varepsilon_t$ for $Q_t$, $s_t$ for the quoted spread and $PI_t$ for the impact term; the relative spread is measured against $m_t$.

**Conditions.** As of the 2002 and 2007 printings for the reporting rules (Rule 605 named by Hasbrouck); definitions timeless. The effective spread equals the quoted spread when every purchase is at the ask and every sale at the bid; it is smaller with price improvement and larger when large orders fill outside the quotes. Realised and effective spreads are equal if the midpoint does not move over the interval; usually it moves with the trade, so realised spreads are smaller [alpha-harris, same chap., p. 425].

When the analyst did not take part in the trades, signs are inferred, usually by the Lee–Ready rule, which overstates costs for traders who do not only take liquidity and cannot see orders filled in several trades [alpha-harris, same chap., p. 423]. Delayed trade reports bias effective costs down, and hidden orders and market-maker improvement distort the inferred sign [s6-hasbrouck, sec. 14.3.3, pp. 150–151].

**Practice.** Quoted minus effective spread measures the price improvement dealers give; effective minus realised spread measures dealers' losses to well-informed traders [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", pp. 425–426]. The realised cost is the revenue of a dealer who sold to the customer at $p_t$ and covered at the later midpoint [s6-hasbrouck, sec. 14.2.2, p. 148]. The effective spread is the least noisy benchmark estimator, because its benchmark is contemporaneous; the realised spread grows noisier the longer the interval [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 427]. The quoted spread measures the cost of immediacy for small orders only; depth and resiliency are other aspects of liquidity [alpha-harris, chap. "Bid/Ask Spreads", p. 312].

**Pitfalls.** Harris also uses *realised spread* for the difference between the prices at which a dealer actually sells and buys on a round trip, which can be negative when quotes move against the inventory [alpha-harris, chap. "Dealers", p. 280]; that is a dealer's outcome, not the midpoint-based estimator above. Mixing doubled and per-side conventions changes figures by a factor of two. Interpretation: effective minus realised spread is the permanent component only if the interval is long enough for prices to adjust and short enough to exclude unrelated news, and quote moves for inventory that have not yet reversed fall into it too; see component identification.

**See also.** spread components (what the measures estimate); Roll's estimator (alternative: a spread from trade prices alone); outside spread (contrast: an undisplayed willingness to trade).

**Result 16.50 Roll's serial-covariance spread estimator**

**Statement.** Let trade prices be $P_t = V_t + \frac{1}{2}SQ_t$, where the fundamental value $V_t$ is a random walk with independent innovations $\varepsilon_t$ of mean zero and variance $\sigma^2$, $S$ is the spread, and the trade sign $Q_t = \pm 1$ is a buy with probability one-half, independently of earlier signs and of $\varepsilon_t$ and $\varepsilon_{t+1}$. Then [alpha-harris, chap. "Volatility", box, p. 415]

$$\operatorname{Var}(\Delta P_t) = \sigma^2 + \tfrac{1}{2}S^2, \qquad \operatorname{Cov}(\Delta P_t, \Delta P_{t-1}) = -\tfrac{1}{4}S^2, \qquad S = 2\sqrt{-\operatorname{Cov}(\Delta P_t, \Delta P_{t-1})},$$

and all autocovariances of order two or more are zero [s6-hasbrouck, sec. 3.4, p. 30]. Roll's estimator replaces the covariance by its sample value [alpha-harris, chap. "Volatility", p. 415].

**Notation.** $S$ is the full spread, in price units; changes are between successive trades. Hasbrouck writes $p_t = m_t + cq_t$ with half-spread $c = S/2$, efficient-price innovation $u_t$ of variance $\sigma_u^2$, and $\gamma_0, \gamma_1$ for the variance and first autocovariance of price changes, so $\gamma_0 = 2c^2 + \sigma_u^2$, $\gamma_1 = -c^2$, $c = \sqrt{-\gamma_1}$ and $\sigma_u^2 = \gamma_0 + 2\gamma_1$ [s6-hasbrouck, sec. 3.4, eqs. (3.3)–(3.5), pp. 29–30]. Harris calls the estimate an effective spread [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 434].

**Conditions.** The spread is constant and noninformational: competitive dealers quote $m_t \pm c$ to cover a cost $c$ per trade [s6-hasbrouck, sec. 3.4, p. 29]. Idea of the proof: $\Delta P_t = \varepsilon_t + \frac{1}{2}S(Q_t - Q_{t-1})$; every cross-product has zero expectation except $\mathbb{E}Q_{t-1}^2 = 1$, which gives $-\frac{1}{4}S^2$ for the first autocovariance [alpha-harris, chap. "Volatility", p. 415]. The variance splits into a fundamental part $\sigma^2$ and a transitory part from bid–ask bounce [alpha-harris, same chap., pp. 415–416].

**Practice.** The estimator is used where quotes are unavailable [s6-hasbrouck, sec. 3.4, p. 30]; Harris advises using it only then [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 434]. For one NYSE stock (PCO, October 2003), $\hat{\gamma}_1 = -0.0000294$ gives $c = \$0.017$ and a spread of \$0.034, against a time-weighted average quoted spread of \$0.032 [s6-hasbrouck, sec. 3.4, p. 30].

**Pitfalls.** In that sample the assumptions fail: the spread ranged from \$0.01 to \$0.49, successive trade signs had correlation about 0.34, and midpoint changes were positively correlated with the latest trade sign [s6-hasbrouck, sec. 3.4, p. 30]. The biases go both ways. Positively autocorrelated signs, $\operatorname{Corr}(q_t, q_{t-1}) = \rho > 0$, give $\gamma_1 = -c^2(1 - 2\rho)$ and bias $\hat{c}$ down; signs positively correlated with the efficient-price innovation, as with informed trading, give $\gamma_1 = -c(c + \rho\sigma_u)$ and bias $\hat{c}$ up [s6-hasbrouck, sec. 4.4, Exercises 4.2–4.3, pp. 38–39]. A positive sample autocovariance is not uncommon, and estimation error alone can produce one when the model is correct; the square root is then undefined [s6-hasbrouck, sec. 4.5, p. 39]. Harris says violations typically make it underestimate the spread [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 434], and that reversals lasting beyond one trade make Roll's variance components understate transitory volatility [alpha-harris, chap. "Volatility", p. 416].

One classification lists informed trading among causes of a less negative covariance; under Hasbrouck's Exercise 4.3 it makes the covariance more negative. Price discreteness and hidden liquidity, also listed there, are not discussed in the source.

**See also.** component identification (generalisation: the generalized Roll model, with an information component); fundamental and transitory volatility (generalisation: random-walk decompositions with a general pricing error); spread measures (when quotes are available); Grossman–Miller liquidity measures (contrast: negative autocovariance from inventory risk).

#### Result 16.51 Separating transitory and permanent spread components: bounce, revisions and identification

**Statement.** The transaction-cost component makes prices bounce between bid and ask, so its price changes reverse; the adverse-selection component moves the dealers' value estimates, so its price changes are permanent and unpredictable; both depend on whether a buyer or a seller initiated the trade, and the usual decomposition explains current and future price changes with the signs of trades [alpha-harris, chap. "Bid/Ask Spreads", pp. 299–300 and 302–303]. In Hasbrouck's generalized Roll model the efficient price moves with the trade and with public news, and trades occur at a cost $c$ either side of it [s6-hasbrouck, sec. 8.2, eq. (8.1), pp. 67–68]:

$$m_t = m_{t-1} + \lambda q_t + u_t, \qquad p_t = m_t + c q_t, \qquad \text{ask} - \text{bid} = 2(c+\lambda),$$

$$\gamma_0 = c^2 + (c+\lambda)^2 + \sigma_u^2, \qquad \gamma_1 = -c(c+\lambda),$$

with all higher autocovariances zero [s6-hasbrouck, sec. 8.3, eq. (8.3), p. 68]. Two moments cannot identify the three parameters $\lambda$, $c$, $\sigma_u^2$: trade prices alone do not separate the components. Only the variance of efficient-price changes, $\lambda^2 + \sigma_u^2 = \gamma_0 + 2\gamma_1$, is identified without restrictions [s6-hasbrouck, same sec., pp. 68–69].

**Notation.** $q_t = \pm 1$ is the trade sign; $\lambda$ is the adverse-selection (information) cost and $c$ the noninformational cost, both per side; $u_t$ is nontrade public information, uncorrelated with $q_t$; $\gamma_0, \gamma_1$ are the variance and first autocovariance of $\Delta p_t$. In Harris's terms the adverse-selection component is $2\lambda$ and the transaction-cost component $2c$ (Interpretation).

**Conditions.** Timing: after the trade at $t-1$ the efficient price is $m_{t-1}$; public news $u_t$ arrives; dealers set bid and ask symmetrically about $m_{t-1} + u_t$; the trade $q_t$ arrives and the efficient price becomes $m_t$ [s6-hasbrouck, sec. 8.2, p. 68]. Setting $\lambda = 0$ gives Roll's model; setting $\sigma_u^2 = 0$ or $c = 0$ identifies the rest, but such restrictions are economically strong [s6-hasbrouck, sec. 8.3, p. 69].

**Practice.** With trade signs, analyses indicate that in most markets the adverse-selection component is larger than the transaction-cost component [alpha-harris, chap. "Bid/Ask Spreads", p. 303]. Inventory control predicts the same short-run quote response as adverse selection: after a customer buys, the dealer's quotes rise either way. The two separate in the long run, since information effects are permanent while inventory effects vanish once offsetting trades restore the dealer's position; for NYSE specialists, most quote dynamics are attributable to trades, with inventories adding little explanatory power [s6-hasbrouck, sec. 11.4, p. 115]. Identification therefore needs more than prints: trade signs and subsequent quote revisions and, to separate inventory, dealer positions.

**Pitfalls.** An estimated permanent component includes any inventory-driven quote change that has not reversed within the horizon studied, and the transitory component includes any mean reversion from uninformed order imbalances (Interpretation). Where the dealer is counterparty to every one-unit trade, $q_t = -\Delta I_t$ for inventory $I_t$ [s6-hasbrouck, sec. 11.3.2, p. 112], and a regression on current and lagged trades and inventories is then collinear; the NYSE data avoid this because the specialist does not take part in every trade [s6-hasbrouck, sec. 11.4, p. 115].

**See also.** Roll's estimator (special case: $\lambda = 0$); spread components (the theory being estimated); spread measures (a model-free split by effective and realised spreads); inventory-based quotes (the inventory alternative); propagator model (impact that decays over many trades rather than one); order-flow impact regression (the Glosten–Harris extension with trade size).

#### Result 16.52 Determinants of bid–ask spreads: primary factors, order-driven equilibrium and observable proxies

**Statement.** Three primary factors set spreads [alpha-harris, chap. "Bid/Ask Spreads", sec. 14.6.1, pp. 311–313]:

- *asymmetric information* widens them, most when informed traders know material information;
- *volatility* widens them: it raises the value of the timing option that standing limit orders and dealer quotes give away when they cannot be adjusted at once, makes diversifiable inventory risk costly to risk-averse dealers, and proxies for asymmetric information;
- *utilitarian trading interest* narrows them: active markets spread dealers' fixed costs over more trades, make inventory easier to lay off, draw public limit orders, and dilute informed flow.

In a continuous order-driven market the equilibrium spread makes the marginal trader indifferent between limit and market orders; it would be zero if traders were identical and equally informed, could cancel instantly and free, and cared neither for time nor risk [alpha-harris, same chap., secs. 14.4.1–14.4.2, pp. 304–310]. Relaxing those assumptions gives Harris's Table 14-2 [alpha-harris, same chap., p. 311]: information asymmetry (increases spreads; great importance), time to cancel limit orders (increases; high), volatility (increases; high), limit-order management costs (increases; moderate), the value of traders' time (increases; moderate), the commission difference between limit and market orders (one for one), and risk aversion (increases; low).

**Conditions.** Timeless (theory); institutional examples from 2002; the European evidence from 2008–2016. With costly limit-order management, the equilibrium spread equals the expected cost of managing limit orders plus twice the value of the timing option [alpha-harris, same chap., p. 307]; with only a commission difference, it equals that difference [alpha-harris, same chap., p. 306]. If all traders were equally uninformed, uncertainty would narrow spreads, because value estimates are less volatile than values; uncertainty widens spreads through asymmetry [alpha-harris, same chap., p. 310]. The primary factors interact: asymmetry widens spreads, which deters uninformed traders and lowers volume, which widens them further [alpha-harris, same chap., p. 312]. Public limit-order traders without dealers' costs can undercut dealers, who survive by exploiting their view of the order flow [alpha-harris, same chap., sec. 14.5, pp. 310–311].

**Practice.** Observable *secondary* determinants stand in for the primary ones [alpha-harris, same chap., sec. 14.6.2, pp. 314–317]. For asymmetric information: disclosure rules, market-condition reports, analyst and press coverage, economy-wide commodity contracts, diversified portfolios and diversified firms, established rather than emerging industries, firm age, enforced insider-trading rules, and expected material news such as earnings. For volatility: financial and operating leverage and uncertain growth for stocks; perishability, storage costs, low inventories and weather for commodities; political, inflation and interest-rate uncertainty for currencies; inflation, rates and credit quality for bonds. For utilitarian interest: trading activity (volume, number of trades), firm size, debt issue size and on-the-run status, and contracts that replicate risks many traders want to hedge.

When asymmetric information is severe or utilitarian interest small, spreads can be so wide that the market fails; small firms then finance through banks and venture capitalists, whose due diligence, restrictions and supervision rights need a private relationship [alpha-harris, same chap., sec. 14.6.3, p. 317]. Empirically, daily spreads rise with daily volatility, the spread rewarding market makers for the risk they bear, and across small-tick instruments spreads follow $\psi_i = a + b\,\sigma_i/\sqrt{N_i}$, with $N_i$ the daily number of trades [alpha-laruelle, chap. 2, secs. 2.2 and 2.3.1, pp. 144–145 and 150–151]; see spread–volatility regression.

When the spread is stuck at one tick, liquidity providers respond by changing size at the best quotes rather than the spread [alpha-laruelle, same chap., sec. 2.2, p. 148]. In crashes liquidity suppliers withdraw: in 1987 buyers waited until they believed all sellers were satisfied, and some Nasdaq dealers took their phones off the hook [alpha-harris, chap. "Bubbles, Crashes, and Circuit Breakers", pp. 561 and 563].

For large-tick instruments, Lehalle and Laruelle add a bounce term inside the same scaling by trading activity, $\psi_i = a + (b_1\sigma_i + b_2\,\eta \cdot \text{tick})/\sqrt{N_i}$, as in the spread–volatility regression (eq. (2.1)). Here $\eta = N_c/(2N_a)$ is half the ratio of price continuations to alternations in traded prices: the more bid–ask bounce, the more alternations and the smaller $\eta$, which is one half when moves are random [alpha-laruelle, chap. 2, sec. 2.2, eq. (2.1), pp. 144–145]. When the spread is stuck at one tick, liquidity providers who cannot step inside it increase the size at the best bid and offer, so a larger book plays the role of a narrower spread (p. 148). Interpretation: the tick is the constraint; bounce and book size are observed outcomes; matching rules (time priority or pro-rata allocation) govern who is filled within the larger queue (CME Globex trading conventions).

Futures often have economically large ticks, with bounce, reversals and larger displayed quantities than comparable equities; the mechanism matches these passages, which are not specific to futures. *Futures evidence.* Dayri and Rosenbaum, on millisecond data for ten futures on four exchanges from 15 May to 31 December 2009, find all of them large-tick: the spread just before a trade was one tick for 72.7 (DAX) to 99.6 (S&P 500) percent of trades.

The bounce statistic $\eta$, the same estimator $N_c/(2N_a)$, averaged 0.035 for the S&P 500 future, 0.087 for the Euro Stoxx 50, 0.12 to 0.14 for the Schatz and Bund, and 0.23 to 0.28 for the DAX, Dow, EUR/USD, five-year Treasury and crude oil contracts, all below the one half of a random walk; they read a small $\eta$ as a large tick [dayri-rosenbaum-2013, sec. 3.2, Table 1, pp. 18–19; $\eta$ estimator, p. 15]. This supports "often" for liquid 2009 futures and the bounce. It contains no equities and no depth measure, so the comparison with equities and the larger displayed quantities remain unchecked.

**Pitfalls.** Secondary factors are correlated with more than one primary factor and with each other, so an observed effect can have the opposite sign to the prediction (emerging industries attract volume), and tests need controls [alpha-harris, same chap., p. 314]. Volatility has a second, opposite effect through gambling and hedging interest, which raise activity [alpha-harris, same chap., pp. 316–317]. Trading activity is the best observable predictor of spreads, but Harris keeps it secondary to preserve the distinction between theoretical and observable determinants [alpha-harris, same chap., p. 312]. These are qualitative predictions; the importance ratings in Table 14-2 are Harris's judgement, not estimates. Harris's related factor is risk replication for hedgers. Among liquidity providers' responses to volatility, the source supports wider spreads and, in crashes, withdrawal, but not smaller displayed size or faster cancellations.

Lehalle and Laruelle report book size moving inversely to the spread over the day [alpha-laruelle, chap. 2, sec. 2.2, p. 148]; that is co-movement, not a measured response to volatility. Interpretation, consistent with Lehalle and Laruelle (pp. 145 and 148) but not stated by them: a low $\eta$ means more of the observed movement is microstructure noise, and a larger book replaces a narrower spread only for orders small enough to fit inside it.

**See also.** spread components; market collapse (a model of market failure); outside spread (the spread of value traders); dealer quote adjustment; spread–volatility regression (an empirical test of the volatility determinant); fundamental and transitory volatility.

#### Definition 16.53 Outside spread: the prices at which value traders will trade

**Statement.** A value trader's *outside spread* is the interval between the prices at which he is willing to buy and to sell; value traders rarely quote these prices, so as not to reveal their value estimates or give free trading options [alpha-harris, chap. "Value Traders", sec. 16.2, p. 340]. Writing $\widehat{V}$ for his value estimate, he buys only below $b^o$ and sells only above $a^o$, and $s^o = a^o - b^o$. Interpretation: if every cost and risk of a one-unit trade could be summarised in an all-in half-cost $c > 0$, the expected surplus would be $\widehat{V} - P - c$ from buying and $P - \widehat{V} - c$ from selling. He would trade only outside the no-trade band $[\widehat{V} - c, \widehat{V} + c]$, so $b^o = \widehat{V} - c$ and $a^o = \widehat{V} + c$. The spread widens with [alpha-harris, same chap., pp. 340–344]:

- adverse selection from news traders who know of value changes he has not yet learnt;
- the winner's curse: the risk that his estimate is wrong, larger when values are hard to estimate and when many others estimate them, with losses from the market impact of trading on a wrong estimate;
- the costs of research, which must be recovered also on the instruments analysed that offer no trade.

**Conditions.** Value trading is profitable only when price differs from fundamental value, which happens in two ways. New information changes the value: news traders, by definition the first to learn it, profit and push the price to the new value. Or uninformed traders push the price away from value: value traders profit and push it back. Value traders therefore profit only by trading against uninformed traders' demands for liquidity [alpha-harris, same chap., sec. 16.1, pp. 338–339]. They do so directly, through limit orders or blocks, or indirectly through dealers whose inventories the uninformed demand has unbalanced [alpha-harris, same chap., secs. 16.1.1–16.1.2, pp. 339–340].

Outside spreads are much wider than dealer spreads: value traders trade when order flow is one-sided and cannot learn their mistakes from it, take larger positions held until prices revert, so bear more inventory risk and financing cost, trade less volume over which to spread fixed costs, and earn at best half the outside spread when price returns to value, while a dealer often earns the whole spread on a round trip [alpha-harris, same chap., sec. 16.3, pp. 344–345].

**Practice.** Value traders buy when dealers lower their quotes to shed inventory taken from uninformed sellers, and so supply depth to the market and resiliency against uninformed price moves; dealers take larger positions where value traders follow the market closely [alpha-harris, same chap., pp. 340, 345–346]. When their value estimates differ enough, value traders' outside spreads overlap and they trade with each other [alpha-harris, same chap., p. 343]. Not buying does not oblige a trader to sell: uncertain estimates mean he should speculate only when he believes an instrument is very mispriced, and the larger the apparent mispricing, the more likely it is a mistake [alpha-harris, same chap., box, p. 343]. In continuous markets the winner's curse bites because traders trade at the prices they quote, and the market aggregates estimates sequentially; in a two-sided single-price auction it matters little [alpha-harris, same chap., pp. 342–343].

**Pitfalls.** The outside spread is not observed in quotes or trades (Interpretation: it is inferred from where value traders act). One classification lists leverage and borrowing among its financing costs; Harris names the cost of financing larger positions, without that detail [alpha-harris, same chap., p. 344]. A single half-cost $c$ understates the problem: adverse selection and the winner's curse are not per-unit costs but depend on what a trade at a given price reveals and on how many others are estimating the value, which is why Harris treats them as risks that widen the spread (Interpretation).

Potential sources of value-trader profit include uninformed liquidity demand and dealer inventory pressure. In Harris, information changes are where news traders profit; value traders gain there only by correcting news traders' under- or overreaction [alpha-harris, chap. "Informed Traders and Market Efficiency", Table 10-1, p. 230, and p. 239]. Dealer inventory pressure is the indirect channel of uninformed demand [alpha-harris, chap. "Value Traders", pp. 339–340].

Dealer and outside spreads differ in valuation target, holding period, risks (inventory and client selection against valuation, financing and the winner's curse), scale and profit capture, as in the Conditions field. Dealer spreads stay tight even when fundamental investors require a wide outside spread, which is Harris's conclusion that outside spreads are much wider than dealer spreads (sec. 16.3, pp. 344–345). Client-selection risk for dealers is Harris's adverse selection from informed traders (dealers and their quotes).

**See also.** spread measures (contrast: displayed and transaction spreads); spread determinants; dealer quote adjustment (value traders restore dealers' inventories); fundamental value; informed trading and aggregation; winner's curse; informed-trader types (value against news traders).

## 16.12 Volatility, liquidity and crashes

#### Definition 16.54 Fundamental and transitory volatility: the random-walk and transitory components of prices

**Statement.** *Volatility* is the tendency of prices to change unexpectedly. *Fundamental volatility* comes from unanticipated changes in instrument values; *transitory volatility* comes from trading by impatient uninformed traders, which pushes prices away from values until they revert. Total

volatility is the sum of the two [alpha-harris, chap. “Volatility”, pp. 410, 413–414]. In statistical form, the trade price is a random-walk efficient price plus a stationary pricing error [s6-hasbrouck, sec. 8.6, p. 72]:

$$p_t = m_t + s_t, \qquad m_t = m_{t-1} + w_t, \qquad \Delta p_t = w_t + \Delta s_t,$$

where $s_t$ has mean zero, is covariance-stationary, and may be serially correlated and correlated with $w_t$. Fundamental volatility is the variance $\sigma_w^2$ of $w_t$. If the price changes have the moving-average representation $\Delta p_t = \theta(L)\varepsilon_t$, then $\sigma_w^2 = \theta(1)^2\sigma_\varepsilon^2$, the same for every structural model consistent with the data, while the variance of the pricing error has only a lower bound [s6-hasbrouck, same sec., eqs. (8.10)–(8.13), pp. 73–74]. Over long horizons $\sigma_w^2 \approx \mathrm{Var}(p_t - p_{t-k})/k$, because microstructure effects matter less as the interval grows [s6-hasbrouck, sec. 8.3, p. 69].

**Notation.** $p_t$ is the trade price, $m_t$ the efficient (fundamental) price, $s_t = p_t - m_t$ the pricing error, $w_t$ the efficient-price innovation; $\theta(1)$ is the sum of the moving-average coefficients. Harris writes $V_t$ for value and $\varepsilon_t$ for its change. An alternative notation is $P_t = M_t + u_t$; Hasbrouck’s $u_t$ is public news in the generalized Roll model, so this entry uses $s_t$.

**Conditions.** Fundamental volatility comes only from *unexpected* changes in the factors that set values. Predictable price changes that pay holders for carrying costs and risk, such as a zero-coupon bond’s creep towards par, are not volatility; an unexpected change in rates is [alpha-harris, chap. “Volatility”, pp. 411–412]. When news is common knowledge prices can change without trading; when few know it, prices move on high volume as the informed trade; and the price revisions that make up the adverse-selection spread component contribute to fundamental volatility [alpha-harris, same chap., p. 411].

Transitory volatility includes bid–ask bounce, produced by the transaction-cost (transitory) spread component, and the price effects of large orders and cumulative uninformed imbalances that reverse when value traders and arbitrageurs act; it includes both the price change and its reversal. Value traders, arbitrageurs and dealers do not cause it but resolve it [alpha-harris, same chap., pp. 413–414]. Roll’s model is the special case $s_t = \frac{1}{2}SQ_t$ with trade signs independent of $w_t$, which gives the two-term variance $\sigma^2 + \frac{1}{2}S^2$ (see Roll’s estimator).

**Practice.** Fundamental factors differ by instrument: supply and demand, interest rates and storage costs for commodities; rates and credit quality for bonds; management, resources and technologies, product and input markets and rates for stocks; inflation, macroeconomic policy and trade and capital flows for currencies. Volatility is higher for commodities that are costly to store or perishable, especially with long production pipelines, variable demand or inelastic demand; for firms whose value depends on unresolved uncertainties, such as research outcomes or the growth behind high price–earnings ratios; for instruments exposed to political risk; and for the equity of highly leveraged firms [alpha-harris, same chap., pp. 411–413]. Because transitory price changes revert, negative serial correlation of price changes is a strong indicator of transitory volatility: over successive trades for bid–ask bounce, and over minutes to months for uninformed order imbalances [alpha-harris, same chap., p. 414].

A variance ratio $V_{M,N} = [\mathrm{Var}(p_t - p_{t-M})/M]/[\mathrm{Var}(p_t - p_{t-N})/N]$ equals one for a random walk; when microstructure effects dominate short horizons, $V_{M,N} > 1$ for $M < N$ [s6-hasbrouck, sec. 8.7, eq. (8.14), pp. 74–75]. Transitory volatility moves with uninformed traders’ transaction costs and is small in liquid markets [alpha-harris, chap. “Volatility”, p. 414]. Regulators can have no lasting effect on fundamental volatility but can change transitory volatility, in either direction [alpha-harris, same chap., p. 410]. Circuit breakers slow fundamental price changes; their net effect on transitory volatility is indeterminate, lowering it when they restrain uninformed traders, order anticipators and bluffers and raising it when they restrain informed traders, dealers and arbitrageurs, and empirical studies have been inconclusive. Margins and transaction taxes cannot discriminate between these trader types [alpha-harris, chap. “Bubbles, Crashes, and Circuit Breakers”, secs. 28.3.2–28.3.4, pp. 575–577].

**Pitfalls.** Negative serial correlation is not proof of transitory volatility: successive spot prices of a perishable commodity are prices of different items, for different delivery dates, so their negative correlation can be fundamental [alpha-harris, chap. “Volatility”, box, p. 413]. Its absence is not proof of a random walk either: lagged price adjustment makes price changes positively autocorrelated [s6-hasbrouck, sec. 8.6, Exercise 8.2, p. 74], and negative autocorrelation at some lags can offset positive autocorrelation at others to give a variance ratio near one [s6-hasbrouck, sec. 8.7, p. 75]. Roll’s variance components understate transitory volatility when reversals take longer than one trade [alpha-harris, chap. “Volatility”, p. 416].

The split of $\mathrm{Var}(\Delta p_t) = \sigma_w^2 + \mathrm{Var}(\Delta s_t) + 2\,\mathrm{Cov}(w_t, \Delta s_t)$ beyond $\sigma_w^2$ is not identified; assigning the covariance term to transitory volatility is a convention, and Harris’s “sum of the two” holds when the components are uncorrelated (Interpretation). Episodic and continuing volatility are not further sources. Harris uses episodic volatility for episodes of large price changes in short intervals [alpha-harris, same chap., p. 410], a description rather than a third source; continuing volatility does not appear in the source.

**See also.** Roll’s estimator (special case and estimator); component identification (the same split for spreads: the generalized Roll model); Garman–Klass volatility (measuring total volatility from ranges); bubbles and crashes (episodes mixing both components); spread determinants (volatility as a determinant); fundamental value (Harris’s value, conditioned on all information, not only public); returns by time scale.

#### Method 16.55 Garman–Klass volatility from open, high, low and close prices

**Statement.** Lehalle and Laruelle estimate intraday volatility over a fixed interval from its open, high, low and close prices $p^O, p^H, p^L, p^C$ and its average price $\bar{p}$ [alpha-laruelle, chap. 2, sec. 2.5.1, p. 172]:

$$\mathrm{Vol}_{GK} = \frac{\sqrt{\frac{1}{2}\left(p^H - p^L\right)^2 - (2\log 2 - 1)\left(p^C - p^O\right)^2}}{\bar{p}}.$$

1. Choose the interval. Lehalle and Laruelle use 5 minutes to show intraday seasonality and 15 minutes to compare markets [alpha-laruelle, same chap., secs. 2.2 and 2.5.1, pp. 146 and 172].
2. Record the open, high, low and close of each interval and apply the formula.
3. Average across intervals or days; the comparison of FTSE 100 and IBEX 35 components used a 20-day moving average of daily averages [alpha-laruelle, same chap., sec. 2.5.1, p. 172].

**Notation.** The superscripts $O, H, L, C$ avoid a clash with the transaction-cost component $C$. In log-ratio form,

$$\sigma_{GK} = \sqrt{\tfrac{1}{2}(\log H/L)^2 - (2\log 2 - 1)(\log C/O)^2}.$$

Since $\log(p^H/p^L) \approx (p^H - p^L)/\bar{p}$ for small ranges, the two forms agree to first order (Interpretation).

**Conditions.** Timeless (method); the applications use 2007–2016 European data. Volatility is not observed: each estimate presumes a model. Under an arithmetic Brownian price the natural estimator sums squared trade-to-trade changes, but on high-frequency data that estimator is polluted by bid–ask bounce, and with jumps they must be removed first. Garman–Klass is a measure not polluted too much by the bounce. Implied volatility measures something else, the uncertainty priced into options [alpha-laruelle, same chap., sec. 2.2, pp. 146–147].

**Practice.** Lehalle and Laruelle use it for the intraday seasonality of volatility, with volume, spread and book size, on Royal Dutch Shell, 2011–2016 (Figure 2.12, pp. 146–148), and to compare the intraday volatility of FTSE 100 and IBEX 35 components after MiFID, finding no significant effect of the UK’s fragmentation on intraday volatility [alpha-laruelle, same chap., sec. 2.5.1, pp. 172–174].

**Pitfalls.** The coefficient $2\log 2 - 1$ and the estimator’s properties come from Garman and Klass (1980); Lehalle and Laruelle give the formula without derivation or assumptions. The estimate is per interval and relative to price; scaling it to a daily or annual figure needs an assumption about

how variance adds across intervals (Interpretation). When the high and low are trade prices they include part of the spread, so on short intervals the range still carries some bounce (Interpretation).

**See also.** fundamental and transitory volatility (what the total contains); spread–volatility regression (a use of interval volatilities); Roll's estimator (the bounce that trade-based estimators pick up); returns by time scale (open-to-close and overnight returns).

**Further reading.** The original estimator [garman-klass-1980].

#### Definition 16.56 Intraday, overnight and close-to-close returns: volatility and correlation by time scale

**Statement.** For day $t$ with opening price $p_t^O$ and closing price $p_t^C$, Lehalle and Laruelle define the *intraday* and *overnight* returns [alpha-laruelle, chap. 2, sec. 2.5.1, eqs. (2.2)–(2.3), p. 175]:

$$r_t^{\text{intra}} = \log p_t^C - \log p_t^O, \qquad r_t^{\text{on}} = \log p_t^O - \log p_{t-1}^C,$$

and compute the correlation between two stocks separately on each series, over a six-month sliding window, averaging the pairwise correlations across an index's components [alpha-laruelle, same sec., pp. 174–175]. Interpretation: the close-to-close (*extraday*) return is exactly their sum, $r_t^{\text{cc}} = r_t^{\text{on}} + r_t^{\text{intra}}$. Volatility likewise has an intraday and an extraday scale: investors with extraday constraints bear extraday volatility, and intraday volatility only through its effect on that scale [alpha-laruelle, same chap., sec. 2.5, p. 169].

**Notation.** Superscripts $O$ and $C$ for open and close, as in Garman–Klass, avoid a clash with the transaction-cost component $C$.

**Conditions.** The two returns come from different market mechanisms: overnight the market is organised as an auction fixing, during the day as a continuous double auction. The authors hold that the opening fixing is less contaminated by microstructure noise than the closing fixing, and so better reflects investors' consensus on the price [alpha-laruelle, same chap., sec. 2.5.1, p. 177]. Interpretation: $\text{Var}(r^{\text{cc}}) = \text{Var}(r^{\text{on}}) + \text{Var}(r^{\text{intra}}) + 2\,\text{Cov}(r^{\text{on}}, r^{\text{intra}})$, so extraday risk mixes the two scales and their covariance.

**Practice.** On the components of the CLES 60, DAX, CAC 40 and IBEX 35, 1991–2012, correlation rises mainly on the overnight side, and overnight correlation explains well the spikes in extraday correlation, to which it is closer; intraday correlation acts as a baseline that overnight correlation keeps bouncing above [alpha-laruelle, same sec., Figures 2.27–2.28, pp. 174–177]. Intraday correlation spiked in summer 2010 and in August 2011, and for three of the indices was in October 2010 at its highest level since 1991 [alpha-laruelle, same sec., pp. 174 and 176]. The authors read intraday correlation as an intermediate marker in a contagion process by which microscopic causes have macroscopic effects [alpha-laruelle, same sec., p. 174].

**Pitfalls.** The authors attribute part of the March–October 2010 rise in extraday correlation to rising intraday correlation, the consequence of a less noisy PFP, and offer growing high-frequency activity as one possible explanation [alpha-laruelle, same sec., p. 177]: a conjecture, not a test. Daily correlation spikes can have different mechanisms at different horizons; the authors find the spikes mainly overnight. An interpretation links overnight correlation to auction and news revaluation, and overnight volatility to information arrival. The auction is in the source; news arriving overnight is an interpretation. A principle said intraday frictions matter for daily risk when they change the closing or auction price or the path used by benchmark investors.

The source asks whether changes in intraday price formation show at the extraday scale [alpha-laruelle, same chap., sec. 2.5, p. 169], and compares the noise in the two fixings (p. 177); benchmark paths are not discussed there. Interpretation: the returns depend on which prices are used, auction prints or last trades, and the open and close of a fragmented market may differ by venue.

**See also.** Garman–Klass volatility (volatility within intraday intervals; the FTSE 100 and IBEX 35 comparison of the same section); fundamental and transitory volatility (variance across horizons); price formation process.

#### Method 16.57 Spread–volatility regression in levels and logs

**Statement.** Lehalle and Laruelle test how the spread moves with volatility in five steps [alpha-laruelle, sec. A.10, pp. 292–298]:

1. Compute the spread $\psi_i$ and the volatility $\sigma_i$ over the same intervals; their example uses 15-minute intervals of Rio Tinto from January 2012.
2. Fit $\psi_i = a_0 + a_1\sigma_i + \epsilon_i$ by ordinary least squares: $\hat{a}_1 = \sum_i(\sigma_i - \bar{\sigma})(\psi_i - \bar{\psi})/\sum_i(\sigma_i - \bar{\sigma})^2$, $\hat{a}_0 = \bar{\psi} - \hat{a}_1\bar{\sigma}$.
3. Inspect the distributions: both variables looked log-normal.
4. Fit $\log\psi_i = a_0 + a_1\log\sigma_i + \epsilon_i$, which in levels is the multiplicative model $\psi = A_0\sigma^{a_1}e^{\epsilon}$, $A_0 = e^{a_0}$.
5. If $\epsilon \sim N(0, \sigma_\epsilon^2)$, then $\mathbb{E}[\psi] = A_0\sigma^{a_1}e^{\sigma_\epsilon^2/2}$, so predict levels with $\hat{\psi} = \hat{A}_0\sigma^{\hat{a}_1}e^{\hat{\sigma}_\epsilon^2/2}$, which they call the unbiased predictor.

A positive $\hat{a}_1$ is consistent with the spread rewarding market makers for the risk they take: the higher the volatility, the more they ask [alpha-laruelle, chap. 2, sec. 2.3.1, p. 150].

**Notation.** $\psi$ is the spread, $\sigma$ the volatility over the interval, $a_0$, $a_1$ the intercept and slope, $\sigma_\epsilon^2$ the residual variance. $A_0$ is the level intercept, not the adverse-selection component $A$.

**Conditions.** As of the 2008–2016 data used; method timeless. The least-squares assumptions are those of the simple linear model: the regressor measured without error; errors with mean zero, constant variance and no correlation with each other or with the regressor; normal errors for the $t$ tests [alpha-laruelle, app. A, sec. A.10.1, pp. 293–295]. Across small-tick instruments, Lehalle and Laruelle report the relation $\psi_i = a + b\,\sigma_i/\sqrt{N_i}$, with $N_i$ the average number of trades per day and $a$, $b$ shared within an asset class; for large-tick instruments, $\psi_i = a + (b_1\sigma_i + b_2\,\eta\,\text{tick})/\sqrt{N_i}$, where $\eta = N_c/(2N_a)$ is half the ratio of price continuations to alternations, smaller the more the price bounces [alpha-laruelle, chap. 2, sec. 2.2, pp. 144–145].

**Practice.** For Rio Tinto, the level fit gave $\hat{a}_0 = 2.4894$, $\hat{a}_1 = 0.0485$, $R^2 = 0.2404$; the log fit $\hat{a}_0 = 0.5609$, $\hat{a}_1 = 0.2252$, $R^2 = 0.1553$; the back-transformed predictor, with $\hat{\sigma}_\epsilon = 0.2509$, explains 0.1097 of the variance of the spread in levels [alpha-laruelle, app. A, Figures A.14, A.17 and A.18, pp. 296–298]. Daily spreads rise with daily volatility on BNP Paribas and BP, 2008–2012; points off the line in August 2011 coincide with the short-selling ban, when market makers could not meet their commitments in financial stocks [alpha-laruelle, chap. 2, sec. 2.3.1, Figure 2.13, pp. 150–151].

*Fit quality*. The total variability splits into explained and residual parts, and $R^2 = \text{EV}/\text{TV} = 1 - \text{RV}/\text{TV}$, equal in the simple model to the squared correlation $r^2$ (sec. A.10.1, p. 295). The equality of the two forms holds for least squares with an intercept; for the back-transformed predictor of step 5 they differ, and Lehalle and Laruelle's $R'^2$ (0.1097) uses the explained form $\sum(\hat{\psi}_i - \bar{\psi})^2/\sum(\psi_i - \bar{\psi})^2$ (p. 298): on simulated data the two forms give 0.2984 and 0.2984 for least squares and 0.2990 against 0.2709 for such a predictor.

*The averaging effect* [alpha-laruelle, app. A, sec. A.14.2, pp. 319–321]. For an index $I_t = \sum_k w_k S_t^k$ the spread aggregates, $\psi^I = \sum_k w_k\psi^k$, "but this relation is not true for the volatility"; so the coefficients of the spread–volatility regression on index quantities need not equal the weighted averages of the stock regressions' coefficients.

On the DAX, April to July 2012, the index regression gave $\hat{a}_0^I = 0.0283$ and $\hat{a}_1^I = 2.6641$ against weighted averages 0.0277 and 2.4734, higher "due to the index volatility which is not the weighted average of the stock volatilities" (Figure A.28, p. 321).

**Pitfalls.** The fit is descriptive: in the example volatility explains a quarter or less of the spread's variance. Unless the comparison controls for them, other variables drive both series. Harris's general warning applies: observable determinants are correlated with each other and with more than one primary factor, so tests need controls [alpha-harris, chap. "Bid/Ask Spreads", p. 314].

Tick size: when the spread is stuck at one tick, liquidity providers add size at the best quotes instead, so in large-tick assets volatility may show in depth rather than in the spread [alpha-laruelle, chap. 2, sec. 2.2, p. 148]. Time of day: spread, volatility, volume and book size share an intraday

seasonality, high after the open, so pooling intervals across the day mixes the seasonal pattern with the relation [alpha-laruelle, same sec., pp. 145–148]. Trading activity: the cross-sectional relation scales volatility by $1/\sqrt{N_i}$ [alpha-laruelle, same sec., pp. 144–145].

The source does not discuss venue and depth as regression controls. Causation runs both ways, since a wider spread adds bounce to trade-based volatility (Interpretation). The predictor is unbiased only with known $a_0$, $a_1$ and $\sigma_\epsilon^2$; with estimates plugged in it is approximately unbiased (Interpretation). The underlying mechanism is that market makers widen spreads when volatility raises adverse-selection and inventory risk, and the relation is cleaner when volatility is scaled by trading activity and the tick is accounted for; Lehalle and Laruelle summarise that "High volume combined with low volatility and a tight bid–ask spread characterize a liquid stock" [alpha-laruelle, chap. 2, sec. 2.3, p. 149].

The spread identity holds for spreads in price units, since the index's best ask and bid are the weighted sums of the components'; for spreads in basis points the index spread is a price-weighted, not a $w$-weighted, average (3.96 against 4.73 bp on simulated components). The text defines $\bar{a}_1 = \sum_k w_k \hat{a}_0^k$ (p. 320), a printing slip for $\sum_k w_k \hat{a}_1^k$.

Index volatility in price units is at most $\sum_k w_k \sigma^k$ for nonnegative weights, with equality only under perfect correlation (1.598 against 2.376 at correlation 0.4). The index regression's slope can therefore be steeper than the stock slopes' average even when every stock follows the same spread–volatility relation: in a simulation with a common relation of slope 2.5 the stock slopes average 2.507 and the index slope is 3.108. If the index volatility were a fixed fraction $c < 1$ of $\sum_k w_k \sigma^k$, the index slope would be the common slope divided by $c$ (derived).

**See also.** spread determinants (the theory being tested: volatility as a primary determinant); Garman–Klass volatility (an interval volatility); spread measures (which spread to regress).

#### Mechanism 16.58 Bubbles and crashes: how mispricing builds and unwinds

**Statement.** A *bubble* is a rise of prices substantially above fundamental values, quickly or over a long period; a *crash* is a very fast fall, often but not always after a bubble. Since fundamental values are not common knowledge, bubbles and crashes can occur at all [alpha-harris, chap. "Bubbles, Crashes, and Circuit Breakers", sec. 28.1, p. 556]. The typical sequence [alpha-harris, same chap., sec. 28.1.1, pp. 556–557]:

1. Buyers grow overly optimistic about new technologies or markets, often without recognising what prices already reflect, and push prices up.
2. Momentum traders and order anticipators buy and accelerate the rise; early buyers grow confident and early sellers buy back to stop losses.
3. Value traders and arbitrageurs cannot sell enough (no position to sell, too little capital to carry a short, short selling hard) or will not (they expect uninformed buying to continue, or doubt their own value estimates).
4. Sellers start trading aggressively: long-term holders, early buyers taking gains, contrarians, value traders, arbitrageurs. Momentum buyers lose interest and optimists lose confidence.
5. Late buyers sell to stop losses, margin calls and stop orders force more selling, and order anticipators sell ahead of them: prices fall quickly. Panic can take prices below value, and a bounce follows.

*The price accelerator* (Harris, after a manuscript by Treynor): rising prices move wealth from pessimists with short positions to optimists with long ones; the richer optimists may buy more, and the losing pessimists may buy to cover margin calls, so prices rise further; the sellers are mild optimists and mild pessimists [alpha-harris, same chap., box, p. 556].

**Conditions.** Timeless (mechanism); the case material is historical, to 2002. Most bubbles and crashes involve both fundamental and transitory volatility. A bubble usually starts on good news, a fundamental change; pseudo-informed traders then overreact, the transitory part. It usually bursts on bad news, but the volume and price response is out of proportion to the news because of the overpricing that preceded it; the reversion towards value counts as transitory volatility, since it occurs only because uninformed traders inflated prices, and panic below value adds more [alpha-harris, same chap., sec. 28.1.2, pp. 557–558].

**Practice.** Harris's 1987 account shows the mechanisms together: portfolio insurance traded like stop orders, and uncertainty about how much of it was waiting kept buyers out; NYSE order printers could not keep up, so traders did not know whether their orders had filled; index futures led the cash market down and traded at a discount of more than 10 percent once index arbitrageurs stopped; and buy-side traders could not reach Nasdaq dealers [alpha-harris, same chap., sec. 28.2.2, pp. 559–563]. Crashes rarely have a single cause: traders become confused about risk, attend to the wrong issues and, when events move fast, panic [alpha-harris, same chap., sec. 28.2.8, p. 571].

**Pitfalls.** A large fall without news of commensurate importance does not show that the market failed. Most crashes correct earlier pricing errors; if they were failures, prices would rebound fully once traders saw them below value, but after most crashes prices rebound little [alpha-harris, same chap., box, p. 559]. Attribution therefore looks at the reversal after the event, at what preceded it, and at overloaded systems, forced selling and broken links between markets, as in 1987 (Interpretation, from the cases on pp. 557–563). Bubbles are hard to identify in real time, and analysts generally understand the causes of extreme volatility better after the fact [alpha-harris, same chap., p. 555]. Among the stories that start a bubble, Harris's general account names technology and new markets, and his Japanese example turns on loose monetary policy [alpha-harris, same chap., sec. 28.2.7, pp. 570–571].

**See also.** fundamental and transitory volatility (the two components in each phase); outside spread (why value traders wait); spread determinants. See also the toy model of the Flash Crash and the May 2010 Flash Crash.

## 16.13 High-frequency trading: strategies, speed races and the costs of other traders

#### Definition 16.59 High-frequency trading: proprietary strategies, the speed tools they share, and why the label names no single mechanism

**Statement.** In the SEC's 2010 concept release the term "high frequency trading" is "relatively new and is not yet clearly defined". It typically refers to professional traders acting in a proprietary capacity whose strategies generate many trades a day, organised as proprietary trading firms, proprietary desks of broker-dealers or hedge funds [sec-concept-release-2010, sec. IV.B, p. 45]. Five characteristics are often attributed to them: (1) high-speed programs for generating, routing and executing orders; (2) co-location and individual data feeds to cut latency; (3) very short holding periods; (4) many orders cancelled shortly after submission; (5) ending the day close to flat (p. 45; restated in [alpha-laruelle, sec. 2.4.3, p. 163]).

Rather than define HFT, the release treats *strategies* and *tools* separately. One firm may run many strategies at once, some generating many trades and some not, and "conceivably" some benefit market quality and long-term investors while others harm them [sec-concept-release-2010, sec. IV.B, p. 46]. To present issues for comment, "without limiting the broad request", it describes four broad types of strategy often associated with proprietary firms [sec-concept-release-2010, sec. IV.B.1, pp. 48–57]:

- *passive market making*: mainly non-marketable resting bids and offers, profiting from the spread and from liquidity rebates, and taking liquidity only to liquidate a position quickly. Layering many orders can give cancellation rates of 90 percent or more, with orders often lasting a second or less (pp. 48–49);
- *arbitrage*: capturing price discrepancies between related products or markets, such as an ETF against its basket, mostly by taking liquidity, with hedged positions that may last several days. The release asks how far such strategies trade price differences between trading centres and depend on latency between direct and consolidated data feeds (pp. 51–52);
- *structural*: exploiting structural vulnerabilities in the market or in other participants. Examples are using the fastest data to trade with participants still offering executions at stale prices, and setting a new NBBO with a small order and then triggering guaranteed-match trades on the other side (pp. 52–53);

- *directional*: a significant, unhedged position on an expected intraday move. Some, such as trading a temporary departure from fundamental value, "often may contribute" to price discovery. The release singles out two: *order anticipation*, detecting large buyers or sellers and trading ahead of them, and *momentum ignition*, orders and trades, perhaps with false rumours, meant to spark a rapid price move or trigger stop orders, followed by unwinding the position (pp. 53–57).

Many of these strategies are not new; what is new is the technology for identifying and executing them (p. 48). The tools discussed separately are co-location and individual trading-centre data feeds. Speed matters "both in the absolute sense of achieving very small latencies and in the relative sense of being faster than competitors" (sec. IV.B.2, pp. 57–58).

**Conditions.** U.S. equities as of January 2010; European venues as of the 2018 printing; Narang's account as of 2024; FINRA Rule 5270 as displayed on 15 October 2026; historical figures are dated as given. Interpretation: speed and automation are means shared by these strategies, not a strategy, and the economic mechanism differs between them. Passive market making is paid the spread for supplying immediacy and bears inventory and adverse-selection risk (Avellaneda–Stoikov, dealer quote adjustment). Arbitrage enforces relations between prices (arbitrage). Trading against stale quotes earns rents from reacting first to public information (latency-arbitrage arms race). Order anticipation profits from information about others' orders rather than about values (order anticipation). Momentum ignition is a form of bluffing (bluffing and manipulation). Effects on liquidity, price discovery and other traders' costs must therefore be judged strategy by strategy.

The release's four types organise a request for comment on U.S. equities in January 2010; they are not a universal classification, and they overlap: trading against stale prices is filed under "structural" and cross-venue price differences under "arbitrage".

**Practice.** Lehalle and Laruelle describe high-frequency market makers in European equities around 2009–2012 as arbitraging between the venues that quote the same stock, which keeps those prices consistent. Maker–taker fee schedules on MTFs reward this, and trading very small orders very often lowers their inventory risk [`alpha-laruelle`, secs. 2.4 and 2.4.3, pp. 158 and 163]. Venues court them with low latency and co-location, "especially since the lower the latency, the smaller the inventory risk"; the authors read one venue's low "flow reactivity" as its high-frequency market makers offering less liquidity when activity rises (p. 158). Probing orders are a tool, not a strategy. The release defines a "pinging" order as an immediate-or-cancel order used to find undisplayed liquidity, and distinguishes its use in a normal search for liquidity from its use to detect and trade in front of large interest [`sec-concept-release-2010`, p. 47, n. 69].

An order anticipator that detects a large buyer first buys, and is largely indifferent to whether the large trader is buying or selling; a long-term investor seeking liquidity wants the opposite side, and both parties benefit from that search [`sec-concept-release-2010`, p. 55]. Liquidity seeking by routers is in smart order routing. Narang groups high-frequency strategies as contractual and non-contractual market making, arbitrage and *fast alpha*. Fast alpha applies the price-based alphas of slower strategies, momentum, mean reversion and technical sentiment, to prices, volumes and order-book information that change during the day; its statistical relationships carry risk because "there is no structural reason" they must persist, and intraday trend following is less common than reversion because impact and slippage work against it [`s7-narang`, chap. 15, "High-Frequency Trading", sec. 15.4, pp. 291–293].

Execution algorithms also adjust their aggressiveness to the order book's imbalance, a "micro price" (chap. 7, pp. 133–134), and one that leaves a detectable footprint lets other participants react "to increase the market impact and slippage" it incurs (p. 130). *Fragmentation and roles*. Lehalle and Laruelle call HFT "the price to pay for fragmentation; it is not possible to put trading venues in competition without agents building high-frequency liquidity bridges across them", and describe high-frequency traders as "acting as market-makers but in most cases without obligations", which blurred the roles of each layer of the market [`alpha-laruelle`, Introduction, pp. 5–6]; HFT updates books faster than a non-co-located observer's round trip, so an order's role as provider or taker is known only in probability (p. 22).

The release's characteristics describe technology and horizon, not whether a strategy is beneficial.

**Pitfalls.** Five strategies are commonly named: passive market making, statistical arbitrage, latency arbitrage, order anticipation and liquidity detection. Compared with the release:

- "statistical arbitrage" on short-horizon relative value or lead–lag signals straddles the arbitrage and directional types;
- "latency arbitrage" corresponds to the structural example of trading against stale prices, and across venues to the release's arbitrage question;
- "liquidity detection" is a tool that order anticipators and ordinary liquidity seekers both use, not a strategy;
- momentum ignition, which the release names, was missing.

The statement, that one firm may run several strategies whose risk and social value differ, matches p. 46, where the difference in social value is a possibility to assess, not a finding. A strategy label does not establish illegality. The release treats order anticipation that involves no breach of duty or misappropriation as a question of policy; front running in breach of a duty, false rumours and manipulative conduct were already prohibited (pp. 53–55 and 57). These legal remarks are as of January 2010 and are not statements of current law. The release's estimate that HFT was 50 percent or more of U.S. equity volume came from press reports (p. 45, n. 67).

Lehalle and Laruelle's "more than 80% of the deals in the US, 40% in Europe, and 35% in Japan" is reported as hearsay ("They are said to be", p. 163) without source or date. Neither is a measured share. *Trade counts as a proxy*. Lehalle and Laruelle read a rising number of trades with a falling average trade size as evidence of growing high-frequency activity: "the number of trades per day is a good proxy to get an idea of how intense high frequency activity is on a stock" (FTSE 100 and CAC 40 rising, IBEX 35 flat, 2007–2012) [`alpha-laruelle`, sec. 2.5.1, pp. 170–171, Figure 2.25; introduction, pp. 20–21, Figure 5].

It is a proxy, not a measure: the same count can reflect genuine liquidity, order slicing or stress-induced churn, as the 2010 staff report warns that "high trading volume is not necessarily a reliable indicator of market liquidity" (the May 2010 Flash Crash). Microstructure alphas on bets of seconds or less correspond to Narang's fast alpha. Neither Narang's third edition nor the other sources here show that machine learning can discern other players' execution patterns, more easily the worse their execution models, or that such patterns are stable.

The "HFT shark strategy", sending very small orders to detect a large iceberg order, then trading in front of it and exiting "with a quick and relatively riskless profit", is also not in the third edition; it is pinging used for order anticipation, as the release describes (p. 47, n. 69; p. 55).

Three qualifications. (i) "Front-run" is used loosely: Narang defines front-running as a fiduciary using knowledge of a customer's order before acting on it, and argues that most high-frequency traders, who see orders only once they reach the market, do not front-run in that sense, except contractual market makers, who see customer flow and for whom it is "theoretically possible" (p. 312) [`s7-narang`, Appendix, sec. A.2.1, pp. 312–314]; FINRA's rule bars members from trading with "material, non-public market information concerning an imminent block transaction" [`finra-rule-5270`, para. (a)], and inferring a hidden order from fills is not obviously that information on the rule's wording (Interpretation; the term is not defined there, and the rule's supplementary material notes that other rules, such as Rules 2010 and 5320, may apply to front-running other orders). Whether pinging for order anticipation is lawful in other jurisdictions is not covered here.

(ii) The profit is not riskless: the anticipator bears the risk that the inferred order is not there, is smaller or ends, and that the price does not move (Interpretation; order anticipation). (iii) The shark strategy is a stylised description, not a supported mechanism.

**See also.** latency-arbitrage arms race; investor cost; order anticipation and bluffing and manipulation; arbitrage; smart order routing.

**Further reading.** Menkveld's survey of high-frequency trading (`wide-menkveld-hft-review`).

**Result 16.60 Latency arbitrage in a continuous limit order book: stale quotes are sniped, the speed race dissipates the rents, and investors pay through the spread (Budish, Cramton and Shim)**

**Statement.** Budish, Cramton and Shim's model [wide-budish-cramton-shim, secs. VI.A–VI.B, pp. 1576–1583]. A security $x$ trades on a continuous limit order book. Its value is perfectly correlated with a public signal $y$, which jumps as a compound Poisson process with rate $\lambda_{\text{jump}}$ and absolute jump size $J$, and $x$ can always be liquidated at that value. Investors arrive at rate $\lambda_{\text{invest}}$, each needing to buy or sell one unit, and take liquidity. Risk-neutral trading firms make and take liquidity, and messages that arrive together are processed serially in random order. In equilibrium exactly one unit is quoted on each side, a bid at $y - s/2$ and an ask at $y + s/2$; the two quotes may belong to one trading firm or to two different firms (Proposition 1(i), p. 1583).

The authors describe play with one firm in the liquidity-provider role for simplicity, noting economically equivalent equilibria in which the role rotates or is random (pp. 1579–1580). When $y$ jumps by more than $s/2$, the firm whose quote is stale sends a cancel while the other $N - 1$ firms send orders to trade against ("snipe") it, so the stale quote is sniped with probability $(N-1)/N$ (Proposition 1(iii), p. 1583). The spread equates investors' payments with the rents from sniping:

$$\lambda_{\text{invest}} \frac{s}{2} = \lambda_{\text{jump}} \Pr\left(J > \frac{s}{2}\right) \mathbb{E}\left[J - \frac{s}{2} \,\middle|\, J > \frac{s}{2}\right]$$

(eq. (3), p. 1582), which does not depend on $N$. The spread is positive with symmetric information and no inventory costs. Sniping costs grow linearly with the quantity quoted, while the benefit of quoting more grows less than linearly, so the spread widens unit by unit and the book is thin (Proposition 2, pp. 1583–1585). This assumes investors trade their whole quantity at once, which the authors note is not optimal for them. When firms can rent speed at $c_{\text{speed}}$ per unit time, the zero-profit conditions for the liquidity provider and the snipers (eqs. (5)–(6), p. 1589) give

$$\lambda_{\text{invest}} \frac{s^*}{2} = N^* c_{\text{speed}}$$

(eq. (7), p. 1590): the total prize, investors' cost through the spread and the total spent on speed are equal, so the rents are dissipated by the race (Proposition 3, pp. 1590–1591). The race is a prisoner's dilemma: welfare would be higher by $N^* c_{\text{speed}}$ if the firms could commit not to invest (Proposition 4, p. 1591). The prize rises with the frequency of jumps and with a mean-preserving spread of their distribution, and does not depend on the cost of speed, the speed of the fastest firms or the speed difference (Proposition 5, p. 1592).

**Notation.** $\lambda_{\text{jump}}, \lambda_{\text{invest}}$ are Poisson arrival rates, not impact slopes; $s$ is the full bid–ask spread in price units; $N$ is the number of fast trading firms; $c_{\text{speed}}$ is a rental cost per unit time. Starred values are equilibrium values with endogenous entry.

**Conditions.** The assumptions are deliberately strong, a "best case" for the continuous book: a perfectly correlated public signal, costless liquidation, no asymmetric information or inventory costs, unit investor demand, prices on a continuum, and equally fast fast firms (pp. 1576–1579). With a discrete tick the liquidity-provider role is strictly preferred and firms race for the top of the book instead (n. 19, p. 1580). The authors do not claim that markets are less liquid than before electronic trading and HFT, which they say the evidence contradicts. Their claim is that markets are less liquid than under a design without sniping, the frequent batch auctions they propose (n. 5, pp. 1554–1555; abstract, p. 1548). Evidence (sec. I, pp. 1552–1553, summarising sec. V): ES futures and SPY, 2005–2011, exchange direct-feed data.

In 2011 the median return correlation was 0.1016 at 10 ms and 0.0080 at 1 ms. The median duration of ES–SPY arbitrage opportunities fell from 97 ms in 2005 to 7 ms in 2011, while the median profit, about 0.08 index points per unit traded, was roughly constant, and their frequency moved with volatility. The ES–SPY prize averaged $75 million a year. The authors call extrapolation to other instruments "common sense" and decline to give a total.

**Practice.** With speed costs that differ across firms, the marginal fast firm earns zero profit and inframarginal firms earn strictly positive profits; total sniping rents then exceed total spending on speed (n. 27, p. 1590). Lehalle and Laruelle describe a related competitive margin. High-frequency market makers extended coverage to less liquid stocks once their hedging was built and their technology, a fixed cost, was paid; growing competition between them "forced them to diversify their sources of revenue". Below a liquidity threshold, hedging and short-selling costs keep them out [alpha-laruelle, sec. 2.4.2, pp. 161–162]. This is their interpretation of coverage statistics for European indexes, 2009–2012, not a test.

**Pitfalls.** Speed does not lose its value once competitors can replicate it. In this model and its ES–SPY evidence, competition does not shrink the arbitrage prize; it raises the speed needed to win it, and under free entry the rents go into speed spending, which investors bear through the spread. What erodes is a firm's *relative* lead: at the margin, profit net of speed costs is zero (Interpretation of eqs. (5)–(7)). Lower costs of speed do sustain profits (n. 27). These sources do not show durable profits from inventory control, better models or privileged order flow. The empirical constancy is for one pair, 2005–2011.

**See also.** HFT strategies; spread against investor cost; Glosten–Milgrom (adverse selection from private information, against sniping on public information); arbitrage; smart order routing (latency and mirrored liquidity).

**Further reading.** The frequent-batch-auction analysis is in secs. VII–VIII of the paper.

**Result 16.61 A narrower spread need not mean lower costs: Lehalle and Laruelle's stylised model of a fast trader quoting inside the book**

**Statement.** Lehalle and Laruelle's stylised book [alpha-laruelle, sec. 2.4.3, pp. 164–165, Figure 2.21]. Two traders, A and B, rest at the best bid and ask, a spread $\psi$ apart, and take turns becoming impatient. Each pays $\psi$ when crossing and nothing when the other crosses, so each pays $\psi/2$ per trade on average. A high-frequency trader then quotes just inside B, so that A, crossing, pays $\psi/2 + \epsilon$; it at once steps in front of A, and B pays $\psi/2 + \epsilon$ too. The effective spread falls to a little over $\psi/2$, but A and B now pay $\psi/2 + \epsilon$ on every trade, because each must cross every time, while the fast trader pays nothing. In the authors' words, "the effective spread is reduced on an overall basis, but … costs for other participants can remain unchanged or slightly increased" (p. 164).

In the calibrated version, $\psi_B$ is the spread before HFT and $\psi_{\text{HFT}}$ the tighter one an HFT quotes. An investor hitting an HFT pays $\psi_B/2 + \epsilon$ with $\epsilon = \psi_{\text{HFT}}/2$, and if HFTs are in a fraction $q$ of trades the spread becomes $\psi_N = q\psi_{\text{HFT}} + (1-q)\psi_B$ (p. 165). An investor posting one order meets an HFT with probability $r$ and then pays $(\psi_{\text{HFT}} + \psi_B)/2$. Otherwise, with equal probability, the investor crosses and pays $\psi_B$ or stays passive and pays 0, so

$$\mathbb{E}[C] = \frac{\psi_B + r\,\psi_{\text{HFT}}}{2}, \qquad \text{Var}(C) = r\left(\frac{\psi_{\text{HFT}} + \psi_B}{2}\right)^2 + (1-r)\frac{\psi_B^2}{2} - \mathbb{E}[C]^2$$

(Figure 2.22, p. 166). The average cost of $N$ independent trades has the same mean and variance $\text{Var}(C)/N$ (p. 168). Meeting HFTs more often raises the expected cost and lowers its dispersion. The dispersion falls a lot for an investor with few trades, such as a retail investor, and little for one with many, such as a pension fund (pp. 167–169, Figures 2.23–2.24).

**Notation.** $\psi$ is a spread in basis points of price; $C$ is one order's cost. Interpretation: in the figure each trader's cost is measured from its own resting quote, so a passive fill costs 0 and a crossing the full spread. $C$ is therefore a cost against waiting at one's own quote, not the half-spread against the midpoint of spread measures.

**Conditions.** Calibration: $\psi_B = 5.41$ bp (April to November 2007) and $\psi_N = 4.73$ bp (April to November 2011), for CAC 40 stocks on the primary market, Euronext Paris, only, with $q = 40$ percent assumed (pp. 165–166). Interpretation: these imply $\psi_{\text{HFT}} = (\psi_N - (1-q)\psi_B)/q = 3.71$ bp. $\mathbb{E}[C]$ then runs from 2.705 bp at $r = 0$ to 4.56 bp at $r = 1$, which matches the ends of Figure 2.24, and is 3.26 bp at $r = 30$ percent and 3.63 bp at

$r = 50$ percent. The model attributes the whole fall in the primary-market spread between 2007 and 2011 to HFT and holds $\psi_B$ fixed, as if the earlier spread would otherwise have persisted.

It fixes $q$ rather than measuring it, ignores trading moved to other venues, and assumes that an investor who meets an HFT must cross while others cross half the time. The two spreads are observations; their attribution to HFT and the cost model are the authors' stylised interpretation, not an estimate of HFT's effect.

**Practice.** Judge fast liquidity provision by what other traders pay (effective and realised spreads, spread measures), by depth, adverse selection and behaviour under stress, not by quoted spreads or message counts. The SEC's 2010 release asked these as open questions. Are orders lasting a second or less "phantom liquidity that disappears when most needed", or do many firms quoting together give a stable market with many size updates but few changes in the NBBO price? Do proprietary firms, which have replaced specialists and market makers with affirmative and negative obligations, supply liquidity in times of stress [sec-concept-release-2010, sec. IV.B.1, pp. 48–51]? In Budish, Cramton and Shim's model the cost of being sniped is priced into the spread and grows with the depth shown (latency-arbitrage arms race).

Lehalle and Laruelle add that HFTs' preference for small trades makes other traders split orders and bear more reporting and technology costs, "fixed losses" unevenly distributed (p. 165); this is an argument, not a measurement.

**Pitfalls.** The text says that raising the probability of meeting an HFT from 30 to 50 percent "increases the expected cost by approximately 0.5bp" (p. 168). The book's own formula and calibration give $0.2 \times 3.71/2 = 0.37$ bp, and Figure 2.24's markers sit at about 3.27 and 3.64 bp, which agrees with 0.37 (Interpretation). The book's "by 20%" means 20 percentage points. That high message rates improve displayed prices but raise cancellation risk and stale-book uncertainty is, in these sources, a stylised mechanism and an open question, not a measured effect. The model's conclusion concerns expected cost per trade under its assumptions; it is not a finding that HFT raised investors' costs.

Interpretation: the effect differs by investor size, since a small investor bears high variance over a few trades, while a large one diversifies it across child orders but still pays the aggregate cost.

**See also.** HFT strategies; latency-arbitrage arms race; spread measures and spread determinants; smart order routing (displayed liquidity is only probable); order anticipation (quote matching steps ahead of passive traders).

#### Result 16.62 The value of queue position: speed buys priority, and in large-tick books a place at the front is worth a sizeable fraction of the spread (Moallemi and Yuan)

**Statement.** Under price–time priority, a resting order's value depends on its place in the first-in-first-out queue at its price. Moallemi and Yuan's model [wide-moallemi-yuan-queue, author working paper, June 2017 revision, abstract and sec. 1, pp. 1–4] identifies two components:

- a *static* component, the trade-off at execution between earning the spread and bearing adverse selection, "and incorporates the fact that adverse selection costs are increasing with queue position": an order at the back of a long queue executes in the next instance only against a large trade, and large trades come more often from informed traders, so "a good queue position acts as a filter on the population of contra-side market orders" (after Glosten 1994; pp. 1 and 3);
- a *dynamic* component, "the optionality associated with the future value that accrues by locking in a given queue position", as the order moves up after trades and cancellations (pp. 1 and 4).

Speed is how positions are won: "orders that are emitted faster will wind up in a better queue position, and this is especially important when traders are reacting to common signals", notably right after a price change, when there is "a race to establish queue positions at the new price" (p. 3).

**Conditions.** The model is for *large-tick* assets, those for which "the bid-ask spread rarely exceeds the minimum tick size" (after Eisler et al., 2012): moving the price by a tick is costly, so traders queue rather than improve the price, and queues are long; for small-tick assets priority can be bought with a better price and queue position matters less (sec. 2, p. 6). It assumes a constant one-tick spread over the model's time scale (p. 6), and it does not apply to pro-rata markets, where queue position is irrelevant (n. 2, p. 2). Evidence: calibrated to Bank of America on Nasdaq over 22 trading days of August 2013 (Table 1, p. 19), and compared with backtests of artificial orders in nine large-tick stocks and ETFs (Table 2, August 2013; Table 3 reports averages over 30 trading days, p. 23).

In every symbol the front of the queue is worth more than the back; the model's gap between an order at the front ("touch value") and one placed at the back ("order value") is 0.22 ticks for BAC, 0.16 for CSCO, 0.17 for INTC, 0.11 for GE and F, 0.14 for EMM and 0.03 to 0.06 for PFE, PBR and EFA (Table 3). The authors call the gap for BAC and CSCO "very large and comparable to the bid ask spread (>0.1 ticks)" (p. 24), and in the abstract "of the same order of magnitude as the bid-ask spread".

**Practice.** Queue position matters when replaying a quoting policy: an order replayed without its queue position overstates its fill probability and understates its adverse selection when it would in fact have joined the back of a long queue (Interpretation of the two components). The authors conclude that queue value "rewards the trading speed of high-frequency firms", disadvantaging traders without fast technology, and ask whether price–time priority suits large-tick assets (sec. 6, p. 25). Budish, Cramton and Shim note the same race for the top of the book when the tick is discrete (latency-arbitrage arms race); Harris's time precedence with a meaningful tick protects liquidity suppliers against quote matchers (order anticipation).

**Pitfalls.** Speed buys queue priority, and priority is most valuable where information matters and after a price change, when the race to the new level begins. It is not most valuable in thin queues: positional value is large where queues are long, in large-tick books, since in a thin queue every position is near the front (Interpretation of pp. 3–6). The paper is an author working paper; figures are from the June 2017 revision.

**See also.** queue imbalance; queue-reactive intensities; order matching and trade pricing (time and pro-rata priority); tick size; make/take fees and the speed race.

**Further reading.** Garriott and others on queuing (w3-garriott-queuing).

#### Model 16.63 Make/take fees and the speed race: high-frequency market makers and investors invest in speed, and the fee split that maximises trading (Foucault, Kadan and Kandel, as summarised by Lehalle and Laruelle)

**Statement.** Lehalle and Laruelle summarise a model of Foucault, Kadan and Kandel [alpha-laruelle, app. A, sec. A.2, pp. 250–253]. There are $M$ high-frequency market makers and $N$ investors (market takers), modelled as buyers. An investor values the asset at $v_0 + \Gamma$; a market maker sells at $v_0 + s\Delta$, a spread of $s$ ticks of size $\Delta$; the venue charges $c_T$ to takers and $c_M$ to makers. Per trade the maker earns $\pi_M = s\Delta - c_M$, the investor $\pi_T = \Gamma - s\Delta - c_T$ and the venue $\bar{c} = c_M + c_T$. Market maker $m$ reaches the book at Poisson rate $\mu_m$ and investor $n$ at rate $\tau_n$; with the aggregate rates

$$\bar{\mu} = \sum_{m=1}^{M} \mu_m, \qquad \bar{\tau} = \sum_{n=1}^{N} \tau_n, \qquad \mathcal{R} = \frac{\bar{\mu}\,\bar{\tau}}{\bar{\mu} + \bar{\tau}}$$

trades per unit time. Market makers race among themselves and investors among themselves, but not across sides. With quadratic costs of speed,

$$\Pi_M(m) = \frac{\mu_m}{\bar{\mu}}\,\pi_M\,\mathcal{R} - \tfrac{1}{2}\beta\mu_m^2, \qquad \Pi_T(n) = \frac{\tau_n}{\bar{\tau}}\,\pi_T\,\mathcal{R} - \tfrac{1}{2}\gamma\tau_n^2, \qquad \Pi_E = \bar{c}\,\mathcal{R},$$

and, once each participant has optimised its speed, the maker fee and taker fee satisfy

$$c_M^\star = s\Delta - \frac{\Gamma - \bar{c}}{1 + \left(\frac{M}{N} \cdot \frac{\gamma}{\beta}\right)^{1/3}}, \qquad \bar{c} = c_T^\star + c_M^\star$$

(eq. (A.4), p. 253). With few market makers the efficient maker fee can be negative (a maker rebate); with many market makers relative to investors, or when investors' speed is costlier ($\gamma$ large), the efficient taker fee can be negative (a taker rebate) (p. 253, Figure A.2).

**Notation.** $s\Delta$ the spread in currency; $\Gamma$ an investor's gain from trade; $\pi_M, \pi_T$ net gains per trade; $\mu_m, \tau_n$ arrival rates (per unit time), $1/\mu_m$ and $1/\tau_n$ times to market; $\bar{\mu}, \bar{\tau}$ *sums* of rates; $\mathcal{R}$ the trade rate; $\beta, \gamma$ cost coefficients of speed; $\mu_m/\bar{\mu}$ and $\tau_n/\bar{\tau}$ shares of the races.

**Conditions.** *Derivation*. Taking $G = \Gamma - \bar{c} = \pi_M + \pi_T$ as fixed and letting each participant ignore its own effect on the aggregate rates (large $M$ and $N$), the first-order conditions give $\bar{\mu}(\bar{\mu} + \bar{\tau}) = M\pi_M\bar{\tau}/\beta$ and $\bar{\tau}(\bar{\mu} + \bar{\tau}) = N\pi_T\bar{\mu}/\gamma$; maximising $\mathcal{R}$ over the split $\pi_M$ then gives $\pi_M/\pi_T = (M\gamma/(N\beta))^{-1/3}$, that is $\pi_M = G/(1 + (M\gamma/(N\beta))^{1/3})$, which is eq. (A.4). Because $\bar{c}$ is fixed, the same split maximises the venue's revenue $\bar{c}\mathcal{R}$. Lehalle and Laruelle do not define "efficient"; on this derivation it means trade-maximising for a given total fee, not welfare-maximising, since speed costs are not netted (Interpretation).

*Against the original.* Foucault, Kadan and Kandel fix the platform's total fee $\bar{c}$ and choose the split to maximise $\Pi_e = (c_m + c_t)\mathcal{R}$ subject to $c_m + c_t = \bar{c}$, so the first-order condition equalises the marginal effects of the two fees on the trading rate (sec. 4, eqs. (18)–(19), p. 23); in the large-market limit ($M, N \to \infty$ with $M/N = q$) Proposition 4 gives $c_m^* = \frac{1}{2}\left(\Delta - 2(L - \bar{c})/(1 + (qr)^{1/3})\right)$, $c_t^* = \bar{c} - c_m^*$ and $\pi_m^* = (L - \bar{c})/(1 + (qr)^{1/3})$, $\pi_t^* = (L - \bar{c})/(1 + (qr)^{-1/3})$, with $r = \gamma/\beta$ the takers' relative monitoring cost (eqs. (22)–(24), p. 25) [fkk-2009, working paper, September 2009].

That is Lehalle and Laruelle's (A.4) with their $s\Delta$ for the paper's $\Delta/2$ and $\Gamma$ for $L$, and it confirms both the large-population assumption and the reading of "efficient" above: the optimum is the trade-maximising (and so revenue-maximising) split for a given total fee. Welfare in the paper is the trading rate times the gains from trade less all monitoring costs, $W = \mathcal{R}L - MC_m - NC_t$, which rises as monitoring costs or fees fall (p. 20), a different objective. The make-take spread $c_m - c_t$ rises with the tick size, with $q$ and with $r$ (Corollary 6, p. 25).

Against exact symmetric equilibria with finite populations ($\beta = 1, \gamma = 2, G = 1$): the trade-maximising maker margin is 0.4463 for $M = N = 5$, 0.4429 for $M = N = 50$ and 0.4425 for $M = N = 5{,}000$, against 0.4425 from the formula. *Printed slips*: $\bar{\tau}$ is printed as $\sum_{m=1}^{M} \tau_m$ and the investor's cost as $\frac{1}{2}\gamma\tau_m^2$, where the sum runs over the $N$ investors; the cost of speed is first written $\frac{1}{2}\beta\mathbf{T}/t^2$ with $\mathbf{T}$ a trading day and then $\frac{1}{2}\beta\mu_m^2$ in the gains (p. 252).

The text's remark that "$\Gamma - \bar{c}$ (the investor gain minus the fees) is obviously larger than $s \cdot \Delta$" needs $\pi_T > c_M$, which the participation conditions $\pi_M, \pi_T > 0$ do not imply ($s\Delta = 1$, $c_M = 0.9$, $c_T = 0$, $\Gamma = 1.05$ gives $\Gamma - \bar{c} = 0.15$).

**Practice.** Lehalle and Laruelle introduce the model to address three questions raised by MiFID's market structure: the roles of high-frequency market makers, "the possible negative externalities linked to their activity", and "the impact of the fee schedules on the efficiency of the PFP (Price Formation Process)" (pp. 250–251). In the model, maker rebates suit scarce or costly market making and taker rebates suit crowded makers and costly investor access, consistent with the text.

The model can be extended to concrete speed costs (co-location, feed handling, quoting engines and hedge automation for makers; broker connectivity, routing latency and algorithm responsiveness for investors) and to a diagnostic (segment by spread, maker count, investor urgency, queue stability, latency cost and whether rebates pay for fleeting quotes). So can the view that fast makers help price discovery when they update quotes and arbitrage stale venues but hurt it when speed mainly earns queue priority or fee rebates without adding durable liquidity. All of this is interpretation: the model has no price discovery, no queues and no cancellations. That fast market makers keep prices consistent across venues is Lehalle and Laruelle's description (HFT strategies); the fee comparison a router makes is fee-adjusted venue choice.

**Pitfalls.** $\bar{\mu}$ and $\bar{\tau}$ are sums, not averages, which is what makes $\mu_m/\bar{\mu}$ a race share and $\mathcal{R}$ the trade rate. Eq. (A.4) is "the efficient maker fee" in the sense derived above: the split that maximises trading for a fixed total fee. Foucault, Kadan and Kandel's results are from the September 2009 working paper (above), not the 2013 *Journal of Finance* version.

**See also.** latency-arbitrage arms race (speed races that dissipate rents); queue-position value; fee-adjusted venue choice; HFT strategies; U.S. equity market structure (access-fee caps).

### Model 16.64 A toy model of the Flash Crash: an overflow of market orders depletes depth, a volume-following seller and a hot-potato game amplify it, and the time term of the solution (Lehalle and Laruelle)

**Statement.** Lehalle and Laruelle model the mechanism they see in the Flash Crash: the usual trading rate at the bid or ask is near an equilibrium that depth renews; an agent consumed far more than depth could renew, emptied one side of the books, and kept its trading rate proportional to traded volume [alpha-laruelle, app. A, sec. A.5, pp. 261–262]. With $F_t$ the arrival rate of sell market orders, traded volume $dV_t = F_t\,dt$, depth refilled at rate $\bar{F}$, overflow $dR_t = (F_t - \bar{F})\,dt$, $r_t = F_t - \bar{F}$, and impact the inverse of depth, $I = 1/D$,

$$\frac{dI_t}{I_t} = \sigma_I\,dW_t + r_t\,dt \qquad \text{(A.12)}, \qquad dP = \sigma^P dW^P + I^+ dR^+ - I^- dR^- \qquad \text{(A.13)}$$

(pp. 262–263). The *hot-potato game*: high-frequency market makers who buy from a large seller hedge in another market against each other, so a fraction $\rho$ of traders re-trades, and a seller who follows market volume trades more; the rate becomes $r_t \simeq \frac{1-\rho^t}{1-\rho}\,r_0$ and the overflow $R_t \simeq \frac{t}{1-\rho}\,r_0$ (pp. 263–264). They then give

$$I_t^- = I_0 \exp\Big(\frac{t^2}{1-\rho}\,\frac{r_0}{2} - \tfrac{1}{2}\sigma_I^2 t + \sigma_I W_t^-\Big) \qquad \text{(A.14)}$$

and a price whose dominant part, $P_0 - c\int \exp(t^2/u)\,dt$, "is highly explosive, driving the price to zero faster than an exponential decay" (p. 265). Their remedies: an obligation for market makers "to refill the market depth quickly enough" and a circuit breaker "not only on price levels but also on market depth break events" (pp. 265–266).

**Conditions.** *The time term*. (i) In $r_t = \frac{1-\rho^t}{1-\rho}r_0 = (1 + \rho + \cdots + \rho^{t-1})\,r_0$, $t$ counts rounds of re-trading: it gives $r_0$ after one round and 0, not $r_0$, at $t = 0$. In clock time on $[0, 1]$, the time axis of Figures A.4–A.5, $\rho^t \geq \rho$ and $r_t \leq r_0$: no amplification. Using it in continuous time needs a round length $\delta$, with $t/\delta$ rounds by time $t$, which the text does not give. (ii) The overflow after $k$ rounds is $\frac{r_0}{1-\rho}\left(k - \frac{\rho(1-\rho^k)}{1-\rho}\right)$; the text's $k\,r_0/(1-\rho)$ is its large-$k$ form and overstates it by up to $\rho\,r_0/(1-\rho)^2$ ($3.75\,r_0$ at $\rho = 0.6$). (iii) The exponent $\frac{t^2}{1-\rho}\frac{r_0}{2}$ of (A.14) equals $\int_0^t R_s\,ds$ with $R_s = s\,r_0/(1-\rho)$: (A.14) solves $dI/I = \sigma_I dW + R_t\,dt$, with the overflow, not (A.12) with the rate, whose solution with $r_t = r_0/(1-\rho)$ has the linear exponent $r_0 t/(1-\rho)$.

The text supports the overflow reading: "when more volume is traded than expected, R is large, thus the impact I is large" (p. 262). By simulation: $E\log(I_1/I_0)$ is 2.459 against 2.455 with the rate and 1.210 against 1.205 with the overflow ($\rho = 0.6$, $r_0 = 1$, $\sigma_I = 0.3$). (iv) $r\,dt$ is a quantity while $dI/I$ is a number, so a scale (per unit quantity for $r\,dt$, per unit quantity and time for $R\,dt$) is implicit. (v) Amplification converges only for $\rho < 1$.

**Practice.** The *liquidity spiral* (a large order consumes depth, widens spreads and raises volatility; providers cut size or cancel; the next order trades through less depth) is the verbal mechanism of this model; the term and its state variable, providers' confidence in replenishment, are interpretations, not the source's. Controls follow from Lehalle and Laruelle's two remedies and from the 2010 staff report (the May 2010 Flash Crash): rate controls on

volume-following execution, depth and price controls, cross-market coordination, real-time detection of overflow and broken references, and pauses such as the CME Stop Logic halt and single-stock circuit breakers (circuit breakers). This combination is a synthesis, not a list from one source.

**Pitfalls.** Equation (A.12) implies that positive overflow makes inverse depth compound upward, but the super-exponential fall of the text needs the overflow reading of (iii). The *hot-potato amplification* is not a fraction of traders mechanically following volume: in the text $\rho$ is "the proportion of traders playing this hot potato game" (p. 264), and the volume follower is the seller. The model is illustrative: its parameters are not estimated, its figures plot depth with a sign convention (bid depth negative) not stated in the text, and the staff report describes the hot potato as high-frequency traders re-trading E-mini contracts among themselves, with cross-market arbitrageurs carrying the pressure to equities, not as market makers hedging across markets with each other.

**See also.** the May 2010 Flash Crash; bubbles and crashes; liquidity and market quality; circuit breakers and trading halts; propagator model.

**Evidence 16.65 The Flash Crash of 6 May 2010: a volume-targeting sell algorithm, a hot-potato volume effect in E-mini futures, withdrawal of liquidity in equities and trades at stub quotes (CFTC and SEC staff report, 2010)**

**Statement.** The CFTC and SEC staffs' findings, reported on 30 September 2010 [sec-cftc-may6-2010, executive summary, pp. 1–7]; the Commissions "expressed no view" on them (cover):

- *Setting.* An unusually turbulent day; by 2:30 p.m. buy-side depth in the E-mini S&P 500 futures had fallen from nearly $6 billion to $2.65 billion and in SPY from about $275 million to $220 million (pp. 1–2).
- *Trigger.* At 2:32 p.m. a large fundamental trader, a mutual fund complex, began selling 75,000 E-mini contracts (about $4.1 billion) as a hedge, through an algorithm targeting 9 percent of the previous minute's volume "but without regard to price or time"; a previous program of that size had taken more than five hours, this one about 20 minutes (p. 2).
- *Absorption and the hot potato.* High-frequency traders and other intermediaries, fundamental buyers and cross-market arbitrageurs, who sold SPY or stocks against the E-mini, absorbed the selling; HFTs, net long about 3,300 contracts, sold about 2,000 between 2:41 and 2:44 p.m. while trading nearly 140,000; the algorithm sped up as volume rose; then, between 2:45:13 and 2:45:27, HFTs traded over 27,000 contracts, about 49 percent of volume, while buying only about 200 net, "a 'hot-potato' volume effect" (p. 3). Buy-side E-mini depth fell to about $58 million, under 1 percent of the morning's, and the E-mini fell more than 5 percent and SPY more than 6 percent between 2:41 and 2:45:27 (pp. 3–4).
- *Pause and recovery.* CME's Stop Logic paused the E-mini for five seconds at 2:45:28; prices stabilised and recovered (p. 4).
- *Equities.* Liquidity providers' automated systems paused; some widened quotes, others cut size or withdrew, weighing data integrity, risk and position limits, P&L and the risk of broken trades; internalisers routed retail orders to exchanges; stop-loss orders met reduced buying interest; with liquidity gone, market orders executed against stub quotes at prices "as low as one penny or as high as $100,000"; over 20,000 trades in more than 300 securities at prices 60 percent or more away from their 2:40 p.m. prices were broken; by about 3:00 p.m. most securities traded near consensus values again (pp. 4–6).

**Conditions.** Lessons drawn by the staff: under stress, the automated execution of a large sell order "can trigger extreme price movements, especially if the automated execution algorithm does not take prices into account"; "high trading volume is not necessarily a reliable indicator of market liquidity"; derivatives and securities markets are interconnected through index products; one participant's withdrawal may not matter, but "a liquidity crisis can develop if many market participants withdraw at the same time"; and pausing a market can let participants reassess (pp. 3 and 6). NYSE's liquidity replenishment points, Nasdaq's self-help declaration against NYSE Arca and delays in NYSE data were examined, and "none of these factors played a dominant role" (n. 10, p. 5).

The arithmetic is consistent: $4.1 billion for 75,000 contracts and $1.9 billion for 35,000 imply an index level near 1,090 at 50 dollars a point; 27,000 contracts at 49 percent imply about 55,000 traded in 14 seconds, 135 times the HFTs' net purchase.

**Practice.** Lehalle and Laruelle's account [alpha-laruelle, chap. 2, sec. 2.5.2, pp. 177–183] agrees on the trigger and adds Procter & Gamble's fall from $61.50 to $39.37 between 14:40 and 14:47 and its volume-weighted spread above $2 (pp. 179–180). It describes the hot potato as high-frequency counterparts hedging in equities and then in futures in a loop, gives NYSE's liquidity replenishment points a possible aggravating role (pp. 180–181) and concludes that "Every layer of the market contributed to the crash" (p. 183). The staff report places the hot potato within the E-mini and found the replenishment points not dominant. The case study (a large automated futures sell program met thin liquidity; HFTs recycled inventory; aggressive flow moved into cash equities; spreads widened; routers or venues paused) matches the report.

Of the *shock propagation channels*, cross-asset hedging, routing changes, quote withdrawal and correlated constraints (risk and position limits, pauses) are in the report; *reference-price dependency* is documented by Lehalle and Laruelle for the Euronext outage of 13 October 2010, when midpoint trading stopped and orders pegged to the primary's quotes were cancelled or paused (p. 184; European fragmentation evidence), not in the report's summary.

*Measured HFT behaviour*. Kirilenko, Kyle, Samadi and Tuzun use CFTC audit-trail transactions in the June 2010 E-mini for 3–6 May 2010; of over 15,000 accounts, intraday intermediaries are accounts that traded at least ten contracts on one of 3–5 May and whose end-of-day positions averaged no more than 5 percent of daily volume, with minute-by-minute deviations from them of no more than 0.5 percent, over 3–5 May, the 16 most active of them being "High Frequency Traders" and the rest "Market Makers", with categories fixed from 3–5 May into 6 May (PDF pp. 5 and 11–14) [w5-kirilenko-flash-crash, accepted manuscript]. High Frequency Traders had 34.22 percent of volume on 3–5 May and 28.57 percent on 6 May, Market Makers 10.49 and 9.00 (Table II, PDF p. 35).

The intermediaries' combined net inventories never exceeded 6,000 contracts over the four days, an order of magnitude below the 75,000-contract sale, and both groups accumulated net long positions as prices fell; High Frequency Traders' second-by-second inventory changes co-move *positively* with contemporaneous and lagged price changes, a relationship that "did not change when prices fell during the Flash Crash", whereas Market Makers' inventory changes are negatively related to price changes and their relationship did change (PDF pp. 5–6). So in this study HFTs neither caused the crash nor changed behaviour during it; the hot-potato trading among them is the staff report's.

*The report's detailed sections* (sec. I, printed pp. 12–17) add: E-mini buy-side resting orders fell below 1,050 contracts by 2:45:28, while SPY kept about 600,000 shares (about 1,200 contract equivalents) of buy depth, and near-inside depth was not affected; the Stop Logic pause is triggered to keep stop-loss cascades inside a "no bust" range of six index points, about 0.6 percent on the day (n. 20, p. 12).

The trader categories behind the summary, classified on 3–5 May, include HFTs as the top 3 percent of intermediaries by number of trades, 16 of over 15,000 accounts trading 1,455,000 contracts on 6 May, almost a third of volume, while rarely holding more than 3,000 contracts either way; the section is "based in part" on Kirilenko et al. (n. 22, pp. 13 and 15).

Of the 35,000 contracts the sell algorithm sold from 2:32 to 2:45, about 18,000 were executed aggressively and 17,000 passively (n. 27), and of the remaining 40,000, sold from 2:45 to 2:51 as prices rose, about 24,000 aggressively (n. 28); over 2:32–2:45 all fundamental sellers sold more than 80,000 contracts net against about 50,000 bought by fundamental buyers, a 30,000 imbalance about 15 times that of the same interval on the previous three days; in the rebound HFTs "did not significantly alter their trading strategy" while nearly half the intermediaries withdrew (pp. 15–16).

Fifteen cross-market firms interviewed, net buyers of more than 100,000 E-minis between 2:00 and 3:00 p.m., saw the E-mini as the primary price-discovery product, bought it against SPY and stock baskets during the fall, and many had stopped their strategies by 2:45 p.m. (sec. I.5, pp. 16–17). Menkveld and Yueshen's account of the broken E-mini–SPY arbitrage link is available in the SSRN working-paper version (Pitfalls).

**Pitfalls.** These are the staffs' findings from their reconstruction. The equities sections add detail: OTC market makers "paused or halted internalization", some only on sell orders as position limits were hit, citing sell pressure, data integrity, latencies and P&L limits, and routed orders to

the exchanges instead [sec-cftc-may6-2010, sec. II.2.e, printed p. 58]; many retail market and stop-loss orders were converted by internalisers into limit orders that followed the price down to stub-level bids; and the "over 20,000 trades representing 5.5 million shares" executed more than 60 percent from their 2:40 p.m. value were broken under the clearly erroneous rules, almost two-thirds of the cancelled shares having traded below $1.00 and about 5 percent above $100 (sec. II.3.a, printed p. 65).

*Cross-market arbitrage* (SSRN version dated 15 November 2017, earlier titled "Anatomy of the Flash Crash"; published in *Management Science*, 2019). Menkveld and Yueshen proxy cross-arbitrage by the dispersion of bids and of asks across venues, using Nanex quote data and Waddell & Reed's own trade records, and find that arbitrage between the E-mini and SPY "weakened at 14:15 and broke entirely at 14:44, one minute before the halt" [w5-menkveld-yueshen-flash, SSRN version, printed pp. 2–3 and 5–7]; in the 50 most crashed S&P 500 stocks, those whose arbitrage broke first crashed first, with controls for volatility and order imbalance, but "We however are unable to identify the root cause for these breakdowns themselves" (pp. 3 and 15).

The large seller traded most of its 75,000 contracts after the breakdown, and at the widest the E-mini ask was more than 100 basis points below the SPY bid (p. 3). A Grossman–Miller calibration, in which a disconnected venue leaves the seller facing half the market makers, puts its overpayment for liquidity at $98.6 million to $229.8 million, two to four times its quarterly operating income (pp. 3–4): a model-based estimate, not an accounting figure. Their online appendix dates the break points that separate cointegrated from non-cointegrated E-mini and SPY prices, so a VECM is fitted before the break and a VAR after [menkveld-yueshen-oa, secs. A–B]. These are associations and a calibration, read alongside the staff's reconstruction, not a causal identification.

The sell algorithm was about half passive: "taking liquidity" describes its effect on net demand, not every fill. The Kirilenko et al. figures are from the accepted manuscript, not the published pages. The rules changed afterwards (single-stock circuit breakers from June 2010, limit up–limit down, revised market-wide circuit breakers, the stub-quote ban: U.S. equity market structure). "Hot potato" names a volume effect among intermediaries, not a measure of HFT's net selling.

**See also.** toy model of the Flash Crash; circuit breakers; HFT strategies; index futures and price discovery; venue competition and fragmentation.

## 16.14 Floor and automated trading systems

**Definition 16.66 Floor against automated trading systems: two technologies for the same order-driven auction, compared on fairness, access, capacity, speed, information and cost (as of 2002)**

**Statement.** Floor-based oral auctions and automated rule-based auctions are both order-driven markets that match buy and sell orders by very similar rules. They differ mainly in technology: in oral auctions traders exchange information in person, in automated markets computers arrange the trades [alpha-harris, chap. "Floor Versus Automated Trading Systems", p. 544]. This entry groups Harris's comparison under six headings; his own sections are fairness, the convenience of distributed access, system capacity, negotiation speed, exchange of information on trading floors, market data reporting and cost (secs. 27.1–27.7, pp. 544–552), and his Table 27-1 lists eleven issues (p. 552):

- *fairness*: automated systems apply their rules exactly and produce complete audit trails; oral auctions depend on traders' skill and honesty and need surveillance. Floor traders see and react first, and in pits even size and voice matter, while automated markets favour traders with computer-generated orders (pp. 544–546);
- *convenience*: distributed access from traders' desks (pp. 546–547);
- *capacity*: electronic matching scales; oral auctions become disorderly "fast markets" when too many traders take part (p. 547);
- *speed*: automated systems complete and report trades faster, but oral negotiation of size is faster than splitting firm orders, which undisclosed orders and negotiation messaging partly address (pp. 547–548);
- *information exchange*: floors win when traders need to know who the counterparty is and whether more size is coming, to avoid informed or price-discriminating traders, and when brokers know of latent interest (pp. 548–549). Automated systems report market data faster and more accurately (p. 549);
- *cost*: both have high initial creation costs; screen-based systems have low operating costs, their main operating costs being telecommunications and data backup. Floor systems are labour-intensive, needing floor brokers, reporters, officials and runners, and well-trained floor brokers are expensive (sec. 27.7, pp. 549–552).

**Conditions.** Automated trading also needs heavy investment in reliability, security and speed (pp. 549–551; operational controls for automated trading). Automated markets exploit economies of scale, which drives exchange mergers. Because of the order-flow externality, the survival of large floor markets does not prove that floor technology is better (p. 553). Harris expects automated systems to serve active markets and small traders, and floors large institutions that need to exchange information (pp. 552–553).

**Practice.** Harris's examples of competition, such as the Paris Bourse against London and the DTB against LIFFE, did not decide the question, because other events, a repealed French stamp tax and German banks repatriating their market, also moved trading (p. 543). *Technology and reporting*. New communications technologies gave traders "instantaneous presence in markets that they formerly could not attend", and computing allowed order-matching features impossible by hand, such as orders shown only to committed traders, linked contingent orders and limit prices indexed to market conditions [alpha-harris, chap. "Competition Within and Among Markets", sec. 26.1, pp. 524–525]. Electronic systems collect market information cheaply because it is already electronic, while floors need market reporters [alpha-harris, chap. 5, sec. 5.4.1, p. 97], and clearing "is a trivial exercise when automated order-matching systems arrange all trades" [alpha-harris, chap. 3, sec. 3.2.2.1, p. 35].

*Capacity*. The condition that $N$ traders sending messages at rate $\lambda$ each need a processing rate $\mu > N\lambda$ is, for the M/M/1 queue it implicitly assumes (Poisson arrivals at total rate $N\lambda$, one server with exponential service at rate $\mu$, first in first out, unlimited buffer), necessary and sufficient for stability: the queue has a stationary distribution if and only if $\rho = N\lambda/\mu < 1$. Stability says nothing about latency or deadlines. In the stationary M/M/1 queue the mean wait in the queue is $\rho/(\mu - N\lambda)$ and the mean time in the system, wait plus service, $1/(\mu - N\lambda)$; with $\mu = 1{,}000$ messages a second these are 1 and 2 ms at $\rho = 0.5$, 9 and 10 ms at 0.9, and 99 and 100 ms at 0.99; the time in the system is exponential, so a message misses a deadline $d$ with probability $e^{-(\mu - N\lambda)d}$, 0.135 for $d = 10$ ms at $\rho = 0.8$.

When the arrival rate varies over time, an average below $\mu$ keeps the backlog from growing from one cycle to the next, but a burst at a rate $\lambda_b > \mu$ lasting $T_b$ builds a backlog of about $(\lambda_b - \mu)T_b$ messages, adding up to $(\lambda_b - \mu)T_b/\mu$ to the delay, which then drains at the rate by which capacity exceeds the arrivals. Whether quotes and time precedence stay reliable therefore depends on how long bursts last, how large a backlog they build, how much buffer there is (a full finite buffer rejects or drops messages) and what deadline matters, not on the average or the peak rate alone.

Simulated with arrivals sampled exactly by thinning: at a constant 800 messages a second the mean wait is 3.94 ms and the mean time in the system 4.94 ms against 4.00 and 5.00 in theory, with 99th percentiles of 20.6 and 21.7 ms against 21.9 and 23.0; with the same average but 1.5 times capacity for the first 100 ms of each second (0.7222 times capacity otherwise; realised average 800.2 a second), the mean wait is 12.53 ms, the mean time in the system 13.53 ms, its 99th percentile 72.9 ms and the share of messages slower than 10 ms 0.343 against 0.131; the backlog peaks near the fluid estimate of 50 messages (a mean wait of 51.7 ms at the end of a burst).

Bursts of 20 and 5 ms at the same peak and average give 99th percentiles of 25.8 and 22.5 ms; a buffer of 20 messages rejects 4.8 percent of messages with the 100 ms bursts against 0.29 percent at a constant rate (0.23 percent in the M/M/1/20 formula). At utilisation 1.05 the wait grows by about 0.05 s per second, to 2.5 and 4.6 s after 50 and 100 s. Drawing each inter-arrival gap from the rate in force at the previous arrival, which is

inexact across rate boundaries; its realised rate, 799.8 a second against a stated 798, happened to be close, and it did not separate the wait from the time in the system.

**Pitfalls.** Both systems have high initial creation costs in absolute terms (p. 549), and Table 27-1 rates the automated system's software costs as relatively lower than the cost of building a floor and training its staff (p. 552); the two statements are compatible. Automated systems have high fixed costs for software, data networks, security and recovery but a low marginal labour cost per order, as the text says. Floors remain valuable when negotiation, discretion and soft information matter more than matching speed (sec. 27.5). The market facts (the NYSE and most U.S. options exchanges still floor-based, the firms and systems named) are as of 2002 and are not current.

**See also.** order exposure decisions (negotiating size and undisclosed orders); block trades; best execution.

## 16.15 Spot FX dealing: market structure, internalisation and last look

### Definition 16.67 Spot FX market structure: venues and access channels, trading protocols, firm and non-firm liquidity, and principal and franchise liquidity

**Statement.** The FX market is fragmented and has no consolidated tape of prices or volumes [exp-fx-algos, sec. 2.2, p. 22]. Cartea, Jaimungal and Walton add that there is no central exchange framework. The most common G10 currencies may trade on more than 20 electronic crossing networks (ECNs), and most major banks also offer trading through their own platforms, applications or APIs [dm-cartea-jaimungal-walton-2019, arXiv version, sec. 1, p. 2]. Four things should be kept apart:

- *Venues and access channels.* The primary electronic venues (EBS Market, Refinitiv Spot Matching, and CME for some pairs) supply the reference prices most algorithms use. Their market share had dwindled as alternative ECNs and liquidity pools emerged; clients reach liquidity by voice or chat, APIs, single-dealer and multi-dealer platforms [exp-fx-algos, pp. 12, 22–23].
- *Trading protocols.* In a "risk transfer" the client requests a price, by request for quote (RFQ) or request for stream (RFS), and trades its full ticket at the price received, paying the spread for immediacy; RFQ systems differ in whether the side is revealed, who receives the request and whether quotes are executable or indicative (pp. 5 and 43). Orders may instead be worked by execution algorithms across venues (sec. 2).
- *Firm and non-firm liquidity.* Anonymous pools based on a central limit order book typically match "firm liquidity", without the option to reject. Disclosed pools allow liquidity to be partitioned by counterparty and typically include "non-firm liquidity", with a short window in which the provider may reject ("last look"); some ECNs offer both (Box D, p. 23).
- *Principal and franchise liquidity.* In principal pools the provider takes the position on its market-making desk. In franchise pools it uses one client's flow to offset another's, *internalising* rather than hedging externally (Box D, p. 23).

Internalisation is a dealer's risk-management choice, not a venue. A dealer may *internalise* a client trade by warehousing the risk until offsetting flow arrives, or *externalise* it by hedging in the open market [cfd-butz-oomen-2019, accepted version, abstract and sec. 1, p. 1]; see FX internalisation horizon.

**Conditions.** Market structure as described in 2018–2020. The BIS figures are a central-bank study group's survey and interviews as of 2020, not measurements of the whole market. The largest dealers report internalisation ratios "as high as 90%", but the term has several definitions, from matching only between users of one platform to including flow routed to a dealer's own desk or to mid-matching venues, so reported ratios are not comparable (Box D, p. 23). Visible depth can mislead: trading interest posted on several venues at once can create a "liquidity mirage", which the report expects last look to exacerbate (p. 27).

**Practice.** Interpretation: routing in FX must weigh firmness and internalisation as well as price; the BIS describes routers that consider quoted size, queue length, fill ratios and fill probability, and that assume lower price impact for internalised orders, an assumption checked through transaction cost analysis (p. 24; compare smart order routing, from European equities). Transaction cost analysis in FX compares traded prices with benchmarks at order arrival, during and after execution, and measures slippage, market impact and the ratio of rejected trades, without a consolidated tape to supply the benchmark (Box B, p. 11).

**Pitfalls.** Prime-brokered access and bilateral relationships are ways of reaching liquidity, not venues. Fixing windows are a matter of benchmark timing, not a venue type. RFQ is plausibly used for size or less liquid pairs, but the source describes it as a protocol for full-size risk transfer without restricting it to those cases. Venue names and market shares are as of 2020. The FX market is often described as "the world's largest and most liquid", open "24 hours per day, 5.5 days per week", mostly OTC in spot, forwards and swaps, with CME futures "a small fraction".

Gliner gives $4 trillion average daily turnover, trading from 8:15 p.m. GMT Sunday to 10 p.m. GMT Friday (about five days, not 5.5), the BIS's division into spot, forwards, swaps and options, with "most foreign exchange … traded through the derivatives market", and CME futures expiring quarterly [s7-gliner, chap. 7, opening and sec. "Trading Currencies", PDF pp. 179 and 182–184]; the BIS's 2025 survey puts turnover at $9.6 trillion a day in April 2025 [bis-triennial-2025, preliminary results]. The participants are central banks, dealers, corporates, asset managers, hedge funds and retail traders.

**See also.** FX internalisation horizon; last look; smart order routing and best execution (equity sources, not FX evidence); dealer inventory risk.

**Further reading.** King, Osler and Rime (wide-king-osler-rime), a survey of FX market structure.

### Model 16.68 FX internalisation horizon: how long a dealer takes to offset client risk internally, and what position skewing costs (Butz and Oomen)

**Statement.** Butz and Oomen model internalisation as a queue: each client trade either increases or decreases the dealer's risk position, and the *internalisation horizon* is the time a risk-increasing trade waits until risk-decreasing trades offset it. To control the build-up of risk the dealer *skews* its prices to attract risk-reducing flow. By Little's law the average horizon is proportional to the ratio of the dealer's risk limit to the rate of client flow [cfd-butz-oomen-2019, accepted version, sec. 1, p. 2]. Precisely, with $\varphi_n$ the stationary distribution of the dealer's position $n$ (in units of trade size) and $\lambda_n^{\pm}$ the arrival rates of buys and sells at position $n$, the expected queue length and the arrival rate of risk-increasing trades are

$$L = \sum_n \varphi_n |n|, \qquad \lambda = \sum_{n\geq 0} \varphi_n \lambda_n^{+} + \sum_{n\leq 0} \varphi_n \lambda_n^{-}$$

(eq. (8), p. 7). The average horizon of a *risk-increasing* trade is $L/\lambda$. Risk-reducing trades are internalised at once, with zero wait, so the average over *all* trades is $W = \frac{1}{2} L/\lambda$ (eq. (9), p. 7). In the skew specifications, $\lambda_0$ is the baseline intensity of buys and of sells, not $\lambda$. Under a binary skew $\lambda_n^{+} + \lambda_n^{-} = 2\lambda_0$, with a risk-reducing tilt $\alpha$ beyond the threshold $R$ (Example 1). Under an exponential skew $\lambda_n^{+} = \lambda_{-n}^{-} = \lambda_0 e^{-n/(2R^2)}$, so that the position is normal with standard deviation $R$ (Example 2, eqs. (5)–(7), pp. 5–6). Proposition 1 then gives the all-trade average

$$W = \frac{R}{\lambda_0}\,\frac{c_R}{4} \text{ (binary)}, \qquad W \approx \frac{R}{\lambda_0}\,\frac{c_R}{\sqrt{2\pi}} \text{ (exponential)}, \qquad c_R \to 1 \text{ as } R \text{ grows},$$

with $c_R$ given in the proposition (eqs. (10)–(11), p. 8). The average is the same whatever rule the dealer uses internally to match offsetting trades, provided its position starts and ends at zero (p. 8).

**Notation.** Positions are counted in unit trade sizes; in the calibration a unit is \$1 million. $R$ is the risk limit (or skew scale) in those units. $\lambda_0$ is the baseline intensity of buys, and of sells, in units per minute, and $\lambda$ the resulting arrival rate of risk-increasing trades. $L/\lambda$ is the conditional horizon of a risk-increasing trade and $W = L/(2\lambda)$ the all-trade average, both in minutes.

**Conditions.** The calibration is top-down, from the BIS 2016 triennial survey rather than dealer data. Spot turnover of \$1,652 billion a day is reduced to customer flow, a representative tier-1 dealer is given a 6.6 percent share, and its risk limit is set so that its position stays within \$25 million 95 percent of the time, $R = \$25\text{mn}/1.96 = \$12.75\text{mn}$. For EUR/USD, $\lambda_0 = \frac{1}{2} \times \$111\text{mn/min} \times 6.6\% = \$3.66\text{mn/min}$ of each type, giving the all-trade average $W \approx 12.75/3.66 \times 1/\sqrt{2\pi} \approx 1.39$ minutes, which the authors report as the average internalisation horizon (pp. 8 and 11). Interpretation: a risk-increasing trade waits about twice as long on average, $L/\lambda \approx 2.78$ minutes.

Summing the exact stationary distribution of Example 2 with these values gives $L/\lambda = 2.774$ and $W = 1.387$ minutes, so the approximation with $c_R \approx 1$ is close. Horizons are single-digit minutes for liquid G10 currencies and tens of minutes for most Asian and emerging-market currencies and some less liquid G10 ones (p. 11, Figure 6), and vary predictably over the day.

**Practice.** Smaller dealers, or those with less risk appetite, can shorten horizons by skewing more aggressively. That has a direct cost, less spread captured on risk-decreasing trades, and an indirect one, "information leakage or signalling risk associated with skewing". Internalisation therefore runs on a continuum from passive to aggressive (p. 2). The authors' message is that a client's transaction costs and liquidity access depend on its own trading and on the dealer's risk management; clients should tell externalisers from internalisers, and passive from aggressive internalisers (abstract). The BIS report gives the user's side: internalisation can reduce information leakage and market impact, though dealers' skewing may reveal information and negate some of that benefit [exp-fx-algos, p. 27].

**Pitfalls.** The authors worked in Deutsche Bank's electronic FX spot division, as the paper discloses in a footnote on its abstract page. The horizons are model outputs from survey aggregates, not observed internalisation times. Generic dealer-inventory theory (dealer inventory risk, dealer quote adjustment, from Harris's equity and floor examples) motivates skewing but is not FX evidence.

**See also.** spot FX market structure; inventory-based quotes; dealer quote adjustment.

### Mechanism 16.69 Last look in spot FX: the provider's final accept or reject window, its validity and price checks, and how it differs from firm liquidity

**Statement.** The FX Global Code defines *last look* as a practice in electronic trading "whereby a Market Participant receiving a trade request has a final opportunity to accept or reject the request against its quoted price". If used, it should be a risk control that checks *validity*, the operational details and sufficient credit, and/or *price*, whether the requested price remains consistent with the current price. The provider has sole discretion, leaving the client with market risk if the request is rejected [exp-fx-code, Principle 17, pp. 22–23]. Under the Code providers should disclose:

- whether and how price changes in either direction affect the decision;
- the expected or typical period of time for making the decision, the duration of the last-look window;
- the purpose of last look.

They should not use it to gather information with no intention of accepting, and should not trade on the information in the request during the window, whether by pricing on electronic platforms or by hedging. The guidance does not apply to an arrangement that has *all* of three features (p. 23):

1. an explicit understanding that the provider will fill the client's request without taking on market risk in connection with it, by first entering into offsetting transactions in the market;
2. the volume traded in the last-look window is passed on to the client in its entirety;
3. this understanding is appropriately documented and disclosed to the client.

Such "cover and deal" arrangements are the Code's only exception to the rule against trading in the window.

**Conditions.** The Code is voluntary: it "does not impose legal or regulatory obligations on Market Participants nor does it substitute for regulation" (p. 1); the edition cited is the update of December 2024. Contractual practice is in each provider's disclosures. In one example, completed on 18 September 2025, Deutsche Bank reports for its wholesale FX business:

- for spot FX, a price check of up to 10 milliseconds, which may be shorter where latency and location matter less, and no additional hold time beyond the time the price and validity checks take; the validity check's own duration is not bounded in the sheet and depends on networks, systems, client connection, liquidity, time of day and product, and the window may differ for products other than spot;
- a validity check covering counterparty-exposure limits, credit checks and currency-pair permissioning;
- symmetric and asymmetric price checks, with the asymmetric check as its default, since it causes fewer rejections;
- that outages or disrupted markets may lengthen the window, and acceptance rates, check durations and pricing may vary by customer;
- no trading during the window except under agreed cover-and-deal arrangements.

[db-fxgc-lp-cover-2025, sec. C.4]. These are one provider's self-reported terms, not market facts.

**Practice.** Keep apart, Interpretation using the sources above:

- a *firm* quote, matched without rejection, as in anonymous order books, from a *conditional* (non-firm) quote subject to last look [exp-fx-algos, Box D, p. 23];
- a *price-check rejection* from a *validity* rejection for credit, exposure or permissioning, and both from a *technical failure* that delays or breaks it;
- the *decision time*, the duration of the last-look window, which the Code asks to be disclosed. Within it, the time spent on the price and validity checks differs from any *additional hold* beyond them, which this provider says it does not apply;
- the client's measures: fill and reject rates, and markouts, prices after the trade compared with the trade price. FX transaction cost analysis includes the ratio of rejected trades and benchmarks after execution [exp-fx-algos, Box B, p. 11].

In theory, with risk-neutral providers who recover losses to latency arbitrageurs from other traders, last look lowers the quoted spread on a single venue. With two venues an equilibrium may keep both or only one, and a last-look venue may even quote wider spreads than a firm one [dm-cartea-jaimungal-walton-2019, arXiv version, abstract]. This is a model prediction, not an empirical finding. Oomen's framework compares "symmetric" and "asymmetric" last-look designs and protocols that do or do not pass price improvement and slippage to the trader.

It finds that the design and protocol set the execution risk, but "the effective transaction costs borne by the trader need not be affected by it". When a trader adversely selects the provider she deals with, the distinction between the designs fades [oomen-2017-lastlook, accepted version, abstract]. It too is a model; it reports no measured fill or reject rates. Measured fill rates, reject rates, hold times and markouts under last look against firm liquidity are not reported in these sources.

**Pitfalls.** Last look reduces stale-quote risk for the provider and creates execution uncertainty and possible information leakage for the client. The first is the model's premise; the second matches the Code's concern with information use and the client's market risk on rejection. Credit, prime-broker limits, rejection rates, hold times, skewed streams and information leakage all affect fill quality, but no source here measures how much each contributes, or shows that they dominate it; credit appears as part of the validity check. The latency-arbitrage evidence of Budish, Cramton and Shim concerns ES and SPY, not FX (latency-arbitrage arms race).

**See also.** spot FX market structure; latency-arbitrage arms race; best execution.

## 16.16 Options dealing: market-maker spreads, cost measurement and retail routing

**Result 16.70 Option market-maker spreads: discrete hedging, volatility uncertainty, inventory and informed trading, priced across linked contracts**

**Statement.** An option's value depends on the underlying price, volatility, rates, dividends, strike and maturity, so the quotes on one underlying form a linked set of contracts, and a market maker who sells one option holds risks that it can hedge in the underlying or in other options. Four findings, each from a different sample or model, separate the risks that this linkage leaves the market maker:

- *Hedging and volatility risk.* For a sample of Chicago Board Options Exchange options, the risks of being unable to rebalance an option hedge continuously and of uncertainty about the underlying's return volatility "each account for a statistically and economically significant proportion" of quoted bid–ask spreads, risks the authors call unique to options [ex-jameson-wilhelm-1992, abstract].
- *Inventory risk.* Decomposing the price impact of option trades into inventory-risk and asymmetric-information components, both are large and the inventory component is larger; order imbalances attributable to inventory risk have "five times larger impact on option prices than previously thought", and past imbalances predict option returns better than other common predictors [wide-muravyev-order-flow, abstract].
- *Informed trading and the hedging instrument.* In Huh, Lin and Mello's model, market makers who hedge against adverse selection widen spreads in both the option and the stock, more in the option, and more when hedging with the underlying than with other options; hedging also changes informed traders' strategies. Their tests are reported as consistent with its key implication [ex-huh-lin-mello-2015, abstract].
- *Who supplies quotes.* In the U.S. listed-options market that Bryzgalova, Pavlova and Sikorskaya study, orders are executed on exchanges, and retail orders are typically filled by a wholesaler's affiliated market maker [ex-bryzgalova-pavlova-sikorskaya-2023, published version, pp. 2 and 11]; the routing is in retail option order routing.

**Conditions.** Findings as of their samples, from the CBOE before 1992 to U.S. options in 2019–2021. The first three are taken from the published papers' abstracts. Jameson and Wilhelm's evidence is for CBOE equity options (published 1992); Huh, Lin and Mello's result is a model prediction whose empirical support is stated only as consistency with one implication. None of them says how large each risk is in today's markets.

**Practice.** Interpretation, keeping the risks apart:

- *hedging cost*: the cost and residual risk of trading the hedge, discretely, in the underlying or other options;
- *volatility risk*: exposure to changes in volatility, which a delta hedge does not remove;
- *inventory risk*: the position a market maker accumulates from net order flow, priced through quote adjustment as in equity dealer models (dealer inventory risk, dealer quote adjustment);
- *information risk*: trading against better-informed traders, in the option or through the hedge (adverse-selection spread component).

A linked set of prices, an implied volatility surface, does not mean liquidity is uniform across it: liquidity is measured contract by contract (option spread measurement), and the liquidity of an option differs from that of the instrument used to hedge it.

**Pitfalls.** The sources here do not show that clearing, exercise, assignment, margin and position-risk management constrain fragmentation: Battalio, Hatch and Jennings's study of multiply listed equity options contrasts June 2000, when quotes were often ignored, with January 2002, when the market more closely resembled a national market; it does not address clearing [ex-battalio-hatch-jennings-2004, abstract].

*What dated filings show.* OCC's rulebook was refused. An SEC notice of an OCC rule change describes two procedures for assigning exercise notices to clearing members: a standard "random" assignment for most option classes, and a pro rata procedure for S&P 500 index options and flexibly structured foreign-currency options. It also describes OCC's assignment directly to short positions in a clearing member's sub-account for a single beneficial owner [fr-occ-2007-assignment, Release 34-56845, 27 November 2007, 72 FR 67991, 3 December 2007, sec. II(A)]. These are clearing-level procedures, dated 2007.

*Current rulebook.* Rule 803 assigns accepted exercise notices "in accordance with the Corporation's procedures" to clearing members with open short positions in the series, may include positions opened that day, excludes a short position once confirmed trade information for a closing purchase has been received, and assigns by 8:00 a.m. Central Time on the next business day; Interpretation .01 assigns to a particular account or sub-account. The rule text itself names neither random nor pro rata assignment, so the 2007 description of OCC's procedures is not confirmed or contradicted as current. Rule 804 leaves the allocation to individual writers to each clearing member's fixed procedures under exchange and self-regulatory rules.

Rule 805(d)(2) deems an expiring option exercised when it is in the money by $0.01 or more unless the clearing member instructs otherwise, and Rule 1803(a) applies Rules 803 and 804 to cash-settled options [occ-rules-2025, Rules 803–805, PDF pp. 88–90, printed pp. 60–62; Rule 1803, PDF pp. 169–170]. No source here links them, or margin, to multiple listing or fragmentation, or measures whether option dealing is dominated by cross-contract risk. Hedging costs pass into option spreads, in direction, by the hedging findings above; high uncertainty widens quotes cross-sectionally, by Jameson and Wilhelm's volatility-uncertainty result; these sources do not show that it reduces quoted size. Generic dealer models from equity markets motivate but do not establish option claims, nor do the futures EFRP (exchange of options for options), portfolio trading or ETF entries.

**See also.** spread determinants; inventory-based quotes; option spread measurement; retail option order routing.

**Method 16.71 Measuring option execution costs: quoted against effective spreads, premium units, execution timing, and segmentation by moneyness, expiry and trade type**

**Statement.** Option spreads are usually measured relative to the option's premium. In Bryzgalova, Pavlova and Sikorskaya's U.S. data, the *quoted spread* is the best ask minus the best bid across all exchanges relative to the midpoint at the time of the trade, and the *effective spread* twice the absolute percentage deviation of the trade price from that midpoint, both frequency-weighted [ex-bryzgalova-pavlova-sikorskaya-2023, published version, Table I, p. 15]. These are the equity measures of spread measures applied per contract. Two corrections apply in options:

- *Benchmark and timing.* Option price changes are predictable at high frequency, and many traders time executions, buying when the option's fair value is close to the ask and selling when it is close to the bid. Effective spreads of such traders are less than 40 percent of conventional measures, the overall average effective spread is one-quarter smaller than conventional estimates, and price-impact measures are also affected [wide-muravyev-pearson, abstract]. Interpretation: the midpoint then overstates the cost of a timed trade.
- *Units.* A spread in percent of premium is not a spread in volatility units: a cheap out-of-the-money option can have a large percentage spread and a small spread in implied volatility (Interpretation).

**Conditions.** The segmentation figures are from Table I of Bryzgalova, Pavlova and Sikorskaya, all U.S. listed-option trades reported to OPRA from 4 November 2019 to 30 June 2021, split into SLIM trades, those executed in single-leg non-ISO price-improvement auctions and used as the authors' retail proxy, and non-SLIM trades, the whole remaining pool, institutional and retail, single- and multi-leg (pp. 7, 11–12 and 15–16):

- *moneyness*: at-the-money trades are about 70 percent of trades in both groups, with quoted spreads of 8.7 percent (SLIM) and 8.4 percent (non-SLIM), far below the 28.7 and 21.3 percent of slightly out-of-the-money trades ($-1$ to $-0.1$); they are not the lowest, since slightly in-the-money trades (0.1 to 1) have 8.6 and 5.9 percent. Table I writes moneyness for calls as $(\text{MidpointPrice} - \text{Strike})/\text{Strike}$, with the opposite sign for puts. Interpretation: since it is compared with the strike, the midpoint here is the underlying's, not the option-premium midpoint used for the spreads; the caption does not say so explicitly;

- *expiry*: options with less than a week to expiration carry about half of SLIM volume and 41.5 percent of non-SLIM volume; their quoted spreads are 12.6 and 13.3 percent and their effective spreads 6.6 and 10.1 percent. Spreads are not monotone in maturity: 3–12 month options have SLIM quoted spreads of 18.5 percent but non-SLIM ones of 10.2 percent;
- *trade size*: SLIM trades below $250 have quoted spreads of 23.5 percent; trades above $20,000 have effective spreads above quoted spreads, consistent with price impact (p. 17);
- *totals*: quoted and effective spreads average 13.7 and 6.7 percent for SLIM and 11.6 and 8.6 percent for non-SLIM trades.

**Practice.** Report spreads per contract with the market, sample period, contract set and units, and separate:

- quoted from effective spreads, and both from timing-adjusted measures against a fair-value benchmark;
- outright from multi-leg trades: Bryzgalova, Pavlova and Sikorskaya restrict their measure to single-leg auctions and note that multi-leg auctions are a noisier retail measure (p. 12), so their results do not isolate the execution costs of multi-leg strategies: the non-SLIM pool mixes multi-leg trades with the rest, and no figure is reported for multi-leg executions as such; where multi-leg orders trade, on Cboe Options, is set by Rule 5.33:
  - a *complex order book* (COB) and a *complex order auction* (COA) for complex strategies;
  - the synthetic best bid or offer (SBBO), "comprised of the best 'net' bid and 'net' offer", formed by netting the legs' best bids and offers;
  - a COA-eligible order may start an auction at a price equal to or better than the SBBO. The auction ends early if, on the same side, an order arrives in a leg that would improve the SBBO beyond the auction price;
  - paragraph (g), "Legging", lets a complex order leg into the simple book under restrictions [fr-cboe-5-33-2021, Release 34-92729, 23 August 2021, public-inspection PDF pp. 3–5 and 10–11].

  *Current Legging text*. Cboe's filing SR-CBOE-2026-081, approved by the exchange on 22 September 2026, reproduces paragraph (g) as now worded. A complex order may execute against the simple book "if it can execute in full or in a permissible ratio and if it has no more than a maximum number of legs (which the Exchange determines on a class-by-class basis and may be up to 16)". The filing adds that legs in different SPX or VIX groups may leg in only if designated a "related group" [cboe-sr-2026-081, Item 2, p. 3; Exhibit 5, Rule 5.33(g) and (g)(6), p. 25]. The rulebook itself (cboe.com) was not reached; the filing is the exchange's own text. These describe mechanisms only; none measures multi-leg costs against outrights;
- the option's liquidity from the hedge's: a narrow option spread can still carry a costly hedge, and the reverse.

**Pitfalls.** That at-the-money contracts are usually deepest is supported only as trade share in one U.S. sample: at-the-money trades are about 70 percent of trades, and their percentage quoted spreads are far below those of slightly out-of-the-money trades but not the lowest of the moneyness bins. Depth, size at the quote, is not measured in the source. That near expiries trade differently is consistent with the expiry split but is not a finding. None of these sources measures complex-order books, so whether spreads and multi-leg trades matter more than outrights is open.

**See also.** spread measures; option market-maker spreads; retail option order routing; best execution.

#### Mechanism 16.72 Retail option order routing: payment for order flow, wholesalers and price-improvement auctions (U.S., 2019–2021)

**Statement.** In U.S. equities wholesalers internalise retail orders off exchange. In the authors' description of U.S. options, whose data are exchange-listed option trades reported to OPRA, "all options orders in the United States must be executed on exchanges", so internalisation was thought specific to equities (p. 11; "all options in the United States trade on exchanges", p. 2). Bryzgalova, Pavlova and Sikorskaya argue that much retail option flow is also effectively internalised [ex-bryzgalova-pavlova-sikorskaya-2023, published version, p. 2]. A retail broker routes the order to a wholesaler in exchange for payment for order flow (PFOF) and a promise of price improvement on the NBBO. The wholesaler's affiliated market maker brings the order, paired with its own contra side, to an exchange's price-improvement auction, where other participants ("responders") may offer a better price within a time window.

Fees are stacked against responders: on most exchanges the affiliated market maker pays $0.05 a contract, while another market maker pays $0.50 to break up a paired order; in that responder-execution case the wholesaler receives a net rebate of $0.30 a contract for bringing the order (p. 11, footnote 15).

**Conditions.** Sample and measure: OPRA trades from 4 November 2019 to 30 June 2021, with the retail proxy built from the single-leg non-ISO price-improvement auction flag that OPRA introduced on that date (pp. 11–12). Some large SLIM trades are likely institutional; the authors check robustness on trades below $20,000 (p. 17). Retail trading reached over 60 percent of total option volume, and nearly 90 percent of PFOF came from three wholesalers (abstract), named as Citadel, Susquehanna and Wolverine, by the second quarter of 2021 (pp. 2 and 11). The fees are as the authors report them for their sample. The exchange-only statement concerns listed options.

Over-the-counter (conventional) options, customised bilateral contracts, also exist, including ones with terms identical or substantially similar to listed options; FINRA's background notes that such trades are not generally centrally cleared nor reported to FINRA, the SEC, an exchange or the Consolidated Audit Trail [finra-rn-22-14, Background and Discussion]. That notice was a request for comment on a proposed regulatory-only reporting requirement, not an effective rule; whether it was later adopted is unverified.

**Practice.** What retail flow pays, in the authors' measures: SLIM trades had an average quoted spread of 13.7 percent and effective spread of 6.7 percent of the premium (Table I), so they received price improvement but paid spreads far above those in equities (p. 14). Aggregate trading costs, measured from trade price to midquote for all SLIM trades, were $6.4 billion over the sample, against about $900 million of commissions; at a 10-day holding horizon the aggregate retail portfolio lost $2.1 billion (pp. 4–5).

**Pitfalls.** That buy-side traders typically consume liquidity because coherent quoting needs specialised models and hedging infrastructure is not established. The routing above shows retail market orders filled by wholesalers' market makers, and Muravyev and Pearson find that many traders time their liquidity-taking [wide-muravyev-pearson, abstract]; neither establishes who supplies or takes liquidity across the options market, or why. Payment for order flow and broker conflicts in equities are in broker conflicts; they do not establish option facts.

**See also.** best execution; broker conflicts; option spread measurement; option market-maker spreads.

# 17 Execution, trading costs and market structure

## 17.1 Execution benchmarks and implementation shortfall

**Definition 17.1 Execution benchmark: a reference price for judging an execution, or the trading target of an algorithm**

**Statement.** The word has two uses.

- A *benchmark price* $P_B$ against which realised execution is measured. With average execution price $\bar{P}$ and Side $= +1$ for a buy and $-1$ for a sell, the cost in basis points is Side $(\bar{P} - P_B)/P_B \times 10^4$, positive when the execution was worse than the benchmark; the profit-and-loss form multiplies it by $-1$ [s7-kissellscience, chap. 3, eqs. (3.11)–(3.14), pp. 105–106]. Harris states it in currency: trade size times the signed difference, trade price minus benchmark for a purchase and benchmark minus trade price for a sale, $C = dq(\bar{P} - P_B)$ with $d = \pm 1$; across all buyers and sellers in a trade the estimated costs sum to zero, so liquidity suppliers tend to show negative costs [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", sec. 21.2, pp. 422–423].
  Common benchmark prices are the quotation midpoint when the manager decided to trade (Perold's implementation shortfall, below), the price when the order arrived in the market, the volume-weighted average price (VWAP) of the day or of the order's own trading interval, a participation-weighted price, and the open, the close, the next day's open or close, or the average of the day's open, high, low and close [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", pp. 424–426]; [s7-kissellscience, chap. 3, pp. 106–108].
- The *target* of an algorithm class. A VWAP algorithm trades in proportion to the intraday volume curve; a TWAP algorithm trades equal amounts in equal time intervals, about 1/390 of a full-day US order per minute, often skipping the opening and closing auctions; a percentage-of-volume (POV) algorithm trades a fixed fraction of market volume until the order is done; arrival-price and implementation-shortfall algorithms balance expected cost against risk at a chosen risk aversion, the latter adding real-time adaptation, and their meaning differs across brokers [s7-kissellscience, chap. 1, pp. 16–18]. A liquidity seeker has no reference price: it is relative-price oriented, captures liquidity across pools and minimises information leakage [alpha-laruelle, sec. 3.4.1, Table 3.2, p. 230].

**Notation.** What each benchmark price fixes (the order's own fills are $s_j$ shares at $p_j$; the unfilled remainder is treated in implementation shortfall):

- *Decision price*: the quotation midpoint when the manager decided to trade [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 426]; in practice most managers use the previous day's close [alpha-harris, same chap., box "The Plexus Iceberg of Transaction Costs", p. 437]. Known before the order is worked.
- *Arrival price* $P_0$: the price when the order is entered in the market, the bid–ask midpoint in market hours, the last trade or official close otherwise [s7-kissellscience, chap. 3, p. 102]. Known when trading starts.
- *VWAP*: total traded value over total volume, each trade weighted by its size [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 425]. Full-day VWAP has no official value; vendors choose the eligible trades and may drop delayed or negotiated ones. Interval VWAP covers the order's own trading horizon, computed from tick data; VWAP to the end of the day judges orders finished early [s7-kissellscience, chap. 3, pp. 107–108]. Known only afterwards, and it includes the order's own trades.
- *Participation-weighted price* at rate $\rho$: the VWAP of the first (shares traded)$/\rho$ shares of market volume from the arrival time [s7-kissellscience, chap. 3, eqs. (3.15)–(3.16), p. 108]. Known only afterwards.
- *Open, close, later closes*: single daily prices; next-day and later prices are used to separate temporary from permanent impact [s7-kissellscience, chap. 3, p. 106].

Kissell's benchmark formulas use the average execution price without fees, so they are gross of explicit costs; his implementation shortfall adds fees [s7-kissellscience, chap. 3, eqs. (3.4)–(3.16), pp. 98–108]. Signs are not standard across the industry: some report a cost, positive when worse, others a gain, positive when better [s7-kissellscience, chap. 3, pp. 97 and 108].

**Conditions.** Definitions timeless; vendor and consultant practice as of 2002 and 2014. Harris compares the benchmark prices on four properties [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", sec. 21.4, pp. 427–432, Table 21-1, p. 433]:

- *noise*, which grows with the time between trade and benchmark: the contemporaneous midpoint is least noisy, open and close most;
- *split orders*: a benchmark moved by the order's own impact understates its cost. A trader who is the only buyer of the day matches VWAP whatever the impact (pp. 428–429);
- *momentum and contrarian trading*, biasing open and, less, VWAP benchmarks; *informed trading*, making the close and, less, VWAP understate costs;
- *gaming*: a broker with discretion over timing can game any benchmark that depends on when it trades, for example by spreading fills through the day to match VWAP or trading at the close. A benchmark fixed before the broker receives the order cannot be gamed, so implementation shortfall cannot (pp. 430–431).

**Practice.** When data cost is no object, implementation shortfall is the best estimator; retail traders use the effective spread, and sponsors who receive only daily average prices use VWAP [alpha-harris, same chap., pp. 426 and 431–432]. VWAP results are not comparable across stocks of different volatility, and converge to zero as the order becomes a large share of volume [s7-kissellscience, chap. 3, p. 107]. Single-trade estimates are noisy; averaging reduces noise but not bias or gaming, and comparisons across brokers are meaningful only for similar trading problems [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 432].

**Pitfalls.** Two interpretations go beyond these sources: that the post-trade midpoint "measures adverse selection" (in Harris the effective minus the realised spread measures dealers' losses to informed traders, spread measures) and that brokers game by "selecting only favourable venues" (Harris's gaming is by timing discretion and by mimicking the benchmark, pp. 430–431, 440). VWAP is an average market price, which includes the order's own trades, not a no-impact counterfactual. Interpretation: a benchmark price is a reference, not an executable price. A midpoint benchmark charges an aggressive order at least the half-spread, as the effective spread does. Decision and arrival prices are known when the order is planned; VWAP, participation-weighted and closing prices are known only after it, so an algorithm can target them only through forecasts, such as a forecast volume curve.

Performance against the benchmark an algorithm targets is not its cost: a VWAP-tracking execution can match VWAP and still have paid spread and impact against arrival. Kissell calls TWAP "a constant participation rate" while describing equal shares per minute, which is a constant rate in time, not in volume [s7-kissellscience, chap. 1, p. 17]. His participation-weighted formula (3.16) has no side factor and is correct as written only for buys. Lehalle and Laruelle's "benchmarks" in Table 3.1 are algorithm classes; PoV and liquidity seekers are not benchmark prices (see benchmark choice). The sources define no TWAP benchmark price.

**See also.** implementation shortfall (the decision-price benchmark with its unfilled remainder); choosing an algorithm's benchmark; spread measures (contemporaneous and post-trade midpoint benchmarks); intraday impact profile (impact measured from the arrival price); mean–variance execution (the schedule behind arrival-price algorithms).

**Definition 17.2 Implementation shortfall: paper portfolio minus real portfolio, with execution, delay and opportunity components**

**Statement.** Implementation shortfall is the paper return of a portfolio assumed to trade the whole order at the manager's decision price, minus the actual return, using actual fills and fees [s7-kissellscience, chap. 3, eqs. (3.1)–(3.4), pp. 97–99]. For an intended order of $S$ shares, fills of $s_j$ shares at $p_j$, decision price $P_d$, price $P_n$ at the end of the measurement period and fees,

$$\text{IS} = \underbrace{\textstyle\sum_j s_j p_j - \sum_j s_j P_d}_{\text{execution}} + \underbrace{\left(S - \textstyle\sum_j s_j\right)(P_n - P_d)}_{\text{opportunity}} + \text{fees},$$

which Kissell calls Perold's formulation [s7-kissellscience, same chap., pp. 100–101, eq. (3.6), p. 104]. With the arrival price $P_0$, Wagner's expanded form splits it into delay, trading and opportunity parts:

$$\text{IS} = S(P_0 - P_d) + \textstyle\sum_j s_j\,(\bar{P} - P_0) + \left(S - \sum_j s_j\right)(P_n - P_0) + \text{fees}, \qquad \bar{P} = \textstyle\sum_j s_j p_j \,/\, \sum_j s_j$$

[s7-kissellscience, same chap., pp. 101–104, eq. (3.7)]. The trading-related part is the arrival cost [s7-kissellscience, eq. (3.9), p. 105]. Harris states the same estimator with the decision-time midpoint: the filled size times the signed gap between average fill and benchmark, plus the unfilled size times the gap between a later price and the benchmark [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 426].

**Notation.** Quantities are signed, $S$, $s_j > 0$ for buys and $< 0$ for sells [s7-kissellscience, chap. 3, eq. (3.9), p. 105]; IS is in currency and positive when the real portfolio did worse. With unsigned quantities and a side $d = \pm 1$: for parent size $Q$, filled $q$ at average $\bar{p}$, decision price $p_0$ and later price $p_T$, IS $= dq(\bar{p} - p_0) + d(Q - q)(p_T - p_0)$ + explicit costs, which is Perold's form with $S = dQ$ and $\sum_j s_j = dq$; on the example below it gives 3,960 for the buy and for the sell. Fees are explicit costs: commissions, taxes, clearing and settlement charges, ticket charges and rebates (p. 98). The spread and impact are inside the fill prices.

**Conditions.** Timeless (accounting); the Plexus figures as of 2002. When the order fills completely, IS reduces to $S(\bar{P} - P_d)$ + fees and does not depend on $P_n$ (pp. 99–100). An unfilled or cancelled remainder is charged at $P_n$, so its cost depends on when it is measured: at the end of trading [s7-kissellscience, chap. 3, p. 102], or a day, a week or a month later; the Plexus Group uses 30 trading days after the decision [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", pp. 436–437]. The remainder counts only for an order that could realistically have been filled: Harris's 4-million-share sell order, 50 days' volume, would show a missed-trade cost of $714,600 that the trader never had the opportunity to avoid (p. 436).

**Practice.** The split shows who is responsible: delay belongs to the manager or desk, trading-related cost to the broker or algorithm, and the arrival cost is the main measure of broker and algorithm performance [s7-kissellscience, chap. 3, pp. 101–104]. When the decision price is unknown, only the market-activity part from $P_0$ is measured (p. 104, eq. (3.8)). Plexus's components, manager timing, trader timing, market impact, commission and missed trades, sum to the total; Plexus estimated its clients' costs at 12 basis points of commission, 20 of impact, 53 of timing and 16 of missed trades [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", box, p. 437].

Interpretation: a buy of 10,000 shares decided at a midpoint of 50.00, arriving at 50.10, filling 3,000 at 50.20 and 3,000 at 50.30, cancelling the rest, with $P_n = 50.60$ and fees of 60, has paper gain 6,000 and real gain 2,040, so IS = 3,960: Perold 1,500 + 2,400 + 60; Wagner delay 1,000, trading 900, opportunity 2,000, fees 60. The mirror-image sell gives the same figures with signed quantities, and Kissell's worked examples ($2,600, and $3,080 split as 1,250, 1,000, 750 and 80) reconcile.

**Pitfalls.** Interpretation: Kissell's intermediate actual return for a partial fill adds the fees instead of subtracting them (p. 100); his final formula is right. His basis-point arrival cost (3.10) divides by the signed $(\sum s_j)P_0$, which reverses the sign for sells; (3.11), with the side factor, does not (in the sell above, −30.06 against +30.06 basis points). Count each cost once: with a midpoint decision or arrival price, the half-spread is already inside the execution component, so adding an effective-spread term double-counts it, and fees enter only as the explicit term. Splitting the trading-related part into spread and impact needs a further benchmark; transaction costs gives an accounting split into half-spread paid and midpoint drift, which is not a causal measure of impact.

The Almgren–Chriss shortfall $X S_0 - \sum n_k \tilde{S}_k$ is the trading-related part for a complete sale measured from the pre-trade price, with half-spread and fees inside the temporary impact; its mean and variance are model forecasts, not a realised shortfall, and it has no opportunity term because the model forces completion. For Perold's paper (`wide-perold-shortfall`), the attribution of the formula to Perold, and of the expanded form to Wagner (1975, 1991), is Kissell's; Harris credits Perold with popularising the decision-midpoint estimator and Treynor, writing seven years earlier, with the paper-portfolio method (p. 424).

**See also.** execution benchmarks; mean–variance execution (expected shortfall and its variance); spread measures; price impact.

**Further reading.** Perold's paper on implementation shortfall, `wide-perold-shortfall`.

**Method 17.3 Choosing an algorithm's benchmark from order size, market context and the risk to hedge (Lehalle and Laruelle)**

**Statement.** Lehalle and Laruelle tie each benchmark of a trading algorithm to the investor needs in a region: in the US, where high-frequency traders are everywhere, stealth; in Asia, comparing trading efficiency across local markets, given their differences; in Europe, both avoiding detection by high-frequency traders and learning how efficiently a set of order books is reached, to adjust routing quickly. "On paper" European investors should therefore target implementation shortfall, Asian ones VWAP or PoV, US ones liquidity seekers. Size and market conditions adjust these "drastic rules": liquidity seekers suit orders without minimum participation constraints, IS small orders, VWAP larger and PoV very large ones [alpha-laruelle, sec. 3.4.1, pp. 227–228]. Their Table 3.1 summarises the remarks (p. 229):

| Benchmark | Region | Order size | Market context | Risk hedged (do not miss) |
|---|---|---|---|---|
| PoV (target) | Asia | more than 10% of ADV | possible negative news | the rapid spread of unexpected news, especially if the investor has the information |
| VWAP/TWAP (target) | Asia, Europe | 5–15% of ADV | unusual volume negligible | the slow spread of information |
| Implementation shortfall (measure) | Europe, US | 0–6% of ADV | possible price opportunities | an unexpected price move |
| Liquidity seeker | US | any | price oscillates around fair value | a liquidity burst or relative price move |

ADV is average daily consolidated volume. Sources: rows from [alpha-laruelle, sec. 3.4.1, Table 3.1, p. 229]; links to the definitions. The table is a summary of the entries, and the risk column is condensed from the source's wording.

**Conditions.** Lehalle and Laruelle's guidance is practitioner judgement: the passage gives no data for it, and the table "should not be read as 'all conditions in the columns have to be fulfilled'", only as possible reasons for a benchmark (p. 228). The size bands overlap as printed (0–6%, 5–15%, above 10%).

**Practice.** The hedged-risk column matters because an algorithm performs a risk–reward arbitrage: once its risk is under control it is free to adjust the prices and quantities of its orders, so optimal execution combines risk control, by following an "optimal" trading rate, with capturing price and liquidity opportunities. The choice also depends on the stock's liquidity and on the investment style behind the trade: hedging trades are more

exposed to execution-specific impact, long-term position building to "systemic" impact that algorithms cannot do much about [alpha-laruelle, sec. 3.4.1, p. 228]. Their Table 3.2 adds the type of stock and of trade suited to each benchmark, and the main features of its algorithm (p. 230):

| Benchmark | Stock | Trade | Main features of the algorithm |
|---|---|---|---|
| PoV | medium to large market depth | long-duration position | follows current market flow; very reactive, can be very aggressive; more price-opportunity driven if the range between maximum and minimum percent is large |
| VWAP/TWAP | any market depth | hedging order; long-duration position; unwinding tracking error (delta hedging of a fast-changing inventory) | follows the "usual" market flow; good if the market moves with unexpected volume in the order's direction; can be passive |
| Implementation shortfall | medium liquidity depth | alpha extraction; hedge of a non-linear position (typically gamma hedging); inventory-driven trade | finishes very fast if the price is good and liquidity sufficient; "cuts losses" if the price moves too far away |
| Liquidity seeker | poor, fragmented market depth | alpha extraction; opportunistic position building; order already split or scheduled | relative-price oriented (between pools or securities); captures liquidity everywhere; stealth, using fragmentation to limit leakage |

Sources: rows from [alpha-laruelle, sec. 3.4.1, Table 3.2, p. 230]. How each algorithm pursues its benchmark, through a trading envelope and tactics, is in trading envelope; what each benchmark measures is in execution benchmarks.

**Pitfalls.** Do not conflate cross-country differences in market conventions with venue fragmentation; the source speaks of differences between local markets, not fragmentation. Table 3.2 also records: for Europe, avoidance of detection by high-frequency traders as well as routing discipline; for PoV, the risk "especially if I have the information" and that it "can be very aggressive"; for VWAP/TWAP, the condition on unexpected volume in the order's direction; and the liquidity seeker's relative-price orientation. A volume-curve schedule describes VWAP, not TWAP. That a benchmark justifies the comparison, not the whole execution policy, is consistent with the risk–reward reading on p. 228 but is not in the source (Interpretation).

The algorithm a trade is given does not settle how its cost is measured; an order worked by a VWAP algorithm can still be judged against arrival price, and matching VWAP does not show low cost (execution benchmarks). The regional pattern is dated by the 2018 printing.

**See also.** execution benchmarks; implementation shortfall; mean–variance execution (the risk-control layer and its trading curve); block trades. Fragmentation, dark trading and venue rules are treated in venue competition and fragmentation.

## 17.2 Execution scheduling and block orders

#### Model 17.4 Mean–variance optimal execution: the Almgren–Chriss trade-off between impact and price risk

**Statement.** An order is split over time to balance two forces: market impact pushes the schedule to trade slowly, price risk pushes it to trade fast [alpha-laruelle, app. A, sec. A.7, p. 274]. In Almgren and Chriss's discrete model a holding $X$ is liquidated over $N$ intervals of length $\tau = T/N$; $x_k$ is the holding at $t_k = k\tau$, with $x_0 = X$ and $x_N = 0$, and $n_k = x_{k-1} - x_k$ is sold in the $k$th interval [alpha-almgren, author preprint of Dec. 2000, sec. 1.1, p. 7]. The price follows $S_k = S_{k-1} + \sigma\tau^{1/2}\xi_k - \tau g(n_k/\tau)$, with permanent impact $g$; the $k$th sale executes at $\tilde{S}_k = S_{k-1} - h(n_k/\tau)$, with temporary impact $h$ that does not enter later prices (same preprint, secs. 1.2–1.3, eqs. (1)–(2), pp. 8–9). The shortfall $X S_0 - \sum_k n_k \tilde{S}_k$ has mean $E(x)$ and variance $V(x) = \sigma^2 \sum_k \tau x_k^2$ (eqs. (4)–(5), p. 10), and an efficient schedule minimises

$$E(x) + \lambda V(x), \qquad \lambda \text{ the risk aversion}$$

(sec. 2.1, eq. (15), p. 13). With linear impact, $g(v) = \gamma v$ and $h = \epsilon\,\mathrm{sgn}(n_k) + (\eta/\tau)n_k$, the unique minimiser for $\lambda \geq 0$ is

$$x_j = \frac{\sinh(\kappa(T - t_j))}{\sinh(\kappa T)} X, \qquad \frac{2}{\tau^2}(\cosh(\kappa\tau) - 1) = \tilde{\kappa}^2 = \frac{\lambda\sigma^2}{\tilde{\eta}}, \quad \tilde{\eta} = \eta - \tfrac{1}{2}\gamma\tau,$$

and $\theta = 1/\kappa$ is the trade's half-life, independent of $T$ [alpha-almgren, same preprint, secs. 2.2–2.3, eqs. (16)–(17), pp. 13–15].

**Notation.** Lehalle and Laruelle's discrete version is for a buy order of $v^\star$ over $N$ slices: $v_n$ shares in slice $n$, market volume $V_n$, volatility $\sigma_n$ in currency per slice, and execution price $\tilde{S}_n(v_n) = S_n + \kappa\sigma_n(v_n/V_n)^\gamma$, with $\kappa$ and $\gamma$ stock-dependent constants [alpha-laruelle, app. A, sec. A.7.1, eqs. (A.16)–(A.19), pp. 275–276]. Here $\kappa$ and $\gamma$ are impact parameters, not Almgren and Chriss's decay rate $\kappa$ and permanent-impact slope $\gamma$; $\lambda$ is risk aversion in both.

**Conditions.** The price is a martingale over the horizon, with no view on its trend [alpha-laruelle, app. A, sec. A.7.2, p. 277]; volumes and volatilities per slice are taken as known in advance (p. 276); the schedule is static, which Almgren and Chriss show is optimal under their price dynamics (preprint, n. 10, p. 13); the trades do not change sign (p. 13). In Lehalle and Laruelle's version:

- *Temporary impact only.* The concession $\kappa\sigma_n(v_n/V_n)^\gamma$ is in currency per share and applies only to slice $n$'s fills: an instantaneous execution cost, not a transient displacement of later prices. The slice costs $v_n$ times it; the expected impact cost of the order is $\kappa \sum_n \sigma_n v_n^{\gamma+1}/V_n^\gamma$, in currency, not per unit time (eq. (A.20), p. 276). There is no permanent-impact term and no spread or fee term, unlike Almgren and Chriss's $g$ and $\epsilon$.
- *Expectation only* ($\lambda = 0$). The schedule $v_n^{\exp} = v^\star\,(V_n/\sigma_n^{1/\gamma}) \,/\, \sum_\ell V_\ell/\sigma_\ell^{1/\gamma}$ (Proposition 2, eq. (A.22), p. 277), uniform if $V_n$ and $\sigma_n$ are constant (Proposition 3), which is then a TWAP schedule.
- *Variance only.* Trade everything in the first slice (Proposition 4, eq. (A.25), p. 278); in Almgren and Chriss this gives zero variance "because of the way that we have discretized time" (preprint, eqs. (12)–(13), p. 12).
- *Mean–variance*, $C = \mathbb{E}(W) + \lambda\,\mathrm{Var}(W)$ (eq. (A.26), p. 279). For $\gamma = 1$ the first-order conditions give a recurrence in the remaining quantity, solved by shooting on the first slice (eq. (A.28), pp. 279–280); larger $\lambda$ trades faster, smaller $\lambda$ more linearly (Figure A.11, pp. 280–281).

**Practice.** A trading algorithm has two layers: a risk-control layer, at a time scale around 10 minutes, that bounds the trading rate from the schedule; and an opportunistic layer, such as smart order routers and liquidity seekers, that chooses the participation rate within those bounds and may react to alpha signals [alpha-laruelle, app. A, sec. A.7, p. 274]. The model's parameters are estimated daily or weekly (p. 274).

Numerical interpretation: the expectation-only schedule equalises marginal impact cost across slices. For $\gamma = 1$ with constant $\sigma_n$ and $V_n$, the mean–variance schedule in this notation equals Almgren and Chriss's sinh form with $\cosh k = 1 + \lambda\sigma V/(2\kappa)$ and Lehalle and Laruelle's recurrence. Lehalle and Laruelle's model is Almgren and Chriss's linear case without the schedule-independent terms (the permanent $\frac{1}{2}\gamma X^2$ and the fixed cost $\epsilon X$ of a one-sided order) and without the $\tilde{\eta}$ correction, with $\kappa\sigma_n/V_n$ in place of $\eta/\tau$ and $\sigma_n^2$ in place of $\sigma^2\tau$.

**Pitfalls.** Indexing. Lehalle and Laruelle's price step (A.16) prices slice $n$ after shock $n$, and their variance (A.24) uses the quantity remaining from slice $n$ on, yet they state the terminal conditions as $x_0 = v^\star$, $x_N = 0$; they also label Proposition 5 as "variance minimization" although it is the mean–variance result, and drop $\kappa$ from the intermediate Lagrange condition, which only rescales the multiplier. With Almgren–Chriss timing: slice $n$ at $S_n$, then $S_{n+1} = S_n + \sigma_n\xi_{n+1}$, so the remaining quantity $x_n$ after slice $n$ bears the next shock and the risk term $\lambda \sum_n x_n^2\sigma_n^2$ is consistent (Interpretation). *Comparative statics*. Higher risk aversion front-loads the schedule; higher impact sensitivity moves it back towards the expectation-only schedule, but not beyond it.

Interpretation: the minimiser depends on $\lambda/\kappa$ only, since $C - v^\star S_0 = \kappa\left[\sum_n \sigma_n v_n^{\gamma+1}/V_n^\gamma + (\lambda/\kappa)\sum_n x_n^2\sigma_n^2\right]$; larger $\kappa$ moves the schedule towards the expectation-only one, not beyond it; in numerical cases the remaining quantity never exceeded the expectation-only path, which is not a proof. *Shape*. The mean–variance path is convex for constant parameters (the sinh form; Lehalle and Laruelle, p. 279), but with time-varying $V_n$ and $\sigma_n$ it follows the volume profile and need not be convex (none of 200 random cases was). The parameters $\kappa$, $\gamma$ of a scheduling model are inputs; the empirical exponents of Almgren et al. (0.6, against size relative to volume during execution, with a 30-minute post-trade horizon) are estimates of a related but not identical quantity. Dark pools and limit orders in a tactical overlay are not in the source.

**See also.** price impact (temporary against transient impact; the Almgren–Chriss impact functions); direct estimation of impact; intraday impact profile (participation and duration); absence of price manipulation (conditions on impact models used in schedules); block trades.

**Further reading.** The journal version of Almgren and Chriss (2001).

**Definition 17.5 Block trade: an order too large to fill by normal procedures, and the four problems of arranging it**

**Statement.** A block trade results from an order too large to fill easily through normal trading procedures, typically more than a day's normal volume; most block traders think of a block as more than a quarter of a day's average volume in an actively traded stock, and in thinly traded instruments a few thousand shares can be one [alpha-harris, chap. "Block Traders", sec. 15.1, p. 322]. Block initiators face four problems [alpha-harris, same chap., sec. 15.2, pp. 323–327]:

- *latent demand*: willing counterparties have not issued orders, because orders are costly to manage or they have not yet thought of trading;
- *order exposure*: shopping the block lets front runners, same-side and opposite-side traders act on it, speeding its impact (Table 15-1, p. 325);
- *price discrimination*: liquidity suppliers fear that more size will follow, so they want the true size before offering liquidity;
- *asymmetric information*: suppliers suspect large, especially anonymous, traders of being informed, and demand concessions, as in adverse selection.

**Conditions.** Exchanges classify blocks by fixed sizes for statistics, the NYSE at 10,000 shares or more regardless of activity or price; Harris suggests classifying by a fraction of average daily volume instead [alpha-harris, same chap., sec. 15.1 and box, p. 323]. Interpretation: typical depth or participation capacity would serve the same purpose.

**Practice.** Solving the problems requires credible information about the initiator: that it wants to trade, exposure only to trustworthy traders, its full size, and evidence that it is uninformed. Since exchanges carry only price and size, most blocks are arranged by telephone upstairs through block brokers and dealers with reputations to protect [alpha-harris, same chap., secs. 15.2.2–15.2.5, pp. 324–327]. Crossing networks, hidden orders and order splitting are the alternatives when orders cannot be exposed (p. 325).

*Sunshine trading* is the opposite: announcing identity, intentions, full size and reasons for trading. It works when the sunshine trader is well known and known to be uninformed and honest, so in large markets at best for the largest traders, and fails when others suspect information or hidden size ("wolves in sheep's clothing"), because such motives cannot be verified on an exchange floor or screen [alpha-harris, same chap., sec. 15.3, pp. 327–328]; by revealing their trades, sunshine traders also give free trading options to order anticipators (p. 328).

Block dealers and brokers audit their clients' motives: uninformed traders have utilitarian reasons unrelated to fundamental values, and block traders are most confident when the client has an easily verified reason to trade that instrument at that time; discretion over which instrument or when suggests strategic choice (Table 15-2: a firm liquidating a diversified portfolio to fund a project, probably uninformed; a legatee buying a single stock, probably informed) [alpha-harris, same chap., sec. 15.4.4, pp. 330–331]. Full and timely trade reporting favours block brokers over block dealers: a dealer who must report quickly can be front-run while liquidating the position, and reports help others infer whether the initiator was informed, which raises the importance of the motive audit; the weakest reporting is in dealer markets (sec. 15.4.3.1, p. 330).

U.S. equity trades were reported within 90 seconds, and the London Stock Exchange had moved from seven days for the largest blocks to 90 minutes (box, p. 331, as of 2002). Harris also describes how blocks met the exchange book in U.S. listed equities in 2002 [alpha-harris, same chap., sec. 15.6.1, pp. 333–334]:

- brokers and dealers who were members of an exchange had to print their block trades in exchange-listed securities at an exchange if the trade took place while the market was open, presenting the matched orders so that the exchange's precedence rules protected all traders;
- a *size precedence* rule let matched crosses of more than 25,000 shares outsize the book at the trade price, jumping orders with time precedence, while orders standing at better prices were filled at the block price, since price priority was enforced universally;
- Harris calls this a compromise between block traders, who want their crosses unbroken, and standing limit-order traders, who want a return on the liquidity they offer;
- block traders often printed crosses at regional exchanges, whose books were thinner.

*Roles in the upstairs market*. Block initiators who cannot convey credible information about their motives use block dealers and block brokers, both of whom investigate whether the client is informed and has revealed its full size [alpha-harris, same chap., sec. 15.4, p. 328]. Block dealers (positioners, facilitators) take the client's position and trade out of it, so they must predict their own liquidation costs; block brokers (assemblers) find the traders who will fill the order and protect their reputations with those liquidity suppliers by knowing their clients (secs. 15.4.1–15.4.2, pp. 328–329).

Dealers have the advantage with impatient traders and small blocks, brokers with the largest blocks that dealers will not hold; most block traders act as both, and a broker willing to take part in a trade it proposes is more credible (sec. 15.4.3, pp. 329–330, box "A Finger in the Guillotine"). The role and skill lists (initiator, liquidity supplier, dealer, broker, upstairs market; latent-demand discovery, motive audit, exposure control, cost analysis, reputation) match these pages.

**Pitfalls.** Disclosing enough to find size, but not enough to be exploited, is a fair reading of the tension between the latent-demand and order-exposure problems (Interpretation), but Harris's own resolution adds disclosure of identity and true size to trusted counterparties. Suppliers' fear of selling too cheaply or buying too richly is the fear that more size will follow. Harris's examples of motives are funding, diversification, covering a short, hedging and currency needs; benchmark changes and the initiator's past behaviour are not among them.

The reporting caveat says that delayed reporting gives informed block traders room to exploit stale prices; Harris's argument is that timely reporting hurts facilitating dealers, and elsewhere that delayed reporting helps traders building or unwinding positions at the expense of other traders (chap. "Buy-Side Traders", p. 389). Some motive audits may be unreliable where prices are too volatile for block traders to build reputations for them (p. 331). The market rule *Equity Block Prints and Size Priority* merged three things that should be kept apart: where a trade is *reported* or printed; its *execution priority* against orders in the book (size against time precedence); and *trade-through protection* of better-priced quotes on other venues, which in the United States is now Rule 611 (best execution).

Trade-reporting facilities and size-priority exceptions belong to current practice; Harris's account is of a 2002 exchange rule. His compromise is a design trade-off: protect the book too much and blocks leave; protect blocks too much and displayed liquidity loses value. *Current U.S. trade-through rule*. Harris's exchange-printing requirement is a 2002 practice. Under Regulation NMS, trading centers must have procedures to prevent trade-throughs of protected quotations except in the nine cases of Rule 611(b): a malfunctioning quoting center, non-regular-way contracts, single-priced opening, reopening or closing transactions, crossed protected quotations, executions of intermarket sweep orders and executions while simultaneously routing them, prices not based on the quote and not determinable when the commitment was made, quotations improved upon within the preceding second, and customer stopped orders [ecfr-17cfr242-611, 17 CFR 242.611(b), eCFR text for 15 October 2026].

None is an exception for block size, so a block printed off-exchange at a price outside the protected quotes needs one of these, typically intermarket sweep orders. *Reporting*. An off-exchange block in an NMS stock is reported to a FINRA trade reporting facility. On the FINRA/NYSE TRF, participants

report “as soon as practicable but no later than 10 seconds after execution”, and transactions not so reported “shall be designated as late” [finra-rules-2026, Rule 6380B(a)(1) and (a)(4), version of 30 March to 5 December 2026]. Modifiers mark among others late reports, trades at an average or other special pricing formula, and stop-stock transactions (6380B(a)(5)).

*How an exchange cross ranks against displayed orders*. This is NYSE’s rulebook compilation dated 8 May 2019; the rules state that they do not apply to UTP securities on Pillar. Rule 127 lets a member cross a block (at least 10,000 shares or $200,000) at a clean-up price outside the quotation. It must first trade with the NYSE best bid (offer), including all reserve interest, and with all book orders limited to better prices than the clean-up price. “The block shall be entitled to priority at the clean-up price”; when the member is positioning to establish or increase a position, it must first fill orders limited to the clean-up price or better [nyse-rules-2019, Rule 127(b)(ii), (c), Supplementary Material .10, PDF pp. 423–424].

Manual crosses under Rule 76 must first be offered to the crowd one minimum variation above the bid, and its electronic cross function allows a print only at or between the protected bid and offer within a 20-second window (Rule 76 and .10, PDF p. 288). Interpretation: price priority is kept, and Harris’s size precedence survives as the block’s priority at its own price, in the 2019 floor rules.

**See also.** order anticipation (front running exposed orders); order exposure decisions; dark pools and crossing networks; spread components (the adverse-selection concession); mean–variance execution (scheduling an order through the open market instead).

## 17.3 Execution algorithms, order exposure and portfolio trading

**Definition 17.6 Automated trading architecture: market data with real-time analytics, two-way venue connections, historical data with recorded messages, and models with an accuracy check**

**Statement.** Lehalle and Laruelle sketch the main components of a broker’s automated trading service (Figure 3.1): tick-by-tick and context databases, a calendar of expiries, expected news and half-days, modelling and model selection, a trading framework with a strategic layer, local trading tactics and listeners at each venue, synchronisation, a trading station, a trading database and TCA [alpha-laruelle, sec. 3.1, pp. 193–194]. The inputs of such tools are four (sec. 3.1.1, pp. 194–197):

- *market data*: real-time feeds, consolidated across venues but quick to disaggregate, because it matters which venue currently offers a price. They hold trades, each with a price, a quantity and a timestamp that is “not always precise enough”, and order books, or the sequence of their modifications, from which the book must be rebuilt on the fly. They are enriched with analytics: the spread, the side that initiated a trade, intraday volatility, an efficient spread and estimates of hidden quantities. Since raw data and analytics are needed together, “the sooner they are computed, the better”, ideally on hardware as soon as raw data arrive (pp. 194–195);
- *connections to venues*, which are not natively synchronised with market data, usually being faster, and are two-way. One direction sends new orders, cancellations and modifications; the other returns acknowledgements, rejections of ill-formed messages, cancellations and partial fills. Incoming messages also yield analytics: dark pools publish only trades, so a dark buy order filled in full at a price shows resting sell quantity at that price or better (pp. 195–196);
- *historical data*: tick-by-tick or order-by-order market data; the analytics already computed in real time; recordings of the messages the firm’s algorithms sent and received, to study past performance; and context such as news, corporate events, circuit breakers and venue outages (p. 196);
- *models*, each with three components. A formula takes parameters and variables: in the impact formula $\kappa\sigma(v/V)^{\gamma}$ the parameters $\kappa, \gamma$ may be estimated once a day and stored, while the variables $\sigma$, $V$ and $v$ are, in the book’s words, “known in real time”. Interpretation: $V$, the market volume during the trading, is available only as it is observed; volume still to come requires a forecast (see the decision-time item in trading envelope). An estimation procedure computes the parameters from historical data. An accuracy computation qualifies the model and its parameters on the immediate past or in real time, so that a model can be deactivated when it is unlikely to be adequate in the current context (pp. 196–197).

**Conditions.** Lehalle and Laruelle’s practitioner description as of the 2018 printing; Narang’s account as of 2024, with his latency figures undated. The inputs feed the components of a trading algorithm: execution parameters, a risk-control layer that sets the trading envelope, and tactics such as smart order routers (trading envelope). For latency, some autonomous tactics sit as close as possible to the venues they address, and the risk-control layer at equal distance from all its robots (sec. 3.1.2, p. 198). This describes a broker’s execution service for fragmented European equities as of the 2018 printing; it is a design, not a standard or a legal requirement.

**Practice.** Interpretation: the recorded messages and context are what later make it possible to replay an execution and explain its performance (decision support, transaction cost analysis). The accuracy computation is a model-monitoring control: governance of models and of the systems that run them is in operational controls for automated trading.

Narang’s version of the same infrastructure: connectivity, a message protocol, and hardware and software, each built or bought [s7-narang, chap. 7, “Execution”, sec. 7.2, pp. 138–140]. Most traders use brokers as trading agents, “due to regulatory and other constraints”, and reach venues by direct market access through the broker’s servers, which he says adds 10 to 30 milliseconds against a fraction of a millisecond from colocated servers (p. 139). The FIX protocol, begun in 1992 between Fidelity and Salomon Brothers, is the standard for order and execution messages; he calls the protocol “free and open source” and distinguishes it from the FIX engines that implement it, which differ in quality and which latency-sensitive traders often build themselves (pp. 139–140). Buying execution algorithms can cost “five or more times as much per share” as running one’s own (pp. 140–141).

**Pitfalls.** The FIX protocol is free and open, but FIX engines vary and need not be open source (Narang; s6-fix). Narang’s chapter does not cover sponsored access; for U.S. broker-dealers market access is governed by Rule 15c3-5 (operational controls). Narang’s latency figures are undated estimates. That the hard part is keeping the decision chain coherent under fragmented venues and live latency, rather than any one model, is a synthesis, not a statement in the source.

Several items are not in the source: holidays, auctions, index events and expected halt days in the calendar (the figure shows expiries, expected news and half-days); book imbalance and queue sizes among the fast analytics; and public book updates as part of the venue connection, where the source keeps the private order messages apart from the market-data feed and notes that the two are not synchronised. Venue names, data and latencies are as of the 2018 printing.

**See also.** trading envelope; decision support; smart order routing (dark probes, latency); operational controls for automated trading.

**Definition 17.7 Trading algorithm: execution parameters, a risk-control layer that sets a trading envelope, and tactics inside it**

**Statement.** Lehalle and Laruelle describe a trading algorithm, instantiated from a class such as VWAP, TWAP, implementation shortfall, target close or liquidity seeking, as three components [alpha-laruelle, sec. 3.1.2, pp. 197–198]:

- *execution parameters*: side and quantity, the benchmark (VWAP, TWAP, percent of volume, entry price, close price), start and end times, bounds on the trading rate (maximum and minimum percent), aggressiveness, access to dark pools;
- a *risk-control layer* that computes the *trading curve*, a minimum and maximum trading rate, from models of market impact, market risk and alpha and from “expected market contexts”, balancing impact against market risk and its adverse selection;
- *trading robots* or liquidity-seeking tactics, such as a smart order router, that navigate inside the envelope set by the risk-control layer using the immediate state of liquidity; the risk-control layer assigns them short-term goals.

Because the analytics behind the curve are noisy, the curve is thick: a *trading envelope*, inside which no trajectory is a priori more efficient than another for risk control. An algorithm trading faster than its curve stops posting; one trading too slowly trades aggressively, paying the

spread [alpha-laruelle, sec. 3.4.1, pp. 231–232]. Envelopes are shown for a VWAP and an implementation-shortfall algorithm at several risk levels (Figures 3.13–3.14, pp. 232–233).

**Notation.** Interpretation: the envelope bounds the cumulative executed quantity, or equivalently its rate, as a function of time, as plotted in Figure 3.2 (p. 199) and Figure 3.13 (p. 232). It applies to the parent order; child-order size, price and venue are the tactics' decisions.

**Conditions.** Practitioner descriptions as of 2014 and 2018; broker definitions vary. How each algorithm class pursues its benchmark:

- *Benchmark against policy.* VWAP and TWAP name both a price statistic and a policy (execution benchmarks). Kissell's VWAP policy trades in proportion to the intraday volume curve and his TWAP policy evenly in time; both are static schedules fixed for the day; his POV policy trades a set fraction of market volume, so it reacts to realised volume but does not guarantee completion, and some brokers add a minimum rate to finish by the end time [s7-kissellscience, chap. 1, pp. 16–17]. Arrival-price and implementation-shortfall algorithms solve a cost–risk optimisation, the latter adding real-time adaptation to volumes and prices; their meaning differs across brokers, as does the scale of the risk-aversion parameter [s7-kissellscience, same chap., pp. 17–19].
- *Information at decision time.* Model parameters, such as $\kappa$ and $\gamma$ in the impact term $\kappa\sigma(v/V)^{\gamma}$, are estimated about once a day from history; the variables $\sigma$, $V$ and $v$ are fed in real time [alpha-laruelle, sec. 3.1.1, p. 197]. Interpretation: $V$, the market volume during the trading, is known only as it occurs, so a schedule fixed at the start uses a forecast of it, and a policy that adapts uses only the volume observed so far.
- *Participation.* Interpretation: neither source says whether the market volume a participation rate $\rho$ applies to includes the order's own fills. If it does, the order trades $\rho/(1-\rho)$ of other traders' volume (25 percent at $\rho = 20$ percent); if not, $\rho$ of it, which is $\rho/(1+\rho)$ of the total (16.7 percent). Kissell's participation-weighted price, the volume-weighted price of the first (shares traded)$/\rho$ shares, counts the order's own fills in the window (p. 108). The measurement window, the denominator and auction volume (often skipped by TWAP, p. 17) need stating.

**Practice.** Benchmarks can be combined by constraints: minimum or maximum participation rates make an algorithm behave like PoV when volume is very high or low; a limit price and a "would level" make it behave like implementation shortfall; a smart order router makes it resemble a liquidity seeker. Such hard combinations are bang-bang controls, applied fully once a threshold is crossed; they can be gamed if the threshold is detected, or adversely selected, finishing too early because of the constraints, if the conditions are not chosen accurately.

Building the logic on envelopes avoids this. Customising then has two steps: inject the desired risk profile and any volume or price constraints into the envelope, and plug in tactics suited to the investment style under defined market conditions, such as "go into dark pools if volatility increases" [alpha-laruelle, sec. 3.4.1, pp. 231–233]. The equations for optimal envelopes are cited there to Bouchard et al. (2011), Lehalle (2012) and Guéant et al. (2012).

**Pitfalls.** Lehalle and Laruelle refer to Appendix A.6 for the trading curve (p. 197), but A.6 is the Harris tick-size model; the scheduling appendix is A.7 (Interpretation, from the table of contents and p. 266). The caption of Figure 3.14 reads "for a VWAP (left) or a (right) trading algorithm", with the second name missing; the text calls the pair a VWAP and an IS (p. 232). In the source a too-fast algorithm stops posting. In its examples the would level turns an algorithm into IS and the SOR into a liquidity seeker.

Orders sent to dark pools come back as probes of their hidden books, and a firm records the messages its algorithms send and receive, with market data and context, to study their performance [alpha-laruelle, sec. 3.1.1, p. 196]. Interpretation: the envelope is a risk-control construct; following it says nothing about the cost against a benchmark measured afterwards.

**See also.** mean–variance execution (a model for the trading curve); choosing an algorithm's benchmark; execution benchmarks; pre-trade and monitoring tools; price impact (the models the risk-control layer uses, developed by `price-impact-execution`).

#### Method 17.8 Decision support for trading algorithms: pre-trade analytics, monitoring indicators and what-if scenarios

**Statement.** Before launch, pre-trade analytics use historical data, models and parameters to estimate and compare expected cost components (bid–ask spread, usual traded volume, usual quantity on the book), expected trading profiles and expected context (news, volatility levels), broken down by sector, currency and country; on portfolios they check consistency and speed some lines up and slow others down from the expected balance of market risk and impact [alpha-laruelle, sec. 3.1.3, pp. 198–199]. During trading, monitoring indicators let a trader, who may watch several hundred algorithms, find where poor performance concentrates; performance indicators estimate the result if trading stopped now or continued under unchanged conditions, from a decomposition of the average price into immediate price, market moves and impact (pp. 200–201).

What-if scenarios give real-time estimates of changed execution conditions, such as finishing now by taking the visible liquidity or converting an implementation-shortfall algorithm to percentage of volume (p. 201). Pre-trade, monitoring and post-trade tools should be compatible: for an indicator such as the bid–ask spread, pre-trade gives its expected average, monitoring its current value, updated expectation and likelihood, and post-trade its value during trading and its influence on performance, with breakdowns by sector, currency, country and side (p. 202). Monitoring indicators are built for human decisions, such as changing execution parameters on the fly, and can also be used by models to identify causes of poor performance (p. 200).

Because users must share the indicators' values with the algorithms, the algorithms themselves publish them as *indicator publishers*, publishing views on their own state in real time to users and companion algorithms (p. 202).

**Conditions.** Practitioner description as of the 2018 printing. Some what-if outcomes are simple formulas, others are estimates (p. 201). Pre-trade figures are forecasts made before the order. The average-price decomposition "follows" the scheduling appendix's (A.26) (p. 201), but (A.26) is an ex-ante objective, $\mathbb{E}(W)+\lambda\,\mathrm{Var}(W) = v^{\star}S_0+\kappa\sum_n \sigma_n v_n^{\gamma+1}/V_n^{\gamma}+\lambda\sum_n x_n^2\sigma_n^2$, labelled immediate buy, market impact and market risk [alpha-laruelle, app. A, sec. A.7.2, p. 279]: its third term is a variance penalty, not a price move.

Interpretation: the realised counterpart, from the book's change of variable (p. 278), is the identity $W = v^{\star}S_0 + \sum_n x_n\sigma_n\xi_n + \kappa\sum_n \sigma_n v_n^{\gamma+1}/V_n^{\gamma}$, dividing by $v^{\star}$ for the average price; it holds exactly inside the model, with $x_n$ the quantity remaining from slice $n$ on, in the book's timing. On real executions the impact term is a model value and the market-move term what is left, so the split is model-dependent. The refined version, adding planned price, unexpected market moves and trading efficiency, is given in words only (p. 201), so its additivity is not shown.

**Practice.** Interpretation: pre-trade estimates belong with the expected cost and risk of a schedule (mean–variance execution); realised performance belongs with implementation shortfall. The average-price decomposition is the post-trade side of transaction cost analysis; its accounting counterpart without a model is the split in transaction costs.

**Pitfalls.** Synchronisation for portfolios with common constraints, such as the same end time or constant exposure to a factor, comes after the pre-trade analysis (p. 200). The pre-trade what-if examples are finishing now and switching from IS to PoV. The immediate price is $S_0$, the price of buying everything at once without impact, and its impact term is the model's value, not an observed causal effect. The pre-, live- and post-trade principle follows p. 202.

The book calls trading algorithms "stateless agents" with introspection (p. 202) while having them publish views on their own state (pp. 200–202): its point is introspection and shared indicators.

**See also.** trading envelope; execution benchmarks; implementation shortfall.

#### Definition 17.9 Order exposure decisions: how a large trader shows, splits and routes an order, and the reactive–proactive continuum

**Statement.** Order submission strategy is the most important determinant of execution quality that traders control: market against limit orders, where to place limits, and when and how to resubmit. Large traders must also decide whether to look for the other side or wait for it, whether to

show their full interest or hide it, whether to split the order over time, whether to use one broker or several to hide their total interest, and whether to trade in one market or many; display decisions are the most important a large buy-side trader makes [alpha-harris, chap. “Buy-Side Traders”, p. 380]. *Reactive* traders trade in response to opportunities others present; *proactive* traders articulate their interests. Proactive traders expose orders to attract reactive ones, and two reactive traders never trade with each other (same chap., sec. 18.3, pp. 383–384).

**Conditions.** *Benefits and costs of exposure.* Display makes buyers and sellers easy to find and attracts reactive traders with latent interest; its costs come from how others respond [alpha-harris, chap. “Buy-Side Traders”, secs. 18.2–18.3, pp. 382–383]. *Parasitic traders*, order anticipators, use the exposed information to profit at the exposed trader’s expense; *defensive liquidity suppliers* use it to step out of the way of a large trader who would otherwise price-discriminate against them, so that the large trader pays more for liquidity (p. 383). Exposure can reveal three things: why the trader trades, the price impact of future trades, and valuable trading options (sec. 18.4, p. 384). Small traders rarely need to care, unless many small orders on one side add up to a large one (p. 383).

Cost is the main reason to be reactive: proactive traders issue instructions before knowing whether an opportunity will arise, which is wasteful when opportunities are rare, but reactive traders risk that proactive ones act first (p. 383). The reactive–proactive continuum is the willingness to bear search costs; the aggressive–passive continuum is the eagerness to trade. They are similar, not the same: aggressive traders tend to be proactive, and traders who can search cheaply tend to be proactive (p. 384).

**Practice.** Evasive strategies: brokers to preserve anonymity, several brokers so that none knows the full interest, anonymous electronic systems, careful choice of whom to show first (the art of block trading), showing only part of the order, undisplayed orders, order indications that carry no commitment, and waiting for another trader’s firm order and taking it [alpha-harris, same chap., sec. 18.5.1, pp. 386–387]. Deceptive and offensive strategies are costly and may be illegal in some jurisdictions (pp. 387–389).

*Narang’s execution algorithms.* Narang frames an algorithm’s decisions as how aggressive to be, which order type to use, what size to send and where to send it, with the goals of completeness and cheapness [s7-narang, chap. 7, “Execution”, sec. 7.1, pp. 129–130]. Market orders and marketable limit orders (a buy at the best offer, a sell at the best bid) are aggressive; a buy at or below the best bid, or a sell at or above the best offer, is passive. A passive order controls its price but may not fill, or fill only in part, and suffers adverse selection when it does fill (pp. 130–133). Liquidity rebates sweeten a passive order only if it executes, and some venues invert the fees (pp. 132–133).

He finds aggressiveness “generally” higher for momentum strategies and lower for mean reversion, and higher for stronger, more confident signals; a limit order inside the spread, possible only when the spread exceeds the minimum tick, is a middle ground (p. 133). Algorithms may also price off a “micro price” that reflects the order book’s imbalance (pp. 133–134). A hidden order “provides no information to the market”, in Narang’s words, meaning that its size and price are not displayed before it trades; it loses priority to displayed orders at the same price on most exchanges, and not all exchanges allow them; iceberging posts most of a large order as hidden pieces (pp. 131 and 134–135).

Harris describes display instructions (undisclosed, hidden, reserve or iceberg orders): the broker fills the displayed part first and then shows more, up to the maximum display; at Euronext in 2002 traders could discover the full size of an undisclosed order only by submitting large marketable fill-or-kill orders [alpha-harris, chap. 4, sec. 4.11.2, pp. 84–85]. Under display precedence, displayed orders rank ahead of undisclosed ones at the same price, the displayed and undisclosed parts of one order being treated separately (chap. 6, p. 117); Bouchaud et al. say an iceberg’s hidden part “usually has lower time priority” [catalogue-0528, chap. 3, sec. 3.2.2, p. 55]. Whether a refreshed slice keeps or loses time priority is a venue rule. Interpretation: hiding removes the pre-trade display, not all leakage.

Executions are still printed and can be read, and the SEC’s 2010 release describes immediate-or-cancel “pinging” orders used to find undisplayed liquidity, including to detect and trade in front of large interest [sec-concept-release-2010, p. 47, n. 69]; that is the order-anticipation risk in order anticipation. A hidden order lowers, rather than removes, what others can infer. Splitting a large order into pieces over time trades impact against the risk of an adverse move, with piece size set from the cost model (pp. 136–137). An algorithm’s detectable “footprint” lets others predict it and raise its costs (pp. 129–130). Routing is in smart order routing.

*How markets protect exposure*. Harris lists the rules: time precedence, with a large enough tick that front runners must pay to step ahead; undisclosed orders, which others discover only by committing to trade, searchable with fill-or-kill orders that reveal nothing if they fail; audit trails and, in some markets, prohibitions on dual trading against broker front running; and delayed trade reporting for traders building or unwinding positions, which “can greatly increase transaction costs for many types of traders” and is commoner in dealer markets [alpha-harris, chap. “Buy-Side Traders”, sec. 18.6, p. 389]. His sidebar reports that at the Paris Bourse and the Toronto Stock Exchange in 1994–1995 traders submitted more undisclosed orders when the tick was a small fraction of price (p. 389).

**Pitfalls.** Proactive execution is not the same as crossing the spread: Harris separates search (reactive–proactive) from eagerness (aggressive–passive), and a proactive trader may post a limit order. Reactive execution lowers information leakage in Harris’s evasive strategy of not exposing until trading (p. 387), but he states it as a strategy, not a regularity. Discretion ranges, protection prices, pegging and minimum fills are not in the source. Systems and venues named (Autex, POSIT, Island and others) are as of 2002. Aggressiveness is side-dependent: a limit order at the current best is fairly aggressive and one beyond it passive, so for a buy the comparison is with the offer and the bid, with the mirror image for a sale.

Fee and rebate levels, hidden-order priority and the availability of order types are venue rules; Narang notes that order types are created and retired often (pp. 134–135). Narang’s third edition treats cancellation as routine queue management and defends high cancellation rates (chap. 14, sec. 14.1.3, p. 270; Appendix, sec. A.2.2, pp. 314–315). The chapter does not discuss placing orders without intending them to execute in order to move others’ view of the market, such as small sell orders entered away from the market and cancelled by a trader who wants to buy.

For trading on or subject to the rules of a registered entity under the Commodity Exchange Act (in practice futures and swaps venues; this gloss is Interpretation), U.S. law makes unlawful any practice “of the character of, or … commonly known to the trade as, ‘spoofing’ (bidding or offering with the intent to cancel the bid or offer before execution)” [usc-7-6c, paragraph (a)(5)(C), 2024 edition]. Its status for securities and in other jurisdictions is not settled here; the passage is kept as a description of conduct, not a technique.

**See also.** block trades (order exposure as one of the four problems; sunshine trading); dark pools and crossing networks; order instructions (display and quantity instructions); order anticipation (the parasitic traders exposure attracts); trading envelope (tactics inside an algorithm). Fragmentation and dark venues are developed by `market-structure-execution-quality`.

#### Definition 17.10 Portfolio trading: program trades, the firm bid/offer market for packages, basket algorithms and index-fund trading

**Statement.** A *program trade* is the simultaneous submission of many orders; for statistics the NYSE and SEC counted trades of 15 or more coordinated transactions worth $1 million or more, about 27 percent of NYSE volume; index arbitrageurs did about 9 percent of the reported program-trade volume, about 2.4 percent of total NYSE volume [alpha-harris, chap. “Index and Portfolio Markets”, p. 489].

In the *firm bid/offer market*, also called the package, basket or portfolio trading market, package dealers quote firm prices for a whole portfolio from a characterisation of it, usually relative to its end-of-day value (for example the close minus 15 cents a share), and the list is revealed only after the close [alpha-harris, same chap., sec. 23.4.1, p. 490]. A *basket algorithm* manages cost against total basket risk, using correlations, rather than each stock’s own risk; constraints include cash balancing, self-financing and minimum and maximum participation [s7-kissellscience, chap. 1, pp. 18–19].

**Conditions.** The characterisation states whether the portfolio is an index portfolio and, if not, summary statistics: quantities, firm sizes, betas, price levels, average trading volumes, volatilities, primary listings and index membership (p. 490). Dealers price portfolios better than single stocks because informed traders are less likely to trade portfolios; clients pay them to assume the trading problem, and dealers may place parts with other clients (pp. 490–491).

Index products (index futures, index options and shares in index funds) let traders trade index risk more cheaply than the components, for three reasons Harris gives: index dealers rarely face well-informed traders, because few have valuable insight into the whole market; index markets are very active, so counterparties are easy to find and dealers turn over inventory quickly; and a single transaction is arranged, cleared and settled instead of many [alpha-harris, same chap., p. 484 and sec. 23.4, p. 489]. Index prices generally lead the cash index, because index traders price only index risk (p. 489).

**Practice.** Index funds rebalance a value-weighted portfolio only when the index list changes, and a price-weighted one also at splits [alpha-harris, same chap., sec. 23.2, p. 486]. They can improve returns by supplying rather than demanding liquidity, substituting non-index stocks for expensive components, rebalancing only with flows and dividends, and holding a subset of the index; each tactic raises tracking error, so many funds use them sparingly (p. 487). A basket algorithm may slow a stock despite good liquidity, or speed one despite poor liquidity, if that reduces residual basket risk (Kissell, pp. 18–19).

Narang describes the same service as a *portfolio bid*: the trader describes a "blind" portfolio by its characteristics, such as the valuation ratios of longs and shorts, sector breakdown and market capitalisations; brokers quote a fee in basis points of the gross market value traded and guarantee the price, and the broker then bears the risk of trading out of the portfolio at future prices [s7-narang, chap. 7, "Execution", p. 128]. "Human" execution of quant portfolios "often looks like" a portfolio bid (p. 128).

**Pitfalls.** Basket, package and portfolio trades are not different markets: Harris uses the three names for one market. Tight index-basket quotes come, in Harris, from less adverse selection; diversified residual risk may also help but is not in the passage (Interpretation). Harris names non-index stocks as substitutes; sector, currency, country, factor loadings and borrow constraints as characterisation inputs, and futures and ETFs as rebalance substitutes, go beyond him, and the timing around reconstitution is in index reconstitution. Index flow is often said to cost less to trade because it is less security-specific, more active, easier to net and easier to hedge with futures, ETFs, options or baskets.

The first two match Harris's adverse-selection and activity reasons; netting and hedging are not among his reasons, and his third is that a package is a single transaction to arrange and settle. "Portfolio transition" is named by Kissell ("transition management", p. 16) but not defined. The NYSE figures are as of 2002.

**See also.** block trades (single-stock size); index reconstitution (price pressure from forced index trading); ETF creation and redemption; trading envelope; pre-trade analytics (portfolio consistency checks).

#### Mechanism 17.11 Positive-feedback hedging: portfolio insurance sells into falls and buys into rises, and its trades reinforce the move

**Statement.** Portfolio insurance is a dynamic trading strategy that replicates a portfolio plus a put option; the formula for how much to trade comes from the Black–Scholes model and depends on the portfolio's volatility, the protection period and the guaranteed minimum value. Insurers buy when prices rise and sell when they fall, the same effect as stop orders, which many used. It works when the money covered is small, but when a large fraction of the market is insured, sellers arrive together and prices can fall substantially; order anticipators sell at the first sign of a fall and buyers withdraw, so expected insurance selling can drive prices down even before it occurs [alpha-harris, chap. "Bubbles, Crashes, and Circuit Breakers", sec. 28.2.2, pp. 560–561].

Interpretation: if a rule holds $q(S)$ shares with $q'(S) > 0$, a fall $dS < 0$ gives $dq = q'(S)\,dS < 0$, a sale; if each trade moves the price by $\lambda\,dq$ with $\lambda > 0$, the induced move has the sign of the original one.

**Conditions.** Interpretation: if the induced move persists until the rule trades again and triggers the same response, the moves form a geometric series with ratio $\lambda q'(S)$, which is dimensionless ($\lambda$ in currency per share, $q'$ in shares per unit of currency); the total is $dS/(1-\lambda q'(S))$ for $\lambda q' < 1$, for example $1.67\,dS$ at $\lambda q' = 0.4$. Purely temporary impact that reverts before the next rebalance gives no such amplification. Harris notes that exchange-listed puts would have made the insured amount visible through open interest and reduced the uncertainty that kept buyers away (p. 561).

**Practice.** Harris's rough sizing for 1987: about $100 billion insured, a typical delta of about 0.4, so a 1 percent fall would require selling about 0.4 percent, roughly 10 million shares at $40, against average daily volume of 160 million shares (box, p. 561). Insurers who raised the protected value as markets rose had to sell more on the way down (box, p. 560).

**Pitfalls.** Interpretation: the size of a rebalancing trade depends on how much the hedge ratio changes, the option's gamma, not on its delta. In a Black–Scholes illustration with zero rates, 20 percent volatility and one year to expiry, a 1 percent fall requires selling about 1.0, 1.6 and 1.8 percent of the insured value for protection at 80, 90 and 100 percent of the current value. Harris's box multiplies the fall by the delta, so its 10 million shares is an order-of-magnitude guess on a different basis. The argument shows only the sign of the effect, not its size or stability.

**See also.** bubbles and crashes (the 1987 crash); price impact; portfolio trading.

## 17.4 Transaction costs and transaction cost analysis

#### Definition 17.12 Transaction costs: explicit, implicit and missed-trade costs, and an accounting split of shortfall

**Statement.** Transaction costs include all costs associated with trading [alpha-harris, sec. 21.1, pp. 421–422]:

- *explicit costs*, which a cost accountant would identify: commissions, exchange fees, taxes, and the resources devoted to trading, such as running a buy-side desk;
- *implicit costs*, which arise because traders affect prices: the bid–ask spread paid by buying at the ask and selling at the bid, and the price impact of large orders;
- *missed trade opportunity costs*, when orders fail to fill, or fill late. Harris's example: a cotton buyer who bid 64.95 cents when the price was 65 and saw it rise to 68 lost 2.75 cents a pound against an aggressive fill at 65.25, $137,500 on 100 contracts.

Explicit costs are easy to measure. Implicit and missed-trade costs need a benchmark: what prices would have been without the trade, or what the unfilled order would have cost to complete (p. 422). Kissell lists ten components, commissions, fees, taxes, rebates, spreads, delay cost, price appreciation, market impact, timing risk and opportunity cost, and classifies them as investment-related, trading-related and opportunity costs, fixed or variable, visible or hidden [s7-kissellscience, chap. 3, pp. 89–95, Tables 3.1–3.2].

Narang names three major components for a quant's cost model [s7-narang, chap. 5, "Transaction Cost Models", sec. 5.1, pp. 86–91]: *commissions and fees*, paid to brokerages, exchanges and regulators, small per trade for quants who use only a bank's infrastructure, with brokers' clearing and settlement fees usually inside the commission (pp. 86–87); *slippage*, "the change in the price between the time a trader (or quant system) decides to transact and the time when the order is actually at the exchange for execution" (p. 87); and *market impact*, how much an order moves the market by its demand for liquidity, "normally defined as the difference between the price at the time a market order enters the exchange and the price at which the trade is actually executed" (pp. 88–89).

His terms map onto the ones above as set out under Pitfalls.

**Notation.** Interpretation: with the signed quantities of implementation shortfall ($S$ intended, $s_j$ filled at $p_j$, decision price $P_d$, arrival midpoint $P_0$, end price $P_n$) and $m_j$ the quotation midpoint just before fill $j$, shortfall splits additively as

$$\text{IS} = \underbrace{S(P_0 - P_d)}_{\text{delay}} + \underbrace{\textstyle\sum_j s_j (p_j - m_j)}_{\text{half-spread paid}} + \underbrace{\textstyle\sum_j s_j (m_j - P_0)}_{\text{midpoint drift}} + \underbrace{\left(S - \textstyle\sum_j s_j\right)(P_n - P_0)}_{\text{opportunity}} + \text{fees}.$$

The second term is the filled size times Harris's liquidity premium, half the effective spread at each fill (spread measures); the middle two terms are Wagner's trading-related part. A buy of 10,000 shares (decision 50.00, arrival 50.10, fills of 3,000 at 50.20 and 50.30 against pre-fill midpoints 50.15 and 50.25, rest cancelled, $P_n = 50.60$, fees 60) gives $1{,}000 + 300 + 600 + 2{,}000 + 60 = 3{,}960$, the paper-minus-real total; the mirror-image sell gives the same.

**Conditions.** Timeless (accounting); Narang's account as of 2024. The split is an accounting identity, not a causal attribution. The midpoint drift mixes the order's own impact with market movement and other traders' impact. Kissell defines market impact as the difference between the price path with the order and without it, which cannot be observed together, so impact cannot be measured exactly [s7-kissellscience, chap. 3, pp. 91–92]; Harris likewise needs a no-trade benchmark (p. 422). The half-spread term measures each fill against its own contemporaneous midpoint, so it misses the cumulative impact of earlier fills, which falls into the drift (Harris's split-order example, pp. 428–429). Kissell's timing risk is the uncertainty of cost, not a realised component (p. 92), and his price appreciation is the price trend that would have occurred anyway (p. 91).

**Practice.** Interpretation: an index or market adjustment (transaction cost analysis) removes the part of the drift predicted by the market's move; what remains is still a residual, which includes impact, stock-specific news and model error, and is impact only under the model used to remove the rest. Of a split into spread, impact, timing and fees, spread, delay (timing before arrival), opportunity and fees are measurable as above, but "impact" is available only as a model estimate or a market-adjusted residual, not as an observed component.

**Pitfalls.** Count each cost once: the spread is inside fill prices, so adding a quoted or effective spread to an arrival-price shortfall counts it twice; fees enter once. Adverse selection and delay are not implicit costs relative to a benchmark: in Harris adverse selection is a cause of the spread, and in Kissell delay is an investment-related cost before the order reaches the market. Borrow charges are not among the source's explicit costs. Kissell describes the spread as the round-trip cost for small orders, not blocks (p. 90).

*Narang's terms.* Interpretation of the mapping, with his claims kept as his:

- *Slippage* runs from the decision to the order's arrival at the exchange, so it is the delay term $S(P_0 - P_d)$ above, signed by side: his sale decided at \$100.00 that reaches the market at \$99.90 costs \$0.10 a share, and a move the other way is a gain (pp. 87–88). The word is not used uniformly (interpretation); name the benchmark and the interval.
- *Market impact* in his definition runs from arrival to fill. It is therefore the trading-related part of shortfall (half-spread paid plus midpoint drift), not impact in the causal sense of price impact. His own example shows the difference: a sale decided at \$100.00 reaches the market at \$100.05 and fills at \$100.20 in a rising market, giving slippage of −\$0.05 and "impact" of −\$0.15. That is a negative arrival-to-fill cost against a benchmark; it does not identify what the order itself did to the price, and he concedes that impact may be impossible to separate from slippage or from the stock's own move (pp. 89–90).
- *Known after execution?* Narang writes that one does not know how much a trade impacts the market "until the trade has already been completed" (p. 89). After completion the fills and the realised cost against a benchmark are known; the impact is not, because the price that would have prevailed without the order is never observed (Kissell, above; price impact). It is estimated by a model or across many orders.
- *Conditional claims.* That trend-following strategies suffer most from slippage and mean-reverting ones least, sometimes gaining (p. 88), is a statement about the expected sign of the price move over the delay relative to the trade's side, and holds only to the extent that prices continue in the direction a strategy trades or revert against it over that interval; he gives no data.
  Lower latency reduces slippage by shortening the delay (p. 88), and higher volatility raises it (pp. 88 and 91); interpretation: with no expected drift over the delay, a longer delay or a higher volatility widens the spread of outcomes rather than raising their mean, and raises the expected cost only together with an adverse drift, such as an accurate short-term forecast that the market starts to price, which is his own condition (p. 88). His GOOG figure, an average daily range 16 times the average day-to-day move, is undated and unsourced.
- *Fees.* The pairing of brokerages with access, exchanges with security and regulators with infrastructure reads his list "respectively"; the text does not pair them, and on p. 87 credits exchanges with access to liquidity and with guaranteeing that both counterparties perform. Interpretation: in current market structure that guarantee is given by a central counterparty, which "interposes itself between counterparties" and ensures the performance of open contracts [exp-pfmi, para. 1.13, p. 9], whether or not the exchange operates it. His fixed charges as a floor "below which the expense of trading cannot go" (p. 91) ignore the liquidity rebates he describes on p. 90 (about 0.2 cents a share to a provider against 0.3 cents charged to a taker, the venue keeping about 0.1).
- *Clearing and settlement.* Narang's clearing, "regulatory reporting and monitoring, tax handling, and handling failure" before settlement, and his settlement, "the delivery of securities in exchange for payment in full" (pp. 86–87), are his descriptions of what brokers charge for, not standard definitions. The CPSS–IOSCO principles describe securities settlement systems that transfer securities "either free of payment or against payment", with delivery versus payment when "delivery of the security occurs if and only if payment occurs", and that may add clearing functions "such as the confirmation of trade and settlement instructions" [exp-pfmi, para. 1.12, p. 8], and central counterparties that net trades multilaterally and take margin [exp-pfmi, para. 1.13, p. 9]. Standard definitions are those of the principles' glossary [exp-pfmi, Annex H, pp. 174–178].
  *Clearing* is "the process of transmitting, reconciling, and, in some cases, confirming transactions prior to settlement, potentially including the netting of transactions and the establishment of final positions for settlement"; the term is "sometimes … also used (imprecisely) to cover settlement", and for futures and options also means "the daily balancing of profits and losses and the daily calculation of collateral requirements" (p. 175).
  *Final settlement* is "the irrevocable and unconditional transfer of an asset or financial instrument, or the discharge of an obligation by the FMI or its participants in accordance with the terms of the underlying contract", "a legally defined moment" (p. 176). *Delivery versus payment* is "a securities settlement mechanism that links a securities transfer and a funds transfer in such a way as to ensure that delivery occurs if and only if the corresponding payment occurs" (p. 176); without it the seller bears *principal risk*, delivering irrevocably "but not receive payment" (p. 178).
  A *central counterparty* "interposes itself between counterparties …, becoming the buyer to every seller and the seller to every buyer and thereby ensuring the performance of open contracts" (p. 174), by *novation*, which replaces the original contract with two new ones, or by *open offer* at the time of the trade (p. 177). So Narang's "clearing" bundles post-trade services (reporting, tax handling, fails) that the standard term does not include, and his settlement is final settlement under DvP; who guarantees performance is the CCP, not the exchange as such or the broker (Interpretation). Harris's description of the same post-trade chain is in market structures and venues.

**See also.** implementation shortfall; execution benchmarks; spread measures; price impact; transaction cost analysis.

### Method 17.13 Transaction cost analysis: measuring costs after the trade, predicting them before, and comparing brokers and algorithms

**Statement.** Traders measure transaction costs to evaluate how well they and their brokers implemented their strategies, whether they traded too aggressively or not enough, and which brokers should get future orders; they predict costs to judge whether strategies are worth implementing. Exchanges, brokers (notably to show best execution when paid for order flow), sponsors (ERISA commission duties) and regulators measure them too [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", pp. 420–421].

Lehalle and Laruelle separate post-trade analysis, usually of one day, which explains what the market context did, from TCA over months of trades, which uses averaging to cancel most market effects and find what part of performance comes from the algorithm's design; its outcome is to match an algorithm or provider to a market context or trading style, so that flow is routed accordingly, using breakdowns by volatility, jumps, liquidity and news, and comparing only stocks of similar liquidity (bid–ask spread, tick size in basis points, quantity on the book, turnover, trade size, free float, trading rate) [alpha-laruelle, sec. 3.1.3, pp. 202–203].

**Notation.** Kissell's measures [s7-kissellscience, chap. 3, pp. 110–113], over the trading horizon, are

$$\text{index cost} = \frac{\text{index VWAP} - \text{index arrival}}{\text{index arrival}} \times 10^4,$$
$$\text{index-adjusted cost} = \text{arrival cost} - \hat{\beta}\,\text{index cost},$$
$$Z = \frac{\text{pre-trade estimate} - \text{arrival cost}}{\text{pre-trade timing risk}}.$$

These are eqs. (3.21)–(3.23); $Z > 0$ means the trade beat its estimate. A cost *measure* is ex post and single-valued; a cost *forecast* is ex ante, a mean with a standard deviation, the timing risk (pp. 95–96).

**Conditions.** Practitioner methods as of 2002, 2014 and 2018. Prediction: explicit information (contracted fees and displayed liquidity, an upper bound on impact for orders smaller than displayed size) and implicit information from past costs, modelled by regression on order variables (size, price placement), contemporaneous market variables (spread, displayed size, recent volume, recent returns, money flow) and general ones (average volume, volatility, market capitalisation); past costs predict future ones only if liquidity conditions are unchanged [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", sec. 21.7, pp. 438–439]. Comparison: estimates from single trades are noisy, averaging removes noise but not bias or gaming, and comparing brokers is meaningful only for similar trading problems (p. 432).

Kissell's $Z$ should be near mean zero and variance one if pre-trade estimates are accurate; the test should be run by order size and strategy, on shares actually traded rather than the full order, on enough orders, and with the investor's own pre-trade model rather than the broker's (pp. 112–113).

**Practice.** Kissell separates cost from performance: comparing the execution price with the day's VWAP is a performance proxy, with the close a tracking-error proxy, while comparison with the open or the arrival price is a cost (p. 96). Harris advises that strategists and traders share why they trade, and that incentives reward joint performance rather than blame (pp. 439–440).

**Pitfalls.** Interpretation: the population, aggregation weights and benchmark must be stated: averaging by order and by value answer different questions, and both depend on which orders reached each broker. Lehalle and Laruelle's averaging cancels market effects only if they are unrelated to which algorithm was chosen; if traders send urgent or difficult orders to one provider, its average cost is worse whatever its quality, so a routing table built on unadjusted averages confounds selection with skill, and outperformance of a benchmark does not show causal superiority. The failure modes (wrong exposure, wrong clock, own-impact contamination, gaming, selection) are consistent with Harris's biases (sec. 21.4) and Kissell's conditions but are a synthesis, not a list in either.

Kissell's index cost is "not adjusted for the side of the trade" (p. 112); with a side-signed arrival cost this is wrong for sells: a sale at 50 basis points below arrival while the index fell 50 basis points and $\hat{\beta} = 1$ shows 100 basis points of adjusted cost instead of 0, so the index term needs the side factor (Interpretation). His approximation "Index VWAP $= \frac{1}{2}R_m$" is an approximation to the index cost, not the index VWAP. Venue variables (hidden fill rates, fee tiers, queue loss, routing latency), limit price and urgency are not among the source's prediction variables.

**See also.** transaction costs; execution benchmarks; pre-trade, monitoring and post-trade tools; aggressiveness trade-off; best execution; the market adjustment parallels relative performance evaluation of managers.

#### Method 17.14 Trading aggressiveness: balancing the transaction cost of trading harder against the missed-trade cost of trading softer

**Statement.** If the additional transaction costs of trading more aggressively are less than the missed trade opportunity costs it would save, traders earn more by trading more aggressively; if the transaction costs saved by trading less aggressively exceed the additional opportunity costs, they earn more by trading less aggressively. Most traders do not estimate these marginal costs but assume they equal average costs, trading harder or softer as their per-unit opportunity costs exceed or fall short of their per-unit transaction costs; Harris advises estimating marginal transaction costs from the costs of the last trades that fill orders [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", sec. 21.6, pp. 437–438].

**Conditions.** Interpretation: the objective is expected total cost, transaction cost plus missed-trade cost, of an order whose size is given; the choice variable is aggressiveness (limit placement, urgency, participation); the rule is the first-order condition of that minimisation and assumes both costs change smoothly with aggressiveness and that the missed-trade cost is measured realistically. Harris states it as advice, with no model or proof. Opportunity costs count only for orders that could have been filled, and depend on the horizon at which they are measured (pp. 435–436).

**Practice.** Well-informed traders, especially on short-term information, should accept substantial transaction costs rather than fail to trade, and strategists should give traders price limits that reflect the value of their information (p. 439). Interpretation: a mean–variance schedule (Almgren–Chriss) is one formal version of the same trade-off, with price risk in place of missed-trade cost.

**Pitfalls.** For the last slice the control variable is marginal cost: Harris's point is that marginal, not average, costs should drive the decision, and that the last fills estimate them. Interpretation: comparing average costs gives the right direction only if marginal and average costs move together.

**See also.** transaction costs; implementation shortfall (the opportunity term); mean–variance execution; order exposure decisions.

## 17.5 Best execution and broker conflicts

#### Definition 17.15 Best execution: three standards, the dimensions of execution quality, and why retail clients see commissions but not prices

**Statement.** A broker taking an order assumes an agency responsibility to obtain best execution, which is not well defined. To unsophisticated clients it means the best possible price for a market order and the fastest execution for a limit order, an absolute standard, which is what the term generally means in U.S. equity markets; to more sophisticated traders, the execution they pay for; to the most sophisticated, the execution they can expect given what they pay and how well they can audit it [alpha-harris, chap. "Brokers", sec. 7.4.2, pp. 160–161].

Price is one dimension of execution quality and speed another; market-order traders care most about speed, limit-order traders about price, and best execution for standing limit orders depends on whether and when they execute [alpha-harris, chap. "Internalization, Preferencing, and Crossing", sec. 25.1, p. 515]. Kissell defines best execution as the process of choosing the strategy most likely to achieve the fund's investment objective, judged on the information available when trading began, not by closeness to a benchmark price [s7-kissellscience, chap. 3, p. 88].

**Conditions.** In U.S. equities as Harris describes them in 2002, dealers claim best execution when they fill marketable orders at the NBBO, a standard that applies only to orders smaller than the displayed size; some markets also expect average price improvement for small market orders (p. 515). Brokers that internalise or accept payment for order flow have a conflict of interest and must regularly and rigorously examine execution quality; SEC Rules 11Ac1-5 and 11Ac1-6 required market centres to publish monthly execution-quality statistics and brokers quarterly routing reports (pp. 517–518, boxes).

Retail traders can audit commissions and response times easily but not price quality, which needs trade and quote data and norms for comparison; so brokers compete on commissions, and in competitive markets the total of spread and commission for small market orders does not depend on how best execution is defined (pp. 517–518). Lehalle and Laruelle contrast two regimes as of 2018: in the U.S. the trade-through rule, they write, makes venues reroute orders marketable at a better price elsewhere, which needs a consolidated pre-trade view; in Europe the executing broker chooses

how to consolidate access to liquidity and must disclose and justify a best-execution policy, raising the cost of fragmentation for intermediaries [alpha-laruelle, sec. 3.4.2, p. 234].

**Practice.** Retail execution quality is measured by price against the NBBO, the midpoint or the effective spread (spread measures), speed, whether and when limit orders execute, and commissions; audit firms compare each night's trades with quotes and report exceptions (Harris, pp. 515–519). Interpretation: these measures depend on order type and size and on the quotes at the time; comparisons between venues need like orders.

*Best execution after fragmentation*. In Lehalle and Laruelle's account MiFID left each broker to define best execution in its own policy, and best execution focused on "immediate price improvement" for aggressive orders from snapshots of visible books; after fragmentation they argue it must cover consolidated access for passive and aggressive orders, with adverse selection, passive splits and reversal as measures judged over the order originator's horizon, and must account for clearing and settlement costs that differ by venue [alpha-laruelle, Introduction, pp. 17–19]; in Europe, best-execution policies took the place of the U.S. trade-through rule [alpha-laruelle, chap. 1, sec. 1.1.2, p. 43]. Execution quality therefore depends on fees, fill probability, latency, hidden liquidity, access, size and horizon as well as displayed price.

**Pitfalls.** Three obligations should not be merged: *quote protection*, *best execution* and *execution-quality reporting*. Quote protection in the U.S. is the order protection rule, 17 CFR 242.611 (eCFR text for 13 October 2026; adopted 70 FR 37620, 29 June 2005; last amended 89 FR 26616, 15 April 2024): it requires a *trading center* to maintain written policies reasonably designed to prevent trade-throughs of protected quotations in NMS stocks, and to surveil them, with exceptions including system failure, non-regular-way trades, single-priced opening, reopening and closing transactions, crossed protected quotations, intermarket sweep orders (executed, or routed simultaneously against the full displayed size of protected quotations), orders whose price is not based on the quote, venues whose best quote within the previous second was equal to or worse than the trade-through price, and stopped customer orders (paragraphs (a)–(b)(9)).

It is a duty of the venue, not the broker's best-execution duty, and it does not oblige a venue to reroute: an intermarket sweep order lets the router take the protected quotations itself. Lehalle and Laruelle's claims that U.S. clearing and settlement fees must be equal across venues, and their description of the European regime, are as of their 2018 printing. U.S. routing leans on protected consolidated quotes, while European brokers choose the consolidation layer (their p. 234). Defining best execution as the best auditable outcome given the client's objective, limit, urgency, available venues and information risk merges Harris's third standard (auditability) with Kissell's process definition, which do not coincide.

Post-fill price movement as a retail adverse-selection measure is a realised-spread idea (spread measures), not part of Harris's best-execution discussion. Rules 11Ac1-5 and 11Ac1-6 are now Rules 605 and 606; the expanded Rule 605 reports required from August 2026, and the 2025 withdrawal of the SEC's best-execution proposal, postdate these statements, which are as of 2002 (`ex-sec-rule605-2024`, `ex-sec-withdrawal-2025`).

**See also.** broker conflicts; transaction cost analysis; spread measures.

#### Definition 17.16 Broker conflicts: the principal–agent problem in order handling, from payment for order flow to front running

**Statement.** Brokers are agents whose interests can differ from their clients'; clients manage the problem with rewards and penalties, which require measuring performance, and measuring execution quality is hard [alpha-harris, chap. "Brokers", secs. 7.4–7.4.1, pp. 159–160]. Harris's conflicts (same chap.):

- *dual trading*: a broker-dealer that internalises a client order wants the opposite price to the client, and may trade ahead of clients on the same side (sec. 7.4.3, p. 161);
- *order preferencing*: routing to a preferred dealer, an own dealer subsidiary, or in exchange for reciprocal flow or payment for order flow, about 1 cent a share in U.S. stocks in 2002, independently of current quotes (sec. 7.4.4, pp. 161–162);
- *soft dollars and directed brokerage*: commissions that buy research or other services, and sponsors directing orders to particular brokers or recapturing commissions (pp. 153–155);
- *dishonest practices*: front running, inappropriate order exposure, fraudulent trade assignment of the best fills to favoured clients, prearranged trading and kickbacks, unauthorised trading and churning (sec. 7.5, pp. 163–167).

Kissell adds venue rebates: when brokers keep maker–taker rebates or pay the fees, routing may serve the broker's net fees rather than the client [s7-kissellscience, chap. 3, p. 90]. Interpretation: if routing to $v$ costs the client $C_v$ in expectation and earns the broker $h_v$, the client's choice is $\arg\min_v C_v$ and a broker maximising its own net would choose $\arg\min_v (C_v - h_v)$; they differ exactly when some $a, b$ have $C_a < C_b$ but $C_a - h_a > C_b - h_b$. This states when a conflict can arise; it is not evidence that brokers route that way, since reputation, regulation and competition for flow also enter the broker's objective.

**Conditions.** Practices and figures as of 2002 and 2014. Payments for order flow resemble kickbacks but generally are not, since competing dealers pass the value of order flow to brokers and brokers to clients through lower commissions (Harris, pp. 166 and 516–517). Dishonesty is likeliest where it is hard to detect; audit trails, supervision and reputation deter it (pp. 163–165).

**Practice.** Interpretation: transaction cost analysis and audit trails turn suspected conflicts into testable comparisons, such as fill quality by venue against the NBBO or price improvement by routing destination, but a comparison shows a conflict's effect only when orders are comparable. *Order-flow arrangements*. Harris adds that brokers who cross client orders internally to earn two commissions and print them at a market with a thin book bypass standing limit orders with time precedence [alpha-harris, chap. "Competition Within and Among Markets", box "Lost Time Precedence", p. 537]. Since the 2024 MiFIR amendments, EU investment firms may not receive payment for forwarding or executing retail clients' orders, subject to a possible national exemption until 30 June 2026 (EU trading venues and transparency).

**Pitfalls.** Order preferencing for service, economics or reciprocal flow is Harris's; reviewing it means comparing all-in fills rather than routing intent, an interpretation consistent with his best-execution standards. Commission recapture is not a broker's activity incentive: Harris treats it as a sponsor's tool to reduce the cost of soft-dollar commissions. Figures and rules are as of 2002.

**See also.** best execution; order anticipation (front running); order exposure decisions.

## 17.6 Smart order routing across fragmented venues

#### Definition 17.17 Smart order routing: splitting a marketable order across venues, and why displayed liquidity is only probable

**Statement.** When several venues trade the same instrument, marketability is venue-specific: with 50 shares offered at 10.00 on one venue and 150 at 11.00 on another, a buy limit at 10.50 is marketable on the first only [alpha-laruelle, sec. 3.4.2, pp. 233–234]. A smart order router (SOR) decides where the child orders of a trading algorithm go; in Lehalle and Laruelle's architecture it is one of the tactics working inside the envelope set by the risk-control layer (trading envelope). Their example (Table 3.3, p. 235) is a market buy of 500 shares against three European books, routed by three brokers (pp. 235–236):

- B1 has Euronext and Chi-X and splits for the best average price: 101.15;
- B2 has all three venues but sends the whole order to the single venue with the best price: the text says 102.50 on Chi-X;
- B3 has all three and splits across them: 100.75, with a free choice of venue, "surely" by fees, for the last 50 shares at 102.00.

With latency or duplicated ("mirrored") liquidity the ranking changes. If a high-frequency trader co-located at Euronext and Chi-X mirrors 150 shares on Turquoise, which the brokers reach 2 ms later than Euronext, it cancels them before B3's order arrives; B3 learns of the 150 unfilled shares about

6 ms later and, with prices rising, ends up worse than B1. What matters is not the visible liquidity but the probability of obtaining it at a price when the order arrives, given latency differences, mirrored orders and hidden liquidity (pp. 238–239).

*Temporal, then spatial.* Large orders are first split through time, balancing market impact against market risk, and in a fragmented market each child order is then split across venues [alpha-laruelle, chap. 1, sec. 1.2.1, pp. 62–64]. Given a child order's limit price and quantity, a split is optimal if it obtains a price at least as good as the limit and finishes as soon as possible, the timing having been decided above it; an SOR is an "agnostic device" that takes a limit or market order and splits it with knowledge of the available liquidity, and under MiFID it must be able to prove that each decision followed the agreed execution policy, which needs records of policies, market snapshots, decisions and executions (pp. 64–66).

Its efficiency is judged by results, not latency: for a given order, at a price at least as good as the main market alone, the more executed on other venues the better (pp. 70–71). Aggregating books lowers the cost of a marketable order, which would otherwise walk one venue's book (Figure 1.16, sell order of 660 shares on Euronext and Chi-X, p. 68).

**Notation.** Interpretation, for the routing decision at the time a child order is released: the inputs are the visible books, the venues the broker's policy allows, the latencies to each, fees, and estimates of hidden liquidity and fill rates; the decision is a quantity per venue (and, for passive orders, a price); the constraints are the child order's size, side and limit and the parent's envelope. The SOR does not decide how much the parent trades when; that is the schedule (mean–variance execution).

**Conditions.** European venues and latencies as of 2012–2018; Harris as of 2002. The example assumes no latency issue for the base case (p. 235) and full access to each venue's displayed depth. Liquidity seeking for a marketable order must allow for hidden liquidity and for the chance that a fast trader cancels when it sees the order reach another venue first; limit orders raise further issues (p. 239).

A liquidity seeker removes liquidity using estimates of hidden liquidity (including dark pools), mirrored orders and latency, and posts limit orders to finish fast. It is an "enhanced limit order" lasting from about 10 minutes in liquid stocks to a few hours in illiquid ones, while adverse selection from finishing fast is left to a scheduling layer at 10 minutes to a few hours. The hard part is estimating fill rates and real liquidity, which cluster (pp. 242–243).

**Practice.** *Duplicate liquidity and transparency.* High-frequency market makers quote on several venues and cancel the rest when one fills, so aggregated pre-trade depth overstates what is executable; with pegged and iceberg orders too, "pre-trade transparency is not very useful for guessing where liquidity really is", and post-trade reports say more, which made small dark pools with post-trade transparency more attractive (chap. 1, pp. 69 and 71). An order's path has latency at each step (consolidating feeds, deciding, reaching venues, matching, returning reports): about 4 ms between Paris and London over a 2012 network backbone (p. 70).

*Average trade size* fell after 2007 while the number of trades rose; because a trade's size is the smaller of the two sides, falling average trade size mostly reflects smaller resting orders spread over more venues, not smaller investor orders (Figure 1.15, FTSE 100, pp. 66–67). *Consolidation.* Harris's three mechanisms join a fragmented market: traders in each segment react to information from others, liquidity demanders and suppliers route to the segments where they do best, and arbitrageurs move liquidity between segments, so that "if an order cannot move to its best market, arbitrageurs will move the best market to the order"; all three need cheap public information about every segment [alpha-harris, chap. 26, sec. 26.4, pp. 533–534].

Interpretation, derived: if an executable ask $a_1$ on one venue and an executable bid $b_2$ on another satisfy $b_2 - a_1 > k$, the all-in cost of buying on one and selling on the other, buying and selling earns $b_2 - a_1 - k$ per unit until prices move or capacity binds; the bound needs both quotes to be executable for the size and the costs to include fees, clearing and inventory. Lehalle and Laruelle allocate across venues from "fill rate curves", the expected fill intensity against distance from the best opposite price, which differ by venue (Figure 3.17, pp. 243–244), citing Laruelle et al. (2011, 2013). Passive splits are in passive order split; fee effects in fee-adjusted venue choice.

*Latency and the state of the book*. Lehalle and Laruelle's routing loop takes $\tau$ for a snapshot of the books to reach the router, $\epsilon$ to decide and $\tau$ for the order to reach the venue, $2\tau + \epsilon$ in all; when books update faster than that, the snapshot "has to be replaced by real-time estimates of the quantity that is really available" [alpha-laruelle, Introduction, p. 22]. They call the threshold a "Nyquist–Shannon" result: an observer "loses information" once the update frequency exceeds half the inverse of the latency, with a "latency bound" of $\frac{1}{2} \cdot 1/f$; 2 ms corresponds to 250 Hz, and a Paris–London round trip of about 8 ms in 2012 to 125 Hz (Proposition 1 and Definition 1, p. 23).

Derived here: the 2 is the round trip, so the condition compares the interval between updates with the round trip, $f > 1/(2\tau)$; it is not the sampling theorem, which concerns reconstructing a band-limited signal from samples. With updates arriving as a Poisson process of rate $f$, the book is unchanged over a round trip $T$ with probability $e^{-fT}$: 0.37 at their threshold $fT = 1$ and 0.14 at $fT = 2$. There is no sharp boundary at which deterministic use of the displayed book fails;

**Pitfalls.** Interpretation, from Table 3.3 by sweeping the cheapest levels first: B1 101.15 and B3 100.75 are right; the best single venue for all 500 shares is Chi-X at 102.05 (Euronext 102.10, Turquoise 102.20), so B2's "102.50" looks like transposed digits. The bracketed quantities for B1, $(25+50+100+100)$ and $(0+25+100+200)$, add to 600 and list depth up to 102.00, not what was bought (200 of the 300 shares at 102.00). Sending to the cheapest visible venue first is not B2's rule, which sends the order to one venue.

Routing logic that weighs queue position, reject history, gateway and matching-engine delays and the information cost of probing, in a loop that updates venue scores from fills, is a reasonable synthesis but not in the source, which names only fees, latency, hidden and mirrored liquidity and fill-rate estimates. Venue-selection criteria (price and depth, execution probability, fees, access, information) and an objective "expected cost = spread/impact + fees − rebates + latency risk + information cost" are likewise a synthesis: the terms are not in common units, and the cost of a missed or late fill is missing. Displayed depth cannot be assumed to stay available while a sequence of child orders is routed. Venue names and the HFT example describe European markets around 2018.

**See also.** trading envelope (routers as tactics); order exposure decisions (whether to show and where); best execution (routing duties and price protection); passive order split; fee-adjusted venue choice; SOR and market-share dynamics; venue market share; European fragmentation evidence; dark pools; HFT strategies (probing for liquidity against order anticipation) and latency-arbitrage arms race.

### Result 17.18 Fee-adjusted venue choice: a worse price with lower fees wins only when the price step is smaller than the fee difference

**Statement.** For a buy of $Q$ shares, taking liquidity on venue $V$ at price $P$ with proportional fee $f$, against venue $V'$ at $P(1+\delta)$ with fee $f - \phi$, $V'$ is cheaper when

$$P(1+\delta)(1+f-\phi)Q < P(1+f)Q \iff \delta < \left(1 - \frac{\phi}{1+f}\right)^{-1} - 1 \simeq \phi,$$

so a worse price is worth taking only if the tick, in relative terms, is about as small as the fee difference: at a price of 100 euros and a fee difference of 0.1 basis point, the tick would have to be below 0.1 cent [alpha-laruelle, sec. 3.4.2, pp. 236–237]. Figure 3.16 plots the critical tick against the fee for $\rho = \phi/f$ (p. 237). Liquidity providers, who pay a fee or receive a rebate when their limit orders fill, would post first on the venue cheapest for them, and market orders would go first to the venue cheapest for takers; if these differ, both cannot persist in equilibrium (pp. 237–238).

**Notation.** $f$ and $\phi$ are fees as fractions of traded value, paid by the taker; a rebate is a negative fee. $\delta$ is the relative price difference, at least one tick.

**Conditions.** Timeless (arithmetic). Interpretation: the threshold is exactly $\delta < \phi/(1+f-\phi)$ for a buy, comparing acquisition cost. For a sale, comparing proceeds, $V'$ at $P(1-\delta)$ with fee $f - \phi$ is better when $P(1-\delta)(1-f+\phi) > P(1-f)$, that is $\delta < \phi/(1-f+\phi)$, also about $\phi$. Illustrative values $P = 100$, $f = 3$ bp, $\phi = 1$ bp: the buy threshold is 0.99980 bp and the sell threshold 1.00020 bp; at $\delta = 0.99$ bp $V'$ wins on both sides, at 1.01 bp $V$ wins. With fees per share, $c$ on $V$ and $c'$ on $V'$, the condition is $P\delta < c - c'$ on both sides: a one-cent tick beats a fee saving of 0.2 cents.

The comparison assumes both quotes are executable for the whole quantity at the time of routing, both orders take liquidity, and the trader actually pays the venue fees; the spread is inside the two prices and is not added again.

**Practice.** *Maker–taker and fee complexity.* MTFs' maker–taker schedules charge liquidity takers and rebate makers, which helps high-frequency market makers be profitable [alpha-laruelle, chap. 1, p. 69; chap. 2, sec. 2.4.3, p. 163]. Fee schedules can defeat a router's arithmetic: clearing and settlement fees computed on netted transactions make an order's cost depend on the member's net inventory, and conditional pricing ties fees to monthly turnover or to volume on another venue, so "the marginal gain due to one order is far from obvious" (chap. 1, pp. 68–69).

Rebates attract passive flow; the economically best venue is the one with the best expected all-in fill, not the best displayed price (interpretation). Interpretation: whether a broker passes venue fees and rebates to the client decides whose cost this is; when it does not, the comparison drives the broker's net, not the client's (broker conflicts). A favourable fee-adjusted quote says nothing about the chance of filling there, or about later price moves.

**Pitfalls.** The inequality compares two takers' costs, not price priority. Access-fee caps change (ex-sec-tick-access-2024).

**See also.** smart order routing; transaction costs (fees as explicit costs); broker conflicts.

#### Method 17.19 Passive order split: ranking venues by queue, distance and trading rate, and the minimum-resting-time split

**Statement.** To post a buy limit of $D$ shares below the best ask across venues, Lehalle and Laruelle compare three criteria on a stylised book (Table 3.4, p. 240): rank in the queue (shares ahead at the limit price), distance of the venue's best ask from the limit, which shows where sellers are aggressive, and the recent selling rate at that price. With $Q_n$ shares ahead and selling rate $r_n$ on venue $n$, an order of size $D$ takes on average

$$\tau_n(D) = \frac{D + Q_n}{r_n}$$

to fill (eq. (3.3)). Splitting $D$ so that every part finishes at the same time $\tau^\star$ gives $f_n D = \tau^\star r_n - Q_n$ and, summing, $\tau^\star = (D + \sum_n Q_n)/\sum_n r_n$ [alpha-laruelle, sec. 3.4.2, pp. 239–241, n. 16]. In their example ($D = 500$; $Q = 450, 100, 200$; $r = 10, 5, 5$ shares a second on Euronext, Chi-X and Turquoise), $\tau(500)$ is 95, 120 and 140 s, $\tau^\star = 62.5$ s, and the split is 35, 42.5 and 22.5 percent, 175, 212 and 113 shares (Table 3.5, p. 241). The criteria rank the venues differently (Table 3.6, p. 242):

| Criterion | 1st venue | 2nd venue | 3rd venue |
|---|---|---|---|
| Rank in the queue | Chi-X | Turquoise | Euronext |
| Distance to the best | Euronext | Turquoise | Chi-X |
| Trading rate | Euronext | Chi-X or Turquoise (tied) | |
| Minimum resting time (split) | Chi-X (212) | Euronext (175) | Turquoise (113) |

Sources: rows from [alpha-laruelle, sec. 3.4.2, Table 3.6, p. 242], with the split sizes from Table 3.5.

**Conditions.** Interpretation: the formula treats the queue ahead and the selling rate as fixed and known, ignores cancellations ahead of the order and later arrivals, and fills each part at the average rate, so $\tau$ is a deterministic approximation; the source points to stochastic methods for accurate splits (Laruelle et al. 2011). Equal finishing times minimise the time to complete the whole order, the latest part, rather than an average.

**Practice.** Interpretation: a venue with $f_n < 0$ cannot finish by $\tau^\star$ even with nothing posted; drop it and recompute $\tau^\star$ over the rest. Setting $f_n$ to zero without recomputing leaves fractions that no longer sum to one. Lehalle and Laruelle's liquidity seeker combines this posting with optimal removal of liquidity and keeps the parent inside a scheduling layer (p. 242; smart order routing).

**Pitfalls.** The book prints eq. (3.4) as $\tau^\star = (D - \sum_n Q_n)/\sum_n r_n$ (p. 241), but its own footnote gives $D = \tau^\star \sum_n r_n - \sum_n Q_n$ and Table 3.5's 62.5 s needs $D + \sum_n Q_n$; with the minus sign the example gives $-12.5$ s. The definition above uses the plus sign. The text calls the Euronext rate "buy orders hit the 95.00 price level"; for a resting buy order the relevant flow is sell orders, as for the other venues. The split sizes are rounded (212.5 and 112.5).

**See also.** smart order routing; order exposure decisions; queue-position value.

## 17.7 Dark trading and fragmentation

#### Definition 17.20 Dark pools and crossing networks: derivative pricing, the kinds of dark venue, integrated and midpoint books, why traders use them, what the evidence says, and their risks

**Statement.** A *dark pool* is a venue that does not publicly display its available liquidity; its main interest is to protect institutions executing large orders from opportunistic short-horizon traders [alpha-laruelle, chap. 1, sec. 1.4, p. 102]. Dark trading existed before as internal crossing, upstairs trading and OTC trading (p. 103). Lehalle and Laruelle list, for European markets after MiFID (pp. 103–104): *crossing networks*, which cross buy and sell orders at a price, typically the mid-price, derived from the primary market; *internalisation pools*, in which a broker crosses its own flow; *electronic market makers*, which trade against the operator's flow and accept or reject incoming orders at once; *exchange-based pools*, hidden liquidity on lit exchanges executed only against sufficiently large contra orders; *dark aggregators*, which split an order across pools and adjust as it fills; and *size-discovery* (large-block) pools created in response to MiFID II's caps.

Crossing networks are the only order-driven markets that are not auctions: all trades take place at prices determined elsewhere, under a *derivative pricing* rule, so they discover only whether traders will trade at those prices, not the prices themselves [alpha-harris, chap. 6, sec. 6.5, p. 132].

**Notation.** Crossing rules: Harris's 2002 U.S. networks were calls; Instinet's after-hours cross and the NYSE's After-hours Session I crossed at the 4 p.m. close, and POSIT crossed eight times a day at the primary midpoint at a random time within seven minutes after each call, allocating excess supply or demand pro rata; some networks operate continuously (pp. 132–134). The main European MTFs' dark books, under MiFID's reference-price waiver, crossed only at the midpoint of a reliable reference market's best bid and offer, with a two-book design: an *integrated book* for displayed orders and hidden orders large in scale (LIS), and a *dark midpoint book* for smaller hidden orders, which match only at the primary midpoint; they accepted mid-pegged dark orders with limit price, minimum acceptable quantity and immediate-or-cancel options (Lehalle and Laruelle, p. 104).

**Conditions.** U.S. crossing networks as of 2002; U.S. dark pools in 2009; European venues and data as of 2010–2018; no current rule stated. *Derivative prices* must be neither stale nor manipulated [alpha-harris, chap. 6, sec. 6.5.2, pp. 134–136]. Informed traders trade eagerly at a stale price and refuse otherwise, an adverse-selection problem: the after-hours crosses received more buy orders when prices had risen after the close, and did not cross stocks with significant after-hours news (p. 135). Anyone who will trade at a price fixed elsewhere later may try to move it: Harris's example saves $15,000 on 500,000 shares by depressing the reference price 3 cents with a 1,000-share sale (p. 135).

*Price ownership.* Primary markets argue that crossing networks skim the easiest orders and use their prices without paying for them; crossing customers reply that they should not pay for price discovery in which they do not take part, and that their own orders in the primary market produce those prices (sec. 6.5.1, p. 134).

*Price discovery.* Lehalle and Laruelle state that dark pools cannot create prices and influence price formation indirectly by attracting activity; the literature they cite disagrees, some empirical studies finding reduced price discovery and others, including Zhu's model, improved informativeness of exchanges (pp. 105–107); see Zhu (wide-zhu-dark-pools) and Buti et al.

**Practice.** *Why use them.* Secrecy, to reduce information leakage and impact, and price improvement, trading at the midpoint without paying the whole spread (Lehalle and Laruelle, p. 108). *Evidence, as reported.* From Buti, Rindi and Werner's U.S. study, as Lehalle and Laruelle summarise it: dark

activity is concentrated in liquid stocks; for a given stock it is higher on days with high volume, higher depth and lower intraday volatility, and lower on days with large order imbalances relative to volume; it is associated with better spreads, depth and volatility and with short-term overreaction (p. 107).

*The working paper* (Dice Center version of 24 June 2010; later published in *Financial Management*, 2022). The data are daily single-counted share volume that eleven dark pools reported voluntarily to SIFMA for calendar 2009, about a third of the roughly 32 active U.S. pools, anonymous by pool, with no public data to check them against and no way to know whether the eleven are a stable fraction across stocks and days [brw-2010, pp. 5–6]. Dark activity is concentrated in large, high-volume, high-price stocks with low spreads, high depth and low short-term volatility, and on a stock's days with high volume and depth and low volatility, and is lower on days with large order imbalances relative to volume and large absolute returns (abstract, p. 2).

On market quality the authors stress that dark activity and market quality are "jointly determined" and claim at best an association: cross-sectionally, more dark activity goes with lower spreads, price impact and short-term volatility; in a matched-pair design (100 NASDAQ and 100 NYSE stocks, matched on size and price, dark activity measured January–June 2009 and market quality July–December), high-dark-activity stocks had quoted spreads 3.51 bp and effective half-spreads 0.53 bp lower (sec. 6, pp. 13–16). This draft's price-efficiency section is marked "TBA" (p. 6), so the "short-term overreaction" finding Lehalle and Laruelle report is not in it.

*Theory* (Zhu, published version). In a model where informed and liquidity traders both choose between an exchange and a midpoint dark pool without guaranteed execution, informed orders cluster on the heavy side and fill less often in the dark, so the dark pool attracts relatively more uninformed flow; "under natural conditions" this raises the exchange's signal-to-noise ratio and lowers pricing error while widening the exchange spread, so better price discovery comes with lower exchange liquidity; the result needs short-lived information (it weakens as the information horizon lengthens), abstracts from pinging, routing and indications of interest, and makes no welfare claim [wide-zhu-dark-pools, pp. 747–750, Propositions 3–4, pp. 763 and 770]. In his numerical example adding a dark pool cuts the market maker's percentage pricing error from 7.1 to 2.6 percent at high volatility (p. 770).

Zhu's result is a model's, and Buti, Rindi and Werner's an association in one year's partial data; neither measures price discovery in current markets. *Their published version* studies 2009 and 2020. Dark pool trading "improves spreads in 2009, but worsens market quality for large caps in 2020"; traders use dark pools to jump the queue for large caps in 2009 and to avoid crossing the spread for small caps in both years, and internalisation is more common for small caps [brw-2022-published, abstract]. So the 2009 association above does not carry over to large caps in 2020.

Lehalle and Laruelle's own one-minute data for CAC 40 and FTSE 100 stocks on Chi-X, BATS and Turquoise dark books in the first three weeks of March 2010 show dark *turnover* concentrated in the most volatile deciles (nearly a quarter of it in the top tenth), but dark *market share* not rising with volatility, and rising strongly with the spread (about three times as high in the widest-spread decile as in the narrowest), most of all for less liquid SBF 120 and FTSE 250 stocks (pp. 108–114).

*Fill rates.* Harris reports that U.S. crossing networks filled less than 10 percent of the order volume submitted, and that unfilled orders were usually sent elsewhere (p. 134); a dark fill rate measured only from executions therefore misses the orders that found nothing. *Bilateral and multilateral.* On a multilateral platform anyone may supply or take liquidity; bilateral trading keeps an asymmetry that protects liquidity providers from toxic flow and is easier to sustain in illiquid instruments (Lehalle and Laruelle, p. 114).

*Broker crossing networks*. Lehalle and Laruelle describe executing brokers, squeezed by the fixed costs of fragmentation and falling assets after the crisis, opening pools marketed as "100% first-hand liquidity"; these BCNs sometimes also hosted proprietary flow of the owning bank, MTFs and some exchanges launched dark pools in response, and some brokers converted BCNs into MTF dark pools to reassure clients [alpha-laruelle, chap. 1, sec. 1.1.4, pp. 55–56]; large investors favoured BCNs "as far as such pools are 'HFT-free guaranteed' " (Introduction, p. 18). In short, dark pools hide pre-trade interest, internalisers match against a broker-dealer's inventory or flow, and crossing networks match natural interest at reference prices; the current EU caps are in EU trading venues and transparency.

**Pitfalls.** *Gaming and toxicity.* Some high-frequency traders detect large dark orders by sending small orders ("fishing") or orders on the opposite side, then move the primary price; protections include large minimum quantities, restricted access, tracking participants who behave like gamers and statistical detection of consistently profitable participants; indications of interest can leak information too (Lehalle and Laruelle, p. 115).

*Internal crossing.* Brokers who cross internally may expose client orders only to their own clients, which makes the search for counterparties harder, and may favour some clients over others; brokers reply that they work harder for two commissions, and exposure to the market protects clients (Harris, chap. 25, sec. 25.4, pp. 520–521; broker conflicts). Dark activity is concentrated in volatile periods only for dark *turnover* in Lehalle and Laruelle's deciles, not for dark *market share*, and Buti et al.'s results as reported point instead to lower-volatility days.

Venue lists, market shares (about 5 percent of non-OTC European trades and 10–12 percent of U.S. share volume, p. 105), the LIS thresholds and the MiFID II caps are dated and not current; see the EU rules entry for the later regime.

**See also.** matching and trade pricing (derivative pricing and pro-rata allocation); order exposure decisions (hidden orders and probing); block trades; smart order routing; best execution; broker conflicts.

**Further reading.** Zhu (published 2014) and Buti, Rindi and Werner (2010 working paper) (Practice); the Rosenblatt dark-pool reports. *Empirical evidence on price discovery*. Comerton-Forde and Putniņš use ASX data from February 2008 to October 2011 and find that "dark trades are less informed than lit trades" and that high levels of dark trading raise adverse selection on the lit exchange.

They find "low levels of non-block dark trading are benign or even beneficial for informational efficiency, but high levels are harmful", with deterioration from about 10 percent of a stock's dollar volume, while "no evidence that block trades in the dark impede price discovery" is found; block trades away from the lit market up to about 40 percent of dollar volume can help [comerton-forde-putnins-2015, author's accepted manuscript of 13 February 2015, abstract and PDF pp. 4–5 and 10]. The thresholds are for one market and period, before the EU's double volume caps and their later revision.

### Method 17.21 Venue market share: value and trade-count shares, auctions and the ex-fixing share, aggregation across stocks, estimation error, and why shares depend on the time of day

**Statement.** For stock $k$ traded on $N$ venues between $t_1$ and $t_2$, with trade $\ell$ at price $P_\ell^k$, size $V_\ell^k$, time $\tau_\ell^k$ and venue $\delta_\ell^k$, Lehalle and Laruelle define the share of venue $n$ in traded value and in trades [alpha-laruelle, chap. 1, sec. 1.1.1, eqs. (1.1)–(1.2), pp. 34–35]:

$$\mathfrak{M}^k(n) = \frac{\sum_{t_1 \le \tau_\ell^k \le t_2} P_\ell^k V_\ell^k \mathbb{1}_n(\delta_\ell^k)}{\sum_{t_1 \le \tau_\ell^k \le t_2} P_\ell^k V_\ell^k}, \qquad \mathfrak{M}_T^k(n) = \frac{\sum_{1 \le \ell \le T} \mathbb{1}_n(\delta_\ell^k)}{T},$$

the value share being close to the share in shares while the price varies little. Across stocks the share can be weighted by turnover, by index weight (free float or capitalisation, not turnover) or equally, and the answers differ (pp. 35–37). The right metric depends on the question (p. 37): the all-included share for a venue's position and revenue; for the probability that a *trade* occurs on venue $n$, the trade-count share, with auctions and continuous trading measured separately; for the probability that a unit of *currency* trades there, the value share, again separately for auctions. Treating market share as a proportion, its estimator for asset $k$ and venue $i$ is the turnover on $i$ over total turnover, $\hat{p}_k^i$, and an index-level share is $\hat{p}^i = \sum_k w_k \hat{p}_k^i$ (app. A, secs. A.8.1–A.8.2, pp. 286–287).

**Notation.** $m(n)$ is venue $n$'s share of auction volume and $p$ the auctions' share of all volume. Interpretation, derived: since venue $n$'s continuous turnover is its total less its auction turnover, its continuous-session share is

$$\mathfrak{M}(n;\text{ex-fixing}) = \frac{\mathfrak{M}(n) - m(n)\,p}{1-p}.$$

Lehalle and Laruelle print $(1-m(n)p)\mathfrak{M}(n)/(1-p)$ (p. 35); the two agree exactly when $m(n)\,p\,(1-\mathfrak{M}(n)) = 0$: when the venue has no auction volume, when there are no auctions, or when the venue has all the trading (a monopolist, share 1 under both); the domain is $0 \le p < 1, 0 \le m(n) \le 1$ and $m(n)p \le \mathfrak{M}(n) \le m(n)p + 1 - p$, and with $p = 1$ the continuous share is undefined (the numerical analysis "agrees only when $m(n) = 0$" was too strong). A primary market with 60 percent of all trading and all the auctions, with auctions 20 percent of volume, has 50 percent of continuous trading, where the printed formula gives 60; the monopolist and no-auction cases agree.

**Conditions.** Methods timeless; data and auction shares as of 2006–2013. *Estimation error.* Lehalle and Laruelle treat trades as independent Bernoulli trials, so $\mathrm{Var}\,\hat{p} = p(1-p)/n$ with a normal confidence interval $\hat{p} \pm z_{1-\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$, adequate when $np > 5$ and $n(1-p) > 5$ (sec. A.8, pp. 284–286), and assuming independence across stocks they derive $\mathrm{Var}(\hat{p}^i_k - \hat{p}^i) = (1-2w_k)p^i_k(1-p^i_k)/n^i_k + p^i(1-p^i)/n^i$ (sec. A.8.3, pp. 287–288), in which the index estimate's variance is replaced by that of a single pooled proportion. Derived: for the fixed-weight estimator $\hat{p} = \sum_j w_j \hat{p}_j$ and independent stock estimates with variances $\sigma_j^2$,

$$\mathrm{Var}(\hat{p}_k - \hat{p}) = (1-w_k)^2\sigma_k^2 + \sum_{j\neq k} w_j^2\sigma_j^2,$$

and with dependent estimates $\mathrm{Var}(\hat{p}_k - \hat{p}) = a^\top \Sigma a$ for $a = e_k - w$ and $\Sigma$ the covariance matrix of $(\hat{p}_j)$. A pooled binomial variance has no basis for an arbitrarily weighted index: with $w = (0.9, 0.1)$, $p = (0.5, 0.5)$ and $n = (100, 100)$ the correct variance is 0.00005 (simulated 0.000050), and the printed formula with 200 pooled observations gives −0.00075, a negative variance. The per-stock variance $\sigma_j^2 = p_j(1-p_j)/n_j$ holds for a *count* share over $n_j$ trades only if each trade goes to the venue independently with the same probability; a *value* share $\sum_\ell v_\ell X_\ell / \sum_\ell v_\ell$ with fixed trade values and independent venue indicators has variance $p_j(1-p_j)/n_{\text{eff}}$, $n_{\text{eff}} = (\sum v_\ell)^2/\sum v_\ell^2$ (92 effective trades out of 2,000 lognormal values, standard deviations 0.0478 both ways).

The textbook's $n^i_k$ and $n^i$ are left undefined, and its formula is an unsupported simplification. Interpretation, by simulation: a value share is not a count of identical trials, and a venue choice repeated over the child orders of one parent breaks independence; with venue choices repeated in runs of 20 trades the standard deviation of a 2,000-trade share is 0.047 against the binomial 0.010. Comparing a stock with an index that contains it is mechanically correlated; a leave-one-out index avoids this (Interpretation).

*Time of day.* Shares differ between the first and last hours (p. 35) and with auctions, which in European markets were mostly a primary-market monopoly; in August 2012 auctions carried 11 to 26 percent of primary-market volume across eleven European indices, and closing prices fed fund NAVs and derivatives' exchange delivery settlement prices (chap. 2, sec. 2.1, Table 2.1, pp. 117–119). Hasbrouck gives the reason auction prices carry weight: closing prices value margin positions, mutual fund shares, cash-settled derivatives and occasionally merger terms, which invites attempts to "mark the close" [s6-hasbrouck, chap. 2, sec. 2.4, p. 18].

*Proportions in general*. Lehalle and Laruelle present the same estimator for any microstructure proportion: "market shares, proportions of HFT in the market, probability to have a transaction at the bid or ask side, etc." (sec. A.8, p. 284). The normal interval "does not always work well, especially for situations with a small sample size and a proportion very close to zero or one", and "fails totally when the sample proportion is exactly zero or exactly one" (pp. 285–286).

Computed exactly from the binomial distribution, the nominal 95 percent Wald interval covers $p$ with probability 0.957 at $n = 30$, $p = 0.5$, but 0.809 at $p = 0.1$ ($np = 3$) and 0.454 at $p = 0.02$ ($np = 0.6$), and 0.908 at $n = 200$, $p = 0.02$ ($np = 4$); Wilson's score interval covers 0.957, 0.974, 0.978 and 0.933 in the same cases; with no successes in 30 the Wald interval is the single point 0 and Wilson's is $[0, 0.114]$. Lehalle and Laruelle state the rule of thumb with the true $p$; in practice it is applied to $\hat{p}$, with an exact or adjusted interval below it, which these coverages support.

The *sampling units* (trade count, traded value, message count, time bucket, security bucket) each define a different estimand, as the count and value shares above show, and the independence needed for the binomial variance must be argued afresh for each unit.

**Practice.** *Volume curves* follow reference times: a linked market's opening (U.S. opening for European stocks; Brazil and South Africa), scheduled macroeconomic releases (the U.S. employment report at 14:30 CET), derivative expiries (the Eurostoxx 50 delivery settlement price, the mean index level from 11:50 to 12:00 CET on the third Friday) and reference-price rules (India's last-half-hour VWAP; Hong Kong's median of five snapshots in the last minute, with 17 percent of volume in the last 15 minutes); the patterns move with daylight saving time and market holidays (sec. 2.1.2, pp. 129–139; Total SA, 1 January 2006 to July 2011). Rules that define a benchmark create volume around it. An average daily profile mixes such days and should not be used unfiltered for a VWAP schedule (p. 139); see intraday deseasonalisation.

**Pitfalls.** In eq. (1.1), a trade-count share replaces traded value by $1_n$, so the denominator becomes the number of trades. Fixing auctions make a primary venue look more competitive only in the all-included share. Market share measured from trades is a rough prior for routing, not a measure of available liquidity (p. 38; smart order routing).

**See also.** fragmentation and concentration metrics; SOR and market-share dynamics; European fragmentation evidence; intraday deseasonalisation (volume models); execution benchmarks.

**Further reading.** Agresti and Caffo (2000) on small-sample intervals, cited by Lehalle and Laruelle.

### Method 17.22 Fragmentation and concentration metrics: the Fragmentation Efficiency Index, coverage, the Gini coefficient and two-sample Kolmogorov–Smirnov comparisons

**Statement.** Fragmentation exists at several levels: market operators, trading venues, order books within a venue, orders split across books, and even connection protocols; market share is the most used metric but each level needs its own [alpha-laruelle, chap. 1, sec. 1.1.1, pp. 33–34]. For shares $q_1, \ldots, q_N$ summing to one, the *Fragmentation Efficiency Index* is normalised entropy,

$$\mathfrak{F}(q_1,\ldots,q_N) = \frac{-\sum_n q_n \log q_n}{\log N} \in [0,1],$$

0 when all liquidity is in one pool and 1 when it is spread evenly (app. A, sec. A.1, Definition 1, pp. 247–249). *Coverage* is the percentage of an index's stocks needed to make 80 percent of its turnover on a venue, 80 percent for a perfectly balanced index; the *Gini coefficient* is $G = 1 - 2B$ with $B$ the area under the Lorenz curve, equivalently half the relative mean difference, 0 for equality and near 1 for concentration (chap. 2, sec. 2.4.1, p. 159; app. A, sec. A.9.1, pp. 288–289). The two-sample Kolmogorov–Smirnov statistic is $D_{n,m} = \sup_x |F^X_n(x) - F^Y_m(x)|$, and equality of the two distributions is rejected at level $\alpha$ if $\sqrt{nm/(n+m)}\,D_{n,m} > k_{1-\alpha}$, the Kolmogorov quantile (sec. A.9.2, pp. 289–291).

**Conditions.** The FEI depends on the venues counted: an empty new venue lowers it from $\mathfrak{F}(t)$ to $\log N/\log(N+1)\cdot\mathfrak{F}(t)$ (about 80 percent for $N = 3$, 90 for $N = 6$), and Lehalle and Laruelle suggest counting a venue only above 1 percent share (p. 250). The same formula measures different things depending on the shares fed to it (trades, value, displayed depth, sessions, universes), and none measures access, connectivity or protocol

costs or hidden liquidity (interpretation, consistent with p. 34). The KS test assumes independent observations from continuous distributions; venue shares across the stocks of one index are neither identically distributed nor independent of the index.

**Practice.** Shares $(0.6, 0.25, 0.15)$ give $\mathfrak{F} = 0.854$, and 0.676 with an empty fourth venue, the predicted ratio; the Lorenz and mean-difference Gini agree (0.565 on a lognormal sample of 101 stocks); Lehalle and Laruelle's Table A.3 values follow from $G = 1 - 2B$ (for example $B = 0.18$, $G = 0.64$). Their application compares SBF 120 market-share distributions on Euronext Paris and Chi-X for 2008–2012 (Tables A.3–A.4, pp. 291–292): the KS statistics are 0.664, 0.235, 0.121 and 0.083, with $p$-values $8 \times 10^{-23}$, 0.003, 0.35 and 0.78. For 2010 and 2012 the test fails to reject equality at 5 percent; that does not show the distributions are the same.

By simulation, samples of 30 from $N(0, 1)$ and $N(0.3, 1.2^2)$ are not rejected in 89 percent of trials. For coverage, they report 45 percent for the FTSE 100 on the LSE, 52 percent for the AEX, 55 for the CAC 40 and 56 for the DAX, January to June 2012 (pp. 160–161).

**Pitfalls.** Lehalle and Laruelle write that for 2010 and 2012 "H0 is accepted, so the market share for the SBF 120 on Euronext Paris and Chi-X have the same distribution" (p. 292); a non-significant test is a failure to reject, not acceptance, and with an index's worth of stocks its power against moderate differences can be low.

They also report Gini values of 0.96 to 0.99 for primary-market turnover across the FTSE 100, AEX, CAC 40 and DAX (p. 160) alongside coverage near 50 percent; with the Lorenz definition of their appendix, coverage of 45 percent is hard to reconcile with a Gini of 0.99, and the chapter's construction (sorted in decreasing order, "two times the area between the repartition curve and the bisector", p. 160) may differ; the values are reported, not relied on. The source applies the Gini to stocks and, in the appendix, to types of actors in general (p. 288); applying it to participant activity or time segments extends it.

**See also.** venue market share; European fragmentation evidence (coverage history); SOR and market-share dynamics.

**Further reading.** Shannon (1948), Gini (1912) and Kolmogorov (1933), cited by Lehalle and Laruelle.

### Model 17.23 Smart order routers and market-share dynamics: Lehalle and Laruelle's toy model of liquidity capture, learning and share feedback, and why variable liquidity loses flow

**Statement.** An SOR splits an order of size $V(\omega)$ across $K$ venues by a key $r = (r_1, \dots, r_K)$, $\sum_k r_k = 1$, to maximise on average the liquidity it captures,

$$C(r, \omega) = \sum_{k=1}^{K} \min\{r_k V(\omega), D_k(\omega)\},$$

with $D_k(\omega)$ the liquidity venue $k$ provides when asked [alpha-laruelle, app. A, sec. A.3.1, eq. (A.5), p. 255]. It learns $r$ by trial and error, reducing the share of a venue that gives less liquidity than average and raising that of one that gives more (eq. (A.7), p. 256). If a fraction $a$ of participants route with SORs, venue $k$'s share evolves as $p_k(t+1) = (1-a)p_k(t) + a\, r_k(t)$ (eq. (A.8), p. 257). Coupled, with independent lognormal draws of $(V, D_1, \dots, D_K)$, the constraint $E[D_k / \sum_\ell D_\ell] = p_k$ (eq. (A.10)), standard deviations proportional to the quantities through an unreported parameter $\nu$, and $E[V] \simeq \sum_k E[D_k]$, their simulation takes a 95/3/2 split, with SORs starting from equal keys, to about 50/25/25 (Figure A.3, pp. 257–258).

Shares end near their start if liquidity is steady and the routers' initial keys are close to the start, and nearer equal splits when liquidity is noisy and keys start near uniform (sec. A.3.5, pp. 259–260).

**Notation.** $\delta_{r_k V < D_k} = 1$ when venue $k$ had more liquidity than was sent to it. $a$ is the share of SOR users and $\gamma$ the learning rate.

**Conditions.** A stylised model of European fragmentation after 2007. The book prints three slips: (A.5) says "Minimize" the expected capture, which the text maximises; (A.8) is printed $p_k(1+1)$ for $p_k(t+1)$; and (A.7) is printed with $\frac{1}{K}\sum_\ell \delta_\ell - \delta_k$, which would lower the key of a venue that had spare liquidity, against the text's description, so the sign is reversed (interpretation). The optimality condition (A.6) equates the venues' average indicators $E[\delta_{r_k V < D_k}]$.

Derived: the marginal value of $r_k$ in $E[\sum_k \min\{r_k V, D_k\}]$ is $h_k(r_k) = E[V 1\{r_k V < D_k\}]$, decreasing in $r_k$; at an optimum $h_k(r_k) = \lambda$ for every venue with $r_k > 0$ (interior) and $h_k(0) \le \lambda$ for any venue left at zero (boundary), which reduces to (A.6) only when $V$ is not random. Counterexample: with two equally likely scenarios, $V = 1$, $D = (0.4, 0.6)$ and $V = 10$, $D = (6, 4)$, the key $(0.5, 0.5)$ equalises the unweighted indicators (both 0.5) but captures 4.95, while $(0.6, 0.4)$ captures 5.4, the grid optimum. A learner consistent with this objective uses the size-weighted indicators as stochastic gradients.

*Illustrative variant, not a reproduction*: with $D_k = \mu_k X_k$, $X_k$ i.i.d. lognormal ($\nu = 0.8$, chosen here), $V$ independent of $D$ with $E[V] = \sum \mu_k$, the means calibrated each period so that $E[D_k / \sum D] = p_k$, $a = 0.3$, and projected stochastic gradient ascent with decreasing steps, the averaged keys agree with an exact solution of the conditions above (largest gap 0.001), and the optimal key is $\mu_k / \sum \mu$ by scale invariance (keys $(0.6, 0.25, 0.15)$ for means $(0.6, 0.25, 0.15)$). Because the calibration gives the largest venue $\mu_1 / \sum \mu > p_1$ (0.971 for $p_1 = 0.95$), shares drift towards it, reaching $(0.998, 0.001, 0.001)$ after 60 periods: under these assumptions the coupled model concentrates rather than fragmenting.

Lehalle and Laruelle's 50/25/25 path is therefore not reproduced; it depends on their unreported $\nu$, their printed update and possibly other choices, and a corrected-sign simulation does not reproduce it (that simulation set $E[D_k] = p_k$, made $V$ depend on total liquidity, and used unweighted indicators). The model is a toy: liquidity is exogenous apart from its mean, there are no prices, fees or latency, and convergence needs the liquidity process to be i.i.d. or ergodic (p. 256).

**Practice.** Lehalle and Laruelle's narrative version (chap. 1, sec. 1.2.2, pp. 71–73): routers adjust splits from the historical performance of their executions; "the fraction of an order to send to a venue with a large variance will thus be minimized by a rational SOR", and venues under-weighted long enough lose share; with shares of 76, 17 and 7 percent in their simulation, the largest market's share fell from 87 to 56 percent with medium variance and to 40 with high variance. They add that "all studies have shown that variance is proportional with market share", so primary markets have more variance and investors send Chi-X more than its share would imply (p. 72); that statement is reported without a source.

*Source search*: no study is cited at the statement, in sec. 1.2.2, or in sec. A.3, whose only citations are Foucault and Menkveld (2008) on fragmentation and Laruelle et al. (2011) on routing across dark pools (p. 253). The book's one formal counterpart is the model's own "fundamental assumption that the standard deviation of the liquidity provided by a venue is proportional to its quantity" (sec. A.3.3, p. 257). Derived: if a venue's liquidity scales with its share, that assumption makes the *variance* proportional to the share squared, not to the share; proportional variance would hold under a Poisson-like count model, which the book does not use. The narrative claim and the appendix's assumption therefore differ, and neither is an empirical finding. The drivers of market share (initial share, liquidity level and variance, SOR adoption $a$, reassessment speed $\gamma$) are the model's parameters.

**Pitfalls.** Volatile fills at an alternative venue lead routers to cut its allocation and push flow back to the primary market. That direction follows from the model for a noisier alternative venue, but it is a simulation and model result, not an empirical finding, and Lehalle and Laruelle's own (unsourced) claim is that the primary is usually the noisier venue, so in their account the same mechanism erodes the primary's share.

**See also.** smart order routing; venue market share; European fragmentation evidence.

**Further reading.** Foucault and Menkveld (2008) and Laruelle et al. (2011), cited by Lehalle and Laruelle.

### Evidence 17.24 European equity fragmentation after MiFID: what moved market share, which stocks fragmented, the Spanish comparison, outages and shared reference prices

**Statement.** Lehalle and Laruelle report, from European equity data around 2008–2013 [alpha-laruelle, chaps. 1–2]:

- *liquidity and fragmentation*: "the more liquid, the more fragmented" (chap. 1, p. 37); MTFs' shares grew faster in liquid stocks, and coverage rose for all venues and indices from 2009 to 2012 but stayed lower for less liquid stocks, which they attribute to costlier hedging and short selling for high-frequency market makers, who extend to less liquid stocks once their hedges and technology are in place, down to a liquidity threshold (chap. 2, sec. 2.4.2, pp. 161–162);
- *spreads and share*: a venue's market share moves with its spread relative to the primary market's, because routers must send flow to the best price; Chi-X's spreads became tighter than the primary's and its share rose; Turquoise's share fell sharply in 2009 when the agreements under which eight market makers had provided constant liquidity ended (sec. 2.3.2, pp. 153–155);
- *resilience*: venues where trading continues as volatility rises take more of the price formation; Chi-X approached the primary market's pattern between 2008 and 2012 (sec. 2.3.3, pp. 156–158);
- *time of day*: on CAC 40 stocks (January–February 2009 and 2012) Euronext's intraday share is highest at the start and end of the day, when its book is the only one filled after the opening auction and when routers are switched on and off (sec. 2.1.3, pp. 139–140);
- *outages*: on 13 October 2010 NYSE Euronext's cash markets halted from 16:42 CET, resuming at 17:20 without market data; MTFs' midpoint books stopped and orders pegged to the primary were cancelled or paused, most large caps kept trading on MTFs near Euronext's last price, but some stocks deviated by about 2 percent, "as if two price formation processes took place in parallel" (sec. 2.5.2, pp. 183–185);
- *Spain*: MTFs had almost no share of IBEX 35 trading from November 2007 to January 2010 (Chi-X first exceeded 0.5 percent in August 2009), because only Iberclear, owned by the primary market BME, could clear and settle until the Titulo V reform of June 2011 (sec. 2.5.1, p. 170).

**Conditions.** The Spanish comparison is a historical contrast, not a control: comparing 15-minute Garman–Klass volatility of FTSE 100 and IBEX 35 stocks from June 2007, Lehalle and Laruelle find no significant effect of UK fragmentation on intraday volatility, and the comparison is confounded from 2010 by the sovereign-debt crisis, to which Spain was more exposed (pp. 171–174). The spread–share and resilience findings are co-movements in their figures, with Schneider Electric's five-minute data for January to July 2012 as one example (p. 156), not estimated causal effects. High-frequency traders' shares of trades ("more than 80% … in the US, 40% in Europe") are reported without a source (p. 163).

**Practice.** *Shared references.* Lehalle and Laruelle argue that European outages showed how hard fragmented trading is without information shared among agents, and that, without a consolidated tape or trade-through rule, MTFs priced off the primary's quotes, so the primary appeared to lead (introduction, p. 5; pp. 184–185). Harris's account of consolidation is the general mechanism: in fragmented markets traders adjust to information from other segments, route to the segments with the best prices, and arbitrageurs move liquidity between segments, which consolidates the market provided information about all segments is public at low cost [alpha-harris, chap. 26, sec. 26.4, pp. 533–534].

*Outages and parallel prices.* When alternative venues keep trading while the primary's data or matching is unavailable, prices may split into parallel formation processes, as in Lehalle and Laruelle's account of the Euronext cash-market halt of 13 October 2010: trading stopped at 16:42 CET and resumed at 17:20 without market data, so that for 40 minutes Euronext resembled a dark pool; most large caps kept trading on MTFs, midpoint trading stopped and orders pegged to the primary's best quotes were cancelled or paused, prices mostly stayed near Euronext's last trade, but some stocks, including EuroStoxx 50 and CAC 40 components, deviated by about 2 percent, and Euronext's later data showed its prices differing from the MTFs' [alpha-laruelle, chap. 2, sec. 2.5.2, pp. 183–185].

This is one episode, described from their own data; how prices on the two sides were reconciled was not quantified. Figures 2.31 and 2.32 (Vivendi and Publicis) are captioned "October 10, 2010", a Sunday; the text's 13 October 2010, a Wednesday, is taken as the date (pp. 185–186).

**Pitfalls.** Spain is not a fragmentation control case: Lehalle and Laruelle use it as a comparison and themselves find no clear effect, and their explanation that primary-market dominance persisted "when alternative venues could not clear and settle efficiently" is for Spain only. Subsidised market share is one episode, Turquoise's in 2009, not a general pattern. Venue names, shares and the regulatory framework are as of 2008–2013.

**See also.** fragmentation metrics; SOR and market-share dynamics; fee-adjusted venue choice; Garman–Klass volatility; HFT strategies; smart order routing; venue market share.

**Further reading.** O'Hara and Ye on fragmentation.

## 17.8 Market regulation, venue competition and circuit breakers

**Definition 17.25 Regulation as market design: who regulates, what market-design questions are, how to evaluate a rule, and the political economy of rule-making**

**Statement.** "Regulators create and enforce rules that facilitate trading"; good rules help traders communicate, prevent fraud and make "all things generally … as they appear", while other rules give favoured traders, brokers or exchanges protection from competition, protect domestic or incumbent markets, express ideology or raise revenue [alpha-harris, chap. 3, sec. 3.5, pp. 59–60]. "The stated purpose of a regulation often is not its true objective" (p. 60).

*Who regulates.* Legislatures set the framework; government agencies (in the United States the SEC and CFTC, with the Federal Reserve setting speculative margins) write and enforce regulations; self-regulatory organisations (exchanges, clearinghouses and trader associations) regulate their members under rules approved by the agency; private bodies set accounting and performance standards; international organisations (IOSCO, the World Federation of Exchanges) coordinate (sec. 3.5.1–3.5.2, pp. 60–66). Harris's summary: "Legislatures provide the regulatory framework, the SROs provide the details, and governmental regulatory agencies provide oversight" (p. 66).

*Design questions* debated around 2002 included a central limit order book, quote- against order-driven systems, internalisation and preferencing, price limits and halts, floor against screen, linkages, trading hours, who sees the book, who owns market data and what may trade [alpha-harris, chap. 9, p. 202].

**Conditions.** *Evaluating a rule.* Harris's normative procedure has four parts: identify all reasonable alternative policies; specify the criteria, a social welfare function with its acceptable trade-offs; evaluate each alternative by positive analysis; and choose the one with the greatest welfare; "When policy becomes the objective rather than the means to the objective, poor results often follow" (chap. 9, box, p. 204). His own objectives, stated as opinion, rank the private interests of utilitarian traders first, the public benefits of liquid, informative markets second, support for profit-motivated traders only when needed for those, and hostility to strategies that exploit other traders, among them manipulators, bluffers, front runners and very fast traders picking off slow dealers (sec. 9.6, pp. 216–217).

The SEC's statutory direction to consider "efficiency, competition, and capital formation" as well as investor protection gives no guidance on trade-offs among them (chap. 3, box, p. 61). *Public and private services.* Markets charge for private services such as order routing; public services such as price continuity and the regulation of insider trading, manipulation and capital structures benefit everyone, so "unregulated competition among market centers produces fewer public services than would be socially desirable" (chap. 26, sec. 26.5.3, p. 538).

**Practice.** *Political economy.* After the 1987 crash, Harris argues, regulators facing public demand for action were better off adopting circuit breakers whether or not another crash followed, and mild rather than severe ones (*regulatory risk*, Tables 28-1 and 28-2); the NYSE's Rule 80A collar on index arbitrage benefited specialists, who outnumbered arbitrageurs among members (470 of 1,366 members were full-time specialists in November 1997, while 20 members did 95 percent of program trades), an example of rent seeking (*regulatory capture*) [alpha-harris, chap. 28, sec. 28.4, pp. 578–581]. "After a crisis, regulators often adopt regulations that have little economic value simply to respond to public demands for action" (p. 582).

*Unintended consequences*: Brazil's 0.38 percent tax on financial transactions of 1997 moved institutional trading in Brazilian stocks to ADRs in New York; Bovespa volume fell from 1.2 billion reais a day in 1997 to 350 million in 2001, partly because of the 2001 crisis, and in September 2001 the government announced an exemption for stock trades (chap. 3, box, p. 66). Exposure protection is a design choice with its own trade-offs: time precedence with a meaningful tick, undisclosed orders, audit trails and delayed reporting (order exposure decisions).

**Pitfalls.** The institutions named are as of 2002 (NASD has since become part of FINRA; national regulators listed in Harris's Table 3-11 have been renamed or merged). Policy is often adopted before transitory and fundamental volatility can be separated: Harris remarks that empirical studies of circuit breakers "generally have been inconclusive" because extreme events are rare (chap. 28, p. 577). A welfare ranking is opinion; the four-step procedure only makes the criteria explicit.

**See also.** what markets do; venue competition and fragmentation; circuit breakers and trading halts; U.S. equity market structure rules; EU trading venues and transparency.

**Further reading.** Harris, chaps. 3, 9, 26 and 28.

**Mechanism 17.26 Venue competition and fragmentation: the order-flow externality, why markets segment, cross-market precedence, the post-MiFID cycle of entry, convergence and consolidation, and technology fragmentation**

**Statement.** A market is *fragmented* when essentially the same thing trades in different market centres and *consolidated* when all traders trade in one; competition among traders for the best price works best in a consolidated market, while competition among market centres to offer services implies fragmentation, "The two competitions therefore are inconsistent" [alpha-harris, chap. 26, p. 524]. *Consolidating force*: each trader who joins a market adds liquidity that attracts others, the *order-flow externality*, a network externality that makes trading winner-take-all and new entry hard (secs. 26.2 and 26.5.1, pp. 526 and 535–536); Hasbrouck likewise calls the liquidity externality a network externality that "favors market consolidation" [s6-hasbrouck, chap. 1, sec. 1.2, p. 5].

*Fragmenting force*: "Markets fragment because traders are not all identical" (sec. 26.3, p. 530): by size (large traders control exposure and value strict time precedence with an economically significant tick; small traders want exposure and small ticks), information (informed traders prefer anonymous consolidated markets; uninformed traders prefer to be identified), patience (impatient traders value dealers; patient traders value order-driven systems with time precedence), access, and creditworthiness (a single standard makes creditworthy traders subsidise others) (secs. 26.3.1–26.3.5, pp. 530–533).

**Conditions.** *Order-flow externality as an inequality.* If a trader's utility at venue $i$ is $U_i = s_i - c_i + \ell(F_i)$ (service, cost, and an increasing function of order flow), an entrant better by $(s_e - c_e) - (s_m - c_m) > 0$ still loses the trader when $\ell(F_m) - \ell(F_e)$ exceeds that margin; this restatement of the proposition is a rearrangement, true whenever the trader chooses the higher utility, and its content lies in the specification (one trader, flows taken as given). With $\ell = \log$, flows of 1,000 against 50 and a 0.25 service advantage, the incumbent wins by 2.75.

*Cross-market precedence.* A market cannot meaningfully enforce time and public-order precedence when other markets trading the same instrument do not: quote matchers trade elsewhere, small markets offer small ticks to attract them, and larger ones follow, as Harris reads the U.S. tick reductions of the 1990s (tick size). His sidebar *Lost Time Precedence* gives the 2002 U.S. example of regional specialists and third-market dealers filling orders at prices first quoted on the NYSE (p. 537). The U.S. answer was a trade-through rule (U.S. equity market structure rules); in Europe, best-execution policies took its place [alpha-laruelle, chap. 1, sec. 1.1.2, p. 43].

**Practice.** *Post-MiFID Europe, as Lehalle and Laruelle describe it.* MiFID, in force from 1 November 2007, abolished concentration rules and introduced brokers' best-execution policies; instead of many venues a few appeared, each running several pools, and "the agents … inherited the crucial role of consolidating the liquidity" [alpha-laruelle, Introduction, p. 17]. Their phases (chap. 1, secs. 1.1.2–1.1.4, pp. 39–62): first attempts (Chi-X, Turquoise, BATS Europe and NASDAQ OMX Europe, opened country by country, Table 1.1, p. 40); convergence towards pan-European offers and a few groups (the LSE Group with Turquoise; BATS and Chi-X; Euronext), which then segmented liquidity into niche pools, "natural" against "opportunistic" (pp. 41–42); and broker crossing networks and MTF dark pools (pp. 54–56).

Their two cautionary cases: Turquoise, founded by nine banks with a market-making agreement whose end made its share drop (pp. 47–48), and NASDAQ OMX's MTF, which "almost never crossed 1% of the market share" and closed (pp. 49–50). Consolidation followed: BATS acquired Chi-X Europe in November 2011, a Deutsche Börse–NYSE Euronext merger was rejected by the European Commission, ICE bought NYSE Euronext in 2013, and Cboe acquired Bats in February 2017 (pp. 60–62); their summary: challengers first offer new services at lower cost, practices converge, then mergers come while "the offer continues to expand" (p. 62). Harris's two consolidation waves, after the telegraph and telephone and from the early 1990s, have the same drivers: the order-flow externality, a falling ratio of variable to fixed costs, and loosened cross-border restrictions (sec. 26.6, pp. 538–539).

*Technology fragmentation*: application protocols (XDP, ITCH, EBS, Pitch, EMDI, RWF), multicast UDP or TCP with venue-specific recovery, and FAST, ZLIB or no compression made a common feed handler hard to build (Lehalle and Laruelle, pp. 53–54). *Shared references*: in the United States the consolidated tape lets participants decide on the same information; Lehalle and Laruelle argue that Europe needs a consolidated post-trade tape rather than reliance on primary markets, citing the Euronext outage of October 2010, during which MTF prices diverged from Euronext's by about 2 percent for some stocks (Introduction, p. 7; chap. 2, sec. 2.5.2, pp. 183–186).

*Price formation across fragmented venues*. With several venues trading one security, Hasbrouck models their prices as cointegrated around a common efficient price (a vector error-correction model) and attributes price discovery by *information shares*, each venue's share of the efficient price's innovation variance, reported as bounds over orderings when innovations are correlated; the venue with the largest share "is in some ways the most important. In a statistical sense, it is the price leader" [s6-hasbrouck, chap. 10, secs. 10.2–10.3, pp. 99–102]. Wider time intervals make events look simultaneous and widen the bounds, and the timestamps and sequencing of different markets' reports are often unreliable (pp. 104–105).

Fragmentation improves prices while venues stay synchronised by routing, arbitrage and consolidated information, and damages price formation when they lose a common reference, primary quotes go stale or participants pause routing under stress. This is consistent with Harris's consolidation mechanisms in this entry and with Lehalle and Laruelle's outage account (European fragmentation evidence), but as a general claim it is an untested synthesis.

**Pitfalls.** Lehalle and Laruelle's dates are not all consistent: the end of Turquoise's market-making agreement is March 2009 in Table 1.1 and on p. 47 but March 2008 on p. 53; NASDAQ OMX Europe runs "from September 2008 to May 2010" on p. 49 but launches in May 2009 and closes in July 2010 in Table 1.1; the LSE's acquisition of Turquoise is placed in February 2011 (pp. 16 and 60); and the Euronext outage is dated 13 October 2010 in the text and 10 October 2010, a Sunday, in the figure captions. These are reported as the book gives them. That segmentation separates "natural" from "opportunistic" liquidity is the authors' interpretation, with Menkveld cited for opportunistic liquidity (p. 42).

Their merger-wave chronology is to 2017; later ownership changes are not covered. Harris's examples (ITS, SOES) are U.S. institutions of 2002. Fragmentation is not systemic risk by itself; it becomes dangerous when participants react to stress in the same direction, related venues desynchronise, or a liquidity withdrawal spreads to hedges, ETFs, futures and correlated stocks. Lehalle and Laruelle frame the question the same way, as a debate between supporters of high-frequency trading, who say it lowers volatility by adding liquidity, and opponents, who cite the Flash Crash, and conclude that the Flash Crash "is not an obvious evidence of an increase in systemic risk" [alpha-laruelle, chap. 2, sec. 2.5, pp. 169 and 177]; the 2010 staff report's lessons on simultaneous withdrawal and interconnected markets are in the May 2010 Flash Crash. These conditions are plausible mechanisms, not established findings.

**See also.** European fragmentation evidence; fragmentation and concentration metrics; smart order routing; dark pools and crossing networks; market structures and venues; fixed-income market structure.

**Further reading.** Harris, chap. 26; Lehalle and Laruelle, introduction and chap. 1, sec. 1.1.

**Rule 17.27 U.S. equity market structure rules: decimal pricing, Regulation NMS and the minimum pricing increment, single-stock and market-wide circuit breakers, stub quotes, market access controls and the consolidated audit trail (status as of 16 October 2026)**

**Statement.** *Decimal pricing.* On 8 June 2000 the SEC ordered the exchanges and the NASD to submit a plan phasing in decimal pricing for stocks and options, starting no later than 5 September 2000 [sec-34-42914, Release 34-42914]; the conversion of consolidated quotations "was successfully completed on April 9, 2001", narrowing the minimum price variation from 1/16 of a dollar to a penny [sec-34-44568, Release 34-44568, p. 38390].

*Regulation NMS* (Release 34-51808, effective 29 August 2005) adopted the Order Protection Rule (policies and procedures to prevent trades at prices inferior to protected quotations, which must be immediately and automatically accessible), the Access Rule (fair access, a cap on access fees, and a ban on patterns of locking or crossing quotations), the Sub-Penny Rule (no orders, quotes or indications in increments below a penny except when priced below $1.00) and amended market data rules [sec-34-51808, p. 37496].

*Minimum pricing increment, current text.* As amended in 2024 [ex-sec-tick-access-2024, Release 34-101070, catalogue entry], 17 CFR 242.612 sets, for orders priced at $1.00 or more, an increment of $0.01 if a stock's time-weighted average quoted spread over the evaluation period (January–March or July–September) exceeded $0.015 and $0.005 if it did not, operative from the first business day of May and of November; $0.0001 below $1.00 [cfr-242-612, text for 14 October 2026].

**Conditions.** *Adoption, stay and compliance of the 2024 amendments.* Adopted 18 September 2024 (Release 34-101070), the amendments also cut the access-fee caps of Rule 610(c) to $0.001 per share for quotations at $1.00 or more and 0.1 percent of the price below, required exchange fees and rebates to be determinable at execution (Rule 610(d)), and accelerated the round-lot and odd-lot information definitions.

On 12 December 2024 the Commission stayed the amendments to Rules 600(b)(89)(i)(F), 610(c) and 612 pending judicial review, not those to 610(d), 603(b) or the round-lot and odd-lot definitions; the D.C. Circuit denied the petitions on 14 October 2025; on 31 October 2025 the Commission exempted compliance with the three stayed rules until the first business day of November 2026; and on 11 June 2026 it extended the exemption from compliance with Rules 600(b)(89)(i)(F) and 612 (the amended minimum pricing increment) and 610(c) (the amended fee caps) to "the first business day of November 2027" [sec-34-105656, 91 FR 36022–36024].

*Effective text against compliance relief.* The amendments took effect on 9 December 2024 [fr-2024-21867, document record], and the amended rule text is in the CFR [cfr-242-612, text for 14 October 2026]; the 2024 stay suspended the effect of three of them pending judicial review, and the October 2025 relief was granted "in anticipation of an end to the partial stay upon the completion of judicial review". What the 2025 and 2026 orders grant is exemptive relief from *compliance*, not a change to the rule text: as of 16 October 2026 the amended Rules 600(b)(89)(i)(F), 610(c) and 612 are effective rules whose compliance is exempted until the first business day of November 2027 [sec-34-105656, pp. 36022–36024]. The June 2026 order cites the competing 2026 initiatives (23/5 trading, Rule 605 compliance) and the systems and rule-filing work the amendments need.

*Odd-lot information.* The adopting release set the first business day of May 2026 as the compliance date for odd-lot information. On 15 January 2026 the Commission exempted the participants of the consolidated-data plans from Rule 600(b)(69)(ii), odd lots priced at or between the national best bid and offer "aggregated at each price level at each national securities exchange and national securities association" (aggregate odd-lot depth), until the first business day of May 2028.

The exemption does not cover Rule 600(b)(69)(iii), the best odd-lot order (the highest odd-lot bid above the national best bid and the lowest odd-lot offer below the national best offer), which the request letter said the exclusive SIPs would begin disseminating in May 2026 [sec-34-104612, pp. 1–5]. Whether that dissemination began on schedule, and whether the round-lot definition was implemented in November 2025 as the letter states, are unverified against the plans' own records.

*Circuit breakers and quoting after 6 May 2010.* Single-stock circuit breakers (five-minute pauses after a 10 percent move in five minutes) were adopted in June 2010 for S&P 500 stocks and extended to all NMS securities in June 2011, per Lehalle and Laruelle [alpha-laruelle, chap. 2, sec. 2.5.2, pp. 181–182].

On 31 May 2012 the Commission approved, as a one-year pilot, the NMS Plan to Address Extraordinary Market Volatility (*limit up–limit down*), which prevents trades outside price bands around a reference price, with bands, for stocks priced above $3.00, of 5 percent for Tier 1 stocks (S&P 500, Russell 1000, certain ETPs) and 10 percent for Tier 2 stocks, 20 percent from $0.75 to $3.00, and the lesser of $0.15 or 75 percent below $0.75, with a proposed implementation date of 4 February 2013 [sec-34-67091, Release 34-67091]; on 11 April 2019 it approved the plan's operation "on a permanent basis" [sec-34-85623, Release 34-85623].

On the same day in 2012 it approved, as a pilot, revised *market-wide circuit breakers*: the S&P 500 replaced the Dow Jones Industrial Average, triggers recalculated daily fell from 10, 20 and 30 percent to 7, 13 and 20 percent, Level 1 and 2 declines halt trading for 15 minutes before 3:25 p.m. and not after, and Level 3 halts trading for the day [sec-34-67090, Release 34-67090]; exchange filings extended that pilot in 2019 and 2020 (to 18 October 2021 in one filing) [sec-fr-2020-23011, MEMX filing, sec. II.A.1]. On 16 March 2022 the Commission approved, on an accelerated basis, NYSE's proposal, as modified by Amendment No. 1, to make its market-wide circuit-breaker rule (Rule 7.12) permanent "in their current form", after a working group's study of the four Level 1 halts of March 2020; the pilot period was to expire on 18 March 2022.

Amendment No. 1 requires participation in industry-wide testing, a report to the Commission after any Level 1, 2 or 3 halt, and a review after 5 percent market declines or changes to reopening processes [sec-34-94441, Release 34-94441, pp. 16286–16296]. *Other venues and later changes.* Nasdaq filed on 12 April 2022 "to adopt the same change to make permanent the MWCB pilot program in Equity 4, Rule 4121", and FINRA on 13 April 2022 for Rule 6121.02 (market-wide circuit breakers in NMS stocks), both immediately effective [fr-2022-mwcb-permanent, 87 FR 23296 and 23291]; the other NYSE, Cboe, Nasdaq, IEX, MEMX and LTSE exchanges made similar filings in April 2022.

The triggers remain 7, 13 and 20 percent of the prior S&P 500 close, with 15-minute Level 1 and 2 halts only between 9:30 a.m. and 3:25 p.m. and a Level 3 halt for the rest of the day; on 4 September 2026 NYSE filed, immediately effective, an amendment to Rule 7.12 so that after a Level 3 halt no SRO resumes before 4:00 a.m. ET the next trading day once 23-hour trading begins (several exchanges planned overnight sessions from 9:00 p.m. to 4:00 a.m. from 6 December 2026), "in coordination with the other SROs" [fr-2026-19125, 91 FR 59278]; Nasdaq filed the same change to Rule 4121 [fr-2026-20605, 91 FR 64439].

*LULD Plan after 2019.* The Commission's plan actions run from the Eighteenth Amendment (permanent operation, 11 April 2019) through the Twentieth (approved 21 April 2020, clarifying the quarterly data the participants report) and amendments of 2020, 2025 and 2026. On 16 September 2024 it disapproved the Twenty-Third Amendment [fr-2024-21508, 89 FR 77203], which would have moved all ETPs but single-stock ETPs into Tier 1, with tighter bands.

On 5 August 2026 it approved the Twenty-Seventh, a first phase of "Overnight Price Bands" for 9:00 p.m. to 4:00 a.m. ET, set by each stock's primary listing exchange, with trading centres open then required to prevent trades outside them [fr-2026-16201, 91 FR 51515]. On 7 October 2026 it put the Twenty-Ninth into effect for up to 120 days, moving the time of the overnight consolidated reference price from 7:45 to 7:30 p.m. [fr-2026-20799, 91 FR 65016].

*Stub quotes.* The Commission approved SRO rules, implemented 6 December 2010, requiring registered market makers to quote within 8 percent of the NBBO for securities under the single-stock circuit breakers (readjusting at 9.5 percent), 20 percent outside the breakers' hours and 30 percent for other securities [sec-34-63255, Release 34-63255].

*Market access.* Rule 15c3-5 (Release 34-63241, dated 3 November 2010; effective 14 January 2011, compliance 14 July 2011) requires brokers with market access to control pre-set credit and capital thresholds and regulatory requirements under their own direct control [cfr-240-15c3-5, paras. (b)–(c)]. *Consolidated audit trail.* Rule 613 (Release 34-67457, effective 1 October 2012) required the SROs to submit an NMS plan for a

consolidated order-tracking system from order inception through routing, cancellation, modification or execution [sec-34-67457, p. 45722, summary and dates]; the CAT NMS Plan was approved on 15 November 2016 [sec-34-79318, Release 34-79318].

*Since 2016.* On 25 July 2025 the Eleventh Circuit vacated the Commission's 2023 order approving the "Executed Share Model" for funding CAT, as arbitrary and capricious because the Commission had not explained why it did not bar SROs from passing their CAT costs through to broker-dealers and had not incorporated actual CAT costs in its economic analysis; the court did not question the model's allocation and stayed its judgment for sixty days after its mandate [fr-2026-05350, secs. I–II]. CAT LLC refiled the model on 5 September 2025 with a promise by each participant not to file pass-through fees, and the Commission approved it, as modified, on 16 March 2026 [fr-2026-05350, sec. I, 91 FR 13410].

On 13 January 2026 it approved, as modified by two participant amendments and by the Commission, an amendment that proposed to codify earlier exemptions and to end the reporting of names, addresses and years of birth for all customers, keeping the anonymised customer identifiers (CCIDs), with the customer database renamed the "Reference Database" in Amendment No. 1 [fr-2026-00762, sec. I and nn. 25–38]. On 27 March 2026 it approved further cost reductions, noting that operating costs had "far exceeded" the $55.8 million a year estimated in 2016 [fr-2026-06255, secs. I–II, 91 FR 16284].

On 20 April 2026 it published a concept release on a "comprehensive review" of CAT and other audit trails, including their funding, with comments due by 22 June 2026 [fr-2026-07651, 91 FR 20945]. The litigation is described from the Commission's and CAT LLC's accounts; see also the technical specifications [wide-sec-cat, web page].

**Practice.** Historical claims to keep apart: Harris's account of trade-through and time-precedence problems before Regulation NMS (venue competition and fragmentation) and his description of market-wide halts at 10, 20 and 30 percent of the DJIA after 1998 [alpha-harris, chap. 28, box, p. 573] predate the 2012 rules. Lehalle and Laruelle date the stub-quote approval 8 November 2010; the approval order in the Federal Register is dated 5 November 2010. They date the vote on Rule 613 11 July 2012, which the Commission's 2026 order also gives as the adoption date; the adopting release is dated 18 July 2012 [fr-2026-05350, sec. II] [fr-2026-00762, n. 3]. That decimalisation narrowed quoted spreads but reduced displayed depth and queue stability is evidence, summarised in tick size and market quality. Reg NMS links U.S. equity venues through protected quotes, but protection covers only automated top-of-book quotations (Rule 611, best execution).

**Pitfalls.** Keep effective rule text and compliance relief apart: the CFR shows the amended Rule 612 and fee caps, but compliance with them is exempted until November 2027; and the odd-lot exemption to May 2028 covers aggregate odd-lot depth, not the best odd-lot order. The compliance status of the 2024 tick and access-fee amendments has changed three times; recheck the SEC's exemptive orders before relying on a date. Lehalle and Laruelle's single-stock circuit-breaker parameters are unverified against the 2010 approval orders. Rule texts are from the eCFR point-in-time edition for 14 October 2026.

**See also.** best execution; tick size; circuit breakers and trading halts; high-frequency trading strategies; EU trading venues and transparency.

**Further reading.** Lehalle and Laruelle, chap. 2, sec. 2.5.2.

### Rule 17.28 EU trading venues and pre-trade transparency: MiFID and MiFID II dates, venue types, waivers, and the double volume cap replaced by a single volume cap (status as of 16 October 2026)

**Statement.** *Dates.* MiFID (Directive 2004/39/EC of 21 April 2004) applied from 1 November 2007, after Directive 2006/31/EC postponed its transposition deadline to 31 January 2007 [eu-dir-2006-31, Art. 1, points (4)–(5)]. MiFID II (Directive 2014/65/EU) and MiFIR (Regulation (EU) No 600/2014) were to apply from 3 January 2017 [eu-dir-2014-65, Art. 93] [eu-reg-600-2014, Art. 55]; Directive (EU) 2016/1034 and Regulation (EU) 2016/1033, both of 23 June 2016, deferred them by twelve months to 3 January 2018 [eu-dir-2016-1034, Art. 1, points (7)–(8)] [eu-reg-2016-1033, recital and Art. 1].

*Venue types* under MiFID II: a *regulated market* and an *MTF* are multilateral systems bringing together third-party interests under non-discretionary rules; an *OTF* is a multilateral system that is neither, for bonds, structured finance products, emission allowances or derivatives; a *trading venue* is any of the three [eu-dir-2014-65, Art. 4(1), points (21)–(24)]. *Systematic internaliser, historical wording.* As adopted in 2014 and applied from 3 January 2018, MiFID II defined a systematic internaliser as an investment firm which, "on an organised, frequent systematic and substantial basis", deals on own account when executing client orders outside a regulated market, an MTF or an OTF without operating a multilateral system.

The frequent and systematic basis was measured by the number of OTC trades executed on own account for clients, the substantial basis by the size of that OTC trading relative to the firm's own or the Union's total trading in the instrument, and the definition applied "only where the pre-set limits … are both crossed or where an investment firm chooses to opt-in" [eu-dir-2014-65, Art. 4(1), point (20)].

*Current wording.* Directive (EU) 2024/790 replaced point (20): a systematic internaliser "means an investment firm which, on an organised, frequent and systematic basis, deals on own account in equity instruments by executing client orders outside a regulated market, an MTF or an OTF, without operating a multilateral system, or which opts in to the status of systematic internaliser" [eu-dir-2024-790, Art. 1, point (3)(b)]. Three things changed. The *substantial-basis* criterion is gone. The status that follows from a firm's own trading covers *equity instruments* only, and the pre-set quantitative limits give way to a qualitative assessment, because the quantitative criteria imposed "an excessive burden" on firms and on ESMA and because Regulation (EU) 2024/791 removes systematic internalisers from the pre-trade transparency requirements for non-equity instruments.

The *opt-in*, already in the 2014 text, remains, and for non-equity instruments it is now the only route: "It should, however, be possible for an investment firm to opt in to become a systematic internaliser for non-equity instruments" [eu-dir-2024-790, recital (7)]. Commission Delegated Regulation (EU) 2026/392 reads the amendment the same way (it "replaced the quantitative criteria … with a qualitative assessment and limited that assessment to equity instruments only, while leaving the possibility … to opt in") and gives that as the reason to delete from Delegated Regulation (EU) 2017/577 the obligation on trading venues, APAs and CTPs to supply the data for the quantitative calculations of Delegated Regulation (EU) 2017/565, Articles 12–15 [eu-reg-del-2026-392, recital (1)].

*Dark trading* uses the *reference price waiver* and the *negotiated trade waiver* of MiFIR Article 4(1)(a) and (b)(i) [ex-eu-mifir-review-2024, recital on the volume cap]. *The four equity waivers and their thresholds.* MiFIR Article 4(1) lets competent authorities waive pre-trade transparency for (a) systems matching at a reference price derived from the first-admission venue or the most relevant market, which must be that venue's current midpoint, or its opening or closing price outside continuous trading; (b) negotiated transactions within the current volume-weighted spread, in illiquid instruments within a set percentage of a reference price, or subject to conditions other than the current market price; (c) orders large in scale; and (d) orders held in an order management facility pending disclosure [eu-reg-600-2014, Art. 4(1)–(2)]; Regulation (EU) 2024/791 removed the volume-cap condition from (b)(i) [ex-eu-mifir-review-2024, Art. 1(3)].

Large in scale for shares and depositary receipts is a minimum order size by the instrument's EU average daily turnover (ADT): €15,000 below €50,000 ADT, then 30,000, 60,000, 100,000, 200,000, 300,000, 400,000 and 500,000, and €650,000 at €100 million ADT or more, with ADT computed from the previous calendar year's EU trading on and off venue [eu-rts1-2017-587, Art. 7 and Annex II, Table 1]; for ETFs it was €1 million and has been €3 million since Delegated Regulation (EU) 2023/944 entered into force on 5 June 2023 [eu-rts1-amend-2023-944, Art. 1(4)].

An order management facility may hold a reserve order of at least €10,000, or another order at least the venue's preset minimum tradable quantity [eu-rts1-2017-587, Art. 8(2)]; Delegated Regulation (EU) 2025/1246 (published 3 November 2025) redefined the reserve order and exempted it from the no-interaction condition, and narrowed the benchmark (VWAP or TWAP) case of negotiated transactions to periods long enough to have "no relation to the current market price" [eu-rts1-amend-2025-1246, Art. 2(4) and (6)]; it left the large-in-scale table for shares unchanged, while replacing the standard-market-size table (Annex V). Its points (2), (3)(a) and (c), (5), (10)(a) and (13) of Article 2 apply from 2 March 2026, the rest from entry into force (Art. 3).

A query of the Publications Office's CELLAR metadata for acts amending the two regulations returned three for 2017/587 (2019/442, 2023/944 and 2025/1246) and two for 2017/588 (2019/443 and 2023/960); an amendment not yet indexed there would be missed.

**Conditions.** *Application dates of the systematic-internaliser change.* Directive (EU) 2024/790 is dated 28 February 2024, was published on 8 March 2024 and entered into force on the twentieth day after publication; Member States had to adopt the measures needed to comply by 29 September 2025 [eu-dir-2024-790, Arts. 2(1) and 3]. As a directive it binds firms through each Member State's transposing law, so the date from which the new definition governs a given firm is the national one.

*National measures* (as of October 2026). The Publications Office's CELLAR index of national implementing measures, queried on 16 October 2026, links 66 notified measures from 24 Member States to Directive (EU) 2024/790 and none from France, Spain or Romania [eu-nim-2024-790, query of national implementing measures]; a listing records a notification, not what the measure says, and several entries (for Czechia, Hungary and Slovakia, for example) are earlier capital-markets laws notified under other directives.

*Ireland*: S.I. No. 436 of 2025, in operation from 29 September 2025, substitutes into the Irish MiFID regulations a definition of systematic internaliser identical to the directive's ("deals on own account in equity instruments … or which opts in to the status of systematic internaliser") [ie-si-436-2025, regs 1 and 4(c)]. *Malta*: L.N. 218 of 2025 (3 October 2025) transposes only Article 1(1) of the directive, deleting regulation 5A of Malta's MiFID regulations [mt-ln-218-2025, regs 1(2) and 2]; where Malta transposed the new definition is unverified. Ireland's dates should not be assumed for other Member States. Delegated Regulation (EU) 2026/392 is dated 20 February 2026, was published on 1 June 2026 and entered into force, directly applicable, on the twentieth day after publication [eu-reg-del-2026-392, Art. 2].

ESMA announced on 10 April 2025 technical standards with a "new qualitative regime for Systematic Internalisers", harmonising the content and procedure of firms' notifications [esma-si-2025-04-10, news item]. It contains *draft* standards, including a new ITS on SI notifications, which ESMA submitted to the Commission on 10 April 2025; the Commission "has three months to decide whether to endorse" them [esma-si-final-report-2025, Next Steps, PDF p. 7]; whether they were adopted is unverified, so they are not cited here as law.

A title search of EU legislation in October 2026 found no implementing regulation on SI notifications under MiFIR Art. 15(5), the ITS's legal basis [esma-si-final-report-2025, Annex II, sec. 6.2.1, PDF p. 43]. It found Delegated Regulation (EU) 2026/482 of 24 November 2025, whose title says it amends Delegated Regulation 2017/567 on, among other things, "the size specific to the instrument for the purposes of obligations for systematic internalisers". A title search can miss an act, so non-adoption is not established.

In the report, whether a firm deals "on a frequent and systematic basis will be determined only by a qualitative assessment", which "should apply only to equity instruments", with opt-in possible for non-equity (para. 8, PDF p. 9); the notification period is extended from two weeks to 20 calendar days, and "All existing SIs will have to resubmit the notification" (paras 21–22, PDF p. 11). On guidance, "ESMA will discuss with NCAs the areas where further guidance is required" and invites firms to ask their NCAs (para. 23): the report itself gives no guidance on applying the qualitative test. ESMA's systematic-internaliser data page redirected to a login.

*From double to single volume cap.* Under the original MiFIR Article 5, trading under those two waivers in an instrument was limited to 4 percent of total EU trading on any one venue and 8 percent across the EU over the previous 12 months; breaching either led the competent authorities to suspend the waivers within two working days for six months, on monthly ESMA data, with warning reports at 3.75 and 7.75 percent [eu-reg-600-2014, Art. 5].

Regulation (EU) 2024/791 of 28 February 2024, published on 8 March 2024 and in force from 28 March 2024, replaced it with a single cap: trading venues suspend their own use of the reference price waiver for three months when trading under it exceeds 7 percent of total EU trading in the instrument, based on quarterly ESMA data covering the previous 12 months; the negotiated trade waiver is no longer capped; monitoring under the new cap "shall start on 29 September 2025"; ESMA reports annually from 29 September 2027 and the Commission may adjust the threshold [ex-eu-mifir-review-2024, Art. 1(4), amending MiFIR Art. 5].

*Transition.* ESMA stated on 27 March 2024 that the double cap would continue, for both waivers, until the single cap applied [esma-transition-statement-2024, p. 2]; its double-volume-cap page records that the double cap was discontinued on 28 September 2025, ending any ongoing suspension [esma-dvc-page, web page]; on 10 October 2025 ESMA confirmed that the single cap "has become operational with the first calculation results published on 9 October 2025" [esma-mifir-review-statement-2025, p. 3].

**Practice.** *Lehalle and Laruelle's account (2018).* MiFID's "few and simple mechanisms" were best execution, three venue statuses and transparency (chap. 1, sec. 1.1.2, p. 42): regulated markets with public books, responsible for circuit breakers; MTFs, often pegged to regulated markets; and systematic internalisers, which few firms chose because of the transparency attached (pp. 43–44); the pre-trade waivers were large in scale, order management facility, negotiated trade and reference price, under which dark pools operated (p. 45). Writing before MiFID II applied, they expected a dark cap of 4 and 8 percent enforced yearly with six-month suspensions and a tick-size regime set by price and trading activity (Introduction, pp. 8–9); the first matched the original Article 5, the second is MiFID II Article 49, which requires tick-size regimes calibrated to liquidity and average spread [eu-dir-2014-65, Art. 49].

*The tick-size regime now*. For shares and depositary receipts, trading venues must apply a tick at least equal to the one in the Annex table for the instrument's liquidity band, set by the average daily number of transactions (ADNT) on its most relevant market (six bands: below 10, 10–80, 80–600, 600–2,000, 2,000–9,000 and 9,000 or more), and for the order's price range (19 ranges from below 0.1 to 50,000 and above); ETFs whose underlyings are solely such equities take the most liquid band; a share whose most relevant market runs only a periodic auction takes the least liquid band [eu-rts11-2017-588, Arts. 2–3 and Annex].

Competent authorities publish each ADNT by 1 March from the previous calendar year's trades, excluding reference-price, negotiated and large-in-scale transactions; venues apply the new band from the first Monday of April (originally 1 April) [eu-rts11-2017-588, Art. 3(1), (2) and (4)] [eu-rts11-amend-2023-960, Art. 1]; since 2019 the authority may adjust the ADNT of a share whose highest-turnover venue is in a third country to include that venue's transactions [eu-rts11-amend-2019-443, Art. 1, new Art. 3(8)–(10)]. The regulation applied from 3 January 2018; the 2019 amendment was published on 20 March 2019 and the 2023 one on 16 May 2023, each in force twenty days later.

Derived from the table: for prices of 0.5 and above the relative tick runs from 1 to 2.5 bp in the most liquid band and from 40 to 100 bp in the least liquid one; a share at €25 has a tick of 0.005 with 12,000 trades a day, 0.05 with 300 and 0.2 with five.

*Systematic internalisers* must quote, improve and execute on the same ticks since MiFIR Article 17a, inserted by Regulation (EU) 2019/2033 and applicable from 26 March 2020, which allowed midpoint matching of orders large in scale [eu-reg-2019-2033, Art. 63(3) and Art. 67(3)(a)]; Regulation (EU) 2024/791 replaced Article 17a so that neither the tick regime nor Article 15(2) prevents systematic internalisers "from matching orders at midpoint", with no size condition and no later application date stated in the amending act [ex-eu-mifir-review-2024, Art. 1(16)]. Directive (EU) 2024/790 lets regulated markets apply a third-country venue's tick to shares with a non-EEA ISIN, or traded in a non-EEA currency, whose most relevant market is outside the EU [eu-dir-2024-790, Art. 1(8), amending Art. 49(2)], through national transposition.

*Payment for order flow*: the 2024 amendments prohibit investment firms from receiving payment for forwarding or executing retail and opted-in professional client orders, with a Member-State exemption possible until 30 June 2026 for firms already doing so [ex-eu-mifir-review-2024, Art. 1, new MiFIR Art. 39a].

**Pitfalls.** Do not quote the 2014 systematic-internaliser definition ("frequent systematic and substantial", all instruments, pre-set quantitative limits) as current, nor describe the opt-in as new; Lehalle and Laruelle's account describe that regime. Do not quote "4 and 8 percent" as current: they applied from January 2018 to 28 September 2025. The single cap is 7 percent, applies only to the reference price waiver, is triggered by venues themselves, and suspends for three months. ESMA's data and suspension files are the operational record. The U.K. regime diverged after 2020. Article 48 of MiFID II (venue circuit breakers) is in circuit breakers and trading halts; best execution is best execution.

**See also.** dark pools; bond transparency rules; venue competition and fragmentation; tick size; U.S. equity market structure rules.

**Further reading.** Lehalle and Laruelle, introduction and chap. 1, sec. 1.1.

**Mechanism 17.29 Circuit breakers and trading halts: halts, price limits, taxes, margins and collars; when a halt helps; the gravitational effect, propagation and related instruments; incidence by trader type**

**Statement.** Harris: "Circuit breakers are trading rules that limit trading activity": trading halts, price limits, transaction taxes, margin requirements and position limits, and collars that restrict access to trading systems [alpha-harris, chap. 28, sec. 28.3, p. 572]. Lehalle and Laruelle classify thresholds as *static* ("a fixed price computed before the opening") or *dynamic* ("a corridor following the recent values of the price"), and breakers as *local* (one instrument) or *global* (a collection, generally all) [alpha-laruelle, chap. 2, sec. 2.5.3, pp. 187–188].

MiFID II requires regulated markets to be able "to temporarily halt or constrain trading if there is a significant price movement in a financial instrument on that market or a related market during a short period", with calibrated parameters reported to ESMA, and a material venue that halts must let authorities coordinate a market-wide response [eu-dir-2014-65, Art. 48(5)]. U.S. rules are in U.S. equity market structure rules.

**Conditions.** *When a halt helps* (Harris, sec. 28.3.1, pp. 572–575). If new fundamental information drives the move, a halt "will merely postpone the inevitable" and makes prices less informative meanwhile; if an imbalance among uninformed traders drives it, stopping them may help. Halts can let liquidity suppliers arrive, publicise imbalances, give time to meet margin calls and remove stop orders, and switch the pricing rule from discriminatory (continuous trading) to uniform (the reopening call), protecting standing limit orders. Against them: the *gravitational effect*, traders who fear a halt "may submit their orders earlier", so a price-triggered rule can raise volatility near its trigger, which discretionary, less predictable halts avoid; and value traders may watch less if they expect halts.

A halt in one market diverts order flow to others trading the same risk, so uncoordinated breakers "may increase volatility". Harris's conclusion: the theoretical net effect on transitory volatility is indeterminate, and empirical studies "generally have been inconclusive" because extreme events are rare (sec. 28.3.4, pp. 576–577).

*One later study*. Hautsch and Horvath study the five-minute single-stock trading pauses introduced in the United States after May 2010 (single-stock pauses from June 2010, a 10 percent move over five minutes; limit up–limit down pauses from April 2013, outside a 5 percent band around the five-minute average price for many stocks) with NASDAQ order-book data: of 1,513 NASDAQ pauses from 1 June 2010 to 30 June 2014 they keep 195 in 166 stocks with enough book activity, and compare them, by difference in differences, with 1,755 moves of more than 5 percent in five minutes in the same stocks from January 2009 to May 2010, before pauses existed (about nine control events per pause) [hautsch-horvath-2017, CFS Working Paper 571, April 2017 draft, secs. 2.2, 3.1 and 4, pp. 7–17].

Around pauses median volatility rose about tenfold, spreads widened before the pause and normalised over the following hour, and top-two-level depth fell by about 25 percent after the pause for at least an hour (sec. 3.2, pp. 14–15). Their conclusion is a trade-off: the pause's existence lets liquidity suppliers lean against price trends, so it works as a safeguard, but volatility and spreads rise before pauses, "in line with the 'magnet-effect"', and stay higher for an hour, so pauses act "as volatility amplifiers" rather than cooling markets (sec. 6, pp. 40–41).

Limits: one exchange's book, a selected subsample of pauses, and a pre-regulation control period of 2009–2010 that they test for similar spreads, volume and volatility; published in *Journal of Financial Economics* 131(2), 2019, pp. 378–403.

*A contrasting estimate*. Kasahara and Yamada estimate the effect of single-stock trading pauses on the Tokyo Stock Exchange, whose triggers are fixed yen thresholds, so that the percentage distance to a pause varies with the price level and serves as an instrument. Their order-book data run from 14 January 2014 to 30 March 2021 for the 500 most valuable firms (chosen on the previous year-end's capitalisation). Pauses "significantly reduce post-event volatility, narrow quoted bid–ask spreads, and facilitate price discovery", with benefits attenuated for highly volatile stocks after negative returns and in broad downturns [jpx-wp-52-2026, JPX Working Paper Vol. 52, 10 April 2026, abstract and sec. 3, PDF pp. 3–8].

They note that their sample is mostly opening-auction pauses triggered by small moves, that they find no magnet effect, and that Hautsch and Horvath's design draws its counterfactual from the pre-regulation period while theirs identifies effects within one regime (PDF p. 4 and conclusion).

Interpretation: the two studies differ in market, pause type and identification, so their opposite volatility findings are not a contradiction to resolve here. *Market-wide halts, March 2020*. In an exchange's filing published by the SEC, the U.S. market-wide circuit breaker produced four Level 1 halts, on 9, 12, 16 and 18 March 2020, and "in each instance, the markets halted as intended upon a 7% drop in the S&P 500 Index, and resumed as intended 15 minutes later". The SRO working group's study of 31 March 2021 concluded that the halts "appear to have had the intended effect of calming volatility in the market, without causing harm", and that the 7, 13 and 20 percent triggers and 15-minute halts were appropriate [fr-cboebzx-mwcb-2021, Release 34-93365, 15 October 2021, public-inspection PDF pp. 4–6].

This is the regulated venues' own evaluation reported by one of them; it is not an independent causal estimate. An earlier version is the University of Osaka Discussion Paper 26-03 of 31 March 2026. Saying that a halt improves price quality if and only if its benefit $B$ exceeds its cost $C$ defines improvement; it is not a result, and its content is the list of benefits and costs above.

*Daily price limits*. Chen, Gao, He, Jiang and Xiong use account-level data from the Shenzhen Stock Exchange, where regular stocks have 10 percent daily limits and special-treatment stocks 5 percent. They find that limits "may lead to unintended, destructive market behavior". Large investors tend to buy on the day a stock hits its upper limit and sell the next day, and their net buying on the limit-hitting day predicts a stronger long-run reversal. Comparing stocks before and after assignment to the tighter limit gives a causal check [chen-gao-he-jiang-xiong-2017, NBER WP 24014, November 2017, abstract and sec. 1]. This is evidence on one market's limit rule.

**Practice.** *Price limits and margin.* Harris's sidebar: a trader with $100,000 margin and $100,000 other assets faces a 25-point fall worth $250,000; without limits the broker immediately loses the margin and must chase $150,000; with a five-point daily limit the trader, unsure whether he is bankrupt, may post margin for two days before defaulting, and owes $50,000 at the end (box "Am I Bankrupt?", p. 574); "Price limits can protect the clearing system by increasing the margin payments that bankrupt traders pay before they default" (p. 582), and markets with halts "can have lower margin requirements" (p. 575).

*Incidence of taxes and margins* (sec. 28.3.2, pp. 575–576): restricting informed traders makes prices less informative and restricting value traders, dealers and arbitrageurs raises transitory volatility, while restricting order anticipators, bluffers and gamblers lowers it; restricting utilitarian traders would be "highly undesirable"; and it is hard to design rules that discriminate among these types, a tax falling hardest on frequent traders, dealers included.

*Design*: Lehalle and Laruelle ask that breakers respond to liquidity imbalance, although "all of them only look at prices", perhaps by combining large moves with poor liquidity; that thresholds reflect the volatility level and its intraday seasonality; and that instruments indexed on halted assets, highly correlated instruments and other venues trading the same instrument be halted in step, which they leave as "a research question"; they cite China's removal of badly calibrated thresholds in January 2016 (pp. 188–189). *Propagation*: in the Flash Crash of 6 May 2010 the absence of a stock-specific breaker was named among aggravating factors (p. 181).

*The exchanges' own review of March 2020*. The SROs' Market-Wide Circuit Breaker Working Group, with industry advisers, concluded that "the MWCB mechanism worked as intended" in the four halts of 9, 12, 16 and 18 March 2020, that the halts "appear to have had the intended effect of calming volatility in the market, without causing harm", that the S&P 500 reference, the 7/13/20 percent triggers and the 15-minute halt are appropriate, and that nothing should stop a halt shortly after the 9:30 open; it recommended making the 2012 pilot rules permanent unchanged, an independent S&P 500 calculation, minimal trade reporting to the SIP after a halt, and annual MWCB tests for Regulation SCI firms (Executive Summary, pp. 1–2).

Its evidence on "no harm" is descriptive: the share of symbols reopening on a trade rather than a quote did not differ from other days, and halt-reopening plus opening auction volume matched January 2020 opening auctions (sec. D, PDF p. 33) [mwcb-wg-2021, Report of 31 March 2021, Executive Summary and sec. D, pp. 1–2 and PDF p. 33]. It is the exchanges' self-assessment, with no counterfactual of the same days without halts.

**Pitfalls.** Harris's examples (the DJIA 10/20/30 percent halts, NYSE Rule 80A collars) are rules of 1998–2002 that have since been replaced (U.S. rules). Collars in Harris restrict access to trading systems, as Rule 80A did for index-arbitrage program trades. That breakers should stop disorderly liquidity demand, not ordinary price discovery, applies Harris's fundamental-versus-transitory distinction; neither source shows how to tell the two apart in real time. Gamblers usually trade on stale information, so discouraging them would reduce transitory volatility (p. 576).

**See also.** bubbles and crashes; fundamental and transitory volatility; regulation as market design; futures margin and settlement; trading sessions and execution systems.

**Further reading.** Harris, chap. 28; Lehalle and Laruelle, chap. 2, sec. 2.5.

### Rule 17.30 Spoofing and layering in securities markets: EU market-abuse law names them; U.S. securities law reaches them through its general manipulation provisions

**Statement.** *EU.* The Market Abuse Regulation, (EU) No 596/2014, defines market manipulation to include placing an order or any other behaviour that "gives, or is likely to give, false or misleading signals as to the supply of, demand for, or price of" a financial instrument (Art. 12(1)(a)(i)), and lists among manipulative behaviour "the placing of orders to a trading venue, including any cancellation or modification thereof, by any available means of trading, including by electronic means, such as algorithmic and high-frequency trading strategies", that has one of those effects by disrupting the trading system, by "making it more difficult for other persons to identify genuine orders …, including by entering orders which result in the overloading or destabilisation of the order book", or by creating a false or misleading signal (Art. 12(2)(c)) [eu-mar-596-2014, Art. 12].

Annex I's indicators include orders cancelled with position reversals in a short period and orders that change the representation of the best bid or offer (Annex I, A(d) and A(f)). The Commission's delegated regulation describes the practice by name: "Submitting multiple or large orders to trade often away from the touch on one side of the order book in order to execute a trade on the other side of the order book. Once the trade has taken place, the orders with no intention to be executed shall be removed — usually known as layering and spoofing" [eu-2016-522, Annex II, sec. 1, point 5(e)].

*U.S. securities.* Section 9(a)(2) of the Exchange Act makes it unlawful "to effect, alone or with 1 or more other persons, a series of transactions" in a security "creating actual or apparent active trading in such security, or raising or depressing the price of such security, for the purpose of inducing the purchase or sale of such security by others" [usc-15-78i, 15 U.S.C. 78i(a)(2), 2023 edition]; Section 9(a)(1) addresses wash and matched orders. Unlike the Commodity Exchange Act's spoofing clause for futures (7 U.S.C. 6c(a)(5)(C), order exposure), these provisions do not use the word.

**Conditions.** The MAR text is the original Official Journal version. A text search of its five amending acts (2016/1011, 2016/1033, 2019/2115, 2023/2869, 2024/2809) found no amendment to Article 12; no consolidated text was compared. MAR applies to instruments admitted to EU trading venues and related instruments. *U.S. rules.* Rule 10b-5 makes it unlawful "to employ any device, scheme, or artifice to defraud" or "to engage in any act, practice, or course of business which operates or would operate as a fraud or deceit upon any person, in connection with the purchase or sale of any security" [ecfr-17cfr240-10b-5, 17 CFR 240.10b-5(a), (c)]. FINRA Rule 2020 forbids members to effect transactions "by means of any manipulative, deceptive or other fraudulent device or contrivance" [finra-rules-2026, Rule 2020].

FINRA names the patterns, in Rule 5210, Supplementary Material .03, "Disruptive Quoting and Trading Activity Prohibited", version in force since 3 April 2017 [finra-rules-2026, Rule 5210.03]:

- *Type 1*: a party enters "multiple limit orders on one side of the market at various price levels", the level of supply and demand then changes, the party's orders on the opposite side are executed, and it then cancels the displayed orders;
- *Type 2*: a party narrows the spread by placing an order inside the NBBO, then executes on the opposite side against a participant "that joined the new inside market".

A "frequent pattern" of these facts constitutes the activity, whether on one venue or across several. FINRA's adopting notice states that the Supplementary Material "does not include an express intent element", that the order of the events does not change its applicability, and that it took effect on 15 December 2016; violations can found an expedited permanent cease-and-desist order under the Rule 9800 Series, against the trader or against a firm providing market access to such a client [finra-rn-17-22, Executive Summary; Background & Discussion]. The effective date differs from the 3 April 2017 start of the version selected on FINRA's rule page; the notice does not explain the difference, which may reflect a later amendment. How SEC and FINRA have applied the provisions in enforcement actions is not covered here.

**Practice.** Interpretation: for a firm's surveillance (RTS 6, Art. 13, RTS 6), the EU texts give testable patterns: orders away from the touch on one side, a trade on the other, then cancellation; high order-to-trade or cancellation ratios around a trade; changes to the displayed best quotes that do not lead to execution. These are indicators, which "shall not necessarily be deemed, in themselves, to constitute market manipulation" (MAR, Annex I, A).

**Pitfalls.** Cancellation as such is not manipulation: market makers cancel most orders. Whether *intent* must be shown differs by regime and by limb.

- *EU administrative law (MAR).* Article 12(1)(a), the limb that covers orders, is framed by effect: an order that "gives, or is likely to give, false or misleading signals" or "secures, or is likely to secure" an abnormal or artificial price, "unless the person … establishes that such transaction, order or behaviour have been carried out for legitimate reasons, and conform with an accepted market practice". Article 12(1)(b) turns on use of "a fictitious device or any other form of deception or contrivance"; the information limb (c) and the benchmark limb (d) require that the person "knew, or ought to have known" that the information was false or misleading [eu-mar-596-2014, Art. 12(1)]. None of these limbs states an intent requirement. ESMA said so in its technical advice: "even situations that were not committed with intent are under the scope of market manipulation of MAR", and an indicator or example that seems to describe manipulative intent "does not imply that, in the absence of the intent that conduct might not fall within the scope of the definition" [esma-2015-224, Annex III, feedback on Q1, para. 12, printed p. 78; sec. 2.2.2, para. 8, printed p. 9].
- *EU criminal law.* Directive 2014/57/EU requires Member States to make manipulation a criminal offence "at least in serious cases and when committed intentionally" (Art. 5(1)); its recital adds that sanctions for breaches of MAR "do not require that intent is proven or that they are qualified as serious" [eu-2014-57-csmad, Art. 5(1); recital 23]. Intent is therefore an element of the criminal offence, not of the administrative one.
- *U.S. securities law.* Section 9(a)(2) has its own purpose element ("for the purpose of inducing the purchase or sale of such security by others", quoted above). Rule 10b-5 requires scienter: the Supreme Court held that "scienter is an element of a violation of § 10(b) and Rule 10b-5, regardless of the identity of the plaintiff or the nature of the relief sought", including SEC injunctive actions, after defining scienter as "intent to deceive, manipulate, or defraud" [us-scienter-cases, Aaron v. SEC, 446 U.S. 680 (1980), syllabus and opinion, pp. 689–695; Ernst & Ernst v. Hochfelder, 425 U.S. 185 (1976), syllabus, pp. 194–214]. Whether recklessness suffices, a question *Hochfelder* left open, was not followed further.
- *FINRA.* Rule 5210.03 has no express intent element and turns on a frequent pattern of the listed facts (above). It is a self-regulatory rule for members, alongside, not an exception to, the regimes above.

The futures-market spoofing clause (7 U.S.C. 6c(a)(5)(C)) is not covered here. Harris's 2002 legality table (bluffing and manipulation) predates both MAR and the Dodd–Frank spoofing clause.

**See also.** bluffing and market manipulation; order exposure decisions; RTS 6.

**Further reading.** SEC and FINRA enforcement releases on layering, and ESMA's Q&A on MAR; ESMA's technical advice (ESMA/2015/224) for the indicators and the role of intent.

## 17.9 Tick size

**Definition 17.31 Tick size: the price grid, tick regimes and relative tick, when the tick binds, how it prices queue jumping, and price clustering**

**Statement.** The tick size defines the price grid on which a limit order must lie: every valid price is a multiple of it, so it is also the smallest difference between two limit prices [alpha-laruelle, chap. 1, sec. 1.3.1, p. 74]. It may apply to every stock whatever its price and liquidity (as in the United States for stocks priced above $1) or depend on price and liquidity bands, a *tick-size regime* (p. 74). The *relative tick* $\vartheta/m(t)$, tick over mid-price, compares instruments: Priceline at about $1,000 in 2014 had a relative tick near $10^{-5}$ with the same one-cent tick as a $1 stock [catalogue-0528, chap. 3, sec. 3.1.5, pp. 48–49]. A *large-tick* instrument has a spread close to one tick most of the time, a *small-tick* one a spread of many ticks (chap. 4, p. 58).

**Conditions.** Rules and figures as of 2002–2015; U.S. and EU tick regimes have changed since. Being the smallest price increment, the tick is a floor on the spread, so a smaller tick lowers spreads only where that floor binds [alpha-laruelle, chap. 1, sec. 1.3.2, p. 77]. Two simple measures of how much it binds (p. 77): the *binding probability*, the share of trades, value traded or quote duration with a one-tick spread, and the *spread leeway*, the spread in ticks minus one. For S&P 500 stocks in July 2012 most stocks below $20 had a one-tick spread most of the time, and stocks above $150 showed no constraint (Figures 1.19–1.20, pp. 78–79).

**Practice.** The tick sets the price of priority: where it is very small, a trader can step ahead of resting orders by one tick for little cost, which leaves little volume at any level and makes the best quotes change very often [catalogue-0528, chap. 3, sec. 3.1.5, p. 49]; Lehalle and Laruelle add that queue jumping with a small tick "makes time priority nearly meaningless" and quotes more fleeting, while with a large tick and a one-tick spread an impatient trader must cross and a liquidity provider must manage its place in the queue [alpha-laruelle, chap. 1, pp. 88–89].

Harris reports that prices cluster on round numbers (whole numbers more than halves, halves more than odd quarters; in decimal markets multiples of 1.00, 0.50, 0.25, 0.20, 0.10 and 0.05), most when the minimum increment is a small fraction of price, the market volatile and the instrument thinly traded, and that traders place orders just above or below round numbers to step ahead of others [alpha-harris, chap. 5, box "Price Clustering", p. 91].

*Ticks, precedence and competition between venues*. Time precedence "is meaningful only when the minimum price increment is not trivially small": too small a tick weakens precedence and price competition, too large a tick makes improving the price expensive [alpha-harris, chap. 6, sec. 6.1.1.2, p. 114]. Large traders who expose orders prefer strict time precedence with an economically significant tick; small market-order traders prefer small ticks, since the spread cannot be smaller than the tick [alpha-harris, chap. 26, sec. 26.3.1, p. 531].

A market cannot keep a meaningful tick when competitors trading the same instrument offer a smaller one: small markets with few limit orders cut ticks to attract quote matchers, and larger ones must follow. Harris's sidebar *The Race to the Bottom* reads the U.S. tick reductions of the 1990s, which "culminated in the full decimalization of the markets in 2001", this way (sec. 26.5.2, pp. 537–538).

In Europe, Lehalle and Laruelle date a "tick war" to June 2009, when BATS and Turquoise cut tick sizes, "leading to a tacit agreement between exchanges under the FESE" (Table 1.1, p. 15); MiFID II then required regulated markets to adopt tick-size regimes calibrated to liquidity and average spread (Art. 49, EU trading venues and transparency). The current U.S. increment rule and the stayed half-penny tick are in U.S. equity market structure rules. Judging who gains from a tick change, and varying ticks with price and liquidity and revising them on spreads, depth, order-to-trade ratios and standard-size costs, are a synthesis of these arguments, not a rule.

**Pitfalls.** Explaining price clustering by communication, search and queueing convenience requires evidence; Harris states when clustering is strongest, not why. Price- and liquidity-banded tick regimes match Lehalle and Laruelle's price ranges and quotation groups, which "theoretically" group stocks by liquidity (p. 74). That further reductions mostly change queue behaviour once the tick stops binding goes beyond the passages: Lehalle and Laruelle report spreads that did not fall, and sometimes rose, for illiquid high-priced stocks after decimalisation (tick size and market quality).

**See also.** Harris's discretised spread; tick size and market quality; spread determinants (large-tick bounce and depth); queue imbalance (large-tick races); order lifecycle (priority rules).

**Further reading.** The U.S. tick-size pilot (Barardehi et al.) and the SEC's 2024 tick and access-fee rule.

**Evidence 17.32 Tick size and market quality: quoted depth, order and trade sizes, order-to-trade ratios, cumulative depth and the evidence from tick reductions**

**Statement.** On FTSE 100 stocks in April 2012, grouped by deciles of the daily spread in ticks, Lehalle and Laruelle find far less volume at the best limits when the tick is small relative to the spread, a nearly threefold fall in the size of orders at the first limit from the first to the last decile, smaller trades, and a 50 percent rise in the number of book updates per trade [alpha-laruelle, chap. 1, sec. 1.3.2, Figures 1.21–1.24, pp. 80–84]. They read the last as a fall in the value of a quote, smaller and shorter-lived (p. 84).

From the decimalisation studies they cite: spreads fell for active low-priced stocks whose spread had been one tick, with "at best no effect" and some increases for illiquid high-priced stocks, and a tick below which further reductions widen spreads (Ahn et al. found the direction of Harris's prediction right but its magnitude overestimated) (pp. 79–80). Harris's prediction was a cross-sectional model projection, not an observed change: for a $1/16 tick he projected, for stocks priced under $10, spreads 38 percent narrower, quotation sizes 16 percent smaller and daily volume 34 percent higher [harris-1994, p. 149].

When the NYSE moved from $1/8 to $1/16 on 24 June 1997, Goldstein and Kavajecz found, for 100 common stocks from the TORQ sample (plus 10 funds) over one four-week period before and two after, quoted spreads $0.03 (14.3 percent) narrower and quoted depth 48 percent lower, while the spread of the limit-order book widened by $0.03 (9.1 percent), its depth at the best prices fell 48 percent and cumulative depth fell throughout the book; small orders gained, large orders lost, and the gains concentrated in frequently traded stocks [goldstein-kavajecz-1998, abstract and results, PDF pp. 2, 5 and 9–15].

**Conditions.** FTSE 100 data of April 2012; decimalisation studies of 1992–2005. Spreads alone do not measure quality: a smaller quoted spread may be available for less volume (p. 81). *Cumulative depth* rebuilt from order-level data measures the cost of a marketable order of any size and includes hidden liquidity, but conditions on how liquidity takers changed their order sizes; Bacidore et al. found NYSE marketable orders cheaper at every size after decimalisation and depth thinner beyond 15 cents from the best quotes, and Goldstein and Kavajecz, with costs in basis points, found the public book less deep and large trades costlier (pp. 85–87). In a fragmented market a single book is not the market, and mirrored orders make aggregation across venues unreliable (pp. 75–76, 85).

These are cross-sectional associations (FTSE 100) and before–after comparisons around single events. Goldstein and Kavajecz's sample is the surviving TORQ stocks, slightly overweight in large stocks (n. 7, PDF p. 9), and the version cited is their September 1998 working paper, not the 2000 journal article. Harris's figures are projections from his 1994 model, whose second sample (second quarter of 1989, tickers EN to MOT, about a quarter of NYSE and AMEX firms) had 48 percent of quotation spreads at $1/8 on average and a third even for stocks priced over $40 (p. 155).

**Practice.** The proxies for tick-size effects, binding probability, spread leeway, depth near the mid-price and message intensity (order-to-trade ratio, cancellations and modifications), are the measures Lehalle and Laruelle use (pp. 77 and 84); see tick size. More messages per trade need not mean more liquidity.

**See also.** tick size; Harris's discretised spread; spread measures; best execution; smart order routing (fragmented books).

**Further reading.** [goldstein-kavajecz-1998] (working paper; journal version 2000) and [harris-1994]. Bacidore et al. (2003), Jones and Lipson (2001), Ahn et al. (1996), Harris (1996) and Aitken and Comerton-Forde (2005), cited by Lehalle and Laruelle.

#### Model 17.33 Harris's discretised spread: a latent continuous spread rounded to the tick, its fit by maximum likelihood, and the counterfactual spread under a smaller tick

**Statement.** Introduced by Harris (1994) to forecast the effect of decimalisation, as presented by Lehalle and Laruelle [alpha-laruelle, app. A, sec. A.6, pp. 266–268]: for stock $j$, trade $t$ on day $d$, the observed spread in ticks is $\Psi_{j,t} = (\text{Ask}_{j,t} - \text{Bid}_{j,t})/\text{tick}_{j,d}$, using only days with a single tick, and $\Psi_{j,t} = \Phi(\tilde{\Psi}_{j,t})$ for a latent continuous spread $\tilde{\Psi}_{j,t}$, where $\Phi$ gives the nearest integer for inputs above 1.5 and 1 otherwise. The latent spread has density $\tilde{f}_{s,\theta}$ with mean

$$E[\tilde{\Psi}_{j,t} \mid T_j, \sigma_{j,d}] = g(T_j, \sigma_{j,d}) \frac{P_{j,d}}{\text{tick}_{j,d}}, \qquad g = \alpha\, T_j^{\beta} \sigma_{j,d}^{\gamma},$$

with $T_j$ mean turnover, $\sigma_{j,d}$ volatility and $P_{j,d}$ the mean traded price; $g$ is the latent relative spread and $P/\text{tick}$ converts it to ticks (pp. 268–269). The discrete probabilities are $f_{s,\theta}(1) = \int_0^{1.5} \tilde{f}_{s,\theta}$ and $f_{s,\theta}(i) = \int_{i-0.5}^{i+0.5} \tilde{f}_{s,\theta}$ for $i \geq 2$, and $(\alpha, \beta, \gamma, \theta)$ maximise $\sum_{j,d,k} N(j,d,k) \ln f(k)$ over the counts $N(j,d,k)$ of $k$-tick spreads (pp. 269–270).

**Notation.** Counterfactual: for a new tick of old tick/$\delta$, the latent spread in *new* ticks is $\delta\tilde{\Psi}$, rounded by the same rule, so $f_{s,\theta,\delta}(1) = \int_0^{1.5/\delta} \tilde{f}_{s,\theta}$ and $f_{s,\theta,\delta}(k) = \int_{(k-0.5)/\delta}^{(k+0.5)/\delta} \tilde{f}_{s,\theta}$ for $k \geq 2$, and the expected spread in *old* ticks is

$$E(\Psi_{j,d} \mid T_j, \sigma_{j,d}, \delta) = \sum_k \frac{k}{\delta} f_{\hat{\alpha} T_j^{\hat{\beta}} \sigma_{j,d}^{\hat{\gamma}} P_{j,d}/\text{tick}_{j,d}, \hat{\theta}, \delta}(k)$$

(pp. 272–273): $k$ counts new ticks and $k/\delta$ converts them to old ones. An observed spread of $k$ old ticks is $k\delta$ new ticks.

**Conditions.** The latent distribution and its dependence on turnover and volatility are held fixed when the tick changes; that is the model's counterfactual assumption, which ignores changes in order exposure, depth and participants (tick size and market quality). Lehalle and Laruelle fit a gamma latent distribution to the 60 most liquid FTSE stocks in April 2012, with standardised $\beta' = -0.46$ and $\gamma' = 2.62$ (more liquidity, lower spread; more volatility, higher), and a pseudo-$R^2$ of 0.8 between observed and expected relative spreads (pp. 270–271).

**Practice.** The expected decrease is largest for stocks whose spread is near one tick (Figure A.9, p. 272). With a gamma latent spread of shape 4 and $\delta = 5$: a latent mean of 0.6 old ticks gives an expected observed spread of 1.010 old ticks now and 0.601 after the change (−41 percent); a mean of 1.2 gives 1.302 and 1.200 (−8 percent); means of 3 and 8 change by less than 0.01 old ticks.

**Pitfalls.** An observed $k$-tick spread does not become roughly $k/\delta$ old-tick units before re-rounding. An observed $k$-tick spread is $k\delta$ new ticks, and $k/\delta$ old ticks is the value of $k$ *new* ticks; nor does the model rescale observed spreads, since it re-rounds the latent spread (for a latent mean of 8 old ticks the expected change is nil, not $8/5 - 8$). Lehalle and Laruelle's caption to Figure A.10 gives the change "in old ticks unit" while the text calls it a relative decrease (p. 273); the figure's axis is a difference in old ticks. Their Figure A.8 fit is in-sample.

**See also.** tick size; tick size and market quality; spread determinants; spread–volatility regression.

**Further reading.** Harris (1994), cited by Lehalle and Laruelle (p. 74). Dayri and Rosenbaum's paper on large-tick assets (arXiv:1207.6325v2) provides futures evidence (spread determinants); its uncertainty-zone model is not summarised here.

## 17.10 Signing trades

#### Method 17.34 Trade-sign classification: the quote rule, the tick rule and Lee–Ready, block prices, signed volume, and what an inferred sign does and does not say

**Statement.** Trade data rarely say which side initiated a trade, so the sign $q_t = +1$ (buyer-initiated) or $-1$ is inferred. If all executions occur at the posted quotes, $q_t = +1$ at the ask and $-1$ at the bid; more generally $q_t = \text{Sign}(p_t - m_t)$ with $m_t$ the prevailing midpoint (the *quote rule*), which leaves midpoint trades unclassified. For those, the trade takes the sign of the most recent price change (the *tick rule*) [s6-hasbrouck, chap. 9, sec. 9.7.1, p. 89].

Harris describes the combination as the Lee and Ready algorithm: above the midpoint a buyer was aggressive, below it a seller; at the midpoint, an uptick or zero uptick means a buyer, otherwise a seller [alpha-harris, chap. 21, p. 423]. For block trades, a print closer to the bid than to the ask was probably seller-initiated, otherwise buyer-initiated, presuming the initiator paid a concession [alpha-harris, chap. 15, sec. 15.5, p. 332].

Signed volume $q_t V_t$, or a concave transform such as $q_t \sqrt{V_t}$, enters impact models; Hasbrouck advises against convex transforms such as $q_t V_t^2$, which amplify extreme volumes (sec. 9.7.2, p. 89).

**Conditions.** Methods timeless; the block statistic as reported in 2002; quote-lag evidence from the NYSE and AMEX in 1987–1988; accuracy evidence from NASDAQ in 2011. Signing needs the quotes and trades in the right order: in limit-order markets the interleaving is usually accurate, but with hybrid markets, different dissemination paths, coarse time stamps or manual steps it is "likely to be suspect" (Hasbrouck, p. 89). The rule at the posted quotes works only where the book can be rebuilt and executions are against visible orders, not hidden or reserve ones (p. 89).

*Lee and Ready's evidence*. On 150 NYSE firms in 1988, for nonopening trades where the last quote revision was more than 5 seconds old (90.9 percent of trades), the tick test called 92.1 percent of trades at the ask buys and 90.2 percent at the bid sells, and outperformed the reverse tick test [lee-ready-1991, sec. I, Table I, pp. 736–737]: agreement with the quote-based classification, not accuracy against known directions.

For isolated trades (the first after 11 a.m. each day with no other trade in a 2-minute window), 59.3 percent of NYSE and 72.4 percent of AMEX quote revisions around the trade were recorded *ahead* of it, with a sharp drop between 5 and 6 seconds before, hence the rule to compare a trade with the quote at least 5 seconds old; this cut the share of trades apparently inside the spread from 35.7 to 30.1 percent (sec. II, pp. 737–739, Fig. 2). The 5-second delay fits their 1988 data; "A different delay may be appropriate for other time periods", and in September and October 1987 most pretrade quotes came within 2 seconds of the trade (n. 10, p. 738).

The lag is a property of the reporting process in a given market and year, not a constant. *Accuracy against known directions*. Jurkatis evaluates the rules on NASDAQ's own book, May to July 2011, over 134 million transactions with nanosecond time stamps, where the direction of the standing order, and so the initiator, is known; he coarsens the time stamps to mimic slower data [jurkatis-2020, secs. 1 and 4, pp. 3 and 12].

With nanosecond stamps all algorithms classify about 98 percent of volume correctly; at one-second stamps the Lee–Ready, EMO and CLNV rules fall to about 90 percent (EMO the best) and his full-information algorithm, which matches each trade to the quote change it leaves in the book, keeps 95; in seconds with many trades the traditional rules drop to 85 percent or lower; and interpolating time stamps within the second, as Holden and Jacobsen propose, *lowers* the traditional rules' accuracy against using the last quote before the recorded time (pp. 3 and 12–14).

His review reports Lee–Ready accuracy of about 85 percent on NASDAQ ITCH and DTAQ data in 2011 (Chakrabarty et al. 2015), 79 percent on Euronext second-stamped data in 2007–2008 and 46 percent on LSE microsecond data in 2017 (Panayides et al. 2019), against 75–93 percent in older studies, and lag recommendations from zero to two seconds, most based on indirect evidence (pp. 1–2 and n. 4). The EMO rule classifies at the ask or

bid and uses the tick test for everything else; CLNV calls trades within 30 percent of the spread of the ask (bid) buys (sells) and uses the tick test in the middle 40 percent (sec. 2.1, pp. 4–5): the three disagree on trades inside the spread. The evidence is for one exchange's book; consolidated-tape data, with more venues and reporting paths, may differ.

**Practice.** Midpoint trades after an uptick, or after a zero tick whose last non-zero change was up, are buys, and after a downtick or a zero downtick sells, for whole-tick midpoints (10.01 in 10.00–10.02) and half-tick ones (10.005 in 10.00–10.01); a midpoint trade with no earlier price change is left unclassified; trades just above or below a half-tick midpoint (10.006, 10.004) follow the quote rule whatever the tick; a trade at the bid is a sell, and at the ask a buy, whatever the previous tick. An implementation must compare prices exactly: in binary floating point $(10.00 + 10.01)/2$ is 10.004999…, below 10.005, so the half-tick midpoint trade after a downtick is wrongly called a buy.

Sign persistence in event time is a stylised fact (stylised facts); Hasbrouck notes that signed size often adds little explanatory power beyond the sign itself (sec. 9.7.2, p. 90). Summed over an interval, signed volume is the trade imbalance of order-flow imbalance, not the event-based OFI.

**Pitfalls.** The sign identifies which side demanded immediacy at that moment, not who was better informed or more eager: a buyer who pegs the bid can accumulate aggressively while the algorithm calls every trade passive (Harris's sidebar, p. 423). Measured as transaction costs, the algorithm overstates the costs of traders who do not only take liquidity, and cannot see that several trades filled one order, so it understates that order's impact cost (p. 423).

Harris reports that "analysts have determined that sellers initiate approximately 80 percent of all large block trades in the U.S. stock markets", but gives no sample or period. He reads the asymmetry as consistent with three of his four block-trading problems: latent demand (buyers can buy only from holders when short selling is hard), price discrimination (sellers can credibly reveal their full size) and asymmetric information (large buyers are presumed informed) (pp. 332–333).

His classification rule is the block price against the prevailing quotes: closer to the bid means seller-initiated (sec. 15.5, p. 332). *Source search*: no citation stands at the statement. The chapter's bibliography lists the candidate studies: Kraus and Stoll (1972), Holthausen, Leftwich and Mayers (1987, 1990), Chan and Lakonishok (1993, 1995), Madhavan and Cheng (1997) and Saar (2001) (Bibliography, "Chapter 15: Block Traders", pp. 607–608). All are in subscription journals.

Saar's open working-paper version (October 1999) is a theory of the price-impact asymmetry and reports no buyer–seller split [saar-1999-wp, abstract and secs. 1–2]. Outside the United States the split need not be lopsided: Alzahrani et al. count 2.3 million block buys and 1.9 million block sales on the Saudi market in 2005–2008 [alzahrani-2010-wp, abstract]. That is another market and definition, not a test of Harris's figure, which remains unsourced. The tick rule is also used at the midpoint. These sources do not discuss false persistence in signs from misclassification and order splitting.

**See also.** spread measures (effective spreads need signs); information-driven bars (the tick rule in bar construction); order-flow impact regression; propagator model; block trade.

**Further reading.** [lee-ready-1991] (sections I–II); Jurkatis's Bank of England working paper for accuracy against known directions (Conditions). Of Ellis, Michaely and O'Hara (2000), only the abstract is used; its sample dates and algorithm are not covered. On a proprietary Nasdaq data set that identifies trade direction, "the quote rule, the tick rule, and the Lee and Ready (1991) rule correctly classify 76.4%, 77.66%, and 81.05% of the trades, respectively". All rules have "only a very limited success in classifying trades executed inside the quotes", which biases accuracy for large trades, high-volume periods and ECN trades. The authors propose a new algorithm for Nasdaq trades [ellis-michaely-ohara-2000, abstract].

## 17.11 Price impact

**Definition 17.35 Price impact: unperturbed, observed and trade prices; reaction, prediction, transient and permanent impact**

**Statement.** *Price impact* is the effect of trading on prices: trades cause price moves that would not otherwise have happened [wide-webster-price-impact, chap. 1, sec. 1.1.2, p. 5]. Webster separates three prices for a trader whose trades $Q$ are part of all trades $L$ [wide-webster-price-impact, chap. 2, Definition 2.2.6, pp. 53–54]:

- the *unperturbed price* $S_t$, the price without the trades $L$ and their impact;
- the *perturbed* or observed price $P_t = S_t + I_t$, where $I_t$ is the impact of $L$;
- the *trade price* $\widetilde{P}_{t-} = S_{t-} + I_{t-} + s_{t-}$, where $s$ is the instantaneous transaction (spread) cost of the trader's own fill, and the subscript $t-$ marks that a fill at $t$ executes against the state just before $t$.

Bouchaud, Bonart, Donier and Gould separate three impacts of a market order executed at $t$, at horizon $\ell$, with $m$ the mid-price and $\mathcal{F}_t$ everything that may have prompted the order [catalogue-0528, chap. 11, sec. 11.2, eqs. (11.1)–(11.4), pp. 210–211]:

$$I^{\text{react}} = \mathbb{E}[\,m_{t+\ell} \mid \text{exec}_t, \mathcal{F}_t\,] - \mathbb{E}[\,m_{t+\ell} \mid \text{no exec}_t, \mathcal{F}_t\,], \qquad I^{\text{obs}} = \mathbb{E}[\,m_{t+\ell} \mid \text{exec}_t\,] - m_t,$$

$$I^{\text{pred}} = \mathbb{E}[\,m_{t+\ell} \mid \text{no exec}_t, \mathcal{F}_t\,] - m_t, \qquad I^{\text{obs}} = I^{\text{react}} + I^{\text{pred}}.$$

Over time, impact has a *transient* part that decays and a *permanent* part. The permanent part combines reaction and prediction: it is higher for traders with predictive power than for noise trades [catalogue-0528, chap. 12, sec. 12.4, p. 240].

**Notation.** $I^{\text{react}}, I^{\text{obs}}, I^{\text{pred}}$ are the reaction, observed and prediction impacts. The source writes $I^{\text{react}}_{t+\ell}(\text{exec}_t \mid \mathcal{F}_t)$ and so on; the time and conditioning arguments are dropped here. The identity $I^{\text{obs}} = I^{\text{react}} + I^{\text{pred}}$ needs every expectation conditioned on $\mathcal{F}_t$; as printed, (11.2) conditions on execution alone, and the authors state that the conditioning on $\mathcal{F}_t$ is always implicit [catalogue-0528, same sec., p. 210] (Interpretation: without it the two sides differ by $\mathbb{E}[\,m_{t+\ell} \mid \text{exec}_t\,] - \mathbb{E}[\,m_{t+\ell} \mid \text{exec}_t, \mathcal{F}_t\,]$). Webster's $P$ is a price, not the $P$ of other entries.

**Conditions.** Timeless (concepts); the empirical statements as of 2018. Two readings of impact frame the literature [catalogue-0528, chap. 11, sec. 11.1, pp. 208–210]. In the efficient-market reading, traders forecast prices and trade accordingly, so trades and later price changes correlate even if trades move nothing; in Hasbrouck's words, quoted there, orders forecast prices. In the order-flow reading, trading itself moves prices, and impact exists even in a model with no information at all, such as the Santa Fe model. Reality probably lies between. If the mid-price were a martingale, observed and reaction impact would coincide; they do not, because the information behind the order raises the expected price even without the trade (p. 211). Studies with random or cash-driven trades find that, over short horizons, mechanical impact is about the same for random and allegedly informed trades; prediction shows only at longer horizons (p. 212).

After a metaorder, the impact path splits into a transient part that vanishes and a permanent part $I_\infty(Q)$, which collects both the prediction behind the order and any permanent reaction to all trades [catalogue-0528, chap. 12, sec. 12.2, eq. (12.5), p. 233]. Bacry, Iuga, Lasnier and Lehalle describe the same divide for permanent impact: a mechanical view, common among econophysicists, in which buying and selling pressure moves prices, and an informational view, favoured by economists after Kyle, in which trades reveal information.

They hold the two as a dual picture rather than choosing one [wide-bacry-life-cycle, arXiv preprint of 9 Dec. 2014, sec. 7.1, pp. 19–20]. Within the mechanical view they report two positions, a "fair pricing" view in which permanent impact is about two-thirds of the temporary impact, and Bouchaud's view that there is no permanent impact, only the long memory of the order flow; both are cited there.

**Practice.** Reaction impact can be estimated by trading at random, by randomising execution of signals, or by conditioning on the rest of the market's order-sign imbalance; prediction impact is very hard to estimate [catalogue-0528, chap. 11, sec. 11.2, pp. 211–212]. The counterfactual

cannot be observed: a trader never sees both outcomes of one trade, which Webster calls the causality challenge [wide-webster-price-impact, chap. 1, sec. 1.1.3, pp. 6–7]. Webster traces impact modelling through four communities: economists' stylised models (Kyle), mathematical finance's control problems (Almgren–Chriss, Obizhaeva–Wang, Gatheral), econophysics' empirical response functions and propagators, and books from quantitative trading desks [wide-webster-price-impact, chap. 1, sec. 1.1.5, pp. 9–15]. The model families in use differ in what they describe:

- *Almgren–Chriss*: the price follows an arithmetic random walk shifted by a permanent impact $g(v)$ of the trading rate $v$; each sale executes at the previous price less a temporary impact $h(v)$ that does not enter the next market price [alpha-almgren, author preprint of Dec. 2000, secs. 1.2–1.3, eqs. (1)–(2), pp. 8–9]. In the linear case $g(v) = \gamma v$ and $h(n_k/\tau) = \epsilon\,\mathrm{sgn}(n_k) + (\eta/\tau)n_k$, with $\gamma$ in (\$/share)/share, $\epsilon$ in \$/share (half the spread plus fees) and $\eta$ in (\$/share)/(share/time) [alpha-almgren, author preprint of Dec. 2000, sec. 1.5, eqs. (6)–(7), pp. 10–11]. The schedule minimises expected cost plus $\lambda$ times its variance [alpha-almgren, author preprint of Dec. 2000, sec. 2, pp. 12–13].
- *Linear impact*: cost or impact proportional to the trading rate (Almgren–Chriss) or to signed size (Kyle).
- *Propagator*: each signed trade moves the price by a decaying kernel (propagator model).
- *Square-root*: metaorder impact concave in size relative to volume (square-root impact law).

Impact during execution is measured from the arrival price, signed by the order's side: in bid–ask spreads by Lehalle and Laruelle [alpha-laruelle, chap. 3, sec. 3.2.1, p. 204], relative to volatility by Almgren et al. (direct estimation); its path is in intraday impact profile.

**Pitfalls.** "Temporary impact" has three meanings. In Almgren and Chriss it is $h(v)$, which lowers the execution price of the trader's own sale in that period but does not enter the next market price [alpha-almgren, author preprint of Dec. 2000, sec. 1.3, eq. (2), p. 9]; their prose describes a short-lived imbalance after which liquidity returns each period (p. 8). Webster accordingly reads it as the instantaneous transaction cost [wide-webster-price-impact, chap. 2, n. 3, p. 53].

Bacry et al. use "temporary market impact" for the impact at the end of a metaorder's execution, distinct from the transient curve during execution and the decay after it [wide-bacry-life-cycle, arXiv preprint of 9 Dec. 2014, sec. 2.2, p. 3]. Neither is a transient displacement that decays over many later trades. Defining temporary impact as the price concession needed to obtain immediacy, expected to decay as liquidity replenishes, mixes the two; say which is meant (Interpretation). Calling a long-run move permanent impact needs care.

Long-horizon impact is hard to measure because price variance grows with the horizon. Correlated metaorders, the same decision continued over days, can look like permanent impact. Separating mechanical from informational impact is difficult [catalogue-0528, chap. 12, sec. 12.4, p. 240]. Bacry et al. remove moves of each stock's reference index (see post-execution impact profile); sector or factor moves are not removed, and Bouchaud et al. suggest conditioning on the rest of the market's order-sign imbalance (p. 212). Removing the information effect before calling the residual permanent impact presumes the effect can be measured; Bacry et al., without the traders' signals, remove the market component and read it as aggregated information (p. 20). Theoretical, reduced-form and practitioner views of impact simplify Webster's four communities.

Orders of the same size and participation can have different impact because of their timing: following a trend, reverting to a mean, or following an index rebalancing [alpha-laruelle, chap. 3, sec. 3.2, p. 204]. Separating execution mechanics from the signal behind the trade is the split between reaction and prediction impact above. The mechanism (temporary impact decays when other traders disagree with the move; permanent impact remains when the order flow reveals information) is the information-based reading of the distinction above, with the caveats on the terms that this entry gives; Harris's resiliency, how quickly prices revert after imbalances initiated by uninformed traders, ties the decay to value traders [alpha-harris, chap. "Liquidity", sec. 19.2, p. 400; sec. 19.3.4, p. 403] (liquidity and market quality).

**See also.** propagator model; absence of price manipulation; price formation process (temporary and permanent impact by consensus, Lehalle and Laruelle); spread component identification (transitory and permanent components of the spread); Kyle single-auction model (permanent impact from information); Grossman–Miller model (a temporary concession for immediacy).

### Model 17.36 Propagator model: transient impact of past trades

**Statement.** In market-order time, with $\varepsilon_n = \pm 1$ the sign of the $n$th market order and $\xi_n$ price changes not induced by trades, the mid-price is [catalogue-0528, chap. 13, sec. 13.2, eq. (13.7), p. 252]

$$m_t = m_{t_0} + \sum_{t_0 \le n < t} G(t-n)\,\varepsilon_n + \sum_{t_0 \le n < t} \xi_n .$$

The *propagator* $G(\ell)$ is the impact at lag $\ell$ of one trade, expected to decrease with $\ell$; $G_\infty = \lim_{\ell\to\infty} G(\ell)$ is the permanent reaction impact. Returns then split into immediate impact, the decay of past impacts and noise, $r_t = G_1\varepsilon_t + \sum_{n<t} K(t-n)\varepsilon_n + \xi_t$ with $K(\ell) = G(\ell+1) - G(\ell)$ (same sec., eq. (13.8)). Equation (13.7) defines the family of linear transient impact models (TIM).

**Notation.** $m_t$ mid-price just before event $t$; $r_t = m_{t+1} - m_t$; $G_1 = G(1)$; $C(\ell) = \mathbb{E}[\varepsilon_t\varepsilon_{t+\ell}]$ the sign autocorrelation; $R(\ell) = \mathbb{E}[\varepsilon_t(m_{t+\ell} - m_t)]$ the response function (observed impact).

**Conditions.** Timeless (model); the empirical motivation as of 2018. Why impact must be transient: if each trade had constant permanent impact $G_1$, then $\mathbb{E}[r_t r_{t+\ell}] = G_1^2 C(\ell)$, and the long memory of order signs would make returns strongly autocorrelated. That contradicts the nearly flat signature plots of real markets, so much of each trade's impact must relax [catalogue-0528, chap. 13, secs. 13.1–13.2, pp. 249–251]. The sign memory comes mainly from order splitting, investors working large orders in pieces because available liquidity is small; on the time scales of the power-law decay of $C(\ell)$, herding is not a relevant factor [catalogue-0528, chap. 10, secs. 10.4–10.5, p. 198].

For a power-law propagator $G(\ell) \approx \Gamma_\infty \ell^{-\beta}$ and signs with $C(\ell) \approx c_\infty \ell^{-\gamma}$, $0 < \gamma < 1$, prices are diffusive at long lags only at the fine balance $\beta = \beta_c = (1-\gamma)/2$; permanent impact ($\beta = 0$) gives trending prices, faster decay ($\beta > \beta_c$) sub-diffusive ones [catalogue-0528, chap. 13, secs. 13.2.2–13.2.3, eq. (13.17), pp. 256–257]. The simple model assumes impact independent of trade volume, time and market conditions, which the authors call gross oversimplifications, relaxed in their chap. 14 [catalogue-0528, same chap., p. 252].

**Practice.** The observed response and the bare propagator differ when signs are correlated: $R(\ell) = G(\ell) + \sum_{0<n<\ell} G(\ell-n)C(n) + \sum_{n>0}[G(\ell+n) - G(n)]C(n)$, which reduces to $G(\ell)$ only for uncorrelated signs. Estimating $G$ from $R$ and $C$ through this relation is sensitive to finite-size effects [catalogue-0528, same chap., sec. 13.2.1, eq. (13.10), p. 253]. Webster calls the Obizhaeva–Wang model a simple form of the propagator model [wide-webster-price-impact, chap. 1, p. 10]; in its generalized form impact follows $dI_t = -\beta_t I_t\,dt + \lambda_t\,dQ_t$ [wide-webster-price-impact, chap. 2, Definition 2.8.1, p. 97]. Interpretation: that is an exponentially decaying kernel in calendar time, $G \propto e^{-\beta\ell}$ for constant $\beta$.

Kernel shapes in use: *non-parametric*, from the lagged sign–return correlation $S(\ell) = R(\ell+1) - R(\ell) = \sum_{n\ge 0} C(|n-\ell|)K(n)$ inverted as $K = \mathbf{C}^{-1}S$ up to a maximum lag [catalogue-0528, chap. 13, eqs. (13.13)–(13.15), p. 254]; *power law*, which the estimates show at long lags, $G(\ell) \approx \Gamma_\infty \ell^{-\beta}$ with $\beta < 1$, decaying to zero but with divergent sum ("long-range resilience", same p. 254); and *exponential*, as in the Obizhaeva–Wang model above.

The original paper by Bouchaud, Gefen, Potters and Wyart writes the price as a sum over past trades with a "bare" propagator $G_0$ weighting each sign by the logarithm of the trade volume, $p_n = \sum_{n'<n} G_0(n-n')\,\varepsilon_{n'}\ln V_{n'} + \sum_{n'<n}\eta_{n'}$, with $\eta$ independent of the signs and uncorrelated in time [wide-bouchaud-propagator, arXiv cond-mat/0307332v2, sec. 3.1, eq. (12), p. 14]. It argues that with power-law sign correlations a constant $G_0$ would make the response grow like $\ell^{1-\gamma}$, so $G_0$ must decay [wide-bouchaud-propagator, same, sec. 3.1, p. 16]; with a power-law ansatz the Hurst exponent satisfies $2H = 2 - 2\beta - \gamma$ [wide-bouchaud-propagator, same, sec. 3.2, eq. (22), pp. 16–17].

**Pitfalls.** In the simple model every trade shares one propagator $G$; trade-dependent decay profiles need the generalised models of chap. 14, which generalise by event type, not by agent. Reading the measured response $R(\ell)$ as the impact of a single trade ignores the sign autocorrelation. Timing and accumulation conventions matter: (a) $m_k$ is the mid-price just *before* trade $k$ when summing over $k' < k$; (b) the noise is the cumulative $\sum \eta$, which makes the price diffusive apart from trading, not a single $\eta_k$, which would make it stationary (Interpretation); (c) the 2004 paper weights trades by $\ln V$, which the book's simple model omits; (d) $R$ and $C$ are measured, and the kernel is estimated from them.

Interpretation: Hasbrouck's generalized Roll model, in spread component identification, is the constant-propagator case applied to the efficient price with independent signs: the trade moves it permanently by $\lambda$.

**See also.** price impact; absence of price manipulation (restrictions on $G$ and on the Obizhaeva–Wang parameters); returns by time scale; fundamental and transitory volatility (the variance ratio and transient components).

**Further reading.** The original paper [wide-bouchaud-propagator], arXiv version.

#### Result 17.37 Absence of price manipulation in impact models

**Statement.** In an impact model, *price manipulation* is a round trip, a strategy that starts and ends with no position, whose expected cost is negative. Webster quotes this definition from Gatheral (2010)[wide-webster-price-impact, chap. 1, n. 3, p. 7]. A usable model must exclude it. Bouchaud, Bonart, Donier and Gould write the expected cost of a closed trajectory with trading rate $j$ and price $x^*$ as $C = \int_0^T j(u)\, x^*(u)\, du$ with $\int_0^T j(u)\, du = 0$. They show that in their latent-liquidity model it is a non-negative quadratic form, so $C \geq 0$ for every schedule, generalising a result for the linear propagator model [catalogue-0528, chap. 19, sec. 19.5, pp. 361–362]. In Webster's generalized Obizhaeva–Wang model, buying $\Delta$ in one jump at $t$ and selling it at $t' > t$, without a spread, earns

$$(S_{t'} - S_t)\Delta - \tfrac{1}{2}\lambda_t(2\beta_t + \gamma'_t)(t' - t)\Delta^2 + o(t' - t),$$

so if $2\beta + \gamma' < 0$, large, quick round trips earn arbitrary expected profits from the trader's own impact. The no-manipulation condition is $2\beta + \gamma' > 0$, after Fruth, Schöneborn and Urusov (2013)[wide-webster-price-impact, chap. 2, sec. 2.7.1, Example 2.7.1, p. 94].

**Notation.** $S$ unperturbed price; $\Delta$ shares; impact $I$ with $dI = -\beta I\, dt + \lambda\, dQ$, decay $\beta$ per unit time, push $\lambda = e^{\gamma}$ in price per share, $\gamma'$ its rate of change. As Webster's profit expression implies, the trader's buy executes at $S_t + \frac{1}{2}\lambda_t\Delta$ on average, the sell at $S_{t'} + e^{-\int_t^{t'} \beta_s\, ds}\lambda_t\Delta - \frac{1}{2}\lambda_{t'}\Delta$.

**Conditions.** Webster's example assumes no spread, no prior impact, jump trades and exact unwinding; the expected profit from $S$ is zero only if $S$ is a martingale. The condition $2\beta+\gamma' > 0$ rules out only the paradoxes that yield arbitrary profits. It does not rule out transaction-triggered manipulation, such as selling first to depress the price for later buying [wide-webster-price-impact, same sec., pp. 94–95]. In a square-root model it bounds forecast volume growth, $v'_t/v_t < 4\beta_t$: trading activity cannot double in less than a quarter of the impact half-life, so rising volume curves can create manipulation opportunities while falling ones cannot [wide-webster-price-impact, same chap., sec. 2.7.3, Example 2.7.2, p. 96].

**Practice.** Using an impact model that admits profitable closed trajectories in execution algorithms leads to instabilities or counter-intuitive trades [catalogue-0528, chap. 19, p. 362], and Webster asks that a model rule out manipulation paradoxes [wide-webster-price-impact, chap. 1, sec. 1.1.4, p. 8]. Interpretation, the accounting behind any check: a trader who buys and sells $Q$ shares at average execution prices $\bar{P}_B$ and $\bar{P}_S$, impact included, and pays explicit costs $F$ in total, ends with $Q(\bar{P}_S - \bar{P}_B) - F$ and no position, hence no mark-to-market term. With a per-share cost $c$ on each side, $F = 2Qc$; with $c$ quoted per round-trip share, $F = Qc$. Spread counted in $\bar{P}$ must not be counted again in $c$.

**Pitfalls.** A realised profitable round trip is not price manipulation: the profit may come from $S_{t'} - S_t$, from information, or from luck. Manipulation concerns expected profit under a model, with no information (Interpretation). Impact that treats buys and sells alike per unit is not by itself a no-manipulation guarantee. Webster's example is linear and symmetric trade by trade, yet admits manipulation when liquidity changes faster than impact decays (Interpretation). That a zero-net sequence cannot mechanically create a higher average sale price holds in Harris's instantaneous, permanent-impact example but is not a general result. Stating the identity as a test needs $c$ defined per side or per round trip. Strategic bluffing, which fools other traders through their inferences, is a different mechanism from manipulation through one's own impact in a model, where nobody is fooled (bluffing and manipulation).

**See also.** propagator model; price impact; bluffing and manipulation (Harris's asymmetric-impact bluff).

**Further reading.** Gatheral, "No-Dynamic-Arbitrage and Market Impact" [catalogue-0501].

## 17.12 Metaorder impact

#### Evidence 17.38 Square-root impact law: metaorder impact concave in size relative to volume

**Statement.** The peak impact of a metaorder of volume $Q$ executed over time $T$ is well described by [catalogue-0528, chap. 12, sec. 12.3.1, eq. (12.6), p. 234]

$$I^{\text{peak}}(Q, T) \cong Y\sigma_T \left(\frac{Q}{V_T}\right)^{\delta}, \qquad Q \ll V_T,$$

with $Y$ of order one ($Y \cong 0.5$ for US stocks), $\delta$ between 0.4 and 0.7, $\sigma_T$ the volatility and $V_T$ the volume traded over the execution horizon. Both sides are in percent of price. In the Capital Fund Management sample of about 500,000 futures metaorders, June 2007 to December 2010, $\delta \simeq 0.5$ for small-tick and $\simeq 0.6$ for large-tick contracts, for $Q/V_T$ from about $10^{-5}$ to a few percent [catalogue-0528, same sec., Figure 12.2, p. 235]. Bacry, Iuga, Lasnier and Lehalle confirm a square-root temporary impact in daily participation on about 400,000 metaorders traded on European markets by one broker in 2010 [wide-bacry-life-cycle, arXiv preprint of 9 Dec. 2014, abstract, p. 1].

**Conditions.** The law is reported across equities, futures, FX, options and Bitcoin, before and after 2005, for small- and large-tick instruments and for different strategies and execution styles [catalogue-0528, same sec., p. 234]. It concerns *peak* impact at the end of execution, not permanent impact; the price reverts after execution (Figure 12.1, p. 234). Impact is not additive: the second half of a metaorder moves the price less than the first, and $Q$ enters relative to the volume traded during execution, not to market capitalisation [catalogue-0528, same chap., sec. 12.3.2, pp. 235–236]. Linear models such as Kyle's predict the opposite [catalogue-0528, same chap., pp. 233 and 241].

Almgren et al., on US orders of 2001–2003, reject an exponent of one-half for temporary impact against size relative to volume during execution at the 95 percent level, estimating $0.600 \pm 0.038$ (direct estimation); Bacry et al. find that adding a duration factor $1/T^{\gamma}$ with $\gamma \simeq 0.25$ reinforces the square-root shape in daily participation [wide-bacry-life-cycle, arXiv preprint of 9 Dec. 2014, abstract, p. 1]. The exponents differ with the regressor (daily participation, trading rate, or size relative to volume during execution) and with the fitting method.

**Practice.** If the impact of the first fraction $\phi$ of a metaorder follows the same law, the execution shortfall per unit volume is two-thirds of the peak impact under square-root impact, against one-half under linear impact [catalogue-0528, same chap., sec. 12.5, eqs. (12.10)–(12.11), p. 241]. Bacry et al. debias post-execution prices with a square-root model of daily participation and report that powers from 0.4 to 0.7 give similar results [wide-bacry-life-cycle, arXiv preprint, sec. 7.2, n. 3, p. 21].

**Pitfalls.** Describing impact only as concave in order size relative to market volume omits the volatility scale $\sigma_T$, the small-participation condition $Q \ll V_T$ and the exponent range; "square-root" names exponents from 0.4 to 0.7. The fitted exponent depends on the estimation method: Bacry et

al. find smaller exponents with robust direct regressions than with log-log regressions (sec. 8, p. 22). The primary study of the futures sample, Tóth et al. (2011), is cited through Bouchaud et al. (wide-toth-square-root).

**See also.** price impact; propagator model; post-execution impact profile; Kyle single-auction model (linear impact, the contrast).

**Further reading.** Tóth et al., the futures study [wide-toth-square-root].

#### Method 17.39 Post-execution impact profile: debiasing later metaorders and removing index moves

**Statement.** To estimate how much of a metaorder's impact remains days after execution, Bacry, Iuga, Lasnier and Lehalle proceed as follows [wide-bacry-life-cycle, arXiv preprint of 9 Dec. 2014, sec. 7.2, pp. 20–22]:

1. For each metaorder executed on day $D$, take daily closes over the 41 days $d = D - 20, \dots, D + 20$.
2. Debias for later flow: from the price moves after day $D$, subtract the temporary impact of the metaorders executed on the following days, using a square-root model of their aggregated daily participation fitted to the data, following Waelbroeck and Gomes (2013), cited there.
3. Split each daily log return into a systematic part and a residual,

$$\log P_d - \log P_{d-1} = \beta(\omega)\,(\log \mathcal{I}_d - \log \mathcal{I}_{d-1}) + \Delta W_d,$$

   with $P_d$ the stock's close, $\mathcal{I}_d$ its reference index and $\beta(\omega)$ the stock's beta over $D - 20$ to $D + 20$ (eq. (21), p. 21).
4. Cumulate from the close of $D - 1$: the idiosyncratic profile $W_d - W_{D-1}$ and the systematic profile $\beta(\log \mathcal{I}_d - \log \mathcal{I}_{D-1})$. Their sum, the post-execution profile, is an estimate of permanent impact (p. 22).

**Notation.** $\mathcal{I}_d$ is an index level, not impact; $P_d$ a close price, not Webster's perturbed price. Impact itself is defined through $\epsilon\,\Delta P_t = \eta_t + \epsilon\,\Delta W_t$ for a metaorder of sign $\epsilon$, with $\Delta W$ the move that would have occurred without it [wide-bacry-life-cycle, arXiv preprint, sec. 2.2, eq. (1), p. 3]; estimation assumes $\mathbb{E}[\epsilon\,\Delta W] = 0$, not $\mathbb{E}[\Delta W] = 0$ (p. 4).

**Conditions.** As of the 2010 European sample. Lehalle and Laruelle restate the procedure, citing the 2015 published paper: each stock's reference index is its home index (CAC 40, FTSE 100, DAX), and $\beta$ is estimated by least squares over $D - 20$ to $D + 20$ [alpha-laruelle, chap. 3, sec. 3.2.2, eq. (3.2), pp. 210 and 213–215].

Interpretation: this matches the preprint's window, so the in-sample estimation of $\beta$ remains an open methodological question rather than a preprint artefact; a restatement by one of the authors is not independent confirmation. The broker's data do not say why each metaorder was placed. The authors therefore assume the clients' aggregate portfolio is close to the market portfolio, with beta one, and treat the systematic component as the information the issuers used in aggregate (p. 20). The participation of metaorders is autocorrelated beyond 20 days, at about 0.3 to 0.4 [wide-bacry-life-cycle, same sec., Figure 10, p. 21], so later executions in the same direction would otherwise be attributed to the first.

**Practice.** Without debiasing, prices keep trending in the direction of the execution-day jump for 20 days, with no reversal. After debiasing, the idiosyncratic profile returns to its initial level within 20 days, and the impact remaining after 20 days is entirely systematic; the authors conclude that no permanent effect is due to the metaorder itself (sec. 7.3, p. 22). They report this as compatible with two studies cited there. Waelbroeck and Gomes (2013) found permanent impact for informed trades but none for cash-driven ones over 60 days. Brokmann et al. (2014) found that, once the issuer's alpha was removed, impact vanished after about a dozen days (p. 20).

**Pitfalls.** Bacry et al. could not classify trades as information- or liquidity-driven, lacking the triggers; that split is Waelbroeck and Gomes's design, reported through Bacry et al. Their method removes only the reference index, not market and sector moves from a factor model. Interpretation:

- The beta is estimated over a window that includes the execution and the days after it, so it is in-sample and can absorb part of the event.
- Removing the systematic part removes any impact the aggregate flow had on the index itself, which the authors read as information.
- The debiasing inherits the error of the square-root model for later flow.

Not charging the execution for a market move is one reading of the decomposition; the authors' own reading is that the systematic part carries the information behind the trades.

**See also.** price impact (permanent against finite-horizon impact; correlated metaorders); square-root impact law (the debiasing model); propagator model (long memory of order flow).

**Further reading.** Waelbroeck and Gomes (2013) and Brokmann et al. (2014), cited by Bacry et al.

#### Evidence 17.40 Intraday metaorder impact: its path during and after execution, and its dependence on participation and duration

**Statement.** Measured as the price move from the arrival price, signed by the order's side and expressed in bid–ask spreads, the average impact of Crédit Agricole Cheuvreux executions on CAC 40 stocks in 2010 rises concavely over the execution: a slice executed early moves the price more than one executed late. After the execution it reverts, to a level below the one reached at the end [alpha-laruelle, chap. 3, sec. 3.2.1, Figure 3.3, pp. 204–206]. Lehalle and Laruelle summarise how impact depends on the participation rate and the duration by

$$\text{Market impact} \propto \text{Duration}^{\alpha} \times \text{Participation}^{\gamma},$$

fitted to averages over buckets of duration (in 5-minute intervals) and participation rate. The fitted surface is concave in each variable separately, with $\gamma \approx 0.5$ [alpha-laruelle, same sec., eq. (3.1), Figure 3.6, pp. 207–209]. Impact rises with both participation and duration, even separately: executions of 25 to 50 minutes lie above those shorter than 25 minutes at every participation rate (Figure 3.5, pp. 206–207).

**Conditions.** The fit is descriptive and the authors do not report $\alpha$ or uncertainty for either exponent. "Participation" there is the period liquidity ratio, the order's share of market volume during its execution. Bacry, Iuga, Lasnier and Lehalle measure the same dependence on another broker sample (European metaorders, 2010) by direct regression of the signed price move at the end of execution [wide-bacry-life-cycle, arXiv preprint of 9 Dec. 2014, sec. 3.2, Table 2, p. 8]:

- on daily participation $R$ with duration, the exponents are 0.54–0.59 for $R$ and −0.23 to −0.35 for duration;
- on the trading rate $r$ with duration, they are 0.37–0.56 for $r$ and +0.15 to +0.24 for duration.

The sign of the duration effect thus depends on what is held fixed. At fixed daily participation a longer execution trades more slowly and costs less; at a fixed participation rate a longer execution trades more and costs more. Interpretation: with impact $\propto R^a T^b$ and $R = rT$, impact $\propto r^a T^{a+b}$, and $a + b \approx 0.2$–$0.36$ from the first rows, consistent with the second and with Lehalle and Laruelle's $\alpha > 0$.

**Practice.** Lehalle and Laruelle list stock characteristics (liquidity, volatility, spread) and order characteristics (aggressiveness, quantity, expected duration) as drivers, and timing as a further one. The same quantity has different impact depending on whether the order follows a trend, reverts to a mean, or follows an index rebalancing, and on whether high-frequency traders can detect it [alpha-laruelle, chap. 3, sec. 3.2, pp. 203–204 and 206]. After one day they take the temporary impact to have vanished, so later moves count as permanent [alpha-laruelle, same sec., p. 209].

**Pitfalls.** Tick size, typical depth, order side, auctions and venue access (lit and dark liquidity, queue priority, routing) are not drivers in the source. On the spread the sources disagree. Bacry et al. find a spread term with exponent 0.28 in a log-log fit (Table 2, (R.3)), while Almgren et al. find no consistent dependence of permanent impact on the spread (direct estimation). In the terms of price impact the decayed part is the decay of the

transient displacement, not temporary impact, which Bacry et al. reserve for the value at the end of execution. Only one fitted participation exponent, $\gamma \approx 0.5$, is reported, so this fit does not establish concavity across datasets.

**See also.** square-root impact law; direct estimation of impact (a different design); post-execution impact profile (the daily horizon); price impact.

**Further reading.** Moro et al. (2009), cited by Lehalle and Laruelle for the concave path.

#### Method 17.41 Direct estimation of permanent and temporary impact from institutional orders (Almgren, Thum, Hauptmann and Li)

**Statement.** From completed orders, Almgren, Thum, Hauptmann and Li estimate two impact functions [wide-almgren-direct-impact, authors' preprint of 10 May 2005, secs. 2.2–4.3, pp. 8–21]:

1. For each order measure three prices (p. 9): the pre-trade midpoint $S_0$, from the last quote before the first transaction; the realised average price $\bar{S}$; and the post-trade midpoint $S_{\text{post}}$, from the first quote at least half an hour after the last execution, carried to the next morning if that falls after the close.
2. Define the permanent impact $I = (S_{\text{post}} - S_0)/S_0$ and the realised impact $J = (\bar{S} - S_0)/S_0$, both signed so that a cost has the sign of the order (p. 10).
3. Measure time in volume time, the fraction of an average day's volume executed so far (p. 10). The order of $X$ shares is executed at a constant rate over $T$, with $V$ the 10-day average daily volume, $\sigma$ an intraday estimate of daily volatility, and $\Theta$ the shares outstanding (pp. 11–13 and 19).
4. Fit, with heteroskedastic weights from the model's error terms (eqs. (7)–(8), p. 19)

$$\frac{I}{\sigma} = \gamma\, T\, \operatorname{sgn}(X) \left|\frac{X}{VT}\right|^{\alpha} \left(\frac{\Theta}{V}\right)^{\delta} + \text{noise}, \qquad \frac{1}{\sigma}\left(J - \frac{I}{2}\right) = \eta \operatorname{sgn}(X) \left|\frac{X}{VT}\right|^{\beta} + \text{noise}.$$

The term $I/2$ removes the effect of the order's own permanent impact on its later fills, so $J - I/2$ is the temporary impact (eq. (2), p. 15).

**Notation.** Here $I$ and $J$ are the paper's impact variables, not the impact process $I_t$ of other entries; $\gamma, \eta, \alpha, \beta, \delta$ are its coefficients and exponents, not the Almgren–Chriss $\gamma$ and $\eta$ of price impact.

**Conditions.** The sample is Citigroup US equity orders from December 2001 to June 2003: 682,562 orders before filtering, 29,509 after (pp. 5–8). The filters keep S&P 500 stocks with daily volatility below 12.5 percent. They exclude market-on-open, market-on-close and VWAP orders and orders with executions after 4:10 pm, and keep orders of at least two transactions, 1,000 shares and 0.25 percent of average daily volume. The median order lasts about half an hour; most are a few percent of daily volume or less (pp. 6–8). The model assumes a constant trading rate in volume time and no cross-impact. It makes permanent impact independent of the schedule, so stealth trading is excluded by construction (p. 13).

**Practice.** The estimates are $\alpha = 0.891 \pm 0.10$, $\delta = 0.267 \pm 0.22$, $\beta = 0.600 \pm 0.038$, one standard deviation (p. 20). The authors fix $\alpha = 1$, which they cannot reject, $\delta = 1/4$ and $\beta = 3/5$, rejecting the square root at the 95 percent level. They then find $\gamma = 0.314 \pm 0.041$ and $\eta = 0.142 \pm 0.0062$ [wide-almgren-direct-impact, same preprint, secs. 4.2–4.3, pp. 20–21]. The fitted model is $I = \gamma\sigma(X/V)(\Theta/V)^{1/4}$ and $J = I/2 + \operatorname{sgn}(X)\,\eta\sigma|X/VT|^{3/5}$ (p. 21). In their example of a purchase of 10 percent of daily volume in IBM ($\sigma = 1.57\%$, $\Theta/V = 263$), permanent impact is 20 bp, and realised cost is 32, 25 and 18 bp for durations of 0.1, 0.2 and 0.5 days (Table 3, p. 22).

Interpretation: recomputing from the fitted coefficients gives 19.9 bp and 32.2, 24.6 and 18.4 bp, matching the table to its rounding; DRI likewise. Temporary impact needs no stock-specific factor beyond volume and volatility, and no consistent dependence on the bid–ask spread was found (p. 19).

**Pitfalls.** The "permanent" impact $I$ is the move to half an hour after the last fill, chosen because the regressed values stop changing with the lag at about that point (p. 9). It is a finite-horizon measurement that can include later flow and information; compare the 20-day profile of the post-execution method. $R^2$ is typically below one percent, because volatility dominates the price moves of individual orders (p. 21). Filling only partly, and the exclusion of block-desk orders, could bias the sample; the authors report nearly identical coefficients with and without partly filled orders (p. 8). The fit reflects US large caps over 2001–2003; the authors expect coefficients, exponents and forms to be re-estimated (pp. 21–22).

They prefer linear permanent impact because, citing Huberman and Stanzl (2004), it is the only form free of arbitrage, while nonlinear temporary impact introduces none (p. 16); see absence of price manipulation for the conditions in other models.

**See also.** price impact (the Almgren–Chriss framework behind the model); square-root impact law (an exponent of 0.6 here); intraday impact profile; post-execution impact profile.

**Further reading.** The paper's published version, *Risk* 18(7), 2005.

#### Method 17.42 Order-flow impact regression: the Glosten–Harris decomposition into permanent and transitory components

**Statement.** Harris presents the Glosten–Harris model as an order-flow transaction-cost estimator [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", box, p. 435]. The trade price is fundamental value plus a transitory component, $P_t = V_t + \theta_t$ with $\theta_t = aq_t + bq_tS_t$. Value changes by a permanent effect proportional to signed size plus other information, $\Delta V_t = \lambda q_t S_t + \varepsilon_t$. Hence

$$\Delta P_t = \lambda\, q_t S_t + a\, \Delta q_t + b\, \Delta(q_t S_t) + \varepsilon_t,$$

estimated by regressing the price change on signed size, the change in sign and the change in signed size. Hasbrouck states the model with a size-independent permanent term as well: $m_t = m_{t-1} + u_t + q_t(\lambda_0 + \lambda_1 V_t)$, $p_t = m_t + q_t(c_0 + c_1 V_t)$ [s6-hasbrouck, sec. 9.8.1, eq. (9.26), p. 91].

**Notation.** $q_t = \pm 1$ the trade sign (house convention); $S_t$ (Harris's Size) or $V_t$ (Hasbrouck) the unsigned size. Interpretation: $a$ is half the fixed transitory spread per side and $\lambda$ the permanent effect per share, as in the spread-component entries.

**Conditions.** Interpretation: differencing Hasbrouck's form gives $\Delta p_t = \lambda_0 q_t + \lambda_1 q_t V_t + c_0 \Delta q_t + c_1 \Delta(q_t V_t) + u_t$. Harris's regression, and the simplified one, are the special case $\lambda_0 = 0$. That follows Harris's statement that information theory suggests a permanent effect proportional to size (p. 435), but it drops a term the general model allows. Glosten and Harris's own estimation treated prices as rounded to the tick and trade signs as latent, using a nonlinear filter, and assumed size and sign exogenous with normal public news [s6-hasbrouck, sec. 9.8.1, pp. 91–92].

**Practice.** Estimated size effects are generally concave, flattening at large sizes, and signed size often adds little beyond the sign. One explanation is time-varying liquidity: traders trade large when impact is low. The index can be trades (event time) or fixed intervals (clock time): event time favours stationarity for one security, clock time comparability across securities. Trade prices suit execution-cost questions; quote midpoints suit long-run impact and are less noisy [s6-hasbrouck, secs. 9.7.2–9.7.4, pp. 89–91]. Conditioning on an index return, or better on recent index-futures returns, matters when attributing price changes to a stock's own trades [s6-hasbrouck, sec. 9.6, p. 87].

**Pitfalls.** Harris writes that "the statistical properties of standard regression methods ensure that the regression error term is uncorrelated with the independent regressor variables" (p. 435). Interpretation: least squares makes the fitted residuals orthogonal to the regressors by construction, but the coefficients estimate the permanent and transitory effects only if the true $\varepsilon_t$ is uncorrelated with them. That is an assumption, Glosten and Harris's exogeneity, not a consequence of the method. Signed order flow is chosen by traders who may trade on information or on the same news.

Coefficients then mix reaction with prediction, as in price impact. Of the usual controls (market moves, time of day, volatility, spread, order splitting), the sources support an index return in the conditioning set (Hasbrouck, p. 87), volume time (Almgren et al.) and volatility and spread terms (Bacry et al.). Controls reduce omitted-variable bias but do not by themselves make the coefficients causal (Interpretation); random trading is the design that does (price impact, Practice).

**See also.** spread component identification (the generalized Roll model, the size-free special case); spread components; price impact; propagator model (impact that decays over many trades rather than one).

**Further reading.** Glosten and Harris (1988).

#### Evidence 17.43 Index reconstitution: price pressure from forced index-fund trading and its partial reversal

**Statement.** Index funds that replicate the Russell indexes rebalance when the indexes are reconstituted each June, buying additions and selling deletions near the same date. In the six years 1996–2001, Russell 3000 additions outperformed deletions by an average of 15 percent in June and underperformed them by 5 percent in July. Harris reports this from Madhavan (2002) [alpha-harris, chap. "Index and Portfolio Markets", box, p. 487].

**Conditions.** Russell reconstitutes at the close of the last trading day in June on capitalisations as of the end of May. The impacts are large because the stocks affected are small [alpha-harris, same box]. Harris notes that changes to Standard & Poor's indexes have similar effects (same box).

**Practice.** The July reversal shows that part of the June move is transitory. Harris offers three explanations for the remainder: investors may value index membership, perhaps for liquidity; a momentum anomaly in small stocks; or reversal in later months [alpha-harris, same box]. The event is predictable demand that is uninformed about the firm, which makes it a natural setting for separating price pressure from information (Interpretation).

**Pitfalls.** Changed ownership, attention or information are not Harris's explanations of the residual, and the box gives no test between his. That the effect is largest when forced demand is large relative to liquidity rests only on Harris's remark that the stocks are small; the box gives no measure of demand relative to liquidity. The figures are averages over 1996–2001 and may not hold under later practice.

**See also.** price impact (transient and permanent components); post-execution impact profile.

**Further reading.** Madhavan, "The Russell Reconstitution Effect" (2002), cited by Harris.

## 17.13 Exchange-traded funds and index products

#### Mechanism 17.44 ETF creation and redemption: authorised participants, in-kind and cash baskets, and arbitrage that tends to keep price near value without guaranteeing it

**Statement.** An open-end mutual fund transacts with its investors at the end of the day at net asset value (NAV), and buys or sells the underlying assets itself when flows do not net. An exchange-traded fund (ETF) also holds a portfolio but, in the authors' description, "does not interact with capital markets directly" (p. 4): secondary trading and in-kind creations and redemptions leave the fund's dealings in the underlying to others, while cash creations and redemptions can require the fund to trade (Practice). Its sponsor contracts with *authorised participants* (APs), typically large financial institutions or specialised market makers, who can create or redeem the fund's shares in large blocks, *creation units*, in exchange for a basket of securities and/or cash [ex-lettau-madhavan-2018, NBER working paper version, pp. 4–5]. Two markets are therefore distinct:

- *creation and redemption* between APs and the fund. The fund publishes its holdings and the basket it will accept for the next business day at the end of each trading day. Transactions are either *in kind*, the AP delivering or receiving a basket identical or very similar to the holdings, or *in cash*; in the authors' description cash creation is "not typical" (p. 5);
- *secondary trading* of ETF shares on exchanges, through brokers, among investors, APs and other liquidity providers. It does not lead to transactions in the underlying securities; across U.S. ETFs secondary volume was about four times creation and redemption activity in 2014, by Investment Company Institute statistics as the authors report them (pp. 5–6).

Creation or redemption happens at the end of the day, but APs typically lock in any profit intraday. If the ETF trades at a premium to an AP's estimate of value, which need not be the NAV, the AP may deliver the creation basket for new ETF shares, which it may sell or keep. The mechanism "works through arbitrage to help keep the price … close to the intrinsic value" of the holdings (p. 6). Harris's 2002 description is the same in kind: ETFs are trusts whose units trade like stocks, and large traders create units by depositing the index portfolio and redeem them for a pro rata share of it [alpha-harris, chap. "Index and Portfolio Markets", sec. 23.5, p. 492].

**Conditions.** The link between price and value is an arbitrage incentive, not a guarantee.

- *Stale reference values.* Deviations from the published NAV need not be arbitrage opportunities. The NAV and the intraday indicative value, disseminated about every 15 seconds, may rest on stale prices, as for funds holding Japanese stocks during U.S. hours or bonds that have not traded [ex-lettau-madhavan-2018, p. 6].
- *No obligation to arbitrage.* APs "are not contractually bound by legal obligations to perform ETF arbitrage" [wide-pan-zeng-etf, ESRB version, p. 1]. Lettau and Madhavan argue that one AP's withdrawal is likely to be covered by others in ETFs with many APs; in an Investment Company Institute study of 931 U.S. ETFs that they cite, the largest averaged 38 APs. Smaller niche ETFs with few APs are the exception they allow for. If all APs withdrew, the fund would trade like a closed-end fund, with possibly wider premiums or discounts (p. 18).
- *Liquidity mismatch in corporate bond ETFs.* Pan and Zeng study the case where a liquid ETF holds illiquid assets, for U.S. corporate bond ETFs only. APs are also bond dealers. Their data are AP lists from two issuers covering on average 83 percent of passive corporate bond ETF assets, 2004–2015, with dealer-identified TRACE bond trades [wide-pan-zeng-etf, abstract and pp. 1–3].
  - *Model predictions* (pp. 2–3). With small bond inventory imbalances, an arbitrage effect: APs correct mispricing, less so as volatility, bond illiquidity and trading costs rise. With large imbalances, an inventory-management effect can dominate, so that creations and redemptions follow the APs' inventories and may run against the mispricing.
  - *Empirical findings* (p. 4). A 1 percent increase in the ETF premium raises AP arbitrage by 50 basis points; a one-standard-deviation rise in market volatility, holding the premium fixed, lowers it by 10 percent; and the fall in sensitivity is larger when the ETF trades at a discount. Under extreme inventory imbalances, creations and redemptions become less sensitive to the mispricing and more sensitive to inventory. Overall, arbitrage still goes in the direction of the opportunity, but a difference-in-differences comparison shows APs doing less (more) net redemption after large positive (negative) inventory shocks despite a discount (premium). The authors "do not find evidence of the complete 'failure' of ETF arbitrage".

  Inventory pressures therefore weaken corrective arbitrage in their sample; that is not evidence that arbitrage fails outright, nor that it never runs the wrong way. The authors contrast equity ETFs, where they see no such mismatch (p. 2); their evidence does not extend to them.

**Practice.** In an open-end fund the fund itself trades the underlying for net flows, and the costs fall on the investors who stay. In an ETF, sellers trade with buyers at a market price, and "net selling does not require the ETF manager to interact with the capital markets": the costs are "externalised" [ex-lettau-madhavan-2018, p. 7]. Interpretation of the mechanism: secondary trading and in-kind creations and redemptions do not themselves require the fund to buy or sell the underlying, so its remaining holders bear none of those costs. Whoever trades the basket, such as an AP, bears them and passes them into the prices it quotes. Cash creations and redemptions, by contrast, can require the fund to trade the underlying and incur costs; the source does not say who bears them.

The authors' illustration: a mutual fund of $100 million, with spreads of 0.20 percent (0.10 percent one way), $5 million of subscriptions and $20 million of redemptions, sells $15 million at the bid and ends at $84,985,000 ($15,000 of costs); the ETF's investors who do not sell "will hold a fund whose assets are valued at $85 million" (p. 7).

Interpretation, reconciling the abbreviated ETF case: pure secondary trading leaves the fund's total assets unchanged at $100 million, with the net $15 million of shares sold now held by whoever bought them, such as an AP. Total assets of $85 million follow if that net $15 million is redeemed in

kind, so that the fund delivers securities rather than selling them. The holders other than the redeeming AP, the original investors who did not sell ($80 million) and the new buyers ($5 million), then hold $85 million without bearing the $15,000.

In-kind transfers can also reduce capital-gains distributions (p. 8; Harris, p. 492, for the same tax point). Among index products Harris also lists open-end index funds, bought and redeemed at closing NAV and not tradable within the day; index futures, cheap to hold on margin but rolled at a cost; and cash-settled index options, used by speculators, hedgers and gamblers (sec. 23.5, pp. 491–492, Table 23-1 as of April 2001).

**Pitfalls.** "Primary market" is used two ways. Harris's Table 23-1 gives the primary market for fund shares as deposit to and redemption from the fund (p. 491), while Lettau and Madhavan use it for the market in the underlying securities in which APs acquire baskets (p. 18). This entry says "creation and redemption" for the first and "secondary trading" for exchange trading in ETF shares.

The creation–redemption link weakens when creation costs, borrow, cash substitutions, taxes or stressed constituent liquidity are material. Stale valuation, illiquid holdings (for corporate bonds) and APs' discretion are in the source; borrow costs, cash substitution and taxes as limits to arbitrage are not.

Other index products compare as follows:

- index futures give leveraged, cash-efficient exposure; Harris names their margins and roll costs, not basis risk;
- index options give convex exposure to the index level and volatility (Interpretation);
- the choice among products turns mostly on speed, funding, tracking error, tax and operations (a synthesis).

Lettau and Madhavan are cited from their NBER working paper of January 2018, not the published *Journal of Economic Perspectives* version. Madhavan was then head of ETF and index research at BlackRock, which the paper discloses (p. 1). The ETF facts are as of 2014–2018 for the United States; current SEC rules on ETF baskets (Rule 6c-11) are not covered here.

**See also.** portfolio trading (index products and program trades); index reconstitution; arbitrage risks; HFT strategies (ETF-basket arbitrage).

## 17.14 Futures: venue conventions, off-exchange trades, margin and price discovery

**Convention 17.45 CME Globex trading conventions: a centralised market with permitted exceptions, matching algorithms, the pre-open and opening match, and the E-mini S&P 500 daily settlement window**

**Statement.** *Centralisation.* Harris reported in 2002 that most futures markets require all trades in their contracts to be arranged within their trading systems, floor or screen. The rules prevent brokers from directing orders to confederates at prices worse than the exchange's, and futures markets had historically had no block-trading procedures or cumbersome ones [alpha-harris, chap. "Block Traders", sec. 15.6.2, p. 334]. CME's current rulebook keeps the principle and names the exceptions: block trades, and exchanges for related positions, which are "permitted to be executed outside of the Exchange's centralized market in accordance with the requirements of this rule" [cme-rulebook-ch5, Rules 526 and 538]; see futures block trades and EFRPs.

*Matching.* On CME Globex a matching algorithm allocates quantity when an aggressor order meets one or more resting orders. The supported algorithms, identified by tag 1142, are Allocation (A), FIFO (F), FIFO with lead market maker (T), FIFO with top order and lead market maker (S), Pro-Rata (C), Configurable (K), Threshold Pro-Rata (O), Threshold Pro-Rata with lead market maker (Q) and an Institutional Prioritization algorithm (V). Under FIFO, price and time are the only criteria; an order loses its priority and is requeued if its quantity is increased, its price changed or its account changed. As one named example, SOFR futures (but not SOFR packs and bundles, which match FIFO) use a pro-rata allocation.

A "top" order that betters the market has priority, then remaining quantity is allocated pro rata to resting orders at the price; orders entered in the pre-open or at the indicative opening price are matched by price and time [ex-cme-matching, page "Supported Matching Algorithms", updated 11 February 2025].

*Opening.* In the Pre-Open and Reserve states orders can be entered or modified, but no matching occurs, so the book can lock or cross. The trading engine computes and publishes an indicative opening price and quantity, "a probable price and quantity at which the market will open or re-open", for outright futures, futures spreads, options spreads and options series [cme-iop-wiki, updated 11 February 2025].

*Daily settlement, one contract, as of 2020.* In the procedure filed for trade dates from 26 October 2020, the E-mini S&P 500 (ES) and the Micro E-mini settle to the daily settlement price of the full-sized S&P 500 futures (SP), rounded to the nearest tradable tick. For SP's lead month, Tier 1 is the volume-weighted average price of trades in the E-mini contract on Globex from 14:59:30 to 15:00:00 Central Time, rounded to the nearest 0.10 index point. The amendment struck the floor trades in the full-sized contract, the multiplier of five and the word "combined".

If no lead-month trades occur in the window, Tier 2 is the midpoint of the bid and ask over it; if no two-sided market is available, Tier 3 derives the price from the cash index by a carry calculation. The second month is derived from the lead month through the calendar spread. Tier 1 settles it to the VWAP of lead–second spread trades in the window "using the same methodology"; Tier 2, if no spread trades occur, applies the last spread trade price to the lead-month settlement to derive the second-month settlement, using the nearer of the spread's bid or ask if the last trade lies outside them; Tier 3, with no spread market information, uses the carry calculation.

Interpretation: in Tier 1, too, the spread price is applied to the lead-month settlement to give the second-month outright, as Tier 2 states (Exhibit B, PDF p. 6). Staff may set another price if the calculation fails or activity is anomalous [cme-es-settlement-2020, Exhibit B, blackline].

**Conditions.** The matching and opening pages are living documentation, as of 15 October 2026; the settlement procedure is stated as filed in 2020, and its current version is unverified. Which algorithm applies to a given contract is set per product; it is stated here for SOFR futures. Other exchanges' and other CME contracts' settlement procedures may differ.

**Practice.** Interpretation: a contract's matching algorithm is needed before estimating passive fill probability. Under FIFO a resting order's fill depends on its queue position; under pro-rata allocation it depends on its size relative to the quantity at the price, which rewards larger posted size (passive order split for the FIFO queue model). When proportional shares are not whole contracts, a remainder must be allocated by a further rule.

**Pitfalls.** Futures markets do use call auctions. On CME Globex the pre-open with an indicative opening price, followed by the opening match, is a call-auction-like uncross. Only at the E-mini S&P 500 close is there no auction: under the 2020 procedure the daily settlement is a 30-second VWAP of continuous Globex trading, with midpoint and carry fallbacks. Margining, clearing, delivery rules and fungibility limit fragmentation in principle; the sources give exchange rules requiring on-exchange trading, with defined exceptions. CME Globex is highly electronic today; Harris's 2002 account still includes floor trading.

**See also.** futures block trades and EFRPs; futures margin and settlement; spread determinants (large ticks); matching and trade pricing (precedence, pro-rata and auction rules in general).

**Convention 17.46 Futures block trades and exchanges for related positions at CME Group: eligibility, the related position, reporting, and the prohibition of transitory EFRPs**

**Statement.** *Block trades* (CME Rule 526, current rulebook [cme-rulebook-ch5, Rule 526]).

- The exchange designates the products in which blocks are permitted and their minimum quantities, and orders may not be aggregated to reach them except by eligible advisers (526.A).

- Each party must be an Eligible Contract Participant under the Commodity Exchange Act; registered or exempt commodity trading advisers and investment advisers with more than $25 million under management, and similar foreign persons, may act for their customers (526.B, I, J).
- A customer must have specified that its order be executed as a block (526.C).
- The price must be "fair and reasonable" given the block's size, other trades in the contract, related markets and the circumstances (526.D).
- Blocks do not trigger conditional orders such as stops or otherwise affect orders in the regular market (526.E).
- The seller or broker reports the block within the period and in the manner the exchange specifies, and the exchange publishes blocks separately from regular-market trades (526.F); blocks are reported to the clearing house, and records are kept (526.G–H).

*Exchanges for related positions* (EFRPs; CME Rule 538 [cme-rulebook-ch5, Rule 538]).

- An EFRP is a privately negotiated off-exchange execution of a futures or options contract together with the simultaneous, opposite execution of an equivalent quantity of the related cash product, by-product, related product or OTC derivative. The types are the exchange of futures for physical (EFP), for risk (EFR, against an OTC swap or other derivative) and of option for option (EOO).
- Each party is buyer on one side and seller on the other, and each side's two legs must be for accounts with the same beneficial ownership (538.A). A third party may facilitate the related position as principal for a customer but must show that it was passed through to the customer, except for immediately offsetting FX EFPs under 538.K: when CTAs, account controllers or others act for a customer in those, the initiating and offsetting cash legs need not be passed through, though the customer bears the profit or loss on the offsetting leg if the futures leg fails to clear (538.A). Opposing accounts must be independently controlled (538.B).
- The related position must be the underlying cash commodity, a by-product, a related product or an OTC derivative with a reasonable degree of price correlation, never a futures or option on futures. It requires a bona fide transfer of ownership or a binding contract (538.C).
- Quantities must roughly match, hedge ratios allowed (538.E). Prices must use the contract's increments and be commercially reasonable (538.F).
- Each EFRP is submitted to the clearing house within the specified time, with the time its terms were agreed, and is accepted only once it has cleared and the first settlement variation and performance bond are confirmed (538.I). Records are kept (538.H).

*Transitory EFRPs.* Rule 538.C forbids an EFRP whose execution is contingent on another EFRP or related-position transaction between the parties, where the transactions offset the related positions without market risk that is material in context. CME Group's advisory notice in force from 7 August 2026 defines these as *transitory EFRPs*. It answers "No" to whether they are permitted in any CME Group product: the time between the transactions is a factor, but the test is whether they "have integrity as independent transactions exposed to market risk" [cme-ra2607-5, pp. 4 and 10, Q14].

Rule 538.K, as amended from 7 August 2026, expressly permits EFRPs in foreign currency futures whose parties immediately offset the related position, but only as EFPs whose related position is a cash transaction or EFRs whose related position is a non-deliverable forward (NDF), in foreign currencies only and never as EOOs; the cash-side confirmations must identify the counterparty, or for a CTA's client its carrying clearing member and account [cme-rulebook-ch5, Rule 538.K]. The exchanges' filing describes these as physical FX cash and FX NDF related positions; the amendment changed only 538.K, leaving 538.A–J unchanged [cme-ra2607-5, pp. 1–3, Exhibit A blackline]. They are not treated as transitory, because the offsetting cash or NDF transaction is not contingent on the EFRP (Q28).

**Conditions.** Product thresholds and reporting deadlines are set in separate exchange notices. The rules are those of CME Group's U.S. exchanges (CME, CBOT, NYMEX and COMEX). History, kept apart from current rules: Harris described futures block rules in 2002 as limited to eligible sophisticated traders at fair and reasonable prices. He described EFPs as originally developed for agricultural hedgers, and as used at CME in currency futures immediately after the close, where exchanges called them "transitory EFPs" because the floor traders bought or sold the cash currency at the same time [alpha-harris, chap. "Block Traders", sec. 15.6.2, p. 334]. CME Group's 2019 notice records that transitory EFRPs had been permitted in CME foreign currency, NYMEX energy and COMEX and NYMEX metals products until 4 August 2014 and have been prohibited since [cme-ra1911-5, p. 1].

**Practice.** Interpretation: an equity block cross interacts with the exchange book under precedence rules (block trades, Harris's 2002 equity rules). A futures block is negotiated away from the book, does not trigger the book's conditional orders and is published separately. The two regimes should not be mixed.

**Pitfalls.** A "transitory EFP", a short-lived EFP used when the book is closed or thin, is historical: it matches Harris's 2002 account, and at CME Group such EFRPs have been prohibited since August 2014. The immediately offsetting FX EFRP of Rule 538.K is a different, permitted structure. An EFP exchanges cash or physical exposure for offsetting futures, as Rule 538 provides in substance, and moves large risk off the book subject to the eligibility and reporting rules of Rules 526 and 538.

**See also.** block trades; CME Globex trading conventions; futures margin and settlement.

### Mechanism 17.47 Futures margin and settlement at CME Clearing: performance bond, settlement variation each cycle, and the default waterfall

**Statement.** At CME Clearing, settlement for outstanding exposure is made with the clearing house in settlement cycles held "at least once each Business Day", at times the clearing house sets and possibly different by contract. Each clearing member, for its own and its customers' positions, pays or collects *settlement variation*: the difference between this cycle's settlement price and the previous cycle's, or the trade price for a new position. Payments are final once the clearing house's bank account is debited or credited [cme-rulebook-ch8, Rule 814]. Separately, clearing members post *performance bond*, initial and additional, as clearing-house staff set it, gross for each omnibus customer account and in approved collateral.

The clearing house retains it "in whole or in part, as Clearing House staff may deem necessary", until the trades it covers have been offset, cash settled, delivered or otherwise closed out; if a clearing member defaults on performance bond, the collateral is taken over and sold, customer collateral being applied to customer requirements and house collateral to house requirements [cme-rulebook-ch8, Rule 820].

On a clearing member's default the Base waterfall runs as follows [cme-rulebook-ch8, Rules 802.A–802.B]:

1. the defaulting member's own resources, with default management run separately for each account class (futures, foreign futures, cleared swaps). A loss in a customer class is met first from that class's customer collateral (performance bond, excess bond and settlement gains). In a defaulted customer futures class this includes the collateral of customers who did not cause the default (802.A.2, 802.G.1). The member's own assets (its Base Guaranty Fund requirement, performance bond, CME shares or cash in lieu and other assets) are applied across account classes with priority to customer classes (802.A.5.i, iii). Collateral left after liquidating the proprietary account may be applied to customer-class losses, pro rata among defaulted customer classes (802.G.6). The restriction runs one way: segregated customer collateral may not be used for the member's proprietary obligations or for another customer account class (802.A.5.iii, 802.G);
2. CME's corporate contribution of $100 million (802.B.1.i);
3. the Base Guaranty Fund, that is, the prefunded contributions of the non-defaulting members, applied in the rule's order of priority: the subordinated amounts of members subordinated after a default-management auction, then the remaining contributions of the other members pro rata, then the seniorized amounts of winning bidders (802.B.1.ii–iii; the fund consists of members' contributions, 802.A.4, so it is counted once here);
4. assessments on the non-defaulting members, capped at 275 percent of each member's Base Guaranty Fund requirement for losses from a single defaulter and 550 percent for all defaults in a cooling-off period, divided pro rata (802.B.1.iv).

**Conditions.** CME rules as of 15 October 2026; Hull's generic account as of 2021. Interpretation, distinguishing the following mechanisms:

- *contract standardisation*, set in each contract's chapter;
- *initial margin*, the performance bond that covers moves between settlements and the time to close out;

- *variation settlement*, the realised gain or loss each cycle;
- *intraday calls*, which Rule 814 permits through more than one cycle a day, though the actual cycle times are unverified;
- the *FCM's* arrangements with its customers, governed by customer agreements and exchange rules, while Rules 814 and 820 bind clearing members;
- *residual default risk*: prices move between cycles, performance bond may fall short, and losses then pass along the waterfall of Rule 802 to CME's contribution, the other members' prefunded contributions and, within the caps, assessments on them.

Daily settlement bounds the exposure that can build between cycles; it does not eliminate credit risk continuously. Harris notes that futures are popular with hedgers and speculators because positions can be taken without posting large margins [alpha-harris, chap. "Index and Portfolio Markets", sec. 23.5, p. 492].

**Practice.** On the customer side the broker or FCM manages client margin calls, liquidation rights and cash balances (Interpretation); the clearing-house rules above are the member side. *A textbook account, generic rather than CME's rules.* Hull illustrates the customer side with a broker's margin account [s7-hulloptions, chap. 2, "Futures Markets and Central Counterparties", sec. 2.4, pp. 51–54]: an initial margin is deposited, the account is settled to the day's gain or loss each day (daily settlement or marking to market), and the daily flow between longs and shorts is variation margin.

Below a maintenance margin, "somewhat lower than the initial margin" and usually about 75 percent of it, the trader receives a margin call to restore the initial level or is closed out; excess over the initial margin can be withdrawn; most brokers pay interest on balances, and Treasury bills (at about 90 percent of value) or shares (about 50 percent) can usually meet initial but not later calls. Clearing members, by contrast, face a maintenance margin equal to the initial margin, margin computed on members' net positions, possible intraday calls, and a guaranty fund; margin is usually set so that the clearing house is "about 99% certain" it covers losses on a member's default (pp. 53–54).

Hull's gold example: two contracts of 100 ounces at \$1,750 lose $200 \times 9 = \$1{,}800$ on a \$9 fall, and the day-7 balance of \$7,980 prompts a call of \$4,020 back to \$12,000 (p. 51; Table 2.1, p. 52; p. 53). These levels, ratios and haircuts are Hull's illustrations, not rules of any clearing house; minimum margins are set by each clearing house and brokers may demand more (p. 53).

**Pitfalls.** Credit risk is not controlled continuously. It is controlled at each settlement cycle, at least daily, with performance bond for the moves in between and a mutualised default waterfall for what remains. The usual features of exchange futures are supported as follows:

- fungibility within an expiry and specification is set by each contract's terms;
- margin and default management built into the market: Rules 802, 814 and 820;
- leverage: Harris p. 492;
- a hedging link to cash baskets, ETFs, options and swaps only in part: Rule 538 allows cash and OTC-derivative positions, and ETFs are not addressed.

**See also.** CME Globex trading conventions (daily settlement prices); futures block trades and EFRPs (acceptance after first settlement variation); arbitrage risks (financing of losing legs).

### Evidence 17.48 Price discovery between index futures and their cash and ETF markets: the E-mini's leadership in 2000, and what explains leadership

**Statement.** Hasbrouck studies price discovery across U.S. equity index markets at up to one-second resolution, using vector error-correction models and information shares, for 1 March to 31 May 2000. He finds:

- for the S&P 500 and Nasdaq-100, whose index markets comprised an ETF, a floor-traded futures contract and an electronic E-mini, most price discovery occurred in the E-mini, whose prices generally led the regular futures and the ETFs; the ETFs' contributions were statistically significant but small;
- for the S&P 400 MidCap, with only an ETF and regular futures, both contributed substantially and the ETF appeared to dominate;
- the S&P 500 sector ETFs contributed little to price discovery in the index.

[ex-hasbrouck-2003, NYU working paper FIN-00-046, draft of 16 November 2000, abstract and pp. 4–5, 13 and 34]. The earlier literature he reviews found that the dominant influence ran from futures to cash, with weaker but measurable effects in the reverse direction (p. 5).

**Conditions.** The evidence is for three U.S. indexes in three months of 2000, when the regular contracts traded on the CME floor during regular hours and on Globex otherwise, and the E-minis on Globex (p. 6). It is from a preliminary working-paper draft. The catalogue describes the published 2003 version's S&P 400 result as shared between the regular futures and the ETF; the working paper differs on this point, and the difference is unresolved. The theoretical treatment of futures–cash price discovery is Garbade and Silber (1983, w5-garbade-silber).

**Practice.** Harris's reason for index-market leadership is that index traders price only index risk, while traders of the components must price every risk of their securities, so index markets discover the price of index risk faster (portfolio trading, p. 489).

**Pitfalls.** Futures are often said to lead when they are cheaper to trade, more leveraged or open during information arrival, and cash markets when constituent-level information dominates. Harris's argument supports only the general direction for index risk. The specific conditions are plausible hypotheses not tested in the source, and the working paper does not attribute the E-mini's leadership to cost or leverage. Leadership in information shares is a statistical measure of where innovations first appear, not a causal claim about who is informed.

**See also.** portfolio trading (why index markets lead price discovery); ETF creation and redemption; arbitrage (basis and delivery arbitrage).

**Further reading.** Garbade and Silber (1983) for the theory.

### Convention 17.49 Government bond futures as Gliner describes them in 2013–2014: U.S. Treasury contract sizes and ticks, physical delivery and cheapest-to-deliver, the German Schatz, Bobl and Bund, and why futures trade below the bond

**Statement.** U.S. Treasury note and bond futures are physically settled, "which lends to the concept of cheapest-to-deliver", and most traders roll before delivery [s7-gliner, chap. 9, sec. "U.S. Fixed Income Futures", PDF p. 264]. Gliner's specifications (pp. 265–266): \$200,000 face for the 2-year (TU) and \$100,000 for the 5-year (FV), 10-year (TY), bond (US, his "15+Year") and ultra bond (WN, "25+Year"); ticks of a quarter of a 32nd (\$15.625) for TU, a quarter (\$7.8125) for FV, a half (\$15.625) for TY and a full 32nd (\$31.25) for US and WN; March, June, September and December months. German futures: Schatz about 2 years (DU), Bobl 4.5 to 5.5 years (OE) and Bund 8.5 to 10.5 years (RX) to maturity (p. 266).

**Notation.** Tick value = face × tick in points /100: $200{,}000 \times (1/128)/100 = \$15.625$; $100{,}000 \times (1/128)/100 = \$7.8125$; $100{,}000 \times (1/64)/100 = \$15.625$; $100{,}000 \times (1/32)/100 = \$31.25$ (2-year, 5-year, 10-year and bond).

**Conditions.** *Why the curve slopes down.* Gliner says Treasuries "have term premium, which implies a positive sloping yield curve, which means that the futures curve on Treasuries is usually in backwardation", unusually among futures (p. 264), and adds that underlying instruments "generally have a higher interest rate than short-term financing costs". The second clause is the reason: with coupon income above the repo rate the forward price is below spot (positive carry). A 4 percent bond at 100 financed at 1 percent for three months has a forward of 99.25 (checks, section 8). Term premium affects the slope of yields, not the futures discount directly; a futures curve in "backwardation" here means lower prices for later contracts.

**Practice.** Contract specifications change, and contracts and tick sizes may have been added or altered since 2014; the specifications above are Gliner's, compared with CBOT's and Eurex's current contract terms below. *Eurex terms*. Eurex's fixed-income futures factsheet gives remaining terms of 1.75 to 2.25 years (Euro-Schatz, FGBS), 4.5 to 5.5 (Euro-Bobl, FGBM) and 8.5 to 10.5 (Euro-Bund, FGBL) on a notional 6 percent coupon, and 24.0 to 35.0 years (Euro-Buxl, FGBX) on a notional 4 percent coupon, with a contract value of EUR 100,000, delivery on the tenth calendar day of the delivery month (or the next exchange day) and last trading two exchange days before [eurex-fi-factsheet, pp. 1–2]; Eurex's contract specifications as of 10 September 2025 define the subject matter of these contracts in the same way [eurex-cs-2025, sec. 1.2.1, p. 1].

Schatz and Buxl bands are therefore current Eurex terms; the factsheet is undated. Two CBOT certifications filed with the CFTC show that the 10-year note contract (ZN, Gliner's TY) is no longer a 6.5-to-10-year contract. *From the September 2023 contract month* the deliverable notes have a remaining term of not less than 6 years 6 months and not greater than 7 years 9 months, and the contract was renamed "U.S. Treasury Note Futures (6½ to 7¾-Year)", partly to separate it from the Ultra 10-Year (TN) when yields exceed 6 percent and the cheapest-to-deliver moves to the long end of the basket [cbot-22-468, Submission 22-468, pp. 1–2].

*As certified in March 2025* Chapter 19 is titled "(6½ to 8-Year)": fixed-principal notes with an original term of at most 10 years and a remaining term of at least 6 years 6 months and less than 8 years, measured from the first day of the delivery month and rounded down to whole quarters; from the March 2026 contract month a reopening that brings an older issue into the band adds it to the basket; the exchange describes the basket as typically eleven notes (six 10-year and five 7-year), and the Ultra 10-Year as the three most recently issued 10-year notes [cbot-25-099, Submission 25-099, pp. 1–4]. With quarter rounding, "less than 8 years" and "not greater than 7 years 9 months" select the same notes (interpretation).

*Current chapters* (official chapter PDFs as of October 2026; the texts carry no amendment dates, and the price-increment footnotes read "Revised March 2008; January 2009"). Chapter 19 has the 25-099 title and basket, a reopening clause, and a minimum fluctuation of half a 32nd ($15.625; a quarter, $7.8125, for intermonth spreads).

The 2-year note is $200,000 face of notes with an original term of at most 5 years 3 months and a remaining term of 1 year 9 months to 2 years, rounded down to whole months, with a minimum fluctuation of one-eighth of a 32nd ($7.8125), spreads included. The 5-year is $100,000 face, at most 5 years 3 months original and at least 4 years 2 months remaining, with a quarter of a 32nd ($7.8125).

The bond (US) is $100,000 face of non-callable bonds with 15 to less than 25 years remaining (or callable bonds not callable for 15 years), with a full 32nd ($31.25; a quarter for spreads). The Ultra Bond (UB, Gliner's WN) is $100,000 face of bonds with at least 25 years remaining, with a full 32nd ($31.25) outright and, for intermonth spreads, "one quarter of one thirty-second of one point (equal to $7.8125 per contract)" (Rule 40102.C) [cbot-treasury-chapters-2026, Rules 18101.A–18102.C, 19102.C, 20101.A–20102.C, 21101.A–21102.C and 40101.A–40102.C].

Gliner's FV, TY, US and UB ticks match the current chapters; his TU tick of a quarter of a 32nd ($15.625) does not: the current rule is an eighth ($7.8125); when it changed is unverified. Spreads to Bunds, such as BTP–Bund, as euro-area risk measures belong to `dm-sovereign-bonds`.

**Pitfalls.** Gliner's first clause, a futures curve typically in backwardation due to term premium, gets the mechanism wrong. Gliner's "15+Year" and "25+Year" labels are his shorthand for the bond and ultra-bond delivery baskets; the bond basket is capped below 25 years (Practice), so "15+" is not open-ended. Gliner's 2-year tick is out of date (Practice).

**See also.** CME Globex trading; futures margin and settlement; cost-of-carry forward price; curve trades; U.S. quotation in 32nds.

## 17.15 Fixed-income and OTC trading

**Definition 17.50 Fixed-income market structure: segments, trading protocols, venues and dealer risk-bearing, kept apart (U.S., as surveyed in 2019)**

**Statement.** Except for a few segments, fixed-income instruments trade in dealer-oriented over-the-counter (OTC) search markets that differ significantly from equity markets [wide-bsv-fi-survey, sec. I, p. 1]. Three things that the word "market" runs together are kept apart here.

- *Segments.* U.S. Treasury securities are "in many ways the simplest" fixed-income instruments: low default risk, high volume, standardised contracts and generally liquid markets (p. 1). Corporate bond trading is fragmented because each issue of an issuer is a distinct contract, not fungible with the others; individual issues did not trade on 52 percent of days in Edwards, Harris and Piwowar's sample, and in 2016 9.3 percent of TRACE bonds never traded and 1.5 percent traded every day (pp. 5 and 14). Structured products split between to-be-announced (TBA) agency MBS, forward contracts on any deliverable pool, the most actively traded structured securities, and specified pools and other structured bonds, where most MBS and ABS did not trade at all in one 21-month sample (pp. 5 and 13–14). Municipal bond trading is also highly fragmented, with over 1.5 million distinct bonds (p. 16).
- *Trading protocols*, the rules by which a client obtains a price. Dealers broadcast indicative bids and offers ("runs"); clients negotiate by phone or chat, or send bid- and offer-wanted lists; in an electronic request for quote (RFQ), as Hendershott and Madhavan describe it on MarketAxess, the client chooses how many dealers to query for a given size and a deadline, sees all quotes at the deadline and may trade with the best (pp. 2–3 and 7–8). Fixed-income quotes are typically indicative, not firm (p. 3).
- *Venues*, the systems on which trading happens: interdealer brokers, about 70 percent of U.S. Treasury interdealer trading being on electronic limit-order-book platforms where principal trading firms also take part; RFQ platforms for customers; and newer all-to-all venues on which institutions trade with each other (pp. 8 and 11).
- *Dealer risk-bearing.* Outside Treasuries and TBA MBS, trading is dominated by customer–dealer OTC trades, traditionally on a principal basis, the dealer absorbing the order into inventory. Dealers have reduced capital commitment in corporate bonds, relying more on prearranged trades in which the dealer has already located the other side, and on shorter intermediation chains; nonbank dealers' share of customer–dealer corporate bond volume rose from 2.4 percent before the 2007–2009 crisis to 12.5 percent in 2014–2016 (p. 4).

**Conditions.** The survey describes U.S. markets with data to about 2017–2018; its electronic-share figures come from industry reports: an industry report published in July 2016 estimated that 19 percent of U.S. investment-grade corporate bond trading was electronically facilitated, mostly by RFQ; the observation period behind that estimate is not given in the passage (p. 2). Hendershott and Madhavan find that periodic one-sided electronic auctions are "a viable and important source of liquidity even in inactively traded instruments", a compromise between bilateral search and continuous double auctions [wide-hendershott-madhavan, abstract]; in the survey's account RFQ use is greatest for recently issued, investment-grade, large-issue bonds (p. 3). O'Hara and Zhou, combining MarketAxess and TRACE data, find electronic RFQ trading "fairly small and segmented" but with wide-ranging effects on transaction costs and execution quality in electronic and voice trading and the interdealer market [wide-ohara-zhou-electronic, abstract]. Both are abstract-level checks.

**Practice.** Interpretation: a dealer's balance sheet is not a venue; the dealer's willingness to hold inventory is a source of liquidity, reached through a protocol (voice, RFQ) on a venue or bilaterally. The survey links the reduction in dealers' capital commitment to post-crisis capital requirements and the Volcker Rule, and notes that counting only bank-affiliated dealers' capital may overstate the effect on markets because nonbank dealers, electronic intermediation and customers' liquidity provision through prearranged trades respond (pp. 4 and 32–33). The generic inventory mechanics are in dealer inventory risk and dealer quote adjustment, from equity and floor examples.

**Pitfalls.** That government bonds are the most standardised and electronically tradable holds for U.S. Treasuries in the survey, whose electronic share is interdealer; customer-to-dealer Treasury trading relies on electronic communication for a large share but was not measured as electronic execution (p. 3). That standard credit indices trade more centrally than bespoke risk, and that swaps often remain dealer-mediated because contracts differ, are not measured here: the survey mentions only that most standardised derivatives, including interest rate swaps, must now be centrally cleared, and a study linking post-trade transparency to lower CDS trading costs (pp. 30 and 32).

*U.S. rules that separate the cases.*

- *Clearing.* The CFTC's clearing requirement covers named classes of fixed-to-floating, basis, FRA and overnight-index swaps by currency, index and maturity range (for example SOFR and Fed Funds OIS), and the North American and European *untranched CDS indices*: CDX.NA.IG and HY, and iTraxx

Europe, Crossover and HiVol, for stated tenors and series. No single-name CDS class is listed [ecfr-17cfr-cftc-swaps, 17 CFR 50.4(a)–(b), last amended 91 FR 57086, 8 September 2026].

- *Execution.* A swap subject to the trade-execution requirement, once a facility has made it "available to trade" by a self-certified determination weighing buyers and sellers, frequency, volume, participants, spreads and resting quotes (37.10), must be executed on a swap execution facility. Unless it is a block trade, it must go through an order book or through a request-for-quote system that sends the request "to no less than three market participants" who are not affiliates, operating with an order book (37.9(a)). Other swaps may trade by any method the facility offers (37.9(c)).

So the rules, not only the contracts, separate standard indices from single names and bespoke risk. *Which swaps are available to trade*. The CFTC's table of MAT filings lists the determinations of October 2013 to March 2014 (Javelin, trueEX, TW SEF, MarketAxess, Bloomberg), for certain interest-rate swaps and CDS. It also lists TW SEF's voluntary submission of 22 May 2023, approved on 7 July 2023 [cftc-smatd-filings, "Swaps Made Available to Trade Determination" table]. That submission recalls its 2013 determination: USD and GBP LIBOR fixed-to-floating swaps in the 2- to 30-year benchmark tenors. It sets out that the CFTC's August 2022 amendment to 50.4 moved the clearing requirement from LIBOR swaps to SONIA and SOFR overnight index swaps of 7 days to 50 years, and it proposes MAT status for SOFR and SONIA OIS [twsef-mat-2023, TW SEF letter of 22 May 2023, pp. 1–3].

It also notes that "on an RFQ platform like TW SEF, there are no resting bids and offers", and measures liquidity there by the number of providers streaming quotes. That is one operator's description of its own protocol. The share of swap trading done by dealer-to-client RFQ is not measured in these sources. RFQ does not dominate U.S. investment-grade corporate bonds in the survey's figure: 19 percent was electronically facilitated, mostly by RFQ, in the estimate of a July 2016 industry report (p. 2); RFQ is one protocol among voice negotiation, runs and lists.

That products trade by RFQ because they are less standardised and active is not a finding: the survey treats why fixed-income trading remains dealer-oriented and OTC as an open question (pp. 37–38). One corporate-bond sample does not describe every fixed-income market.

**See also.** bond transparency rules; bond transparency and trading costs; dealer inventory risk; ETF creation and redemption (corporate bond ETFs and dealer inventories); portfolio trading.

### Rule 17.51 Bond transparency rules: U.S. TRACE reporting and dissemination, and EU post-trade deferrals (as of October 2026)

**Statement.** Transparency rules separate *reporting* to a regulator or its facility from *public dissemination*, and *post-trade* publication of completed trades from *pre-trade* publication of quotes or orders.

- *U.S. reporting.* Under FINRA Rule 6730 as displayed on 15 October 2026, each member that is a party to a transaction in a TRACE-eligible security must report it "as soon as practicable, but no later than within 15 minutes of the Time of Execution", with stated exceptions, some of which allow 60 minutes [finra-rule-6730, Rule 6730(a)].
- *U.S. dissemination.* Under Rule 6750, FINRA disseminates information on all transactions in TRACE-eligible securities, Rule 144A transactions included, immediately upon receipt, except that certain collateralised mortgage obligation trades are disseminated in aggregate weekly and monthly; individual trades in on-the-run nominal coupon Treasuries are disseminated at the end of the day; and other Treasuries, foreign sovereign debt, CMBS, CDOs, some CMO trades and certain affiliate, list-offering and takedown transactions are not disseminated, FINRA being allowed to publish aggregated statistics [finra-rule-6750, Rule 6750(a)–(d) and .01].
- *EU post-trade deferrals.* ESMA's manual of 12 May 2026 gives worked examples of the bond regime that applies from 2 March 2026 under amended MiFIR Article 11 and RTS 2 Article 8a. For a trade in a group-1 (most liquid sovereign) bond on a trading venue, a medium-size trade in category 1 is published within 15 minutes, and a large one in category 3 has its price published within one trading day and its volume within one week; competent authorities may add supplementary deferrals for sovereign bonds, omitting volume for up to six months beyond the initial deferral, or publishing weekly aggregates [esma-transparency-manual-2026, secs. 4.2.8 and 4.2.8.2, pp. 262–272].

**Conditions.** Dates and status:

- *History, United States.* TRACE reporting of corporate bonds began in July 2002 within 75 minutes, with immediate public dissemination only for about 500 large investment-grade and 50 high-yield bonds; dissemination was widened in 2003–2005, and the reporting lag cut to 45 (October 2003), 30 (October 2004) and 15 minutes (July 2005), with all registered corporate bonds disseminated by January 2006. ABS and MBS reporting began in May 2011 without dissemination, which was phased in for TBAs (2012), specified pools (2013), Rule 144A bonds (2014), ABS (2015) and CMOs (2017). Treasury reporting began in July 2017, with no public dissemination at the time [wide-bsv-fi-survey, Table 2, p. 9]. Disseminated sizes were capped, for example "5MM+" for investment-grade and "1MM+" for high-yield corporate trades (p. 10, footnote 10); current caps may differ.
- *Changes and proposals, United States.* FINRA's filing SR-FINRA-2024-004 proposed reducing the 15-minute reporting limit to one minute, with exceptions for firms with de minimis activity and for manual trades; SR-FINRA-2025-008, by its title, maintains "the currently effective 15-minute outer limit" for the securities covered by the earlier filing, and the rulebook lists both as effective 8 June 2026 [finra-sr-2024-004, filing pages]. SR-FINRA-2026-009, approved by the SEC on 24 July 2026, expands the indicator that suppresses from dissemination qualifying same-day, same-price, same-security transactions between a member and its *non-member* affiliate to transactions between *member* affiliates, renaming it the Affiliate–Principal Transaction indicator; FINRA said it would set an implementation date at least 12 months after approval and allow voluntary earlier use [sec-34-105991, pp. 47909–47911]. The rulebook lists it as effective 2 August 2027.
  Until then the presently effective exclusion is the non-member-affiliate one; the displayed text of Rule 6750(d)(1), which already reads "Affiliate—principal transaction indicator", is the future wording. The proposal of the SEC's Fixed Income Market Structure Advisory Committee to delay dissemination of very large trades by 48 hours was a proposal, not a rule (survey, p. 29).
- *European Union.* Fixed-income securities traded without pre- or post-trade transparency in Europe until MiFID II applied in January 2018, which required reporting of bond transactions to a national regulator and real-time publication for liquid, non-block trades, and pre-trade publication of bids and offers for small trades in a limited set of about 200 sovereign bonds (survey, p. 30; historical). On 10 October 2025 ESMA stated that the Commission's amending RTS 1 and RTS 2 would apply most revised bond transparency requirements from 2 March 2026 following publication in the Official Journal, which had not yet happened, and invited market participants to anticipate them [esma-mifir-review-statement-2025, p. 4]. *Level 1 and RTS 2 as amended*.
  - *Pre-trade.* Regulation (EU) 2024/791 confines venue pre-trade transparency for bonds, structured finance products and emission allowances to central limit order books and periodic auctions. Venues using those systems "shall make public current bid and offer prices and the depth of trading interests" continuously during trading hours, calibrated by trading system [eu-2024-791, Art. 1, point (5), replacing MiFIR Art. 8(1)–(2) and deleting Art. 8(3)–(4)]. Request-for-quote and voice systems are thus no longer named for bonds.
  - *Deferral categories.* Bond deferrals are organised in five categories: medium size liquid (1), medium size illiquid (2), large liquid (3), large illiquid (4) and very large (5). When a deferral lapses, "all the details of the transactions on an individual basis shall be published". Sovereign-debt supplementary deferrals are limited to the issuing Member State's instruments, with volume omitted for up to six months (Art. 1, point (9), amending MiFIR Art. 11(1a) and (3)).
  - *Maximum durations.* The amending RTS 2 sets the maxima [eu-2025-1246, Art. 1, point (7), inserting Art. 8a(1) into Delegated Regulation 2017/583]: 15 minutes for price and volume in category 1; end of the trading day in category 2; price by the end of the next trading day and volume within one week in category 3; price by the end of the second trading day and volume within two weeks in category 4; four weeks for both in category 5. It also requires post-trade publication "within five minutes" (amended Art. 7(4)). It was published in the Official Journal on 3 November 2025 (OJ L 2025/1246), and its Article 1 applies from 2 March 2026. The size and liquidity thresholds of Annex III, Table 2.6, were not transcribed.

These agree with ESMA's worked examples above.

- *United Kingdom*. The FCA's PS24/14 (November 2024) replaces the inherited regime, whose "overly lengthy publication deferrals" it criticises, with four size groups for bonds: real-time publication up to the first threshold, then deferrals of price and size together to the end of the trading day after execution, two weeks, and three months for the largest trades; it considered publishing price after four weeks and volume after three months for the largest group but chose "both after three months" (PDF pp. 42–43); most of its rules take effect on 1 December 2025 (PDF p. 14) [fca-ps24-14, PS24/14, chaps. 2–5, PDF pp. 12, 14 and 42–43]. The thresholds by issuer, issue size and maturity (Table 8) and the Handbook text (MAR 11) were not transcribed.

**Practice.** Interpretation: a TRACE report is a regulatory obligation of the dealer; what the public sees is set by Rule 6750, so the reported and the disseminated sets of U.S. trades differ, most visibly for Treasuries. In the EU the deferral category, not only the instrument, decides when price and volume become public. Trade-size caps and deferrals let dealers "hedge or sell a portion of [their] position before disclosure", the reason the survey gives for U.S. regulators' initial delays (p. 29).

**Pitfalls.** TRACE is FINRA's reporting facility for all TRACE-eligible securities, of which corporate bonds are one class; reports go to FINRA, and public dissemination is a separate step with exceptions. Whether TRACE improved post-trade transparency is an evidence question (bond transparency and trading costs). No EU statement of aims supports the view that European transparency rules aim to make fixed-income liquidity easier to measure without forcing all products into a CLOB. The 2002–2017 history is from a survey, not from the rule filings themselves.

**See also.** fixed-income market structure; bond transparency and trading costs; best execution.

### Evidence 17.52 Bond transparency and trading costs: U.S. post-trade transparency evidence, the open question of large trades, and what data allow clients to measure

**Statement.** Edwards, Harris and Piwowar estimate average transaction costs by trade size for every U.S. corporate bond that traded more than nine times between January 2003 and January 2005, from a complete record of OTC secondary trades. Costs fall significantly with trade size; highly rated, recently issued and near-maturity bonds cost less; costs are lower for bonds with transparent trade prices and drop when TRACE starts to disseminate their prices; and the results "suggest that public traders benefit significantly from price transparency" [wide-edwards-harris-piwowar, abstract]. In the survey's account, their estimates fall from 75 basis points for $5,000 trades to under 10 basis points for $1 million or more, the reverse of the equity pattern [wide-bsv-fi-survey, p. 5].

Three studies with different designs, a stratified random sample of BBB bonds phased in in April 2003 (Goldstein, Hotchkiss and Sirri), the disseminated-against-undisseminated comparison of Edwards, Harris and Piwowar, and a difference-in-differences on insurance-company trades (Bessembinder, Maxwell and Venkataraman), all conclude that public reporting through TRACE is associated with significant reductions in customer execution costs; the last also suggests a liquidity externality to bonds not yet reported. Similar associations are reported for CDS, TBA MBS and Rule 144A bonds (survey, p. 30).

**Conditions.** U.S. evidence from 2002–2017 samples; Rule 2232 as displayed on 15 October 2026; other regulatory facts as of the survey (2019). These are associations in U.S. corporate bond, CDS, TBA and Rule 144A samples from the phase-in periods, as summarised by a survey. The survey states the limits: much of the evidence concerns execution costs of completed trades, which may reflect order splitting and do not capture the cost and difficulty of executing large orders; the optimal degree of post-trade transparency "has not been established"; dealers, and in some cases large buy-side investors, are often critical of greater transparency; and the regime optimal for one fixed-income market may not be optimal for another (pp. 38–39).

Back, Liu and Teguia's model, as the survey reports it, has trade disclosure conveying the dealer's information and reducing its rents when it disposes of inventory (p. 29), a model prediction. Pre-trade transparency in bonds is a matter of open questions and proposals in the survey, such as the public display of customer interest advocated by Harris (2015), not of evidence (p. 37).

**Practice.** What data let clients measure, as of the survey: TRACE and MSRB transaction data; an enhanced TRACE file for researchers with masked dealer identities and actual sizes; aggregators that combine trades, dealer runs and platform quotes for buy-side institutions (pp. 7–8 and 10). Fixed-income markets have no analogue of SEC Rules 605 and 606 on execution quality and order routing, and without consolidated quotations many participants cannot see the best available prices or identify trade-throughs; FINRA and the MSRB adopted best-execution rules (pp. 31–32).

Since 14 May 2018 FINRA Rule 2232(c) requires a confirmation to show the mark-up or mark-down when a member trades as principal in a corporate or agency debt security with a non-institutional customer and makes offsetting trades of at least that size on the same trading day, looking through non-arm's-length affiliate trades; (d) exempts functionally separate trading desks and same-day fixed-price-offering sales [finra-rule-2232, paras. (c)–(f)]; [finra-rn-17-08, effective date and MSRB paragraph]. FINRA's notice reports a parallel MSRB rule for municipal securities, with the same effective date. Measure costs by trade size, rating and segment, and state which trades are visible: completed trades only, capped sizes, deferred or undisseminated classes.

**Pitfalls.** Publishing pre-trade interest in illiquid bonds can reveal inventory needs and increase dealer risk, so policy balances monitoring against exposing liquidity. The sources give this as dealers' argument and an open question, and as the survey's reason for U.S. reporting delays; none measures the effect of pre-trade transparency on bond liquidity. That electronification changes execution quality and the governance of trading desks is not established causally: O'Hara and Zhou report effects of electronic RFQ trading in their abstract, and nothing here concerns desk governance. Transparency is not shown here to help or to harm liquidity in general.

**See also.** bond transparency rules; fixed-income market structure; spread measures; best execution; transaction cost analysis.

# 18 Commodity futures: storage, curves, hedging and options

## 18.1 Hedging with futures

**Definition 18.1 Futures hedges and basis: short and long hedges, the basis, basis risk, cross hedging, contract choice and stack and roll**

**Statement.** Hull treats *hedge-and-forget* strategies: a futures position is taken at the start and closed at the end, with no adjustment in between [s7-hulloptions, chap. 3, "Hedging Strategies Using Futures", p. 70]. A *short hedge*, a short futures position, suits a hedger who owns or will own an asset and expects to sell it; a *long hedge* suits one who will buy an asset and wants to lock in a price now (sec. 3.1, pp. 71–72). The *basis* is "spot price of asset to be hedged − futures price of contract used"; an increase is a strengthening, a decrease a weakening (sec. 3.3, p. 76). With spot $S_i$, futures $F_i$ and basis $b_i = S_i - F_i$ at the start $t_1$ and end $t_2$, the effective price obtained or paid with a hedge is

$$S_2 + F_1 - F_2 = F_1 + b_2$$

for both short and long hedges (p. 77); $F_1$ is known at $t_1$, and the uncertainty in $b_2$ is *basis risk*. *Cross hedging* uses a futures contract on a different asset (p. 77; sec. 3.4, p. 79); with $S_2^*$ the price at $t_2$ of the asset underlying the futures, Hull writes the hedged price $S_2 + F_1 - F_2$ as

$$F_1 + (S_2^* - F_2) + (S_2 - S_2^*),$$

whose two last terms are "the two components of the basis": $S_2^* - F_2$ is the basis that would exist if the hedged asset were the futures' underlying (the futures-underlying basis), and $S_2 - S_2^*$ is "the basis arising from the difference between the two assets" (p. 77). Both expressions assume a hedge ratio of 1.0, one futures unit per unit of exposure, which Hull calls natural when the two assets are the same and "not always optimal" in a cross hedge (p. 79). Interpretation, derived here: with a hedge ratio $h$, the size of the futures position per unit of exposure taken in the direction that offsets it (short for a seller, long for a buyer), the effective price received or paid per unit is

$$S_2 + h(F_1 - F_2),$$

which reduces to $F_1 + b_2$ at $h = 1$; $h = 0$ is no hedge, and a negative $h$ is a futures position in the same direction as the exposure (long futures against a long exposure). Whether that adds or removes risk depends on how the two prices co-move, not on the sign of $h$: when $\mathrm{Cov}(\Delta S, \Delta F) < 0$ the minimum-variance ratio of the next entry is negative and reduces variance. Prices are per unit of each asset, and $h$ is in futures units per unit of exposure, so it carries any conversion between the two assets' units (none is needed in Hull's jet-fuel and heating-oil example, both in gallons).

**Conditions.** Concepts timeless; contract examples as of 2021. Hedges are imperfect because the hedged asset may differ from the futures' asset, the date of purchase or sale may be uncertain, and the futures may have to be closed before its delivery month (pp. 75–76). Basis changes can help or hurt: an unexpected strengthening improves a short hedger's position and worsens a long hedger's (p. 77). Hull defines the basis as spot minus futures but notes the reverse definition is sometimes used, especially for financial futures (p. 76, n. 2); the sign convention must be stated. As a rule of thumb, Hull chooses the delivery month closest to, but after, the hedge's end, because basis risk grows with the gap and prices can be erratic in the delivery month; the rule assumes enough liquidity (p. 78).

When the hedge outlasts all liquid contracts, *stack and roll* closes one contract and opens a later one, repeatedly; Metallgesellschaft's rolled hedges in the early 1990s produced cash outflows before the offsetting gains on its supply contracts, a liquidity problem (sec. 3.6, pp. 89–90).

**Practice.** Hull's examples: an oil producer shorting August futures at \$49 against a spot of \$50 realises about \$49 a barrel whether spot ends at \$45 (a \$4 futures gain) or \$55 (a \$6 loss) (p. 71); $S_1, F_1, S_2, F_2 = 2.50, 2.20, 2.00, 1.90$ give $b_1 = 0.30$, $b_2 = 0.10$ and an effective price of 2.30 (p. 77); the yen hedge gives $1.0800 + (-0.0050) = 1.0750$ cents and \$537,500 for 50 million yen (Example 3.1, pp. 78–79); the oil purchase gives $48.00 + 0.90 = 48.90$ and \$978,000 (Example 3.2, p. 79); the rolled oil hedge earns $0.80 + 0.50 + 0.40 = 1.70$ a barrel against a \$3 fall in spot (p. 90).

**Pitfalls.** Hull's oil example is a short hedge only; his long-hedge example is copper (pp. 71–72). A mirror-image long hedge on the oil figures, with a loss of 4 and a gain of 6, is arithmetically consistent (a buyer paying the August spot and losing or gaining on the futures also ends near \$49), but it is not Hull's. The three causes of imperfect hedges are Hull's (pp. 75–76). Whether hedging adds value to a firm, and time-varying hedge ratios, are outside these passages.

**See also.** minimum-variance hedge ratio; derivative contracts (convergence); cost-of-carry forward price (why futures and spot differ before delivery); arbitrage (Harris's basis).

**Further reading.** *Why firms hedge*. Froot, Scharfstein and Stein's working paper argues that "if external sources of finance are more costly to corporations than internally generated funds, there will typically be a benefit to hedging: hedging adds value to the extent that it helps ensure that a corporation has sufficient internal funds available to take advantage of attractive investment opportunities" [ec-froot-scharfstein-stein-1993, NBER WP 4084, May 1992, abstract]. With increasing marginal costs of external finance and fixed investment opportunities, firms "should always fully hedge" (sec. 4, p. 12). When investment opportunities move with the hedged risk, the optimal hedge ratio $h^*$ falls below one if they are positively correlated with internal funds. It exceeds one, overhedging, if they are negatively correlated, and can be negative when investment is very sensitive to the risk (eq. (20), p. 14).

So the variance-minimising ratio of the next entry is the value-maximising one only when investment needs are unrelated to the hedged price (Interpretation). *Evidence, abstracts only*. Tufano's database of North American gold miners finds "little empirical support" for theories that see risk management as maximising shareholder value. Firms whose managers hold more options manage less gold-price risk, and those whose managers hold more stock manage more, suggesting that managerial risk aversion plays a part [ec-tufano-1996, abstract]. Jin and Jorion, for 119 U.S. oil and gas producers in 1998–2001, verify that hedging lowers stock-price sensitivity to oil and gas prices but find that it "does not seem to affect" market values in that industry [ec-jin-jorion-2006, abstract]. So, on their abstracts, the two industry studies do not support the value channel of the theory above. Dynamic hedge ratios require separate estimation.

**Model 18.2 Minimum-variance hedge ratio: derivation, hedge effectiveness, number of contracts, daily settlement and tailing, and beta hedges with index futures**

**Statement.** The *hedge ratio* is the size of the futures position over the size of the exposure [s7-hulloptions, chap. 3, "Hedging Strategies Using Futures", sec. 3.4, p. 79]. Ignoring daily settlement, with $\Delta S$ and $\Delta F$ the changes in spot and futures prices over the life of the hedge and $\Delta S = a + b\,\Delta F + \epsilon$ approximately linear, a hedger long the asset and short $h$ futures per unit holds $\Delta S - h\Delta F = a + (b - h)\Delta F + \epsilon$, whose variance is minimised at $h = b$ (p. 80):

$$h^* = \rho\,\frac{\sigma_S}{\sigma_F}, \qquad N^* = \frac{h^* Q_A}{Q_F},$$

with $\sigma_S, \sigma_F$ the standard deviations of $\Delta S, \Delta F$, $\rho$ their correlation, $Q_A$ the units hedged and $Q_F$ the units per contract (eqs. (3.1)–(3.2), pp. 80–81). Hedge effectiveness, the share of variance removed, is the regression $R^2$, equal to $\rho^2$ (p. 81).

**Notation.** $\Delta S, \Delta F$ price changes per unit, in the same price units; $h$ dimensionless; $N^*$ contracts. Interpretation, derived here: for a hedged position $V(h) = \Delta S - h\Delta F$, $\operatorname{Var} V(h) = \sigma_S^2 - 2h\rho\sigma_S\sigma_F + h^2\sigma_F^2$, a convex quadratic minimised where $d \operatorname{Var} V(h)/dh = 0$, at $h^* = \rho\sigma_S/\sigma_F = \operatorname{Cov}(\Delta S, \Delta F)/\operatorname{Var} \Delta F$, the OLS slope; the minimum is $\sigma_S^2(1-\rho^2)$. A short futures position is $h > 0$ against a long exposure; a hedger short the asset buys $h^*$. The sign of $h^*$ is the sign of $\rho$: with $\operatorname{Cov}(\Delta S, \Delta F) < 0$, $h^* < 0$ is a long futures position against a long exposure, and it still reduces the variance to $\sigma_S^2(1-\rho^2)$. With $\sigma_S = \sigma_F = 1$ and $\rho = -0.6$, $h^* = -0.6$ leaves a variance of 0.64 against 1 unhedged, while $h = +0.6$ raises it to 2.08.

**Conditions.** The estimate assumes the future resembles the past; $\rho, \sigma_S, \sigma_F$ are usually estimated from equal non-overlapping intervals, ideally as long as the hedge (p. 81). The $R^2 = \rho^2$ result is in-sample: it describes the estimation period and assumes the linear relation and its parameters hold over the hedge. *Daily settlement.* Futures make a series of one-day hedges; with $\hat\sigma_S, \hat\sigma_F, \hat\rho$ for percentage one-day changes, the hedge from a regression of percentage changes is $\hat h = \hat\rho\hat\sigma_S/\hat\sigma_F$ and $N^* = \hat h V_A/V_F$, with $V_A = SQ_A$ and $V_F = FQ_F$; equivalently the one-day price-change hedge is $\hat\rho\hat\sigma_S S/(\hat\sigma_F F)$ (eq. (3.3), p. 83). Price-change and return regressions therefore differ by the ratio $S/F$ and must not be mixed.

*Tailing*: allowing for interest on daily settlement, Hull divides $N^*$ by 1.05 for a hedge with one year left at 5 percent (p. 83); interpretation: the general factor is the growth of \$1 over the remaining life at the relevant rate, which equals $1 + 0.01r$ only for one year at simple rate $r$ percent.

**Practice.** Recomputing Hull's Example 3.3 from his Table 3.2 data: $\sigma_F = 0.0313$, $\sigma_S = 0.0263$, $\rho = 0.928$, $h^* = 0.778$ and $N^* = 0.778 \times 2{,}000{,}000/42{,}000 = 37.0$, so 37 contracts (p. 82). The same $h^*$ results from minimising the sample variance of $\Delta S - h\Delta F$ numerically and equals the OLS slope; the variance falls to 13.8 percent of unhedged, so 86.2 percent is removed, equal to $\rho^2$; a ratio of 1 leaves 20.9 percent (Figure 18.1). Hull's one-day example gives $0.8 \times 2{,}200{,}000/54{,}600 = 32.2$ contracts (p. 83).

*Index futures*: to hedge a portfolio of value $V_A$ and beta $\beta$, short $N^* = \beta V_A/V_F$ contracts, assuming the futures matures close to the hedge; to move beta from $\beta$ to $\beta^*$, short $(\beta - \beta^*)V_A/V_F$ if $\beta > \beta^*$, otherwise go long $(\beta^* - \beta)V_A/V_F$ (eq. (3.5), pp. 86 and 88). Hull's example: $1.5 \times 5{,}050{,}000/252{,}500 = 30$ contracts, gaining $30 \times (1{,}010 - 902) \times 250 = \$810{,}000$ if the index falls to 900; the hedged position grows at about the risk-free rate (Table 3.4, pp. 86–87).

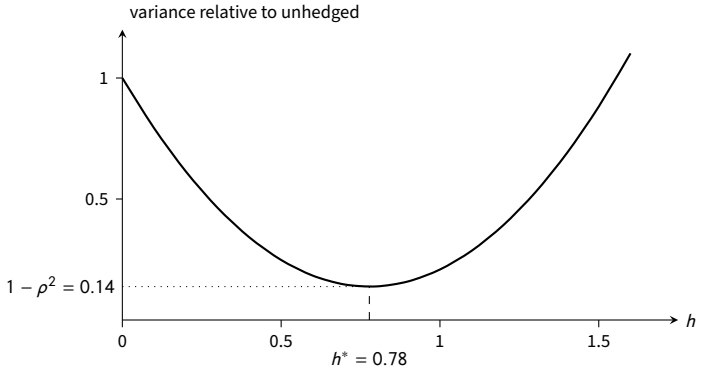



Figure 18.1: Variance of $\Delta S - h\Delta F$ relative to the unhedged variance for Hull's jet-fuel and heating-oil data (Table 3.2, p. 82), computed for this entry; the minimum is at $h^*$, as in Hull's schematic Figure 3.3 (p. 81).

*Static against rolling ratios out of sample.* A U.S. Gulf Coast jet-fuel spot exposure (EIA daily, from April 1990) is hedged weekly with NYMEX heating-oil or WTI contract 1, dropping the weeks in which contract 1 rolls; a ratio estimated once on the first 260 usable weeks is compared with one re-estimated on the previous 104 weeks. After week 260, the out-of-sample variance reduction is 49.3 percent static against 53.8 percent rolling with heating oil (1997–2024), and 25.3 against 28.1 percent with WTI. Rolling helps in both halves of the evaluation period, more after 2010 with heating oil (73.2 against 79.8 percent); the first five years' static ratio (1.31 for heating oil) was high relative to the full-sample 1.02 [eia-jet-fuel-spot, spot series EER_EPJK_PF4_RGC_DPG]. Bivariate GARCH ratios are untested, and weekly spot changes are not an airline's actual exposure.

**Pitfalls.** The simplified tailing rule "$N^*/(1+0.01r)$" holds only for one year remaining, as in Hull's example. Setting $h = b$ minimises the standard deviation only with $\epsilon$ uncorrelated with $\Delta F$, which OLS guarantees in sample. Hull's index-hedge table assumes a predictable dividend yield, a constant risk-free rate and perfect correlation between portfolio and index returns, so "in practice … the hedge works rather less well" (p. 87, n. 5); CAPM supplies an expected return there, not a guarantee. Hull's reasons for index hedging (confidence in stock selection, short-term protection without selling) are motivations, not evidence that such hedges add value (pp. 87–88).

**See also.** futures hedges and basis; factor risk model (beta as a factor exposure); duration and convexity.

## 18.2 Curves and inventories

**Definition 18.3 Contango, backwardation and ending stocks: storage and financing costs, scarcity premia, seasonal crop curves, the stocks-to-use ratio, and roll yield stated carefully**

**Statement.** Gliner: commodities are driven by supply (producers, costs, yield, weather, politics) and demand (population, development, alternative energy) [s7-gliner, chap. 10, secs. "Supply Drivers" and "Demand Drivers", PDF pp. 293–296]. *Ending stock* "effectively measures the inventory of a particular commodity"; crops are harvested once or twice a year while demand is continuous, and the stock carried over becomes the next season's beginning balance (sec. "Ending Stock", p. 296, Figure 10.1).

*Contango* is near-term futures trading below later ones, from storage, transport and the opportunity cost of capital (gold in a vault). *Backwardation* is near-term futures at a premium, as in a supply shortage (cocoa during an Ivory Coast export stoppage) or just before a harvest when stocks are depleted. A crop curve can show both, contango after harvest and backwardation before the next (sec. "Contango and Backwardation", pp. 297–298, Figure 10.2, corn as of 30 September 2013).

**Notation.** Balance sheet: beginning stocks + production + imports − consumption − exports = ending stocks; stocks-to-use = ending stocks / total use. Full carry: $F = Se^{(r+u-y)T}$ with storage cost $u$ and convenience yield $y$ (cost-of-carry forward price).

**Conditions.** The identity "ending stocks = beginning stocks + production − consumption" holds for the world, not for a country, which also has trade; with imports of 5 and exports of 12 on a balance of 10, 100 and 95, ending stocks are 8, not 15. Low stocks relative to use support prices and make them sensitive to small supply changes, as in the storage model; calling them the most important driver of commodity prices overstates that relation.

**Practice.** The monthly WASDE report [physical-usda-wasde-2026, catalogue entry] gives USDA's U.S. and world supply-and-demand balances. Contango as the normal state fits monetary metals with large above-ground stocks, such as gold; it is not a general rule for storable commodities, which can be backwardated when inventories are scarce.

**Pitfalls.** Contango gives a negative and backwardation a positive roll yield only for the roll component holding the curve fixed, not for total returns, and the sign says nothing on its own about a risk premium: whether backwardation reflects scarcity or compensation for risk is a separate question. *Own bounded test*. Data: EIA's public NYMEX series (contracts 1, 2 and 4, which EIA stopped updating on 5 April 2024) and EIA's weekly inventories. Backwardation is $B_t = \ln(C_{1,t}/C_{2,t})$ at complete month-ends. The inventory signal is the log deviation from the same week's mean over the previous five years, taken from the latest weekly report treated as public by settlement on $t$.

The return is next month's return from holding contract 2, and each pair of month-ends is kept only if contract 2 at $t$ is contract 1 at $t+1$ under NYMEX's last-trading-day rules. The hypotheses and pass rules were written into the script before its first run. They were not preregistered.

- *Scarcity: supported.* Backwardation rises as inventories fall. WTI, August 1987 to February 2024 (438 months): $\beta = -0.111$, Newey–West $t = -5.9$ ($C_1/C_4$: $t = -6.4$). Henry Hub, January 2015 to February 2024 (110 months): $\beta = -0.114$, $t = -4.3$. Both survive a two-week reporting lag.
- *Premium: not supported.* On the full price-only samples (WTI, January 1985 to February 2024, 470 months; Henry Hub, January 1994 to February 2024, 362 months), the slope of next-month returns on $B_t$ is +0.02 ($t = 0.1$) for WTI and −0.05 ($t = -0.3$) for gas. Inventories add nothing. Out-of-sample $R^2$ against the historical mean is −1.8 percent for WTI (Clark–West $t = -0.4$) and −0.8 percent for gas ($t = -1.4$, from January 2004).

Settlement-to-settlement figures are closing-price associations, since the signal and the entry use the same settlement. An *executable variant* forms the signal at settlement $t$ and trades the next day's settlement in the same contract, with two ticks per round trip (fills at settlement and the cost are assumptions; for Henry Hub, NYMEX's TAS order type covered the spot and the second through seventh months by May 2019, NYMEX Submission 19-158; whether TAS was available for Henry Hub earlier in the sample is unverified). Exploratory only, added after the first run and not part of the test: in this variant WTI's long-backwardation, short-contango rule earned +0.92 percent a month (Newey–West $t = 2.3$; +1.91 against −0.16 percent), while gas earned +0.58 ($t = 0.9$). Two WTI and three gas months exceed ±50 percent and are retained in every estimate.

*A third commodity*. The same code, imported unchanged, was applied to NYMEX heating oil (EIA's "New York Harbor No. 2 Heating Oil Future", contracts 1 and 2) with U.S. distillate stocks (WDISTUS1, same Wednesday release as crude, so the crude lag of six days). The hypotheses, timing rules and acceptance criteria were recorded before the data were downloaded, though not preregistered. The expiry rule assumed (last business day of the month before delivery) is EIA's own statement of the contract it reports: "For gasoline, heating oil, and propane each contract expires on the last business day of the month preceding the delivery month.

Thus, the delivery month for Contract 1 is the calendar month following the trade date" [`eia-futures-definitions`, Futures Prices (NYMEX), definitions, "Contract 1"]. This is EIA's description, unconfirmed against NYMEX's rule text. The roll check agrees with it in six of seven five-year blocks (the seventh, 2024, has one roll), but it runs on the dates on which EIA reports both contracts, so it is consistent with the mapping and does not establish the exact expiry date.

*Month-ends are the last common observation, not always the last trading day*: before about 2007 EIA reports contract 2 on only 100–190 days a year, so 79 of the 361 selected dates fall before the last date in the month on which EIA reports either contract, 12 by more than five days (at most 21, 9 August 1996); the gap is in the panel.

- *Scarcity: supported.* February 1994 to February 2024 (361 months), $\beta = -0.110$, $t = -5.3$, and −0.104 ($t = -5.3$) with a two-week lag.
- *Premium: not supported.* With the log held-contract return the protocol specifies, the slope is +0.33, $t = 1.55$, below the threshold of 2. Out-of-sample $R^2$ is −0.95 percent from February 2004, with Clark–West $t = 0.31$.

*Return definition*. A simple-return specification gives slope +0.36 ($t = 1.67$) and out-of-sample $R^2 = -0.9$ percent (Clark–West $t = 0.37$) from simple returns, which the imported WTI and gas code uses, although the heating-oil protocol had recorded log returns. Scarcity does not use returns and is unchanged. An exploratory regression, chosen after the results were seen and not part of the verdict, uses only the 243 month pairs whose two dates are both the month's last observed date: its slope is +0.07, with $t = 0.20$. The executable variant does not apply: heating oil expires at the month-end itself, so the contract that is $C_2$ at $t$ is already $C_1$ the next day. EIA's series keeps one label across the 2013 change from heating oil to ultra-low-sulphur diesel; whether and how EIA spliced the two contracts is unverified.

*Hedging pressure*. The protocol was written before the positions data were downloaded. Hedging pressure is commercial traders' net short share in the CFTC's legacy futures-only report for WTI (contract market code 067651, whose name changed in 2000 and to "WTI-Physical" in February 2022), taken from the latest report whose Tuesday positions date is at least three days before the month-end [`cftc-cot-history`, legacy futures-only historical files, 1986–2024]. Over 458 months, January 1986 to February 2024, it averaged +0.08 (net short) and does not predict the next month's held-contract return: slope −0.008, Newey–West $t = -0.17$; out-of-sample $R^2 = -1.1$ percent, Clark–West $t = -0.28$. It is nearly uncorrelated with the basis (−0.07), and adding it to the basis changes neither.

Hedging pressure is not supported on this rule; release delays such as government shutdowns are not modelled. Limits: three U.S. energy contracts; a monthly horizon; no metals or agriculture. EIA's release tables cover only 2025–2026 [`eia-weekly-release-schedules`, holiday tables], so availability rests on a conservative convention: six days after the week ending (crude), seven (gas), and eleven over Christmas and New Year. Henry Hub's pre-1997 expiry rule is inferred from the data, not from a rulebook.

**See also.** cost-of-carry forward price; futures hedges and basis; commodity indices; agricultural markets.

# 19 Physical commodity markets: delivery, price discovery and trading

## 19.1 Crude oil, metals and agricultural markets

**Definition 19.1 Crude oil markets: grades, WTI and Brent, OPEC and spare capacity, chokepoints and pipelines, and the 3:2:1 crack spread, as of 2012–2013**

**Statement.** Oil supplied 33 percent of world energy demand in 2012, coal 30 and natural gas 24 [s7-gliner, chap. 10, sec. “Energy”, PDF p. 301]. Quality is measured by API gravity (higher means lighter; below 10 it sinks in water; 25 or less is “heavy”) and sulphur content (sweet against sour), light sweet grades being the most valuable (pp. 301–302). OPEC has 12 members producing over 35 percent of world output and holding more than 70 percent of reserves; Saudi Arabia has the largest spare capacity, so “the Call on OPEC” usually means Saudi production rises (pp. 302–305, Table 10.4, OPEC-12 quota 30 million barrels a day as of September 2013).

WTI, the U.S. benchmark, is delivered at Cushing, Oklahoma, and trades on NYMEX (CL), 1,000 barrels a contract; Brent, from North Sea fields, “is used to price more than half of the world’s internationally traded crude oil” and trades on ICE (CO), 1,000 barrels; Brent’s premium over WTI in Gliner’s time reflected North Sea depletion, seaborne supply risk and pipeline constraints at Cushing (pp. 305–307). World consumption was 90 million barrels a day in 2012; the Strait of Hormuz carried 17 million barrels a day in 2011, “about 20 percent of global traded oil and 35 percent of seaborne oil”, Malacca 14 million, Bab el-Mandeb 3 million and the Suez Canal 2 million; the Druzhba pipeline runs over 2,300 miles and carries up to 1.4 million barrels a day (pp. 308–310).

The *crack spread* is the refining margin; the 3:2:1 spread (three barrels of crude against two of gasoline and one of heating oil) is the most popular because gasoline output is almost double heating oil’s, with product prices in cents a gallon converted at 42 gallons a barrel (sec. “Crack Spread”, pp. 311–312).

**Notation.** 3:2:1 crack per barrel $= \left[2 \times (\text{RBOB} \times 0.42) + 1 \times (\text{HO} \times 0.42) - 3 \times \text{CL}\right]/3$, products in cents per gallon and crude in dollars per barrel; Gliner’s worked example gives $27.95 a barrel (p. 312), and crude at $80, gasoline at $2.40 and heating oil at $2.70 a gallon give $25.00.

**Conditions.** All shares, capacities, membership and contract details are Gliner’s, as of 2011–2013. Benchmark composition and trade flows may have changed since. *Membership.* OPEC’s home page lists twelve Member Countries in its navigation: Algeria, Congo, Equatorial Guinea, Gabon, Iran, Iraq, Kuwait, Libya, Nigeria, Saudi Arabia, the United Arab Emirates and Venezuela [opec-home-2026, home page navigation].

*The UAE.* Four kinds of evidence, kept apart. (i) The announcement: the UAE’s state news agency reported on 28 April 2026 that the UAE “announced its decision to exit the Organisation of the Petroleum Exporting Countries (OPEC and OPEC+), effective 1 May 2026”, having joined in 1967 through Abu Dhabi [wam-uae-opec-2026, pp. 1–2].

(ii) EIA’s classification: its analysis of 23 June 2026 treats the UAE as having left effective 1 May, and puts OPEC’s 2025 crude output, UAE included, at 28.0 million barrels a day, 35 percent of the world total, or 31 percent without the UAE, and OPEC+ at about 46 percent, about 42 without it [eia-uae-opec-2026, web page]; its live glossary lists eleven current members without the UAE [eia-opec-glossary, glossary entry].

(iii) OPEC’s own pages: the site’s navigation, on the home page and on a press release of 3 May 2026, still lists the UAE among twelve Member Countries, and that release, on a June production adjustment of 188 thousand barrels a day by seven OPEC+ countries (Saudi Arabia, Russia, Iraq, Kuwait, Kazakhstan, Algeria and Oman), says nothing about membership [opec-pr-2026-05-03, release and site navigation].

(iv) The statutory rule: under Article 8A of OPEC’s Statute a withdrawal notice “shall take effect at the beginning of the next calendar year after the date of its receipt by the Conference, subject to the Member having at that time fulfilled all financial obligations arising out of its membership” [opec-statute-2021, 2021 edition, Art. 8A, PDF p. 12], which on its face would put the legal effect at 1 January 2027, not 1 May 2026, unless the Conference agreed otherwise.

No OPEC statement on the date or legal effect of the UAE’s withdrawal was found, so whether the UAE is a member in law on 16 October 2026 is open; for market data, EIA already counts OPEC without it. EIA also notes the conflict in Iran from 28 February 2026 and the effective closure of the Strait of Hormuz, so Gliner’s chokepoint flows above are historical.

**Practice.** The crack spread is a margin before operating costs; refiners’ actual yields and product slates differ from 3:2:1. Political risk in producers and chokepoint disruptions move Brent first, in Gliner’s account.

**Pitfalls.** *Terminology.* Gliner writes that “kerosene is used for diesel” (p. 301); kerosene is jet fuel’s base and diesel is a separate distillate. Gliner calls Saudi Aramco “private” and lists it among the “biggest IOCs” (pp. 303–304); it is a state-owned national oil company. Gliner describes Brent as “deliverable to Sullom Voe” (p. 306). ICE now describes its Brent Crude futures as “a deliverable contract based on EFP delivery with an option to cash settle against the ICE Brent Index price for the last trading day”, 1,000 barrels, trading to the last business day of the second month preceding the contract month [ice-brent-spec, contract specifications]: delivery is by exchange for physical, not to a terminal, and there is a cash-settlement option.

*NYMEX WTI, in part.* NYMEX’s certification to the CFTC of 27 January 2023 (Submission 23-057) reprints Rules 200101 and 200104 of the Light Sweet Crude Oil futures (CL) for contract months from April 2023: deliverable domestic crude is a pipeline’s common Domestic Sweet stream of WTI-type light sweet crude with, among other limits, sulphur at most 0.42 percent by weight, API gravity from 37 to 42 degrees and Reid vapour pressure below 9.5 psi (the test method moving to ASTM D-6377), delivered free on board at any pipeline or storage facility in Cushing, Oklahoma with access to Enterprise, Enbridge or Plains storage, each “including any successor in such capacity” [nymex-23-057, Exhibits A–B].

*Current chapter* (official PDF as of October 2026, undated). Chapter 200 now names ASTM D-6377 for vapour pressure and keeps the 23-057 grade and Cushing terms; the trading unit is 1,000 U.S. barrels (42,000 gallons), with a delivery tolerance of 2 percent either way except for book-outs and in-tank or in-line transfers; the minimum fluctuation is $0.01 a barrel ($10 a contract); and trading ends on the third business day before the 25th calendar day of the month preceding delivery, or, if the 25th is not a business day, the third business day before the last business day preceding the 25th [nymex-ch200-2026, Rules 200101 and 200102.B–F]. Gliner says only that WTI is the higher grade; that it historically traded at a premium for that reason is an interpretation.

**See also.** natural gas; contango and backwardation; commodity indices.

**Further reading.** Benchmarks, price reporting and the negative WTI settlement of April 2020.

**Definition 19.2 Metals markets for macro: gold and real rates, silver, platinum, copper and aluminium, with their main producers, uses and contracts as of 2013**

**Statement.** *Gold* (XAU) [s7-gliner, chap. 10, sec. “Precious Metals”, PDF pp. 319–321]: jewellery “more than 40 percent of demand” and investment the second driver; central banks held 31,920 tonnes in October 2013, the United States 8,134 tonnes, 73 percent of its reserves (Table 10.6); “China has overtaken” South Africa “as the leading producer”, and “China, the United States, Australia, and South Africa collectively account for more than 40 percent of global gold production” (p. 320); India, the largest consumer, used over 500 tonnes for jewellery and 300 for investment in 2011, “of which 50 percent is used for weddings”; gold is preferred to fiat money when real interest rates are negative (Barsky and Summers’s Gibson’s paradox); GLD trades at a tenth of an ounce and COMEX gold (GC) is 100 troy ounces.

*Copper* (sec. “Industrial Metals”, pp. 325–327): “Dr. Copper”, a leading indicator; production led by Chile and Peru; China the second producer and largest importer, about 40 percent of imports; uses equipment manufacturing 52 percent, building construction 33 and infrastructure 15; LME copper in dollars a tonne, 25 tonnes a contract, and COMEX copper (HG) in cents a pound, 25,000 pounds. *Aluminium* (pp. 327–328): the most abundant metal in the crust; China over 40 percent of both production and consumption and self-sufficient; energy-intensive to produce; LME aluminium 25 tonnes.

**Notation.** Troy ounce 31.1 grams; LME 3-month rolling forward quoted in dollars per tonne.

**Conditions.** Figures are Gliner’s for 2011–2013, from the World Gold Council, GFMS, ICSG, USGS and the World Copper Bulletin as he cites them; production shares are compared with USGS in Pitfalls. *Demand shares now*. Gold: the World Gold Council’s full-year 2025 figures give total demand including OTC of 5,002 tonnes, investment 2,175 tonnes and central banks 863, and jewellery at a five-year low of 1,542 tonnes (India 430.5, mainland China 360.1) [wgc-gdt-fy2025, press release and jewellery section, 29 January 2026].

Jewellery is thus about 31 percent of demand, not Gliner’s “more than 40 percent”, and investment, at about 43 percent, is now the largest use. Silver: the *World Silver Survey 2026* gives 2025 total demand of 1.13 billion ounces and industrial demand of 657.4 million, about 58 percent, so most silver demand is still industrial; mine supply was 846.6 million ounces [silver-institute-wss-2026, press release of 15 April 2026].

*Platinum now*. Metals Focus’s figures for the World Platinum Investment Council put 2025 total demand at 8,671 koz: automotive (autocatalyst) 3,039, jewellery 2,214, industrial 2,266 and net investment 1,152, and refined mine production at 5,557 koz, of which South Africa 3,957, Russia 677, Zimbabwe 515 and North America 212 [wpic-pq-2026q2, Table 1, p. 5]. Those are shares of 35.0, 25.5, 26.1 and 13.3 percent of demand and 71.2 percent of refined mine supply from South Africa; Gliner’s catalytic converters at 38 percent and jewellery at 31 percent are 2013 figures on another basis. The same table forecasts a 265 koz surplus for 2026 after deficits in 2023–2025, a forecast, not data.

*Copper and aluminium now*. The ICSG’s *World Copper Factbook 2025*, on preliminary 2024 figures, gives mine output of 23 million tonnes (Chile 24 percent), smelter output of 23.3 million (China 51 percent), refined output of 27.5 million (China 45 percent) and refined usage, by semis plants and other first users, of 27.4 million, of which China 58 percent [icsg-factbook-2025, p. 5]: Gliner’s “roughly 40 percent of imports” for China is not comparable with a usage share.

By end use the Factbook’s 2024 chart, sourced to the International Wrought Copper Council, has building construction 26 percent, infrastructure (power and telecom) 17, transport 13, industrial equipment 12, consumer and general products, cooling and electronics 23, and others 9 (p. 48); the ICA’s 2023 dataset, for 2022 and the copper content of semis by end use (world total 30,582.6 thousand tonnes), groups its markets as Gliner does, into building construction 26.2 percent, infrastructure 17.3 and equipment manufacture 56.5 (industrial 11.5, transport 12.9 and other equipment 32.0) [ica-semis-enduse-2023, World sheet, rows 48–51], against Gliner’s 33, 15 and 52; the Factbook’s 2024 chart, grouped the same way, gives 26, 17 and 57.

Gliner’s year and basis are not stated, so the gap is a difference between his figures and the 2022 and 2024 data, not a measured trend. For aluminium, the IAI and Mission Possible Partnership’s April 2023 transition strategy says “Approximately 50% of global aluminium demand today comes from the transport and construction sectors (26% and 24%, respectively)”, with electrical and machinery equipment 11 percent each, foil stock and packaging 8 percent each and consumer durables 6 (94 percent named); it credits the passage on p. 15 to a 2020 World Economic Forum and Accenture report and the same figures on p. 26 to the IAI Material Flow Model [iai-mpp-2023, PDF pp. 15 and 26, endnotes 3 and 16], so they describe a model baseline of about 2020, not a dated statistical release, and give no country shares.

*Contracts.* COMEX: copper (HG) 25,000 pounds of Grade 1 electrolytic cathode (ASTM B115), quoted in multiples of five hundredths of a cent a pound ($12.50 a contract); silver (SI) 5,000 troy ounces of at least 999 fineness, $0.005 an ounce outright ($25) and $0.001 for spreads; gold (GC) 100 troy ounces of at least 995 fineness, as one 100-ounce bar or three kilo bars, $0.10 an ounce ($10); aluminium 25 metric tons of P1020A, $0.25 a ton ($6.25) [comex-metals-chapters-2026, Rules 107102, 111101–111102, 112101–112102 and 113101–113102].

LME: Copper Grade A is refined electrolytic copper to BS EN 1978:2022 Cu-CATH-1, GB/T 467-2010 or ASTM B115-10 (2021) Grade 1, in full-plate cathodes of LME-listed brands, and High Grade Primary Aluminium is P1020A (revised January 2018) or GB/T 1196-2018 Al99.70, both in lots of 25 tonnes, 2 percent more or less [lme-rulebook-2026-10, Part 6, pp. 6-1 and 6-11]; minimum price fluctuations are “at such levels as the Exchange may from time to time prescribe” (Rule 11.3.7), so they are not in the rulebook.

*Aluminium production by country*. USGS’s *Mineral Commodity Summaries* (February 2026) estimate world smelter production at 74.0 million tonnes in 2025 (72.8 in 2024), of which China 45.0 (44.0), about 61 percent in both years, India 4.2, Russia 3.9, Canada 3.3 and the United Arab Emirates 2.7; world year-end capacity 79.7 million tonnes [usgs-mcs-2026-aluminum, Aluminum, p. 2]. Gliner’s “over 40 percent” of production is therefore well below the current share; USGS gives no consumption by country, so his consumption share is unverified. Gliner’s contract sizes match these texts.

**Practice.** Gold’s negative relation to real rates is an empirical tendency with long breakdowns, not a fixed correlation.

**Pitfalls.** Gliner writes that platinum “is almost twice has [sic] heavy as gold” with densities of 21 and 19 (p. 323); the ratio is about 1.11 (checks, section 14). He writes that Chile represents “more than 40 percent of consumption” of copper (p. 325), which cannot be right for Chile, a producer and exporter; it probably means China, which he calls the largest importer in the same paragraph, but the text does not say so; Chile at 27 percent of production and China at 43 percent of consumption were not matched to his table. “India largest consumer (50% used for weddings)” compresses his statement about India’s 2011 demand. “China overtook South Africa as largest producer” is stated by Gliner (p. 320).

*Production shares against the data*. *Gold*: USGS’s *Mineral Commodity Summaries 2013* gives 2011 and 2012 (estimated) mine output of 362 and 370 tonnes for China, 234 and 230 for the United States, 258 and 250 for Australia and 181 and 170 for South Africa, against world totals of 2,660 and 2,700 tonnes: 38.9 and 37.8 percent for the four [usgs-mcs-2013, gold, p. 2], a little under Gliner’s “more than 40 percent”, whose source for that sentence is not stated (his Table 10.7 cites the World Gold Council, GFMS and the LBMA).

*Silver*: Gliner’s Table 10.9 (World Silver Survey 2013 and GFMS) is internally inconsistent. Its ounces (Mexico 162.2, China 117.0 million) sum, with Other 120.0, to its stated total of 787.0 million, on which Mexico and China are $279.2/787.0 = 35.5$ percent and the top five 62.6 percent; but its printed shares (27.3 and 19.7 percent, and 120.0 as 20.2 percent) imply a denominator near 594 and add to 132.5 percent; his “47 percent” and “83 percent” for the top five follow the printed shares. USGS’s 2013 summary gives 2012 estimates of 4,250 tonnes for Mexico and 3,800 for China against a world total of 24,000, 33.5 percent [usgs-mcs-2013, silver, p. 2], close to the 35.5 percent the table’s own ounces imply.

Gliner’s figures are quoted as he prints them, with the discrepancy noted. *Current shares* (USGS, *Mineral Commodity Summaries 2025*, 2024 estimates): gold, China 380 of 3,300 tonnes, and China, Russia, Australia, Canada and the United States together 41 percent; silver, Mexico 6,300 and China 3,300 of 25,000 tonnes (38 percent); copper mine output led by Chile (5,300 of 23,000 thousand tonnes) and refined output by China (12,000 of 27,000) [usgs-mcs-2025, gold, silver and copper, pp. 1–2]; these are production shares, not consumption shares.

**See also.** contango and backwardation; real yields; commodity indices.

**Further reading.** Metal inventories, exchange delivery, loco settlement and leases.

### Definition 19.3 Agricultural markets for macro: corn, wheat, soybeans, cotton and coffee, crop calendars and sensitive stages, CBOT and ICE contracts, ethanol demand, and weather with the livestock cascade

**Statement.** Gliner: agricultural supply depends on weather and planting, and “some of the best agriculture traders have full-time meteorologists on staff”; crop calendars show the sensitive periods [s7-gliner, chap. 10, sec. “Agriculture”, PDF p. 330]. *Corn* (pp. 330–335): the United States produces over 35 percent and consumes over 30 percent of the world’s corn and is the largest exporter; Iowa, Illinois and Nebraska produce 47 percent of U.S. corn; uses are feed, ethanol (27 percent of U.S. corn demand in 2011, after the Energy Policy Act of 2005) and food; Chinese feed

imports rose from nothing in 2007 to over 5 percent of world imports; silking from June is the critical period; CBOT corn (C) is quoted in cents a bushel, 5,000 bushels a contract, 56 pounds a bushel, for March, May, July, September and December.

*Cotton* (pp. 341–344): character, grade and staple; China, India and the United States 65 percent of production in 2013, China the largest producer and consumer, the United States the largest exporter with almost 30 percent; ICE cotton (CT) in cents a pound, 50,000 pounds, roughly 104 bales of 480 pounds. *Coffee* (pp. 344–348): arabica about 60 percent of production, robusta the rest; Brazil the largest producer (37.2 percent in 2012–13) and Vietnam second (16.6 percent), in bags of 60 kilograms (Table 10.21); the United States and the European Union almost half of demand; ICE arabica (KC). *Weather and livestock* (p. 333): when grain prices rise sharply, feeding livestock becomes expensive and farmers slaughter early, so near-term livestock prices fall while later prices rise.

**Notation.** Futures month codes F (Jan), H (Mar), K (May), N (Jul), Q (Aug), U (Sep), V (Oct), X (Nov), Z (Dec); one bushel of corn 56 pounds, of wheat and soybeans 60.

**Conditions.** Shares, contract terms and delivery months are Gliner's for 2012–2013 (USDA tables as he reproduces them); CBOT grain terms are compared with the December 2024 filing below. A corn contract at 500 cents is $25,000 notional (checks, section 14).

**Practice.** Translate a view on a crop's balance sheet (ending stocks) into a specific contract month, grade and delivery location; the basis between local cash and futures then matters.

**Pitfalls.** *Beyond these sources*: El Niño and La Niña, NOAA drought monitors, relatively inelastic demand, Russian wheat export bans, coffee's biennial cycle and 20-year tree life are plausible influences that these sources do not cover. Two against ICE: Coffee "C" (KC) is 37,500 pounds, quoted in cents a pound, for March, May, July, September and December, physically delivered; Cotton No. 2 (CT) is 50,000 pounds net weight, for March, May, July, October and December, matching H, K, N, V, Z [ice-softs-specs, contract specifications].

*CBOT grain terms*. Each contract is 5,000 bushels with a minimum fluctuation of a quarter cent a bushel ($12.50 a contract). Corn: No. 2 yellow at par, No. 1 at 1½ cents over and No. 3 at 2 to 4 cents under, in September, December, March, May and July. Soybeans: No. 2 yellow at par, No. 1 at 6 cents over and No. 3 at 6 cents under, in September, November, January, March, May, July and August. Wheat: No. 2 Soft Red Winter, Hard Red Winter, Dark Northern Spring and Northern Spring at par and No. 1 at 3 cents over, in July, September, December, March and May.

All three stop trading on the business day before the 15th calendar day of the delivery month, and daily price limits are reset twice a year at 7 percent of a 45-day average settlement, rounded to 5 cents with a 20-cent floor for corn, expanding by 1.5 times after a limit settlement [cbot-24-443, Exhibit, Rules 10101–10104, 11101–11102 and 14101–14102]. The filing itself raised load-out premiums on shipping certificates from 6 to 9 cents a bushel and the St. Louis–Alton corn and soybean differential from 16.25 to 24 cents over par, effective after the 2027 expiries.

The current Chapters 10, 11 and 14 (official PDFs, undated) agree with the reprint on these terms, and carry the location differentials in two schedules, for corn up to December 2027 and from March 2028, and for soybeans up to November 2027 and from January 2028 [cbot-grain-chapters-2026, Rules 10101–10102, 10105, 11101–11102, 11105 and 14101–14102]; no change after the January 2025 effective date was found in the terms above. Gliner's months for corn and soybeans match; his cotton contract size "50,00 0 lbs" is a typo for 50,000.

**See also.** contango and ending stocks; futures hedges and basis; commodity indices.

**Further reading.** WASDE [physical-usda-wasde-2026].

# 20 Commodity risk premia and investing: factors, allocation and index products

## 20.1 Commodity indices

**Definition 20.1 Commodity indices and their record: the CRB, S&P GSCI and Bloomberg indices, commodity ETFs and their energy weight, and the 1970–2006 risk, return and correlation figures Gliner reproduces**

**Statement.** The Commodity Research Bureau (CRB) Index "is composed of 22 commodities and dates back to the 1950s"; the Deutsche Bank ETFs DBA (agriculture: corn, soybeans, sugar, cattle, coffee, cocoa and more) and DBC (energy, precious metals, agriculture and industrial metals) give basket exposure, the caution with DBC being that "more than 50 percent of the index is energy weighted" [s7-gliner, chap. 10, sec. "CRB Index", PDF p. 299]. "Commodities have a higher volatility than foreign exchange, equities, and fixed income", with "an attractive Sharpe ratio as a relative comparison" and, before 2008, negative correlation with equities; Table 10.1, from Goldman Sachs's *Commodities as an Asset Class* (July 2007), gives annualised monthly continuously compounded returns, volatilities and correlations of equities, bonds, cash and the S&P GSCI for 1970–2006, with an equity–GSCI correlation of −0.04 (sec. "Return and Volatility", p. 300).

**Notation.** A production-weighted index (S&P GSCI) weights commodities by world production value; the Bloomberg Commodity Index (BCOM, formerly DJ-UBS) uses liquidity and production with caps.

*BCOM*: target weights combine two-thirds liquidity and one-third production percentages, using five-year averages, then apply annual diversification limits (no commodity above 15 percent, no commodity with its derivatives above 25 percent, no group such as energy above 33 percent, and at least 2 percent per commodity "as liquidity allows"); the 2026 target weights are energy 29.44, grains 21.15, precious metals 18.84, industrial metals 15.76, softs 9.17 and livestock 5.64 percent, across 23 commodities; BCOM itself is an excess-return index, BCOMTR adds the return on Treasury-bill collateral (the 13-week bill auction high rate) and BCOMSP is a spot version; the roll shifts 20 percent a day over the sixth to tenth business days of the month; values before the launch on 14 July 1998 are "historical estimations" [bcom-methodology-2026, secs. 1.2 with Table 1, 2.2, 2.6, 3.1–3.2 and app. B, pp. 4–6, 11, 23–25, 31–33 and 40].

*S&P GSCI*: weights are contract production weights built from five-year world production averages, subject to total-dollar-value trading and reference-percentage-dollar-weight thresholds (at least 1.00 percent to enter, 0.10 percent to stay); the excess-return index (S&P GSCI ER) "is … calculated based on the contract daily return" and the total-return index "reflects the performance of a total return investment in commodities—contract daily return plus the daily interest on the funds hypothetically committed to the investment"; the roll runs from the fifth to the ninth business day of the month at 20 percent a day; the headline index has base date 31 December 1969 (value 100) and launch date 11 April 1991 [spdji-gsci-methodology-2026, PDF pp. 7, 9, 13 and 18, printed pp. 6, 8, 12 and 17].

**Conditions.** Figures to 2006; ETF details as of 2013. The 1970–2006 statistics are one sample from a bank's 2007 note. *Live and back-tested history*. S&P's education note states: "The S&P GSCI was launched April 11, 1991. … All data prior to such date is back-tested hypothetical data" [spdji-why-gsci-2024, PDF p. 6, notes to the exhibits], and the methodology's disclosure says that information before a launch date "is hypothetical back-tested performance, not actual performance, and is based on the index methodology in effect on the launch date" [spdji-gsci-methodology-2026, PDF p. 39]. The investable index "arguably began in 1991" and later indices were launched "with backfilled history" [dunsby-nelson-2010, p. 36].

So Gliner's Table 10.1 (1970–2006) combines back-tested values from January 1970 to 10 April 1991 with live values from 11 April 1991; the boundary falls inside calendar 1991, which is neither wholly back-tested nor wholly live, and the monthly return for April 1991 straddles it. *Variants.* A *spot* index tracks futures prices only; an *excess-return* index adds the roll from rolling futures; a *total-return* index adds interest on fully collateralising the futures (S&P's ER and TR definitions above; the education note separates the "S&P GSCI" and the "S&P GSCI Excess Return (ER)" by the roll, PDF p. 9).

Gliner's table labels the returns "Annualized Monthly Returns (Continuously Compounded)" and does not say which GSCI variant Goldman Sachs used, so its 1970–2006 figures should not be read as either total-return or spot returns. Whether correlations with risk assets rose significantly after 2008 is the financialisation question and is untested here.

**Practice.** Read an index's rulebook for weights, roll schedule and rebalancing before using it as a benchmark; ETF returns differ from spot prices by the roll and fees (contango and backwardation).

**Pitfalls.** Gliner's caution about energy exposure above 50 percent concerns DBC; the GSCI's energy weight has declined since 2008 (education note, Exhibit 8, PDF p. 8), while BCOM's 33 percent group cap holds its 2026 energy target at 29.44 percent. Whether commodities are particularly useful as inflation hedges is an empirical question, to be measured by horizon and regime.

**See also.** contango and backwardation; regime map; crude oil.

# 21 Gas and power markets: hubs, storage, networks and market design

## 21.1 Natural gas

**Definition 21.1 Natural gas for macro traders: producers, shale, flaring, LNG and the Henry Hub futures contract as Gliner describes them in 2013**

**Statement.** Gliner calls natural gas "the cleanest and safest fossil fuel", producing "30 to 50 percent less carbon dioxide than gasoline or coal"; it is used in power generation, heating, transport and to make ammonia for fertiliser [s7-gliner, chap. 10, sec. "Natural Gas", PDF pp. 313–316]. Shale gas, produced with horizontal drilling and hydraulic fracturing, is expected to supply 50 percent of U.S. gas by 2020 (p. 314). Flaring burns unwanted gas, abundant in the United States where it is produced alongside oil even when there is no demand (p. 315). Cooled to −260°F, gas becomes LNG, taking "600 times less space" (p. 315). The United States and Russia account for 38 percent of world production (Table 10.5, p. 316).

Gas is priced in dollars per million British thermal units (MMBtu), roughly one thousand cubic feet (p. 317); the NYMEX contract (NG) is for delivery at Henry Hub, Louisiana; the front of the curve is "predominantly a speculators' market" and the back a producers' hedging market; the UNG ETF tracks the contract, with contango and rolls a caution (pp. 317–318).

**Notation.** MMBtu million British thermal units; Mcf thousand cubic feet; LNG liquefied natural gas.

**Conditions.** Figures are Gliner's for 2012–2013. Emissions comparisons depend on whether methane leakage is counted. Gliner gives the contract as "10,000 million British thermal units" (10,000 MMBtu, p. 317), as of 2013, which the current chapter confirms (below). *Delivery, as amended in 2016.* NYMEX Rule 220104 of the Henry Hub Natural Gas futures (NG), as certified to the CFTC on 11 July 2016 with effect from 26 July 2016, provides that "Delivery shall be made free-on-board ('F.O.B.') at the buyer's interconnection point at the Henry Hub", is complete when gas passes that point "via physical flow or displacement", and recognises the Intra Hub Transfer service as an acceptable interconnection point; the filing describes the hub as operated by Sabine Pipe Line [nymex-16-080, Submission 16-080, pp. 1–3 and Appendix A].

*Current chapter* (official PDF as of October 2026; no amendment date in the text, and its page footer reads "© Copyright 2009"). The trading unit is 10,000 MMBtu with a delivery tolerance of 2 percent either way; the minimum fluctuation is $0.001 per MMBtu ($10 a contract), with $0.00025 for inter-commodity spreads on Globex; trading ends on the third business day before the first day of the delivery month; and Rule 220104 reads as certified in 2016, with Sabine Pipe Line LLC as the hub operator [nymex-ch220-2026, Rules 220101–220104]. Whether Chapter 220 was amended after 2016 in respects not covered here cannot be told from the undated text.

Filings (not an exhaustive list): NYMEX Submission 17-048 of 23 February 2017 certified "administrative, non-substantive amendments" to ten natural gas contracts, Chapter 220 among them, effective for trade date 13 March 2017, which "will, in no manner, impact the structure of the Contracts" [nymex-17-048, pp. 1–2]; Submission 18-465 of 15 November 2018 amended Rule 589's special price-fluctuation limits for NG and seven associated contracts [nymex-18-465, cover sheet], a Chapter 5 rule, not the Chapter 220 text. Two 2019 filings are relevant. Submission 19-158 expanded trading at settlement (TAS) for Henry Hub futures from "the spot (except on the last trading day) and second through seventh contract months" to the spot and "second through twelve contract months", for trade date 20 May 2019 [nymex-19-158, pp. 1–2].

Submission 19-198 standardised the daily settlement procedure documents of six Henry Hub contracts, Chapter 220 included, from trade date 4 September 2019 [nymex-19-198, pp. 1–2]. Both are Chapter 5 or procedure changes, not amendments to the Chapter 220 text. *After 2019.* The CFTC's listing of NYMEX product terms-and-conditions filings from September 2019 to October 2026 holds 281 filings. Submission 20-166 of 20 March 2020 raised the special price fluctuation limits of Rule 589 for NG and five associated contracts "from 7% to 15% in light of recent market volatility", effective for trade date 6 April 2020 [nymex-20-166, pp. 1–2]. That is again a Chapter 5 change. The other natural-gas filings in the period concern options, Platts and TTF contracts, or other look-alikes.

None amends the Chapter 220 text after 2019. The listing covers product terms only, not organisation-rule filings, and the current chapter is undated, so an unlisted change cannot be excluded.

**Practice.** Henry Hub is one hub; regional hubs and LNG-linked prices can diverge from it. *U.S. LNG exports since Gliner*: EIA's annual series of liquefied U.S. natural gas exports, in million cubic feet, runs 2,924 in 2013 and 16,255 in 2014, 186,841 in 2016, 1,819,547 in 2019 and 5,508,984 in 2025, about 15.1 billion cubic feet a day [eia-lng-exports-2026, series N9133US2], by the release of 30 September 2026. So the market Gliner described in 2013, with LNG essentially absent from U.S. exports, has become one in which LNG exports link Henry Hub to world gas prices (Interpretation; the link itself is not measured here).

**Pitfalls.** Attributing U.S. flaring to regulatory requirements is misleading; Gliner attributes it to gas produced where there is no demand or infrastructure. That shale turned the U.S. from importer into exporter goes beyond Gliner's 2013 account.

**See also.** crude oil markets; contango and backwardation.

**Further reading.** Hub formation and gas–oil price links.

# 22 Crypto settlement, venues and derivatives

## 22.1 Crypto venues, perpetuals and trading risk

**Definition 22.1 Crypto venue and instrument structure: fragmented executable prices, reference prices and best execution, and spot, dated futures, perpetuals (linear, inverse, quanto), options and pools**

**Statement.** Crypto assets trade on many independently owned venues. Makarov and Schoar, using tick data from 34 exchanges in 19 countries, find that individual exchanges mostly run centralised order books like equity markets, but that, unlike regulated equity markets, the crypto market "lacks any provisions to ensure that investors receive the best price", contrasting the U.S. NBBO requirement [s6-makarov, discussion-paper version of December 2018, pp. 1–2 and footnote 1]. Kept apart here:

- *executable prices*: the bids and offers on each venue, in its own quote currency, custody and fee terms;
- *reference prices*: indices a venue or benchmark administrator computes from several venues, against which derivatives are marked or settled (Deribit marks positions with a Mark Price and computes funding against a Deribit Index, perpetual funding);
- *fair-value models*: a trader's own estimate of value, which need not equal either;
- *legal best-execution obligations*: duties under a jurisdiction's rules (best execution), which Makarov and Schoar found absent across crypto venues as a market-wide price-protection rule (p. 1).

*Instruments. Spot* trades exchange one asset for another; on Deribit spot trades are brokerage services settled and delivered immediately [gapclose-deribit-rules, Rulebook v2.0, Rules 1.2(a) and 11.3(a)]. *Dated futures* expire, and convergence to spot at expiry limits the basis [roadmap-perpetual-pricing, arXiv v2, p. 2]. *Perpetual futures* have no expiry and are anchored by periodic funding payments (perpetual funding). Ackerer, Hugonnier and Jermann distinguish three contract types (pp. 2–3):

- *linear*: contract size in units of the base asset (1 BTC), margined and settled in the quote currency (USD, or in practice a stablecoin such as USDT, p. 7);
- *inverse*: the base asset itself (BTC) is the margin and settlement asset, and the contract size is in the quote currency (for example 10,000 USD);
- *quanto*: a third currency (for example ETH) is used for margining and settlement.

*Options*: Deribit lists European-style cash-settled options and linear options; its European options are priced in BTC or ETH and cannot be exercised before expiry (Rulebook, Appendix A, 2(d)). *Listing snapshot, 17 October 2026* (Deribit public API, one venue on one day; open interest as the API reports it, in units of the underlying, the contract size being one coin for both BTC listings; units not confirmed from the API documentation). BTC and ETH options are listed as inverse ("reversed", premium and settlement in the coin): 938 BTC instruments over 12 expiries, a median of 42 strikes per expiry. USDC-settled linear options are listed on BTC, ETH, SOL, XRP, AVAX, HYPE and TRX. Inverse BTC options carry 390,301 of open interest against 2,781 for USDC-settled BTC options (0.7%).

Open interest is concentrated: the top 10% of inverse BTC instruments hold 67% of it, 202 of 938 hold none, and two expiries (30 October and 25 December 2026) hold 72% [deribit-api-2026, `get_instruments` and `get_book_summary_by_currency`, 17 October 2026]. *On-chain pools* (automated market makers) are outside this entry's scope.

**Conditions.** The fragmentation evidence is historical. Over December 2017 to February 2018 the daily average price ratio between the United States and Korea exceeded 15 percent and reached 40 percent on several days; Japan–U.S. differences were about 10 percent and U.S.–Europe about 3 percent, while differences within a country typically did not exceed one percent on average (pp. 1–2). The authors estimate at least $2 billion of potential arbitrage profit over that period from executed trades (p. 2, footnote 2). Spreads co-moved across countries and opened during rapid appreciation; between cryptocurrencies on the same exchanges spreads were an order of magnitude smaller, and pairwise capital-control tightness was positively related to the correlation of spreads, which the authors read as evidence that capital controls contribute (pp. 2–3). These are 2017–2018 findings; current spreads were not measured.

**Practice.** Interpretation: a crypto price is a price on a venue, in a quote currency, with that venue's custody, collateral and fees. Quote-currency differences (USD, USDT, USDC, KRW) and settlement currency (linear against inverse) change the units of P&L and margin. Cross-references: fragmentation without a consolidated tape in spot FX is in spot FX market structure, routing across venues in smart order routing, limits to arbitrage in arbitrage risks.

*A second venue on the same day*. On OKX's public API at about 11:52 UTC, OKX lists 1,466 BTC options: 786 inverse (settled in BTC) over 14 expiries and 680 linear USD-settled ("_UM") over 10, each with a 0.01 BTC multiplier. Of $2.55 billion of open interest, 99.2 percent is in the inverse contracts; the three largest expiries hold 64.7 percent (25 December and 30 October 2026 and 26 March 2027) and the ten largest instruments 21.1 percent [okx-api-2026, `public/instruments` and `public/open-interest`, BTC-USD options, 17 October 2026]. So the Deribit pattern, inverse contracts dominant and open interest concentrated in the quarterly and year-end expiries, is repeated at a second venue on the same day; two venues on one day are not a time series.

**Pitfalls.** Crypto has no universal consolidated tape, but that does not make fair value venue-dependent: venue prices differ, fair value does not. The sources support the absence of a market-wide best-price rule in 2018 (Makarov and Schoar) and say nothing on consolidated tapes. Funding keeps perpetual futures near spot only loosely (perpetual funding). Option exposure is concentrated in a few strikes and expiries on the one venue sampled above; concentration across venues is not shown. Deribit's terms are one venue's.

**See also.** perpetual funding; crypto trading risk layers; best execution; spot FX market structure.

**Mechanism 22.2 Perpetual funding: premium and interest terms, sign convention, accrual and settlement, damper and caps (Deribit as one venue), and why convergence is not guaranteed**

**Statement.** A perpetual future has no expiry, so it cannot rely on convergence at maturity; instead, longs and shorts exchange periodic *funding* payments. Ackerer, Hugonnier and Jermann describe the payment as generally the sum of a *premium* part $\kappa_t(f_t - x_t)$, paid by longs when the perpetual price $f_t$ is above spot $x_t$, and an *interest* part $\iota_t x_t$ set by the exchange to reflect the interest-rate differential between the two currencies; for a linear contract the long's cash flow over one period is

$$(f_{t+1} - f_t) - \kappa_t(f_t - x_t) - \iota_t x_t$$

in the quote currency [roadmap-perpetual-pricing, arXiv v2, sec. 3, pp. 7–8, eqs. (4)–(6)]. Funding is commonly computed every 8 hours (p. 2). In their frictionless no-arbitrage model the exchange can choose the interest factor so that the perpetual price equals spot for any sufficiently large premium rate (p. 4; Corollary 1, p. 12). That is a model result. He, Manela, Ross and von Wachter state that perpetuals "are not guaranteed to converge to the spot price", derive bounds with trading costs, and find empirical deviations larger than in traditional currency markets, co-moving across currencies and diminishing over time [roadmap-perpetual-fundamentals, arXiv v7 of 17 September 2026, abstract].

**Conditions.** *One venue's rule: Deribit, Funding Specifications page edited 27 July 2026* [gapclose-deribit-rules, Funding Specifications]:

- *sign*: when the funding rate is positive, longs pay shorts; when negative, shorts pay longs; Deribit charges no fee on funding, so payments net to zero between holders;
- *rate*: premium rate $P = (\text{Mark Price} - \text{Deribit Index})/\text{Deribit Index}$; a damper sets funding to zero for $|P| \leq 0.025\%$ and otherwise moves $P$ towards zero by 0.025%; a cap then bounds the rate, ±0.5% for BTC, ±1.0% for ETH and ±5.0% for the USDC- and USDT-margined perpetuals:

$$\text{rate} = \min\big(c,\ \max\big(-c,\ \max(0.025\%, P) + \min(-0.025\%, P)\big)\big);$$

- *units and cadence*: the rate is expressed per 8 hours but accrues continuously; the payment is rate × position size × (period / 8 hours), in the settlement currency of the contract: BTC in Deribit's BTC example, where a USD 100,000 position is 1 BTC, an inverse contract in Ackerer, Hugonnier and Jermann's sense (Interpretation), and USDC for a USDC-margined linear one;
- *settlement*: accrued funding shows at once in account equity (realised session P&L) and can be used for trading, but moves to the cash balance only at the daily settlement at 08:00 UTC, after which it can be withdrawn (also Rulebook v2.0, Rule 11.4).

**Practice.** Worked example (Deribit's): BTC perpetual mark USD 100,075 against index USD 100,000 gives $P = 0.075\%$ and a rate of 0.05% per 8 hours; a long of USD 100,000 (1 BTC) pays $0.05\% \times 1\ \text{BTC} \times 1/480 = 0.000001041667$ BTC per minute and 0.0005 BTC over 8 hours. A USDC-margined SOL perpetual at mark 100.10 against index 100 gives 0.075%, so a USDC 100,000 long pays 0.15625 USDC per minute and 75 USDC over 8 hours. The printed max/min formula agrees with the prose rule on a grid of premia and in the other examples (damper, caps, offsetting minutes). Interpretation: funding is a carry cost or income of holding the perpetual against spot, in the contract's settlement currency.

*What the funding gap measures*. He, Manela, Ross and von Wachter derive no-arbitrage perpetual prices in frictionless markets and bounds with trading costs; empirically "The mean absolute deviation is about 52% to 90% per year across different cryptocurrencies", larger than covered-interest-parity deviations, comoving across coins and shrinking by about 22 percentage points a year; trading the deviation earns a Bitcoin Sharpe ratio of 3.35 under retail-level costs (3.27 after effective spreads) (sec. 1, printed pp. 3–4). They explain the gap by end users who "demand perpetual futures for leveraged exposure" against arbitrageurs with limited balance-sheet capacity, and find deviations compressed after Binance introduced portfolio margining in April 2022 (printed pp. 4–5) [roadmap-perpetual-fundamentals, arXiv v7, sec. 1, printed pp. 3–5]. On that account the funding gap measures leverage demand meeting scarce arbitrage capital; the paper tests capacity, not positioning concentration.

**Pitfalls.** Funding is the periodic payment that ties perpetual futures to spot, with longs paying when it is positive; the sign convention matches Deribit's. The tie to spot holds as a design intent and a model result, not a guarantee. Funding is a carry term; that it is also a crowding signal is not supported. The dated-futures evidence in crypto carry ties the basis to small traders' demand for leverage and to limited arbitrage capital, which supports a leverage-demand reading.

Venue rules differ: other venues may set cadence, interest term, clamps and index construction differently, and Deribit's own page contains slips in its ETH examples (an index "at USD 10,000" and "a 1 BTC long position" in the ETH column of Example 3; 7.5% inside the ETH cap formula of Example 5), whose stated results are nevertheless correct.

**See also.** crypto venue and instrument structure; crypto trading risk layers; futures margin and settlement; lending fees and rebates.

**Definition 22.3 Crypto trading risk layers: liquidation, insurance fund, socialised loss and auto-deleveraging at one venue, kept apart from custody, stablecoin and network-settlement risk**

**Statement.** The losses a crypto trader can bear beyond market risk come in layers, governed by different documents and failing differently:

- *liquidation*: forced closing of positions when margin falls short. Under Deribit's rulebook, if a member fails to meet maintenance margin, Deribit may close positions, cancel orders, delta-hedge with futures and perpetuals, apply auto-deleveraging where applicable, or close the whole account, together the Liquidation Process, and may charge a liquidation fee [gapclose-deribit-rules, Rulebook v2.0, Rules 10.1 and 10.5];
- *insurance fund*: if an account's margin balance in a currency is negative after liquidation, Deribit transfers assets from its insurance fund, funded mainly by liquidation fees and excess collateral from liquidations, to bring it to zero; if the fund in that currency is depleted, Deribit may, in its absolute discretion, convert another currency of the fund (Rules 10.6–10.7; definition of Insurance Fund);
- *socialised loss*: Deribit may, in its absolute discretion, trigger the Socialised Loss Mechanism if the insurance fund is depleted in all currencies, or if it is depleted in the bankrupt account's currency and Deribit decides not to convert other currencies (Rule 10.8), so the mechanism can be triggered while other currencies remain in the fund, and need not be triggered when the fund is exhausted. Once it is, Deribit may deduct from all members with positive profit and loss in the same day's session, in any currency, their pro-rata share of the negative balance, halt trading, cancel orders and close positions at the mark price; members agree to this and may dispute only a manifest error in their share (Rules 10.9–10.11);
- *auto-deleveraging*: a step in the liquidation waterfall in which, once liquidity-support providers' capacity is used, eligible positions of a distressed account are transferred to a deleveraged counterparty at a preferential price and closed against that member's existing positions (definition of Auto-Deleveraging; the full waterfall is in the Exchange Membership Terms, clause 12);
- *custody*: Deribit holds members' assets in its own wallets or through a third-party custodian of its choice, or acts as limited agent over a member-chosen custodian for margining, liquidation and settlement (Rule 2, guidance);
- *stablecoin, network and bridge risk*: redemption and depeg of stablecoin collateral, chain congestion, reorganisations and transfer delays. *One depeg documented*. In March 2023 Silicon Valley Bank entered receivership on Friday 10 March at 11:37 a.m. ET. Circle announced around 10 p.m. that it could not withdraw $3.3 billion of USDC reserves held there, about 8 percent of the total. Redemptions had already surged before the announcement. Primary issuance and redemption, "constrained by working hours of the U.S. banking systems", largely stopped over the weekend, and USDC fell to 86 cents on secondary markets. Dai's one-to-one Peg Stability Modules against USDC were drained, and stablecoins with no SVB exposure also lost their pegs. The depegs eased only after the FDIC, Treasury and Federal Reserve announced full protection of SVB depositors [feds-2025-svb-stablecoins, FEDS Notes, 17 December 2025, summary and sec. 1].
  So a fully backed stablecoin used as collateral can lose its peg when its primary market is closed and its reserves are frozen, and smart-contract links can pass the loss to other coins.

**Conditions.** All rules above are Deribit FZE's, a virtual-asset service provider regulated by Dubai's VARA, under Rulebook version 2.0 applicable 13 August 2026 (Rule 1.1); other venues' waterfalls may differ. Venue insolvency is a separate legal process from a venue's socialised-loss mechanism.

**Practice.** Interpretation: keep apart a trader's own liquidation (its positions closed), a loss to the venue's backstop (insurance fund), a loss charged to other members (socialised loss), a position closed against a counterparty's will (auto-deleveraging) and a loss of assets held at the venue (custody, insolvency). Cross-venue strategies add transfer and settlement paths between venues, during which capital cannot respond. Margin and default waterfalls in a regulated futures clearing house are in futures margin and settlement.

*Settlement finality and reorganisations*. For a systemically important stablecoin arrangement, CPMI and IOSCO apply Principle 8: clearly define "the point at which a transfer of a stablecoin through the operational method used becomes irrevocable and unconditional", give it a clear legal basis, and prevent "any misalignment between the state of the ledger and legal finality". They note that with probabilistic settlement finality "can never be achieved with absolute certainty" and forks can revoke transactions, the risk growing with the number of transactions added (secs. 2 and 3.4, PDF pp. 5 and 16) [cd-cpmi-iosco-stablecoin-pfmi-2022, July 2022, secs. 2 and 3.4]. Under Principle 9 a stablecoin used for money settlements "should have little or no credit or liquidity risk", judged by the holder's legal claim, the reserves and timely conversion at par. This frames chain reorganisations and transfer delay as settlement-finality questions.

**Pitfalls.** Venue risk (outage, withdrawal halt, auto-deleveraging or insolvency), liquidation risk, stablecoin risk, bridge and settlement risk, and fragmentation are distinct layers. Auto-deleveraging and liquidation are described for Deribit only; fragmentation is a market-structure fact (crypto venue and instrument structure), linked by Makarov and Schoar to capital controls rather than to slow crypto transfers.

**See also.** perpetual funding; crypto venue and instrument structure; futures margin and settlement.

#### Evidence 22.4 Crypto carry on dated futures: size, drivers (small traders' demand for leverage) and limits to arbitrage (spot access, margin and cross-margining), 2019–2024

**Statement.** Schmeling, Schrimpf and Todorov measure the annualised basis of one- and three-month constant-maturity bitcoin and ether futures over spot, using data from Binance, OKEx, FTX, Huobi, BitMEX, Deribit and the CME (the CME from August 2020) [roadmap-carry, revision of 1 October 2025, abstract, sec. 1 and data section, PDF pp. 3–9].

- *Size.* Carry "can reach exceptionally high levels, sometimes exceeding 40% per annum". It averaged about 7 percent a year across exchanges from April 2019 to July 2024 (pp. 3 and 5).
- *Not fundamentals.* In $F_t = S_t e^{(r_t+u_t-y_t)(T-t)}$, interest rates bear "little to no relation" to the carry and storage costs are negligible. That implies a large negative convenience yield: investors prefer being long futures to holding spot (p. 5).
- *Demand.* Higher carry goes with larger net long positions of smaller traders in the CFTC's Commitments of Traders data, who add positions in strong trends and periods of attention. Dealers and leveraged funds take the short side. Introducing CME micro bitcoin futures raised CME carry relative to other venues in a difference-in-differences test (pp. 6–7).
- *Limits to arbitrage.* Regulation long kept many professionals from holding spot, and margin frictions bind. With no cross-margining on traditional exchanges, a long spot position cannot collateralise a short futures position. The futures leg suffers severe drawdowns, and carry predicts liquidations of short futures. *Scale, from Table 7*: in OKEx data, a monthly regression of next-month sell liquidations, "scaled by open interest", on the standardised one-month basis, with implied volatility, realised skew and open interest as controls, has a basis coefficient of 2.22 (standard error 0.79) and an intercept of 10.84 (1,619 observations, adjusted $R^2$ 0.29; Table 7, col. 2, printed p. 41, PDF p. 44).
  Since the text expresses liquidations "as a percentage of open interest" (PDF p. 25), a one-standard-deviation rise in carry goes with sell liquidations about 2.2 percentage points of open interest higher, against a level of about 10.8 percent when the regressors are at their means; buy liquidations are not predicted (0.41, s.e. 0.45). In the subsample built to exclude profit-taking (basis widening on at least 70 percent of days in the two weeks before expiry), the coefficient is 2.02 on an intercept of 9.33 (211 observations).
  The prose states the effect in two ways: "an increase in the standardized carry by 10% predicts an increase in liquidations of short futures positions by 22% of total open interest" (PDF p. 7) and "a rise in standardized carry by 10% predicts a 22% increase in total sell liquidations (relative to total open interest)" (PDF p. 25). Neither maps cleanly onto the table: a 22 percent rise relative to the intercept is close to $2.22/10.84 \approx 20$ percent, while "22% of total open interest" is ten times the coefficient, and what "10%" of a standardised variable means is not stated. The earlier "liquidations of 22 percent of open interest" in this entry is withdrawn.
  The liquidation measure "capture[s] both forced liquidations by the exchange due to missed margin calls and voluntary liquidations by the trader" (PDF p. 25), so it is contract closing, not forced liquidation alone. After spot bitcoin ETFs were introduced in 2024, carry fell by about 3 percentage points across exchanges and 5 more on the CME, 36 and 97 percent of the mean (p. 8).

**Conditions.** The evidence is for fixed-date futures, which converge to spot at expiry; perpetual futures are discussed only through other papers (p. 8). The causal claims rest on two difference-in-differences designs around specific events. The exchange sample includes FTX, which failed in November 2022.

**Practice.** A cash-and-carry trade that "appears risk-free" in the absence of frictions (p. 5) carries funding and liquidation risk before convergence. Size it against the margin path and the venue's liquidation rules, not against the annualised basis alone (crypto trading risk layers). For perpetuals, the paper restates other work: He et al. find that perpetuals "typically trade at a positive funding rate, that is, the long side pays the short side", and Christin et al. that a carry trade short perpetual futures earns high Sharpe ratios (p. 8).

**Pitfalls.** Table 5 (printed p. 39, PDF p. 42) regresses the one-month basis on treatment and post dummies in two-month windows: around the spot-ETF launch, $Post = -0.03$ and $Treat \times Post = -0.05$ (both *s.e.* 0.01; intercept 0.13; 291 observations), and around CME's micro futures $Treat \times Post = +0.11$ (intercept 0.28; 292 observations). The basis is in decimals of an annual rate, so the coefficients are the text's "around 3%" and "5%" in annualised percentage points of basis, matching the Size bullet's statement.

**See also.** perpetual funding; cost-of-carry forward price (the pricing equation); convenience yield in commodities (opposite sign).

**Further reading.** roadmap-perpetual-fundamentals for funding rates; s11-r15.

#### Definition 22.5 Automated market maker pools: the constant-product rule and its fee, price impact and the cost of providing liquidity, and concentrated liquidity in Uniswap v3

**Statement.** A Uniswap pair pools reserves $x$ and $y$ of two tokens. Uniswap v1 and v2 maintain "the invariant that the product of the reserves cannot decrease"; "traders pay a 30-basis-point fee on trades, which goes to liquidity providers" [uniswap-v2-2020, sec. 1, p. 1]. Version 2 enforces the fee through $(x_1 - 0.003\,x_{in})(y_1 - 0.003\,y_{in}) \geq x_0 y_0$, with $x_0$, $y_0$ the balances before and $x_1$, $y_1$ after the swap (sec. 3.2.1, eq. (10), p. 7). Uniswap v3 keeps $x \cdot y = k$ but lets a liquidity provider confine liquidity $L = \sqrt{k}$ to a price range $[p_a, p_b]$, a *position* [s6-uniswap, sec. 2, eqs. (2.1)–(2.2), p. 2]. Its real reserves satisfy

$$\left(x + \frac{L}{\sqrt{p_b}}\right)\left(y + L\sqrt{p_a}\right) = L^2,$$

so it trades like a constant-product pool with larger "virtual reserves" inside the range. Outside the range it holds one asset only, "is no longer active, and no longer earns fees". Fee tiers are set per pool, initially 0.05, 0.30 and 1 percent (sec. 1, p. 1).

**Notation.** Derived, with price $p = y/x$ and fee $f$, $\gamma = 1 - f$: selling $\Delta x$ returns $\Delta y = y\gamma\Delta x/(x + \gamma\Delta x)$. A position's real reserves at price $p$ inside its range are $x = L(1/\sqrt{p} - 1/\sqrt{p_b})$ and $y = L(\sqrt{p} - \sqrt{p_a})$.

**Conditions.** The formulas are the protocols' design as described in their 2020 and 2021 whitepapers. Deployed pools, fee switches, later versions (UniswapX, v4) and other AMM designs differ. Gas costs, transaction ordering and the maximal extractable value they enable are outside these papers.

**Practice.** Numerical checks:

- *Price impact.* Selling 10 ETH into a 1,000 ETH and 2,000,000 USDC pool returns 19,743.16 USDC, an execution price of 1,974.32 against a pool price of 2,000 before and 1,960.65 after. Before the fee the shortfall is $\gamma\Delta x/(x + \gamma\Delta x)$, 0.99 percent for a trade of 1 percent of the reserve. The product $k$ rises by the fee retained.
- *Cost of providing liquidity.* With arbitrageurs moving the pool to an external price $p_0 r$, a full-range position is worth $2\sqrt{r}/(1 + r)$ times the buy-and-hold value before fees: $-5.7$ percent for $r = 2$ or $1/2$, and $-20$ percent for $r = 4$. That is the "impermanent loss" that fee income must cover.
- *Concentration.* The same $L$ on $[1600, 2500]$ at a price of 2,000 needs 94,427 of capital against 894,427 on the full range, 9.5 times less. Below $p_a$ the position is all X (50 units); above $p_b$, all Y (100,000).

*Fees, block times, gas and pool design*.

- *Fees and blocks* (roadmap-amm-fees, arXiv v2 of 23 July 2025). With a fee $\gamma$ and Poisson block arrivals at rate $\lambda$, the probability that a block contains a profitable arbitrage is $P_{\text{trade}} = 1/(1 + \sqrt{2\lambda}\,\gamma/\sigma)$; for small fees and fast blocks, arbitrage profit is approximately loss-versus-rebalancing times $P_{\text{trade}}$ and fees paid by arbitrageurs approximately LVR times $(1 - P_{\text{trade}})$, so "faster blockchains will result in reduced LP losses"; with fixed gas costs added, "lower gas fees lead to smaller losses for LPs" (abstract and sec. 1, pp. 1–4) [roadmap-amm-fees, arXiv v2, abstract and sec. 1, pp. 1–4]. That is the gas-cost question in theory.
- *Reported prices* (s6-cfmm, arXiv v4). A CFMM's reported price is the slope of a supporting hyperplane of its reachable set; an arbitrageur facing reference prices $c$ is "always incentivized" to move the pool there, so the pool tracks the reference market. A fee of $1 - \gamma$ on inputs makes a path-independent CFMM path-deficient: a trader must put in $1/\gamma$ more input (sec. 3, pp. 13–16 and 20–21) [s6-cfmm, arXiv v4, sec. 3, pp. 13–21]. With a fee, a constant-product pool's price is pushed only into a band around the reference price (own reading of the path-deficiency result, not a quoted bound). Angeris, Kao, Chiang, Noyes and Chitra show that under common conditions constant-product markets "must closely track the reference market price" and simulate stability under a wide range of conditions [s11-r12, arXiv v7, abstract].
- *Many assets* (roadmap-cfmm-convex, arXiv v1, July 2021): choosing a multi-asset trade against a CFMM can be posed as a convex optimisation problem and solved reliably (abstract) [roadmap-cfmm-convex, arXiv v1, abstract and sec. 1].
- *Uniswap v4* (exp-uniswap4, August 2024): pools keep v3's concentrated liquidity, but a pool's creator can attach a "hook contract" run at set points of a call, which can "manage the swap fee of the pool dynamically, implement custom curves, and adjust fees"; all pools sit in one singleton contract with "flash accounting", which nets token balances within a lock to cut gas on multi-pool trades (abstract and sec. 1, pp. 1–2) [exp-uniswap4, whitepaper, August 2024, pp. 1–2].
- *Auctioning the pool* (exp-amamm, arXiv v4 of 12 February 2025): an on-chain auction for the right to be "pool manager" of a constant-product pool, who sets the fee, receives fees and can capture some arbitrage; they "prove that under certain assumptions" this pool has higher equilibrium liquidity than any fixed-fee AMM (abstract) [exp-amamm, arXiv v4, abstract]. A design proposal with proofs under assumptions, not evidence of deployment.

**Pitfalls.** The whitepapers are the developers' design descriptions, not audits of deployed code. The impermanent-loss expression assumes the pool is arbitraged to an outside price and ignores fees and gas; both are own derivations, not statements of the sources.

**See also.** crypto venues and instruments; crypto trading risk layers; crypto carry.

**Further reading.** roadmap-uniswapx; roadmap-cfmm-routing-efficient.

### Mechanism 22.6 DeFi lending liquidations: fixed-spread and auction designs, close factors and health factors, and how much becomes liquidatable when prices fall

**Statement.** Qin, Zhou, Gamito, Jovanovic and Gervais study Aave, Compound, MakerDAO and dYdX, over 85 percent of Ethereum lending, from their launch to April 2021. Two designs dominate: auctions (MakerDAO) and fixed-spread liquidation (Aave, Compound, dYdX), where a liquidator repays debt and claims collateral worth the repaid debt times $(1 + LS)$, with LS the liquidation spread, up to a *close factor*, the maximum share of the debt repayable in one liquidation (sec. 2.3, eq. (1), PDF p. 3).

Over the period, liquidation proceeds were $807.46 million in 28,138 events by 2,011 liquidator addresses; 641 auction liquidations were unprofitable for liquidators; Aave V2 had accumulated $87.4 thousand of bad debt by the end of April 2021; and a 43 percent fall in ETH, like 13 March 2020, would make $1.07 billion of collateral liquidatable on MakerDAO. Fixed-spread designs "favor liquidators over borrowers" because they allow more collateral than necessary to be sold (sec. 1, PDF p. 2) [roadmap-defi-liquidations, arXiv v2, secs. 1–2, PDF pp. 1–3]. Perez, Werner, Xu and Livshits find on Compound to September 2020 that "variations of only 3% in an asset's dollar price can result in over 10m USD becoming liquidable", with over 70 percent of liquidatable positions liquidated immediately (abstract) [oc-perez-knife-edge-2021, arXiv v6, abstract].

In Aave V3, the V2 rule that only half a position could be liquidated at once was relaxed: a position "approaching insolvency (HF $< 0.95$)" can be liquidated in full; on layer-2 chains a grace period applies after a sequencer outage for $0.95 < HF < 1$ (PDF pp. 7–8 and 14) [al-aave, Aave V3 Technical Paper, 27 January 2022, PDF pp. 7–8 and 14].

**Notation.** HF the health factor, below one liquidatable; LS liquidation spread (bonus); CF close factor; LT liquidation threshold.

**Conditions.** The figures are for 2019–2021 protocol versions and parameters, which governance changes; the HF definition in terms of collateral, liquidation thresholds and debt is the protocols' (not transcribed here).

**Practice.** To assess a liquidation, record the protocol version and parameters at the pinned block, compute the maximum repay amount from the close factor and the HF band, and net the bonus of gas and DEX slippage before calling a liquidation profitable.

**Pitfalls.** The Aave paper is the developers' description; parameters such as the e-mode example (97 percent LTV, 98 percent threshold, 2 percent bonus for stablecoins, PDF p. 6) are examples, not current values.

**See also.** AMM pools; crypto trading risk layers; liquidity spirals.

**Further reading.** oc-heimbach-huang-2024; oc-lehar-parlour-2022; oc-eskandari-oracles-2021.

### Evidence 22.7 Stablecoin runs and flights to safety, and stablecoin flows in Treasury bill yields

**Statement.** Anadu et al. document net flows "from riskier to safer stablecoins on days of crypto-market stress" and estimate a break-the-buck threshold at $1: a coin priced at or above $1 has flows uncorrelated with its price, while below $1 its daily outflow rate "jumps by 34 basis points"; the 2022 (Terra) and 2023 runs show the same flight to safety, and flows tend to stay within a blockchain, as money-fund flows stay within fund families (abstract and sec. 1, PDF pp. 2 and 4–5) [cd-anadu-2023, Staff Report 1073, September 2023, revised April 2024, abstract and sec. 1].

Ahmed and Aldasoro estimate, with instrumented local projections on daily data from 2021 to 2025, that "a 2-standard deviation inflow into stablecoins lowers 3-month Treasury yields by 2–2.5 basis points within 10 days", with little effect at longer tenors, and that outflows raise yields two to three times as much as inflows lower them (abstract) [rt-ahmed-aldasoro-stablecoins, BIS WP 1270, May 2025 version, abstract].

**Conditions.** Safe and risky are classified from issuers' reported reserves, whose reporting quality varies (Anadu et al., n. 6). The yield estimate is for 2021–2025 and depends on the instrument.

**Practice.** A depeg can be dated as the first close below $1, the threshold at which these authors find redemptions accelerate. *Discrepancies noted*: the guide's reading note for this paper says "the USD 0.99 break-the-buck threshold"; the paper estimates $1. The paper's introduction calls the money-fund threshold "95-cent" while its later section defines breaking the buck as a market-price NAV below $0.995 (PDF p. 5 against the money-fund background section); the latter is the usual definition.

**Pitfalls.** Figures are from the BIS paper's May 2025 version; the catalogue records a later revision with data to March 2026.

**See also.** money-fund runs; international stablecoin standards; crypto trading risk layers.

**Further reading.** wide-lyons-stablecoin; wide-liu-makarov-schoar-terra.

### Rule 22.8 International standards for stablecoins: the FSB's redemption, stabilisation and reserve recommendations (July 2023)

**Statement.** The FSB's high-level recommendations for "global stablecoin" arrangements require that authorities ensure "a robust legal claim to all users against the issuer and/or underlying reserve assets" and "timely redemption"; "For GSCs referenced to a single fiat currency, redemption should be at par into fiat"; and an effective stabilisation mechanism, clear redemption rights and prudential requirements (Recommendation 9, PDF pp. 13–14) [cd-fsb-stablecoin-2023, Final Report, 17 July 2023, Recommendation 9, printed pp. 9–10]. Because Recommendation 9 requires

an effective stabilisation mechanism, "so-called algorithmic stablecoins do not meet the High-Level Recommendations" (n. 2, printed p. 1). CPMI and IOSCO apply the PFMI's settlement-finality and money-settlement principles to systemically important arrangements (crypto trading risk layers).

**Conditions.** These are international recommendations to authorities, not binding law; jurisdictions implement them in their own regimes (for example the EU's MiCA).

**Practice.** For a design check, ask in order: who holds the legal claim and on what, at par and how fast; what reserves back it and how they are managed; and what makes a transfer final on the chain used.

**Pitfalls.** A *global* stablecoin is a category defined by potential reach across jurisdictions; the recommendations apply in full to that category.

**See also.** stablecoin runs.

**Further reading.** `cd-cpmi-iosco-stablecoin-pfmi-2022`; `cd-davernas-maurin-vandeweyer-2026`.

#### Mechanism 22.9 Auto-deleveraging on perpetual venues: last-resort loss socialisation, its trilemma, and the 10 October 2025 cascade

**Statement.** Chitra describes auto-deleveraging (ADL) as "a last-resort loss socialization mechanism for perpetual futures venues", triggered "when solvency-preserving liquidations fail", after insurance funds are exhausted, and proves a trilemma: "no policy can simultaneously satisfy exchange solvency, revenue, and fairness to traders". On Hyperliquid on 10 October 2025, ADL closed $2.1 billion of positions in 12 minutes; against transparent benchmark allocations the production algorithm "overshot the minimum trader profit haircut", costing profitable traders an estimated $45.0 to $51.7 million, about $653.6 million of positions, and the comparison suggests Binance "overutilized ADL far more" (abstract and sec. 1) [`cd-chitra-adl-2025`, arXiv v3, abstract and sec. 1].

**Conditions.** The excess-haircut estimates depend on the author's benchmark allocations and on reconstructed venue data; the author is at Gauntlet, a risk-management firm for crypto protocols.

**Practice.** To analyse a liquidation cascade, write each venue's waterfall (liquidation, insurance fund, ADL or socialised loss) from its rules, and record which positions ADL closed and at what price; compare with crypto trading risk layers, which describes one venue's rules.

**Pitfalls.** The paper's numbers are one event; whether Hyperliquid's later open-interest loss is due to ADL is the author's conjecture (sec. 1).

**See also.** perpetual funding; liquidity spirals.

**Further reading.** `cd-soska-2021`; `exp-brunnermeier-pedersen-liquidity`.

#### Model 22.10 Market and funding liquidity spirals: when margins are destabilising (Brunnermeier and Pedersen)

**Statement.** Speculators who provide liquidity are financed against margins. Brunnermeier and Pedersen call margins *destabilising* "if they can increase in illiquidity". When the financier knows the fundamental value and that prices will return to it, margins are stabilising: illiquidity lowers the margin on the side that buys low or sells high, a "cushioning" effect (Proposition 1). When the financier is uninformed and fundamentals have ARCH-type volatility, margins rise with observed price moves, $m = \bar{\sigma} + \bar{\theta}|\Delta p|$, including moves caused by liquidity shocks (Proposition 2).

With destabilising margins or large existing positions, liquidity can be fragile, with multiple equilibria and "a discontinuous drop of market liquidity" after small losses. A *margin spiral* (tighter margins force selling) and a *loss spiral* (losses on existing positions force selling) reinforce each other (sec. 1, printed pp. 2–4; Propositions 1–3, printed pp. 19–21) [`exp-brunnermeier-pedersen-liquidity`, NBER WP 12939, February 2007, sec. 1 and Propositions 1–3].

**Notation.** $m$ margin; $\bar{\sigma}$, $\bar{\theta}$ parameters of the volatility process; $\Delta p$ the observed price change; $\Lambda$ the gap between price and fundamental.

**Conditions.** Model of 2007–2009. Destabilising margins need financiers who cannot tell fundamental from liquidity shocks; the model is stylised (three periods), and the empirical motivation is margins on S&P 500 futures in 1987, 1990 and 1998.

**Practice.** A crypto liquidation cascade fits the margin-spiral reading when venues raise maintenance requirements or mark against falling prices and forced sales move the mark; whether it does is an empirical question.

**Pitfalls.** Propositions follow the 2007 working paper, which may differ from the *Review of Financial Studies* version in common-investor contagion.

**See also.** auto-deleveraging; common-investor contagion; DeFi liquidations.

**Further reading.** `exp-brunnermeier-pedersen-liquidity` (published version).

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